sixth edition

POLICY ANALYSIS

CONCEPTS AND PRACTICE

DAVID L. WEIMER
AND AIDAN R. VINING

eRESOURCES
ROUTLEDGE

Policy Analysis: Concepts and Practice is not only an excellent introduction to the field and the practice—a textbook that provides the conceptual foundations and craft skills enabling its readers to produce policy analyses competently and quickly—but also the most influential and successful attempt to organize the vast, diverse, and interdisciplinary domain of policy analysis around a core analytical vision, and the powerful theoretical apparatus associated to it.

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- A chapter dedicated to distinguishing between policy analysis, policy research, stake-holder analysis, and research about the policy process.
- An extensively updated chapter on policy problems as market and governmental failure that explores the popularity of Uber and its consequences.
- The presentation of a property rights perspective in the chapter on government supply to help show the goal tensions that arise from mixed ownership.
- An entirely new chapter on performing analysis from the perspective of a public agency and a particular program within the agency's portfolio: public agency strategic analysis (PASA).
- A substantially rewritten chapter on cost–benefit analysis, to better prepare students to become producers and consumers of the types of cost–benefit analyses they will encounter in regulatory analysis and social policy careers.
- A new introductory case with a debriefing that provides advice to help students immediately begin work on their own projects.

Policy Analysis: Concepts and Practices remains a comprehensive, serious, and rich introduction to policy analysis for students in public policy, public administration, and business programs.

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Policy Analysis
Concepts and Practice

Sixth Edition

David L. Weimer and Aidan R. Vining

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To Rhiannon, Rory, and Eirian

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Preface

When we began our study of policy analysis at the Graduate School of Public Policy (now the Goldman School), University of California at Berkeley, the field was so new that we seemed always to be explaining to people just what it was that we were studying. It is no wonder, then, that there were no textbooks to provide us with the basics of policy analysis. More than a dozen years later, we found ourselves teaching courses on policy analysis but still without what we considered to be a fully adequate text for an introductory course at the graduate level. Our experiences as students, practitioners, and teachers convinced us that an introductory text should have at least three major features. First, it should provide a strong conceptual foundation of the rationales for, and the limitations to, public policy. Second, it should give practical advice about how to *do* policy analysis. Third, it should demonstrate the application of advanced analytical techniques rather than discuss them abstractly. We wrote this text to have these features.

We organize the text into five parts. In Part I we begin with an example of a policy analysis and then emphasize that policy analysis, as a professional activity, is client oriented, and we raise the ethical issues that flow from this orientation. In Part II we provide a comprehensive treatment of rationales for public policy (market failures, broadly defined) and we set out the limitations to effective public policy (government failures). In Part III we set out the conceptual foundations for solving public policy problems, including a catalogue of generic policy solutions that can provide starting points for crafting specific policy alternatives. We also offer advice on designing policies that will have good prospects for adoption and successful implementation, and how to think about the choice between government production and contracting out. In Part IV we give practical advice about doing policy analysis: structuring problems and solutions, gathering information, and measuring costs and benefits. Furthermore, there is an entirely new chapter on performing analysis from the perspective of a public agency and a particular program within the agency's portfolio: public agency strategic analysis (PASA). Part V briefly concludes with advice about "doing well and doing good."

We aim our level of presentation at those who have had, or are concurrently taking, an introductory course in economics. Nevertheless, students without a background in economics should find all of our general arguments and most of our technical points accessible. With a bit of assistance from an instructor, they should be able to understand the remaining technical points.

We believe that this text has several potential uses. We envision its primary use as the basis of a one-semester introduction to policy analysis for students in graduate programs in public policy, public administration, and business. (Thorough treatment of all topics covered, including cost–benefit analysis, would probably require two semesters.) We believe that our emphasis on conceptual foundations also makes it attractive for courses in graduate programs in political science and economics. At the undergraduate level, we think our chapters on market failures, government failures, generic policies, and cost–benefit analysis are useful supplements to, and perhaps even replacements for, the commonly used public finance texts that do not treat these topics as comprehensively.

New to This Edition

Faculty and students will find that a great many substantive changes have been made throughout the text since the fifth edition was published. Among the more notable changes are:

- A new introductory case on the shortage of kidneys for transplantation (Chapter 1);
- A sketch of policy analysis steps that serves as an introductory overview of the craft of policy analysis (Chapter 1);
- A postscript added to the Madison taxicab case describing the impact of Uber on the market and its regulation (Chapter 9);
- More explicit treatment of important roles in the implementation process (Chapter 12);
- Introduction of a property rights framework for assessing hybrid organizations (Chapter 13);
- An updated British Columbia salmon fishery case (Chapter 16);
- A substantially revised and more accessible cost–benefit analysis chapter (Chapter 17);
- A new chapter that sketches public agency strategic analysis (Chapter 18).

Acknowledgments

A reviewer of the first edition of this text told us that we had expounded what he takes to be the "Graduate School of Public Policy approach to public policy." His comment surprised us: we had not consciously attributed our peculiar views of the world to any particular source. But in retrospect, his comment made us realize how much our graduate teachers contributed to what we have written. Although they may wish to disavow any responsibility for our product, we nevertheless acknowledge a debt to our teachers, especially Eugene Bardach, Robert Biller, Lee Friedman, the late C. B. McGuire, Arnold Meltsner, William Niskanen, Philip Selznick, and the late Aaron Wildavsky.

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Part I
Introduction to Public Policy Analysis

1 Preview

After earning your policy degree last May, you immediately went to work on Mary Smith's campaign for the U.S. Senate. She won!

Your enthusiastic efforts during the campaign earned you a position on her transition team, which the senator-elect created to identify policies for addressing problems she views as particularly important. Senator-elect Smith hopes to be appointed to the Senate Committee on Health, Education, Labor & Pensions because of her interest in health policy. One specific problem she hopes to address is the long and growing waiting list for kidney transplants. She asks the transition team to consider policies that might reduce the length of the transplant waiting list by facilitating, even encouraging, more people to make living donations of kidneys.

Neither you nor your colleague tasked with preparing the policy analysis addressing the kidney shortage have any background in health issues. Fortunately, you had a course on policy analysis as a student that gave you experience in gathering relevant information quickly from already-available research and organizing it effectively to provide useful advice. You work very hard on the report during the one week you have available, not only because you hope it will earn you a position on the senator-elect's legislative staff in Washington, but also because once you start learning about the topic you find it interesting and you realize that your advice might actually contribute to better public policy. Your report follows.

Box 1.1 Reducing the U.S. Kidney Transplant Shortage by Increasing the Number of Live-Donor Kidneys

To: Senator-elect Mary Smith
From: Transition staff
Date: November 30, 2016
Concerning: Reducing the U.S. Kidney Transplant Shortage by Increasing the Number of Live-Donor Kidneys

Executive Summary

Kidney transplants generally improve both the quality of life and longevity of patients with end-stage renal disease (ESRD). By substituting for the costly alternative treatment, dialysis, they also reduce Medicare expenditures over the lifetime of patients. However, a shortage of kidneys has resulted in a large and growing waiting list for kidney transplants. Potential increases in the number of deceased-donor kidneys are relatively small because the pool of feasible donors is small. Potential increases in the number of live-donor kidneys are large because the pool of feasible donors is very large. Financial incentives, ranging from greater compensation for non-medical expenses to removal of the current ban on sale and purchase, can increase the availability of live-donor kidneys. The transition team recommends that you introduce legislation to establish expanded reimbursement with relative listing, a policy that would increase federal reimbursement for non-medical expenses incurred in making a live donation to the transplant waiting list, allow nonprofit organizations to reimburse non-medical expenses incurred by any living donor, and create a priority on the transplant list for living donors and one relative of their choice.

The National Organ Transplant Act of 1984 currently bans commercial transactions in transplant organs. An unregulated market for live kidneys would eliminate the shortage of transplant kidneys. However, it would have inherent inefficiencies as well as raise ethical concerns about exploitation of the poor, undue advantage for the wealthy, and commodification of the human body. In light of these disadvantages, this analysis considers three other alternatives to current policy that would make live donation either more financially attractive or less financially prohibitive: (1) brokered auctions would impose an intermediary between sellers (vendors) and purchasers (patients); (2) expanded reimbursement with relative listing would offset more of the non-medical costs for living donors and give priority to deceased-donor kidneys to living donors and their designated relatives, if they become patients; and (3) purchase with demand-side chain auctions would allow purchase of vendor kidneys by a chartered entity at an administered price and allocation of kidneys to the transplant centers that can use them to initiate the largest number of transplants.

The transition team assesses these alternatives in terms of five policy goals: (1) protection of human dignity; (2) equity; (3) efficiency; (4) favorable fiscal impact; and (5) political feasibility. Although current policy protects the human dignity of U.S. residents, provides equity in access to transplantation, and is politically feasible, it inefficiently forgoes potential increases in the supply of kidneys and favorable fiscal impacts offered by the other alternatives.

Brokered auctions would substantially increase the availability of kidneys by thousands or even tens of thousands annually and generate very large savings for the federal government. However, these gains in efficiency and favorable fiscal impact come with an increased threat to human dignity through exploitation of the poor, greater inequity in access to transplant organs in terms of wealth, and very poor prospects for adoption.

Expanded reimbursement with relative listing would increase the availability of kidneys by hundreds or perhaps thousands annually and produce savings for the federal government. It would have small impacts on human dignity and equity, and would likely be politically feasible.

Purchase with demand-side auctions would likely increase the availability of kidneys by the tens of thousands, offering the potential for eventually effectively eliminating the kidney transplant waiting list. It would also have favorable fiscal impacts for the federal government and not increase inequity in allocation. However, it does risk exploitation of the disadvantaged; moreover, as a radical departure from current policy, it would have poor prospects for adoption.

Considering these trade-offs, expanded reimbursement with relative listing offers the best prospects for a desirable and feasible improvement relative to current policy. The transition team recommends that you ask the Congressional Research Service to draft legislation for this alternative.

Policy Problem: Too Few Kidneys for Transplant

End-stage renal disease (ESRD) can currently be treated with either dialysis or transplantation of a replacement kidney. Most ESRD patients prefer a transplant because it generally provides a higher quality of life and greater longevity than dialysis. However, transplantation currently requires available human kidneys, which come from either deceased or living donors. In the United States, the demand for kidney transplants currently far exceeds the supply of donated kidneys. The waiting list for kidney transplants has over 100,000 patients.[1] In 2015, available kidneys allowed only 17,821 transplants, with 69 percent coming from deceased donors and 31 percent from live donors. That year, 35,037 new patients entered the waiting list and 4,295 left it because they died. Recent increases in rates of obesity suggest that the demand for kidney transplants will increase faster than population growth in the future, because obesity contributes to diabetes and hypertension which are risk factors for ESRD. Increased prevalence of obesity also reduces the pool of potential donors of healthy kidneys.

The shortage of kidneys has implications for federal expenditures as well as the lives of ESRD patients. Medicare covers much of the cost of dialysis and transplantation not covered by private insurers. Its expenditures pay over 80 percent of the costs of treating ESRD––$30.9 billion in 2013, or about 7.1 percent of total Medicare paid claims.[2] Over the expected lives of patients, a transplant involves lower cost to Medicare than does dialysis. A recent study estimates that each kidney transplant saves taxpayers on average about $146,000.[3]

Increasing the supply of deceased-donor kidneys could reduce, although not eliminate, the shortage. Organ procurement organizations (OPOs) receive authorizations for donations, either post-death from family members or pre-death from donors, for approximately 73 percent of eligible deaths (brain death in a hospital by someone 70 years old or younger and without exclusionary medical conditions).[4] If authorizations were received for all the eligible deaths, then approximately another 4,500 kidney transplants could be done each year. Currently, about 15 percent of donations come from those suffering cardiac rather than brain death, although some transplant centers have increased their rate of use to approximately 30 percent.[5] If all centers reached this level of use, then the total availability of kidneys could increase by almost 6,000. Although an increase of this magnitude would be extremely valuable if it actually could be obtained, it would be too small to slow the growth of the kidney transplant waiting list.

Changes in the allocation rules for deceased-donor kidneys implemented in December 2014 by the Organ Procurement and Transplantation Network (OPTN) include provisions that may affect future kidney availability.[6] A new classification system for the quality of donated kidneys is likely to reduce refusals by patients to accept kidneys from older or sicker donors. Better matching of expected kidney graft life with expected patient life is likely to reduce the need for second transplants that would otherwise result when patients outlive their grafts. Nonetheless, the new rules are likely to have only a modest impact on the length of the waiting list.

Kidneys donated by living people could substantially reduce or even eliminate the kidney waiting list. Humans are born with two kidneys but only need one kidney to live a healthy life under normal circumstances. Live donation involves a small surgical risk: a 0.03 percent chance of death and less than 1 percent chance of a major complication.[7] Although some questions about long-term impacts have been raised by a recent Norwegian study,[8] long-term follow-up of living donors in the United States finds neither increased risk of renal disease nor reduced longevity.[9] Although living kidney donation poses very small medical risk, it often involves travel and subsistence costs as well as potentially the loss of wages by donors and their caregivers. A recent study found that, although some donors received financial assistance from government or private sources, 89 percent suffered a net financial loss: average costs for donors included direct expenses of $1,157 and lost wages of $1,660, with their caregivers incurring an additional $377 in lost wages.[10]

The number of living donors peaked at 6,647 in 2004.[11] Less than one-third of transplanted kidneys now come from living donors, a decline since the peak. Most of these donors are relatives or close friends of the ESRD patient. Often people who are willing to donate to a specific person cannot do so because of biological incompatibilities that would result in the patient's immune system rejecting the donated kidney. Although evidence is accumulating that patients can often be desensitized to increase the range of kidneys feasible for transplantation, outcomes appear less favorable than for compatible kidneys.[12] A more common approach to donor incompatibility is the arrangement of *paired donations*, which may enable incompatible donors to facilitate transplants for their intended patients by an exchange of donations across pairs: the donor in the first incompatible pair gives to the patient in the second incompatible pair, and the donor in the second incompatible pair donates to the patient in the first incompatible pair. Exchanges can also involve three or more pairs, though they are more difficult to arrange.

A *transplantation chain* begins with an altruistically motivated nondirected kidney donation––approximately 180 per year or about 3 percent of all kidney donations in the last five years were nondirected.[13] The kidney is transplanted to the patient in an incompatible donor–patient pair. The donor in that pair donates a kidney to the patient in another incompatible pair. The process continues until no patient in an incompatible pair can be found to receive the donation. Instead, the donor in the last pair receiving a kidney donates a kidney to the compatible person on the waiting list with highest priority. By seeding a chain, a single nondirected donation can facilitate multiple transplants that would not otherwise occur. For example, a chain in 2015 involved 34 donors and patients.[14]

The National Organ Transplant Act of 1984 (PL 98–507) established the legal framework for the regulation of solid organ transplantation in the United States. Section 274c(a) strictly bans the sale of organs: "It shall be unlawful for any person to knowingly acquire, receive, or otherwise transfer any human organ for valuable consideration for use in human transplantation if the transfer affects interstate commerce." The provision has been interpreted to allow insurers, transplant recipients, states, or others to reimburse living kidney donors for their medical, travel, and subsistence costs. Further, the Organ Donation and Recovery Improvement Act of 2004 (PL 108-216) authorized the Secretary of Health and Human Services to make grants to public or private entities to reimburse live kidney donors and up to two caregivers for their travel, subsistence, and incidental expenses. It also allows reimbursement to those who incur these expenses in good faith but are actually unable to make donations. The grants may fund payments only for costs not covered by state programs, insurers, or organ recipients. Federal employees may take up to 30 days of paid leave to make organ donations. Many states have similar programs for their employees, and some allow their residents to deduct donation-related expenses from taxable income.

Senator Smith, at your request, this analysis proposes and analyzes changes in federal policy that could substantially reduce the kidney transplant waiting list by encouraging more live kidney donation. Where possible, it provides quantitative estimates of the impacts of the effects of alternatives to current policy on the increase in the number of kidneys available for transplantation. However, because of the deadline for preparing this report and the scarcity of available information, these estimates should be viewed only as indicating likely orders of magnitude.

Problem Framing: Unregulated Market Unacceptable, Market Ban Questionable

When goods are supplied in competitive markets with many buyers and sellers, shortages cannot occur: if demand exceeds supply, prices rise to clear the market by either eliciting greater supply, reducing the quantity demanded, or both. In the case of live-donor kidneys in the United States, law prohibits the operation of a market so that a shortage of kidneys for transplant persists. Those seeking a kidney transplant can wait for a deceased donation to become available, seek a live donation from a friend or relative, or engage in “transplant tourism” by buying a kidney in an illicit market in a country that does not enforce bans on kidney markets as vigorously as does the United States.

Three questions require answers to frame the kidney shortage as a public policy problem. First, in the absence of a ban on kidney markets, would a market for kidneys operate? Second, if an unregulated market did operate, would it efficiently allocate scarce kidneys in ways consistent with other important social values? Third, if not, could a regulated market provide greater efficiency, be more consistent with important social values, or both?

First, it is very likely that a market for live-donor kidneys would exist if it were not prohibited. The ban on commercial transactions was included in the 1984 law in response to the announced intentions of a physician to create an international kidney exchange.[15] The existence of markets for human eggs, which imposes an irreplaceable loss on the donor, and surrogate motherhood, which involves risks to health, suggest that live-donor kidney markets would arise. So too does the existence of illicit markets enabling transplant tourism.

Second, would an unregulated market be efficient and desirable? There are at least two "market failure" reasons to believe it would not. One failure is seller–buyer information asymmetry. Live-donor kidneys are very heterogeneous in medically relevant qualities. Some of these qualities can be assessed through medical tests. Other qualities can be imperfectly imputed from medical histories, age, and perhaps reported or observed lifestyles. Nonetheless, it is likely that kidney sellers would enjoy an informational advantage relative to buyers about the quality of the kidneys they offer. This information asymmetry results in inefficiency if buyers purchase kidneys that they would otherwise not buy if they were fully informed about quality.

Additionally, buyers may enjoy an informational advantage (or asymmetry) about the quality of the medical services that will be provided to sellers before, during, and after the kidney is extracted and transplanted. This information asymmetry on the supply side of the market may also result in inefficiency if people sell kidneys that they would not have sold if they had been fully informed about the quality of the services they would receive––the “regretted sales” outcome that is reported to often occur in transplant tourism markets.[16] Thus, an unregulated market would almost certainly suffer from dual-sided information asymmetry (this can occur whenever the good has more than one quality dimension that determines value).

The second market failure is closely related. Because of the heterogeneity of organs and their values, any market for kidneys that sought to match buyers and sellers would unfold using an intermediate or so-called platform organization. As seen in many information markets, the outcome of this market development process would almost certainly be a single intermediary platform with monopoly power or a few platform providers with market power. Platform concentration tends to occur because sellers naturally want access to the maximum number of buyers and buyers naturally want access to the maximum number of sellers. Ironically, the search by suppliers for many buyers and by buyers for many sellers (that is, competition) can result in a platform provider with monopoly power over both. It is hard for the market to solve this inefficient outcome via market forces because the platform has declining average cost that makes successful entry by other providers difficult.

In addition to inefficiencies resulting from information asymmetry and platform concentration, an unregulated market would pose a number of ethical concerns. These concerns often serve as a rationale to support market prohibition. First, an unregulated market poses a threat to human dignity because it would risk exploitation of the poor. Financial stress may economically coerce some people into selling kidneys that they would otherwise not sell.[17] Second, an unregulated market would obviously favor those with the resources to purchase kidneys. Not only would they be able to outbid other patients for offered kidneys, but they would be better able to hire agents to search out potential sellers, further increasing the risk of exploitation of the poor or less informed. Third, many ethicists argue that allowing the commodification of kidneys in markets intrinsically violates human dignity by treating the human body as having instrumental value and undercutting the exercise of altruism.[18] The first two ethical concerns are related to the outcomes of a market; the third to its very operation.

Third, the market failures, exploitation of the poor, and undue advantage for the wealthy could potentially be addressed, at least to some extent, through either a regulated market or market-like policy, but the objection to commodification would remain. However, the once-almost-universal rejection of commodification by ethicists has eroded somewhat in recent years.[19] One counter argument recognizes that the preservation of life is an important value that conflicts with a rejection of commodification. Another sees bans on commodification as reducing the autonomy of those who might want to sell kidneys for economic gain.

In summary: An unregulated live-donor kidney market could potentially eliminate the shortage of transplant kidneys. However, it would be subject to several market failures that would prevent it from efficiently allocating live-donor kidneys. Its operation and likely outcomes would also raise serious ethical concerns. However, in view of the large losses in quality of life and longevity of patients that result from the prohibition of commercial transactions, alternative policies involving commodification that substantially mitigate the disadvantages of an unregulated market deserve consideration.

Policy Goals

Policy alternatives that affect the availability of kidneys from live donors should be assessed in terms of several goals: impact on the preservation of human dignity, impact on equity, impact on efficiency, impact on the fiscal state of the federal government, and political feasibility.

Impact on preservation of human dignity

The ideal policy should ensure the dignity of potential and actual living donors. Potential living donors, including those who are financially or otherwise disadvantaged, should have autonomy over the decision to donate. The health of actual donors should be guarded. Although the primary impact of policies on human dignity will be on U.S. residents, the impacts of the policies on residents of countries who are exploited by transplant tourism should also be considered.

Impact on equity

The ideal policy should promote more-equal access to transplantation. Access to this valuable medical treatment should not depend primarily on wealth.

Impact on efficiency

The ideal policy should maximize the number of successful kidney transplants and more generally promote a socially efficient transplant system. Transplantation generally improves the quality of life and longevity of ESRD patients. A cost–benefit analysis estimated that each additional transplant produces an average time-discounted gain of about 4.7 quality-adjusted life years relative to dialysis.[20] The greater the availability of healthy kidneys for transplant, the greater is the efficiency gain as measured by the number of patients who can realize these benefits.

Favorable fiscal impact

The ideal policy should minimize the net fiscal cost to the federal government of the ESRD Program. Funds expended on this program are not available for other uses. Reducing the net costs of the ESRD Program allows for other uses of the freed-up funds.

Political feasibility

The policy should be sufficiently acceptable to be adopted into law. For a policy to produce its anticipated benefits, it must be adopted. Ideally, a recommended policy would be adopted soon with certainty. More realistically, political feasibility can be assessed in terms of the likelihood of adoption sometime in the future.

Policy Alternatives

This analysis considers four policy alternatives: (1) current policy; (2) brokered auctions, which would established a regulated market; (3) expanded reimbursement with relative listing, which would encourage donation by reducing out-of-pocket costs borne by donors and allowing donors to place a relative at the top of the transplant waiting list; and (4) purchase with demand-side chain auctions, which would create an entity to purchase kidneys and distribute them to promote chain donations. A description of the key elements of each of the alternatives follows.

Current Policy

Under current policy, the sale or purchase of live-donor kidneys is strictly prohibited. Protocols in place require substantial efforts by transplant centers to ensure that those making live donations are fully informed and their donations freely made. To avoid relational coercion, centers report medical incompatibility for potential donors at their request, even if they are compatible.[21]

Those making altruistic directed or nondirected kidney donations may receive reimbursement for travel and subsistence costs related to the donation from insurers, state programs, transplant centers, or kidney recipients. Those not compensated fully from these sources may receive compensation from the National Living Donor Assistance Center if their annual income is less than 300 percent of the official poverty line or they can document extreme financial hardship from the donation. Payments from the center are limited to $6,000 and the center is constrained by an annual operating budget of about $3.5 million. Currently, the center does not provide compensation for lost wages by either donors or their caregivers.

Brokered Auctions [22]

This alternative creates a regulated market that does not allow direct exchanges between buyers and sellers of kidneys. Patients would bid for kidneys offered by potential donors (called *vendors* in this and the other alternative involving sale of kidneys) through intermediary organizations. These intermediary organizations would serve as brokers, or platforms, with fiduciary duty to patients and vendors in terms of disclosure and confidentiality of information, liability for harm to patients or vendors, and transparency.

Brokers would have responsibility for fully disclosing the medically relevant characteristics of kidneys offered by vendors, and they would have responsibility for fully informing potential vendors of all the costs and risks that they would bear and the payments that they would receive. Brokers would be required to protect the anonymity of patients and vendors by restricting the release of information not needed specifically for medical purposes related to transplantation. To facilitate competition, brokers would be required to make public on a timely basis the prices paid to vendors and data on the kidneys supplied, such as the age and blood type of the vendor. Brokers would have to make public the screening rules they apply in determining how far to evaluate potential vendors. The broker organizations and their chief executive officers would have legal liability for failure to discharge these responsibilities and requirements.

Several additional features would help protect the interests of vendors. First, brokers would be prohibited from imposing any cost, such as charges for evaluation or pre-transplant preparation, on vendors who withdraw from the process, so that potential vendors who at some point agree to sell kidneys would have the right to decide not to do so at any later time without financial penalty––at no time would kidneys be removed without current consent. Second, the broker would have responsibility for arranging for annual medical examinations of vendors and for submitting data on the health of these vendors on a regular basis to the Scientific Registry of Transplant Recipients, which has begun to collect data on live donors. The broker would be required either to provide funds for these tasks in escrow or to arrange for these tasks to be carried out by a transplant center under a long-term contract. Third, vendors would be given the option of directing payments to charitable organizations. Thus a vendor could participate in the exchange on a purely altruistic basis, providing a kidney to a patient and a monetary payment to a charity.

Expanded Reimbursement with Relative Listing

The Save Lives, Save Money Now Act, which is currently being advocated by Stop Organ Trafficking Now!,[23] would provide financial support for those making non-directed kidney donations. The proposal has four key elements. First, the federal ESRD Program would pay up to $10,000 of the non-medical costs incurred by those who donate kidneys for transplant to the compatible patient highest on the kidney transplant waiting list in the donor service area (DSA), or geographic locale, in which the donation is made. Second, the ESRD Program would provide $20,000 per donated kidney to compensate local organ procurement organizations for the costs of testing potential donors and providing their relevant information to the allocation system. Third, the donor would be listed in a living-donor registry in the DSA in which the donation was made. At a future time, the donor could add the name of one relative to the list––defined in this analysis as a spouse, child, parent, or sibling. At any time, those on the living-donor list could elect to move to the front of the kidney transplant waiting list in the DSA. The opportunity to advance a relative to the top of the waiting list in the future creates an insurance benefit for donors.[24] Fourth, 501(c)(3) nonprofit charitable organizations would be allowed to reimburse non-medical donation expenses up to $10,000 for any nondirected donation to a transplant center to facilitate the initiation or extension of a transplantation chain.

Purchase with Demand-Side Chain Auctions

Unlinking demand and supply requires methods for setting prices and allocating supply administratively. This alternative would create an organization to implement these tasks, the Kidney Exchange (KEX). It would have the following features. First, it would be a Congressionally chartered nonprofit organization with a legislated monopoly over the purchase of live-vendor kidneys. Second, like brokers in the brokered auction alternative, it would have responsibility for ensuring the anonymity and privacy of vendors, for specifying screening rules (the procedures, from initial screening to subsequent testing to possible donation, it would follow in securing kidneys), and for fully informing transplant centers about the characteristics of offered kidneys and vendors about costs, risks, and payments. Third, the purchase price would be determined primarily as a function of net expenditure savings to the federal government from a single additional transplant. The Centers for Medicare and Medicaid Services (CMS) would annually determine the financial breakeven price that Medicare could pay for a live-vendor kidney transplant for a patient who would otherwise remain on dialysis. The costs of screening potential vendors would be subtracted from this amount to set the price. Fourth, it would be funded through a start-up grant and then through a fee on facilitated transplantations. Fifth, it would be overseen by the Department of Health and Human Services or an organization that it designates, such as the OPTN.

Allocation of vendor kidneys would involve an “auction” among transplant centers—or more likely, associations of transplant centers like the currently operating New England Program for Kidney Exchange and the Alliance for Paired Donation—in which the bidding currency would be the number of transplants that would result if the kidney were used to seed a proposed chain. The process would operate as follows: First, the transplant center that identifies a potential vendor would submit the kidney characteristics to the KEX. Second, the KEX would post information about the availability of the kidney to participating transplant centers and alliances. Third, transplant centers and networks with patients with willing but incompatible donors would bid for the kidney to create single or multiple paired exchanges that would be executed within a specified time period. Fourth, the KEX would allocate the kidney based on the length of the transplant chain the kidney could initiate. Fifth, the last donor in an incompatible dyad in the chain would donate to the cadaveric kidney waiting list––patients at the transplant center that originated the vendor who seeded the chain would receive first priority for this kidney. Sixth, if two transplant centers propose chains of equal length, then the KEX would break the tie based on the sum of sensitivity levels of the patients in the chains. Seventh, failure to implement fully the chain that was bid would result in a penalty for the bidder: reduction in bid lengths by the shortfall for the auctions during the next six months.

Assessment of Alternatives

This section draws on available research to assess the policy alternatives in terms of the policy goals.

Current Policy

Human Dignity

The prohibition of the market for live kidneys prevents exploitation of potential vendors in the United States. Additionally, the practices surrounding live donation reasonably protect potential donors from making uninformed or relationally coerced decisions about donation. However, the prohibition on sales denies vendors full autonomy over the use of their own bodies, a restriction not applied in a number of other medical areas, such as human egg donation or surrogate motherhood. Also, the prohibition on commodification in the United States has induced some U.S. residents to engage in transplant tourism, which, because it occurs in illicit or unregulated markets, poses a high risk of exploitation of vendors.[25] Despite almost universal prohibition against commodification, perhaps 5–10 percent of all kidney transplants worldwide involve a vendor.[26] Demand from U.S. residents almost certainly accounts for some of these transactions that are detrimental to human dignity of the poor in other countries.

Equity

Wealthier patients enjoy some advantage in the current system because they can more easily provide compensation to facilitate a directed donation. So too do younger patients, because they are more likely to have living relatives and perhaps because they elicit more sympathy, and to those with greater social capital, which may increase their chances of finding a willing donor. Those who secure live donations indirectly share their good fortune by taking themselves off the transplant waiting list, thereby increasing the chances that some other patients receive a kidney.

Efficiency

Current policy is inefficient in three ways. First, it precludes vendors and patients from making mutually beneficial exchanges that would increase the total number of transplants. Second, it prevents some otherwise willing donations from those who cannot afford the associated uncompensated non-medical costs. Third, despite efforts by transplant centers to facilitate paired and chain donations, many potential donors cannot make the directed donations they wish to make because of incompatibility with patients. The relatively small number of available nondirected donations limits the use of chain donations.

Favorable Fiscal Impact

The cost of the ESRD Program is large and likely to continue to grow under current policy with increasing demands for dialysis.

Political Feasibility

Although Congress has been sympathetic to facilitating improvements in the live-donation system, as evidenced by the Organ Donation and Recovery Improvement Act, further legislation is unlikely without political leadership. Maintaining current policy is highly feasible.

Brokered Auctions

Human Dignity

Even with the vendor protections in place concerning disclosure of information, the brokered auctions would pose some threat to human dignity through the risk of exploitation of the financially desperate. Brokers would have an incentive to purchase kidneys at the lowest possible price, which would tend to be relatively attractive to potential vendors of limited financial means. Brokers, especially for-profit entities, would also have a strong incentive to find vendors, perhaps adding to the risk of exploitation through aggressive recruitment. Also, the linking of specific buyers to sellers would create a strong incentive for brokers to press potential vendors with rare but demanded characteristics. Somewhat offsetting these concerns, potential vendors would gain the opportunity to choose to sell their kidneys if they so wished. Also, surveys generally do not find a strong link between income and stated willingness to donate for compensation, suggesting that vendors might be less concentrated among those with low income than commonly assumed.[27]

The brokered auction would almost certainly reduce the number of U.S. residents who engage in transplant tourism. Obtaining a kidney in the domestic market would substantially reduce transaction costs in arranging for a kidney and verifying its quality. This is especially so as, currently, foreign transactions are often illegal or of questionable legality and therefore done sub rosa. The costs of finding and monitoring brokers would be much lower. Further, the brokered auction would eliminate the need for costly, and perhaps risky, foreign travel. It would also allow the patient to have the transplant done at a regulated center that would provide the full range of pre- and post-transplant services. As the interests of vendors—including protection against coercion—are likely to be stronger in the regulated domestic market than in the developing countries that now provide kidneys to transplant tourists, the net impact on vendors worldwide may be small.

Equity

Participating in the market as a patient would require wealth. The higher the equilibrium prices that result, the wealthier one would have to be to obtain a kidney from a vendor. Further, wealthier patients would be able to bid for higher-quality kidneys from younger vendors. This is precisely the sort of inappropriate advantage raised by opponents of commodification. However, someone who vacates the waiting list through purchase of a vendor kidney and would otherwise have received a deceased-donor kidney, enables another patient remaining on the waiting list to have a transplant.

Efficiency

The impact on the availability of kidneys for transplant is potentially very large. In a recent survey, 29 percent of respondents indicated that financial incentives would increase their likelihood of donating a kidney.[28] With respect to donations to strangers by those amenable to financial incentives, payments as low as $10,000 would begin to induce some respondents to donate and payments of $100,000 would be considered by most as undue influence to donate, suggesting that they would indeed do so. Survey evidence also suggests that payments would be unlikely to reduce altruistic donation to relatives and friends;[29] further, those who would otherwise make altruistic nondirected donations would have the option of becoming a vendor but directing their payment to a charity. Thus, the supply side of the market seems quite strong. With a pool of potential U.S. donors in the tens of millions, it is quite possible that prices below $100,000 would supply a sufficient number of kidneys from vendors eventually to eliminate the waiting list for kidney transplants.

Is there sufficient wealth among patients on the waiting list to support demand for large numbers of kidneys from vendors? Median household wealth in the United States was $68,828 in 2011, rising with age from $6,676 for householders younger than 35 years to $170,516 for householders 65 years or older.[30] These figures are likely to overestimate the wealth of householders suffering from ESRD. Nonetheless, because 66 percent of patients on the kidney transplant waiting list are 50 years or older, it appears that household wealth would support substantial demand for vendor kidneys. Additionally, it is likely that in some cases friends and relatives would raise money to contribute to bids.

Based only on the fragmentary evidence about the potential supply of and demand for vendor kidneys, a confident prediction of the number of additional kidneys that would be made available to patients is not possible. Nonetheless, it is likely that thousands, and possibly tens of thousands, of additional kidneys would be added to supply each year. Brokered auctions would thus have potential for substantially reducing the length of the kidney transplant waiting list.

Favorable Fiscal Impact

Based on an analysis that takes account of the costs of dialysis and transplantation, each additional transplant facilitated by a purchase from a vendor would provide a fiscal benefit of $146,000 to the ESRD Program.[31] Consequently, the brokered auction would potentially generate hundreds of millions of dollars each year in savings for the federal government.

Political Feasibility

Despite the potential gains to patients and large fiscal savings, concerns about threats to human dignity, equity in access to transplants, and commodification itself would make brokered auctions politically unfeasible. Indeed, even proposing a brokered auction would likely be politically risky, especially for a first-term senator.

Expanded Reimbursement with Relative Listing

Human Dignity

Helping overcome financial barriers to donation would promote human dignity by enabling more potential donors to exercise choices to act altruistically. The federal support for donations to the waiting list would pose no risks of exploitation. The private support from charitable organizations would allow patients to reimburse directed donors for wage losses, raising some concern about increased relational coercion. A greater risk of relational coercion would arise from the opportunity for donors to place a relative at the top of the waiting list. Potential donors, who are able to claim incompatibility under the current system to avoid making a donation, would be vulnerable to pressure to make a nondirected donation to move a relative to the front of the waiting list. The narrow definition of "family member" limits those potentially vulnerable to relational coercion. Also, continued progress in desensitization will gradually make the marginal impact of this alternative on relational coercion less important. It would not have a direct effect on transplant tourism.

Equity

The opportunity for private charities to provide compensation would improve equity by increasing access to kidneys for lower-income patients. However, wealthier patients may be able to pass compensation to potential donors by making contributions to private charities. On net, there would likely be a small improvement in equity in terms of facilitating transplants for the less wealthy. Those with large immediate families would potentially have more access through nondirected donations by close relatives.

Efficiency

The number of kidneys available for transplant would increase through two mechanisms: financial incentives and gaining priority for deceased-donor kidneys for relatives. Survey evidence suggests that financial compensation would likely increase the likelihood of donating to a stranger but not to a friend or family member; specifically, compensation appeared to increase the percent of respondents with a willingness to donate to strangers from about 2 percent to about 37 percent.[32] The median amount of compensation required to elicit consideration of such donation was $10,000, suggesting that half of the 37 percent would not find the program limit adequate, reducing the percentage of willing donors from 37 percent to about 19 percent. Because only 29 percent of respondents listed paying for non-medical expenses as their intended use of the compensation, rather than using it for some other purpose such as to pay off debt (38 percent) or increase savings (14 percent), it is reasonable to assume that only 29 percent of the 19 percent expressing a willingness to donate to strangers would actually do so if payments were limited to compensation for non-medical expenses. This suggests that the upper bound on the percentage of respondents willing to make nondirected donations would be about 6 percent. Thus, compensation would potentially increase the willingness to donate from about 2 percent to about 6 percent.

Applying this threefold increase to the current annual number of nondirected donations of about 180, the compensation induces an additional 360 nondirected donations. The assumptions made in moving from the survey results to this number suggest that it should be treated as an upper bound. However, because nondirected donations induced by compensation provided by charities could be used to facilitate chain donations, it is possible that this number underestimates the increase in the number of transplants that would be induced by the compensation provisions.

Researchers estimated that in 2005 there were 4,000 patients on the waiting list who had willing but incompatible donors, and 750 more join the list each year.[33] Scaling these estimates using the ratios of the 2015 to 2005 waiting list numbers and new additions to the list—1.2 and 1.7, respectively—leads to estimates of about 6,700 for the current number of patients with willing but incompatible donors and 900 for annual additions of such patients to the waiting list. In 2015, 578 of these patients were accommodated through paired donations and another 209 through anonymous donations. With some additional patients made compatible through desensitization, probably about 800 per year are being accommodated so that the stock of unaccommodated can be estimated to be growing at about 100 per year. If the listing of relatives were fully effective in converting willing to actual donors, it would eliminate the stock of such patients over several years then result in a steady-state addition of about 100 transplants per year.

Favorable Fiscal Impact

The fiscal impact would be substantial and reduce expenditures. Even if all of the additional compensation-induced donations were covered fully by the federal government, the savings for the ESRD Program would be more than $100,000 per donated kidney. The listing-induced donations would save the full $146,000 difference between the expected cost of dialysis and transplantation. The long-run annual savings would be on the order of approximately $400 million.

Political Feasibility

Several features of this alternative make it politically attractive. It increases the number of transplants, reduces costs for the federal government, and incrementally extends the existing federal compensation program. It would likely draw some opposition from the transplant community because the priority given to patients on the living-donor list would override priorities that position patients on the deceased-donor waiting list.

Purchase with Demand-Side Chain Auction

Human Dignity

As with a market-determined price in brokered auctions, the administered price offered by the KEX would pose some threat to human dignity through the risk of exploitation of financially desperate vendors. However, this nonprofit intermediary would not have as strong an incentive to recruit vendors as would competitive brokers, so that the risks of exploitation would likely be lower than with brokered auctions. However, unlike the brokered auctions, this alternative would not have a direct effect on transplant tourism because it does not provide a domestic alternative for individuals to purchase kidneys.

Equity

This alternative does not involve inappropriate advantage beyond that under current policy. Each chain seeded by a purchased kidney would ultimately provide a kidney to the waiting list from the willing but incompatible donor for the last patient in the chain. However, as is currently the case, those who can find willing but inappropriate donors would enjoy increased access to transplantation because they would be more likely to be included in a chain.

Efficiency

The administered price paid to vendors would likely be between $50,000 and $100,000, based on savings of $146,000 per transplant and the current reimbursement of $50,000 for a delivered deceased-donor kidney paid to organ procurement organizations. Actual costs of procuring a kidney would likely be lower than this amount because vendor kidneys could be procured in more controlled circumstances. However, costs would likely be incurred by the KEX in screening potential vendors who do not progress all the way to delivery of kidneys.

Survey responses suggest that a price near $100,000 would elicit substantial supply.[34] The cost–benefit analysis of payments for kidneys previously cited assumed that a payment of $45,000 would elicit sufficient supply to double the number of transplants per year, so that after about five years the waiting list would disappear. As this alternative would pay a much higher price as well as likely multiply the number of transplants for each vendor kidney through its use in seeding chains, it is reasonable to predict that it would generate sufficient supply to eliminate the waiting list.

Favorable Fiscal Impact

The vendor kidney itself would not produce a savings for the ESRD Program. However, each vendor kidney would likely result in one or more additional transplants. Consequently, a reasonable lower bound on annual federal savings in the steady state would be about half of the 18,000 additional transplants times $146,000, or about $1.3 billion.

Political Feasibility

The substantial increase in the number of transplants and the federal savings would make this alternative attractive for some stakeholders. Because it would not raise as serious concerns about equity as would brokered auctions, it would likely be somewhat more acceptable. However, as it involves commodification and concerns about exploitation of the disadvantaged, it would be viewed as a radical change in policy and therefore have little chance of immediate adoption.

Considering the Trade-offs

As summarized in Appendix A, none of the policies dominate in terms of all five policy goals. All three alternatives to current policy would increase efficiency by expanding the supply of live-donor kidneys and provide a fiscal advantage by reducing the costs of the ESRD Program. Purchase with demand-side chain auctions would have the best prospects for sufficiently increasing supply to eliminate eventually the kidney transplant waiting list. Brokered auctions would substantially increase supply, perhaps sufficiently to eliminate the waiting list. Expanded reimbursement with relative listing would likely increase the annual supply by hundreds per year through its financial provisions and possibly even more through its provision effectively allowing live donors to move a relative to the top of the deceased-donor waiting list.

The brokered auction would pose the greatest threat to human dignity. Purchase with demand-side chain auction would pose a similar if somewhat less severe threat; expanded reimbursement with relative listing would pose some threat by encouraging patients to pressure close relatives to donate so that those patients increase their chances of receiving a deceased-donor kidney.

Neither expanded reimbursement with relative listing nor purchase with demand-side chain auction would substantially change the level of equity in the current transplant system. However, brokered auctions would substantially advantage patients wealthy enough to bid successfully for vendor kidneys.

The brokered auction would have little prospect of being adopted by Congress because, in addition to eliciting objections to its commodification of kidneys, it would raise concerns about human dignity and equity. Although purchase with demand-side auctions would likely fare somewhat better, it is very unlikely that it could gain sufficient support for adoption, especially with a freshman senator as its sponsor. Expanded reimbursement of donation costs with relative listing would have much better prospects than either of the other two alternatives involving commodification of kidneys.

Recommendation

The transition team recommends that you support expanded reimbursement with relative listing. It would increase the supply of kidneys for transplant, reduce the costs of the ESRD Program, pose only small threats to the exploitation of potential donors, and have good prospects for garnering political support. Your first steps should be to request drafting assistance from the Congressional Research Service to craft legislation to authorize the elements of this alternative. You should also coordinate with Stop Organ Trafficking Now! to take advantage of its efforts to promote the similar Save Lives, Save Money Now Act. The process of building support for this alternative and your eventual success in achieving it, would enable you to determine if a more efficient alternative, like purchase with demand-side chain auction, could be designed to be ethically acceptable and politically feasible.

Appendix A

Summary Comparison of Policy Alternatives for Addressing Kidney Shortage

	Current Policy	**Brokered Auction**	**Expanded Reimbursement with Relative Listing**	**Purchase with Demand-Side Chain Auctions**
Human Dignity	Protects U.S. residents; encourages transplant tourism	Substantial threat of exploitation of disadvantaged; reduces transplant tourism	Some threat of relational coercion	Moderate threat of exploitation of disadvantaged
Equity	Small advantage to wealthy and those with social capital	Substantial inequality in terms of wealth	Increased equity in terms of wealth; decrease in terms of available relatives	Small decrease in terms of added advantage for those with willing donors

	Current Policy	Brokered Auction	Expanded Reimbursement with Relative Listing	Purchase with Demand-Side Chain Auctions
Efficiency	Inefficient: long waiting list for transplants	Annual increase in thousands or tens of thousands; possible elimination of waiting list	Annual increase in hundreds or thousands	Annual increase in tens of thousands; likely elimination of waiting list
Favorable Fiscal Impact	Cost of ESRD Program will continue to increase as more patients use dialysis	Large reductions in net costs for federal government	Moderate reductions in net costs for federal government	Large reductions in net costs for federal government
Political Feasibility	High	Very low	Moderate	Low

Debriefing

Your decision to read this book suggests that you have not already had a course in policy analysis, so that you probably have not previously encountered a report like the one you just read––it almost certainly differs in format and content from undergraduate papers you wrote. This entire book provides you with the conceptual foundations and craft skills to enable you to produce policy analyses competently and quickly. If you work as a policy analyst, then the advice you provide will take many forms, ranging from brief discussions in the hallway to substantial reports prepared by large teams using original data and working over an extended period. Nonetheless, mastering the canonical template displayed in this chapter will help you give good advice when using these other formats as well.

Before we embark on a systematic treatment of the generic template in the chapters that follow, we briefly highlight some of its most important features, both to help you put your journey through the chapters into perspective and to provide you with some upfront help in case your instructor wants you to perform a trial policy analysis along the way.

Write to Your Client

Policy analysis as a professional activity is done for a client. Whether as employees or consultants, professional policy analysts earn their keep by giving reasoned advice to their clients. Typically, the client is a person, like Senator-elect Smith (hereafter, just "Senator"), who makes or influences public policy decisions. Many policy analysts have public administrators as their clients. Some have nongovernmental clients, such as the leaders of organizations that seek to improve public policy in specific substantive areas or promote particular policy approaches. Clients can also be governing organizations, such as legislatures, regulatory bodies, or international organizations, that seek advice about the decisions they must make. Although as a policy analyst you should always seek to make society better, as a professional you do so through service to a specific client with authority or influence. If you become an elected official or public administrator, then at times you may be your own client.

Senator Smith presented you with a specific policy problem, the long waiting list for kidney transplants, and asked you to consider a class of alternatives, policies that would increase the availability of kidneys for transplantation from living donors. She also gave you a deadline. In this case, the deadline was dictated by her convenience. However, if she were seeking advice in advance of an upcoming Senate committee or floor vote, then the deadline would be inflexible––your analysis would be much less valuable, perhaps worthless, and your tardiness possibly career ending, if you delivered it after the vote!

In addition to the two fundamental responsibilities to the client of answering the question posed and doing so on time, you also have to be concerned about effectively communicating your advice. Your report should be written so that your client, and those with whom she chooses to share your report, can easily understand its content. You had to work hard to convey simply the quite complicated substance of kidney transplantation and relevant economic concepts, such as information asymmetry, so that Senator Smith, an educated lay person, could follow your analysis without difficulty. However, even a relatively short policy analysis may be too long for a busy client to read in its entirety. That is why you include a brief executive summary, an essential component of any policy report more than a few pages in length.

Your executive summary has several important features in addition to brevity. Its first paragraph sets out the issue and ends with your recommendation. The subsequent paragraphs appropriately rehearse your complete analysis, including a statement of the relevant policy goals and assessed alternatives, and the reason for the recommendation. Many busy clients would know the basic approach of your analysis just from reading the executive summary. Once you have a track record for sound analysis and judgment, some clients may read only the first paragraph to see the recommendation. Write your executive summary as if that will be the case.

You divided the body of your report into clearly labeled sections. This aspect of templating helps your client follow the logic of your analysis and makes it easy for her to read selectively specific sections of interest. For example, Senator Smith may be inclined to accept your recommendation based on the executive summary alone, but she may want to read your description and follow your assessment of the recommended alternative before making a final decision. Adding a table that summarizes your assessment of alternatives in terms of the relevant policy goals also makes it easier for your client to understand how you reached your recommendation.

Understand the Policy Problem

Policy analyses often start with a client expressing concern about some undesirable condition. Senator Smith wants to do something about the long waiting list for kidney transplants. Sometimes you will be asked to assess a policy proposal rather than an undesirable condition. In the policy proposal case, your task usually involves stepping back to identify the undesirable condition the proposal intends to address, and then stepping forward with alternatives that might be even better than the proposed alternative that prompted the request for analysis. So, for example, imagine that one of Senator Smith's constituents had asked her to allow people to sell kidneys. You would still want to identify the undesirable condition that kidney sales are intended to address and what other policy alternatives deserve attention, as you actually do in responding to the senator.

Understanding the policy problem involves at least two tasks. The first task is to diagnose the undesirable condition and, if possible, put it into quantitative perspective. Your diagnosis usually requires you to learn quickly about the subject. If you have been working in a policy area for a long time, then doing so may be relatively easy, even for newly emerging issues, because you know a lot about the context and likely have a network of colleagues to consult. If instead of being new to transplantation policy you had previously analyzed alternatives for increasing the availability of deceased-donor kidneys, then you would likely already know a lot about transplantation. Starting from scratch is daunting, but if you build your skill in intelligently using electronic sources, you can often get started quickly. Once you establish that a condition is worthy of concern––people forgo quality of life and often die waiting for kidney transplants––it is usually useful to quantify the condition and put it into perspective. You were able to find data showing that the waiting list for kidneys is long and that it is growing––there is an undesirable condition and it is getting worse. It may be useful to put other conditions into perspective by making comparisons across policy jurisdictions. For instance, if you were advising a state legislature on policies to increase the availability of deceased-donor kidneys, then you might compare donation rates for your state with those of other states. If your state is lagging, then you might look at deceased-donor policies in the more successful states to begin to get ideas about policy alternatives. Such comparisons are important, because it is the possibility that your client could influence policies offering improvement that turns an undesirable condition into a policy problem.

The second task involves framing the policy problem as a market or government failure (or maybe both) or other circumstance in which individual choices lead to undesirable social outcomes. Conventionally defined, market failures are situations in which individual choices do not lead to an efficient allocation of resources. In considering the possibility of an unregulated market for kidneys, you identify several market failures. You argue that an unregulated market for living-donor kidneys would lead to inefficient transactions because of information asymmetry, one of the classic market failures. Government failures are the ways that collective action leads to inefficient allocations of resources. Your report identifies that current policy produces an inefficient allocation by not allowing a market in kidneys. However, the market ban reasonably reflects other important values and does not inherently result from a collective action problem, such as the disproportional weight representative government tends to place on organized (relative to unorganized) interests that often results in restrictions on competition or other questionable limits on individual choices.

Market failures produce inefficiency. However, even efficient markets can produce socially undesirable allocations of resources. We usually want floors on consumption, ensuring that people have the basic necessities of life. We also usually want some minimum levels of participation in the political processes that choose levels of goods like environmental quality. We often want less unequal access to the consumption of goods, either generally through a more equal distribution of wealth, or more specifically through access to specific types of goods, such as schooling and medical care. You address concerns about exploitation of the poor and undo advantage for wealthy patients in your assessment of kidney commodification.

A benefit of framing a policy problem as a particular market failure is that it allows you to take advantage of the substantial theoretical and empirical knowledge available across many policy areas about its consequences and ways of mitigating these consequences. Framing as a market failure often provides an initial model for identifying and assessing policy alternatives. Your alternatives involving kidney vendors respond to information asymmetry by imposing an intermediary—brokers in one case and the KEX in the other. Often you will need other models as well. So, for example, analyses of alternatives for increasing kidneys from deceased donors might also usefully draw on models of altruistic behavior from psychology or sociology.

Another aspect of understanding the problem involves deciding on the type of alternatives that you will consider. Quite regularly, opinion pieces appear in major newspapers arguing that shortages of transplant organs could be substantially reduced by replacing our opt-in system (in which people explicitly commit to donating organs upon their death) to an opt-out system (in which it is assumed that people will donate unless they specifically object while living). Such recommendations usually ignore the possible magnitude of supply from deceased donors and their advocates often fail to take account of the importance of the effectiveness of the organizations that identify potential deceased donors and facilitate the removal of donated organs for achieving high donation rates. Even though you were asked to consider policies to increase the availability of kidneys from living donors, you make clear in your analysis why policies to increase deceased donation could not stop the kidney transplant waiting list from growing, an important point in light of the greater ethical concerns raised by the alternatives aimed at increasing live donations.

Be Explicit About Values

Public policy should promote outcomes that people value. We try to clarify these values by setting out and justifying goals relevant to the policy problem being addressed. As we want to get the most out of available resources, efficiency is almost always a relevant goal. In many policy areas, however, we also have goals involving the distribution of the goods produced by the available resources. These are substantive goals––things we want for their own sake. We may also have instrumental goals, such as sustainable public finance or political feasibility, that facilitate achievement of the substantive goals.

Your analysis identifies five goals: efficiency, human dignity (which could be treated as a universalistic version of the equity goal), equity, and two instrumental goals. You operationalize efficiency as the number of kidney transplants per year. With more information for making predictions, you might operationalize efficiency as the number of quality-adjusted life years gained, a potentially better measure of value. You also might add a second dimension for assessing the impact of alternative policies on efficiency: the resources expended in screening donors or vendors, including the value of their time. Choosing how to operationalize a goal often involves a trade-off between capturing the relevant values and one's capacity for predicting impacts of policies on promotion of those values.

Your two distributional concerns, which you label protection of human dignity and equity, capture impacts of policies on both vendors and patients. Although you could have treated both as versions as equity, in this analysis that would only hide information. All the policy alternatives except the brokered auction have quite similar effects on equity. Had you not made brokered auctions one of your alternatives, you might not have included equity among your goals because it would not help in distinguishing much among the alternatives. Conversely, if you had included some other alternative that introduced yet another dimension of distribution, say an alternative that specifically affected some ethnic group, then you should add it as a new goal or a second dimension, or impact category, for equity. More generally, your set of goals should allow you to take account of all of the major consequences of your alternatives.

A favorable fiscal impact and political feasibility are your two instrumental goals. You include fiscal impact because spending on one public program involves higher taxes, less spending on other programs, or a bigger deficit. Note that fiscal impact differs from efficiency: cutting back on the maintenance of public infrastructure may be inefficient, even if its immediate fiscal impact is favorable. You also include political feasibility as a goal to indicate how likely it is that any of the policies would actually be adopted. Often fiscal impact directly affects political feasibility. In your analysis, however, the alternatives to current policy all have favorable fiscal impacts but two of them are unlikely to be politically acceptable. Keeping the two instrumental goals separate is appropriate in terms of choosing goals that help distinguish meaningfully among the particular policy alternatives being analyzed.

Specify Concrete Policy Alternatives

Policy alternatives must be specified in sufficient detail to allow you to predict their consequences. You appropriately include current policy as an alternative to avoid a bias toward change for its own sake. Your introduction of the problem sets out the key elements of current policy: a ban on commercial kidney transactions and incomplete compensation to living donors and their caregivers for non-medical expenses. You base your alternative policies on proposals made by academic policy researchers and an advocacy organization, focusing on those details most relevant to predicting the impacts of the alternatives relevant to the policy goals. Most of the details were specified in your sources. However, the original proposal for expanded reimbursement and relative listing did not define "relative." As you have some concerns about relational coercion, you limit the definition of relatives for purposes of this policy to include only immediate family members. In doing so, you were following a common procedure for developing policy alternatives––borrowing and tinkering. You first borrow a policy concept from some source, either a proposal or a working model, and then tinker with its details to make it more attractive for the circumstances of the policy problem you are addressing. In other situations, you might translate a generic policy instrument, such as an excise tax on a good whose full social cost is not captured in its price, into a specific alternative, or even design an alternative from scratch. In these cases, you must be especially careful to specify details needed for prediction.

Either more thorough analysis or transition from policy alternative to actual policy proposal can necessitate even greater specification of detail. For example, assessing the impacts of purchase with demand-side chain auctions on access to transplants by disadvantaged ethnic minorities or highly sensitized patients would require you to be more specific about the outreach efforts of the KEX. If this alternative were to be accepted, then greater detail about the governance and funding of the KEX would have to be specified sufficiently for legislation to provide an adequate authorization for its implementation.

Predict and Value Impacts

Finding information relevant to prediction and valuation almost always poses a challenge to analysts. Short deadlines and limited availability of directly relevant data usually require that you draw on research that has already been done. You were able to find some survey evidence to help you predict the impact of financial payments on willingness to be a living kidney donor. However, the surveys did not ask the exact questions you would have asked had you designed your own survey. Consequently, you had to be creative in squeezing relevant information for purposes of prediction from the data. You were more fortunate with respect to valuation because you found a high-quality cost–benefit analysis of the financial and social benefits of live donation that was recently published in a peer-reviewed journal. You also tapped into a few sources of descriptive data, such as waiting list information from the Organ Procurement and Transplantation Network.

With more time and resources, you might be able to gather data and evidence that would enable you to make more confident predictions or valuations. For example, perhaps you could conduct a survey that would allow you to ask questions directly relevant to predicting the impact of financial incentives on willingness to donate. You might also be able to request specific data or relevant simulations from the Scientific Registry of Transplant Recipients, which collects comprehensive longitudinal data on patients, transplant recipients, and donors. Perhaps it would be possible to operate a pilot program within existing authority to gather evidence about impact. For instance, perhaps the Secretary of Health and Human Services could be convinced to direct the National Living Donor Assistance Center to fund fully the reimbursement of non-medical donation expenses in a particular jurisdiction. An evaluation of the impact of full reimbursement could then be used to predict the impact of expanded reimbursement nationwide.

Although cross-national comparisons may sometimes be helpful, differences in constitutions and cultures often make prediction based on evidence about the impact of a similar policy in another country tenuous. You are likely to have more success in drawing on subnational comparisons. For example, there may be dozens of evaluations of locally implemented programs, such as interventions aimed at reducing criminal recidivism. Indeed, some types of programs may have a sufficient number of evaluations so that you can use or even conduct a meta-analysis that combines their results to produce more confident assessments of impacts. In these cases, it may be possible to engage in highly touted evidence-based policy making. Of course, this avenue will not be open if you are analyzing a novel policy alternative. Unlike academic researchers, who rarely make out-of-sample predictions, your predictions of the impacts of a unique policy alternative must necessarily be guided by logic and theory, rather than systematic empirical evidence.

Consider the Trade-Offs

As in your analysis of the kidney shortage, you will almost always identify multiple values relevant to the policy problem, which in turn call for multiple goals. Your policy alternatives will almost always involve trade-offs among the goals. All three of your alternatives to current policy offer increased efficiency. However, brokered auctions would also result in substantial inequality, and both it and purchase with demand-sided chain auctions would likely not be politically feasible. Expanded reimbursement with relative listing would provide the smallest increase in efficiency of the three alternatives to current policy, but it would be the most politically feasible. No alternative dominates the others in terms of all five goals−−if one alternative did dominate, then you would be wise to ask yourself if you had constructed serious alternatives and had set out goals that captured their full ranges of consequences.

Certain types of trade-offs commonly arise. Market-like policies, such as brokered auctions, often involve a trade-off between efficiency and equity. More radical policies offering greater efficiency, such as purchase with demand-side chain actions, often have poorer prospects for political feasibility. More efficient programs sometimes require more public spending, although your alternatives to current policy do not. You should expect trade-offs. Most clients will expect you to acknowledge and analyze them. Nonetheless, sometimes you can find a policy that hits a sweet spot and advances most, if not all, of the relevant goals.

Although sometimes tedious for analyst and client alike, considering trade-offs requires each of the alternatives to be considered in terms of each of the goals. You do so by moving systematically through each of the five goals for each alternative, with headings and subheadings to make clear which assessment is being made in each subsection. Your table in the appendix summarizing your assessment of the alternatives in terms of the goals provides a useful discipline, because an empty cell would make clear that you had failed to assess one of the alternatives in terms of one of the goals.

Make a Recommendation

Unless your client asks you not to do so, you should explicitly recommend one policy, either current policy or one of the alternatives to it that you analyzed. You recommend expanded reimbursement with relative listing because it offers gains in terms of efficiency and fiscal impact without raising serious concerns in terms of human dignity, equity, and political feasibility. Your recommendation appropriately follows from your analysis, especially your discussion of trade-offs.

Was it a waste of your effort to consider the other alternatives to current policy? No—each of these alternatives was worth considering because each would provide substantial improvements in terms of efficiency and fiscal impact. If political feasibility were not one of the relevant goals, then your analysis might have led you to recommend purchase with demand-side chain auctions instead. Raising this alternative may be useful in alerting your client to an alternative that could eliminate the waiting list for kidney transplants. Further, circumstances can change to make an alternative more politically feasible: sometimes events temporarily open so-called policy windows through which well-developed policy alternatives can be pushed to adoption.

What if your client rejects your recommendation? Although you may be disappointed, your analysis may still contribute to a better decision by your client. If you applied your craft well, then your client will have been better informed about relevant goals, alternatives, and trade-offs. She may accept your predictions of the impacts of policies on achievement of the goals, but simply place different weights on the goals than do you in your assessment. Your analysis facilitates a more informed decision, albeit not the one that you would have made.

For Discussion

1. Think back to the research papers you have written for undergraduate courses. In what ways were your papers similar in content and structure to the kidney shortage report? In what ways did your papers differ?
2. Now think about articles you have read in scholarly journals. How does the sample policy analysis differ in content and structure from published academic research?

Notes

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2
What is Policy Analysis?

The product of policy analysis may be advice as simple as a statement linking a proposed action to a likely result: passage of bill *A* will result in consequence *X*. It may also be more comprehensive and quite complex: passage of bill *A*, which can be achieved with the greatest certainty through legislative strategy *S*, will result in aggregate social costs of *C* and aggregate social benefits of *B*, but with disproportionate costs for group 1 and disproportionate benefits for group 2. At whatever extremes of depth and breadth, policy analysis is intended to inform some decision, either implicitly (*A* will result in *X*) or explicitly (support *A* because it will result in *X*, which is good for you, your constituency, or your country).

Obviously, not all advice is policy analysis, so to define it we must be more specific. We begin by requiring that the advice relate to public decisions and that it be informed by social values. That is not to say that policy analysts do not work in private organizations. Businesses and trade associations often seek advice about proposed legislation and regulations that might affect their private interests—when their employees or consultants consider the full range of social consequences in giving such advice they are providing policy analysis. Of course, the majority of policy analysts work in government and nonprofit organizations where day-to-day operations inherently involve public decisions, as well as in consultancies that serve these public and private organizations. Because our interest centers on policy analysis as a professional activity, our definition requires that policy analysts, in either public or private settings, have clients for their advice who can participate in public decision making. With these considerations in mind we hazard the following simple definition: *policy analysis* is client-oriented advice relevant to public decisions and informed by social values.

One reason for introducing this definition is that it helps us keep our focus on the purpose of this book: developing the practical approaches and conceptual foundations that enable the reader to become an effective producer and consumer of policy analysis. We emphasize development of a professional mind-set rather than the mastering of technical skills. If we keep central the idea of providing useful advice to clients, then an awareness of the importance of learning the various techniques of policy analysis and of gaining understanding of political processes that produce policy will naturally follow.

Another reason is that this definition emphasizes the importance of social values in policy analysis. Social values can come into play even when advice seems purely predictive. By looking at consequences of policies beyond those that affect the client, the analyst implicitly places value on the welfare of others. Good policy analysis takes a comprehensive view of consequences and social values. As will become clear in subsequent chapters, we believe that economic efficiency deserves routine consideration as a social value, not only because it corresponds fairly well to aggregate welfare, but also because it tends to receive inadequate weight in the deliberations of representative governments. However, other values related to the distribution of goods within society are also commonly important.

We begin this introduction to policy analysis by identifying its place within the creation and application of public policy knowledge. We then consider where policy analysts can be found and what sorts of things they do. Finally, we discuss the sorts of skills most essential for success.

Policy Analysis in Perspective

Two questions help us classify policy knowledge for the purpose of putting policy analysis into perspective. First, is the primary focus on the content of policy or the process that produces it? Content is the "what" of policy; process is the "how." The former is about the knowledge that informs our understanding of the substance and implications of adopted or potentially adopted policies; the latter is knowledge that informs our understanding of the process through which public policies are adopted within any particular political system.[1] Second, who is the primary audience for the knowledge? We distinguish between clients, either individuals or organizations that play a part in the policymaking process, and epistemic communities defined as "network[s] of professionals with recognized expertise and competence in a particular domain and an authoritative claim to policy-relevant knowledge within that domain or issue-area."[2]

Table 2.1 divides policy knowledge by these two dimensions into four activities. The upper right cell defines *policy research* as having its primary focus on the "what" of policy and its primary audience is the community of researchers and others, such as policy analysts and advocates, interested in some substantive policy issue. Policy research often employs the methods of the social science disciplines. While academic research in these disciplines looks for relationships among the broad range of variables describing behavior, policy research focuses on relationships between variables that reflect social problems and other variables that can be manipulated by public policy. The desired product of policy research is a more-or-less verified hypothesis of the form: if the government does *X*, then *Y* will result.

For example, academic research into the causes of crime might identify moral education within the family as an important factor. Because our political system places much of family life outside the sphere of legitimate public intervention, however, there may be little that the government can do to foster this kind of learning within the home. The policy researcher, therefore, may take family values as a given and focus instead on factors at least partially under government control, such as the certainty, swiftness, and severity of punishment for those who commit crimes. The policy researcher may then be willing to make a prediction (a hypothesis to be tested by future events), say, that if the probability of arrest for a certain crime is increased by 10 percent, then the frequency of that crime will go down by 5 percent.

A fine line often separates policy research and policy analysis. The strength of client orientation distinguishes them in our scheme. Policy researchers are less closely tied to public decision makers. While decision makers may be interested in their work, policy researchers usually view themselves primarily as members of a network that consists primarily of other researchers. Sometimes their primary motivation for doing policy research is personal financial gain or the excitement of seeing their work influence policy; perhaps more often they do it to gain resources or attention for their broader research programs. Because they place primary importance on having the respect of others in their network, policy researchers are often as concerned with the publication of their work in professional journals as they are with its use by decision makers. Thus, the pace of their work is more often dictated by the time needed to meet academic standards than by the opportunity to influence particular policy decisions.

Table 2.1 Creation and Application of Public Policy Knowledge

		Primary Audience	
		Client	*Epistemic Community*
Primary Focus	*What*	Policy analysis (narrowly defined)	Policy research
	How	Stakeholder analysis	Policy process research

The well-defined disciplinary orientation of most academic policy research tends to limit its direct applicability. The translation of research findings into implementable policies requires attention to practical considerations of little general interest to other disciplinary researchers. Returning to our example, a prediction that an increase in the probability of arrest will decrease the crime rate is only the first step in developing and predicting the consequences of a policy option. How can the arrest rate be increased? How much will it cost? What other impacts will result? How can it be determined if the predicted reduction in the crime rate has actually occurred? The answers to such questions require information of a specific nature, often of little general interest and even less disciplinary interest. Consequently, policy researchers often leave these sorts of questions to policy analysts, who will actually craft policy options for decision makers.

The lower right cell of Table 2.1 defines *policy process research* as having its primary focus on the how of policy making, and its primary audience consists largely of academics. While policy research draws most heavily from substantively oriented professional fields like public health and environmental science, and from the academic disciplines of economics and sociology, policy process research draws most heavily from political science. Its frameworks and theories explain why and how the content of public policy changes or remains the same. It identifies generic processes that operate across, or repeatedly within, particular policy areas.

Two features of policy process research distinguish it from other fields within political science. First, it emphasizes the relationship between the substance of policy and political processes. Although the direction of interest is usually from politics to substance, it has a long tradition of also considering how the substance of policy affects the nature of politics.[3] For example, lobbying and other forms of active political participation are more likely when small numbers of individuals or organizations bear large shares of the costs or receive large shares of the benefits from policy change.[4]

Second, policy process research looks inclusively across political institutions. Rather than focusing on political behavior within particular institutions, such as the executive, legislative, or judicial branches, it considers their collective effect on the substance of public policy. So, for example, while scholars of the U.S. Congress typically focus primarily on its processes and outputs, a policy process researcher is interested in the final policy that results after all other relevant political institutions have exerted their authorities and influences. Often this interest extends beyond adoption to the various processes through which policies are implemented. When attention to implementation includes assessment of the outcomes of policies, policy process research on the "how" merges into policy research concerned with the "what." Policy process research can assist policy analysts in thinking about the bigger picture of policy adoption. It may also provide more directly useful resources for stakeholder analysis.

The lower left cell of Table 2.1 defines *stakeholder analysis* as being concerned with the "how" for a specific client. This is often called political and organizational analysis because it usually requires an understanding of the interests, beliefs, and resources of relevant social, political, and organizational actors, or stakeholders, and the arenas in which they influence, or potentially could influence, the adoption and implementation of policy.[5] Various ways of systematically organizing and analyzing information about stakeholders have been developed.[6] Although clients usually conduct their own informal stakeholder analyses, they often seek advice from those who either have knowledge about particular arenas where policy decisions are made or understanding of stakeholders relevant to the crafting of strategies for getting things done.

Stakeholder analysis as a distinct activity is instrumental rather than normative. That is, it is concerned with achieving adoption or implementation of some policy rather than with the social value outcomes the policy is intended to produce. As advice about strategy, it is most often provided confidentially to clients. However, rather than done as a distinct activity, it can also be integrated into policy analysis through consideration of instrumental goals like political feasibility or prospects for successful implementation.

The upper left cell of Table 2.1 brings us to *policy analysis*: a focus on the "what" of policy for a specific client. Within this narrow perspective of policy analysis, its focus is exclusively normative: what policies promote socially valuable outcomes? It typically requires the identification of relevant social values and the development of appropriate policy goals for promoting them. It also typically requires the development of policy alternatives, prediction of their impacts on policy goals, and consideration of trade-offs across policy alternatives. However, in practice policy analysis usually considers instrumental as well as substantive goals. Almost always, analysts must consider the prospects for successful implementation; they usually also consider political feasibility. In doing so, they incorporate stakeholder analysis in varying degrees, resulting in the more broadly defined policy analysis that we consider in this book.

Policy analysis has similarities to familiar professions like medicine, law, and journalism. Like medicine, it involves diagnosis of problems and prescription of solutions for a specific client, leading some to classify it as a "clinical" profession.[7] Like both doctors and lawyers, policy analysts owe substantial responsibility to their clients. As we discuss in the next chapter, however, policy analysts also have duties to adhere to analytical integrity and promote the good society. Unlike lawyers who need only to argue their clients' side of cases, policy analysts have a responsibility to present all sides. If they argue only for their clients, then they become advocates rather than analysts. Even as advocates, however, like good lawyers, their preparation for arguing for their clients should include a balanced assessment if for no other reason than anticipating counter-arguments.

The comparison of policy analysis with journalism may at first seem strange. Journalists typically concern themselves with recent events; they are rarely called upon to make predictions about the future. When they write about public policy, the need to attract a wide readership often leads them to focus on the unusual and the sensational rather than the routine and the mundane. Narratives with victims, heroes, and villains catch readers' interest more effectively than nuanced discussions of competing social values. Their contribution to the political process, therefore, is more often introducing policy problems to the public agenda than providing systematic comparisons of alternative solutions. Nevertheless, policy analysts and journalists share several goals and constraints.

Tight deadlines drive much of journalists' work, especially in this electronic age. Because news quickly becomes stale, journalists often face the prospect of not being able to publish unless they make the next edition. Similarly, the advice of policy analysts, no matter how sophisticated and convincing, will be useless if it is delivered to clients after they have had to vote, issue regulations, or otherwise make decisions. Rarely will it be the case of better late than never.

Tight deadlines lead both journalists and policy analysts to develop similar strategies for gathering information. Files of background information and networks of knowledgeable people are often extremely valuable resources because they enable journalists to put events quickly in context. These resources play a similar role for policy analysts, but may also provide information useful for assessing the technical, political, and administrative feasibility of policy alternatives when time does not permit systematic investigation. Policy analysts, like journalists, wisely cultivate their information sources.

Finally, communication is a primary concern. Journalists must be able to put their stories into words that will catch and keep the interest of their readers. Policy analysts must do the same for their clients. Effective communication requires clear writing—analysts must be able to explain their technical work in language that can be understood by their clients. Also, because the attention and time of clients are scarce resources, writing must be concise and convincing to be effective.

Policy Analysis as a Profession

Until the 1980s, few of those actually doing policy analysis would have identified themselves as members of such a profession; even fewer filled positions labeled "policy analyst." Many employees who do policy analysis held, and continue to hold, positions as economists, planners, program evaluators, budget analysts, operations researchers, or statisticians. Over the past 30 years, however, policy analysis has emerged as an established profession.[8] Positions called "policy analyst" are now more common in government agencies, and are often filled by people who have been trained in schools of policy analysis. Many practicing analysts trained in a variety of disciplines are members of the Association for Public Policy Analysis and Management.[9] However, its members represent only a fraction of those actually practicing the craft of policy analysis.

Practicing policy analysts work in a variety of organizational settings, including federal, state, and local agencies and legislatures; consulting firms; research institutes and think tanks; trade associations and other organizations representing interest groups; and business and nonprofit corporations. Policy analysts can now be found in all the major industrialized countries, and schools of public policy can now be found in Europe and Asia as well as North America.[10] The way analysts practice their craft is greatly influenced by the nature of their relationships with their clients and by the roles played by the clients in the political process. Because these relationships and roles vary greatly across organizations, we should expect to see a wide range of analytical styles. Let us look at a few examples of organizational settings in which policy analysts ply their craft.

First, consider the U.S. federal government. Where are the policy analysts? Beginning with the executive branch, there are small but influential groups of analysts in the National Security Council and Domestic Policy staffs within the White House. As presidential appointees in politically sensitive positions, they generally share closely the philosophy and goals of their administration. Their advice concerns the political, as well as the economic and social, consequences of policy options. They often coordinate the work of policy analysts in other parts of the executive branch.

The Office of Management and Budget (OMB) and, to a lesser extent, the Council of Economic Advisers (CEA), also play coordinating roles within the federal government. Analysts in OMB are responsible for predicting the costs to the federal government of changes in policy and overseeing the issuing of major rules by federal regulatory agencies. They also participate in the evaluation of particular programs. The major role that OMB plays in the preparation of the administration budget gives its analysts great leverage in disputes with the federal agencies; it also often leads the analysts to emphasize budgetary costs over social costs and benefits. Analysts on the CEA do not play as direct a role in the budgetary process and, therefore, retain greater freedom to adopt the broad perspective of social costs and benefits. Without direct leverage over the agencies, however, their influence derives largely from the perception that their advice is based on the technical expertise of the discipline of economics.[11]

Policy analysts work throughout all federal agencies. In addition to small personal staffs, agency heads usually have analytical offices reporting directly to them.[12] These offices have a variety of names that usually include some combination of the words "policy," "planning," "administration," "evaluation," "economic," and "budget." For example, at various times the central analytical office in the Department of Energy has been called the Office of the Assistant Secretary for Policy and Evaluation and the Policy, Planning, and Analysis Office. Often, the heads of agency subunits have analytical staffs that provide advice and expertise relevant to their substantive responsibilities. Later in this chapter we briefly consider policy analysis in the Department of Health and Human Services (DHHS) to illustrate the sorts of functions analysts perform in federal agencies.

Policy analysts are also common in the legislative branch. Both Congress as a whole and its individual members serve as clients. Policy analysts work for Congress in the Government Accountability Office (GAO),[13] the Congressional Budget Office (CBO), and the Congressional Research Service (CRS). The analytical agendas of these offices are set primarily by the congressional leadership but sometimes also by individual members of Congress. Of course, members have their own personal staffs, including legislative analysts. Most of the analysis and formulation of legislation, however, is done by committee staffs that report to committee chairs and ranking minority members.[14] Committee staffers, often recruited from the campaign and personal staffs of members, must be politically sensitive if they are to maintain their positions and influence.

How influential is policy analysis in policy formation and choice in Congress? Based on his detailed study of communication surrounding four policy issues in the areas of health and transportation, David Whiteman concludes: “The results … indicate that policy analysis clearly does flow through congressional communication networks. In three of the four issues examined, analytic information played a significant role in congressional deliberations.”[15] Much of the communication takes place through discussions between congressional staffers and analysts in government offices and think tanks rather than as formal written reports.

Turning to state governments, we find a similar pattern. Governors and agency heads usually have staffs of advisers who do policy analysis.[16] Most states have budget offices that play roles similar to that of OMB at the federal level. Personal and committee staffs provide analysis in the state legislatures; in some states, such as California, the legislatures have offices much like the CBO to analyze budgetary and other impacts of proposed legislation. The Washington State Institute for Public Policy (WSIPP) is exemplary in providing the Washington State legislature with sophisticated analyses, often in the form of cost–benefit analyses.[17] A number of other states have adopted the WSIPP cost–benefit analysis tool with the assistance of a program sponsored by private foundations.[18]

At the county and municipal levels, legislative bodies rarely employ persons who work exclusively as policy analysts. Executive agencies, including budget and planning offices, usually do have some personnel whose major responsibility is policy analysis. Except in the most populous jurisdictions, however, most analysis is performed by employees with line or managerial duties. Consequently, they often lack the time, expertise, and resources for conducting analyses of great technical sophistication. Nevertheless, because they often have direct access to decision makers, and because they can often observe the consequences of their recommendations firsthand, policy analysts at the local level can find their work professionally gratifying despite the resource constraints they face.

What do public agencies do if their own personnel cannot produce a desired or mandated analysis? If they have funds available, then the agencies can purchase analysis from consultants. Local and state agencies commonly turn to consultants for advice about special issues, such as the construction of new facilities or major reorganizations, or to meet evaluation requirements imposed by intergovernmental grant programs. Federal agencies not only use consultants for special studies, but also as routine supplements to their own staff resources. In extreme cases, consulting firms may serve as “body shops” for government offices, providing the services of analysts who cannot be hired directly because of civil service or other restrictions.

The importance of the relationship between client and analyst is extremely apparent to consultants. Usually, consultants are paid to produce specific products. If they wish to be hired again, then they must produce analyses that the clients perceive as useful. Consultants who pander to the prejudices of their clients at the expense of analytical honesty are sometimes described as “hired guns” or “beltway bandits.” Consultants best able to resist the temptation to pander are probably those who have a large clientele, provide very specialized skills, or enjoy a reputation for providing balanced analysis—they will not suffer greatly from the loss of any one client and they will be able to find replacement business elsewhere if necessary.

Researchers in academia, think tanks, and policy research institutes also provide consulting services. Although their work is usually not directly tied to specific policy decisions, researchers at places like the RAND Corporation, the Brookings Institution, the American Enterprise Institute for Public Policy Research, the Cato Institute, the Urban Institute, Resources for the Future, the Institute for Defense Analyses, and the Institute for Research on Public Policy (Canada) sometimes do produce analyses of narrow interest for specific clients. It is often difficult in practice to determine whether these researchers better fit the policy analysis or the policy research paradigms presented above. The last two decades have seen a substantial growth in the number of think tanks––there are more than 1,800 think tanks in the United States with almost 550 located in or near Washington.[19] Many of the newer think tanks with strong ideological identifications, however, have predispositions toward particular policies that often interfere with the professional validity of the analyses they provide.

Finally, large numbers of analysts neither work for, nor sell their services to, governments. They often work in profit-seeking firms in industries heavily regulated by government, in trade associations and national labor unions concerned with particular areas of legislation, and in nonprofits that have public missions in their charters. For example, consider a proposal to make health insurance premiums paid by employers count as taxable income for employees. Private firms, trade associations, and labor unions would seek analysis to help determine the impact of the proposed change on the pattern and cost of employee benefits. The American Medical Association would seek analysis of the impact on the demand for physician services. Health insurance providers, such as Blue Cross and Blue Shield, commercial insurers, and managed care organizations, would want predictions of the effect of the change on the demand for their plans and the cost of medical care. These interests might also ask their analysts how to develop strategies for supporting, fighting, or modifying the proposal as it moves through the political process.

It should be obvious from our brief survey that policy analysts work in a variety of organizational settings on problems ranging in scope from municipal refuse collection to national defense. What sorts of functions do analysts actually perform in their organizations?

A Closer Look at Analytical Functions

We pointed out earlier that the nature of policy analysis varies widely. In subsequent chapters we set out a framework for doing comprehensive policy analysis—how an individual analyst should go about producing a structured analysis that assesses problems presented by clients and systematically compares alternatives for solving them. We think this is the best way to learn how to do policy analysis because it covers the range of functions that analysts commonly perform. By mastering it, analysts not only prepare themselves for performing the inclusive functions but also gain a useful framework for putting what they are doing into perspective.

As an example, we present a brief overview of some of the analytical functions identified by the U.S. Department of Health and Human Services (DHHS). We single out DHHS because it is a very large federal agency with responsibilities that demand the full range of analytical functions, and because DHHS has codified what it sees to be the important functions of its policy analysts.

DHHS is very large by any measure. It oversees many specialized agencies, such as the Food and Drug Administration, the National Institutes of Health, the Centers for Medicare & Medicaid Services, and Centers for Disease Control and Prevention, to name just a few. In fiscal year 2016, it administered spending of over $1 trillion, issued more grants than any other federal agency, and employed more than 80,000 full-time equivalents nationwide in its constituent units. As such, it is one of the largest and most complex bureaus in the world. DHHS is of such size and scope that the Office of the Secretary (OS), the central coordinating office for DHHS, itself employs approximately 2,400 people. The purpose of the OS includes providing independent advice and analysis concerning program issues, analyzing trade-offs among programs, and developing common policies across agencies. While much of what the OS does involves administration and monitoring, there is no clear separation of these tasks from policy analysis.

Although policy analysts can be found throughout DHHS, we focus on the Office of the Assistant Secretary for Planning and Evaluation (ASPE), because it has the clearest and most direct mandate for doing policy analysis.[20] (The Office of the Assistant Secretary, Management and Budget has closely related policy analysis responsibilities but with greater emphasis on budgetary and cost issues; the two offices often work together on policy analysis projects.)

ASPE analysts perform a variety of functions. An ASPE orientation document specifically alerts new analysts to four major functions that they will be likely to perform.[21] First, analysts play a “desk officer” function that involves coordinating policy relevant to specific program areas and serving as a contact for the line agencies within DHHS that have responsibilities in these areas. For example, a desk officer might cover biomedical research issues and work closely with analysts and other personnel at the National Institutes of Health. Desk officers serve as the eyes and ears of the department, "going out to the agency, talking with the staff about issues and options before they reach decision points, and knowing what issues are moving and what are not."[22] Desk officers are also expected to reach outside of DHHS to identify concerns and ideas from academics and those who deal with the programs in the field. By staying on top of issues, desk officers can provide quick assessments of proposed policy changes in their areas.

Second, analysts perform a policy development function. This is important to DHHS because ASPE resources "constitute some of the few flexible analytic resources in the Department."[23] Policy development often involves special initiatives within DHHS but can also be done through task forces that include personnel from other departments. These initiatives often result in policy option papers or specific legislative proposals.

Third, analysts perform a policy research and oversight function. "ASPE spends approximately $20 million a year in both policy research and evaluation funds" to carry out this core function.[24] It is important to emphasize that DHHS, like many other government agencies, contracts out a considerable amount of policy-relevant research. Therefore, analysts at ASPE are both consumers and producers of policy research and analysis. ASPE analysts also participate in reviews of the research plans of other agencies, help formulate and justify plans for allocating evaluation funds, and serve on agency panels that award research contracts and grants.

Fourth, analysts perform a "firefighting" function. Fires can be "anything from a request from the White House to review the statement of administration accomplishments on welfare reform … to preparing an instant briefing for congressional staff because a key committee is preparing to mark up a bill, to helping … [the] Office of the Secretary prepare for a meeting with a key outside group tomorrow."[25] The term *firefighting* conveys the urgency of the task—analysts drop whatever else they are doing until the fire is put out!

These four categories of functions illustrate the great variety of tasks that analysts are routinely called upon to perform. Some of these tasks are ongoing, while others are more episodic; some have short deadlines, others extend for long periods. Some are internal to the analysts' organization, others require interaction with external analysts and decision makers. Some involve topics of great familiarity, others present novel issues. What sorts of basic skills help analysts prepare for this diversity of tasks?

Basic Preparation for Policy Analysis

Policy analysis is as much art and craft as science.[26] Just as the successful portraitist must be able to apply the skills of the craft of painting within an aesthetic perspective, the successful policy analyst must be able to apply basic skills within a reasonably consistent and realistic perspective on the role of government in society. In order to integrate effectively the art and craft of policy analysis, preparation in five areas is essential.

First, analysts must know how to gather, organize, and communicate information in situations in which deadlines are strict and access to relevant people is limited. They must be able to develop strategies for quickly understanding the nature of policy problems and the range of possible solutions. They must also be able to identify, at least qualitatively, the likely costs and benefits of alternative solutions and communicate these assessments to their clients. Chapter 1 previewed these basic informational skills; Chapters 14 and 15 develop them in depth.

Second, analysts need a perspective for putting perceived social problems in context. When is it legitimate for government to intervene in private affairs? In the United States, the most basic normative answer to this question has usually been based on the concept of efficiency and more specifically *market failure*—a circumstance in which the pursuit of private interest does not lead to an efficient use of society's resources or a fair distribution of society's goods. But market failures, or widely shared normative claims for the desirability of social goals other than efficiency, such as greater equity in the distributions of economic and political resources, should be viewed as the only necessary conditions for appropriate government intervention. Sufficiency requires that the form of the intervention not involve consequences that would inflict greater social costs than social benefits. Identification of these costs of intervention is facilitated by an understanding of the ways collective action can fail. In other words, the analyst needs a perspective that includes *government failure* as well as market failure. The chapters in Parts II and III provide such a perspective. Chapters 4–7 analyze the various market failures and other rationales that have been identified; Chapter 8 discusses the systematic ways that government interventions tend to lead to undesirable social outcomes; Chapter 9 considers the interaction between market and government failures; and Chapters 10–13 set out conceptual foundations for developing policies to correct market and government failures. These chapters provide a "capital stock" of ideas for categorizing and understanding social problems and proposing alternative policies for dealing with them.

Third, analysts need technical skills to enable them to predict better and to assess more confidently the consequences of alternative policies. The disciplines of economics and statistics serve as primary sources for these skills. Although we introduce some important concepts from microeconomics, public finance, and statistics in the following chapters, those readers who envision careers in policy analysis would be well advised to take courses devoted to these subjects.[27] Even an introduction to policy analysis, however, should introduce the basics of cost–benefit analysis, the subject of Chapter 17.

Fourth, analysts must have an understanding of political and organizational behavior in order to predict, and perhaps influence, the feasibility of adoption and the successful implementation of policies. Also, understanding the worldviews of clients and potential opponents enables the analyst to marshal evidence and arguments more effectively. We assume that readers have a basic familiarity with democratic political systems. Therefore, practical applications of theories of political and organizational behavior are integrated with subject matter throughout the text, but particularly in the context of thinking about policy adoption and implementation (Chapters 11 and 12), organizational design (Chapter 13), and government failure (Chapter 8).

Finally, analysts should have an ethical framework that explicitly takes account of their relationships to clients. Analysts often face dilemmas when the private preferences and interests of their clients diverge substantially from their own perceptions of the public interest. We consider approaches to professional ethics for policy analysts in the next chapter.

For Discussion

1. The Legislative Analyst's Office, which functions as the "eyes and ears" of the California legislature, was founded in 1941. It served as a model for the federal Congressional Budget Office. Visit its Website (www.lao.ca.gov) to view its history and samples of its products. Would you expect the analysis produced by the Legislative Analyst's Office to be more or less politically neutral than analytical offices within the California executive branch?
2. Think tanks differ in a variety of ways, including their areas of specialization and the degree to which they advocate specific policies. Characterize the following think tanks after visiting their Websites: Cato Institute (www.cato.org), Fraser Institute (www.fraserinstitute.ca), Progressive Policy Institute (www.ppionline.org), RAND Corporation (www.rand.org), and Resources for the Future (www.rff.org).

Notes

1 Lasswell makes the distinction between "knowledge in" and "knowledge of" the policy process. Harold D. Lasswell, *A Pre-View of the Policy Sciences* (New York: American Elsevier, 1971), 1.

2 Peter M. Haas, "Introduction: Epistemic Communities and International Policy Coordination," *International Organization* 46(1), 1992: 1–35, at 3.

3 Theodore J. Lowi, "Four Systems of Policy, Politics, and Choice," *Public Administration Review* 32(4) 1972, 298–310.

4 James Q. Wilson, *The Politics of Regulation* (New York: Basic Books, 1980).

5 Ronald K. Mitchell, Bradley R. Agle, and Donna J. Wood, "Toward a Theory of Stakeholder Identification and Salience: Defining the Principle of Who and What Really Counts," *Academy of Management Review* 22(4) 1997, 853–86.

6 John M. Bryson, Gary L. Cunningham, and Karen J. Lokkesmoe, "What to Do when Stakeholders Matter: The Case of Problem Formulation for the African American Men Project of Hennepin County, Minnesota," *Public Administration Review* 62(5) 2002, 568–84.

7 Iris Geva-May, ed., *Thinking Like a Policy Analyst: Policy Analysis as a Clinical Profession* (New York: Palgrave Macmillan, 2005).

8 For an excellent overview, see Beryl A. Radin, *Beyond Machiavelli: Policy Analysis Reaches Midlife*, 2nd ed. (Washington, DC: Georgetown University Press, 2013).

9 Information about membership and annual conferences can be obtained at www.appam.org.

10 The Policy Press has launched a very informative series of edited volumes that describe and assess the practice of policy analysis in Australia, Brazil, Germany, Japan, Taiwan, and a growing number of other countries. For example, see F. K. M. van Nispen and Peter Scholten, eds., *Policy Analysis in the Netherlands* (Bristol UK: Policy Press, 2015).

11 Herbert Stein, "A Successful Accident: Recollections and Speculations about the CEA," *Journal of Economic Perspectives*, 10(3) 1996, 3–21.

12 For example, on the role of analysis at the State Department, see Lucian Pugliaresi and Diane T. Berliner, “Policy Analysis at the Department of State: The Policy Planning Staff,” *Journal of Policy Analysis and Management* 8(3) 1989, 379–94. See also Robert H. Nelson, “The Office of Policy Analysis in the Department of the Interior,” 395–410 in the same issue.

13 The General Accounting Office, the prior name of the GAO, and the Bureau of the Budget, the forerunner of OMB, were established in 1921 with the creation of an executive budget system. During much of its history, GAO devoted its efforts primarily to auditing government activities. In the late 1960s, however, GAO became a major producer of policy analysis in the form of program evaluations with recommendations for future actions. Because GAO must serve both parties and both legislative chambers, and because its reports are generally public, it faces stronger incentives to produce politically neutral analyses than OMB. For a comparative history of these “twins,” see Frederick C. Mosher, *A Tale of Two Agencies: A Comparative Analysis of the General Accounting Office and the Office of Management and Budget* (Baton Rouge: Louisiana State University Press, 1984).

14 See Carol H. Weiss, “Congressional Committees as Users of Analysis,” *Journal of Policy Analysis and Management* 8(3) 1989, 411–31. See also Nancy Shulock, “The Paradox of Policy Analysis: If It Is Not Used, Why Do We Produce So Much of It?” *Journal of Policy Analysis and Management* 18(2) 1999, 226–44.

15 David Whiteman, *Communication in Congress: Members, Staff, and the Search for Information* (Lawrence: University of Kansas Press, 1995), at 181.

16 For an excellent assessment of influence, see John A. Hird, *Power, Knowledge, and Politics: Policy Analysis in the States* (Washington, DC: Georgetown University Press, 2005).

17 See, for example, Steve Aos, Roxanne Lieb, Jim Mayfield, Marna Miller, and Annie Pennucci, *Benefits and Costs of Prevention and Early Intervention Programs* (Olympia: Washington State Institute for Public Policy, 2004).

18 Pew-MacArthur Results First Initiative, www.pewtrusts.org/en/projects/pew-macarthur-results-first-initiative.

19 Think Tanks and Civil Society Program, *2013 Global Go To Think Tank Index Report*. (Philadelphia: University of Pennsylvania, 2014), 21–23.

20 For a long-term view of analysis at ASPE, see George D. Greenberg, "Policy Analysis at the Department of Health and Human Services," *Journal of Policy Analysis and Management* 22(2) 2003, 304–7.

21 Assistant Secretary, Policy and Evaluation, "All about ASPE: A Guide for ASPE Staff," no date.

22 Ibid. E-1.

23 Ibid. E-2.

24 Ibid.

25 Ibid.

26 For a strong statement of this viewpoint, see Aaron Wildavsky, *Speaking Truth to Power: The Art and Craft of Policy Analysis* (Boston: Little, Brown, 1979), 385–406.

27 There are three reasons why a solid grounding in economics and statistics is important for the professional policy analyst: (1) the techniques of these disciplines are often directly applicable to policy problems; (2) researchers who use economic models and statistical techniques are important sources of policy research—the ability to interpret their work is therefore valuable; and (3) analytical opponents may use or abuse these techniques—self-protection requires a basic awareness of the strengths and limitations of the techniques.

3
Toward Professional Ethics

The policy analyst with a philosopher king as a client would be fortunate in several ways. The analyst could prepare advice with the knowledge that it would be thoughtfully evaluated on its merits by a wise leader who placed the welfare of the kingdom above considerations of private or factional interest. Good advice would be adopted and implemented solely on the word of the king, without resort to complicated political or organizational strategies. Thus, as long as the king truly had wisdom, benevolence, and power, the analyst could expect that only reasoned and reasonable differences of opinion would come between recommendations and action. In other words, the analyst would not have to fear conflict between the professional ideal of promoting the common good and the practical necessity of serving a client.

Although we often discuss policy analysis as if all clients were philosopher kings, reality is never so kind. History offers many examples of despots and few examples of consistently benevolent autocrats. Representative government seeks to limit the damage that can be done by abusive leaders by providing opportunities to replace them without resort to revolution. Yet, in regimes of representative government, many cooks contribute to the policy broth. The distribution of authority by constitution or tradition among elected officials, bureaucrats, legislators, and magistrates guarantees many their place at the kettle. Prevailing norms of democratic participation ensure that they receive a variety of advice and demands from their fellow citizens to whom they are accountable, either directly or indirectly, at the ballot box. Presidents and prime ministers may enjoy more favored positions than other participants but, except for very mundane or exceptional circumstances, even they generally lack authority to select and to implement policies by simple directive.[1] Even in political systems where the authority of the chief executive verges on the dictatorial, the limits of time and attention imposed by nature necessitate the delegation of discretion over many routine decisions to other officials.

Analysts must expect their clients to be players in the game of politics—players who not only have their own personal conceptions of "the good society" but who also must acknowledge the often narrow interests of their constituencies if they hope to remain in the game. The reality that, outside the classroom, policy analysis cannot be separated from politics has important practical and ethical implications. Analysis that ignores the interests of the client may itself be ignored; recommendations that ignore the interests of other key players are unlikely to be adopted or successfully implemented. In the extreme, if efficacy were the only professional value, "good" analysts would be those who helped their clients become better players in the game of politics. But other values, not always explicitly stated, lead to broader ethical considerations. Analysts should not only care that they influence policy but also that they do so for the better.

Much of the literature on ethics and public policy concerns the values that we should consider in attempting to select better policies.[2] It reminds analysts that no single value, such as economic efficiency, can provide an adequate basis for all public decision making. We focus in this chapter on professional ethics rather than the comparative merits of substantive policies, which is the subject of the remainder of this book. Our objective in the following sections is to sketch a framework for thinking about the ethical responsibilities of the professional policy analyst. To do so, we must pay attention to the nature of the relationships between analysts and clients and the various contexts in which they evolve.[3] Even absent explicit and universally accepted ethical guidelines, we will at least become acquainted with the most common analytical environments and the dilemmas they sometimes raise for practitioners.

Analytical Roles

Policy analysis, like life itself, forces us to confront conflicts among competing values. Often conflicts arise inherently in the substantive question being considered. For example, should a policy that would yield a great excess of benefits over costs for society as a whole be selected even if it would inflict severe costs on a small group of people? Our answers will depend on the relative weights we give to the values of efficiency (getting the greatest aggregate good from available resources) and equity (fairness in the way it is distributed). These values, along with others such as the protection of human life and dignity and the promotion of individual choice and responsibility, provide criteria for evaluating specific policy proposals.

Rather than focus on values from the unique perspectives of particular policy issues, here we consider values relevant to the general question of how analysts should conduct themselves as professional givers of advice. Three values seem paramount: *analytical integrity, responsibility to client*, and *adherence to one's personal conception of the good society*. Conflicts among these values raise important ethical issues for analysts.

To understand the nature of these values and the contexts in which they become important, we consider three conceptions of the appropriate role of the analyst.[4] Each role gives priority to one of the three values, relegating the remaining two to secondary status. We can anticipate, therefore, that none of the three roles provides an adequate ethical standard in its pure form in all circumstances. Our task will be to search for appropriate balance.

Objective technicians hold analytical integrity as their fundamental value. They see their analytical skills as the source of their legitimacy. The proper role for the analyst, in their view, is to provide objective advice about the consequences of proposed policies. Objective technicians feel most comfortable applying skills within recognized standards of good practice. Therefore, they often prefer to draw their tools from the disciplines of economics, statistics, and operations research, all of which employ well-established methods. They realize that they must often work under severe time constraints and data limitations. Nevertheless, they want to believe that researchers in the disciplines would approve of their work as methodologically sound under the circumstances.

Table 3.1 Three Views on the Appropriate Role of the Policy Analyst

	Fundamental Values		
	Analytical Integrity	*Responsibility to Clients*	*Adherence to One's Conception of Good*
Objective Technician	Let analysis speak for itself. Primary focus should be predicting consequences of alternative policies.	Clients are necessary evils; their political fortunes should be secondary considerations. Keep distance from clients; select institutional clients whenever possible.	Relevant values should be identified but trade-offs among them should be left to clients. Objective advice promotes good in the long run.
Client's Advocate	Analysis rarely produces definitive conclusions. Take advantage of ambiguity to advance clients' positions.	Clients provide analysts with legitimacy. Loyalty should be given in return for access to privileged information and to political processes.	Select clients with compatible value systems; use long-term relationships to change clients' conceptions of good.

	Fundamental Values		
	Analytical Integrity	*Responsibility to Clients*	*Adherence to One's Conception of Good*
Issue Advocate	Analysis rarely produces definitive conclusions. Emphasize ambiguity and excluded values when analysis does not support advocacy.	Clients provide an opportunity for advocacy. Select them opportunistically; change clients to further personal policy agenda.	Analysis should be an instrument for progress toward one's conception of the good society.

As asserted in Table 3.1, *objective technicians* view clients as necessary evils. Clients provide the resources that allow objective technicians to work on interesting questions. In return, clients deserve the most accurate predictions possible. The political fortunes of clients should take second place behind analytical integrity in the preparation, communication, and use of analyses. Analysts should try to protect themselves from interference by not becoming too closely associated with the personal interests of their clients. In general, they should select institutional clients, because such clients are likely to provide greater opportunities for preparing and disseminating objective analyses. For example, one is likely to face less interference with analytical integrity working for the Congressional Budget Office, which must be responsive to Congress as a whole as well as anticipate changes in partisan control, than working directly for a member of Congress who must run for reelection every two years.

The objective technician believes that values relevant to the choice of policies should be identified. When no policy appears superior in terms of all the relevant values, however, trade-offs among competing values should be left to the client rather than being implicitly imposed by the analyst. The analyst contributes to the good society, at least in the long run, by consistently providing unbiased advice, even when it does not lead to the selection of personally favored policies.

The *client's advocate* places primary emphasis on his or her responsibility to the client. He or she believes that analysts derive their legitimacy as participants in the formation of public policy from their clients, who hold elected or appointed office or who represent organized political interests. In return for access, clients deserve professional behavior that includes loyalty and confidentiality. Like physicians, analysts should "do no harm" to their clients; like attorneys, they should vigorously promote their clients' interests.

To some extent the client's advocate views analytical integrity in the same way attorneys view their responsibility in an adversarial system. Analysts have a primary responsibility never to mislead their clients through false statements or purposeful omissions. Once clients have been fully informed, however, analysts may publicly interpret their analyses in the best possible light for their clients. Because analysis rarely produces definitive conclusions, analysts can emphasize the possible rather than the most likely when doing so favors their clients. The client's advocate believes that analytical integrity prohibits lying, but it requires neither full disclosure of information nor public correction of misstatements by clients.

Client advocates must relegate their policy preferences to a secondary position once they make commitments to clients. Therefore, their selection of clients matters greatly. When analysts and clients share similar worldviews, less potential exists for situations to arise that require analysts to help promote policies that are inconsistent with their own conceptions of the good society. Upon discovering that their clients hold very different worldviews, analysts may nevertheless continue such relationships if they believe that they will be able to make their clients' outlooks more like their own over periods of extended service. Indeed, they may believe that they have a responsibility to try to change their client's beliefs before switching to new clients.

Issue advocates believe that analysis should be an instrument for making progress toward their conception of the good society. They focus on values inherent in policy outcomes rather than on values, like analytical integrity and responsibility to the client, associated with the actual conduct of analysis. They see themselves as intrinsically legitimate players in the policy process. They may also see themselves as champions for groups or interests, such as the environment, the poor, or the victims of crime, that they believe suffer from underrepresentation in the political process.

Issue advocates select clients opportunistically. Clients unable or unwilling to promote the advocate's personal policy agendas should be abandoned for clients who can and will. Analysts owe their clients only those duties spelled out in the contractual arrangements defining the relationships; loyalty to one's conception of the good society should take priority over loyalty to any particular client.

Like the client's advocate, the issue advocate believes in taking advantage of analytical uncertainty. When analysis does not support one's policy preferences, the issue advocate questions the simplifying assumptions that must inevitably be employed in dealing with complex issues, or challenges the choice of criteria used to assess alternatives. (The latter will almost always be a possible strategy when one does not agree with conclusions.) Though issue advocates desire the respect of other analysts, especially when it contributes to effectiveness, they may be willing to sacrifice respect to obtain important policy outcomes.

Value Conflicts

One can imagine each of these extreme roles being ethically acceptable in specific circumstances. For example, analysts on the White House staff enjoy privileged positions with respect to information and political access. An important factor in their selection was, undoubtedly, their perceived loyalty to the president. In accepting these positions, they were implicitly if not explicitly committing themselves to a high degree of discretion in confidential matters. Except in the most extreme cases where failure to act would lead with reasonable certainty to significant violations of human rights or constitutional trust, honoring confidentiality and otherwise behaving as a client advocate seem to be ethically defensible. In contrast, a consultant hired by the Nuclear Regulatory Commission to analyze the risks associated with alternative policies for nuclear waste disposal might appropriately act as an objective technician, placing analytical integrity above the political interests of the commission. One might even argue that the consultant has an ethical duty to speak out publicly if the commission were to misrepresent the study to obtain a political outcome radically different from that which would otherwise have resulted.

In general, however, the analyst need not adopt any of the three roles in its extreme form. Rather than select one of the three fundamental values as dominant and sacrifice the other two as circumstances demand, the analyst should attempt to keep all three under consideration. The ethical problem, then, involves deciding how much of each value can be sacrificed when conflicts arise.

In any situation, the range of ethical behavior will be bounded by the minimal duties the analyst owes to each of the values. The development of professional ethics, either collectively or individually, may be viewed as an attempt to discover these minimal duties. In the discussion that follows, we consider some of the common situations in which value conflicts arise and minimal duties must be determined. We begin by considering the range of actions the analyst has available for responding to severe conflicts in values.

Responses to Value Conflicts: Voice, Exit, and Disloyalty

The most serious ethical conflicts for policy analysts usually pit responsibility to the client against other values. A variety of factors complicate ethical judgment: continued access to the policy issue, the status of current and future employment, the personal trust of the client, and the analyst's reputation. Many of these factors involve implications that go well beyond the particular ethical issue being considered. For example, loss of employment directly affects the economic and psychological well-being of analysts and their families, as well as the sort of advice that will be heard on the issue at hand. It may also affect the advice about similar issues that will be offered in their organizations in the future. We must be careful, therefore, to look for consequences beyond the particular issue at stake.

So far we have spoken of the analyst as if he or she were the direct employee of the client. Some analysts, such as consultants reporting directly to project managers or political appointees on the personal staffs of administrators and legislators, have clearly defined persons as clients. Analysts usually have immediate supervisors who can generally be thought of as clients. These supervisors, however, often operate in organizational hierarchies and, therefore, often have their own clients, who will also be consumers of the analysts' advice. Limiting the definition of the client to the immediate supervisor would unreasonably absolve analysts from responsibility for the ultimate use of their products. At the same time, we do not want to hold analysts accountable for misuse totally beyond their control. For our purposes, we consider the client to be the highest-ranking superior who receives predictions, evaluations, or recommendations attributable to the analyst. Thus, an analyst working in a bureau may have different persons as clients at different times. Sometimes the client will be the immediate supervisor; other times the client will be a higher-ranking official in the bureau.

Note that we have purposely adopted a narrow, instrumental conception of the client. There is some temptation to look for an ultimate client: analysts themselves as moral persons, the social contract as embodied in the constitution, or the public interest as reflected in national laws.[5] To do so, however, would assume away the essence of the professional role. Instead, we see personal morality, the constitution, and laws as the sources of other values that often conflict with responsibility to the client.[6]

What are the possible courses of action for analysts when demands by their clients come in conflict with their sense of analytical integrity or their conception of the good society? We can begin to answer this question by considering the concepts of voice and exit developed by Albert O. Hirschman. In his book *Exit, Voice, and Loyalty*, Albert Hirschman explores how people can react when they are dissatisfied with the organizations in which they participate.[7] They may exercise voice by working to change the organization from within, or they may simply exit, leaving the organization for another. For example, parents dissatisfied with the quality of education provided by their local school district might exercise voice by attending school board meetings or by standing for election to the board. Alternatively, they may put their children in private schools or move to another community with a better school district. In Hirschman's framework, loyalty helps determine how much voice is exercised before exit is chosen. Attachment to the community and commitment to public education, for example, will influence parents' choice between voice and exit.

We find it useful to use Hirschman's concepts of voice and exit and to add a third concept: disloyalty. An action is disloyal when it undercuts the political position or policy preferences of the client. Note that we thus abandon Hirschman's use of "loyalty." Rather than being a contributing factor to the choice between voice and exit, we specify loyalty as another dimension of action.

Analysts can exercise various combinations of voice, exit, and disloyalty when they confront value conflicts. The logical possibilities are presented in Figure 3.1, where voice, exit, and disloyalty are represented by circles. Actions involving more than one of the dimensions are represented by intersections of circles. For example, we label voice alone as "protest," while "leak" combines protest with disloyalty. We specify seven different actions for purposes of discussion.

Consider the following situation: You work in a government agency as a policy analyst. You have just been assigned the job of developing an implementation strategy for a policy that you believe is bad. After careful deliberation, you decide that the policy is sufficiently bad that you believe it would be morally wrong for you simply to follow orders. Under what conditions might you feel ethically justified in choosing each of the actions listed in Figure 3.1?

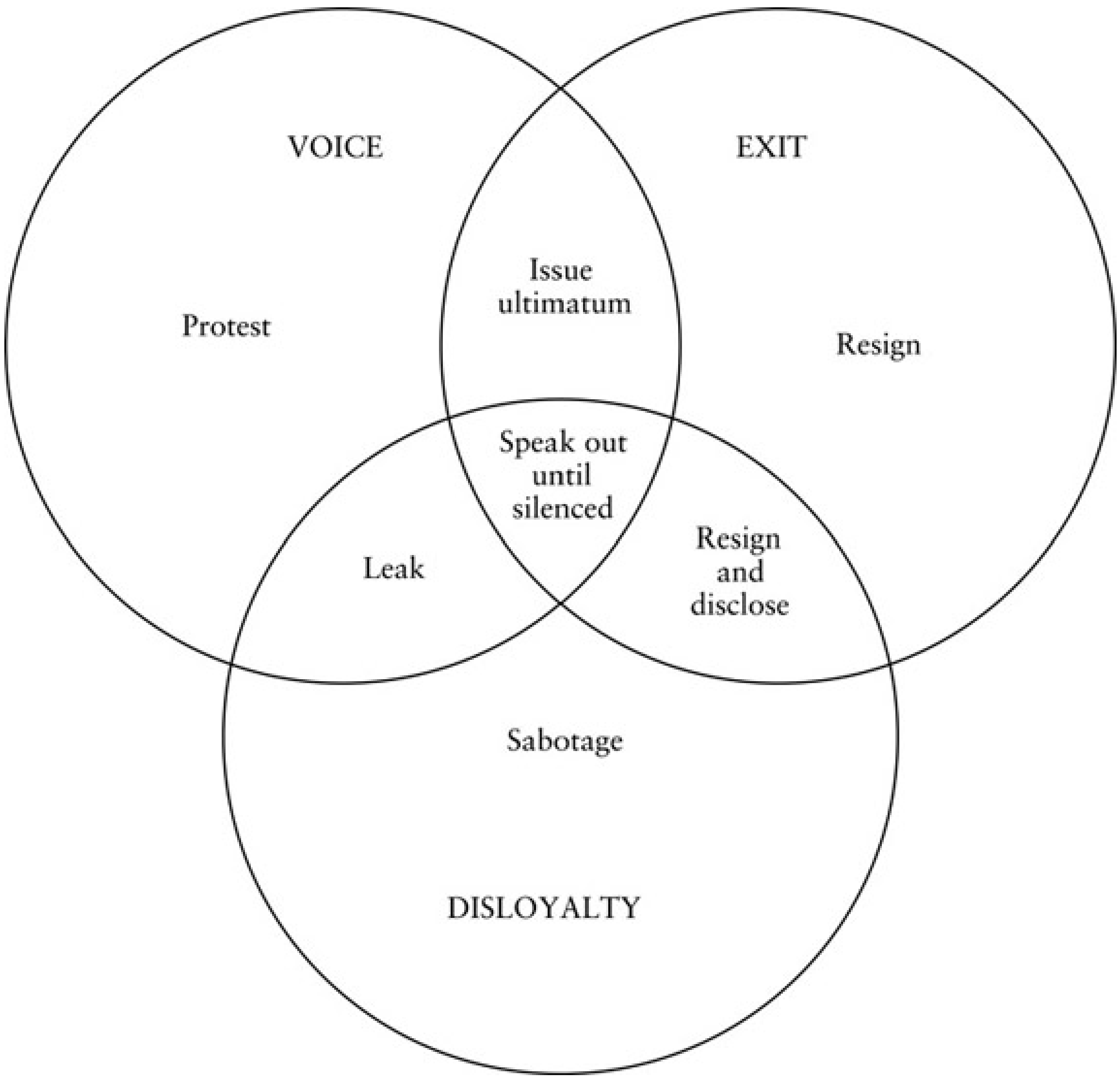


Figure 3.1 Alternative Responses to Value Conflicts

You might try to change the policy through *protest* within the agency. You would probably begin by informally discussing your objections to the policy with your supervisor. If your supervisor lacks either the inclination or the authority to reverse the policy, then you might next make your objections formally through memoranda to your supervisor, your supervisor's supervisor, and so forth, until you reach the lowest-ranking official with authority to change the policy. At the same time, you might speak out against the policy at staff meetings whenever you have the opportunity. You might also request that the assignment be given to someone else, not only because you would feel morally absolved if someone else did it, but because the request helps to emphasize the intensity of your objections. At some point, however, you will have exhausted all the avenues of protest within the agency that are recognized as legitimate.

Although you remained loyal to the agency, your protest probably involved personal costs: the time and energy needed to express your opinions, the personal offense of your superiors, perhaps the loss of influence over the policy in dispute, and the potential loss of influence over other issues in the future. If you were successful in reversing the policy, then at least you can take comfort in having disposed of your ethical problem. If you were unsuccessful, then you must make a further assessment by comparing the values you can achieve by remaining loyal to the agency with the values you give up by participating in the implementation of the bad policy.

A very different approach would be to *resign* when asked to prepare the implementation strategy.[8] You decide that contributing to implementation would be so ethically objectionable that it justifies your leaving the agency. Your personal costs will depend largely on your employment opportunities outside the agency. If you are highly skilled and have a good reputation, then it may be possible for you to move directly to a comparable position. If you are less skilled and not well regarded in the professional network, then you may face unemployment or underemployment for a time.

But what are the ethical implications of your action? Only if your skills were essential to implementation would resigning stop the bad policy. If the policy goes forward, then the ethical value of your resignation is questionable. Although you were able simultaneously to remain loyal to the agency and to avoid contributing directly to the implementation, you forfeited whatever influence you might have had within the agency over the policy. You also may have betrayed the terms of your employment contract as well as some of the personal trust placed in you by your superiors and colleagues, and you may have jeopardized other worthy projects within the agency by withdrawing your contributions. If you believe either that the policy is very bad or you would have had a good prospect of overturning it from within the agency, then running away by resigning seems to lack fortitude.

Combining voice with the threat of exit by *issuing an ultimatum* is likely to be ethically superior to simply resigning. After employing the various avenues of protest within the agency, you would inform your superior that if the policy were not reversed you would resign. Of course, you must be willing to carry out your threat, as well as bear the costs of greater personal animosity than you would face from simple resignation. You gain greater leverage in your protest against the policy, but you lose influence over future decisions if you actually have to execute your threat.

Beyond your personal decision to resign, there may be a larger social issue at stake. Why do you find the policy so objectionable while others approve of it? Perhaps the answer is that you have a better developed ethical sense; you are more principled. Is it good for society in the long run to have people such as you leave important analytical positions?[9] On the other hand, the reason for disagreement may be that both you and your superiors hold morally justifiable values that happen to conflict. Although we may have some concern about maintaining diversity within our public agencies, the danger of selective attrition seems less serious when it results from legitimate differences of moral opinion rather than from a clash between expediency, say, and basic principles.

Now consider actions that involve disloyalty to your client. You might *leak* your agency's plans to a journalist, congressman, interest group leader, or other person who can interfere with them.[10] You are taking your protest outside the agency, and doing so surreptitiously. Even if you are not a pure Kantian, anytime you do not act openly and honestly you should scrutinize closely the morality of your actions. Further, an important moral tenet is that one take responsibility for one's actions.[11] By acting covertly, you hope to stop the bad policy without suffering any adverse personal consequences from your opposition beyond the moral harm you have done to yourself by betraying the trust of your client and by acting dishonestly.

You should not view the violation of confidentiality, by the way, solely as a betrayal of personal trust. Confidentiality often contributes to organizational effectiveness. The expectation of confidentiality encourages decision makers to seek advice beyond their closest and most trusted advisers and to consider potentially desirable alternatives that would attract political opposition if discussed publicly.[12] Your decision to violate confidentiality has implications not just for your client but also for the expectations others have about the confidentiality they will enjoy when considering good as well as bad policies.

You can at least avoid the dishonesty by speaking out publicly. One possibility is that you *resign and disclose* your former client's plans to potential opponents. Although you are being honest and taking responsibility for your actions, disclosure, by violating the confidentiality you owe to your client, is still disloyal. You also forfeit the opportunity of continuing your protest within the agency. Another possibility is that you *speak out until silenced.* Such whistle-blowing keeps you in your agency for at least a while. Your agency will probably move quickly to exclude you from access to additional information that might be politically damaging. You must expect that eventually you will be fired or, if you enjoy civil service protection, exiled to some less responsible assignment that you will ultimately wish to leave.[13] Therefore, your approach combines voice and disloyalty with eventual exit.

When is whistle-blowing likely to be ethically justified? Peter French proposes four necessary conditions: First, you must exhaust all channels of protest within your agency before bringing your objections to the attention of the media, interest groups, or other units in the government. Second, you must determine that "procedural, policy, moral, or legal bounds have been violated." Third, you must be convinced that the violation will "have demonstrable harmful immediate effects upon the country, state, or citizens."

Fourth, you must be able to support specific accusations with "unequivocal evidence."[14]

French's conditions limit justifiable whistle-blowing to fairly exceptional circumstances. We might question whether they should all be viewed as necessary. For example, if great harm were at stake, we might view whistle-blowing as ethical even if the evidence brought forward falls short of unequivocal. We should also recognize that the conditions call for great judgment on the part of the potential whistle-blower, especially with respect to anticipating the harmful effects of inaction, and therefore constitute only general guidelines. Nevertheless, they seem appropriately to demand a careful weighing of all values, including loyalty.

Consider again the appropriateness of leaking. In addition to whatever conditions you believe justify whistle-blowing, you must also have a moral reason for acting covertly. In some extreme situations, perhaps involving the reporting of criminal acts in democracies or the supporting of human rights in totalitarian states, you might feel justified in acting covertly because your life or that of your family would be jeopardized by open protest. You might also be justified acting covertly if you were convinced that you could prevent serious harm that might occur in the future by remaining in your position.

Finally, you might consider *sabotage*—disloyalty without voice or exit. In designing the implementation plan for the policy you abhor, you might be able to build in some subtle flaw that would likely force your agency to abandon implementation at some point. For example, you might select a pilot site in the district of a powerful congresswoman who will strongly oppose the policy once it becomes apparent.[15] But such sabotage is morally suspect not only because it involves covert action but also because it operates through obstruction rather than persuasion. Only the most extreme conditions, including all those needed to justify leaking plus the absence of any reasonable avenues for protest, justify sabotage. It is hard to imagine situations in democratic regimes that produce these conditions.

Some Examples of Value Conflicts

Clients, because they have political interests related to their own policy preferences, the missions of their agencies, or their own personal advancement, may refuse to accept the truthful reports of their analysts. In some situations, clients may put pressure on analysts to "cook up" different conclusions or recommendations. In other situations, the clients may simply misrepresent their analysts' results to other participants in the decision-making process. What are the minimal duties of analysts in these situations?

Demands for Cooked Results

Most analysts desire at least the option to act as objective technicians. Faced with the task of evaluating alternative courses of action, they want the freedom to make reasonable assumptions, apply appropriate techniques, report their best estimates, and make logical recommendations. Unfortunately, clients sometimes hold strong beliefs that lead them to reject their analysts' findings, not on the basis of method but solely on the basis of conclusions. If the client simply ignores the analysis, then the analyst will undoubtedly be disappointed but generally faces no great ethical problem—the analyst is simply one source of advice and not the final arbiter of truth. The ethical problem arises when the client, perhaps feeling the need for analytical support in the political fray, demands that the analyst alter the work to reach a different conclusion. Should the analyst ever agree to "cook" the analysis so that it better supports the client's position?

A purist would argue that analytical integrity requires refusal; issue an ultimatum and resign if necessary. Can less-than-complete refusal ever be ethical?

We should keep in mind that, because analysis involves prediction, analysts rarely enjoy complete confidence in their conclusions. Careful analysts check the sensitivity of their results to changes in critical assumptions and convey the level of confidence they have in their conclusions to their clients. We can imagine analysts developing plausible ranges of results. For example, although the analyst believes that the cost of some program is likely to be close to $10 million, the most conservative assumptions might lead to an estimate of $15 million and the most optimistic assumptions to an estimate of $5 million. After making the range and best estimate clear to the client, would it be ethical for the analyst to prepare a version of the analysis for public distribution that used only the most optimistic assumptions?

Analysts who view themselves as a client's advocate might feel comfortable preparing the optimistic analysis for public use; those who see themselves as issue advocates might also if they share their client's policy preferences. After all, analysis is only one of many political resources and it rarely encompasses all the relevant values. For analysts viewing themselves as objective technicians, however, the question is more difficult. Limiting the analysis to optimistic assumptions violates their conception of analytical integrity: in honest analysis, the assumptions drive the results rather than vice versa. Nevertheless, objective technicians may feel justified in going along if they are confident that their client's political opponents will draw attention to the slanted assumptions.[16] When objective technicians believe that the aggregate of analysis reaching the political forum will be balanced, their acquiescence appears less serious in terms of ethics and more serious in terms of professional reputation.

Indeed, if we focus solely on consequences, might not analysts have a responsibility to slant their analysis to counter the slanted analyses of others? Imagine that the person making the final decision lacks either the time or the expertise to evaluate the technical validity of the analyses that are presented. Instead, the decision maker gives the results of each analysis equal weight. In our example, the final decision would be based on the average of the cost estimates presented by the various analysts. If one analyst gives a pessimistic estimate and another gives a realistic estimate, then the final decision will be biased toward the pessimistic. If the second analyst gives an optimistic estimate instead, then the final decision may be less biased. The broader consequences of compromising the procedural value of analytical integrity, however, may be to increase the professional acceptability of slanted analyses, and thus make it less likely that analysts will adopt the role of neutral technician in the future. Perhaps attacking the methodology of the slanted analysis directly rather than counter slanting, even if less effective for the issue at hand, would be better in the long run from the perspective of the social role of the professional policy analyst.

Misrepresentation of Results

Analysts have less ethical room in which to maneuver when their clients try to force them out of the range of the plausible. Defense of analytical integrity would seem generally to require protest backed up with the threat of resignation. The analysts' predicament, however, becomes much more complicated when their clients do not actually try to force them to cook up results but, rather, misrepresent what they have already done.

An analyst facing such misrepresentation is in a position similar to that of the defense attorney in a criminal case in which the client insists on being given the opportunity to commit perjury as a witness. By actively participating in the perjury, the attorney would be clearly violating his or her responsibility as an officer of the court. A more interesting problem arises if the client switches attorneys, conceals the truth, and then commits the perjury. Hearing of the testimony, the first attorney knows that perjury has been committed. Must he or she inform the court? One value at stake is the integrity of judicial fact finding. Another is the confidentiality of the communication between defendant and attorney that encourages defendants to be truthful so that their attorneys can give them the most vigorous defense. Although there seems to be a consensus in the U.S. legal profession that actually participating in perjury is unethical, there does not appear to be a consensus about the responsibility of attorneys when they know that former clients are committing perjury.[17]

Confidentiality probably plays a more important social role in the relationship between defense attorney and defendant than between analyst and client. The former contributes to a system of justice that rarely convicts or punishes the innocent; the latter to more inquisitive and open public officials. Further, the public official's obligation to honesty arises from a public trust as well as private virtue so that public dishonesty, unjustified by other overriding values, lessens the force of confidentiality. Therefore, the analyst's ethical burden seems to go beyond refusal to participate actively in the misrepresentation of the analysis.

Before taking any action, however, the analyst should be certain that the misrepresentation is intentional. Usually this involves confronting the client privately. Upon hearing the analyst's concern, the client may voluntarily correct the misrepresentation through private communication with relevant political actors or other remedial action. The client might also convince the analyst that some other value, such as national security, justifies the misrepresentation. If the analyst becomes convinced, however, that the misrepresentation is both intentional and unjustified, then the next step (following the guidelines for whistle-blowing) should be to determine the amount of direct harm that will result if the misrepresentation is left unchallenged. If little direct harm is likely to result, then resignation alone may be ethically acceptable. If the direct harm appears substantial, then the analyst bears a responsibility to inform the relevant political actors as well.

Ethical Code or Ethos?

Professions often develop ethical codes to guide the behavior of their members. The codes typically provide guidelines for dealing with the most common ethical predicaments faced by practitioners. The guidelines usually reflect a consensus of beliefs held by members of professional organizations.[18] Established and dominant professional organizations with homogeneous memberships enjoy the best prospects for developing ethical codes that provide extensive and detailed guidance.[19] Although a professional organization for policy analysts exists (the Association for Public Policy Analysis and Management), it is still relatively small and seeks to serve a very diverse membership with strong ties to other, more established, professions. Not surprisingly, it has not yet tried to develop an ethical code. Even when it becomes more established, the great diversity of its members and the organizational contexts in which they work suggest the difficulty of developing a code that directly addresses a wide range of circumstances.[20]

Students of the policy sciences, however, have suggested some general guidelines that deserve consideration. For example, Yehezkel Dror proposes that policy scientists not work for clients who they believe have goals that contradict the basic values of democracy and human rights, and that they should resign rather than contribute to the realization of goals with which they fundamentally disagree.[21] Obviously, the analyst who chooses only clients with similar worldviews and value systems is less likely to face conflicts between the values of responsibility to client and adherence to one's conception of good than the analyst who is less selective. Unfortunately, analysts often find themselves in situations where selectivity is impractical. All analysts face the problem of inferring the values and goals of potential clients from limited information; in addition, analysts employed in government agencies may find themselves working for new clients when administrations change. We have already discussed the reasons why resignation is not always the most ethical response to value conflicts between analysts and clients.

Most of Dror's other proposals seem relevant to policy analysis. For example, he proposes that clients deserve complete honesty, including explicated assumptions and uncensored alternatives, and that analysts should not use their access to information and influence with clients to further their own private interests. But these sorts of admonitions would follow from the moral system most of us would accept as private persons. In fact, some would argue that the moral obligations in most professions are not strongly differentiated from those of the nonprofessional.[22] A reasonable approach to professional ethics for policy analysts, therefore, may be to recognize responsibility to the client and analytical integrity as values that belong in the general hierarchy of values governing moral behavior.

Rather than wait for a code of ethics, perhaps we should, as Mark Lilla argues, work toward an ethos for the policy analysis profession.[23] As teachers and practitioners of policy analysis, we should explicitly recognize our obligations to protect the basic rights of others, to support our democratic processes as expressed in our constitutions, and to promote analytical and personal integrity.[24] These values should generally dominate our responsibility to the client in our ethical evaluations. Nevertheless, we should show considerable tolerance for the ways our clients choose to resolve difficult value conflicts, and we should maintain a realistic modesty about the predictive power of our analyses.

For Discussion

1. Imagine that you are a policy analyst working as a budget analyst for your state's department of education. You have made what you believe to be a fairly accurate prediction of the cost of a statewide class-size reduction program for elementary schools. You just discovered that your supervisor is planning to testify before a committee of the state assembly that the cost would be less than half of what you predicted. What factors would you consider in deciding on an ethical course of action?
2. Imagine that you are an analyst in a public agency. Think of a scenario in which you would feel compelled to resign. Avoid extreme scenarios; instead, think of a situation that would just push you over the edge. (Thanks to Philip Ryan, "Ethics and Resignation: A Classroom Exercise," *Journal of Policy Analysis and Management* 22(2) 2003, 313–18.)

Notes

1 Commenting on the U.S. executive, Richard E. Neustadt concludes, "Command is but a method of persuasion, not a substitute, and not a method suitable for everyday employment." Richard E. Neustadt, *Presidential Power* (New York: John Wiley & Sons, 1980) at 25.

2 See, for example: Charles W. Anderson, "The Place of Principles in Policy Analysis," *American Political Science Review* 74(3) 1979, 711–23; Robert E. Goodin, *Political Theory & Public Policy* (Chicago: University of Chicago Press, 1982); Peter G. Brown, "Ethics and Policy Research," *Policy Analysis* 2(2) 1976, 325–40; Joel L. Fleishman and Bruce L. Payne, *Ethical Dilemmas and the Education of Policymakers* (New York: Hastings Center, 1980); Douglas J. Amy, "Why Policy Analysis and Ethics Are Incompatible," *Journal of Policy Analysis and Management* 3(4) 1984, 573–91; and William T. Bluhm and Robert A. Heineman, *Ethics and Public Policy: Method and Cases* (Upper Saddle River, NJ: Prentice Hall, 2006).

3 For the contrast between a genuine discourse about values in a consensual environment and the potential manipulation of this discourse in adversarial processes, see Duncan MacRae, Jr., "Guidelines for Policy Discourse: Consensual versus Adversarial," in Frank Fischer and John Forester, eds., *The Argumentative Turn in Policy Analysis and Planning* (Durham, NC: Duke University Press, 1993), 291–318.

4 Our approach here benefits from Arnold J. Meltsner, *Policy Analysts in the Bureaucracy* (Berkeley: University of California Press, 1976), 18–49, who developed a classification of styles to understand better how analysis is actually practiced, and from Hank Jenkins-Smith, "Professional Roles for Policy Analysts: A Critical Assessment," *Journal of Policy Analysis and Management* 2(1) 1982, 88–100, who developed the three roles we use.

5 Many writers have chosen to approach professional ethics with the question: Who is the real client? See, for example, E. S. Quade, *Analysis for Public Decisions* (New York: American Elsevier, 1975), 273–75.

6 John A. Rohr argues that public officials have a responsibility to inform their actions by studying the constitutional aspects of their duties through relevant court opinions and

the substantive aspects through legislative histories. John A. Rohr, "Ethics for the Senior Executive Service," *Administration and Society* 12(2) 1980, 203–16; and *Ethics for Bureaucrats: An Essay on Law and Values* (New York: Marcel Dekker, 1978).

7 Albert O. Hirschman, *Exit, Voice, and Loyalty* (Cambridge, MA: Harvard University Press, 1970).

8 For an overview of the practical and ethical issues, see J. Patrick Dobel, "The Ethics of Resigning," *Journal of Policy Analysis and Management* 18(2) 1999, 245–63.

9 For an elaboration of this point, see Dennis F. Thompson, "The Possibility of Administrative Ethics," *Public Administration Review* 45(5) 1985, 555–61.

10 In our discussion, *leaking* refers to the sharing of confidential information with the intention of undermining the client's decision or political position. The sharing of confidential information can also be instrumental in good analysis and to furthering the client's interest in systems where information is decentralized. Analysts may be able to increase their efficacy by developing relationships with their counterparts in other organizations—the exchange of information serves as the instrumental basis of the relationships. For a discussion of the importance of these professional relationships in the U.S. federal government, see William A. Niskanen, "Economists and Politicians," *Journal of Policy Analysis and Management* 5(2) 1986, 234–44. Even when the analyst believes that the revelation is instrumental to the client's interests, however, there remains the ethical issue of whether the analyst should take it upon his or her self to break the confidence.

11 We might think of leaking as a form of civil disobedience, a type of protest often considered ethically justified. The classification fails, however, because most definitions of civil disobedience include the requirement that the acts be public. See Amy Gutmann and Dennis F. Thompson, eds., *Ethics & Politics* (Chicago: Nelson-Hall, 1984), 79–80. For a thoughtful discussion of personal responsibility in the bureaucratic setting, see Dennis F. Thompson, "Moral Responsibility of Public Officials: The Problem of Many Hands," *American Political Science Review* 74(4) 1980, 905–16.

12 For a discussion of this point and whistle-blowing, see Sissila Bok, *Secrets: On the Ethics of Concealment and Revelation* (New York: Pantheon, 1982), 175, 210–29.

13 Whistle-blowing is likely to be personally and professional costly. Federal programs intended to provide financial compensation to whistle-blowers may not fully compensate these costs and, further, may induce socially undesirable behaviors in the organizations covered. See Peter L. Cruise, "Are There Virtues in Whistleblowing? Perspectives from Health Care Organizations," *Public Administration Quarterly* 25(3/4) 2001/2002, 413–35.

14 Peter A. French, *Ethics in Government* (Englewood Cliffs, NJ: Prentice Hall, 1983), 134–37.

15 See Rosemary O'Leary, "Guerrilla Employees: Should Managers Nurture, Tolerate, or Terminate Them?" *Public Administration Review* 70(1) 2010, 8–19; and Janet P. Near and Marcia P. Miceli, "After the Wrongdoing: What Managers Should Know about Whistleblowing," *Business Horizons* 59(1) 2016, 105–14.

16 More generally, this example suggests that the appropriate role for the analyst will depend on the policy environment. In closed fora where the analysis is most likely to be decisive, the role of neutral technician seems most socially appropriate. In more open fora where all interests are analytically represented, advocacy may be the most socially appropriate role. For a development of this line of argument, see Hank C. Jenkins-Smith, *Democratic Politics and Policy Analysis* (Pacific Grove, CA: Brooks/Cole, 1990), 92–121.

17 For general background, see Phillip E. Johnson, *Criminal Law* (St. Paul, MN: West, 1980), 119–32.

18 For empirical assessments of the degree of consensus over what constitutes ethical behavior within two policy-related professions, see Elizabeth Howe and Jerome Kaufman, "The Ethics of Contemporary American Planners," *Journal of the American Planning Association* 45(3) 1979, 243–55; and James S. Bowman, "Ethics in Government: A National Survey of Public Administrators," *Public Administration Review* 50(3) 1990, 345–53.

19 The American Society for Public Administration adopted a general set of moral principles that evolved into a code of ethics for members in 1984. The code, which was revised in 1994, can be found on the back cover of *Public Administration Review*. It provides specific admonitions relevant to policy analysts under five general headings:

Serve the Public Interest, Respect the Constitution and the Law, Demonstrate Personal Integrity, Promote Ethical Organizations, and Strive for Professional Excellence.

20 For a discussion of some of the problems of developing an ethical code, see Guy Benveniste, "On a Code of Ethics for Policy Experts," *Journal of Policy Analysis and Management* 3(4) 1984, 561–72, which deals with the conduct of scientists and others who provide expert advice on policy questions.

21 Yehezkel Dror, *Designs of Policy Science* (New York: American Elsevier, 1971), 119.

22 See, for example, Alan H. Goldman, *The Moral Foundations of Professional Ethics* (Totowa, NJ: Rowman and Littlefield, 1980).

23 Mark T. Lilla, "Ethos, 'Ethics,' and Public Service," *Public Interest* (63) 1981, 3–17.

24 See J. Patrick Dobel, "Integrity in the Public Service," *Public Administration Review* 50(3) 1990, 354–66, for a discussion of commitments to regime accountability, personal responsibility, and prudence as moral resources for exercising discretion.

Part II
Conceptual Foundations for Problem Analysis

4
Efficiency and the Idealized Competitive Model

Collective action enables societies to produce, distribute, and consume a great variety and abundance of goods. Much collective action arises from voluntary agreements among people—within families, private organizations, and exchange relationships. The policy analyst, however, deals primarily with collective action involving the legitimate coercive powers of government: public policy encourages, discourages, prohibits, or prescribes private actions. Beginning with the premise that individuals generally act in their own best interest, or at least what they perceive as their self-interest, policy analysts have to provide rationales for any governmental interference with private choice. The task applies to the evaluation of existing policies as well as to new initiatives. As our case chapters show, it is an essential step in any comprehensive analysis and will often provide the best initial insight into complex situations.

Our approach to classifying rationales for public policy begins with the concept of a perfectly competitive economy. One of the fundamental bodies of theory in modern economics deals with the properties of idealized economies involving large numbers of *profit-maximizing firms* and *utility-maximizing consumers*. Under certain assumptions, the self-motivated behaviors of these economic actors lead to patterns of production and consumption that are efficient in the restricted sense that it would not be possible to change the patterns in such a way so as to make some person better off without making some other person worse off.

Economists recognize several commonly occurring circumstances of private choice, referred to as *market failures*, that violate the basic assumptions of the idealized competitive economy and, therefore, interfere with efficiency in production or consumption. The traditional (or standard) market failures, which we discuss in Chapter 5, provide widely accepted rationales for such public policies as the provision of goods and the regulation of markets by government agencies. Economists, until recently, have paid less attention, however, to the plausibility and appropriateness of some of the more fundamental assumptions about the behavior of consumers. For example, economic models usually treat the preferences of consumers as stable and basically unchanging. Is this reasonable? Do consumers always make the right calculations when faced with decisions involving such complexities as risk? Negative answers to any of these questions, which we consider in Chapter 6, can also provide rationales for public policies.

Of course, efficiency is not the only social value. Human dignity, distributional equity, economic opportunity, and political participation are values that deserve policy consideration along with efficiency. On occasion, public decision makers or private individuals as members of society may wish to give up some economic efficiency to protect human life, make the final distribution of goods more equitable, or promote fairness in the distribution process. As analysts, we have a responsibility to address these multiple values and the potential conflicts among them. We discuss distributional and other values as rationales for public policies in Chapter 7.

The Efficiency Benchmark: The Competitive Economy

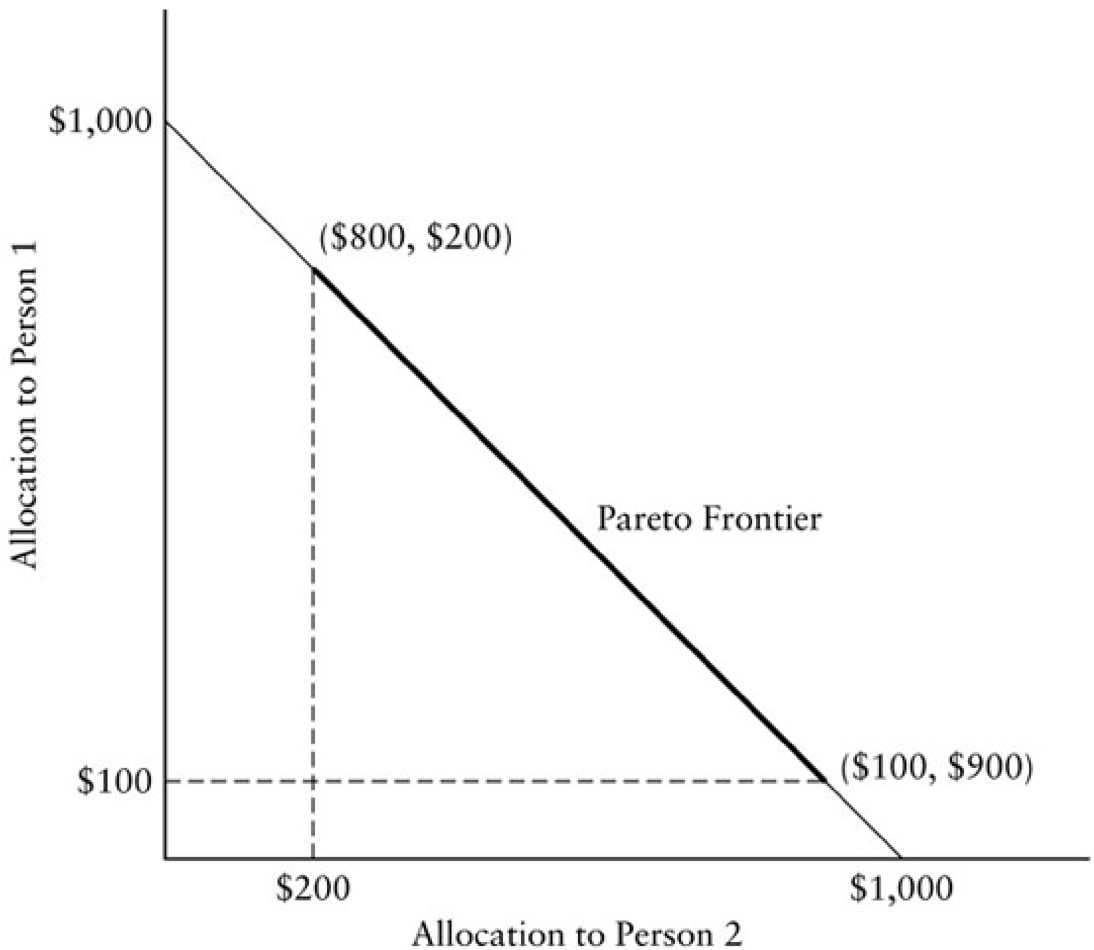
Imagine a world where each person derives *utility* (each person's perception of their own well-being) from personally consuming various quantities of all possible goods, including things, services, and leisure. We can conceptualize each person as having a utility function that converts the list of quantities of the goods consumed into a utility index such that higher numbers imply greater well-being. This framing makes several basic assumptions. First, other things equal, the more of any good a person has, the greater that person's utility. (We can incorporate unpleasant things such as pollution within this framework by thinking of them as "goods" that decrease utility or result in *disutility*.) And, second, additional units of the same good give ever-smaller increases in utility; in other words, they result in *declining marginal utility.*

Analysis of the competitive model makes the following assumptions about the production of goods: Firms attempt to maximize profits by buying factor inputs (such as labor, land, capital, and materials) to produce goods for sale. The technology available to firms for converting factor inputs to final goods is such that, at best, an additional unit of output would require at least as many units of inputs to produce as the preceding unit; in other words, it becomes more costly in terms of resources to produce each additional unit of the good. Firms behave competitively in the sense that they believe that they cannot change the prices of factor inputs or products by their individual actions.

Each person has a budget derived from selling labor and his or her initial endowments of the other factor inputs (their wealth), such as capital and land. Individuals maximize their well-being by using their incomes to purchase the combinations of goods that give them the greatest utility.

In this simple and idealized world, a set of prices arises that distributes factor inputs to firms and goods to persons in such a way that it would not be possible for anyone to find a reallocation that would make at least one person better off without making at least one person worse off.[1] Economists refer to such a distribution as being *Pareto efficient.* It is a concept with great intuitive appeal: wouldn't, and shouldn't, individuals and society collectively be dissatisfied with an existing distribution if we could find an alternative distribution that would make someone better off without making anyone else worse off? Although we would need additional criteria for choosing between two distributions that were each Pareto efficient, we should, unless we are malevolent, always want to make Pareto-improving moves from inefficient distributions to efficient ones.

Figure 4.1 illustrates the concept of Pareto efficiency involving the allocation of $1,000 between two people. Assume that the two people will receive any mutually agreed upon amounts of money that sum to no more than $1,000. The vertical axis represents the allocation to *person 1* and the horizontal axis represents the allocation to *person 2*. An allocation of all of the money to *person 1* would appear as the point on the vertical axis at $1,000; an allocation of all of the money to *person 2* would appear as the point on the horizontal axis at $1,000. The line connecting these two points, which we call the *potential Pareto frontier*, represents all the possible allocations to the two persons that sum exactly to $1,000. Any point on this line or inside the triangle it forms with the axes would be a technically feasible allocation because it gives shares that sum to no more than $1,000.



Status quo: point ($100, $200)
Potential Pareto frontier: line segment connecting ($1,000, $0) and ($0, $1,000)
Pareto frontier: line segment connecting ($800, $200) and ($100, $900)

Figure 4.1 Pareto and Potential Pareto Efficiency

The potential Pareto frontier indicates all the points that fully allocate the $1,000. Any allocation that does not use up the entire $1,000 cannot be Pareto efficient, because it would be possible to make one of the persons better off by giving her the remaining amount without making the other person worse off. The actual Pareto frontier depends on the allocations that the two people receive if they reach no agreement. If they each receive nothing in the event they reach no agreement, then the potential Pareto frontier is the *actual Pareto frontier* in that any point on it would make at least one of the persons better off without making the other person worse off.

Now suppose that, if these two people reach no agreement about an allocation, *person 1* receives $100 and *person 2* receives $200. This point ($100, $200) can be thought of as the *status quo point*—it indicates how much each person gets in the absence of an agreement. The introduction of the status quo point reduces the Pareto frontier to the line segment between ($100, $900) and ($800, $200). Only moves to this segment of the potential Pareto frontier would actually guarantee that each person is no worse off than under the status quo.

Note that whether a particular point on the potential Pareto frontier is actually Pareto efficient depends on the default allocations that comprise the status quo. More generally, Pareto efficiency in an economy depends on the initial endowments of resources to individuals.

The idealized competitive economy is an example of a *general equilibrium model*—it finds the prices of factor inputs and goods that clear all markets in the sense that the quantity demanded exactly equals the quantity supplied. Although general equilibrium models can sometimes be usefully applied to policy problems, limitations in data and problems of tractability usually lead policy analysts to evaluate policies in one market at a time, that is, with a *partial equilibrium model*.[2] Fortunately, a well-developed body of theory enables us to assess economic efficiency in the context of a single market.

Market Efficiency: The Meaning of Social Surplus

We need a yardstick for measuring changes in efficiency. Social surplus, which measures the net benefits consumers and producers receive from participation in markets, serves as that yardstick. In the context of the ideally competitive economy, the Pareto-efficient allocation of goods also maximizes social surplus. When we look across markets, the set of prices and quantities that give the greatest social surplus is usually Pareto efficient. Moreover, differences in social surplus between alternative market allocations approximate the corresponding sum of differences in individual welfares. As *social surplus is the sum of consumer surplus and producer surplus*, we consider each in turn.

Consumer Surplus: Demand Schedules as Marginal Valuations

Imagine that you have the last five tickets to the World Cup soccer final match. You walk into a room and announce to those present that you own all the remaining tickets to the event and that you will sell these tickets in the following manner: Starting with a high stated price, say, $500, you will keep lowering it until someone agrees to purchase a ticket. You will continue lowering the stated price until all five tickets are claimed. (An auction such as this with declining prices is known as a *Dutch auction*; in contrast, an auction with ascending prices is known as an *English auction*.) Each person in the room decides the maximum amount that he or she is willing to pay for a ticket. If this maximum amount must be paid, then each person will be indifferent between buying and not buying the ticket. Figure 4.2 displays the valuations for the persons in descending order from left to right. Although the stated prices begin at $500, the first acceptance is at $200. The purchaser obviously values this ticket at least at $200. Value in this context means the maximum amount the person is willing to pay, given his or her budget and other consumption opportunities. You now continue offering successively lower stated prices until you sell a second ticket, which a second person accepts at $180, the second-highest valuation. Repeating this process, you sell the remaining three tickets at $160, $140, and $120, respectively.

You were happy as a seller to get each person to pay the amount that he or she valued the ticket. If, instead, you had set the price at $100 so that five people wanted to buy tickets, then the purchasers would get tickets at a price substantially lower than the maximum amounts that they would have been willing to pay. For example, the person with the highest valuation would have been willing to pay $200 but only has to pay your set price of $100. The difference between the person's dollar valuation of the ticket and the price that he or she actually pays ($200 minus $100) is the surplus value that the person gains from the transaction. In a similar way, the person with the second-highest surplus gains $80 ($180 minus $100). The remaining three purchasers receive surpluses of $60 ($160 minus $100), $40 ($140 minus $100), and $20 ($120 minus $100). Adding the surpluses realized by these five purchasers yields a measure of the *consumer surplus* in this market for World Cup final tickets of $300.

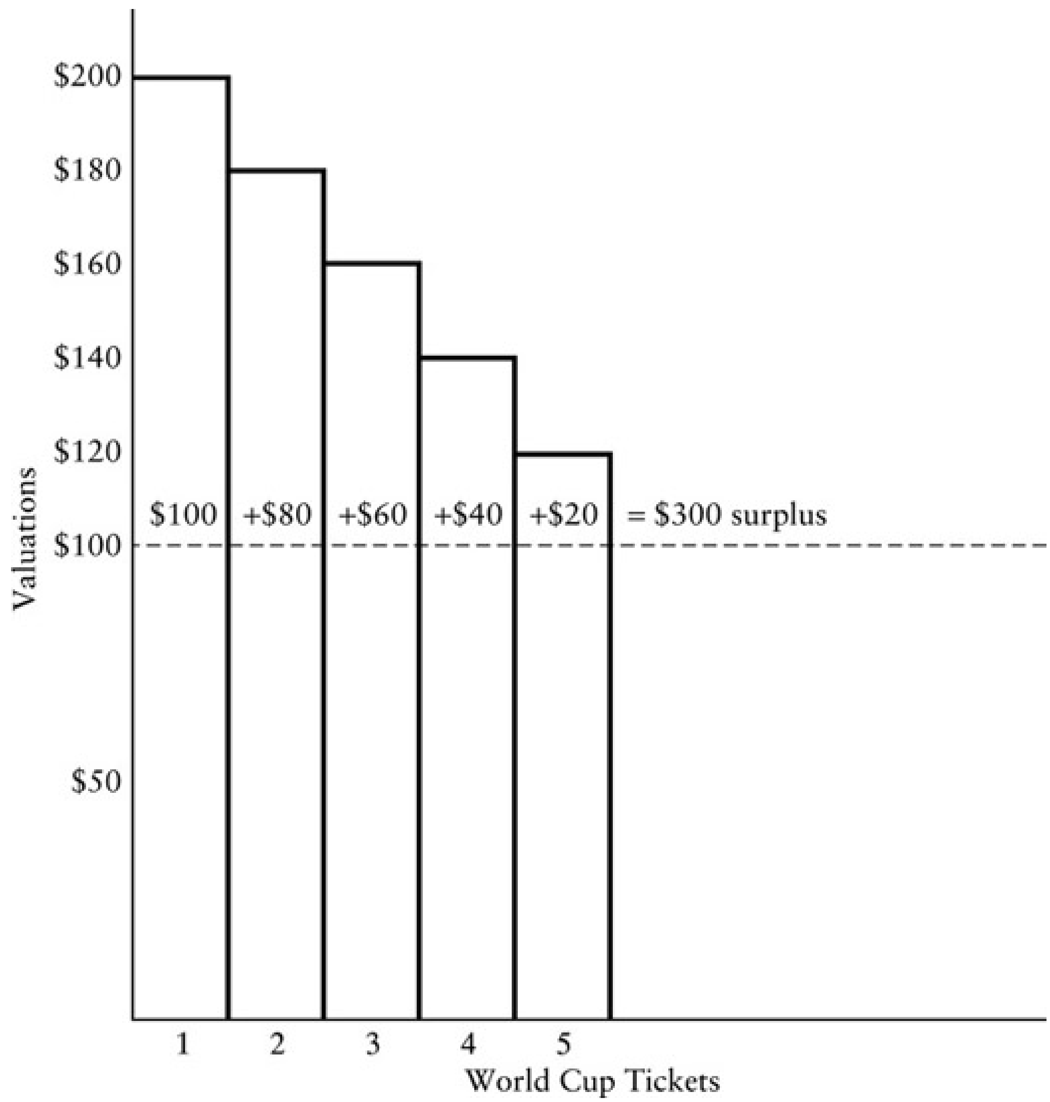


Figure 4.2 Consumer Values and Surpluses

The staircase in Figure 4.2 is sometimes called a *marginal valuation schedule* because it indicates how much successive units of a good are valued by consumers in a market. If, instead of seeing how much would be bid for successive units of the good, we stated various prices and observed how many units would be purchased at each price, then we would obtain the same staircase but recognize it as a *demand schedule*. Of course, we would also get a demand schedule by allowing individuals to purchase more than one unit at the stated prices. If we had been able to measure our good in small enough divisible units, or if demanded quantities were very large, then the staircase would smooth out to become a curve.

How do we move from this conceptualization of consumer surplus to its measurement in actual markets? We make use of demand schedules, which can be estimated from observing market behavior.

Line D in Figure 4.3 represents a person's demand schedule for some good, X. (Later we will interpret the curve as the demand schedule for all persons with access to the market.) Note that this consumer values all units less than *choke price*, P_c, the price that "chokes off" demand. The horizontal line drawn at P_0 indicates that she can purchase as many units of the good as she wishes at the constant price P_0. At price P_0 she purchases a quantity Q_0. Suppose, however, she purchased less than Q_0; then she would find that she could make herself better off by purchasing a little more because she would value additional consumption more than its cost to her. (The demand schedule lies above price for quantities less than Q_0.) Suppose, on the other hand, she purchased more than Q_0, then she would find that she could make herself better off by purchasing a little less because she would value the savings more than the lost consumption. At given price P_0, the quantity Q_0 is an *equilibrium* because the consumer does not want to move to an alternative quantity. The area of the triangle under the demand schedule but above the price line, $P_c a P_0$, represents her consumer surplus from purchasing Q_0 units of the good at price P_0.

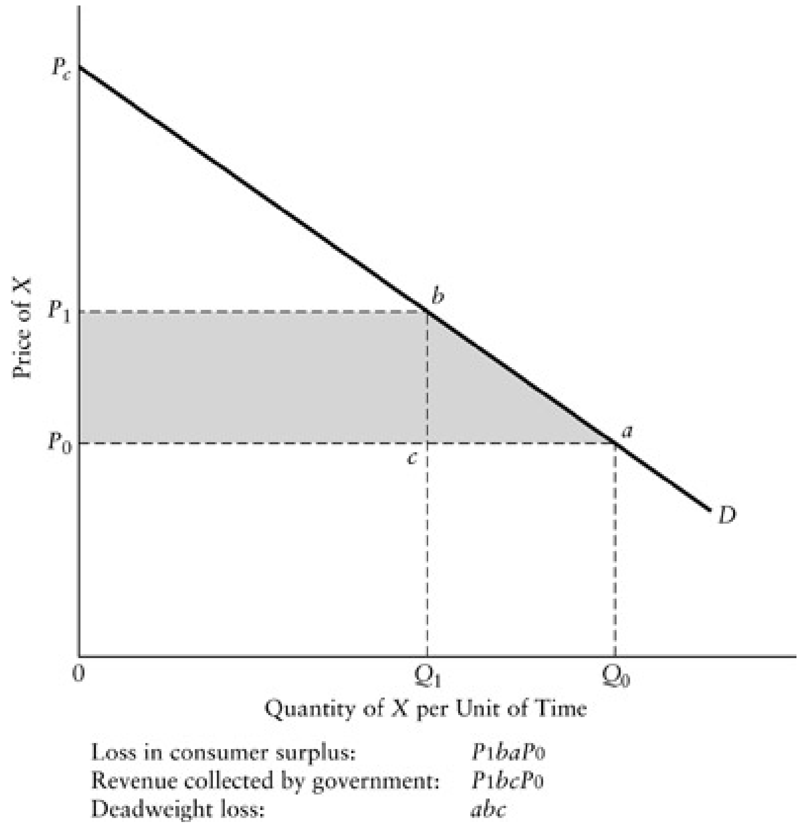


Figure 4.3 Changes in Consumer Surplus

Changes in consumer surplus are often the basis for measuring the relative efficiencies of alternative policies. For example, how does consumer surplus change in Figure 4.3 if a government policy increases price from P_0 to P_1? The new consumer surplus is the area of triangle P_cbP_1, which is smaller than the area of triangle P_caP_0 by the area of the trapezoid inscribed by P_1baP_0 (the shaded region). We interpret the area of the rectangle P_1bcP_0 as the additional amount the consumer must pay for the units of the good that she continues to purchase and the area of the triangle *abc* as the surplus the consumer gives up by reducing consumption from Q_0 to Q_1.

As an example of a government policy that raises price, consider the imposition of an excise (commodity) tax on each unit of the good in the amount of the difference between P_1 and P_0. Then the area of rectangle P_1bcP_0 corresponds to the revenue raised by the tax, which, conceivably, could be rebated to the consumer to offset exactly that part of the consumer surplus lost. The consumer would still suffer the loss of the area of triangle *abc*.

Because there are no revenues or benefits to offset this reduction in consumer surplus, economists define this loss in surplus due to a reduction in consumption as the *deadweight loss* of the tax. The deadweight loss indicates that the equilibrium price and quantity under the tax are not Pareto efficient —if it were possible, the consumer would be better off simply giving the tax collector a lump-sum payment equal to the area of P_1baP_0 in return for removal of the excise tax and its associated deadweight loss.

The loss of consumer surplus shown in Figure 4.3 approximates the most commonly used theoretical measure of changes in individual welfare: *compensating variation*. The compensating variation of a price change is the amount by which the consumer's budget would have to be changed so that he or she would have the same utility after the price change as before. It thus serves as a dollar measure, or *money metric*, for changes in welfare. If the demand schedule represented in Figure 4.3 is derived by observing how the consumer varies purchases as a function of price, holding utility constant at its initial level (it thus would be what we call a *constant utility demand schedule*), then the consumer surplus change would exactly equal the compensating variation.

Figure 4.4 illustrates how compensating variation can be interpreted as a money metric, or proxy, for utility. The vertical axis measures expenditures by a person on all goods other than good X; the horizontal axis measures how many units of X she consumes. Suppose initially that she has a budget, B, but that she is not allowed to purchase any units of X, say, because X is manufactured in another country and imports of it are prohibited. She will, therefore, spend all of B on other goods. The *indifference curve* I_0 indicates all the combinations of expenditures on other goods and units of X that would give her the same utility as spending B on other goods and consuming no units of X. Now imagine that the import ban is lifted so that she can purchase units of X at price P_x. She can now choose any point on the line connecting B with the point on the horizontal axis indicating how many units of X she could purchase if she spent zero on other goods, B/P_x. Once this new budget line is available, she will maximize utility by selecting a point on the highest feasible indifference I_1, by purchasing x_1 units of X and spending her remaining budget on other goods. Once *X* is available, it would be possible to return her to her initial level of utility by reducing her initial budget by the distance between *B* and *C* on the vertical axis. This amount is the compensating variation associated with the availability of *X* at price P_X. It is a dollar value, or money metric, for how much value she places on being able to consume *X*.

Instead of asking how much money we could take away to make the person as well off after introduction of imports of *X* as before, we could ask how much money we would have to give her if *X* were not available, so that she is as well off as she would be with imports of *X*. This amount, called *equivalent variation*, is shown as the distance between *A* and *B* on the vertical axis—if her budget were increased from *B* to *A*, then she could reach indifference curve I_1 without consuming *X*.

In practice, we usually work with empirically estimated demand schedules that hold constant consumers' *income* (rather than utility) and all other prices. This *constant income*, or *Marshallian*, demand schedule involves decreases in utility as price rises (and total consumption falls) and increases in utility as price falls (and total consumption rises). In comparison with a demand schedule holding utility constant at the initial level, the Marshallian demand schedule is lower for price increases and higher for price reductions. Fortunately, as long as either the price change is small or expenditures on the good make up a small part of the consumer's budget, the two schedules are close together and estimates of consumer surplus changes using the Marshallian demand schedule are close to the compensating variations.[3]

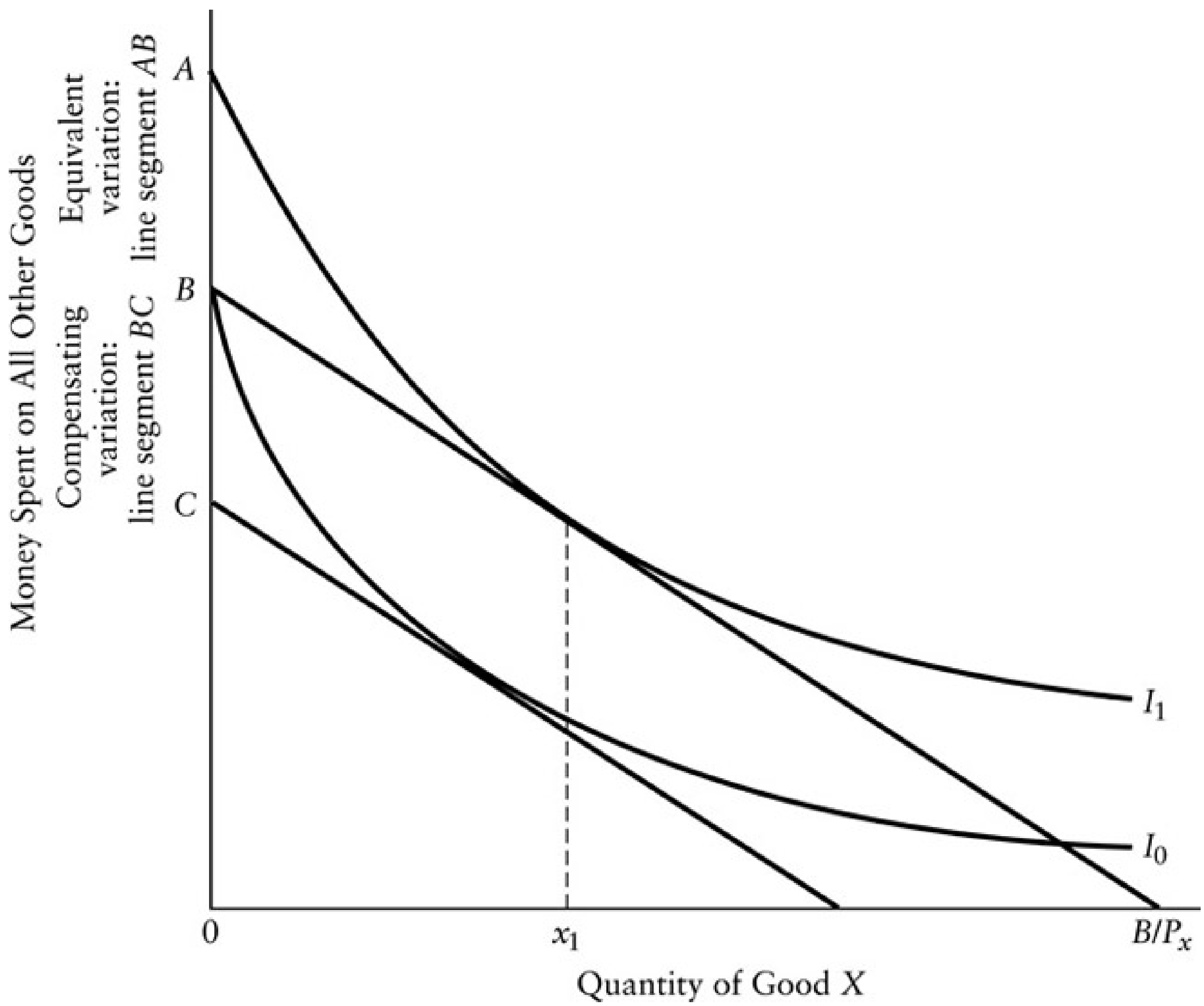


Figure 4.4 Money Metrics for Utility

Now, moving from the individual to society, consider the relationship between price and the quantity demanded by all consumers. We derive this *market demand schedule* by summing the amounts demanded by each of the consumers at each price. Graphically, this is equivalent to adding horizontally the demand schedules for all the individual consumers. The consumer surplus we measure using this market demand schedule would just equal the sum of the consumer surpluses of all the individual consumers. It would answer these questions: How much compensation would have to be given in aggregate to restore all the consumers to their original utility levels after a price increase? How much could be taken from consumers in the aggregate to restore them all to their original utility levels after a price decrease?

Thus, if we can identify a change in price or quantity in a market that would produce a net positive increase in social surplus, then there is at least the potential for making a Pareto improvement. After everyone is compensated, there is still something left over to make someone better off. Of course, the change is not actually Pareto improving unless everyone is given at least his or her compensating variations from the surplus gain.

Our primary use of consumer surplus in Chapter 5 is to describe the inefficiencies associated with the various market failures. For this purpose, an exclusive focus on the potential for Pareto improvement is adequate. In the context of cost–benefit analysis, the Kaldor–Hicks compensation principle advocates a similar focus on net positive changes in social surplus as an indication of the potential for Pareto improvements. When we consider cost–benefit analysis as a tool for evaluating policies in Chapter 17, we discuss the implications of focusing on potential rather than actual improvements in the welfare of individuals.

Producer Surplus: Background on Pricing

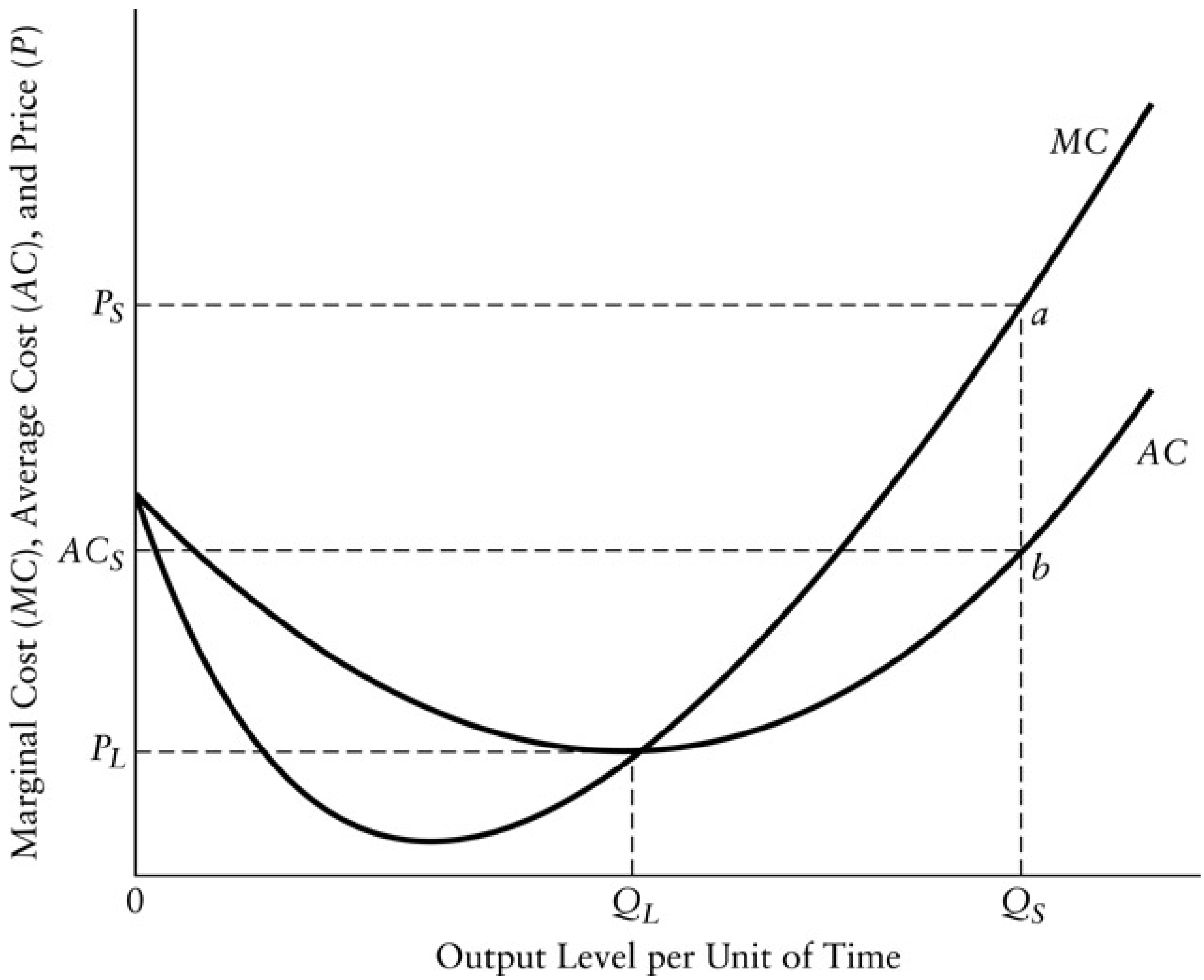


Figure 4.5 Average and Marginal Cost Curves

In the ideal competitive model, it is standard to assume that marginal cost of production for each firm rises with increases in output beyond equilibrium levels. Because firms have some fixed costs that must be paid before any production can occur, the average cost of producing output first falls as the fixed costs are spread over a larger number of units, then rises as the increasing marginal cost begins to dominate, because the use of some input, such as labor, becomes less efficient. Consequently, some output level minimizes the average cost for the firm. The curve marked *AC* in Figure 4.5 represents a U-shaped *average cost* curve for the firm. The *marginal cost* curve is labeled *MC*. Note that the marginal cost curve crosses the average cost curve at the latter's lowest point. When marginal cost is lower than average cost, the latter must be falling. When marginal cost is higher than average cost, the latter must be rising. Only when marginal cost equals average cost does average cost remain unchanged. This is easily grasped by thinking about your average score for a series of examinations—only a new score above your current average raises your average.

The total cost of producing some level of output, say, Q_S, can be calculated in one of two ways. First, because average cost is simply total cost divided by the quantity of output, multiplying average cost at this quantity (AC_S in Figure 4.5) by Q_S yields total cost as given by the area of rectangle $AC_S b Q_S 0$. Second, recalling that marginal cost tells us how much it costs to increase output by one additional unit, we can approximate total cost by adding up the marginal costs of producing successive units from the first all the way up to Q_S. The smaller our units of measure, the closer will be the sum of their associated marginal costs to the total cost of producing Q_S. In the limiting case of infinitesimally small units, the area under the marginal cost curve (*MC* in Figure 4.5) from zero to Q_S exactly equals total cost. (Those familiar with calculus will recognize marginal cost as the derivative of total cost, so that integrating marginal cost over the output range yields total cost; this integration is equivalent to taking the area under the marginal cost curve between zero and the output level.)

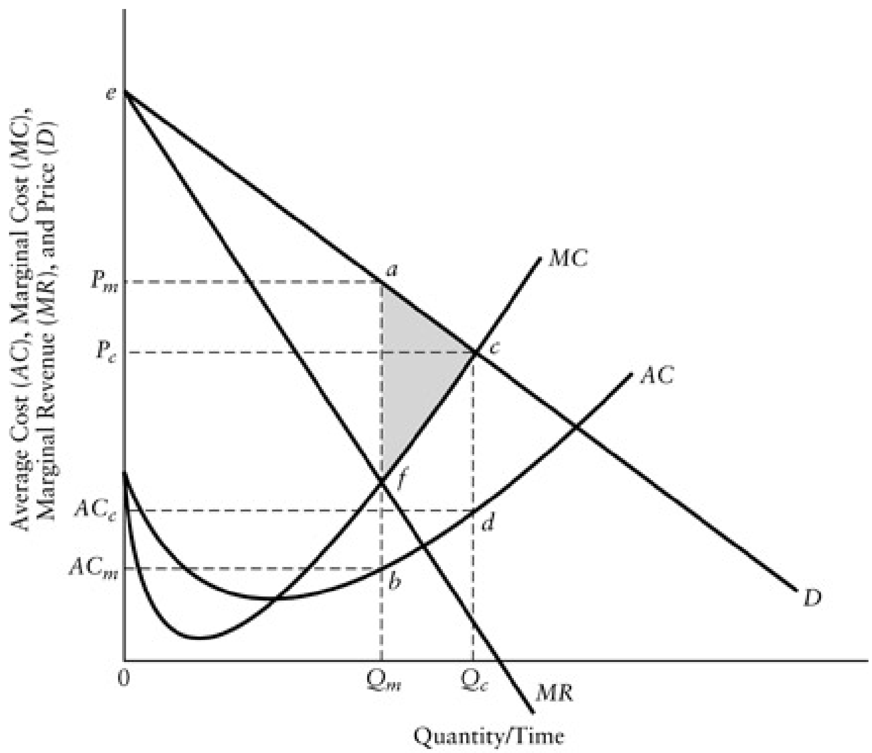
Now suppose that the market price for the good produced by the firm is P_S. The firm would maximize its profits by producing Q_S, the quantity at which marginal cost equals price. Because average cost is less than price at output level Q_S, the firm would enjoy a profit equal to the area of rectangle $P_S ab AC_S$. Profit equals total revenue minus total cost. (Total revenue equals price, P_S, times quantity, Q_S, or the area of rectangle $P_S a Q_S 0$; the total cost of producing output level Q_S is the area of rectangle $AC_S b Q_S 0$.) In the competitive model, profit would be distributed to persons according to their initial endowments of ownership. But these shares of profits would signal to other producers that profits could be made simply by copying the firm's current technology and inputs. As more firms enter the industry, however, total output of the good would rise and, therefore, price would fall. At the same time the new firms would bid up the price of inputs so that the entire marginal and average cost curves of all the firms would shift up. Eventually, price would fall to P_L, the level at which the new marginal cost equals the new average cost. At P_L profits fall to zero, thus removing any incentive to enter the industry.

With no constraints on the number of identical firms that can arise to produce each good, the Pareto-efficient equilibrium in the idealized competitive model is characterized by zero profits for all firms. (Note that we are referring to economic profits, not accounting profits. *Economic profit* is total revenue minus payments at competitive market prices to all factors of production, including an implicit rental price for capital owned by the firm. *Accounting profit* is simply revenue minus expenditures.) If the firm does not make an explicit payment to shareholders for the capital it uses, then accounting profits may be greater than zero even when economic profits are zero. To avoid confusion, economists refer to economic profits as *rents*, which are defined as any payments in excess of the minimum amounts needed to cover the cost of supply. Rents can occur in markets for inputs such as land and capital as well as in product markets.

In the real economic world, unlike our ideal competitive model, firms cannot be instantaneously replicated; at any time some firms may thus enjoy rents. These rents, however, attract new firms to the industry over time, so that over the long run we expect rents to disappear. Only if some circumstance prevents the entry of new firms will the rents persist. Therefore, we expect the dynamic process of profit seeking to move the economy toward the competitive ideal.

To understand better the concept of rent, it is useful to contrast pricing in a monopolistic industry with one that is competitive. To begin, consider the case of an industry with a single firm that does not have to worry about future competition. This monopoly firm sees the entire demand schedule for the good, labeled *D* in Figure 4.6. It also sees a *marginal revenue curve* (*MR*), which indicates how much revenue increases for each additional unit offered to the market. The marginal revenue curve lies below the demand schedule because each additional unit offered lowers the equilibrium price not just for the last but for all units sold. For example, imagine that increasing supply from 9 to 10 units decreases the market price from $100/unit to $99/unit.

Revenue increases by \$90 (10 times \$99 minus 9 times \$100). The height of the marginal revenue curve above 10 units is thus \$90, which is less than the height of the demand schedule, \$99. As long as marginal revenue exceeds marginal cost (*MC*), the firm can increase profits by expanding output. The profit-maximizing level of output occurs when marginal cost equals marginal revenue (where *MC* intersects *MR*). In Figure 4.6, this output level for the monopoly firm, Q_m, results in a market price, P_m, and profits equal to the area of rectangle P_mabAC_m: total revenue (P_m times Q_m) minus total cost (AC_m times Q_m).



	Competitive pricing	Monopoly pricing
Consumer surplus:	P_cce	P_mae
Total revenue:	P_ccQ_c0	P_maQ_m0
Total cost:	AC_cdQ_c0	AC_mbQ_m0
Rent:	P_ccdAC_c	P_mabAC_m
Deadweight loss:	0	*acf*

Figure 4.6 Monopoly Pricing, Rents, and Deadweight Loss

In contrast to the case of monopoly, consider the production decisions of one of the firms in a competitive industry. Because it provides a small part of the total industry supply, it ignores the effects of its supply on market price and, therefore, equates marginal cost with price (the intersection of *MC* and *D*), yielding price P_C and profits $P_C cdAC_C$. The difference in profit between monopoly and competitive pricing is the *monopoly rent*, a type of economic rent.

Remembering that the profits of the firm go to persons, we should take account of these rents in our consideration of economic efficiency. A dollar of consumer surplus (compensating variation) is equivalent to a dollar of distributed economic profit. If we set the price and quantity to maximize the sum of consumer surplus and rent, then we will generate the largest possible dollar value in the market, creating the prerequisite for a Pareto-efficient allocation.

The largest sum of consumer surplus and rent results when price equals marginal cost. A comparison in Figure 4.6 of the sums of consumer surplus and rent between the competitive and monopoly pricing cases illustrates this general proposition. The sum in the monopoly case, where marginal cost equals marginal revenue ($MC = MR$), is the area between the demand schedule and the marginal cost curve from quantity zero to Q_m. The sum in the competitive case, where marginal cost equals price ($MC = P$), is the area between the demand schedule and the marginal cost curve from quantity zero to Q_C. Obviously, the sum of consumer surplus and rent under competitive pricing exceeds that under monopoly pricing by the shaded area between the demand schedule and the marginal cost curve from Q_m to Q_C. This difference, the area of triangle *acf*, is the deadweight loss caused by monopoly pricing. That this area is the deadweight loss follows directly from the observation that the marginal benefit (vertical height of *D*) exceeds the marginal cost (height of *MC*) for each unit produced in expanding output from Q_m to Q_C.

Producer Surplus: Measurement with Supply Schedules

We usually deal with markets in which many firms offer supply. Therefore, we desire some way of conveniently summing the rents that accrue to all the firms supplying the market. Our approach parallels the one we used to estimate compensating variations. First, we introduce the concept of a supply schedule. Second, we show how the supply schedule can be used to measure the sum of rents to all firms supplying the market.

Imagine constructing a schedule indicating the number of units of a good that firms offer at each of various prices. Figure 4.7 shows the common case of firms facing increasing marginal costs. Firms offer a total quantity Q_2 at price P_2. As price increases, firms offer successively greater quantities, yielding an upward-sloping *supply schedule.* The schedule results from the horizontal summation of the marginal cost curves of the firms. (For example, refer to *MC* in Figure 4.5.) Each point on the supply curve tells us how much it would cost to produce another unit of the good. If we add up these marginal amounts one unit at a time, beginning with quantity equal to zero and ending at the quantity supplied, then we arrive at the total cost of producing that quantity. Graphically, this total cost equals the area under the supply curve from quantity zero to the quantity supplied.

Suppose the market price is P_3 so that the quantity supplied is Q_3. Then the total cost of producing Q_3 is the area P_1aQ_30. The total revenue to the firms, however, equals price times quantity, given by the area of rectangle P_3aQ_30. The difference between total revenue and total cost equals the total rent accruing to the firms. This difference is called *producer surplus.* It is measured by the total shaded area in Figure 4.7 inscribed by P_3aP_1.

Producer surplus need not be divided equally among firms. Some firms may have unique advantages that allow them to produce at lower cost than other firms, even though all firms must sell at the same price. For instance, a fortunate farmer with very productive land may be able to realize a rent at the market price, while another farmer on marginal land just covers total cost. Because the quantity of very productive land is limited, both farmers face rising marginal costs that they equate with market price to determine output levels. The general point is that unique resources—such as especially productive land, exceptional athletic talent, or easily extractable minerals—

can earn rents even in competitive markets. Excess payments to such unique resources are usually referred to as *scarcity rents.* Unlike monopoly, *scarcity rents do not necessarily imply economic inefficiency.*

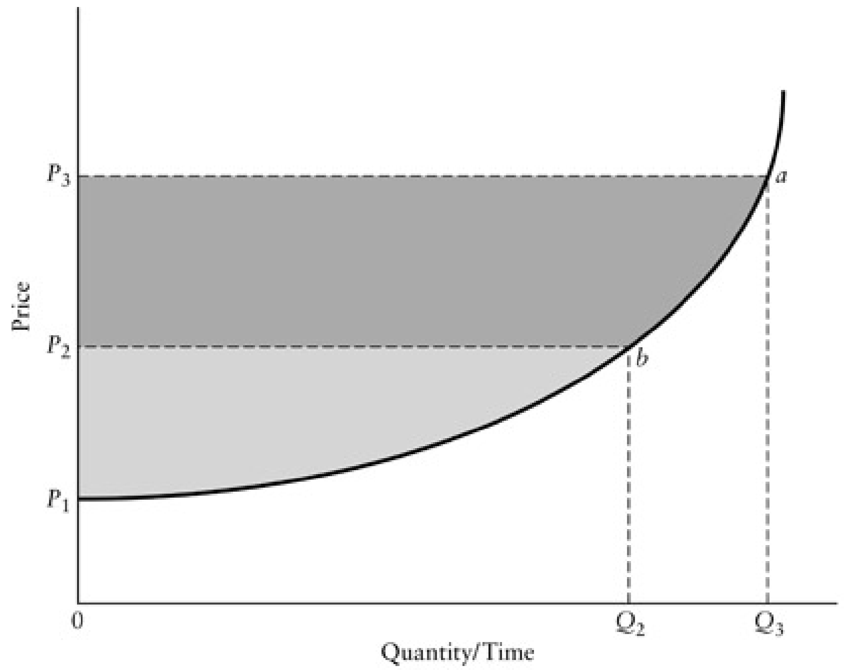


Figure 4.7 A Supply Schedule and Producer Surplus

Changes in producer surplus represent changes in rents. For example, if we want to know the reduction in rents that would result from a fall in price from P_3 to P_2, then we compute the change in producer surplus in the market. In Figure 4.7 the reduction in rents equals the darker shaded area P_3abP_2, the reduction in producer surplus.

Social Surplus

We now have the basic tools for analyzing efficiency in specific markets. A necessary condition for Pareto efficiency is that it should not be possible to increase the sum of compensating variations and rents through any reallocation of factor inputs or final products. We have shown how changes in consumer surplus measure the sum of compensating variations and how changes in producer surplus measure changes in rents. The sum of consumer and producer surpluses in all markets is defined as *social surplus*. Changes in social surplus, therefore, measure changes in the sums of compensating variations and rents. For evaluating the efficiency implications of small changes in the price and quantity of any one good, it is usually reasonable to limit analysis to changes in social surplus in its market alone.

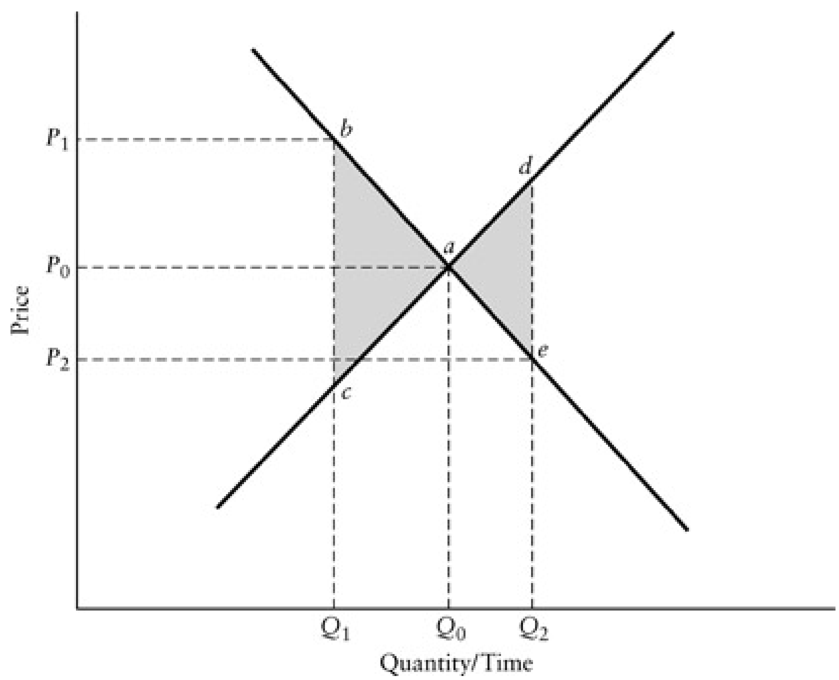


Figure 4.8 Inefficiencies Resulting from Deviations from the Competitive Equilibrium

Figure 4.8 reviews the inefficiencies associated with deviations from the equilibrium price and quantity in a competitive market in terms of losses of social surplus. The efficient competitive equilibrium occurs at price P_0 and quantity Q_0, the point of intersection of the supply (S) and demand (D) schedules. A policy that reduces quantity to Q_1 involves a loss of social surplus given by the area of triangle *abc*—each forgone unit between Q_1 and Q_0 yields marginal value (as given by the height of the demand schedule) in excess of marginal cost (as given by the height of the supply schedule). Consequently, social surplus can be increased by changing policy so that the quantity supplied and demanded moves closer to Q_0. A policy that increases quantity to Q_2 involves a loss of social surplus given by the area of triangle *ade*—each additional unit supplied and demanded between Q_0 and Q_2 yields marginal cost (as given by the height of the supply schedule) that is in excess of marginal benefit (as given by the height of the demand schedule). Consequently, changing policy to move the quantity supplied and demanded closer to Q_0 increases social surplus.

Caveats: Models and Reality

The general equilibrium model helps us understand a complex world. As is the case with all models, however, it involves simplifications that may limit its usefulness. It is worth noting three such limitations.

First, the general equilibrium model is static rather than dynamic. Real economies constantly evolve with the introduction of new goods, improvement in technologies, and changes in consumer tastes. An amazing feature of the price system is its capacity for conveying information among decentralized producers and consumers about such changes, which Friedrich Hayek refers to as "the particulars of time and place."[4] The equilibria of the competitive framework give us snapshots rather than videos of the real world. Usually, the snapshot is helpful and not too misleading. Nevertheless, policy analysts should realize that substantial gains in social welfare result from innovations that were not, and perhaps could not, have been anticipated. Policy analysts should be careful not to take too static a view of markets. Large rents that seem well protected by barriers to entry, for example, may very well spur innovative substitutes. In discussing each of the market failures in the next chapter, we consider some of the market responses that may arise to reduce social welfare losses.

Second, the general equilibrium model can never be complete: modelers simply do not have enough information to incorporate all goods and services. If they could, then it would be unlikely they could solve the model for its equilibrium. Our switching from the general equilibrium model to models of individual markets is a purposeful restriction of the model so that it can be usefully applied. In most applications it is a reasonable approach, though sometimes goods are such strong complements or substitutes that it is not reasonable to look at them separately.[5]

Third, the assumptions of the general equilibrium model are often violated in the real world. In the two chapters that follow, we consider the most important of these violations of assumptions. We do so in the context of specific markets, acknowledging that doing so may not fully capture all their implications in the wider economy.[6] Nonetheless, we see this analysis as highly valuable in helping policy analysts get started in understanding the complexity of the world in which they work.

Conclusion

The idealized competitive economy provides a useful conceptual framework for thinking about efficiency. The tools of applied welfare economics, consumer and producer surplus, give us a way of investigating efficiency within specific markets. In the next chapter, we explicate four situations, the traditional market failures, in which equilibrium market behavior fails to maximize social surplus.

For Discussion

1. Assume that the world market for crude oil is competitive, with an upward-sloping supply schedule and a downward-sloping demand schedule. Draw a diagram that shows the equilibrium price and quantity. Now imagine that one of the major oil exporting countries undergoes a revolution that shuts down its oil fields. Draw a new supply schedule and show the loss in consumer surplus in the world

oil market resulting from the loss of supply. What assumptions are you making about the demand for crude oil in your measurement of consumer surplus?

2. Now assume that the United States is a net importer of crude oil. Show the impact of the price increase resulting from the loss of supply to the world market on social surplus in the U.S. market.

Notes

1 For the history of general equilibrium theory, see E. Roy Weintraub, "On the Existence of a Competitive Equilibrium: 1930–1954," *Journal of Economic Literature* 21(1) 1983, 1–39.

2 For a review of the use of general equilibrium models in education, see Thomas J. Nechyba, "What Can Be (and What Has Been) Learned from General Equilibrium Simulation Models of School Finance?" *National Tax Journal* 54(2) 2003, 387–414.

3 In any event, the consumer surplus change measured under the Marshallian demand curve will lie between compensating variation and equivalent variation. See Robert D. Willig, "Consumer Surplus without Apology," *American Economic Review* 66(4) 1976, 589–97. For a more intuitive justification of consumer surplus, see Arnold C. Harberger, "Three Basic Postulates for Applied Welfare Economics," *Journal of Economic Literature* 9(3) 1971, 785–97.

4 F. A. Hayek, "The Use of Knowledge in Society," *American Economic Review* 35(4) 1945, 519–30, at 522.

5 See Anthony E. Boardman, David H. Greenberg, Aidan R. Vining, and David L. Weimer, *Cost–Benefit Analysis: Concepts and Practice*, 3rd ed. (Upper Saddle River, NJ: Pearson Prentice Hall, 2006), chapter 5.

6 R. G. Lipsey and Kelvin Lancaster, "General Theory of the Second Best," *Review of Economic Studies* 24(1) 1956–1957, 11–32.

5
Rationales for Public Policy

Market Failures

The idealized competitive model produces a Pareto-efficient allocation of goods. That is, the utility-maximizing behavior of persons and the profit-maximizing behavior of firms will, through the "invisible hand," distribute goods in such a way that no one could be better off without making anyone else worse off. Pareto efficiency thus arises through voluntary actions without any need for public policy. Economic reality, however, rarely corresponds perfectly to the assumptions of the idealized competitive model. In the following sections we discuss violations of the assumptions that underlie the competitive model. These violations constitute market failures, that is, situations in which decentralized behavior does not lead to Pareto efficiency. Traditional market failures are shown as circumstances in which social surplus is larger under some alternative allocation to that resulting under the market equilibrium. Public goods, externalities, natural monopolies, and information asymmetries are the four commonly recognized market failures. They provide the traditional economic rationales for public participation in private affairs.

Public Goods

The term *public*, or *collective, good* appears frequently in the literature of policy analysis and economics. The blanket use of the term, however, obscures important differences among the variety of public goods in terms of the nature of the market failure and, consequently, the appropriate public policy response. We begin with a basic question that should be raised when considering any market failure: Why doesn't the market allocate this particular good efficiently? The simplest approach to providing an answer involves contrasting public goods with private goods.

Two primary characteristics define private goods: rivalry in consumption and excludability in ownership and use. *Rivalrous consumption* means that what one person consumes cannot be consumed by another; a perfectly private good is characterized by complete rivalry in consumption. *Excludable ownership* means that one has control over use of the good; a perfectly private good is characterized by complete excludability. For example, shoes are private goods because when one wears them no one else can (rivalrous consumption) and, because when one owns them, one can determine who gets to wear them at any particular time (excludable ownership).

Public goods, on the other hand, are, in varying degrees, *nonrivalrous* in consumption, *nonexcludable* in use, or *both.* In other words, we consider any good that is not purely private to be a public good. A good is nonrivalrous in consumption when more than one person can derive consumption benefits from some level of supply at the same time. For example, a particular level of national defense is nonrivalrous in consumption because all citizens benefit from it without reducing the benefits of others—a new citizen enjoys benefits without reducing the benefits of those already being defended. (Each person, however, may value the uniformly provided level of defense differently.) A good is nonexcludable if it is impractical for one person to maintain exclusive control over its use. For example, species of fish that range widely in the ocean are usually nonexcludable in use because they move freely among regions such that no individual can effectively exclude others from harvesting them.

In practice, a third characteristic related to demand, the potential for *congestion*, or *congestibility*, is useful in describing public goods. (The term *crowding* is often used interchangeably with congestion.) Levels of demand for a good often determine the extent to which markets supply public goods at inefficient levels. *A good exhibits congestion if the marginal social cost of consumption exceeds the marginal private cost of consumption.* For example, a considerable number of people may be able to hike in a wilderness area without interfering with each other's enjoyment of the experience (a case of low demand) so that the marginal social cost of consumption equals the marginal private cost of consumption and there is no congestion. However, a very large number of hikers may reduce each other's enjoyment of the wilderness experience (a case of high demand) such that the marginal social cost of consumption exceeds the marginal private cost of consumption and, therefore, the wilderness is congested. The key point to recognize is that some goods may only be nonrivalrous over some range of usage, but at some higher level of use consumers begin to impose costs on each other. Later we interpret the divergence between social and private marginal costs as an externality.

Excludability and Property Rights

Excludability implies that some individual can exclude others from use of the good. In most public policy contexts in developed democracies, power to exclude others from use of a good is dependent on property rights granted and enforced by the state and its (judicial) organs. *Property rights* are relationships among people concerning the use of things.[1] These relationships involve claims by rights holders that impose duties on others. In contrast to political systems in which institutional arrangements create and enforce property rights, there are anarchic, or Hobbesian, situations with no government and no constraining norms or conventions, where physical force alone determines the power to exclude.

Property rights to a good can be partitioned in multiple ways beyond what we think of as simple ownership.[2] For example, a farmer may have a property right that allows using water from a river only during specific months of the year. However, for the purposes of a discussion of excludability, we treat goods as being controlled or “owned” by a single actor. Effective property rights are characterized by clear and complete allocation of claims and high levels of compliance by those who owe the corresponding duties. In this context, where the claim is to exclusive use of the good, compliance simply means accepting exclusion. *De jure* property rights, which are granted by the state, are typically clear though sometimes incomplete. These de jure property rights, however, may be attenuated, or in some cases superseded, by extra-legal behaviors such as trespass, squatting, poaching, or custom. Behaviors such as these may give rise to *de facto* property rights, the claims that actually receive compliance from duty bearers. Sometimes de jure property rights do not exist because changes in technology or relative prices create new goods that fall outside of existing allocations. For example, advances in medical technology have made stem cells valuable, but the right to their use is yet unclear.[3] Note that de facto property rights may or may not present excludability problems, though often they involve substantial costs to the claimants if they must employ physical protection systems, vigilance, stealth, or retaliation to enforce them. If these costs become too high, then individuals may abandon attempts to exclude others from use of the good. In these cases, the good is effectively nonexcludable.

Nonrivalrous Goods

As we concluded in our discussion of efficient pricing, the production of a private good, involving rivalrous consumption, will be in equilibrium at a level where price equals marginal cost ($P = MC$). On the demand side, the marginal benefits consumers receive from additional consumption must equal price ($MB = P$) in equilibrium. Therefore, marginal benefits equal marginal cost ($MB = MC$). Essentially the same principle applies to nonrivalrous goods. Because all consumers receive marginal benefits from the additional unit of the nonrivalrous good, however, it should be produced if the sum of all individual consumers’ marginal benefits exceeds the marginal cost of producing it. Only when output is increased to the point where the sum of the marginal benefits equals the marginal cost of production is the quantity produced efficient.

The sum of marginal benefits for any level of output of a purely nonrivalrous public good (whether excludable or not) is obtained by vertically summing the marginal benefit schedules (demand schedules) of all individuals at that output level.[4] Repeating the process at each output level yields the entire social, or aggregate, marginal benefit schedule. This process contrasts with the derivation of the social marginal benefit schedule for a private good: individual marginal benefit schedules (demand schedules) are added horizontally because each unit produced can only be consumed by one person.

Figure 5.1 illustrates the different approaches. Panel (a) represents the demand for a rivalrous good; panel (b) the demand for a nonrivalrous good. In both cases the demand schedules of the individual consumers, Jack and Jill, for the good appear as D_1 and D_2, respectively. The market demand for the rivalrous good, the downward-sloping dark line in panel (a), results from the horizontal addition of individual demands D_1 and D_2 at each price. For example, at price P_0, Jack demands Q_1 and Jill demands Q_2, so that the total quantity demanded is Q_0 (equal to $Q_1 + Q_2$). Repeating this horizontal addition at each price yields the entire market demand schedule. Note that at higher prices—above P_{c1}, Jack's *choke price*—only Jill is willing to purchase units of the good, so that the market demand schedule follows her demand schedule. If there were more consumers in this market, then the quantities they demanded at each price would also be included to obtain the market demand schedule.

Panel (b) presents the parallel situation for the nonrivalrous good, with the caveat that we are unlikely to be able to observe individual demand schedules for public goods (for

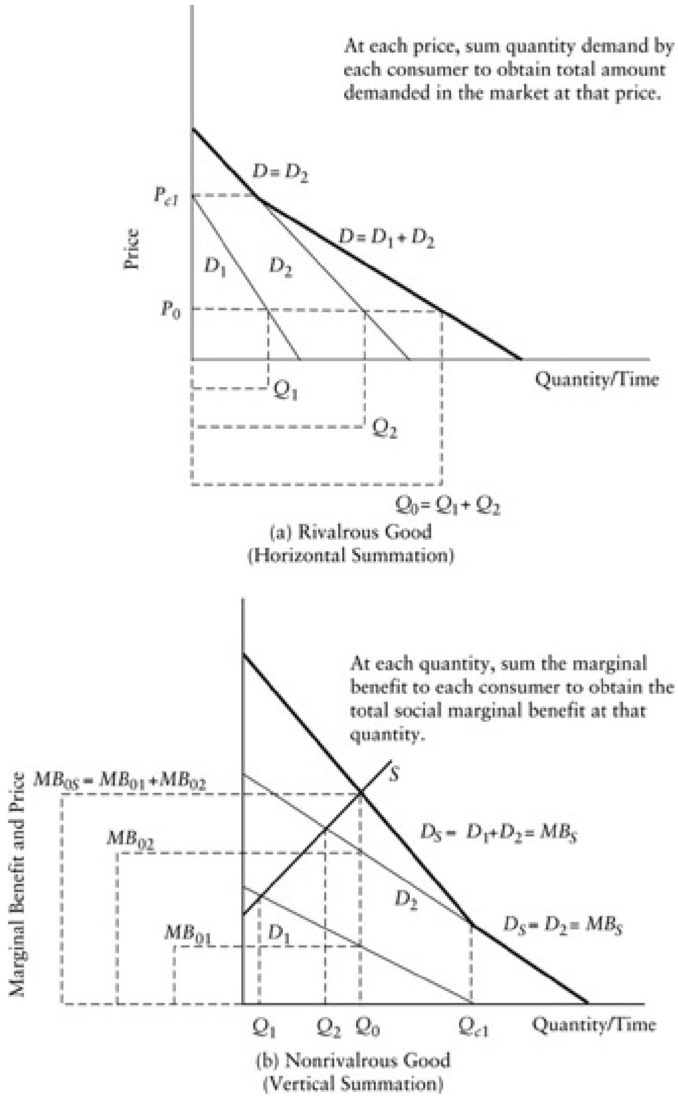


Figure 5.1 Demand Summation for Rivalrous and Nonrivalrous Goods

reasons we discuss below). Here, the social marginal benefit at Q_0 (MB_{0S} corresponding to the point on D_S above Q_0) is the sum of the marginal benefits enjoyed by Jack (MB_{01} corresponding to the point on D_1 above Q_0) and Jill (MB_2 corresponding to the point on D_2 above Q_0). The social marginal valuation at this point is obtained by summing the amounts that Jack and Jill would be willing to pay for the marginal unit at Q_0. The entire social marginal valuation schedule (MB_S) is obtained by summing Jack and Jill's marginal valuations at each quantity. Note that for quantities larger than Q_{c1}, Jack places no value on additional units, so that the social demand (social marginal benefit) schedule corresponds to the demand schedule of Jill.

Notice that in this example the upward-sloping supply schedule indicates that higher output levels have higher marginal costs (a brighter streetlight costs more to operate than a dimmer one); it is only the marginal cost of consumption that is equal, and zero, for each person (they can each look at what is illuminated by the light without interfering with the other). In other words, there are zero marginal social costs of consumption but positive marginal costs of production at each level of supply.

A crucial distinction between rivalrous and nonrivalrous goods is that the valuations of individual consumers cannot directly tell us how much of the nonrivalrous good should be provided—only the sum of the valuations can tell us that. Once an output level has been chosen, every person must consume it. Therefore, the various values different persons place on the chosen output level are not revealed by their purchases, as they would be in a market for a rivalrous good. *Thus, price neither serves as an allocative mechanism nor reveals marginal benefits as it does for a rivalrous good.*

Why wouldn't individuals supply the level of output that equates marginal cost with the sum of individual marginal benefits as they would in a market for a rivalrous good? Return to panel (b) of Figure 5.1. There, the supply schedule for the good, labeled *S*, indicates the social marginal cost of producing various levels of the good. If Jill, who has demand D_2, for the nonrivalrous good makes her own decision, she will purchase a quantity Q_2, where her own marginal benefit schedule crosses the supply schedule, which is less than the socially optimal quantity, Q_0, which equates social marginal benefit with social marginal cost. Jack, who has demand D_1, would purchase Q_1 units of the good if he were the only demander in the market. But if Jack knew that Jill would make a purchase, he would not find it in his own self-interest to purchase any of the nonrivalrous good because, as the marginal benefit schedules are drawn, Jill's purchase quantity would exceed the amount that Jack would want to purchase on his own. He would have an incentive to *freeride* on Jill's consumption, which he gets to enjoy because of nonrivalry. In other words, once Jill purchases Q_2 units, Jack would not find it in his personal interest to purchase any additional units on his own because each unit he purchases gives him less individual benefit than its price.

A market could still generate the optimal amount of the nonrivalrous good if all consumers would honestly reveal their marginal valuations. (Notice that one of the great advantages of the market for rivalrous goods is that consumers automatically reveal their marginal valuations with their purchases.) *Consumers do not normally have an incentive to reveal honestly their marginal valuations, however, when they cannot be excluded from consumption of the good.*

Congestibility: The Role of Demand

An economically relevant classification of public goods requires attention to more than their physical characteristics. At some level of demand, consumption of a good by one person may raise the marginal costs other persons face in consuming the good, so that the marginal social cost of consumption exceeds the marginal private cost. Later we will define the divergence between social and private costs as an externality. In the context of congestion, or crowding, we are dealing with a particular type of externality—an externality of consumption inflicted only on other consumers of the good.

Whether or not a particular good is congested at any particular time depends on the level of demand for the good at that time. Changes in technology, population, income, or relative prices can shift demand from levels that do not involve externalities of consumption to levels that do. For example, a road that could accommodate 1,000 vehicles per day without traffic delays might sustain substantial traffic delays in accommodating 2,000 vehicles. In some cases, seasonal or even daily shifts in demand may change the externalities of consumption, making a good more or less congested. For example, drivers may face substantial traffic delays during rush hour but no delays during midday.

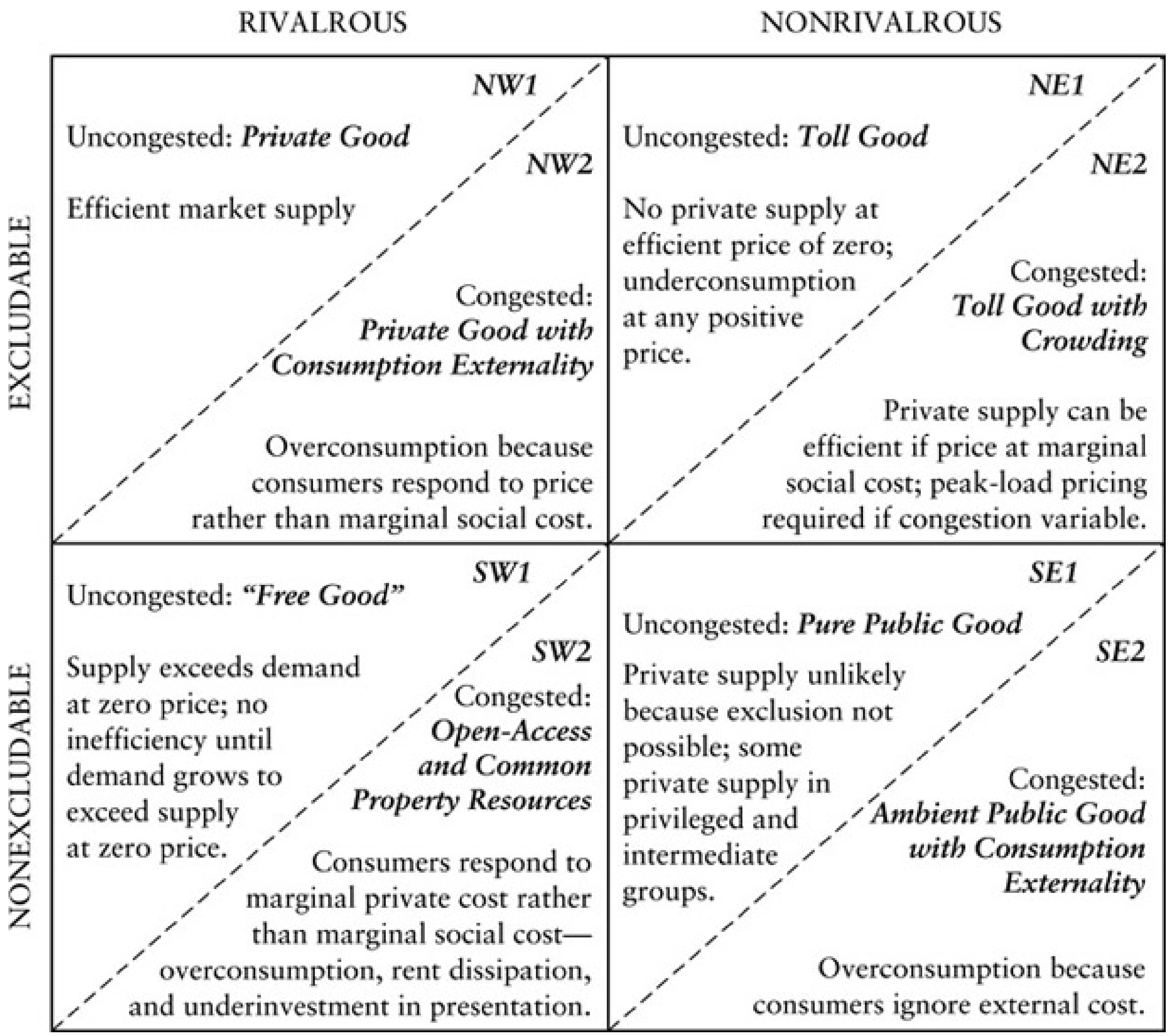
We must be careful to distinguish between the marginal social cost of consumption and the marginal cost of production. A purely nonrivalrous and noncongestible good exhibits zero marginal costs of consumption. Yet increments of the good (unless they occur naturally) require various factor inputs to produce. For instance, one way to increase the level of defense is to increase readiness, say, by shooting off more ammunition in practice. But it takes labor, machines, and materials to produce and shoot the ammunition, things that could be used to produce other goods instead. Thus, the marginal cost of production of defense is not zero; the marginal cost of consumption is likely to be zero, however, for a given level of supply.

Some care is required in thinking about congestion in cases in which supply cannot be added in arbitrarily small units. Many goods that display nonrivalry in consumption also display *lumpiness* in supply. For example, one cannot simply add small units of physical capacity to an existing bridge. To provide additional units of capacity, one must typically either build a new bridge or double-deck the existing one. Either approach provides a large addition to capacity. But this lumpiness is irrelevant to the determination of the relevance of congestion. The important consideration is whether or not the external costs of consumption are positive at the available level of supply.

To summarize, three characteristics determine the specific nature of the public good (and hence the nature of the inefficiency that would result solely from market supply): the degree of rivalry in consumption; the extent of excludability, or exclusiveness, in use; and the existence of congestion. The presence of nonrivalry, nonexcludability, or congestion arising from changes in levels of demand can lead to the failure of markets to achieve Pareto efficiency. The presence of either nonrivalry or nonexcludability is a necessary condition for the existence of a public good market failure.

A Classification of Public Goods

Figure 5.2 presents the basic taxonomy of public goods with the rivalrous/nonrivalrous distinction labeling the columns and the excludability/nonexcludability distinction labeling the rows. Additionally, the diagonals within the cells separate cases where congestion is relevant (congested) from those cases where congestion is not relevant (uncongested). By the way, note that the definitions of public goods that we provide differ starkly from a common usage of the term "public good" to describe any good provided by government. In fact, *governments, and indeed markets, provide both public and private goods* as defined in this taxonomy.



[Figure 5.2](#) A Classification of Goods: Private and Public

Rivalry, Excludability: Private Goods

The northwest (NW) cell defines private goods, characterized by both rivalry in consumption and excludability in use: shoes, books, bread, and the other things we commonly purchase and own. In the absence of congestion or other market failures, the self-interested actions of consumers and firms elicit and allocate these goods efficiently so that government intervention would have to be justified by some rationale other than the promotion of efficiency.

When the private good is congested (that is, it exhibits externalities of consumption), market supply is generally not efficient. (In competitive markets the marginal social cost of the good equals not just the price—the marginal cost of production—but the sum of price and the marginal social costs of consumption.) Rather than consider such situations as public goods market failures, it is more common to treat them under the general heading of externality market failures. Consequently, we postpone consideration of category NW2 in Figure 5.2 to our discussion of externalities.

Nonrivalry, Excludability: Toll Goods

The northeast cell (NE) includes those goods characterized by nonrivalry in consumption and excludability. Such goods are often referred to as *toll goods*. Prominent examples include bridges and roads that, once built, can carry significant levels of traffic without crowding. Other examples include goods that occur naturally, such as wilderness areas and lakes. Because exclusion is in many cases economically feasible, a private supplier might actually come forward to provide the good. For example, an enterprising individual might decide to build a bridge and charge a crossing toll that would generate a stream of revenue more than adequate to cover the cost of construction—hence, the label toll good. Clearly, in these situations the problem is not supply, per se. Rather, the problem is twofold. First, the private supplier may not efficiently price the facility that is provided. Second, the private supplier may not provide the correct facility size to maximize social surplus (Q_S in Figure 5.1[b]).

With respect to the pricing problem, first consider the case of nonrivalrousness and no congestion (NE1), one in which we can be certain that the private supplier will not charge the efficient price, which is zero. In the absence of congestion, the social marginal cost of consumption is zero, so that any positive price inappropriately discourages use of the bridge. The shaded triangular area *abc* in panel (a) of Figure 5.3 represents the deadweight loss that results from a positive price (or toll) P_1 with the demand schedule for crossings given by *D*. From the diagram, we can see that any positive price would involve some welfare loss. The reason is that if a positive price is charged, some individuals who, from the social perspective, should cross the bridge will be discouraged from doing so. Why? Because those individuals who would receive marginal benefits in excess of, or equal to, the marginal social cost of consumption should cross. As the marginal social cost of consumption is zero (that is, lies along the horizontal axis), any individual who would derive any positive benefit from crossing should cross. Those who obtain positive marginal benefits less than the price, however, would not choose to cross. The resulting deadweight loss may seem nebulous—the lost consumer surplus of the trips that will not take place because of the toll—but it represents a misallocation of resources that, if corrected, could lead to a Pareto improvement.

The analysis just presented is complicated if the good displays congestion over some range of demand (NE2). Return to the bridge example. We assumed that capacity exceeded demand—but this need not be the case. Consumption is typically uncongested up to some level of demand, but as additional people use the good, the marginal social costs of consumption become positive. Goods such as bridges, roads, and parks are potentially subject to congestion because of physical capacity constraints. Whether or not they are actually congested depends on demand as well as capacity. Panel (b) of Figure 5.3 shows the marginal social costs of consumption for goods that are congested. At low demand levels (for example, D_L), the marginal cost of consumption is zero (in other words, consumption is uncongested), but at high levels of demand (for example, D_H), consumption at a zero toll imposes marginal costs on all users of the good. Line segments 0*f*, *fg*, and *gh* trace out the marginal social cost of consumption for the good over the range of possible consumption. Notice that these costs are imposed by the marginal consumer, not by the good itself. In the case of the bridge, for instance, the costs appear as the additional time it takes all users to cross the bridge. If additional users crowd onto the bridge, then quite conceivably marginal costs could become almost infinite: gridlock prevents anyone from crossing the bridge until some users withdraw. The economically efficient price in the case of high demand is shown in panel (b) as P_C.

Let us take a closer look at the incentives that lead to socially inefficient crowding. Suppose that there are 999 automobiles on the bridge experiencing an average crossing time of ten minutes. You are considering whether your auto should be number 1,000 on the bridge. Given the capacity of the bridge, your trip will generate congestion—everyone crossing the bridge, including yourself, will be slowed down by one minute. In other

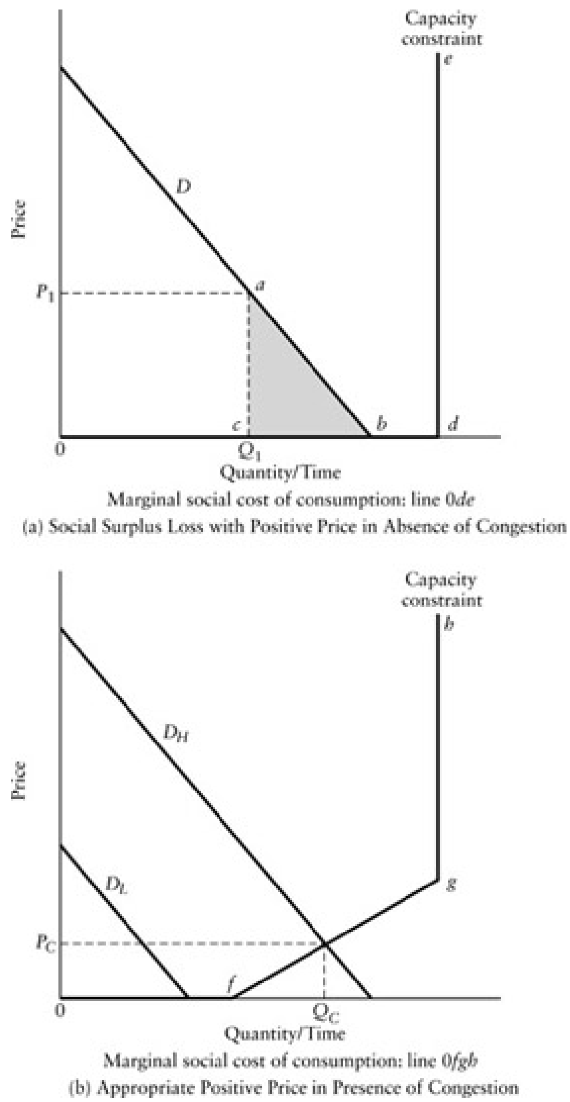


Figure 5.3 Toll Goods

words, the average crossing time of the users rises by one minute to 11 minutes. Your personal calculus is that you will cross if your marginal benefits from crossing exceed your marginal costs. In this case your cost is 11 minutes (the new average consumption cost of 11 minutes for the group of users). If you decide to cross, however, the marginal social costs of your decision will be 1,010 minutes (the 11 minutes you bear plus the 999 additional minutes your use inflicts on the other users). Thus, from the social perspective, if everyone places the same value on time as you, then you should cross only if the benefits that you receive from crossing exceed the cost of 1,010 minutes of delay!

In practice, nonrivalrous excludable public goods that exhibit congestion involve quite complex pricing problems; however, the basic principle follows readily from the above discussion. Let us assume that the congestion occurs at regular time periods as demand shifts over times of the day (roads at rush hour, for instance) or seasons of the year. Efficient allocation requires that the price charged to users of the good equals the marginal costs imposed on other users during each period of the day, implying a zero price during uncongested periods (more generally, price should equal the marginal cost of production in the absence of congestion) and some positive price during congested periods (the so-called peak-load price).

Private firms provide many toll goods. The firms must pay the cost of producing the goods with revenues from user fees—that is, the stream of revenues from the fees must cover the cost of construction and operation. To generate the necessary revenues, firms must usually set the tolls above the marginal social costs of consumption. Indeed, as profit maximizers, they set tolls at levels that maximize their rent (the area of rectangle P_1aQ_10 in Figure 5.3[a]), based on the demand schedule they face. Thus, market failure results because the fees exclude users who would obtain higher marginal benefits than the marginal social costs that they impose. The magnitude of the forgone net benefits determines the seriousness of the market failure.

The problem of inefficient scale arises in the case of private supply because firms seeking to maximize profits anticipate charging tolls that restrict demand to levels that can be accommodated by smaller facilities. The result is that the facilities built are too small from a social perspective. Further, the choice of facility size determines the level of demand at which congestion becomes relevant.

Nonrivalry, Nonexcludability: Pure and Ambient Public Goods

We now turn to those goods that exhibit nonrivalrous consumption and where exclusion is not feasible—the southeast (SE) quadrant of Figure 5.2. When these goods are uncongested, they are *pure public goods*. The classic examples of such public goods are defense and lighthouses. One of the most important public goods in modern societies is the generally available stock of information that is valuable in production, consumption, or exchange. With certain exceptions, to be discussed below, *pure public goods will not be supplied at all by markets* because of the inability of private providers to exclude those who do not pay for them. Contrast this with the NE quadrant, where there is likely to be market provision, but at a price that results in deadweight losses.

The number of persons who may potentially benefit from a pure public good can vary enormously, depending on the good: ranging from a particular streetlight, with only a few individuals benefiting, to national defense, where all members of the polity presumably benefit. Because benefits normally vary spatially or geographically (that is, benefits decline monotonically as one moves away from a particular point on the map), we commonly distinguish among local, regional, national, international, and even global public goods.[5] While this is a convenient way of grouping persons who receive benefits, it is only one of many potential ways. For example, persons who place positive values on wilderness areas in the Sierras may be spread all over North America—indeed, all over the world. Some, or even most, of those who actually reside in the Sierras may not be included in this category because their private interests depend upon commercial or agricultural development of the area rather than upon preservation.

Potentially valuable exchange may not occur because necessary information has public-good characteristics. Market provision often is facilitated through information intermediaries.[6] Indeed, *information intermediaries*, or *platform providers*, such as Facebook or Google Search, have become so ubiquitous that their intermediation seems somehow a natural part of the marketplace. However, beneficial exchange is unlikely to be realized if the required information is a pure public good. For example, exclusion may not be feasible in matching ex-prisoners with potential employers because the prisoners do not have adequate resources to pay for inclusion and private firms do not have a specific demand for hiring prisoners. When exclusion is possible, the information has the characteristics of a toll good. As with toll goods generally, there is likely to be too little private supply and pricing above marginal cost. Platforms can also evolve to have the characteristics of natural monopolies that we describe later in the chapter.[7]

We have already touched briefly on the major problem in the SE quadrant. People who would actually receive some level of positive benefits, if the good is provided, often do not have an incentive to reveal honestly the magnitude of these benefits: *if contributions for a public good are to be based on benefit levels, then individuals generally have an incentive to understate their benefit levels*; if contributions are not tied to benefit levels, then individuals may have an incentive to overstate their benefit levels to obtain a larger supply. Typically, the larger the number of beneficiaries, the less likely is any individual to reveal his or her preferences. In such situations, private supply is unlikely. As Mancur Olson has pointed out, however, two specific situations can result in market supply of pure public goods.[8] He labels these situations the *privileged group* and the *intermediate group* cases.

The privileged group case is illustrated in Figure 5.4, where three consumers receive benefits from the good according to their marginal benefit schedules. For example, the three might be owners of recreational facilities in a valley and the good might be spraying to control mosquitoes. The marginal benefit schedule of *person 3* (D_3) is high relative to the marginal benefit schedules of *persons 1* and *2*. (By relatively high, we mean that at any quantity, *person 3* places a much higher marginal value on having an additional unit than do the other two persons.) In fact, it is sufficiently high that *person 3* will be willing to purchase Q_3 of the good if the other two persons purchase none. (*Person 3*'s demand schedule intersects with the supply schedule at quantity Q_3.) Of course, once *person 3* purchases Q_3, neither *person 1* nor *person 2* will be willing to make additional purchases. In effect, they free ride on *person 3*'s high demand. (We can speak of *person 3*'s marginal benefit schedule as a demand schedule because she will act as if the good were private, revealing her demand at various prices.) Despite the provision of Q_3 units of the public good, compared to the socially efficient level Q_0, social surplus is lower by the area of triangle *abc*.

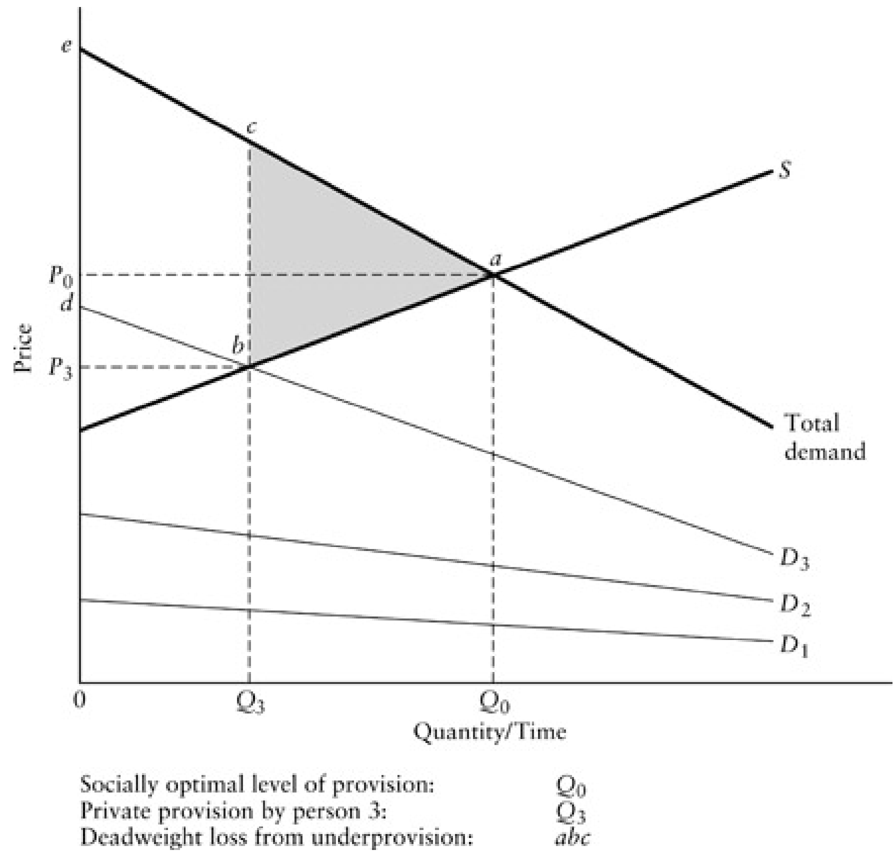


Figure 5.4 Private Provision of a Public Good: Privileged Group

In this case the demand of *person 3* makes up such a large fraction of total demand that the amount purchased (Q_3) is fairly close to the economically efficient level (Q_0). In this sense, the three persons form a privileged group. Even when no one person has a sufficiently high demand to make a group privileged, however, *some* provision of the good may result if the group is sufficiently small so that members can negotiate directly among themselves. We recognize such a group as "intermediate" between privileged and unprivileged: two or more members may, for whatever reasons, voluntarily join together to provide some of the good, although usually at less than the economically efficient level. Intermediate groups are generally small, or at least have a small number of members who account for a large fraction of total demand.

The situation just described closely resembles a positive externality (benefits accruing to third parties to market transactions), which we discuss in detail in the next section. Clearly, if *person 1* and *person 2* did not agree to finance jointly the public good at the efficient level (Q_0), they would nevertheless receive benefits from Q_3, the amount purchased by *person 3*. Through the purchase of Q_3, *person 3* receives the private consumer surplus benefit given by the area of triangle P_3bd and confers an external (to herself) consumer surplus benefit of area *bced* on the other group members. For analytical convenience, however, we maintain a distinction between public goods and positive externalities. We restrict the definition of an externality to those situations where provision of a good necessarily requires the joint production of a private good and a public good; we reserve the public good classification for those cases where there is no joint production. So, for example, mosquito control poses a public-good problem, while chemical waste produced as a by-product of manufacturing poses an externality problem.

In many situations, large numbers of individuals would benefit from the provision of a public good where no small group receives a disproportionate share of total benefits. Such cases of large numbers raise the free-rider problem with a vengeance. In situations involving large numbers, each person's demand is extremely small relative to total demand and to the cost of provision. The rational individual compares his individual marginal benefits and costs. Take as an example national defense. The logic is likely to be as follows: My monetary contribution to the financing of national defense will be infinitesimal; therefore, if I do not contribute and everyone else does, the level of defense provided, from which I cannot be effectively excluded, will be essentially the same as if I did contribute. On the other hand, if I contribute and others do not, national defense will not be provided anyway. Either way I am better off not contributing. (As we will discuss shortly, free-riding arises in other contexts besides the market provision of public goods; it can also occur when attempts are made to supply the economically efficient quantity of the public good through public-sector mechanisms.)

To summarize, then, the free-rider problem exists in the large-numbers case because it is usually impossible to get persons to reveal their true demand (marginal benefit) schedules for the good (it is even difficult to talk of individual demand schedules in this context because they are not generally observable). Even though all would potentially benefit if all persons agreed to contribute to the financing of the good so that their average contributions (in effect, the price they each paid per unit supplied) just equaled their marginal benefits, self-interest in terms of personal costs and benefits discourages honest participation.

The concept of free-riding plays an important role in the theory of public goods. There has been considerable debate among economists about the practical significance of free-riding. Both observational and experimental evidence, however, suggest that free-riding occurs, but perhaps not to the level predicted by theory.[9] In the context of relatively small groups that permit face-to-face contact, such as neighborhood associations, social pressure may be brought to bear, ensuring that failure to contribute is an unpleasant experience.[10] More generally, we would expect a variety of voluntary organizations with less individual anonymity to arise to combat free-riding. There has also been considerable theoretical interest in pricing mechanisms that encourage people to reveal their preferences truthfully, though they have as yet been little used in practice.[11]

Thus far we have not considered the issue of congestion in the SE quadrant in Figure 5.2. Some goods are simply not congestible and can be placed to the left (SE1) of the diagonal within the SE quadrant. For instance, mosquito control, a local public good, involves nonexcludability, nonrivalry in consumption, and noncongestibility—no matter how many people are added to the area, the effectiveness of the eradication remains the same. Similarly, national defense, a national or international public good, in general is not subject to congestion. In contrast, nature lovers may experience disutility when they meet more than a few other hikers in a wilderness area.

We label nonrivalrous and nonexcludable goods that exhibit congestion, and thus are appropriately placed into SE2, as *ambient public goods with consumption externalities.* Air and large bodies of water are prime examples of public goods that are exogenously provided by nature. For all practical purposes, consumption (use) of these goods is nonrivalrous—more than one person can use the same unit for such purposes as disposing of pollutants.[12] In other words, consumption of the resource (via pollution) typically imposes no Pareto-relevant impact until some threshold, or ambient-carrying capacity, has been crossed. Exclusion in many market contexts is impossible, or at least extremely costly, because access to the resources is possible from many different locations. For example, pollutants can be discharged into an air shed from any location under it (or even upwind of it!).

Relatively few goods fall into category SE2. The reason is that as congestion sets in, it often becomes economically feasible to exclude users so that goods that might otherwise be in SE2 are better placed in NE2 instead. For example, to return to wilderness hiking: once the density of hikers becomes very large, it may become economically feasible for a private owner to issue passes that can be enforced by spot checks. Wilderness in this context is an ambient public good only over the range of use between the onset of crowding and the reaching of a density where the pass system becomes economically feasible. The efficiency problems associated with many of the goods that do fall into category SE2 can alternatively be viewed as market failures due to externalities. So, for example, the carrying capacity of a body of water can be viewed as an ambient public good that suffers from overconsumption. Alternatively, and more conventionally, the pollution that the body of water receives can be viewed as an externality of some other good that is overproduced.

Rivalry, Nonexcludability: Open Access, Common Property, and Free Goods

Consider goods in the SW quadrant, where consumption is rivalrous, but where exclusion is not economically feasible; in other words, there is *open access* to the good. We should stress that in this quadrant we are dealing with goods that are rivalrous in consumption. Trees, fish, bison, oil, and pastureland are all rivalrous in consumption: if I take the hide from a bison, for instance, that particular hide is no longer available for you to take. Specifically, open access means that anyone who can seize units of the good controls their use. We normally use the term open access to describe unrestricted entry of new users. However, the same kind of property right problem arises with unrestricted use by a fixed number of individuals who already have access. Yet in these circumstances it is often plausible to talk in terms of ownership in the sense that a fixed number of users may share collective property rights to the good. These owners may still engage in *open effort*. Nonetheless, holding the property rights gives them an incentive and possibly the means to reduce or eliminate overconsumption.

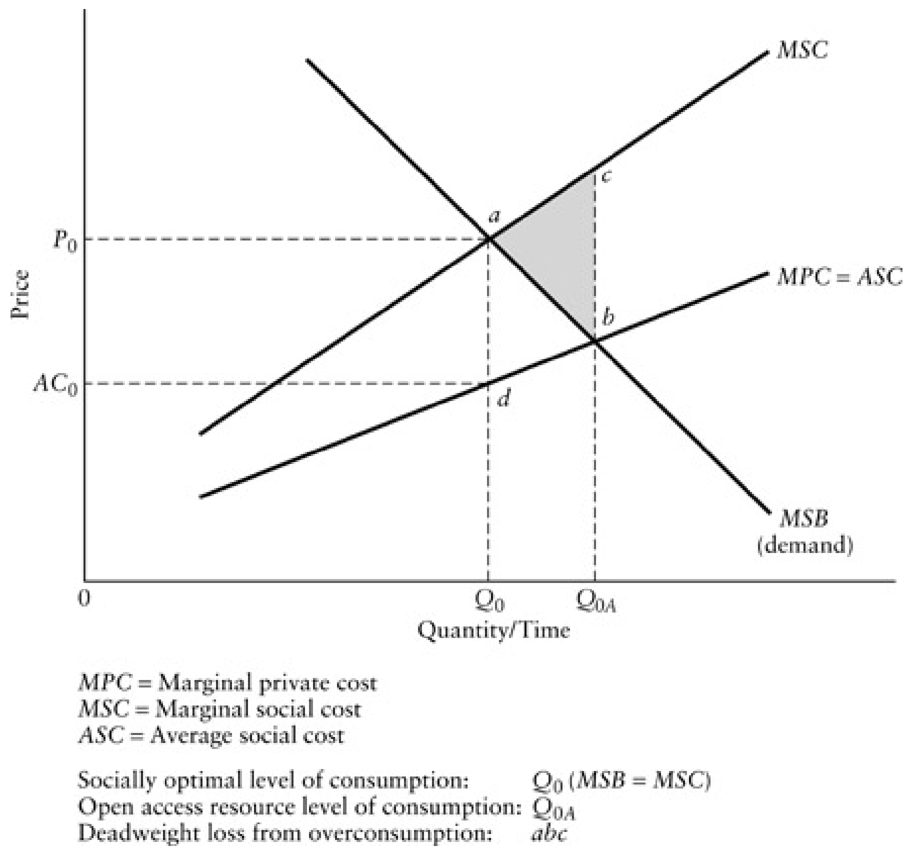
No immediate market failure appears in those cases in which the good is naturally occurring and where supply exceeds demand at zero price. As anyone can take these goods without interfering with anyone else's use, we refer to them as *free goods* (SW1). Thus, although they are theoretically rivalrous in consumption, from an efficiency perspective they are not because of the excess supply.

In situations in which demand is higher so that there is no excess supply at zero price, we have what is usually referred to as an *open-access resource* (SW2). When access is limited to a defined group of potential users (in other words, entry is restricted), and the users own the good in common, there is *common property ownership*. The distinction between open-access and common property situations requires some elaboration. In the case of common property, the limiting of access to a defined set of persons opens the possibility of self-governance among them that reduces or eliminates open-effort inefficiencies.[13] In the case of open access, however, the threat of new entrants effectively eliminates the possibility of self-governance. Even in cases of common property, individually rational behavior by members of the defined group can lead to inefficiency in a way that may end up being indistinguishable from open access—in such cases common property results in a *common property resource problem*. Consequently, although much of the following discussion is framed in terms of open access, it is generally relevant to common property and open effort as well.

Market failure arises in the open-access case from the "infeasibility" of exclusion. Why the quotation marks? Because, as we will see, open-access problems often occur in situations in which institutional features, rather than the inherent nature of the goods, make exclusion infeasible. For example, in most countries oil does not suffer from open access because their governments have adopted laws that keep and enforce exclusive property rights to subsurface resources for the governments themselves. In the United States, however, oil has often suffered from the open-access problem because of the "rule of capture," a legal doctrine that gives most subsurface rights to the owner of the surface property. When different people own separate pieces of property over the same pool of oil, the rule of capture prevents exclusion. Unless all the owners agree to treat their combined property as a unit, the common reservoir of oil will be extracted too quickly.

Nonexcludability leads to economically inefficient overconsumption of rivalrous goods. It can also lead to underinvestment in preserving the stock of goods or to over-investment in capital used to capture the good.[14] Naturally occurring resources are especially susceptible to the open-access problem. Persons with access to the resource realize that what they do not consume will be consumed by someone else. Each person, therefore, has an incentive to consume the resource at a faster rate than if he or she had exclusive ownership. For instance, deforestation often results when a population relies on a forest as a source of firewood. Further, the availability of underpriced firewood does not give users the appropriate incentive to invest in stoves that use less wood, and the fact that anyone can cut and gather wood discourages individuals from replanting or nurturing trees.

Figure 5.5 illustrates the efficiency losses associated with overconsumption when there is open access to a resource. The marginal social benefit schedule (*MSB*) represents the horizontal summation (remember, we are dealing with a rivalrous good) of all the demand schedules of individuals. The economically efficient level of consumption, Q_0, results when marginal social cost (*MSC*) equals marginal social benefit (*MSC* = *MSB*). Each individual, however, takes account of only the costs that he or she directly bears—that is, marginal private costs (*MPC*). This private marginal cost turns out to be the average cost for all demanders (*ASC*) if marginal social cost is borne equally (that is, averaged) among consumers. With everyone rationally treating average group cost as their marginal cost, equilibrium consumption will be at Q_{0A}, which is greater than the economically efficient level Q_0. The shaded triangle *abc* measures the loss in social surplus that results from the overconsumption.



[*Figure 5.5*] Overconsumption of Open-Access Resources

An example may help clarify why individuals in open-access and open-effort (common property) situations have an incentive to respond to marginal private cost rather than marginal social cost. Imagine that you are in a restaurant with a group of ten people who have agreed to split the bill evenly.

If you were paying your own tab, then you would not order the fancy dessert costing \$10 unless you expected to get at least \$10 worth of value from eating it. But because the actual cost to you of the dessert will be \$1 (the average increase in your bill, and for the bills of everyone else in the group—\$10 divided by ten people), you would be rational (ignoring calories, your remaining stomach capacity, and social pressure) to order it as long as it would give you at least one more dollar in value. You might continue ordering desserts until the value you placed on one more fell to \$1, your marginal private cost and the group average cost. In other words, your out-of-pocket cost for an additional dessert is the increment to your bill—it is the cost of the dessert split equally among the members of the group, or the average cost for members of the group. But this result is clearly inefficient because you and everyone else in the group could be made better off if you refrained from ordering the last dessert in return for a payment from the others of any amount between \$1 (your marginal private cost) and \$9 (the difference between the marginal social cost and your marginal private cost). Remember that the problem arises here because you have access to items on the menu at below their real social (in this case, group) cost.

Note two things about this example. First, as the number of people in the group is restricted to ten, this is an open-effort rather than open-access situation in that membership in the group is closed. However, remember that we initially defined open access broadly as including "unrestricted use by those who already have access." If anyone could freely join the group and join in the tab splitting, then it would be a pure open-access situation. As group size increases, the divergence between marginal private and marginal group (social) costs increases, so that overordering increases. Second, in this example, overordering is mitigated by the fact that participants must consume the food at the table. If the participants could take doggie bags, or even resell ordered food, then the incentives to overorder would be even greater. Natural resource extractors, who typically sell seized units rather than consume them, are thus like those who are able to use doggie bags.

		Rancher 2	
		50 Head	100 Head
Rancher 1	50 Head	($1,000, $1,000)	($600, $1,200)
	100 Head	($1,200, $600)	($700, $700)

Figure 5.6 Choice of Herd Size as a Prisoner's Dilemma

Modeling common property as a game between two potential users makes clear how individual incentives lead to overexploitation from a social perspective. Imagine that two ranchers have access to a grassland. Each must decide how many head of cattle to graze on the grassland not knowing how many head the other will graze. For simplicity, imagine that the choices are either 50 or 100 head of cattle. These choices are the *strategies* available to the ranchers in this game. Each pair of strategies, shown as the cells in Figure 5.6, yields pairs of *payoffs* to the two ranchers. If each chooses to graze 50 head, then each earns a profit of $1,000 because the field can accommodate the total herd size of 100. If each chooses to graze 100 head, then each earns a profit of only $700 because the total herd size is much too large for the field so that the cattle gain too little weight. If *rancher 1* grazes 100 head and *rancher 2* grazes 50 head, then *rancher 1* earns a profit of $1,200 and *rancher 2* a profit of only $600. If *rancher 1* grazes 50 and *rancher 2* grazes 100, then it is *rancher 2* who earns the $1,200 and *rancher 1* the $600. In these strategy combinations, the herd size is too big for the field, but the lost weight per head does not offset the advantage to the more aggressive rancher of grazing the extra 50 head.

A prediction of behavior in a game is the *Nash equilibrium. In a two-person game, a pair of strategies is a Nash equilibrium if, given the strategy of the other player, neither player wishes to change strategies.* In the game at hand, it is clear that each rancher restricting his herd size to 50 head is not a Nash equilibrium: each could raise his profit from $1,000 to $1,200 by switching to 100 head. But if one switched to 100 head, the other could raise his profits from $600 to $700 by also switching. Only when both choose 100 head would neither player have an incentive to move back to 50 head. Thus, the only Nash equilibrium in this game is for both to choose herds of 100 head. That this equilibrium is Pareto inefficient is clear—each would be made better off if they chose herds of 50 head.

Games with similar structure, called *prisoner's dilemmas*, are widely used by social scientists to model problems of cooperation.[15] They are called *noncooperative games* because of the assumption that the players cannot make binding commitments to strategies before they must be chosen. If it were possible to make binding commitments, then we could imagine the ranchers cooperating by agreeing to limit their herd sizes to 50 head before strategies are chosen. As this game is only played one time, it is called a *single-play game*. One could imagine that the ranchers faced the problem of choosing herd sizes each year. It might then be reasonable to model their interaction as a *repeated game* consisting of successive plays of the single-play, or *stage*, game. If the repeated game consists of an infinite, or uncertain, number of repetitions, and players give sufficient weight to future payoffs relative to current payoffs, then cooperative equilibria may emerge that involve each repeatedly choosing strategies that would not be equilibria in the stage game. (We return to these ideas when we discuss corporate culture and leadership in Chapter 12.)

Natural resources (both renewable and nonrenewable) have the potential for yielding scarcity rents, or returns in excess of the cost of production. With open access or open effort, these rents may be completely dissipated. In Figure 5.5, for instance, consumption at the economically efficient level Q_0 would yield rent equal to the area of rectangle P_0adAC_0. The economically efficient harvesting of salmon, for example, requires catch limits that keep the market price above the marginal costs of harvesting. In the absence of exclusion, however, these rents may well be completely dissipated.[16] (At consumption level Q_{0A} in Figure 5.5, there is no rent.) The reason is that fishers will continue to enter the industry (and those already in it will fish more intensively) as long as marginal private benefits, the rents they can capture, exceed the marginal private costs. Just as in the restaurant example, each fisher will ignore the marginal costs that his behavior imposes on other fishers. If every fisher is equally efficient, then his marginal private cost equals the average cost for the fishers as a group.

The question of how rent should be distributed is usually one of the most contentious issues in public policy. For example, how should the catch limits for salmon be divided among commercial, sport, and Aboriginal fishers? Nevertheless, from the perspective of economic efficiency, someone should receive the scarcity rent rather than allowing it to be wasted. Indeed, economic efficiency as measured by reductions in the dissipation of rent was one of the goals of the regulatory alternatives for the British Columbia salmon fishery analyzed in Chapter 16.

Note that demand for a resource may be sufficiently low so that it remains uncongested at zero price. Increases in demand, however, may push the resource from a free good (SW1) to an open-access good (SW2). A number of free goods, including such resources as buffalo, forests, aquifer water, and rangeland, have been historically important in North America. Typically, however, the free goods eventually disappeared as demand increased and excess supply was eliminated.[17] Open access often led to rapid depletion and, in some cases, near destruction of the resource before effective exclusion was achieved. For example, open access permitted destruction of the Michigan, Wisconsin, and Minnesota pine forests.[18] Nonexcludability continues to be at the heart of many water use problems in the western United States.[19] When animal populations are treated as open-access resources, the final result of overconsumption may be species extinction; such a result has already occurred with certain birds and other animals with valuable organs or fur.

Thus far we have not specified the meaning of "feasible" exclusion. Many of the goods we have given as examples appear not to be inherently nonexcludable. Indeed, one of the most famous historical examples of open-access resource—sheep and cattle grazing on the medieval English pasture—was "solved," willy-nilly, without overt government intervention, by the enclosure movement, which secured property rights for estate holders. Similar enclosures appear to be currently taking place on tribal, historically open access, lands in parts of Africa.

It is useful to dichotomize open-access and common property problems into those that are *structural* (where aspects of the goods preclude economically feasible exclusion mechanisms) and those that are *institutional* (where economically efficient exclusion mechanisms are feasible but the distribution of property rights precludes their implementation). The averaging of restaurant bills, which we previously discussed, serves as an excellent illustration of an institutional common property problem. We can imagine an obvious exclusion mechanism that we know from experience is economically feasible: separate bills for everyone. Institutional common property resource problems usually do not fundamentally inhere in the nature of the resource. Rather, they most often result from the failure of government to allocate enforceable property rights (again, the type of situation explored in the salmon fishery example in Chapter 16).

Typically, the crucial factor in making a distinction between structural and institutional problems is whether or not the good displays *spatial stationarity*. Trees are spatially stationary, salmon are not, and bodies of water may or may not be. When resources are spatially stationary, their ownership can be attached to the ownership of land. Owners of the land are usually able to monitor effectively all aspects of their property rights and, consequently, ensure exclusive use. Given exclusion, common property resources become private resources that will be used in an economically efficient manner. Without spatial stationarity, ownership of land is not a good proxy for low monitoring costs and the viability of enforcing exclusion. It does not necessarily follow that some form of private ownership could not deal with the open-access or common property problem, but it does suggest that ownership of a defined piece of land or water will not be adequate to ensure exclusion. Allocating fishing rights to specific water acreage where the fish stock moves over considerable distances, or associating the rights to oil extraction to ownership of land when the oil pool extends under a large number of parcels, illustrate the difficulty of creating effective property rights for nonstationary resources.

In summary, a stationary good may have common property resource characteristics simply because its ownership is not well defined, perhaps because of the historical accident that at one time supply exceeded demand at zero price. Nonstationary goods generally require more complex policy interventions to achieve efficiency because the linking of property rights to landownership will not serve as an effective proxy for exclusive ownership of the resource.

Reprise of Public Goods

Returning to Figure 5.2, we summarize the efficiency implications of the various types of market failures involving public goods. To reprise, the major problem with toll goods (NE quadrant—nonrivalry, excludability) is underconsumption arising from economically inefficient pricing rather than a lack of supply per se. Congestion usually further complicates these problems by introducing the need for variable pricing to achieve efficiency. In the case of pure goods (SE quadrant—nonrivalry, nonexcludability), the pervasiveness of free-riding generally leads to no market supply at all. In specific circumstances (a privileged or intermediate group in which one or a few persons account for a large fraction of demand), however, there may be some, and perhaps even nearly efficient, market supply. In the case of open-access resources (SW quadrant—rivalry, nonexcludability), inefficiency results because individuals do not equate marginal social costs with marginal benefits but, rather, marginal private costs with marginal benefits. Hence, they inefficiently overconsume and inefficiently underinvest in enhancing open-access resources.

Externalities

An *externality* is any valued impact (positive or negative) resulting from any action (whether related to production or consumption) that affects someone who did not fully consent to it through participation in voluntary exchange. Price changes in competitive markets are not relevant externalities because buyers and sellers are engaging voluntarily in exchange. We have already encountered a variety of externalities in our discussion of public goods: private supply of nonrivalrous goods by privileged and intermediate groups (a positive externality) and the divergence between marginal private and marginal social costs in the use of congested resources (a negative externality). We reserve the label *externality problem* for those situations in which the good conveying the valued impact on nonconsenting parties is the by-product of either the production or consumption of some good.

As is the case with open-access resources and ambient public goods, externality problems involve attenuated property rights because either the rights to exclusive use are incompletely specified or the costs of enforcing the rights are high relative to the benefits. Secure and enforceable property rights often permit private transactions to eliminate the economic inefficiency associated with an externality by opening up the possibility for markets in the external effect. Indeed, one can think of an externality problem as a *missing market*. We return to this point after discussing a few examples.

Common examples of negative externalities include the air and water pollution generated by firms in their production activities, the cigarette smoke that nonsmokers must still breathe in public places in some countries, and the unsightliness generated by a dilapidated house in a well-kept neighborhood. Persons who suffer these externalities place different values on them. For instance, you may really fear the health consequences of second-hand cigarette smoke, but I may be too old to worry about it. Although I would be willing to pay only a small cost to avoid sitting near a smoker, say, waiting an extra ten minutes for a restaurant table in the nonsmoking section, you might be willing to pay considerably more, say, waiting 30 minutes or leaving the restaurant altogether. Note that we can think of placing a value on these externalities in the same way we do for the goods we voluntarily consume.

Common examples of positive externalities include vaccinations that reduce everyone's risk of infectious disease (so-called herd immunity) and the benefits that neighbors receive from a homeowner's flower garden and nicely painted house. An important category of positive externality arises in the context of communication networks. One more person connecting to a Web-based digital music exchange provides marginal social benefits that exceed marginal private benefits because everyone already on the network has one more interface for potential exchange.[20] Such positive externalities are usually referred to as *network*, or *adoption, externalities*.[21]

Table 5.1 Examples of Externalities

	Positive	*Negative*
Producer-to-Producer	Recreational facilities attracting people who give custom to nearby businesses	Toxic chemical pollution harming downstream commercial fishing
Producer-to-Consumer	Private timber forests providing scenic benefits to nature lovers	Air pollution from factories harming lungs of people living nearby
Consumer-to-Consumer	Immunization by persons against contagious disease, helping reduce risk to others	Cigarette smoke from one person reducing enjoyment of meal by another
Consumer-to-Producer	Unsolicited letters from consumers providing information on product quality	Game hunters disturbing domestic farm animals

Externalities can arise in either production or consumption. Production externalities affect either firms (producer-to-producer externalities) or consumers (producer-to-consumer externalities); consumption externalities may also affect the activities of firms (consumer-to-producer externalities) or those of other consumers (consumer-to-consumer externalities). In this context, the category of consumers that are the recipients of externalities includes everyone in society. Table 5.1 provides simple examples of each type of externality. (Keep in mind that sometimes the same activity may constitute a positive externality for some but a negative externality for others.) Classifying a situation that potentially involves an externality is often a good way to begin considering its efficiency implications, distributional impacts, and, most important, possible remedies.

Efficiency Losses of Negative and Positive Externalities

Figure 5.7 illustrates the resource-allocation effects of a negative externality in production. In the presence of a negative externality, firms will produce too much of the private good that generates the externality. The market supply schedule for the private good, *MPC*, indicates the marginal costs borne directly by the firms producing it. For example, if the private good were electricity, *MPC* would represent the marginal amounts firms have to pay for the coal, labor, and other things that show up in their ledgers. But *MPC* does not reflect the negative impacts of the pollution that results from burning the coal. If somehow we could find out how much each person in society would be willing to pay to avoid the pollution at each output level, then we could add these amounts to the marginal costs actually seen by the firms to derive a supply schedule that reflected total social marginal costs. We represent this more inclusive supply schedule as *MSC*.

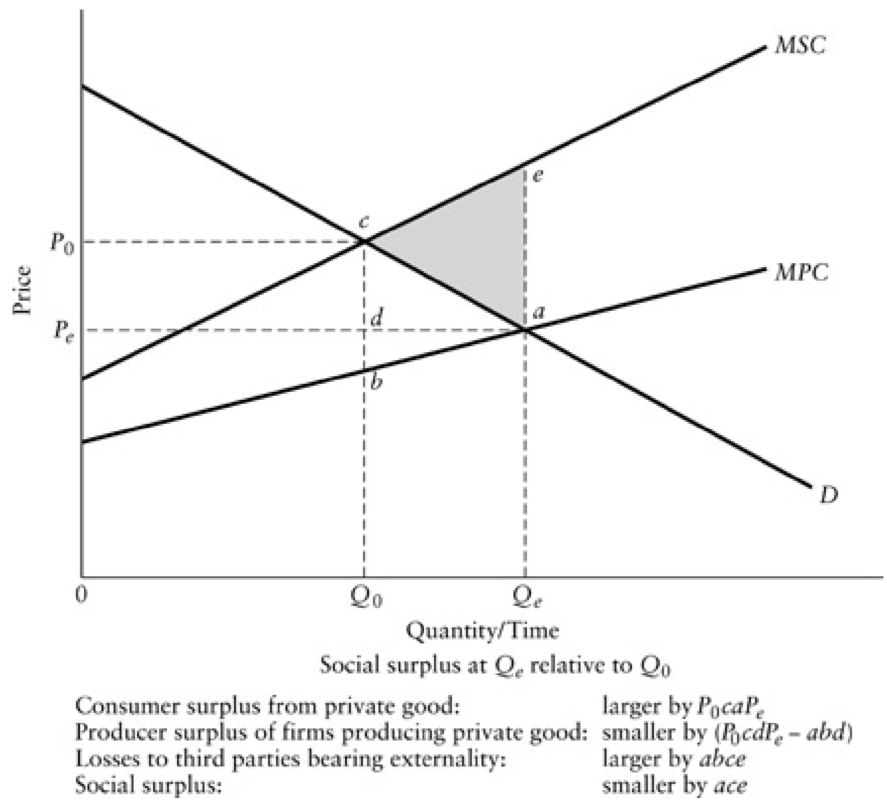


Figure 5.7 Overproduction with a Negative Externality

Economic efficiency requires that social marginal benefits and social marginal costs be equal at the selected output level; this occurs at quantity Q_0, where marginal social cost (*MSC*) and demand (*D*) intersect. But because firms do not consider the external costs of their output, they choose output level Q_e at the intersection of *MPC* and *D*. Relative to output level Q_0, consumers of the private good being produced gain surplus equal to area P_0caP_e (because they get quantity Q_e at price P_e rather than quantity Q_0 at price P_0) and producers lose surplus equal to area P_0cdP_e minus area *abd* (the first area captures the effect of the lower price, the second the effect of greater output). Those who bear the external costs, however, suffer a loss given by the area *abce* (the area between the market and social supply curves over the output difference; remember, the vertical distance between the supply curves represents the external marginal cost). The net loss in social surplus is the area of triangle *ace*, the algebraic sum of the surplus differences for the consumers and producers of the private good and the bearers of the externality. (This net social surplus loss is simply the deadweight loss due to the overproduction—the area between *MSC* and *D* from Q_0 to Q_e.) In other words, Pareto efficiency requires a reduction in output from the equilibrium level in the market (Q_e) to the level at which marginal social costs equal marginal social benefits (Q_0).

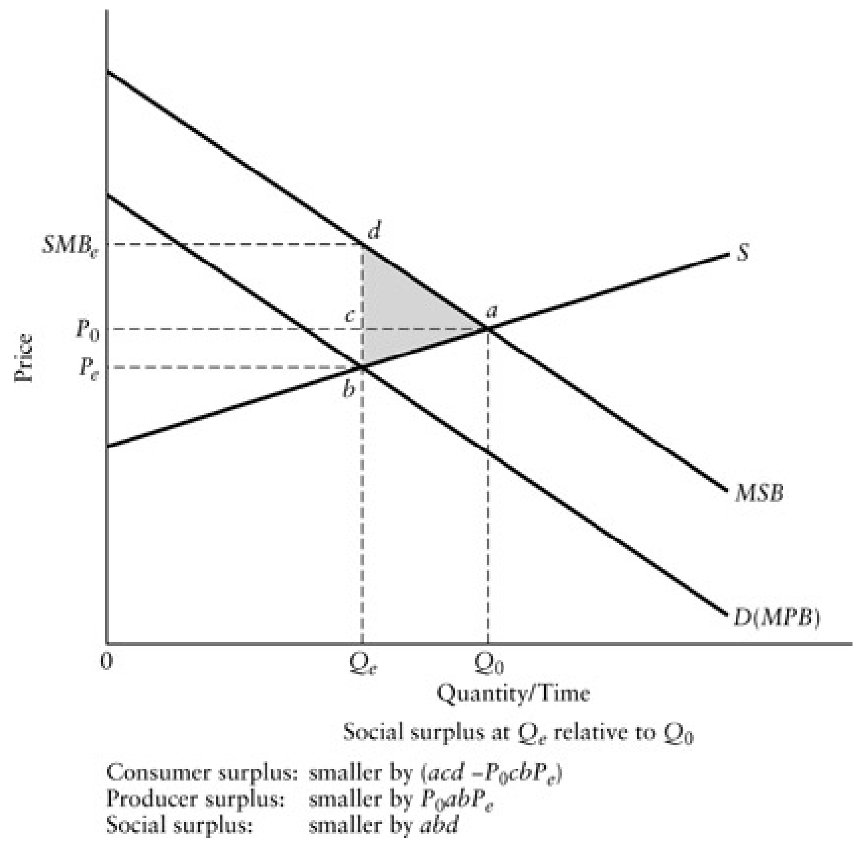


Figure 5.8 Underproduction with a Positive Externality

Turning to positive externalities, we can think of the private good generating benefits that cannot be captured by the producer. For example, firms that plant trees for future harvesting may provide a scenic benefit for which they receive no compensation. In Figure 5.8 we illustrate the demand schedule for the private good (trees for future harvest) as *D*, which also gives the marginal private benefits (*MPB*). At each forest size, however, the social marginal benefit schedule would equal the market demand plus the amounts that viewers would be willing to pay to have the forest expanded by another unit. *MSB* labels the social marginal benefit schedule, which includes both the private and external marginal benefits. Market equilibrium results at output level Q_e, where the market demand schedule and the supply schedule intersect. But if output were raised to Q_0, consumer surplus would increase by area *acd* (resulting from increased consumption from Q_e to Q_0) minus area P_0cbP_e (due to the price rise from P_e to P_0), and producer surplus would increase by area P_0abP_e. The net increase in social surplus, therefore, would be area *abd.* Again, we see that in the case of an externality, we can find a reallocation that increases social surplus and thereby offers the possibility of an increase in efficiency.

Market Responses to Externalities

Will the market always fail to provide an efficient output level in the presence of externalities? Just as pure public goods will sometimes be provided at efficient, or nearly efficient, levels through voluntary private agreements within intermediate groups, so too may private actions counter the inefficiency associated with externalities. Ronald Coase first pointed out the relevance of such private responses in a seminal article on the externality problem.[22] He argued that in situations in which property rights are clearly defined and costless to enforce, and utility is linear in wealth, costless bargaining among participants will lead to an economically efficient level of external effect. Of course, the distribution of costs and benefits resulting from the bargaining depends on who owns the property rights.

Before exploring the issue further, we must stress a very restrictive assumption of Coase's model that limits its applicability to many actual externality situations. Namely, Coase assumes zero transaction costs in the exercise of property rights.[23] In many real-world situations, transaction costs are high, usually because those producing and experiencing the externality are numerous. With large numbers of actors, the bargaining that produces the Coasian outcome becomes impossible because of the high costs of coordination in the face of opportunities for a free ride.

Nevertheless, Coase's insight is valuable. He pointed out that in the case of small numbers (he assumes one source and one recipient of the externality), the allocation of property rights alone would lead to a negotiated (that is, private) outcome that is economically efficient in the sense of maximizing the total profits of the parties. Assuming that individuals make decisions based on the dollar value rather than the utility value of external effects, efficiency results whether a complete property right is given to the externality generator (one bears no liability for damages caused by one's externality) or to the recipient of the externality (one bears full liability for damages caused by one's externality). Either rule should lead to a bargain being reached at the same level of externality; only the distribution of wealth will vary, depending on which rule has force. Assuming that individuals maximize utility rather than net wealth, however, opens the possibility for different allocations under different property rights assignments.[24]

A moment's thought should suggest why large numbers will make a Coasian solution unlikely. Bargaining would have to involve many parties, some of whom would have an incentive to engage in strategic behavior. For example, in the case of a polluter with no liability for damages, those experiencing the pollution may fear free-riding by others. In addition, firms may engage in opportunistic behavior, threatening to generate more pollution to induce payments. Under full liability, individuals would have an incentive to overstate the harm they suffer. Other things equal, we expect that the greater the number of parties experiencing the externality, the greater will be the costs of monitoring damage claims.

Nevertheless, private cooperation appears effective in some situations. For instance, neighborhood associations sometimes do agree on mutually restrictive covenants, and individual neighbors occasionally do reach contractual agreements on such matters as light easements (which deal with the externalities of the shadows cast by buildings).

Moreover, there is an important case where Coase-like solutions do arise, even with large numbers of parties—namely, where: (1) property rights become implicitly established by usage; (2) the value of the externality (whether positive or negative) is captured by (more technically, "capitalized into”) land values; (3) considerable time has passed such that the initial stock of external parties has “rolled over”; and (4) externality levels remain stable. The relevance of these conditions can best be explained with an example. Suppose that a factory has been polluting the surrounding area for many years without anyone challenging its owner’s right to do so. It is probable that the pollution will result in lower property values.[25] Residents who bought before the pollution was anticipated will have to sell their houses for less—reflecting the impact of the pollution. New homeowners, however, will not bear any Pareto-relevant externality, because the negative impact of the pollution will be capitalized into house prices. The lower prices of the houses will reflect the market’s (negative) valuation of the pollution. In other words, through the house price it is possible to get a proxy dollar measure of the disutility of pollution. Notice that a second generation of homeowners (that is, those who bought houses after the pollution was known and capitalized into prices) would receive a bonus if the existing allocation of property rights were changed so that the factory had to compensate current homeowners for existing levels of pollution. Of course, if there are unexpected changes in the level of pollution (or new information about the harmful impacts of the pollution—see our discussion of information asymmetry below), there will be, in effect, new (either positive or negative) impacts; in these situations, considerable argument is likely to occur over who has rights to compensation for the changes.

Natural Monopoly

Natural monopoly occurs when average cost declines over the relevant range of demand. Note that this definition is in terms of both cost and demand conditions. In the case of natural monopoly, a single firm can produce the output at lower cost than any other market arrangement, including competition.

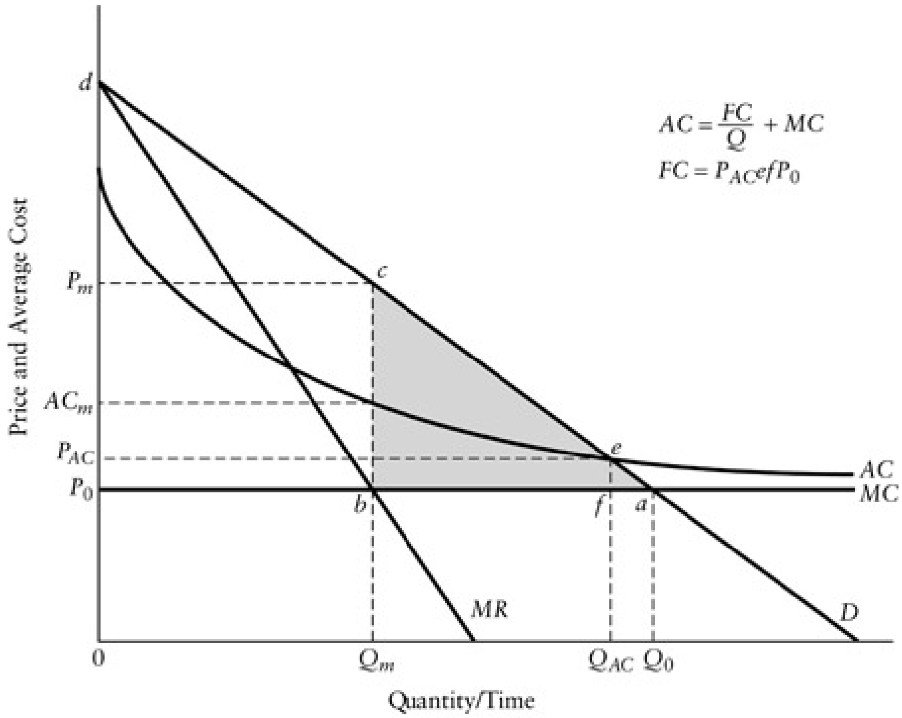
While the cost-and-demand conditions establish the existence of a natural monopoly situation, the *price elasticity of demand* determines whether or not the natural monopoly has important implications for public policy. The price elasticity of demand measures how responsive consumers are to price changes. Specifically, the price elasticity of demand is defined as the percentage change in the quantity demanded that results from a 1 percent change in price.[26] If the absolute value of the price elasticity of demand is less than one (a 1 percent change in price leads to less than a 1 percent reduction in the quantity demanded), then we say that demand is inelastic and an increase in price will increase total revenue. A good is unlikely to have inelastic demand if there are other products that, while not exactly the same, are close substitutes. In such circumstances, the availability of substitutes greatly limits the economic inefficiency associated with natural monopoly. For example, although local cable in some television markets may have the cost-and-demand characteristics of natural monopolies, many substitute products, including over-the-air television, satellite TV providers, and DVD players, may prevent cable television companies from realizing large monopoly rents in their local markets.

We should also keep in mind that, although we stated the basic definition of natural monopoly in static terms, markets in the real world are dynamic. Technological change may lead to different cost characteristics, or high prices may bring on close or even superior substitutes. The result may be the elimination of natural monopoly or the supplanting of one natural monopoly technology by another.[27] For example, especially in developing countries, cellular phones are direct substitutes for land-line phones, providing competition for traditional telephone systems.

Allocative Inefficiency under Natural Monopoly

We show a cost structure leading to natural monopoly in Figure 5.9. Marginal cost (*MC*) is shown as constant over the range of output. Fixed cost (*FC*), which is not shown in the figure, must be incurred before any output can be supplied. Because of the fixed cost, average cost (*AC*) starts higher than marginal cost and falls toward it as the output level increases. ($AC = FC/Q + MC$, where Q is the output level.) Although marginal cost is shown as constant, the same analysis would apply even if marginal cost were rising or falling, provided that it remains small relative to fixed cost.

Figure 5.9 shows the divergence between the firm's profit-maximizing behavior and economic efficiency. Let us first consider the economically efficient price and output. Efficiency requires, as we discussed in the previous chapter, that price be set equal to marginal cost ($P = MC$). Because marginal cost is below average cost, the economically efficient output level Q_0 results in the firm suffering a loss equal to *FC*. Obviously, the firm would not choose this output level on its own. Instead, it would maximize profits by selecting output level Q_m, which equates marginal revenue to marginal cost ($MR = MC$). At output level Q_m, the market price will be P_m and profits equal to the area of rectangle $P_mcbP_0 - FC$ will result. Relative to output level Q_0, consumer surplus is lower by the area of P_mcaP_0, but the profit of the firm is larger by P_mcbP_0. The net loss in social surplus due to the underproduction equals the area of the shaded triangle *abc*, the deadweight loss to consumers. (As noted in our discussion of Figure 4.5, units of forgone output would have offered marginal benefits in excess of marginal costs—a total loss equal to the area between the marginal benefit, or demand schedule, and the marginal cost curve. Again, in Figure 5.9, this loss is the area of triangle *abc*.)



	Monopoly pricing (P_m)	Efficient pricing (P_0)	Average cost pricing (P_{AC})
Consumer surplus:	P_mcd	P_0ad	$P_{AC}ed$
Total revenue:	P_mcQ_m0	P_0aQ_00	$P_{AC}eQ_{AC}0$
Total cost:	$FC + P_0bQ_m0$	$FC + P_0aQ_00$	$FC + P_0fQ_{AC}0$
Producer surplus:	$P_mcbP_0 - FC$	$- FC$	0
Social surplus:	$P_0dcb - FC$	$P_0ad - FC$	$P_{AC}ed$

Net social surplus loss of monopoly pricing relative to efficient pricing: *abc*
Net social surplus loss of average cost pricing relative to efficient pricing: *aef*

Figure 5.9 Social Surplus Loss from Natural Monopoly

Suppose that public policy forced the natural monopolist to price at the economically efficient level (P_0). The monopolist would suffer a loss of *FC*. Consequently, the monopolist would go out of business in the absence of a subsidy to offset the loss. Notice the dilemma presented by natural monopoly: *forcing the monopolist to price efficiently drives it out of business; allowing the monopolist to set price to maximize profits results in deadweight loss.*

Briefly consider what would happen if the firm were forced by public policy to price at average, rather than marginal, cost—that is, if the firm priced at P_{AC} and produced Q_{AC} in Figure 5.9. Clearly, under these circumstances the firm can survive because its costs are now being just covered. (Note that *FC* equals $P_{AC}efP_0$.) Although the deadweight loss under average cost pricing is much lower than under monopoly pricing (area *aef* versus area *abc*), it is not eliminated. Therefore, *average cost pricing represents a compromise to the natural monopoly dilemma.*

Restraints When Markets Are Contestable

We can imagine circumstances in which the natural monopoly firm might, in fact, be forced to price at average cost because of the threat of competition from potential entrants. The crucial requirement is that entry to, and exit from, the industry be relatively easy. Whether or not the firm has in-place capital that has no alternative use—capital whose costs are *sunk*—usually determines the viability of potential entry and exit. If the established natural monopoly has a large stock of productive capital that cannot be sold for use in other industries, it will be difficult for other firms to compete because they must first incur cost to catch up with the established firm's capital advantage. The in-place capital serves as a barrier to entry; the greater the replacement cost of such capital, the higher the barrier to entry. For example, once a petroleum pipeline is built between two cities, its scrap value is likely to be substantially less than the costs a potential competitor would face in building a second pipeline. The greater is the difference, the greater is the ability of the established firm to fight off an entry attempt with temporarily lower prices. Of course, keep in mind that the ability of the firm to charge above the marginal cost level will be influenced by the marginal costs of alternative transportation modes, such as truck and rail, which can serve as substitutes.

Much economic research considers industries with low barriers to entry and decreasing average costs, which, because of the threat of potential entry, are said to be in *contestable markets.*[28] We expect markets that are contestable to exhibit pricing closer to the efficient level. One of the most important debates in the literature arising from the contestable market framework concerns the empirical significance of in-place capital as effective barriers to entry.[29] Most natural monopolies appear to enjoy large advantages in in-place capital, raising the question of whether they should be viewed as being in contestable markets.

As we have seen, the "naturalness" of a monopoly is determined by the presence of decreasing average cost over the relevant range of output. In many situations, it appears that average cost declines over considerable ranges of output but then flattens out. In other words, initial economies of scale are exhausted as output increases. What happens if demand shifts to the right (for example, with population increases) such that the demand curve intersects the flat portion of the average cost curve? There may be considerable room for competition under these circumstances. Figure 5.10 illustrates such a situation. If demand is only D_1 (the "classic" natural monopoly), only one firm can survive in the market. If demand shifts outward beyond D_2 to, say, D_3, two or more firms may be able to survive because each can operate on the flat portion of the average cost curve. Simply because two firms could survive at D_3 does not mean that two competing firms will actually emerge as demand expands. If the original natural monopoly firm expands as demand moves out, it may be able to forestall new entrants and capture all of the market. Nevertheless, we are again reminded of the importance of looking beyond the static view.

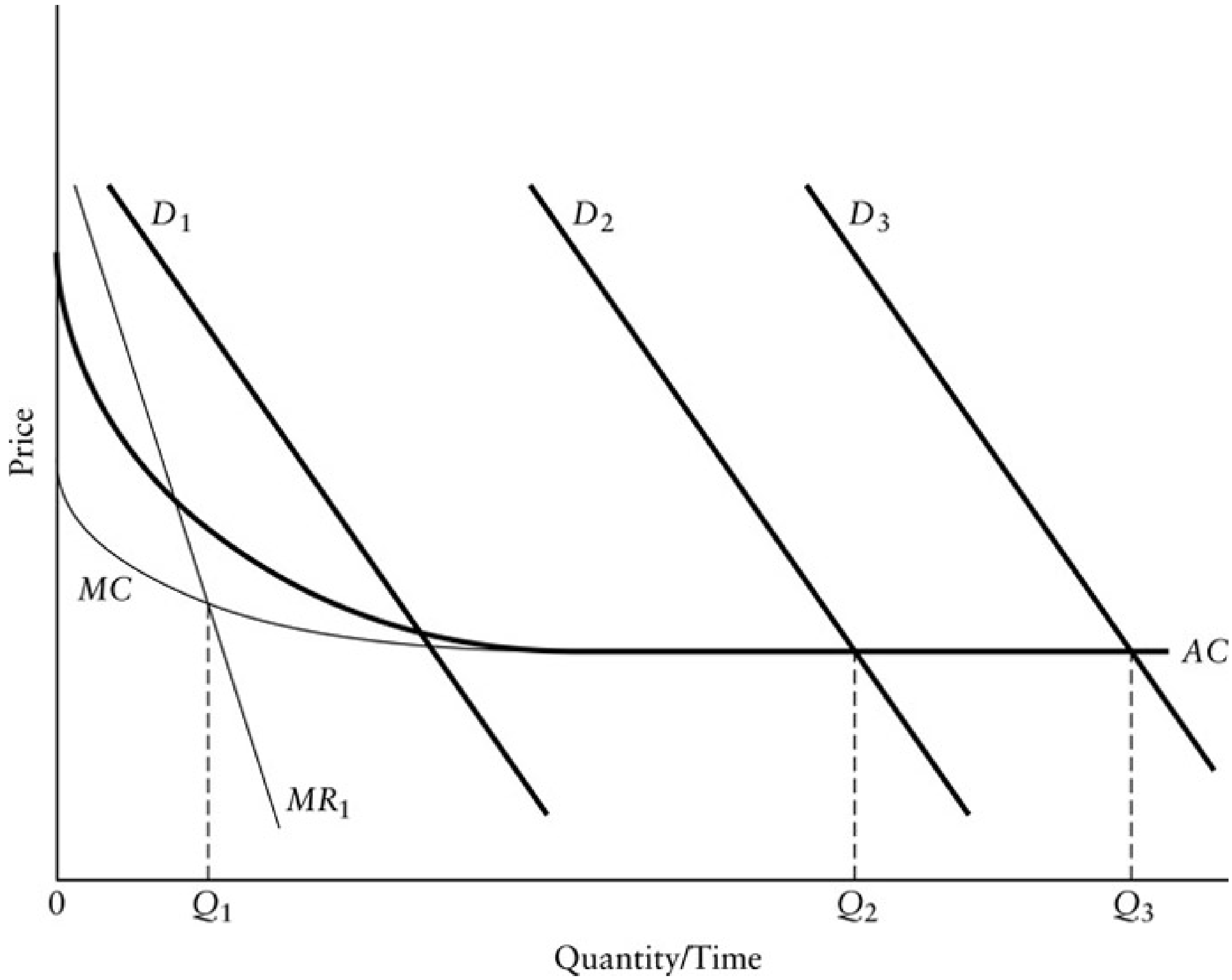


Figure 5.10 Shifting Demand and Multiple Firm Survival

Source: William G. Shepherd, *The Economics of Industrial Organization* (Englewood Cliffs, NJ: Prentice Hall, 1979), Fig. 3.4, 59.

When considering natural monopoly from a policy perspective, we often find that legal and regulatory boundaries do not correspond to the boundaries delineating natural monopoly goods. Most discussion of industrial issues tends to be within the framework of product "sectors," such as the electric and the telephone industries. Unfortunately, the economic boundaries of natural monopolies are not likely to conform to these neat sectoral boundaries. Historically, regulation has often not recognized this unpleasant fact. In addition, the existence, or extent, of a natural monopoly can change as production technology or demand changes.

The telecommunications industry illustrates these definitional problems. In 1982, the U.S. Justice Department and AT&T agreed on a plan for the breakup of the corporation. The agreement called for a division into two parts: the part that could be expected to be workably contestable and the part that had strong natural monopoly elements. Both long-distance services (contestable) and equipment manufacture and supply (competitive) were largely deregulated, while local telephone exchanges were deemed to be regional natural monopolies.[30] Similar problems with sectoral definitions have not been well recognized in other contexts, however. For example, electricity generation and transmission have typically been treated as part of an electrical utility natural monopoly, although the evidence suggests that in many circumstances only transmission has the required cost and demand characteristics to be considered a natural monopoly.[31] (Running multiple sets of transmission lines across the countryside would generally be inefficient; having multiple firms generating electricity would not be.)

Another dimension, apart from the sectoral, where the problem of defining natural monopoly arises is the spatial. Over what spatial area does the natural monopoly exist? Almost all natural monopoly regulation corresponds to existing city, county, state, and federal boundaries; but the economic reality of a natural monopoly knows no such bounds—it is purely an empirical question how far (spatially) the natural monopoly extends. Again, the appropriate spatial boundaries of a natural monopoly can change with changes in technology.

X-Inefficiency Resulting from Limited Competition

Thus far we have described the potential social costs of natural monopoly and whether it prices to either maximize profits or cover cost, in terms of deadweight loss caused by allocational inefficiency. The social costs, however, may be larger, because natural monopolies do not face as strong incentives as competitive firms to operate at minimum cost. One of the greatest advantages of a competitive market is that it forces firms to keep their costs down; in other words, the whole average and marginal cost curves are as low as they can possibly be. In the absence of competition, firms may be able to survive without operating at minimum cost. Harvey Leibenstein coined the phrase *X-inefficiency* to describe the situation in which a monopoly does not achieve the minimum costs that are technically feasible.[32] (As we shall see, *X*-inefficiency is not fully descriptive because transfers as well as technical inefficiencies are often involved.) One also sometimes finds the terms *cost inefficiency, operating inefficiency*, or *productive inefficiency* used to describe the same phenomenon.

In Figure 5.11 we incorporate *X*-inefficiency into the basic analysis of monopoly pricing. But ignore *X*-inefficiency for the moment. The deadweight loss associated with a profit-maximizing natural monopoly is the darkly shaded triangular area *abc* (already discussed in Figure 5.9). If we take *X*-inefficiency into account, however, the minimum possible marginal cost curve is lower than that observed from the behavior of the firm. The actual deadweight loss, therefore, is at least equal to the larger triangular *aeg*—the additional loss of *cbeg* results because output is Q_{mx} rather than Q_m.

Some or all of the very lightly shaded area $MC_{mx}fgP_0$ represents unnecessary cost that should be counted as either producer surplus or deadweight loss. If marginal costs are higher than the minimum because the managers of the firms employ more real resources, such as hiring workers who stand idle, then the area $MC_{mx}fgP_0$ should be thought of as a social surplus loss. If, on the other hand, costs are higher because the managers pay themselves and their workers higher-than-necessary wages, we should consider this area as rent—it represents unnecessary payments rather than the misuse of real resources.[33] If, however, potential employees and managers spend time or other resources attempting to secure the rent (say, by enduring periods of unemployment while waiting for one of the overpaid jobs to open up), then the rent may be dissipated and thus converted to deadweight loss.

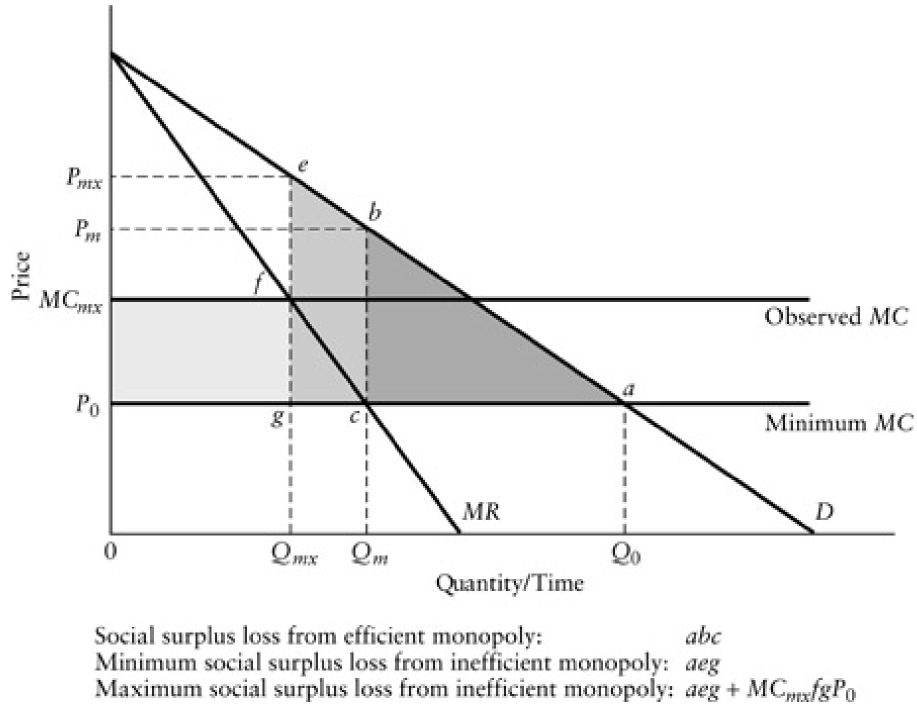


Figure 5.11 X-Inefficiency under Natural Monopoly

Note that in the case of an unregulated natural monopoly, one would not expect *X*-inefficiency to occur unless there were a divergence of information and interests between owners of the firm and its managers. We put this type of problem into the perspective of agency theory in our discussion of government failures in Chapter 8.

In summary, natural monopoly inherently involves the problem of undersupply by the market. The extent of undersupply depends on the particular cost-and-demand conditions facing the monopolist and the extent to which the market can be contested. Natural monopoly may involve additional social surplus losses because the absence of competition permits production at greater-than-minimum cost to persist.

Information Asymmetry

Please note that we do not use the title "information costs" or "imperfect information" in this section. The reason is that information is involved in market failure in at least two distinct ways. First, information itself has public-good characteristics. Consumption of information is nonrivalrous—one person's consumption does not interfere with another's; the relevant analytical question is primarily whether exclusion is possible. Thus, in the context of lack of supply of public goods, we are interested in the production and consumption of information itself. Second, and the subject of our discussion here, there may be situations in which the amount of information about the characteristics of a good varies in relevant ways across persons. The buyer and the seller in a market transaction, for example, may have different information about the quality of the good being traded. Similarly, there may be differences in the amount of information relating to the attributes of an externality between the generator of the externality and the affected party. Workers, for instance, may not be as well informed about the health risks of industrial chemicals as their employers. Notice that *in the information asymmetry context we are not primarily interested in information as a good, but in differences in the information that relevant parties have about any good*. We thus distinguish between information *as* a good (the public good case) and information *about* a good's attributes as distributed between buyer and seller or between externality generator and affected party (the *information asymmetry* case).

Figure 5.12 illustrates the potential social surplus loss associated with information asymmetry.[34] D_U represents the quantities of some good that a consumer would purchase at various prices in the absence of perfect information about its quality. It can, therefore, be thought of as the consumer's uninformed demand schedule. D_I represents the consumer's informed demand schedule—the amounts of the good that would be purchased at various prices if the consumer were perfectly informed about its quality. The quantity actually purchased by the uninformed consumer is determined by the intersection of D_U with the supply schedule, S. This amount, Q_U, is greater than Q_I, the amount that the consumer would have purchased if fully informed about the quality of the good. The darkly shaded area *abc* equals the deadweight loss in consumer surplus resulting from the overconsumption. (For each unit purchased beyond Q_I, the consumer pays more than its marginal value as measured by the height of the informed demand schedule.) This excess consumption also results in a higher equilibrium price (P_U), which transfers surplus equal to the area $P_U baP_I$ from the consumer to the producer of the good. Figure 5.12 signals the presence of information asymmetry if the producer could have informed the consumer about the true quality of the good at a cost less than the deadweight loss in consumer surplus resulting when the consumer remains uninformed. More generally, *we have market failure due to information asymmetry when the producer does not supply the amount of information that maximizes the difference between the reduction in deadweight loss and the cost of providing the information.*

The same sort of reasoning would apply if the consumer underestimated rather than overestimated the quality of the good. The consumer would suffer a deadweight loss resulting from consuming less than Q_I. The incentives that the producer faces to provide the information, however, can be quite different in the two cases. In the case where the consumer overestimates quality, providing information results in a lower price and, therefore, a smaller transfer of surplus from the consumer to the producer, an apparent disincentive to provide the information. In the case where the consumer underestimates quality, providing information results in a higher price that increases producer surplus; the prospect of this gain may encourage the producer to supply information. As we discuss later, however, this incentive to provide information can be muted if producers are unable to get consumers to distinguish their products from those of competitors.

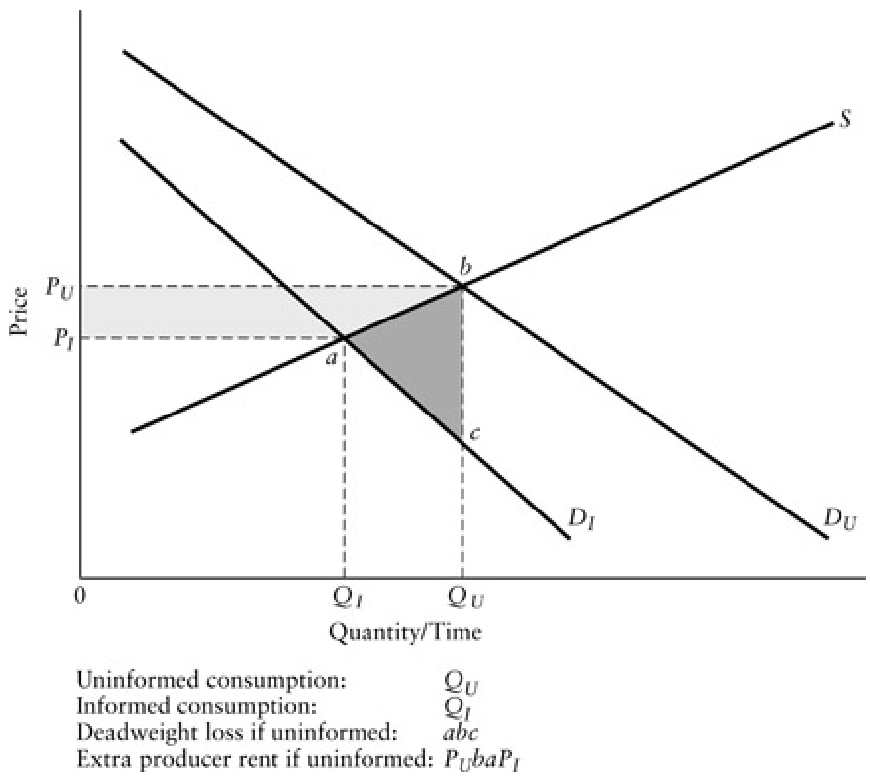


Figure 5.12 Consumer Surplus Loss from Uninformed Demand

Diagnosing Information Asymmetry

Our first task in deciding when information asymmetry is likely to lead to market failure is to classify goods into useful categories. Economists have generally divided goods into two categories: *search goods* and *experience goods.*[35] A good is a search good if consumers can determine its characteristics with certainty prior to purchase. For example, a chair in stock in a store is a search good because consumers can judge its quality through inspection prior to purchase. A good is an experience good if consumers can determine its characteristics only after purchase; examples include meals, hairstyling, concerts, legal services, and used automobiles. We add a third category, which we call *post-experience goods*, to distinguish those goods for which it is difficult for consumers to determine quality even after they have begun consumption. For example, people may fail to associate adverse health effects with drugs that they are consuming. Experience goods and post-experience goods differ primarily in terms of how effectively consumers can learn about quality through consumption. After some period of consumption, the quality of the experience goods generally becomes apparent; in contrast, continued consumption does not necessarily reveal to consumers the quality of the post-experience good.[36]

Within these three categories, a number of other factors help determine whether information asymmetry is likely to lead to serious market failure. The effectiveness of any information gathering strategy, other things equal, generally depends on the *variance in the quality* of units of a good (heterogeneity) and the *frequency* with which consumers make purchases. The potential costs of the information asymmetry to consumers depend on the extent to which they perceive the *full price* of the good, including imputed costs of harm from use.[37] The *cost of searching* for candidate purchases and the full price determine how expensive and potentially beneficial it is for consumers to gather information.

Search Goods

Searching can be thought of as a sampling process in which consumers incur costs to inspect units of a good. A consumer pays a cost C_S to see a particular combination of price and quality. If the price exceeds the consumer's marginal value for the good, no purchase is made and the consumer either again pays C_S to see another combination of price and quality or stops sampling. If the consumer's marginal valuation exceeds price, then the consumer either makes a purchase or pays C_S again in expectation of finding a more favorable good in terms of the excess of marginal value over price. When C_S is zero, the consumer will find it advantageous to take a large sample and discover the complete distribution of available price and quality combinations so that the pre-search information asymmetry disappears. For larger C_S, however, the consumer will take smaller samples, other things equal, so that information asymmetry may remain. Additionally, because the range in price for a given quality is likely to be positively correlated with price, optimal sample sizes will be smaller the larger the ratio of *Cs* to expected price.

The more heterogeneous the available combinations of price and quality, the more likely that the consumer will fail to discover a more favorable choice for any given sample size. In contrast, even small samples will eliminate information asymmetry if the price and quality combinations are highly homogeneous. Once consumers realize that nearly identical units are offered at the same price, the optimal sample size falls to one.

Going beyond a static view, the frequency of purchase becomes important in determining whether information asymmetry remains. *If the frequency of purchase is high relative to the rate at which the underlying distribution of combinations of price and quality changes, then consumers accumulate information over time that reduces the magnitude of the information asymmetry.* If the frequency of purchase is low relative to the rate of change in the underlying distribution, then accumulated information will not necessarily lead to reductions in information asymmetry. In either case, however, frequent purchasers may become more experienced searchers so that *Cs* falls and larger samples become efficient.

Thus, if search costs are small relative to the expected purchase price *or* the distribution of price and quality combinations is fairly homogeneous *or* the frequency of purchase is high relative to the rate of change in the distribution of price and quality combinations, then information asymmetry is unlikely to lead to significant inefficiency. *In the case where search costs are high relative to the expected purchase price, the distribution of price and quality combinations is very heterogeneous, and the frequency of purchase is relatively low, information asymmetry may lead to significant inefficiency.*

If it is possible for producers to distinguish their products by brand, however, then they have an incentive to undertake informative advertising that reduces search costs for consumers. When brands are difficult to establish, as in the case of, say, farm produce, retailers may act as agents for consumers by advertising prices and qualities. Because the veracity of such advertising can be readily determined by consumers through inspection of the goods, and because retailers and firms offering brand name products have an incentive to maintain favorable reputations, we expect the advertising generally to convey accurate and useful information. Reputation is likely to be especially important in emerging electronic commerce markets because consumers cannot directly examine goods before purchasing them. For example, a study found that sellers' good reputations had a statistically significant, albeit small, positive impact on price for an Internet auction good.[38]

In summary, search goods rarely involve information asymmetry that leads to significant and persistent inefficiency. When inefficiency does occur, it takes the form of consumers forgoing purchases that they would have preferred to ones that they nevertheless found beneficial. From the perspective of public policy, intervention in markets for search goods can rarely be justified on efficiency grounds.

Experience Goods: Primary Markets

Consumers can determine the quality of experience goods with certainty only through consumption. To sample, they must bear the search costs, C_S, and the full price, P^* (the purchase price plus the expected loss of failure or damage collateral with consumption).[39] The full price of consuming a meal at an unfamiliar restaurant, for instance, is the sum of purchase price (determined from the menu) and the expected cost of any adverse health effects (ranging from indigestion to poisoning) from the meal being defective. Of course, even when the expected collateral loss is zero, prior to consumption, the marginal value that the consumer places on the meal is not known with certainty.

In contrast to search goods—where, holding search costs constant, consumers optimally take larger samples for more expensive goods—they optimally take smaller samples for more expensive experience goods. Indeed, for all but the very inexpensive experience goods, we expect sampling (equivalent to the frequency of purchase for experience goods) to be governed primarily by durability. For example, sampling to find desirable restaurants will generally be more frequent than sampling to find good used automobiles.

As is the case with search goods, *the more heterogeneous the quality of an experience good, the greater is the potential for inefficiency due to information asymmetry.* The consumption of an experience good, however, may involve more than simply forgoing a more favorable purchase of the good. Once consumption reveals quality, the consumer may discover that the good provides less marginal value than its price and therefore regret having made the purchase regardless of the availability of alternative units of the good. The realized marginal value may actually be negative if consumption causes harm.

Learning from the consumption of experience goods varies in effectiveness. If the quality of the good is homogeneous and stable, then learning is complete after the first consumption—consumers know how much marginal value they will derive from their next purchase, including the expected loss from product failure or collateral damage. If the quality is heterogeneous or unstable, then learning proceeds more slowly. Unless consumers can segment the good into more homogeneous groupings, say, by brands or reputations of sellers, learning may result only in better ex ante estimates of the mean and variance of their ex post marginal valuations. When consumers can segment the good into stable and homogeneous groupings, repeated sampling helps them discover their most-preferred sources of the good in terms of the mean and variance of quality. For example, national motel chains with reputations for standardized management practices may provide travelers with a low-variance alternative to independent motels in unfamiliar locales.

When can informative advertising play an important role in reducing information asymmetry? Generally, informative advertising can be effective when consumers correctly believe that sellers have a stake in maintaining reputations for providing reliable information. A seller who invests heavily in developing a brand name with a favorable reputation is more likely to provide accurate and useful information than an unknown firm selling a new product or an individual owner selling a house or used automobile.

When consumers perceive that sellers do not have a stake in maintaining a good reputation, and the marginal cost of supply rises with quality, then a "lemons" problem may arise: consumers perceive a full price based on average quality so that only sellers of lower-than-average-quality goods can make a profit and survive in the market.[40] In the extreme, producers offer only goods of low quality.[41]

When reliability is an important element of quality, firms may offer warranties that promise to compensate consumers for a portion of replacement costs, collateral damage, or both. The warranties thus provide consumers with insurance against low quality: higher purchase prices include an implicit insurance premium, and the potential loss from product failure is reduced. The quality of a warranty may itself be uncertain, however, if consumers do not know how readily firms honor their promises. Further, firms may find it impractical to offer extensive warranties in situations where it is costly to monitor the care taken by consumers. Nevertheless, warranties are a common device for reducing the consequences of information asymmetry for experience goods.

Experience Goods: The Role of Secondary Markets

Producers and consumers often turn to private third parties to help remedy information asymmetry problems. Certification services, agents, subscription services, and loss control by insurers are the most common market responses that arise.

Certification services "guarantee" minimum quality standards in processes or products. Professional associations, for instance, often set minimum standards of training or experience for their members—the board-certified specialties in medicine are examples. Perhaps closer to most people's experience, the Better Business Bureau requires members to adhere to a code of fair business practices. Underwriters Laboratories test products against minimum fire safety standards before giving its seal of approval.[42] When such services establish their own credibility, they help producers to distinguish their goods satisfying the minimum standards from goods that do not.

Agents often sell advice about the qualities of expensive, infrequently purchased, heterogeneous goods. These agents combine expertise with learning from being participants in a large number of transactions between different pairs of buyers and sellers. For example, most people do not frequently purchase houses, which are expensive and heterogeneous in quality. Because owners typically enjoy an informational advantage by virtue of their experiences living in their houses, prospective buyers often turn to engineers and architects to help assess structural integrity. Art and antique dealers, jewelers, and general contractors provide similar services.

The problem of excluding nonpaying users limits the supply of agents for goods that are homogeneous because one consumer can easily pass along information that the agent provides to prospective purchasers of similar units of the good. Inexpensive goods are unlikely to provide an adequate incentive for consumers to pay for the advice of agents. (Note that the full price is the relevant measure: one might very well be willing to pay for a visit to a doctor to get advice about a drug with a low purchase price but a high potential for harm to one's health.) Finally, agents are less likely to be relatively attractive for goods with high frequencies of purchase because consumers can often learn effectively on their own.

Consumers often rely on the experiences of friends and relatives to gather information about the quality of branded products. They may also be willing to pay for published information about such products. But such subscription services, which have public-good characteristics, are likely to be undersupplied from the perspective of economic efficiency because nonsubscribers can often free ride on subscribers by interrogating them and by borrowing their magazines. (The very existence of subscription services such as *Consumer Reports* suggests that a large number of consumers see the marginal costs of mooching or using a library as higher than the subscription price.)

Insurers sometimes provide information to consumers as part of their efforts to limit losses. For example, health maintenance organizations, which provide medical insurance, often publish newsletters that warn members of potentially dangerous and unhealthy goods such as diet products and tanning salons. Casualty insurers may signal warnings through their premiums as well as through direct information. Fire insurance underwriters, for instance, set premiums based on the types of equipment that businesses plan to use; they also often inspect commercial properties to warn proprietors of dangerous equipment and inform them about additions that could reduce their risks of fire.

Experience Goods: Summary

By their very nature, experience goods offer the potential for serious inefficiency caused by information asymmetry. Secondary markets, however, limit the inefficiency for many experience goods by facilitating consumer learning and providing incentives for the revelation of product quality. Nevertheless, problems are likely to remain in two sets of circumstances: first, when quality is highly heterogeneous, branding is ineffective, and agents are either unavailable or expensive relative to the full price of the good; and second, where the distribution of quality is unstable, so that consumers and agents have difficulty learning effectively. In these fairly limited circumstances, market failure may justify public intervention on efficiency grounds.

Post-Experience Goods

Consumption does not perfectly reveal the true quality of post-experience goods.[43] Quality remains uncertain because the individual consumer has difficulty recognizing the causality between consumption and some of its effects. For example, consumers may not recognize that the fumes of a cleaning product increase their risks of liver cancer because they do not expect the product to have such an effect. Often, the effects of consumption only appear after delay, so that consumers do not connect them with the product. Young smokers, for instance, may fully appreciate the addictive power of tobacco only after they are addicted, although the manufacturer did know, or should have reasonably known, of this effect. Beyond drugs, many medical services appear to be post-experience goods because patients often cannot clearly link the state of their health to the treatments that they have received.

In some cases it is extremely difficult to distinguish between the lack of supply of a public good (the research to find out about the long-term effects of a product) and information asymmetry. An example is DES, a drug that increases the risk of cancer in mature daughters of women who used it during pregnancy. Many other drugs require extended periods before all their effects are manifested. Did the seller know, or could be reasonably expected to know, of the effect at the time of sale? If so, then it is an information asymmetry problem.

In terms of our earlier discussion, the consumer of a post-experience good may not have sufficient knowledge of the product to form reasonable estimates of its full price even after repeated consumption. In other words, although experience goods involve risk (known contingencies with known probabilities of occurrence), post-experience goods involve uncertainty (unknown contingencies or unknown probabilities of occurrence). Further, repeated consumption of a post-experience good does not necessarily lead to accurate conversion of uncertainty to risk.

Obviously, the potential for substantial and persistent inefficiency due to information asymmetry is greater for post-experience goods than for experience goods. Generally, *frequent purchases are not effective in eliminating information asymmetry for post-experience goods.* In contrast to the case of experience goods, information asymmetry can persist for an extended period for even homogeneous post-experience goods.

Turning to private responses to the information asymmetry inherent in the primary markets for post-experience goods, we expect secondary markets in general to be less effective than they are for experience goods because of learning problems. Nevertheless, secondary markets can play important roles.

For example, consider the services that are or have been privately offered to provide information about pharmaceuticals. The *Medical Letter on Drugs and Therapeutics*, one of the few medical periodicals that covers its costs through subscriptions rather than advertising, provides independent assessments of all new drugs approved by the Food and Drug Administration. Many hospitals, and some insurers, have formulary committees that review information about drugs for participating physicians. Between 1929 and 1955, the American Medical Association ran a seal of acceptance program that subjected all products advertised in the *Journal of the American Medical Association* to committee review. The *National Formulary* and the *United States Pharmacopeia* were early attempts at standardizing pharmaceutical products. Of course, most of these information sources are directed at physicians and pharmacists, who commonly serve as agents for consumers of therapeutic drugs. Although their learning from direct observation may not be effective, they can often determine whether advertising claims for products are based on reasonable scientific evidence.

Reprise of Information Asymmetry

The potential for inefficiency due to information asymmetry between buyers and sellers can be found in many markets. There is some evidence that Web-based, or e-commerce, markets are particularly prone to information asymmetry on some dimensions; for example, consumers cannot touch or feel goods and are, therefore, less likely to be able to assess directly their quality.[44] The potential is rarely great for search goods, often great for experience goods, and usually great for post-experience goods. The extent to which the potential is actually realized, however, depends largely on whether public goods problems hinder the operation of secondary market mechanisms that provide corrective information. Thus, market failure is most likely in situations where information asymmetry in primary markets occurs in combination with public goods problems in secondary markets.

Conclusion

Traditional market failures involve circumstances in which the "invisible hand" fails to produce Pareto efficiency. They thus indicate possibilities for improving the efficiency of market outcomes through collective action. Along with the pursuit of distributional values, which we consider in Chapter 7, they constitute the most commonly advanced rationales for public policy. However, other, less easily accommodated, discrepancies between the competitive framework and the real world also suggest opportunities for advancing efficiency through collective action. These discrepancies, the subject of the next chapter, provide additional rationales for public policy.

For Discussion

1. Suppose that you and two friends are going to share a house. You have to decide whether each of you will purchase your own groceries and cook your own meals, or whether you will purchase groceries and cook meals as a group. What factors might make you more or less likely to choose the group option? Would you be more or less likely to choose it if there were five in the group rather than only three?
2. Consider a developing country with a single railroad between the primary port and the most populous city. Under what conditions might the railroad have natural monopoly characteristics?
3. Under what conditions is the market for vaccination against a communicable disease likely to be inefficient?
4. Show the social surplus loss in a situation in which a consumer's uninformed demand schedule lies below her informed demand schedule.

Notes

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3 See Patti Waldmeir, "The Next Frontier: The Courtroom," *Scientific American* 293(1) 2005, 17.

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6 Marc Rysman, "The Economics of Two-Sided Markets," *Journal of Economic Perspectives* 23(3) 2009, 125–43.

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32 See Harvey J. Leibenstein, *Beyond Economic Man* (Cambridge, MA: Harvard University Press, 1976); and Roger S. Frantz, *X-Efficiency: Theory, Evidence and Applications* (Boston: Kluwer, 1988).

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36 See Aidan R. Vining and David L. Weimer, "Information Asymmetry Favoring Sellers: A Policy Framework," *Policy Sciences* 21(4) 1988, 281–303.

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38 Mikhail I. Melnik and James Alm, "Does a Seller's Ecommerce Reputation Matter? Evidence from eBay Auctions," *Journal of Industrial Economics* 50(3) 2002, 337–49.

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6
Rationales for Public Policy
Other Limitations of the Competitive Framework

Although the four traditional market failures (public goods, externalities, natural monopolies, and information asymmetries) represent violations of the ideal competitive model, we nevertheless are able to illustrate their efficiency consequences with the basic concepts of producer and consumer surplus. The consequences of relaxing other assumptions of the competitive model, however, cannot be so easily analyzed with the standard tools of microeconomics. This in no way reduces their importance. They often serve as rationales, albeit implicitly, for public policies.

We begin by considering two of the most fundamental assumptions of the competitive model: first, participants in markets behave competitively, assuming their decisions don't affect prices; and second, individual preferences can be taken as fixed, exogenous, and fully rational. We next look at the assumptions that must be made to extend the basic competitive model to an uncertain and multi-period world. We then consider the assumption that the economy moves from one to another equilibrium without substantial transaction costs as circumstances change. Finally, we address briefly the role of macroeconomic policy in managing aggregate economic activity.

Thin Markets: Few Sellers or Few Buyers

Participants in markets behave competitively when they act as if their individual decisions do not affect prices, thus taking prices as given. This is a reasonable assumption when no seller accounts for a noticeable fraction of supply and no buyer accounts for a noticeable fraction of demand, which is usually the case when there are many buyers and many sellers. In *thin markets*, that is markets with few sellers or few buyers, imperfect competition can lead to prices that differ from the competitive equilibrium and hence result in Pareto-inefficient allocations of inputs and goods.

We have already considered one important example of noncompetitive behavior: *natural monopoly*. In the case of natural monopoly, declining average cost over the relevant range of demand makes supply by more than one firm inefficient. Yet the single firm, if allowed to maximize its profits, will produce too little output from the perspective of economic efficiency. Whether or not a firm is a natural monopoly, it can at least temporarily command monopoly rents when it is the only firm supplying a market. The monopoly rents, however, will attract other firms to the market.[1] Unless technological, cost, or other barriers block the entry of new firms, the monopolist will eventually lose its power to set price and command rents.

Noncompetitive behavior can also occur on the demand side of the market. In the case of pure *monopsony*, a single buyer faces competitive suppliers and, therefore, can influence price by choosing purchase levels. When supply is not competitive, the monopsonist may be able to extract some of the rents that would otherwise go to suppliers. The resulting allocation of rents, however, will not necessarily be more efficient than that without the exercise of monopsony power.

Intermediate between cases of perfect competition and monopoly is *oligopoly*. In an oligopolistic industry, two or more firms account for a significant fraction of output. Both theory and evidence suggest that welfare losses primarily depend on the number of firms in a market.[2] Welfare losses appear large in cases of three or fewer firms but are likely to be very small for larger numbers. The belief that oligopoly often leads to monopoly pricing, either through collusive agreements to limit output (*cartelization*) or through cutthroat competition to drive firms from the industry to obtain an actual monopoly, has motivated much of antitrust law. While defining the market for assessing monopoly power is relatively straightforward for goods with low transportation costs, it becomes more complicated for locally provided services such as hospital care.[3] A full evaluation of antitrust policies requires attention to such measurement issues, as well as a critical review of the various models of oligopoly, tasks beyond the scope of this book.[4] It is sufficient to note here that imperfect competition often serves as a rationale for public intervention.

The Source and Acceptability of Preferences

The basic competitive model assumes that each person has a fixed utility function that maps his or her consumption possibilities into an index of satisfaction.[5] The model makes no assumption about how the particular utility functions arise, just that they depend only on the final bundles of consumed goods. However, the preferences underlying the utility functions must come from somewhere. Either people have fully developed preferences at birth or preferences are formed through participation in society.[6] If the latter, then they either arise in a sphere of activity totally independent of all economic exchange, or, in violation of the assumption of fixed preferences, depend on what goes on in the economic world.

Endogenous Preferences

Historically, the perception that preferences can be changed has been the basis for many public policies. The specific rationales usually involve arguments that, without instruction or correction, certain persons would engage in behavior that inflicts costs on others. In other words, the choices based on certain preferences had negative external effects. Such policies as universal education, which supplements the formation of values in the home and community, and criminal rehabilitation, which alters the willingness of some to harm others, were responses to the external effects of preferences. Although we should be skeptical that particular education and rehabilitative programs will actually impart values in the intended ways, we may nevertheless view these public efforts as conceptually valid.

The question of preference stability arises in debates over the social implications of advertising. Does advertising actually change preferences, or does it simply provide information that helps people better match available goods to the preferences they already hold? If advertising actually does change preferences, then we can imagine situations in which private advertising leads to economic inefficiency. For example, if advertising convinced me that, holding other consumption and my income constant, I now additionally need virtual reality goggles to maintain my level of happiness, I have been made worse off by the advertising. Further, the gains to the makers of these devices cannot be large enough to make the advertising even potentially Pareto improving.

What does the evidence say about whether advertising can change preferences? At one time, economists generally viewed the empirical evidence as suggesting that advertising primarily affects market shares for different brands, rather than the total quantity of the good demanded.[7] However, empirical evidence suggesting that advertising has quantity effects has been accumulating for goods that, like tobacco and alcohol, have particular relevance to public policy.[8] More generally, in cases in which advertising does appear to increase market demand, however, we often have no empirical basis for clearly distinguishing between decisions that result from better information about available choices from those that result from changes in preferences. Only the latter potentially involves economic inefficiency.

Preferences may also change as a result of the consumption of addictive goods−−utility at any time depends on both past and current consumption.[9] Repeated use of cocaine and tobacco, for example, produce physical dependencies that increase the relative importance of these goods in people's utility functions. People who do not fully foresee the consequences of the dependency may overconsume to the detriment of their future happiness. Although information asymmetry provides a plausible explanation for the initiation of addictive consumption of recreational substances by adolescents, or addictive consumption of prescription drugs by adults, it does not explain why persons wishing to stop harmful addictive consumption have such great difficulty doing so. Although individuals vary greatly in terms of the likelihood that their current consumption of harmful addictive substances will lead to addiction as conventionally understood, some people experience significant neurological change that makes self-control in the face of temptation to consume immediately very costly.[10] Policies that reduce the costs of self-control can potentially enable some people to reduce addictive consumption that improves their well-being from their own perspective.[11] Thus, policies that address addiction may enhance efficiency.

Utility Interdependence: Other-Regarding Preferences

Going beyond the question of how preferences arise, we next confront the implications of relaxing the assumption that the utility of individuals depends only on the goods that they personally consume. Such preferences are *self-regarding*. Most of us are also *other-regarding* in the sense that we do care about the consumption of at least some others: we give gifts to those we know and donations to charities that help those we do not know. This sort of altruistic behavior is most obvious in the context of the family.

Parents typically care deeply about the goods their children consume. We can accommodate this interdependence in the competitive model by taking the household, rather than the individual, as the consuming unit. As long as we are willing to think of the household as having a utility function that appropriately reflects the preferences of household members, the competitive equilibrium will be economically efficient. Unfortunately, parents are not always adequately paternal. In cases of abuse, for instance, most would agree that public intervention is justified to ensure that the utilities of the children are being adequately taken into account in the distribution of goods (including especially safety).

Interdependence causes much greater conceptual difficulty when utilities depend not only on the absolute quantities of goods people consume but also on the relative amounts.[12] In the basic competitive model, increasing the goods available to one person without decreasing the goods available to anyone else is always a Pareto-efficient redistribution. The redistribution may not be Pareto improving, however, if some people care about their relative consumption positions. For instance, if your happiness depends to some extent on your income being higher than that of your brother, then, when your brother receives a salary increase that puts his income above yours, you will feel worse off even though your consumption possibilities have not changed. Although many people undoubtedly do care to some extent about their relative standings among colleagues, relatives, or some other reference group, the implications of such interdependence for the economic efficiency of the competitive economy is unclear. If nothing else, preferences based on relative consumption limit the straightforward, intuitive interpretation of the Pareto principle.

Taste for Fairness: Process-Regarding Preferences

In addition to the possibility that people are other-regarding in important ways, they may be *process-regarding* as well.[13] People often care not just about the distribution of goods but also about how the distributions were produced. Both individuals waiting for transplant organs and the general public, for example, almost certainly have a preference for allocation by medical criteria rather than by celebrity status.

Laboratory experiments involving *ultimatum games* often suggest the presence of process-regarding preferences.[14] In ultimatum games, one player proposes a split of a fixed amount of money and the other player either accepts the split or rejects it. As rejecting the split results in a zero payment for the proposer and the rejecter, a prediction of the equilibrium when both players are only self-regarding and do not expect to interact again in the future is that the proposer offers the smallest possible positive amount and the other player accepts the proposal. Often, however, the proposer offers substantially more than the minimum amount, and, if offered a very small share, the other player rejects it. Unfortunately, the stakes in these experiments, typically involving college students, tend to be low, leaving open the question if process-regarding preferences would dominate self-regarding preferences in more realistic situations. For example, you might prefer to reject a split that gives you only $1 out of a possible $10 but not to reject a split that gives you only $1,000 out of a possible $10,000.

Some evidence of the existence of relevant process-regarding has been found in public policy applications. For example, a study of attitudes toward the choice of sites for nuclear waste repositories in Switzerland found that, in addition to expected economic impacts and risks, respondents' perceptions of the acceptability of the rules for choosing sites affected their willingness to accept facilities near their hometowns.[15] Perceived fairness was the major determinant of acceptability.

Legitimacy of Preferences

The basic competitive model treats all preferences as legitimate, all consumers as sovereign. Almost everyone would agree, however, that certain preferences leading to behaviors that directly harm others should not be considered legitimate from the social perspective. For example, our criminal laws prohibit us from taking out our anger on others through assault on their persons. Of course, we can think of this as simply prohibiting one from inflicting involuntary transactions, which directly violate the Pareto principle, on others.

Our laws and customs, however, also prohibit the exercise of other preferences that do not seem to harm others directly. For instance, some people get pleasure from sexual intercourse with animals. This activity, if carried out in private with the individual's own animal, would seem not to influence directly the utility of others. Why is it not, therefore, Pareto inefficient to prohibit bestiality? Although the reason may not be obvious from the efficiency perspective, in most societies there is a consensus favoring prohibition. Sometimes we can identify external effects that seem to justify such prohibitions. For example, rationales presented for restrictions on prostitution include preventing such external effects as the undermining of family stability, the exploitation of prostitutes by pimps, and the spread of venereal diseases. Perhaps disease and family stability are also relevant to the prohibition of bestiality. It may also be, however, that the prohibition rests as much on some widely shared conception of what behaviors violate human dignity. Of course, even accepting the undesirability of the behavior, the question still remains as to whether public restriction or private suasion is more appropriate.

Reprise of Preference Problems

Preferences in the real world are neither as stable nor as simple as assumed in the competitive model. The extent to which this divergence keeps the economy from achieving Pareto efficiency remains unclear, however. Yet, while we recognize the limitations of the standard assumptions, our relatively poor understanding of preferences should lead us to tread carefully in using perceived problems with them to justify public policies.

The Problem of Uncertainty

Without great conceptual difficulty, the basic competitive model can be extended to incorporate multiple periods and uncertainty. Instead of assuming that people have utility functions defined over the goods consumed in a single period with complete certainty, we assume that they have utility functions over all goods in all periods and under all possible contingencies. In effect, we simply distinguish goods by time of consumption and "state of nature" (or *contingency*) as well as by physical characteristics. In a two-period world (this year versus next) with two possible states of nature (heavy versus light snow), for example, a physical commodity such as rented cross-country skis enters the utility functions of consumers as four different goods: skis this year if snow is light, this year if snow is heavy, next year if snow is light, and next year if snow is heavy. With the crucial assumption that efficient markets exist for all these goods at the beginning of the first time period, the resulting equilibrium allocation of goods will be Pareto efficient.[16]

Availability of Insurance

The assumption that efficient markets exist for all goods under all contingencies implies that it is possible to buy actuarially fair insurance so that each person's utility will remain constant, no matter what state of nature actually occurs. Insurance is actuarially fair when the premium exactly equals the expected payout. Of course, to know the expected payout, one must know the probabilities of each of the possible contingencies, as well as their associated payouts. In standard terminology, *risk* involves contingencies with known probabilities, and *uncertainty* involves contingencies with unknown probabilities or unknown contingencies.

Efficient and complete insurance markets do not exist in the real world. For a start, insurers must routinely charge something more than the actuarially fair price to cover their administrative costs (the transaction costs of insurance). More importantly, however, two sets of factors limit the range of contingencies that can be insured at even approximately actuarially fair prices: characteristics of the contingencies themselves and behavioral responses to the available insurance. In order to set actuarially fair prices, insurers must know the probabilities of the covered contingencies. The most common types of casualty insurance protect against contingencies that occur frequently enough to allow reliable estimation of their probabilities from experience. For example, by observing the accident records of a large number of drivers, insurers can make fairly accurate predictions of the probability that anyone with specific characteristics (such as gender, age, and driving environment) will have an accident over the next year. *Experience rating* of this sort, however, is not possible, or at least not accurate, for very rare events such as major earthquakes. Consequently, insurers are unable to offer actuarially fair insurance against many rare events. Further, the price of the insurance they do offer will likely include a *risk premium*, which is an addition to the estimated actuarially fair price to reflect their lack of confidence in the estimates of probabilities.

An additional premium above the actuarially fair price may also be demanded in situations in which individual risks are not independent because, after pooling, some collective, or social, risk remains.[17] If the probability that any one policyholder will have an accident is independent of the probabilities that other policyholders will also have accidents—a reasonable assumption with respect to automobile accidents, for instance—then as the number of policyholders gets larger, even though the variance in total loss increases, the variance in the average loss per policy falls so that smaller risk premiums per policy are required.[18]

The convergence between observed frequencies and underlying probabilities, which is known as the *law of large numbers*, breaks down when individual probabilities are not independent. In the absence of independence, insurers should demand larger risk premiums to guard against the probability that losses in any period will greatly exceed expected losses. For example, the probabilities of individual losses from flooding are clearly not independent if all policyholders live in the same river valley—an insurer would have to hold very large reserves to avoid bankruptcy in years with floods. In order to build these reserves, however, the insurers would have to price their insurance well above actuarially fair levels.

In all but extreme situations, insurers generally are able to mitigate social risk through some sort of diversification. For example, insurers may be able to spread coverage over several different floodplains. When geographic diversification is not possible, insurers may be able to shift some of the risk to reinsurers, who can balance it against other types of risk. If the probability of loss correlates negatively with other risky holdings, then the total risk held by the reinsurer will fall. Some social risks, however, such as nuclear war or dramatic global climate change, would involve sufficiently large and widespread negative impacts so that diversification would not be possible.

Incomplete Insurance Markets: Adverse Selection, Moral Hazard, and Underinsuring

Turning to the behavioral responses to available insurance, we consider adverse selection, moral hazard, and underinsuring of irreplaceable commodities.

We have already encountered the concept of *adverse selection* (of goods) in our discussion of information asymmetry in goods markets. In the insurance context, it pertains to insured individuals and arises when insurers cannot costlessly classify potential policy-holders according to risk categories. It therefore involves what is sometimes referred to as *hidden information*. Within each category, some individuals will have either higher-than-average or lower-than-average probabilities of loss. Those with higher-than-average probabilities will tend to find the insurance attractive; those with lower-than-average probabilities will not. As more of the former buy policies and more of the latter decline to buy, the average for the group must rise. This in turn drives the insurance price higher above the actuarially fair level for the group as a whole. Eventually, only those with the highest probabilities choose to remain covered.

Insurers can sometimes limit adverse selection by offering inclusive policies to groups such as company employees, for which insurance is only one attraction of membership. In general, however, the presence of information asymmetries tends to keep the prices of many types of insurance well above actuarially fair levels for many potential buyers. One rationale for public insurance programs is that they can be made mandatory so that adverse selection can be avoided.

Moral hazard refers to the reduced incentive that insurees have to prevent compensable losses.[19] If fully insured, then they can make themselves better off, and perhaps society in the aggregate worse off, by spending less of their own resources on loss prevention than they would in the absence of insurance.[20] Actual reductions in loss protection are more likely the more costly it is for insurers to monitor behavior. It is thus sometimes referred to as the problem of *hidden action*. Insurers often try to limit moral hazard by requiring insurees to pay a fraction of the suffered loss through copayments.

Note that moral hazard can also be useful in understanding increases in risk in many policy-relevant arenas outside of conventional insurance markets. For example, consider the implications of changes in the origination and holding of mortgages during the housing boom that ended in 2007. Historically, banks originated most mortgages and kept them as assets for an extended period. Consequently, they had a strong incentive to avoid mortgages with high risks of default because they bore the entire risk. However, the more recent introduction of pools of mortgages as financial instruments created a moral hazard that encouraged both banks and mortgage brokers to originate as many mortgages, including so-called subprime mortgages, as possible, because they could earn the origination fees and immediately pass along all their risk to the pool. As ownership of the pools was dispersed widely, and the risk of default was low as long as housing prices were rising, monitoring the quality of mortgages in the pool was a public good problem and undersupplied. This combination of moral hazard in origination and the public-good nature of monitoring set the stage for a severe financial crisis triggered by declining housing prices.

People may underinsure for contingencies involving the loss of irreplaceable goods that cannot be exactly replicated. For many plausible utility functions, economically rational people will only wish to purchase actuarially fair insurance to cover the monetary part of their losses.[21] Thus, in a world consisting of irreplaceable goods such as good health, beloved spouses, and unique works of art, people would not necessarily purchase enough insurance to give them constant utility under all contingencies. Even if they wished to, given experience rating and moral hazard problems, insurers would not be willing or able to offer actuarially fair insurance for all irreplaceable goods anyway. For example, insurers would consider the dangers of moral hazard if you claimed that you needed $1 million to compensate yourself fully for the supposed accidental death of your dog.

Subjective Perception of Risk

So far we have assumed that people effectively evaluate and use information to arrive at rational decisions in situations involving risk. However, a large body of experimental research in cognitive psychology and economics suggests that people tend to make several systematic errors in assessing probabilities.[22] In order to deal economically with a great variety of information in a complex world, people tend to employ *heuristics* (rules of thumb) that sometimes lead to correct decisions, but nevertheless involve predictable biases. For instance, it appears that people often estimate the probabilities of events by the ease with which instances of occurrence can be brought to mind.[23] Recall, however, depends on a number of factors, such as the personal salience of events, that tend to bias estimates of relative probabilities systematically. For example, in a study of flood and earthquake insurance coverage, Howard Kunreuther and colleagues found that knowing someone who suffered flood or earthquake loss was the single most important factor for distinguishing between purchasers and nonpurchasers.[24]

The growing dissatisfaction among economists with use of the expected utility hypothesis for predicting individual behavior also raises questions about our understanding of the way people make risky choices, and hence shakes our confidence somewhat in the rationality of private behavior involving risk.[25] The *expected utility hypothesis* states that people choose among alternative actions to maximize the probability-weighted sum of utilities under each of the possible contingencies. For example, if a person's utility, $U(w)$, depends only on his wealth (w), and the probability that his wealth will be w_A is p and the probability that his wealth will be w_B is $(1 - p)$, then his expected utility is $pU(w_A) + (1 - p)\, U(w_B)$. The expected utility hypothesis is consistent with the competitive model and lies at the heart of most economic analysis of individual responses to risk.

Laboratory experiments, however, have identified quite a few situations in which the expected utility hypothesis seems to be systematically violated. Although individuals tend to underestimate the probabilities of very infrequent events, for instance, they tend to be overly sensitive to small changes in the probabilities of rare events relative to the predictions of the expected utility hypothesis. These findings have led cognitive psychologists and economists to begin to explore other formulations of behavior under uncertainty. As yet, the full implications of the alternative formulations for our evaluation of economic efficiency remain unclear.

More General Issues Related to the Rationality of Decision Making

In discussing uncertainty, we briefly touched on the issue of decision making and judgment biases that arise in the way people perceive risks. But risk perception is not the only context in which individuals behave in ways that appear to be inconsistent with the tenets of classical utility theory. Considerable experimental evidence suggests that people make very different valuations of gains and losses.

These differences underlie *prospect theory*. Daniel Kahneman and Amos Tversky developed prospect theory, which suggests that individuals deviate from expected utility maximization in a number of systematic ways: they value gains and losses from a reference point, which is, in turn, based on some notion of the status quo (hence this is sometimes called the *status quo effect* or, more commonly, the *endowment effect*). Also, they place more weight on losses than on gains—a simultaneous loss and a gain of the same size would leave an individual worse off. In a context of risky choices, they are risk averse toward gains and risk seeking toward losses. They prefer a smaller certain gain to a larger probable gain when the expected values of the two alternatives are the same, but they exhibit *loss aversion* in that they prefer a larger probable loss to a smaller certain loss when the expected value of the two alternatives is the same. Figure 6.1 illustrates the shape of the utility function assumed by prospect theory.

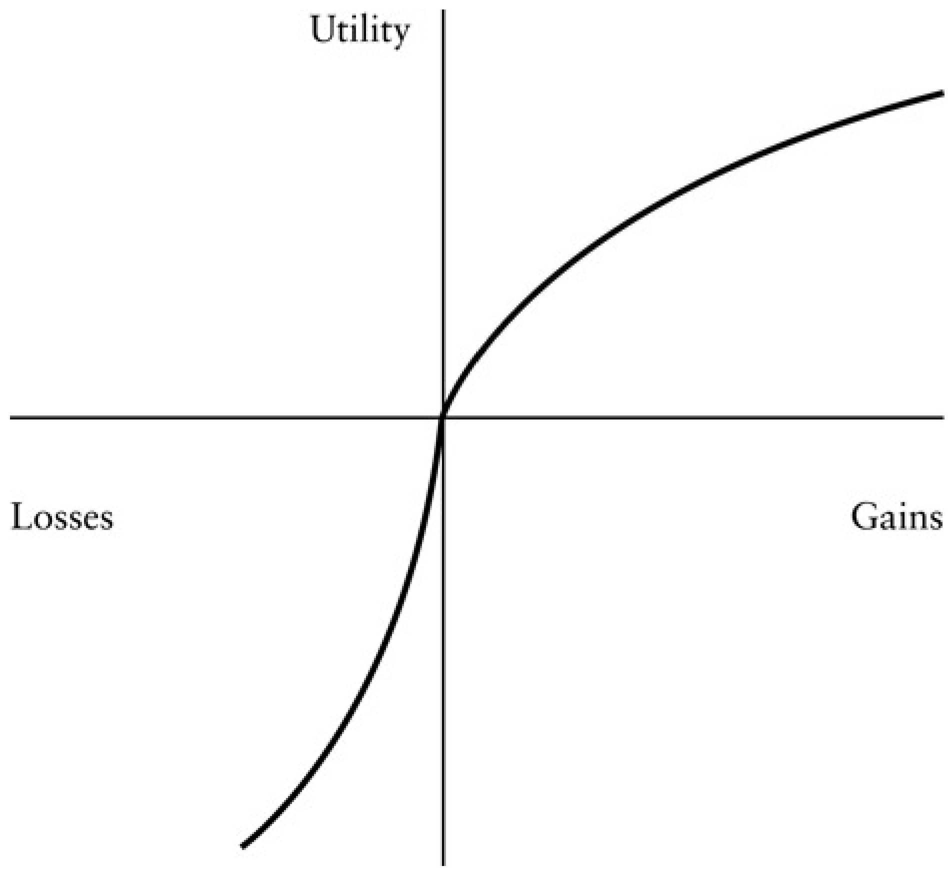

Figure 6.1 Utility Function Showing Loss Aversion

Loss aversion is inconsistent with the standard competitive framework. To what extent does it pose a problem for public policy? One response is that it does not (at least, not much). Individuals simply value losses more highly than gains; even though this is inconsistent with a simple utility maximization model, it is not irrational. Unfortunately, however, many public policy issues can be framed as resulting in *either* a loss *or* a gain to a given individual (such as a survey respondent), especially in the context of political rhetoric, which we discuss in Chapter 8. Most relevant to our discussion of market failures, however, the asymmetry of gains and losses undermines the usefulness of the Coase theorem (Chapter 5), even in the context of small numbers.

More generally, *behavioral economics* seeks to incorporate the findings of cognitive psychology, neuroscience, evolutionary biology, and other related disciplines into more realistic models of economic behavior.[26] For example, the standard economic approach would predict little difference in participation in company savings plans between those that require employees to *opt in* (affirmatively make a decision to participate) and those that require employees to *opt out* (affirmatively make a decision not to participate). Empirical evidence strongly suggests, however, that participation is much higher in the opt-out plans than in the opt-in plans.[27]

Reprise of Uncertainty Problems

The foregoing observations about insurance markets and individual responses to risk suggest that public policies may have potential for increasing economic efficiency in circumstances involving uncertainty. Public assessments of risks, for example, may be an appropriate response when people make important systematic errors in their private assessments, and public insurance may be justified when private coverage is significantly incomplete. Until we have a better understanding of how individuals actually deal with uncertainty, however, our evaluation of economic efficiency in circumstances involving uncertainty must itself remain uncertain.

Intertemporal Allocation: Are Markets Myopic?

As we have already indicated, the competitive model can be extended to the intertemporal allocation of goods. If we assume that it is possible to make contracts in the current period for the production and delivery of goods in all future periods (in other words, forward markets exist for all goods), then the competitive equilibrium will be Pareto efficient.[28] However, such an assumption is clearly a significant oversimplification.

Capital Markets

In equilibrium, the demand for borrowing must just equal the supply of loans. The resulting rate of interest is the price that clears this *capital market*. Of course, in the real world the financial institutions that make up the capital market cover their administrative expenses by lending at higher rates than at which they borrow. They may also demand premiums for riskier loans.

Just as the theoretical efficiency of the competitive equilibrium depends on the existence of complete insurance markets to accommodate uncertainty, it depends on complete, or perfect, capital markets to accommodate intertemporal allocation. Perfect capital markets allow anyone to convert future earnings to current consumption through borrowing. But in reality, the anticipation by lenders of moral hazard greatly reduces the amount people can actually borrow against future earnings. Borrowers who owe large percentages of their incomes may be tempted to reduce their work effort because they realize smaller net returns to their labor than if they were debt free. With the now almost universal prohibition against slavery and the existence in most countries of bankruptcy laws, borrowers have no means of convincing lenders that they will maintain full work effort. Therefore, at least with respect to labor income, capital markets appear to be imperfect. It is at least conceivable that some public policies, such as guaranteed loans for education, might be justified on the grounds of imperfect capital markets.

At a more fundamental level, scholars have long debated whether capital markets lead to appropriate levels of saving and investment for future generations. No one lives forever. How, then, can we expect the decisions of current consumers to take adequate account of the unarticulated preferences of those yet to be born? One response recognizes that generations overlap. Most of us consider the welfare of our children and our grandchildren, who we hope will outlive us, in our economic decisions. Rather than attempting to consume all of our resources before the instant of death, most of us plan on bequeathing at least some of our savings to family or charity. Even when we do not intentionally plan to leave unconsumed wealth behind, we may do so unintentionally through untimely death.

Despite intentional and unintentional bequests, positive market interest rates prevail in most societies. Some philosophers and economists object on normative grounds to interpreting positive interest rates as appropriate prices for trading consumption across time periods.[29] They argue that consumption by someone in the future should be given the same weight from the social perspective as consumption by someone today. Their argument implies that any investment yielding a positive rate of return, no matter how small, should be undertaken. Because investment in the competitive economy demands rates of return at least as high as the rate of interest, they generally advocate public policies to increase investment.

The underinvestment argument seems to ignore the legacy of capital from one generation to the next. Net investments increase the available capital stock. Most forms of capital do, of course, depreciate over time. A bridge, for instance, needs maintenance to keep it safe, and eventually it may have to be abandoned or rebuilt. But perhaps the most important element of our capital stock is knowledge, which does not depreciate. Improvements in technology not only increase current productive options but those in the future as well. This legacy helps explain the growing wealth of successive generations in countries, like the United States, that rely primarily on private investment. The underinvestment argument loses force, therefore, if the total capital stock, including intangible capital such as knowledge, can be reasonably expected to continue to grow in the future.

One variant of the underinvestment (or overconsumption of current wealth) argument deserves particular attention. It concerns the irreversible consumption of natural resources such as animal species, wilderness, and fossil fuels. When an animal species becomes extinct, it is lost to future generations. Likewise, fossil fuels consumed today cannot be consumed tomorrow. Although it may be technically possible to restore wilderness areas that have been developed, the cost of doing so may make it practically unfeasible. Will markets save adequate shares of these resources for future generations?

The answer is almost certainly "yes" for resources, such as fossil fuels, that have value as factor inputs rather than final goods. As long as property rights are secure, owners of resources will hold back some of the resources in anticipation of future scarcity and the higher prices the scarcity will command.[30] As higher prices are actually realized, substitutes will become economically feasible. Although generations in the distant future will be consuming less fossil fuel, they undoubtedly will produce energy from alternative sources and use energy more efficiently to produce final goods. There is no particular reason to believe, therefore, that any collective decision about how much to save for future generations will be any better than allocations made by private markets.

Resources that are themselves unique goods, such as wilderness areas and animal species, raise more serious doubt about the adequacy of market forces to ensure preservation. When owners have secure property rights and can anticipate selling their resource as a private good in the future, they will base their decisions about the extent of development on their estimates of future demand. But this can happen only if a market mechanism exists for articulating the demand. This is often not the case for natural areas, even when exclusion is possible, because there is usually no way to secure contributions from people who would be willing to pay something to keep the resource in place so as to retain the option of visiting it sometime in the future. This willingness of nonusers to contribute to maintenance of the resource for their own possible use in the future is called *option demand*; their willingness to pay something for the preservation of the resource for use by future generations or for its own intrinsic worth is called *existence value*.[31]

A final point, which we address more fully in Chapter 17 in the context of time discounting in cost–benefit analysis, concerns the validity of interpreting the market interest rate as the social marginal rate of time preference, even if we accept use of the latter as appropriate when comparing current and future consumption. Lenders generally charge higher rates of interest for riskier investments. But when we look across the full set of investment projects, the aggregate risk will be much less than the risk associated with any individual project because the circumstances that tend to make some projects fail tend to make others succeed. For example, the low energy prices that, ex post, make synthetic fuel plants appear to be bad investments may make recreational facilities requiring long road trips appear to be good investments. The market interest rate, therefore, will tend to exceed the social rate of time preference, which should only reflect the social risk and not the risk to individual lenders. Although a problem caused by public policy rather than inherent in markets, a similar wedge is introduced by taxes on profits. If, for whatever reason, the market interest rate exceeds the social rate of time preference, then private investment will fall short of the economically efficient level.

Reprise of Intertemporal Problems

In summary, specific reservations about the efficiency of capital markets, and general concerns about the adequacy of the weight given to the preferences of future generations, may serve as plausible rationales for public policies intended to improve the intertemporal allocation of resources and goods.

Adjustment Costs

The standard competitive model is static. For any given set of assumptions about utilities, production functions, and factor endowments, a unique allocation of factor inputs and goods constitutes an equilibrium in the sense that, at the prevailing prices, no consumers can make themselves better off by changing the quantities of goods they purchase and no firms can increase their profits by changing input mixes or output levels. If any of the assumptions change, then a new equilibrium distribution will arise. We can assess welfare changes by comparing allocations in the two equilibria (economists call this *comparative statics*). In doing so, we implicitly assume, however, that the economy can move costlessly from one equilibrium to another.

In reality, the economy is never static. Changing incomes, the introduction of new products, the growing workforce and capital stock, good and bad harvests, and numerous other factors necessitate continual adjustments toward efficient allocation. In most circumstances, as long as prices are free to move in response to changes in supply and demand, the adjustment process itself occurs costlessly: persons change their consumption to maximize utility and firms change their factor inputs to maximize profits in response to the new prices. Some persons may be worse off under the new allocation, but this will be fully reflected in the comparison of social surpluses before and after the reallocation. For example, an epidemic that decimates the population of chickens and drives up the price of eggs will, other things equal, make egg consumers worse off. The consumer surplus loss measured in the egg market serves as a good approximation for the reduction in welfare they suffered.

The picture changes if prices are *sticky* because institutional or psychological factors constrain their free movement. Constraints on price movements keep the economy from immediately reaching the new Pareto-efficient equilibrium. As a result, comparing the old and new equilibria may overstate social surplus gains and understate social surplus losses. The larger the needed adjustment and the more rigid are prices, the greater are the adjustment costs that do not show up in the comparative statics.

Consider, for example, the effects of an oil price shock of the sort experienced during the Arab oil embargo (1973–1974) and the Iranian revolution (1979–1980).[32] A dramatic rise in oil prices makes petroleum a much more expensive factor input. The supply schedules for goods that require petroleum as a factor input will shift up so that marginal costs of production rise. The market clearing product price will be higher and the market clearing quantity will be lower than before the price shock. Therefore, at any wage rate, firms will demand less of all factor inputs, including labor, than before the shock. If all factor prices were flexible, then we would expect this shift to result in reductions in both the quantity of labor demanded and the wage rate. But contracts and custom often keep firms from immediately lowering wage rates. Firms, therefore, will reduce the quantity of labor they use more than they would if wages were free to fall. Either underemployment (employees cannot work as many hours as they want at the prevailing wage rate) or, more likely, again for reasons of contract and custom, involuntary unemployment will result.

Wage rigidities due to either implicit or explicit contracts should not necessarily be interpreted as market failures.[33] They may be the result of attempts by firms and workers to share the risks associated with fluctuations in the demands for products; workers may accept lower-than-market-clearing wages during good times to secure higher-than-market-clearing wages during bad times. The picture is complicated, however, not only by all the limitations to complete insurance markets that we have already discussed, but also by the prevalence of collective bargaining that tends to give greater protection to workers with longer tenure. Incomplete markets for risk spreading make it at least possible that public programs, such as unemployment insurance, may contribute to greater economic efficiency.

Macroeconomic Dynamics

The business cycle poses a sort of social risk that may justify public stabilization policies.[34] The economy is dynamic and tends to go through cycles of expansion and recession. During recessions, unemployed labor and underutilized capital indicate inefficiency in the economy-wide use of factor inputs. The government may be able to dampen recessions through its fiscal and monetary policies. *Fiscal policies* involve taxation and expenditures.

During recessions, for instance, the government may be able to stimulate demand by spending in excess of revenues. *Monetary policy* refers to manipulation of the money supply. In general, the money supply must continually increase to accommodate the growing economy. During recessions, such as that in 2008–2009, the government may increase the money supply at a faster rate to lower interest rates, at least temporarily, thereby stimulating investment and current consumption.

Unfortunately, there is no consensus among economists about either the causes of business cycles or the most appropriate policies for reducing their adverse effects. Although fiscal and monetary policies undoubtedly have potential for improving the dynamic efficiency of the economy, we must await advances in the field of macroeconomics before we can have confidence in anyone's ability to use them for anything approaching fine tuning. Fortunately, in almost all situations, policy analysts can take monetary and fiscal policies as given and still provide good advice about economic efficiency.

Conclusion

Table 6.1 summarizes the traditional market failures presented in Chapter 5 and the other limitations of the competitive framework discussed in this chapter. The table also summarizes circumstances in which public policy interventions can potentially increase efficiency. Our understanding of their implications varies greatly, however. At one extreme, economics provides well-developed theory and empirical evidence to help us understand and apply the traditional market failures. At the other extreme, economics hardly deals at all with the questions of the origins of preferences and their stability over time. Other disciplines provide many insights relevant to preferences, but do not provide a comprehensive and normatively grounded theory to guide public policy intervention.

*Table 6.1*A Summary of Market Failures and Their Implications for Efficiency

Traditional Market Failures (Chapter 5)	
Public Goods	Pure public goods (undersupply) Open access/common property (overconsumption, underinvestment) Toll goods (undersupply)
Externalities (Missing Markets)	Positive externalities (undersupply) Negative externalities (oversupply)
Natural Monopoly	Declining average cost (undersupply) With costly monitoring (undersupply, X-inefficiency)
Information Asymmetry	Quality overestimation of experience, post-experience goods (overconsumption) Quality underestimation of experience, post-experience goods (underconsumption)
Other Limitations of the Competitive Framework (Chapter 6)	
Thin Markets	Cartelization (undersupply)
Preference Problems	Endogenous preferences (typically overconsumption) Utility interdependence (distributional inefficiency) Unacceptable preferences (overconsumption)
Uncertainty Problems	Moral hazard, adverse selection, unique assets (incomplete insurance) Misperception of risk (violation of expected utility hypothesis)
Intertemporal Problems	Nontraded assets, bankruptcy (incomplete capital markets)
Adjustment Costs	Sticky prices (underemployed resources)

Traditional Market Failures (Chapter 5)	
Macroeconomic Dynamics	Business cycles (underemployed resources)

Despite these disparities of development, the list in Table 6.1 provides an important stock of ideas for policy analysts. Policies that do not find a basis in one of these market failures or related problems must be justified by values other than efficiency. It is to these other values that we next turn.

For Discussion

1. Why is the premium for health insurance provided by an employer typically much lower than the premium for the same coverage purchased by an individual?
2. Why is it that public subsidy programs often result in much higher levels of expenditures than initially estimated?

Notes

1 Of course, firms are always seeking opportunities to price above competitive levels. See, for example, David A. Besanko, David Dranove, Mark Shanley, and Scott Schaefer, *Economics of Strategy,* 4th ed. (New York: John Wiley & Sons, 2007).

2 Robert A. Ritz, "On Welfare Losses Due to Imperfect Competition," *Journal of Industrial Economics* 62(1) 2014, 167–90.

3 Martin S. Gaynor, Samuel A. Kleiner, and William B. Vogt, "A Structural Approach to Market Definition with an Application to the Hospital Industry," *Journal of Industrial Economics* 61(2) 2013, 243–89.

4 For an overview, see W. Kip Viscusi, Joseph E. Harrington, Jr., and John M. Vernon, *Economics of Regulation and Antitrust*, 4th ed. (Cambridge, MA: MIT Press, 2005).

5 Some psychologists argue that, rather than being stable, preferences are constructed within the particular contexts of choice. See Sarah Lichtenstein and Paul Slovic, eds., *The Construction of Preferences* (New York: Cambridge University Press, 2006).

6 Gary Becker proposes that persons do have fairly stable and uniform utility functions over the consumption of basic "household goods," such as health and recreation, that result when market goods and time are combined according to individual production functions; see Gary Becker, *The Economic Approach to Human Behavior* (Chicago: University of Chicago Press, 1976). This formulation preserves the properties of a competitive equilibrium but begs the question of how the different preference functions arise. On the normative implications of Becker's approach, see Alexander Rosenberg, "Prospects for the Elimination of Tastes from Economics and Ethics," *Social Philosophy & Policy* 2(2) 1985, 48–68.

7 William S. Comanor and Thomas A. Wilson, "The Effect of Advertising on Competition: A Survey," *Journal of Economic Literature* 17(2) 1979, 453–76; Mark S. Albion and Paul W. Farris, *The Advertising Controversy: Evidence on the Economic Effects of Advertising* (Boston: Auburn House, 1981).

8 For example, see Henry Saffer and Dhaval Dave, "Alcohol Consumption and Alcohol Advertising Bans," *Applied Economics* 34(11) 2002, 1325–34; and Wei Hu, Hai-Yen Sung, and Theodore E. Keeler, "The State Antismoking Campaign and Industry Response: The Effects of Advertising on Cigarette Consumption in California," *American Economic Review* 85(2) 1995, 85–90.

9 Gary S. Becker and Kevin M. Murphy, "A Theory of Rational Addiction," *Journal of Political Economy* 96(4) 1988, 675–700.

10 See Alan I. Leshner, "Addiction Is a Brain Disease, and It Matters," *Science* 278(5335) 1997, 45–47; B. Douglas Bernheim and Antonio Rangel, "Addiction and Cue-Triggered Decision Processes," *American Economic Review* 94(5) 2004, 1558–90; and Faruk Gul and Wolfgang Pesendorfer, "Harmful Addiction," *Review of Economic Studies* 74(1) 2007, 147–72.

11 Assessment of efficiency within the market for potentially addictive substances is complex because some people may consume without becoming addicted. For an application to smoking, see David L. Weimer, Aidan R. Vining, and Randall K. Thomas, "Cost–Benefit Analysis Involving Addictive Goods: Contingent Valuation to Estimate Willingness-to-Pay for Smoking Cessation," *Health Economics* 18(2) 2009, 181–202.

12 On the implications of interdependence, see Robert H. Frank, *Choosing the Right Pond: Human Behavior and the Quest for Status* (New York: Oxford University Press, 1985). For an empirical application, see Fredric Carlsson, O. L. O. F. Johansson-Stenman and Peter Martinsson, "Do You Enjoy Having More than Others? Survey Evidence of Positional Goods," *Economica* 74(296) 2007, 586–98.

13 Avner Ben-Nur and Louis Putterman, "Values and Institutions in Economic Analysis," in Avner Ben-Nur and Louis Putterman, eds., *Economics, Values, and Organization* (New York: Cambridge University Press, 1998), 3–69.

14 See Colin Camerer and Richard H. Thaler, "Ultimatums, Dictators, and Manners," *Journal of Economic Perspectives* 9(2) 1995, 209–19.

15 Bruno S. Frey and Felix Oberholzer-Gee, "Fair Siting Procedures: An Empirical Analysis of Their Importance and Characteristics," *Journal of Policy Analysis and Management* 15(3) 1996, 353–76.

16 This extension is in the spirit of Kenneth J. Arrow and Gerard Debreu, "Existence of an Equilibrium for a Competitive Economy," *Econometrica* 22(3) 1954, 265–90.

17 On the limitations of insurance markets, see Richard J. Zeckhauser, "Coverage for Catastrophic Illness," *Public Policy* 21(2) 1973, 149–72; and "Resource Allocation with Probabilistic Individual Preferences," *American Economic Review* 59(2) 1969, 546–52.

18 J. David Cummins, "Statistical and Financial Models of Insurance Pricing and the Insurance Firm," *Journal of Risk and Insurance* 58(2) 1991, 261–301.

19 For a theoretical treatment of moral hazard, see Isaac Ehrlich and Gary S. Becker, "Market Insurance, Self-Insurance, and Self-Protection," *Journal of Political Economy* 80(4) 1972, 623–48.

20 Moral hazard also describes risky actions by the insured to qualify for compensation. See, for example, the behavior of air traffic controllers under the Federal Employees

Compensation Act by Michael E. Staten and John R. Umbeck, "Close Encounters in the Skies: A Paradox of Regulatory Incentives," *Regulation*, April/May 1983, 25–31.

21 See Philip J. Cook and Daniel A. Graham, "The Demand for Insurance and Protection: The Case of Irreplaceable Commodities," *Quarterly Journal of Economics* 91(1) 1977, 141–56.

22 For overviews, see Amos J. Tversky and Daniel Kahneman, "Judgment under Uncertainty: Heuristics and Biases," *Science* (185) 1974, 1124–31. For an explanation of these misperceptions in terms of Bayesian updating, see W. Kip Viscusi, "Prospective Reference Theory: Toward an Explanation of the Paradoxes," *Journal of Risk and Uncertainty* 2(3) 1989, 235–63.

23 Tversky and Kahneman call this the *heuristic of availability* (ibid., 1127). They also describe the *heuristic of representativeness*, when people misestimate conditional probabilities, and the *heuristic of anchoring*, the tendency of people not to adjust their initial estimates of probabilities adequately as more information becomes available (ibid., 1124 and 1128).

24 Howard Kunreuther, *Disaster Insurance Protection: Public Policy Lessons* (New York: John Wiley, 1978), 145–53.

25 For a review, see Chris Starmer, "Developments in Non-Expected Utility Theory: The Hunt for a Descriptive Theory of Choice Under Risk," *Journal of Economic Literature* 38(2) 2000, 332–82.

26 Matthew Rabin, "Psychology and Economics," *Journal of Economic Literature* 36(1) 1998, 11–46; Jessica L. Cohen and William T. Dickens, "A Foundation for Behavioral Economics," *American Economic Review* 92(2) 2002, 335–38.

27 Brigitte C. Madrian and Dennis F. Shea, "The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior," *Quarterly Journal of Economics* 116(4) 2001, 1149–87. More generally, see Richard H. Thaler and Cass R. Sunstein, *Nudge: Improving Decisions About Health, Wealth, and Happiness* (New Haven, CT: Yale University Press, 2008).

28 We distinguish three types of markets: *spot markets* encompass transactions involving immediate delivery of goods, *forward markets* encompass contracts that specify future

delivery at a specified price, and *futures markets* are specially organized forward markets in which clearinghouses guarantee the integrity and liquidity of the contracts.

29 For a review of the objections, see Robert E. Goodin, *Political Theory & Public Policy* (Chicago: University of Chicago Press, 1982), 162–83.

30 Harold Hotelling first set out the basic model of *exhaustible resources* in "The Economics of Exhaustible Resources," *Journal of Political Economy* 39(2) 1931, 137–75.

31 See Burton A. Weisbrod, "Collective Consumption Services of Individual Consumption Goods," *Quarterly Journal of Economics* 78(3) 1964, 471–77; John V. Krutilla, "Conservation Reconsidered," *American Economic Review* 57(4) 1967, 777–86; and Aidan R. Vining and David L. Weimer, "Passive Use Benefits: Existence, Option, and Quasi-Option Value," in Fred Thompson and Mark T. Green, eds., *Handbook of Applied Public Finance* (New York: Marcel Dekker, 1998), 319–45.

32 For a discussion of how the economy adjusts to oil price shocks, see chapter 2 of George Horwich and David L. Weimer, *Oil Price Shocks, Market Response, and Contingency Planning* (Washington, DC: American Enterprise Institute, 1984).

33 See Sherwin Rosen, "Implicit Contracts: A Survey," *Journal of Economic Literature* 23(3) 1985, 1144–75.

34 For an overview of economic thought on the business cycle, see Victor Zarnowitz, "Recent Work on Business Cycles in Historical Perspective: A Review of Theories and Evidence," *Journal of Economic Literature* 23(2) 1985, 523–80.

7
Rationales for Public Policy
Distributional and Other Goals

The traditional market failures and other limitations of the competitive framework discussed in Chapters 5 and 6 identify circumstances in which private economic activity may not lead to Pareto efficiency. Thus, they indicate the potential for Pareto improvements through public policy. As we discuss in the next chapter, correcting government failures also offers opportunities for making potential Pareto improvements.

Values other than efficiency, however, warrant consideration in our assessments of the extent to which any particular combination of private and public activity achieves the "good society." As individuals, we turn to philosophy, religion, and our moral intuition to help ourselves develop systems of values to guide our assessments. Absent a consensus on the values to be considered and their relative importance when they conflict, our political institutions must select the specific values that will have weight in collective decision making. Nevertheless, we sketch here some of the more commonly employed rationales for public policies that are based on values other than efficiency.

Social Welfare beyond Pareto Efficiency

Before we discuss distributional and other important *substantive values* as distinct goals that compete with Pareto efficiency, we present two perspectives for viewing these values as contributing to economic efficiency, more broadly defined.[1] One approach involves the specification of a *social welfare function* that aggregates the utilities of the individual members of society. Another approach shifts the focus from the valuation of particular allocations of goods to the valuation of alternative *institutional arrangements* for making allocations.

Explicit Social Welfare Functions

The concept of Pareto efficiency offers a way of ranking allocations of goods without making explicit comparisons of the utilities of individuals. The avoidance of interpersonal utility comparisons is achieved at the cost of an implicit acceptance of the initial distribution of endowments of productive factors among individuals. The specification of a social welfare function, which converts the utilities of all individuals into an index of social utility, provides an alternative approach to defining economic efficiency and aggregate welfare. Rather than define efficiency as the inability to make someone better off without making someone else worse off (the Pareto principle), efficiency is defined as the allocation of goods that maximizes the social welfare function (the "greatest good" principle).

To clarify the difference between these definitions, consider a society consisting of three people. Assume that each of these people has a utility function that depends only on wealth, assigns higher utilities to higher levels of wealth, and assigns smaller increments of utility to successive increments of wealth (that is, they exhibit positive but declining marginal utility of wealth). Note that, for a fixed quantity of wealth, any allocation among the three people would be Pareto efficient: it would not be possible to shift wealth to increase one person's utility without decreasing the utility of at least one of the others. Imagine, however, that these people agree that wealth should be allocated to maximize a social welfare function specified as the sum of the utilities of the three persons. If we assume that the three people have identical utility functions, then the allocation of wealth that maximizes social welfare is exactly equal shares of wealth for each person. (Only when each person has exactly the same marginal utility of wealth would opportunities for increasing the social welfare function disappear−−the assumption of identical utility functions implies that the equality of marginal utility can occur only when wealth is equal.) Thus, this social welfare function would identify one of the many Pareto-efficient allocations as socially optimal.

Table 7.1 illustrates socially optimal policy choices under three different social welfare functions for a three-person society. The social welfare function labeled "Utilitarian" simply sums the utilities of the three persons ($U_1 + U_2 + U_3$); it would be maximized by policy C. The social welfare function labeled "Rawlsian" maximizes the utility received by the person deriving the lowest utility, a *maximin principle, maxi*mizing the *mini*mum utility realized by anyone. It would identify policy B as the social optimum. The social welfare function labeled "Multiplicative" is proportional to the product of the utilities of the three persons ($U_1 \times U_2 \times U_3$); it indicates policy A as the social optimum. In this example, each of the three social welfare functions identifies a different policy as socially optimal.

Utilitarianism is an influential welfare function developed by Jeremy Bentham and, especially, John Stuart Mill.[2] It also has attracted modern adherents, such as the Nobel Prize winner John Harsanyi, who argued for a utilitarian-based rule of maximizing the expected average utility of all individuals.[3] Utilitarianism is *consequentialist*. Such a philosophy asserts that actions are to be evaluated in terms of the preferences of individuals for various consequences, which, in turn, can be aggregated. Utilitarianism is sometimes described as the "greatest good for the greatest number," but that is inaccurate. As Table 7.1 demonstrates, utilities are added up; therefore, utilitarianism is simply the "greatest good." Broadly, utilitarianism underlies the Kaldor−Hicks principle described in Chapter 4 and is the foundation for cost−benefit analysis.

Although utilitarianism does not differentially weight anyone's utility, whether rich or poor, its founders perceived it to be both egalitarian and democratic. Utilitarianism is egalitarian in spirit because, as John Stuart Mill argued, individuals exhibit declining marginal utility with respect to wealth, which justifies some level of redistribution from rich to poor. Utilitarianism is democratic in spirit because it posits that the utility of everybody

Table 7.1 Alternative Social Welfare Functions

	Individual Utilities from Alternative Policies			*Alternative Social Welfare Functions*		
	Person 1 U_1	*Person 2* U_2	*Person 3* U_3	*Utilitarian* $U_1 + U_2 + U_3$	*Rawlsian Minimum* (U_1, U_2, U_3)	*Multiplicative* $(U_1 \times U_2 \times U_3)$ / *1,000*
Policy A	80	80	40	200	40	**256**
Policy B	70	70	50	190	**50**	245
Policy C	100	80	30	**210**	30	240
				Choose policy C	Choose policy B	Choose policy A

counts, the commoner as well as the king. A common criticism of utilitarianism is that it offers weak protection for fundamental individual rights, because it does not guarantee minimal allocations to individuals.

Rawlsianism, as shown in Table 7.1, is a highly equalizing social welfare function. John Rawls applies the maximin principle—"the greatest benefit of the least advantaged members of society"—on the basis of a provocative thought experiment. This thought experiment is intended to free us from our particularistic values that may be more driven by self-interest than we are ready to admit.[4] He asks us to imagine people who must decide on a system of institutions for society without knowing what their own endowments (such as race, intelligence, and wealth) will be in that society. Behind this *veil of ignorance*, in an *original position*, people would deliberate as equals. Not knowing what one's endowment will be encourages one to consider the overall distribution of opportunities and outcomes. Rawls argues that people would unanimously exhibit risk aversion and would therefore select a social welfare function that raises the position of the least advantaged, leading to greater equality of outcomes. As Rawls posits people agreeing to institutions behind the veil of ignorance, his philosophical approach is in the spirit of the *social contract* developed by John Locke and Jean-Jacques Rousseau.[5]

Rawlsianism has been criticized on several grounds.[6] One is that it proposes extreme redistribution that reduces incentives to create wealth.

Another criticism is that, in practice, people in the original position would not be either as risk averse or as consensual as Rawls suggests. Researchers have tried to address these kinds of questions in a number of experiments that attempt to replicate the original position and the veil of ignorance to some degree. One set of experiments, for example, found that most subjects strongly reject Rawls's maximin principle. However, some also reject a Harsanyian "maximizing the average utility" welfare function (although fewer and less strongly). Most subjects preferred an allocation that maximizes average utility with some floor constraint.[7] Notice that these empirical finding are broadly consistent with the *muliplicative* social welfare function shown in Table 7.1, as it avoids allocations with very low levels of utility to any individuals.

The existence of a universally accepted social welfare function would greatly reduce the complexity of the task that policy analysts face in ranking policy alternatives by eliminating the necessity of making trade-offs among incommensurate values. Analysts would design policy alternatives and predict their consequences, but the social welfare function would mechanically rank the alternatives. The policy analyst would hardly carry a normative burden. A number of conceptual and practical problems, however, prevent the social welfare function from rescuing policy analysts from having to confront trade-offs between efficiency and other values.

First, social welfare functions that directly aggregate the utilities of individuals assume that these utilities can be observed. However, utility is a subjective concept that defies direct measurement.[8] Indeed, most economists eschew any explicit interpersonal comparisons of utilities for this reason. Instead, they usually rely on the principle of *no envy* to define equity: a social allocation is equitable if no one would prefer anyone else's personal allocation to their own.[9] In this formulation, people compare allocations across individuals in terms of their own utility functions. For example, splitting an estate between two heirs by allowing one to divide the assets into two shares and the other to have first choice of shares will result in an equitable allocation in the sense that the chooser will select the allocation that he or she most prefers and, therefore, have no envy, and the divider, not knowing which allocation will be selected, will divide so as to be indifferent between the two allocations and, therefore, also have no envy.

The principle of no envy does not provide a practical basis for constructing social welfare functions, however. The problem of utility measurement can be sidestepped somewhat by relating social welfare to the consumption that produces utility for the members of society.[10] So, instead of social welfare being specified as some function of personal utilities, it is given as a function of either the quantities of goods consumed by each person or the economic resources, such as income or wealth, available to each person for purchasing goods of choice. When the social welfare function depends on the consumption of specific goods, society places values on consumption patterns without direct regard for individual preferences. For example, the social welfare function might place maximum weight on some target level of clothing consumption, even though some persons would prefer to consume more, and some persons less, than the target level.

Second, whether the social welfare function depends on individual utilities, wealth levels, incomes, or consumption patterns, it must somehow be specified. Absent unanimity among the members of society over the appropriate social welfare function, no fair voting procedure can guarantee a stable choice.[11] However, unanimity is unlikely because individuals can anticipate how they will personally fare under alternative social welfare functions. How, then, would the social welfare function come to be specified? Elevating the preferences of those who happen to be governing at the moment to the status of social welfare function seems presumptuous, especially if current choices will have implications beyond their likely tenures in office. Thus, rather than being an objective starting point for analysis, the social welfare function is more likely to be the product of value judgments made as part of the analysis.

As a society, we never explicitly choose the aggregate level of equity. Redistributions that have implications for equity are selected sequentially and often delivered through programs bundled with policies seeking other goals, such as efficiency. If we did so, then the process would be subject to some of the preference aggregation problems mentioned above. Therefore, it is extremely difficult to deduce what level and kind of redistribution society would choose if a perfect revelation mechanism did exist. Experiments, however, do provide some hints. An experimental game is especially useful for this purpose. In the *ultimatum* game, introduced in Chapter 6, a proposer offers a division of a fixed sum of money (X) to a responder, who can either accept or reject it. If the responder accepts the offer, then both the proposer and the responder receive their designated percentage of X. If the responder rejects the offer, then both receive nothing. How often do responders reject? "Dozens of studies with these games establish that people dislike being treated unfairly, and reject offers of less than 20 percent of X about half the time, even though they end up receiving nothing."[12] In contexts where proposers can observe acceptances or rejections and respond, they typically offer 40 to 50 percent of X. The ultimatum game tells us about "fairness" rather than equity. The extent of pure redistributive altruism is more clearly revealed in the *dictator game*. Here, the proposer can dictate the allocation because the responder cannot refuse it. Proposers offer considerably less than in the ultimatum game, but still generally offer an average of 20 to 30 percent of the sum to be divided.[13]

Third, even if a social welfare function were somehow to be validly specified, limitations in information and cognition would hinder its practical use. Predicting the impacts of policies on the millions of individuals who make up society boggles the mind. It would almost certainly be necessary to divide society into groups to make the application of the social welfare function more tractable, but such grouping risks obscuring important differences among individuals within groups. For example, grouping solely by income level might not be appropriate if people in different life stages, locations, or conditions of health face very different costs in maintaining households.

Fourth, when analysts specify hypothetical social welfare functions to rank alternative policies, perhaps to make explicit the consideration of distributional issues, they tend to make the social welfare functions depend only on immediate consequences so as to keep the burden of prediction manageable. Such myopia can lead to rankings other than those that would be made with greater consideration of consequences occurring in the more distant future. For example, an exclusive focus on the current distribution of income ignores the impacts of policies on investment decisions in physical and human capital that significantly affect the size and distribution of income in the future. Using a myopic social welfare function as an expedient thus risks diverting attention from the full range of appropriate values.

Choosing Institutions versus Choosing Allocations

Several intellectual streams argue for a recasting of the "social welfare problem," from one of choosing distributions of goods to one of choosing the social, economic, and political institutions from which the distributions will actually emerge. The constitutional design perspective focuses on the choice of the fundamental procedural rules governing political decision making.[14] The property rights perspective considers the implications of the rules governing ownership and economic activity, recognizing that they affect how much is produced as well as how it is distributed.[15] Others have sought to understand the significance of social values, norms, habits, conventions, and other informal forces that influence interaction among members of society.[16] An appreciation of the importance of institutions, the relatively stable sets of formal and informal rules that govern relationships among members of society, unifies these otherwise very different perspectives.

The distinction between valuing alternative distributions and valuing alternative institutions is relevant no matter what normative framework underlies the assessment of social welfare. It arises most explicitly, however, in the distinction between *act-utilitarianism* and *rule-utilitarianism*.[17] Under act-utilitarianism, the rightness of an act depends on the utility that it produces. Under rule-utilitarianism, the rightness of an act depends on its adherence to general rules or principles that advance social utility; it thus provides a moral basis for the creation and adherence to rights and duties that restrain the expediency of act-utilitarianism.[18] So, for example, giving public relief to people who suffer severe property loss from a flood might be judged under act-utilitarianism as desirable because it results in a more uniform distribution of wealth. Under rule-utilitarianism, however, the giving of public relief might be viewed as undesirable according to the principle that public policy should not encourage moral hazard––if compensation is given, then others will be encouraged to live in floodplains in anticipation that they, too, will be compensated for losses. Note that if utility interdependence among the population makes relief to the current flood victims a public good that would be underprovided by private charity, act-utilitarianism rather than rule-utilitarianism would be consistent with the common (myopic) interpretation of Pareto efficiency.

Rule-utilitarianism encourages consideration of a variety of values that contribute indirectly to social utility through more effective political, social, and economic institutions. By increasing the chances that dissident views will be heard, for instance, largely unbridled freedom of speech reduces the risks of consistently bad political decisions being made by misguided majorities. Policies that give a major role to parents in the socialization of children help preserve the family as the most fundamental social institution. Maintaining fair procedures for resolving disputes over the enforcement of contracts facilitates efficiency in economic exchange among private parties. Moral and political considerations hinder promotion of institutional values when these values involve immediate consequences that are, in isolation, undesirable: some speech is highly offensive; even intact families sometimes grossly fail children; fair procedures for dispute resolution do not always favor those who would benefit most from compensation. Nonetheless, the focus on institutions deserves consideration because it encourages a broader and less myopic perspective.

Substantive Values Other Than Efficiency

Viewing market failure as a necessary condition for government intervention implies that Pareto efficiency is the only appropriate social value. Strong arguments can be made for always *including* efficiency as a substantive value in policy analysis: only the malevolent would oppose making someone better off without making anyone else worse off; the pursuit of efficiency produces the greatest abundance of goods for society, thereby facilitating the satisfaction of wants through private actions, both self-interested and altruistic; and, given the decision to pursue other substantive values through public policy, doing so efficiently preserves the greatest potential for satisfying material wants. Here we sketch the rationales for some of the more important substantive values that often compete with efficiency.

Human Dignity: Equity of Opportunity and Floors on Consumption

We begin with the premise that all people have intrinsic value, which derives from the very fact that they are human beings rather than from any measurable contribution they can make to society. Our own dignity as human beings requires us to respect the dignity of others. Although the meaning of dignity ultimately rests with the individual, it involves, at least to some extent, one's freedom to choose how one lives. A good society must have mechanisms for limiting the extent to which any person's choices interfere with the choices of others. It also should facilitate broad participation in the institutions that determine the allocation of private and public goods.

We have focused on market exchange as a mechanism for translating the preferences of individuals into allocations of private goods. To participate in markets, however, one must have something to exchange. In the competitive framework one has a set of endowments, ownership of labor and other factor inputs. Someone who has no endowments would be effectively barred from participating in the market process. With very small endowments, people have very little choice; without any endowments, people cannot express their choices of private goods at all. As survival depends on the consumption of at least some private goods, such as food and protection from exposure, we might find that a Pareto-efficient allocation involves the premature deaths of some people.

Almost all of us would consider such extreme allocations to be inappropriate. In many societies, people take it upon themselves to mitigate the most drastic consequences of extreme allocations through voluntary contributions to charities or to disadvantaged individuals. For example, until the role of the federal government expanded greatly during the 1930s, private charity was a major source of assistance for widows, orphans, and the disabled in the United States.[19] Concern that charity may not adequately reach these vulnerable populations is a widely accepted rationale for public assistance.

Note that these preferences do not necessarily favor the Rawlsian social welfare function presented in Table 7.1. They might very well be consistent with a utilitarian social welfare function that places lower bounds on consumption levels or on some version of the multiplicative social welfare function. The essential concern is the absolute, rather than the relative consumption of those worse off.

More generally, we might agree that viable participation in market exchange requires some minimal endowment of assets. Because most people have potential to sell their own labor, one approach to increasing the number of people with the minimal endowment is to increase their effectiveness and marketability as workers. Public policies intended to provide remedial education, job training, and physical rehabilitation may play an important role in increasing effective participation in the private sector.[20] So, too, might policies that protect against discrimination by employers on the basis of factors not relevant to performance on the job. Direct public provision of money or in-kind goods, however, may be the only way to ensure minimal participation of those, such as the severely disabled, who have little or no employment potential.

Note that while transfers to ensure floors on consumption would increase the equality of the distribution of goods in society by increasing consumption by the least wealthy, this is only a side effect of preserving human dignity. Once everyone reaches some minimum level of consumption, preservation of human dignity does not necessarily call for further redistribution to increase equality.

Increasing participation in decisions over the provision and allocation of public goods also merits consideration as an appropriate value in the evaluation of public policies. As we previously discussed, markets generally do not lead to efficient levels of provision of public goods. Actual choices are made through political processes. Public policies that broaden political participation may be desirable, therefore, in terms of respect for individual preferences for public goods. We support universal adult suffrage, for instance, not because we expect it necessarily to lead to greater efficiency in the provision of public goods but, rather, because it recognizes the inherent value of allowing people some say over the sort of society in which they will live.

Finally, we must consider situations in which people are incapable of rationally exercising choice. For example, most of us would not consider preventing either young children or mentally impaired individuals from seriously harming themselves to be a violation of their human dignity (although we may disagree on the definitions of "young" and "impaired"). On the contrary, we might embrace an affirmative duty to act paternalistically. The problem, from the perspective of public policy, is deciding the point at which individual choices should be overridden.

Increasing the Equality of Outcomes

Respect for human dignity seems to justify public policies that ensure some minimum level of consumption to all members of society. Most of us would agree that the minimum level should be high enough to ensure the commonly recognized needs for decent or dignified survival. The level is theoretically absolute, but in practice it is not fixed. It is absolute in the sense that it does not explicitly depend on the wealth, income, or consumption of others in society: individuals need some level of food and shelter to survive, no matter how others live. Yet, this level is not fixed because the collective assessment of what constitutes *dignified* survival undoubtedly reflects the aggregate wealth in society. The U.S. government regularly estimates how much income families must have to consume a set of basic goods that were considered necessary for decent survival in 1965. The number of persons in families raised above this *poverty line* often serves as a measure of success of government programs.[21] For instance, in 2015 the official poverty line for a family of four with two related children under age 18 years was $24,036. (Since 2010, the U.S. Census has also estimated a supplemental poverty line that takes account of taxes, work expenses, and government transfers.) As the United States has become wealthier, however, there is a growing tendency to use multiples, such as 125 percent of the poverty line, in evaluations, suggesting a change in our collective assessment of the decent minimum.

Rather than look just at the absolute consumption levels of the poorest members of society, we might also consider the entire distribution of wealth. Personal attributes, family circumstance, and chance lead to a wide dispersion in the levels of wealth enjoyed as outcomes in a market economy. Several conceptual frameworks seem to support greater equality in outcomes as an important social value.

The Rawlsian social welfare function shown in Table 7.1 clearly leads to greater equality of outcomes. As previously noted, even the utilitarian social welfare function shown in Table 7.1 provides an argument for viewing equality of outcomes as a social value if we assume that all individuals have identical utility functions exhibiting declining marginal utility with respect to wealth. That is, individuals get less additional happiness from additional units of wealth as their levels of wealth increase. Under these assumptions, the more equal the distribution of any given level of wealth, the higher the ranking of that distribution by the social welfare function. In terms of tax policy, for instance, this argument leads to the notion of *vertical equity*—– those with greater wealth should pay higher taxes so that everyone gives up the same amount of utility.[22] (The related notion of *horizontal equity* requires that those in similar circumstance be treated alike.)

An important consideration arises in thinking about equality of outcomes as a social value. The discussion above implicitly assumes that the amount of wealth to be distributed among individuals is constant. This may be true for a one-time, unanticipated reallocation. Once people anticipate reallocation, however, they will begin to change their behaviors in ways that generally reduce the total amount of wealth that will be available. In the extreme, where everyone received the same share of wealth, the incentives for work, investment, and entrepreneurship would be greatly reduced. Costly and intrusive control mechanisms would have to be put in place to try to get people to work and invest as much as they would have in the absence of the redistribution. Undoubtedly, the result would be a much less wealthy society, perhaps resulting in lower absolute levels of wealth for most members of society.

One might expect that, as national and global wealth increase over time (as they generally have over the past 200 years), wealth and income inequality might naturally decrease—in other words, one might expect that "a rising tide raises all boats." However, this does not appear to be the case in the United States. Thomas Piketty and Emmanuel Saez show that the income share of the top income decile in the United States fell from about 45 percent in 1930 to under 35 percent by 1970, but since has increased to above 45 percent.[23] Globalization is often advanced as an explanation for the dramatic return of inequality since 1970. However, this explanation is inconsistent with the pattern of income inequality over time in Europe and Japan. For example, the income share of the top decile in Europe fell to about 30 percent in 1970 but has since risen to only 35 percent.[24] Piketty and Saez note that much of the rise in the share of income going to the top income decile arises from increases in executive compensation.[25] For example, the ratio of chief executive officer (CEO) to worker compensation rose in the United States, from 20 to 1 in 1965, to 295 to 1 in 2013.[26] The large ratio suggests that, within current institutional arrangements, CEOs enjoy an advantage in bargaining with other stakeholders over corporate profits.[27]

Two public goods problems may be contributing to the imbalance between wages and executive compensation in this bargaining process. First, labor unions provide a public good to their members in negotiating for a share of corporate profits through higher wages. However, a combination of changes in industrial composition and so-called “right to work” laws that allow workers to free ride on union wage agreements have contributed to declining union membership and therefore a weaker position in terms of wage bargaining. Second, monitoring executive compensation is also a public good that is likely undersupplied because of the diffuse ownership of stocks.

Unfortunately, we should generally expect the total wealth in society to shrink more, the greater the amount of redistribution attempted. Arthur Okun uses the analogy of transferring water with a “leaky bucket.”[28] If we try to transfer a little water, we will lose a little; if we try to transfer a lot, we will lose a lot.[29] Of course, as discussed in Chapter 4, using taxation to correct market failures increases efficiency so that these taxes offer the potential for increasing efficiency and redistributing wealth. For example, carbon taxes could be combined with rebates to provide progressive redistribution without decreasing economic efficiency.[30] Nonetheless, the key question is commonly how much current and future wealth are we as a society collectively willing to give up to achieve greater equality in distribution? In practice, we must rely on the political process for an answer.

Preserving Institutional Values

Historical accident, as well as purposeful individual and collective choices, divides the world into distinct polities. National governments play a dominant role in establishing the terms of political and economic interaction. They draw their legitimacy from *constitutions*, both formal and informal, that define the rights and duties of those who reside within their boundaries, and the procedures their officials must follow in the creation and enforcement of laws. Political jurisdictions subordinate to national governments generally have their own constitutions, though often in the form of charters granted by the superior level of government. Whether at national or subnational levels, these constitutions provide essential rules for organizing both collective and private decision making.

The constitutions themselves derive legitimacy from a number of sources: the perception that their content provides order in a reasonable and fair way; the nature of their establishment, especially the extent to which they enjoyed the consent of those whom they initially governed; and the ease with which people can choose alternative polities through immigration and emigration.[31] Because constitutions maintain legitimacy through adherence, keeping public policy within the bounds of recognized constitutional principles is a social value. There is also social value in protecting legitimate constitutions from threats external to the polity. Thus, policies that promote national security may be justified by a social value other than increasing efficiency through the provision of a public good.

Within constitutional frameworks, societies benefit when people voluntarily comply with laws. Voluntary compliance reduces enforcement costs. It also reflects, and almost certainly enhances, the legitimacy of the political system. One factor that seems to contribute to voluntary compliance is the perception of fairness. For example, surveys and laboratory experiments suggest that taxpayers who view the tax system as fair are more likely to comply than those who view it as unfair.[32] Perceptions of fairness also affect compliance with traffic laws,[33] and people who view processes for determining sites as fair are more likely to be willing to accept noxious facilities near their hometowns.[34] More generally, there is social value in making policies correspond to common perceptions of fairness. These perceptions can relate to the procedures embedded in the policies for their implementation, as well as the process through which the policies are adopted.

As we previously noted, the perspective of rule-utilitarianism suggests that there is likely to be social value in preserving or strengthening important social institutions such as the family. A difficulty arises, however, because people hold divergent views about the intrinsic value of existing social institutions. For example, some people note the subservient role often played by women in the traditional family and, therefore, would object to viewing its preservation as a social value.

Some Cautions in Interpreting Distributional Consequences

Before relating the distributional consequences of alternative policies to substantive values, we must answer this question: "What is being distributed to and from whom?"[35] Even putting aside the ever-present difficulty of prediction, answering this question raises many conceptual issues that can lead to dubious assessments of distributional consequences. We outline here some of the more important issues that arise in distributional analysis.

Measurement Issues

In a market economy, money gives access to private goods. Personal income, the flow of payments to persons for use of the input factors (labor, land, capital) they own, provides a conceptually attractive measure of purchasing power. Income as commonly measured, however, deviates from purchasing power in several important ways that complicate distributional analysis.

First, not all wealth is fully reflected in measured income.[36] Consider ownership of housing, for example. When owners rent their houses to others, the values of the houses are reflected in their incomes. When they live in the houses instead, their incomes do not reflect the values of their houses, even though they derive consumption benefits from living in them. A conceptually correct measure of actual total income would impute income to those who live in houses that they own based on the rent they could have charged for their property. Not imputing income to such assets can cloud the interpretation of income distributions. For example, many elderly have low incomes but nevertheless have considerable wealth invested in their homes and other assets: in 2011 the ratios of median household net worth to median household income by age of the head of household in the United States were as follows: 35–44 years, 0.57; 45–54 years, 1.32; 55–64 years, 2.57; and 65 years and older, 5.15.[37] These large differences in ratios across age groups shows dramatic differences between income and wealth.

Second, government tax and transfer programs alter the amount of "disposable" income that people have available to purchase private goods. After-tax income can usually be measured or predicted with relative confidence. Yet, deductions and other adjustments made to arrive at taxable income imply that taxpayers with the same pretax income may have very different *disposable incomes*.

With respect to transfer programs, cash benefits are more easily figured into disposable income than in-kind benefits, such as subsidized housing and health care, that must be assigned an imputed dollar value. Taking account of in-kind benefits can substantially change assessments of inequality.[38] Measuring or predicting the cumulative effect of transfer programs is especially complicated when eligibility depends on income from other sources. Such *means-testing* can produce situations in which those with pre-transfer incomes just too large to qualify are worse off than those with lower pre-transfer incomes.

Third, individuals usually consume as members of households. Not all members of a household necessarily contribute income toward the consumption. More importantly, the cost per person of providing such basic goods as housing is generally smaller for multi-member households than for single-person households. This suggests that comparisons of income better reflect basic consumption opportunities if they are done on the basis of households, rather than per capita. The difference can be substantial: in 2014, median household income in the United States was $53,657 and per capita income for those over 15 years old was $28,757.[39] But changing family composition and problems of definition make household income a less-than-ideal basis for measuring economic well-being.[40] Consider an economy with a single household of two wage earners. Simply by splitting into two households—say, because of divorce—the average household income of the economy would fall by half (per capita income would remain unchanged). Such a dramatic fall almost certainly overstates the reduction in consumption opportunities (though perhaps not the overall reduction in social welfare!).

Less hypothetically, changing demographic patterns, such as reductions in average family size and the rate of labor force participation by women, affect comparisons of household income over time. Similarly, these differences may affect comparisons of family income between different groups in a population. With respect to definitional issues, consider a college student with part-time employment who spends most of the year living away from parents: at what point should this student be considered a new household?

Index Issues

Comparing distributions poses the problem of choosing appropriate metrics. Ideally, we would like an index that ranks distributions according to appropriate distributional values. Difficulties arise because no single index can fully summarize a distribution. They also arise because of ambiguity in operationalizing distributional values.

Consider first the problem of characterizing a distribution. The commonly used measures of central tendency, the median (the middle value) and the arithmetic mean (the average), do not convey the spread of distributions. The median of a distribution of income would remain unchanged if everyone with income below the median had his or her income reduced by, say, half. The mean would remain unchanged by taking money from the poorest and giving it to the richest. A measure such as sample variance, based on squared deviations from the mean, comes closer to measuring equality, but does not distinguish between distributions with many people above the mean and distributions with many people below the mean, a difference certainly relevant to almost any assessment of equality.

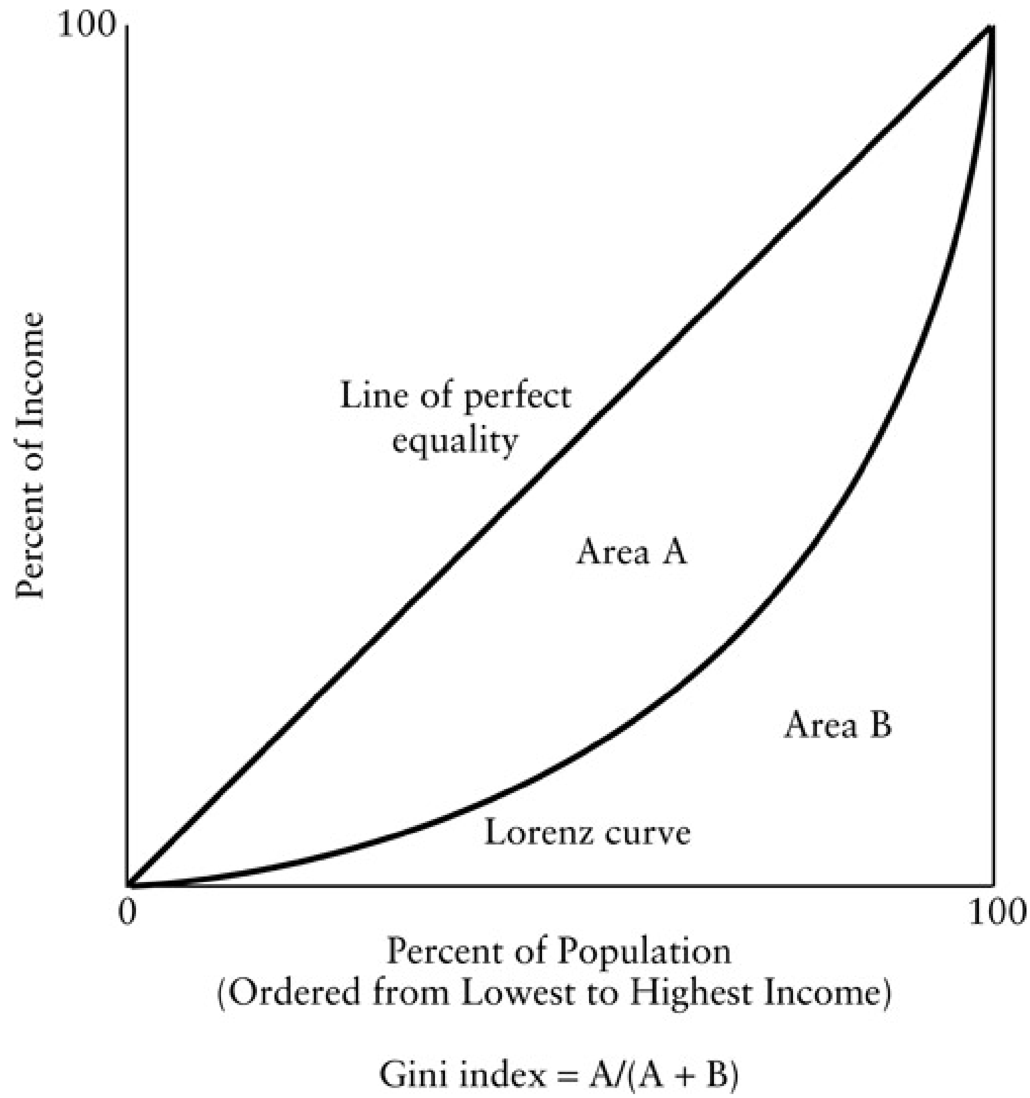

Figure 7.1 Lorenz Curve and Gini Index of Income Inequality

A commonly used distributional measure is the *Gini index of relative inequality*, which is related to the *Lorenz curve*.[41] A Lorenz curve, shown as the curved line in Figure 7.1, is constructed by ranking the population by income and asking: What percentage of income goes to the poorest *X* percent of the population? With percentage of income measured on the vertical axis and percentage of population measured on the horizontal axis, the income distribution will trace out a curve going from the origin to the point representing 100 percent of the income going to 100 percent of the population. A straight line connecting these points represents a perfectly equal distribution of income. Any distribution with imperfect equality would lie entirely below this straight line. The Gini index of relative inequality is proportional to the area in a Lorenz diagram between the line of perfect equality and the Lorenz curve for the distribution. A Gini coefficient of zero implies perfect equality; coefficients closer to one correspond to less equality. The Gini coefficient for the United States increased from 0.397 in 1967 to 0.480 in 2014, indicating a substantial increase in inequality.[42]

The Gini coefficient provides an attractive measure for comparing the equality of distributions over the entire range of income. If the distributional value of concern is the income of the poorest members of society, however, then we may prefer an alternative measure that looks only at the lower end of the distribution. For example, one possibility is the percentage of income going to the lowest income quintile, the 20 percent of households with the lowest incomes. Alternatively, taking an absolute rather than a relative perspective leads to measures such as the sum of payments needed to bring all poor households up to the poverty line. Choosing appropriately among the possible indices that could be constructed requires an explicit examination of the values underlying the distributional concern.

Categorization Issues

Indices can be used to provide measures of equality within groups. Yet, distributional analysis often deals with equality across identifiable groups defined by characteristics such as gender, race, age, and region. A common approach is to compare mean, or median, income between groups. For example, income comparisons are often made between men and women, whites and nonwhites, and those adults under 65 years of age and those older. Care is needed in drawing inferences from such intergroup comparisons, because the groups almost always differ in important ways other than their defining characteristics.

Consider, for instance, the observed differences in median household incomes between whites and nonwhites. If the groups were similar with respect to educational achievement, family structure, and age profiles, then the difference would be strong prima facie evidence of racial discrimination in employment. Yet, if the groups differ markedly in terms of these factors, then the inference of discrimination in employment is much weaker. A more appropriate approach would be to compare incomes for households of similar composition and educational achievement. Of course, educational differences themselves might have resulted from racial discrimination, either through the resources made available to schools with different racial compositions or discrimination in housing markets that affects access to public schools.

Silent Losers Issues

Distributional analyses often fail to identify all the groups affected by policies. Especially likely to be missed are *silent losers*, those who fail to voice protest against the policies causing their losses. In most democratic political systems, complaining increases the chances that losses will at least be considered. By remaining silent, the losers risk being overlooked by politicians and analysts alike.

One reason losers fail to protest is that they unexpectedly suffer losses as individuals. At the time of adoption of the policy, they do not anticipate being losers; when they do suffer losses, protest offers little prospects for personal gain. For example, consider residential rent control. The distributional analysis is often cast in terms of current residents versus current landlords. But another group that loses is potential renters, people who sometime in the future desire to live in the community yet cannot because below-market rents "lock in" long-time residents to rent-controlled units, discourage investors from expanding the rental stock, and encourage landlords to convert rental units to condominiums. Individuals who cannot find apartments have little prospect of personal gain from protest and, therefore, little incentive to bear the cost of protest.

Losers may fail to protest because they do not connect their losses to the policy. For example, petroleum product price controls in the United States and Canada during the oil price shocks of the 1970s were most often justified on distributional grounds, the petroleum industry should not be allowed to gain windfall profits, and higher petroleum product prices hurt low-income more than high-income consumers. This distributional characterization, however, ignores the group that suffers the largest personal losses: those who lose their jobs because of the economic inefficiency caused by the price controls.[43] Yet, these people, like many policy analysts at the time, generally failed to make the connection between the inefficiency caused by the price controls and the higher rate of unemployment during oil price shocks.

Losers may be silent because they are not yet born! Policies affecting the environment, for instance, may have significant effects on the quality of life of future generations. Usually these effects will begin to occur in the more immediate future so that they have some political saliency. Yet, some effects may only become apparent after a considerable period of time. The increasing concentration of carbon dioxide in the atmosphere may be contributing to substantial global climate change in the future through the greenhouse effect. The interests of future generations seem to be receiving consideration in discussions of global climate change, though not necessarily as fully as they deserve. Advocates of public policies to reduce carbon dioxide emissions dramatically tend to note future environmental effects; advocates of a more cautious approach raise the implications of current policies for the wealth of future generations.

Choosing Distributional Values

The distributional values relevant to any particular policy analysis depend on the nature of the problem being addressed. Often a more equal distribution of wealth is a relevant distributional value, though, for practical reasons, a more equal distribution of income is an expedient proxy. However, in many situations relevant distributional values of concern have non-economic dimensions, such as race, gender, age, or other demographic characteristics. In some situations, distributional values may even be related to culpability. For example, in some U.S. jurisdictions adolescents involved in "sexting" have been prosecuted under child pornography laws. In assessing changes in such laws, one might argue that the harshness of punishment should distinguish among the motives and vulnerabilities of originators, recipients, and propagators of sexually explicit photographs. The choice of distributional values should be both relevant to the policy problem being addressed and also help distinguish among the alternatives considered.

As an illustration of selecting distributional values related to context, consider analysis of alternative allocation rules for deceased-donor kidneys. As noted in Chapter 1, the Organ Procurement and Transplantation Network (OPTN) implemented a new allocation system in 2014. In the ten preceding years, the OPTN committees considered a variety of alternatives to the system then in place. One approach involved allocating kidneys to patients to maximize life years following transplant (LYFT). Allocation based on LYFT would increase the efficiency of allocation as measured by the total lifespans of those receiving transplants. However, there was also concern about fairness in access to transplants for African Americans, and also for highly sensitized patients who were incompatible with large fractions of available kidneys.

Table 7.2 compares simulation results for the basic LYFT alternative, a LYFT alternative modified to give higher priority to highly sensitized patients, and the then-current allocation system. As expected, LYFT increased the total number of years of life after transplant. It also increased the percentage of kidneys going to African Americans and the highly sensitized as measured by a panel reactive antibody (PRA) score greater than or equal to 80, indicating less than one chance in five that the patient would be compatible with any donated kidney. The modified LYFT showed further small gains in racial and sensitized access, but at a cost of a smaller increase in the total number of years of life after transplant. Although it is quite common to consider the impact of public policies on racial or ethnic minorities, the consideration of transplant access for the highly sensitized is specific to kidney allocation.

Table 7.2 Comparison of Alternative Deceased-Donor Kidney Allocation Systems

	Selected Alternatives		
	Current System	*LYFT*	*Modified LYFT*
Total Life Span After Transplant (years)	107,865	122,140	118,133
Increase in Life Span (%)	-	13.2	9.5
African-American Recipients (%)	32.5	36.5	37.7
Sensitized (PRA 80+) (%)	13.1	13.4	14.0

Source: Abstracted from *Report of the OPTN/UNOS Kidney Transplantation Committee to the Board of Directors,* February 20–21, 2008, Exhibit A, Simulations 1, 18f, and 28.

Instrumental Values

The ultimate goal of public policy is to advance the substantive values that characterize a conception of the good society. Achieving political feasibility and operating within various resource constraints are important as *instrumental values*, things we desire not necessarily for their own sake but because they allow us to obtain policies that promote substantive values.

Political Feasibility

Public policies cannot directly contribute to substantive values unless they are adopted and successfully implemented. In Chapter 11, we consider in some detail how analysts can assess and increase political feasibility in designing policies. Here we simply outline some of the factors generally relevant to political feasibility that often are treated as values by policy analysts.

The distributional consequences of policies are of fundamental concern to participants in the political process. Public officials draw support from constituencies with strong interests. They tend to support policies favored by their constituencies and oppose policies not favored by them. This often leads to distributional considerations that do not necessarily correspond to substantive values. For example, representatives may be concerned with the quality of some category of public spending across the districts that they represent, even if less equal distributions would contribute more to such values as efficiency or a more equal distribution of wealth across households nationally. Consequently, even when distributional values do not seem substantively relevant, they nevertheless may be instrumentally relevant to political feasibility.

Revenues and Expenditures

Levels of public expenditure, usually determined through the budgetary process, have political relevance in several ways. First, raising public revenue is generally economically and politically costly, so that, other things equal, policies involving less direct public expenditure tend to receive greater general political support. Second, because the substantive effects of policies are often difficult to predict, expenditure levels often serve as proxies for the level of effort the government is making to solve specific social problems. So, for example, the level of federal expenditure on the "drug war" has symbolic value to many that only very roughly, if at all, corresponds to reductions in drug abuse and the negative externalities associated with it.

Some public decision makers, such as managers of public agencies, face strict budget constraints that they must satisfy in making decisions. For example, a legislature may allocate a fixed level of funding to an agency for launching a new program. Absent a request to the legislature for a supplement, policies for implementing the new program must meet the instrumental goal of costing no more than the allocated budget. The same reasoning would apply to specific categories of resources, such as personnel levels and borrowing authority, that are constrained from the perspective of the decision maker. Expending no more than the amounts of the resource available is instrumental to achieving the substantive values sought by the policy.

Conclusion

We have set out two broad classes of rationales for public policies: the correction of market failures to improve efficiency in the production and allocation of resources and goods, and the reallocation of opportunity and goods to achieve distributional and other values. We have devoted the majority of our attention to market failures, not because distributional values are any less important than efficiency but, rather, because the basic analytics of welfare economics serve as useful tools for positive prediction as well as normative evaluation. That is, the various market failures can often serve as models for understanding how unsatisfactory social conditions arise and how they might be remedied.

Market failures and unsatisfied distributional goals are necessary but not sufficient grounds for public intervention. We must always consider the costs of the proposed intervention. Just as markets fail in fairly predictable ways, we can identify generic government failures that either contribute indirectly to the costs of policy interventions or cause them to fail outright. We discuss these limits to public intervention in the next chapter.

For Discussion

1. In many public facilities, waiting lines are longer for women's than men's rest rooms. Is gender equity in this context important? How could it be operationalized?
2. Is there inherent social value in favoring small over larger firms?

Notes

1 If people are willing to pay for more equal distributions of wealth in society, then these amounts should be taken into account in assessing efficiency. In practice, however, we face the problem of reliably eliciting these amounts through surveys. See David L. Weimer and Aidan R. Vining, "An Agenda for Promoting and Improving the Use of CBA in Social Policy," in David L. Weimer and Aidan R. Vining, eds., *Investing in the Disadvantaged: Assessing the Benefits and Costs of Social Policies* (Washington, DC: Georgetown University Press, 2009), 249–71.

2 Jeremy Bentham, *An Introduction to the Principles of Morals and Legislation* (1789) (Buffalo, NY: Prometheus Press, 1988); John Stuart Mill, *Utilitarianism* (1861) in Alan Ryan, ed., *Utilitarianism and Other Essays: J. S. Mill and Jeremy Bentham* (New York: Penguin, 1987), 272–338.

3 John C. Harsanyi, "Morality and the Theory of Rational Behavior," *Social Research* 44(4) 1977, 623–56.

4 John Rawls, "Justice as Fairness: Political Not Metaphysical," *Philosophy & Public Affairs* 14(3) 1985, 223–50, at 227; John Rawls, *A Theory of Justice* (Cambridge, MA: Harvard University Press, 1971).

5 John Locke, *The Works of John Locke* (1700) (Westport, CT: Greenwood, 1989); Jean-Jacques Rousseau, *Of the Social Contract* (New York: Harper & Row, 1984).

6 See, for example, Harsanyi, "Morality and the Theory of Rational Behavior." Importantly, under some assumptions about the nature of preferences, the two rules become indistinguishable. See Menahem E. Yarri, "Rawls, Edgeworth, Shapley, Nash: Theories of Distributive Justice Re-Examined," *Journal of Economic Theory* 24(1) 1981, 1–39.

7 Norman Frohlich, Joe A. Oppenheimer, and Cheryl L. Eavey, "Choices of Principles of Distributive Justice in Experimental Groups," *American Journal of Political Science* 31(3) 1987, 606–36. We must be cautious, however, in interpreting these results because subjects' responses to redistributive questions are sensitive to the framing of questions and whether redistributive changes are seen as gains or losses. See Ed Bukszar and Jack L. Knetsch, "Fragile Redistribution Choices Behind a Veil of Ignorance," *Journal of Risk and Uncertainty* 14(1) 1997, 63–74.

8 For a brief overview, see Amartya Sen, *On Ethics and Economics* (New York: Basil Blackwell, 1987). For a more detailed discussion, see Robert Cooter and Peter Rappaport, "Were the Ordinalists Wrong about Welfare Economics?" *Journal of Economic Literature* 22(2) 1984, 507–30.

9 For an overview, see H. Peyton Young, *Equity: In Theory and Practice* (Princeton, NJ: Princeton University Press, 1994).

10 See Abram Bergson (as Burk), "A Reformulation of Certain Aspects of Welfare Economics," *Quarterly Journal of Economics* 52(2) 1938, 310–34.

11 This point has been raised as a fundamental criticism of Bergson's approach to social welfare functions. See Tibor Scitovsky, "The State of Welfare Economics," *American Economic Review* 51(3) 1951, 301–15. In the next chapter, we discuss this objection more generally as a fundamental problem inherent in democracy.

12 Colin Camerer, "Progress in Behavioral Game Theory," *Journal of Economic Perspectives* 11(4) 1997, 167–88, at 169.

13 Ibid.

14 See, for example, Geoffrey Brennan and James M. Buchannan, *The Reason of Rules: Constitutional Political Economy* (New York: Cambridge University Press, 1985).

15 For a discussion of institutional arrangements from the perspective of welfare economics, see Daniel W. Bromley, *Economic Interests and Institutions: The Conceptual Foundations of Public Policy* (New York: Basil Blackwell, 1989).

16 Some examples: Amitai Etzioni, *The Moral Dimension: Toward a New Economics* (New York: Free Press, 1988); James C. March and Johan P. Olsen, *Rediscovering Institutions:*

The Organizational Basis of Politics (New York: Free Press, 1989); and Adam Cureton, "Making Room for Rules," *Philosophical Studies* 172(3) 2015, 737–59.

17 On rule-utilitarianism, see Richard B. Brands, "Toward a Credible Form of Utilitarianism," in Michael D. Bayless, ed., *Contemporary Utilitarianism* (Garden City, NY: Anchor Books, 1968), 143–86. For a fuller development, see Russell Hardin, *Morality within the Limits of Reason* (Chicago: University of Chicago Press, 1988).

18 See John C. Harsanyi, "Rule Utilitarianism, Equality, and Justice," *Social Philosophy & Policy* 2(2) 1985, 115–27. Harsanyi argues that rule-utilitarianism provides a rational basis for deontological principles, such as rights and duties, by relating them to the maximization of social utility.

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20 Robert Haveman considers ways of redesigning the U.S. system of redistribution so that it is directed more toward equalizing opportunity. For example, he proposes the establishment of capital accounts that youths could use for education, training, and health services. Robert Haveman, *Starting Even: An Equal Opportunity Program to Combat the Nation's New Poverty* (New York: Simon & Schuster, 1988).

21 For an overview of the official and supplemental poverty lines, see Robert Haveman, Rebecca Blank, Robert Moffitt, Timothy Smeeding, and Geoffrey Wallace, "The War on Poverty: Measurement, Trends, and Policy," *Journal of Policy Analysis and Management* 34(3) 2015, 593–638.

22 For a demonstration of the application of various measures of vertical and horizontal equity, see Marcus C. Berliant and Robert P. Strauss, "The Horizontal and Vertical Equity

Characteristics of the Federal Individual Income Tax," in Martin David and Timothy Smeeding, eds., *Horizontal Equity, Uncertainty, and Economic Well-Being* (Chicago: University of Chicago Press, 1985), 179–211. See also David F. Bradford, ed., *Distributional Analysis of Tax Policy* (Washington, DC: AEI Press, 1995).

23 Thomas Piketty and Emmanuel Saez, "Inequality in the Long Run," *Science* 344(6186) 2014, 838–43. See also Kevin Bryan and Leonardo Martinez, "On the Evolution of Income Inequality in the United States," *Federal Reserve Bank of Richmond Economic Quarterly* 94(2) 2008, 97–120.

24 Ibid., 838.

25 Ibid., 842.

26 Lawrence Mishel and Alyssa Davis, "CEO Pay Continues to Rise as Typical Workers are Paid Less," Issue Brief No. 389, Economic Policy Institute, June 2014.

27 Gurupdesh Pandher and Russell Currie, "CEO Compensation: A Resource Advantage and Stakeholder Bargaining Perspective," *Strategic Management Journal* 34(1) 2013, 22–41.

28 Arthur M. Okun, *Equality and Efficiency*: *The Big Tradeoff* (Washington, DC: Brookings Institution, 1975), 91–100.

29 Charles L. Ballard, "The Marginal Efficiency Cost of Redistribution," *American Economic Review* 78(5) 1988, 1019–33; and Edgar K. Browning, "The Marginal Cost of Redistribution," *Public Finance Review* 21(1), 1993, 3–32.

30 Jared Carbone, Richard D. Morgenstern, Roberton C. Williams III, and Dallas Burtraw, *Deficit Reduction and Carbon Taxes: Budgetary, Economic, and Distributional Impacts* (Washington, DC: Center for Climate and Electricity Policy, Resources for the Future, August 2013).

31 For example, public support for the democratic regime in the Federal Republic of Germany has grown steadily since 1951; see Russell J. Dalton, *Politics in West Germany* (Boston: Scott, Foresman, 1989), pp. 104–106. Economic success under the regime surely contributed to its legitimacy. Probably the declining percentage of the population who lived through imposition of the regime at the end of the Second World War, and the growing percentage of the population that selected it through immigration, also contributed to growing legitimacy.

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36 For an overview, see Eugene Steuerle, "Wealth, Realized Income, and the Measure of Well-Being," in David and Smeeding, eds., *Horizontal Equity, Uncertainty, and Economic Well-Being*, 91–124.

37 Calculation based on Carmen DeNavas-Walt, Bernadette D. Proctor, and Jessica C. Smith, "Income, Poverty, and Health Insurance Coverage in the United States: 2011," U.S. Census Bureau, September 2012, Table 1; and Marina Vornovitsky, Alfred Gottschalck, and Adam Smith, *Distribution of Household Wealth in the U.S.: 2000 to 2011*, U.S. Census Bureau, July 2013, Table A1.

38 Alari Paulus, Holly Sutherland, and Panos Tsakloglou, "The Distributional Impact of In-Kind Public Benefits in European Countries," *Journal of Policy Analysis and Management* 29(2), 2010, 243–66.

39 Carmen DeNavas-Walt and Bernadette D. Proctor, "Income and Poverty in the United States: 2014," U.S. Census Bureau, September 2015, Table 1; and U.S. Census Bureau and U.S. Bureau of Labor Statistics, "Current Population Survey (CPS) Annual Social and Economic (ASEC) Supplement: 2015," Table PINC-01.

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William Alonso and Paul Starr, eds., *The Politics of Numbers* (New York: Russell Sage Foundation, 1987), 83–131.

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42 DeNavas-Walt and Proctor, "Income and Poverty in the United States: 2014," Table A2.

43 See George Horwich and David L. Weimer, *Oil Price Shocks, Market Response, and Contingency Planning* (Washington, DC: American Enterprise Institute, 1984), pp. 104–107.

8
Limits to Public Intervention

Government Failures

Every society produces and allocates goods through some combination of individual and collective choice. Most individual choice, expressed through participation in markets and other voluntary exchanges, furthers such social values as efficiency and liberty. But some individual choice, like that arising in situations we identify as market failures, detracts from social values in predictable ways. Collective choice exercised through government structures offers at least the possibility for correcting the perceived deficiencies of individual choice. Yet, just as individual choice sometimes fails to promote social values in desired and predictable ways, so, too, does collective choice. Public policy, therefore, should be informed not only by an understanding of market failure but of *government failure* as well.

The social sciences have yet to produce a theory of government failure as comprehensive or widely accepted as the theory of market failure. In fact, we must draw on several largely independent strains of research to piece one together.[1] From social choice theory, which focuses on the operation of voting rules and other mechanisms of collective choice, we learn of the inherent imperfectability of democracy. From a variety of fields in political science, we consider the problems of representative government. Public choice theory and studies of organizational behavior help us understand the difficulties of implementing collective decisions in decentralized systems of government and of using public agencies to produce and distribute goods.

Combined, the insights from these fields suggest that even governments blessed with the most intelligent, honest, and dedicated public officials will not promote the social good in all circumstances.

Policy analysts should exercise caution in advocating public intervention in private choice. Some minor market failures are simply too costly to correct; some distributional goals are too costly to achieve. More fundamentally, we cannot predict with certainty just how government interventions will work out. It is clear that government actions often fail to advance the social good, but the theory of government failure is neither as comprehensive nor as integrated as the theory of market failure. It can, though, raise important warnings about the general problems likely to be encountered in pursuing social values through public policies. Enthusiasm for perfecting society through public intervention, therefore, should be tempered by awareness that public intervention arises from imperfect collective choice and may require challenging implementation efforts.

We classify government failures into four general features of political systems: direct democracy, representative government, bureaucratic supply, and decentralized government. They have varying degrees of practical relevance for policy analysts. At one extreme are the characteristics of direct democracy, which simply warn analysts to be skeptical of claims that the results of elections and referenda provide unambiguous mandates for efficient policies. At the other extreme are the characteristics of bureaucratic supply and decentralized government, which directly help analysts anticipate the problems likely to be encountered during the implementation of public policies. An understanding of the characteristics of representative government can help analysts determine and improve the political feasibility of their preferred policy alternatives. Although we note some practical implications of particular government failure along the way, our primary purpose is to provide a conceptual framework useful for analysts who must operate in a politically complex world.

Problems Inherent in Direct Democracy

Over the course of history, societies have employed a variety of mechanisms for making social choices. Monarchies and dictatorships, in which the preferences of one or a small number of people dominate social choice, have given way in many countries to systems with broader bases of participation. Universal adult suffrage has come to be viewed as an essential element of democracy, which itself is an important social value in many national traditions. In democracies, voting serves as the mechanism for combining the preferences of individuals into social choices.

If voting were a perfect mechanism for aggregating individual preferences, then the job of the policy analyst would be much easier. Questions about the appropriate levels of public goods provision, redistribution, and public regulation of private activity could be answered either directly, through referenda, or indirectly, through the election of representatives who serve as surrogates for the members of society. The vast number of issues arising in a large and complex society, however, makes reliance on referenda impractical for important issues. Even as improvements in communication technology greatly reduce the logistical costs of voting, the time and other costs citizens face in learning about issues limit the attractiveness of direct referenda. Further, reliance on referenda for the revelation of social values suffers from a more fundamental problem: no method of voting is both fair and coherent.

The Paradox of Voting

Imagine that a school board holds referenda to determine the size of the budget for its schools. It proposes to base the budget on the results of pairwise voting among three possible budget alternatives: *Low* (no frills), *Medium* (similar to other public schools in the region), and *High* (best of everything). Suppose that three groups of voters can be identified according to their preferences over the three alternative budgets. "Moderates" are people who have children in school but also see property taxes as burdensome. Their first choice is *Medium* and their second choice is *High*; *Low* is their least preferred budget. "Fiscal Conservatives" think that the schools are wasting their tax dollars. They prefer *Low* to *Medium* and *Medium* to *High*. The third group, "Effective Schoolers," want their children to receive the best possible schooling. Therefore, they prefer *High*. If *High* is not selected, however, then they will send their children to private schools. Therefore, their second choice is *Low* and their third choice is *Medium*.

Table 8.1 summarizes the voters' policy choices under three different agendas. Agenda A offers the voters *High* versus *Low* in round 1. As both Moderates and Effective Schoolers favor *High* over *Low, High* wins with 65 percent of the vote. In round 2, *High*, as winner of round 1, now faces *Medium*. As Moderates and Fiscal Conservatives favor *Medium* over *High, Medium* wins with 80 percent of the vote and is, therefore, the selected alternative. Agenda B pits *Medium* versus *Low* in round 1. *Low* wins with 55 percent of the vote but loses against *High* in the second round. Consequently, *High* is the selected budget alternative. Finally, under agenda C, *Medium* defeats *High* in the round 1, but *Low* defeats *Medium* in round 2, so *Low* is the selected budget alternative. Thus, each agenda results in a different policy choice!

Table 8.1 An Illustration of the Paradox of Voting

	Preferences over School Budget Levels			
Voter Groups	First Choice	Second Choice	Third Choice	Percent of Voters
Moderates	Medium	High	Low	45
Fiscal Conservatives	Low	Medium	High	35
Effective Schoolers	High	Low	Medium	20

	Preferences over School Budget Levels			
Voter Groups	First Choice	Second Choice	Third Choice	Percent of Voters
Agenda A (Result: Medium)				
Round 1: Medium versus Low	High wins 65% to 35%			
Round 2: Medium versus High	Medium wins 80% to 20%			
Agenda B (Result: High)				
Round 1: Medium versus Low	Low wins 55% to 45%			
Round 2: Low versus High	High wins 65% to 35%			
Agenda C (Result: Low)				
Round 1: High versus Medium	Medium wins 80% to 20%			
Round 2: Medium versus Low	Low wins 55% to 45%			

The situation is even more complicated if we recognize the possibility that some people will be tempted to be opportunistic in voting. For example, consider how Fiscal Conservatives might try to manipulate the outcome under agenda B. If they have reasonable estimates of the percentages of voters who are Moderates and Effective Schoolers, then they can anticipate that *Low* will lose overwhelmingly to *High* in the second round. If, instead, *Medium* were paired with *High* in the second round, then *Medium* would win. The Fiscal Conservatives, who prefer *Medium* to *High*, might decide to vote for *Medium* in the first round over *Low*, even though they prefer *Low*. If the other voters continued to vote their true preferences, then *Medium* would be selected in the first round with a majority of 80 percent and defeat *High* in the second round with a majority of 80 percent. Consequently, by voting in this opportunistic manner, the Fiscal Conservatives would be able to avoid their least preferred alternative, *High*, which otherwise would have resulted. Because this opportunistic voting requires one to realize that a more favorable final outcome can sometimes be achieved by voting against one's true preferences in the preliminary rounds, political scientists usually refer to it as *sophisticated voting*.

Of course, other voters may also attempt to manipulate the outcome by voting against their true preferences in the first round. The final social choice would depend, therefore, not only on the agenda but also on the extent to which people engage in sophisticated voting. At the heart of this indeterminacy lies what is often referred to as the *paradox of voting*. It brings into question the common interpretation of voting outcomes as "the will of the people."

The French mathematician and philosopher Condorcet first raised the paradox of voting in the eighteenth century. Although later rediscovered by Charles Dodgson (Lewis Carroll) and others, its relevance to voting in democracies was not widely recognized until after the Second World War.[2] As long as the paradox of voting was viewed as a peculiar result of a particular voting scheme, scholars could treat it as a curiosity. But, in 1951, Kenneth Arrow proved that any voting rule that satisfies a basic set of fairness conditions could produce illogical results.[3]

Arrow's *general possibility theorem* applies to any rule for ranking alternatives in which a group of two or more persons must select a policy from among three or more alternatives. It assumes that any such scheme must satisfy at least the following conditions to be considered fair. First, each person is allowed to have any transitive preferences over the possible policy alternatives (*axiom of unrestricted domain*). *Transitivity* requires that if alternative 1 is preferred to alternative 2, and alternative 2 is preferred to alternative 3, then alternative 1 is preferred to alternative 3. Second, if one alternative is unanimously preferred to a second, then the rule for choice will not select the second (*axiom of Pareto choice*). Third, the ranking of any two alternatives should not depend on what other alternatives are available (*axiom of independence*). That is, if the group ranks alternative 1 above alternative 2 in the absence of alternative 3, then it should rank alternative 1 above alternative 2 when alternative 3 is also available. Fourth, the rule must not allow any one person's dictatorial power to impose his or her preferences regardless of the preferences of others (*axiom of nondictatorship*).

Arrow's theorem states that any fair rule for choice (one that satisfies the four axioms above) will fail to ensure a transitive social ordering of policy alternatives. In other words, cyclical (intransitive), or incoherent, social preferences like those appearing in the paradox of voting can arise from *any* fair voting system.[4] Also, any voting system that is designed to prevent intransitive orderings must violate at least one of the axioms.

What are the implications of Arrow's theorem for the interpretation of democracy? First, because cycles can arise with any fair voting scheme, those who control the agenda will have ample opportunity for manipulating the social choice. Referring back to Table 8.1, note that even though only 20 percent of the voters prefer *High* over the other alternatives, *High* will result if voting follows agenda B. More generally, it appears that, in a wide range of circumstances involving policy alternatives with more than one dimension (for example, choosing tax rates and deductions in a tax bill), once a cycle arises anywhere in the social ordering, a person with control of the agenda can select a series of pairwise votes to secure any of the alternatives as a final choice.[5] Thus, one would have to know how a final vote was reached before one could evaluate the extent to which it reflects the will of the majority.

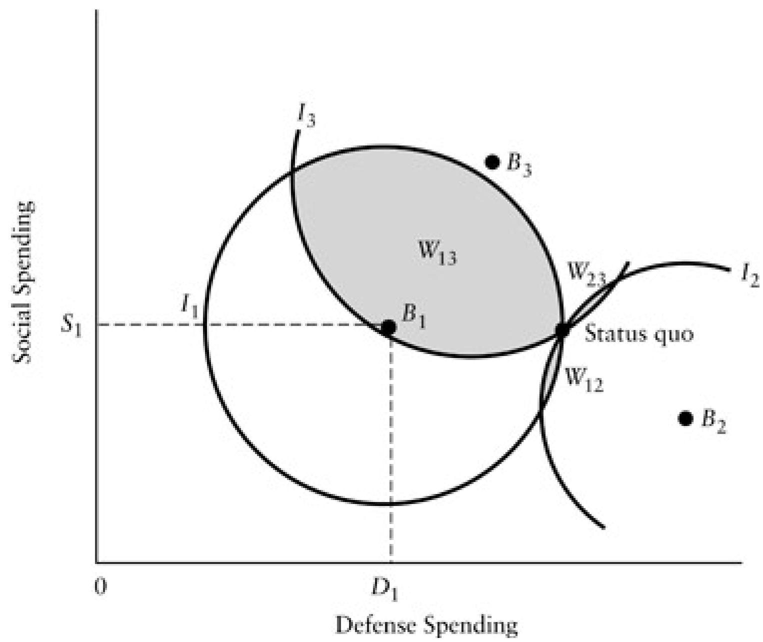


Figure 8.1 Agenda Control in a Two-Dimensional Policy Space

Figure 8.1 provides an illustration of agenda manipulation using a model that shows policy preferences for three legislators over alternative combinations of social and defense spending.[6] The points labeled B_1, B_2, and B_3 are the *ideal points*, or *bliss points*, for the three legislators. For example, legislator 1 most prefers that S_1 be spent on social programs and D_1 on national defense. If we assume that each legislator prefers points closer to his or her bliss point than farther away, then each will have circular indifference curves. The indifference curves shown in the figure pass through the status quo of S_{SQ} and D_{SQ}, and give all the combinations of spending that each legislator finds equally satisfactory to the status quo. Any point inside a voter's indifference curve will be closer to the bliss point than any point on the indifference curve and, therefore, preferred to the status quo. Note that the lens-shaped areas called *win sets* and labeled W_{13} and W_{23} represent points that would be favored by a majority of legislators (two out of three) over the status quo. Imagine that legislator 1 has agenda power in the sense that only she can propose an alternative to the status quo. If she proposes her bliss point, B_1, it will win a majority of votes because it is within W_{13}.

Consequently, by controlling the agenda she can obtain the spending alternative that she most prefers.

Second, the introduction of alternative policies that create cycles is often an attractive strategy for voters who would otherwise face an undesirable social choice. For instance, William Riker argues persuasively that the introduction of the status of slavery as an important national issue in the United States after 1819 can be interpreted as the ultimately successful method in a series of attempts by northern Whigs (later Republicans) to split the nationally strong Democrat Party.[7] Riker presents a fundamental generalization: persistent losers have an incentive to constantly introduce new issues in an attempt to create voting cycles that offer opportunities to defeat dominant coalitions during the resulting periods of disequilibrium. Therefore, we should expect voting cycles to be more frequent than if they simply resulted from chance combinations of individual preferences. In other words, political disequilibria, and the attendant problem of interpreting the meaning of social choices, are especially likely when the stakes are high.

Preference Intensity and Bundling

Imagine a society that decides every public policy question by referendum. If people vote according to their true preferences on each issue, then social choices may result that are both Pareto inefficient and distributionally inequitable. For example, consider a proposal to build an express highway through a populated area. If we were able to elicit truthful answers from everyone about how much they would value the road, then we might find that a majority would each be willing to pay a small amount to have the road built, while a minority, perhaps those living along the proposed route, would require a large amount of compensation to be made as well off after the construction of the road as before. If the total amount the majority would be willing to pay falls short of the total amount needed to compensate the minority, then the project is neither Pareto efficient, nor even potentially Pareto efficient. Furthermore, most would view the project as inequitable because it concentrates high costs on a minority. Nevertheless, the project would be adopted in a majority rule referendum if everyone voted his or her true preferences.

Of course, not everyone in the majority will necessarily vote according to strictly private interests. Some may vote against the project out of a sense of fairness. Others may foresightedly fear setting a precedent that might be used to justify unfair policies when they are in the minority. Absent these constraints, however, direct democracy can lead to *tyranny by the majority*, whereby either a permanent majority consistently inflicts costs on a permanent minority or a temporary majority opportunistically inflicts very high costs on a temporary minority.

Majorities may also inadvertently inflict high costs on minorities because voting schemes do not allow people to express the intensity of their preferences. No matter how much someone dislikes a proposed project, he or she gets to cast only one vote against it. Even if those in the majority wish to take the interests of the minority into consideration, they have no guarantee that the minority will be truthful in revealing the intensity of their preferences. In fact, those in the minority would have an incentive to overstate the intensity of their dislikes, perhaps checked only by the need to maintain some level of credibility. We can contrast this characteristic of collective choice with markets for private goods that allow people to express the intensity of their preferences by their willingness to purchase various quantities at various prices.

The danger of tyranny by the majority makes democracy by referenda generally undesirable; the complexity of modern public policy makes it impractical. Perhaps the closest a society could come to direct democracy would be the election of an executive who would serve subject to recall. The executive would at least have some opportunity to fashion policies that take into consideration strong preferences of minorities, and would have an incentive to do so if the composition of minorities with strong preferences changed with issues. Otherwise, two or more minorities with strong preferences might form a new majority in favor of recall.

Candidate executives would stand for office on platforms consisting of positions on important policy issues. Voters would have to evaluate the bundle of positions offered by each candidate. Different voters may vote for the same candidate for different reasons. In fact, a candidate holding the minority position on every important issue may still win the election. For example, free-traders consisting of one-third of the electorate may vote for a candidate because of her position on trade policy, which they believe to be most salient, even if they dislike her positions on all the other issues. Voters consisting of another one-third may dislike her position on trade policy but vote for her because of her position on defense spending, which they view as the most important issue. Thus, she may win the election even though a majority of voters opposes her positions on all issues, including trade and defense policies. The general implication for democratic systems is that whenever people must vote on a bundle of policies, it is not necessarily the case that any particular policy in the winning bundle represents the will of a majority. Even a landslide victory may not represent a "mandate from the people" for the winner's proposed policies!

Democracy as a Check on Power

The paradox of voting, the possibility of minorities with intense preferences, and the problem of bundling all show the imperfection, and imperfectability, of democracy as a mechanism of social choice. Sometimes consistent social values do not exist; other times voting does not discover them.

Despite these inherent problems, direct democracy offers several advantages. The opportunity for participation encourages citizens to learn about public affairs. Actual participation may make citizens more willing to accept social choices that they opposed because they had an opportunity to be heard and to vote. Indeed, referenda may provide a means of putting divisive political issues at least temporarily to rest. For example, the 1980 and 1995 Quebec referenda on sovereignty defused, for at least a number of years, the issue of Quebec independence. The 1975 referendum in the United Kingdom on the question of membership in the European Economic Community (the European Union) appeared to settle the issue in favor of continued participation despite strong opposition.[8]

The use of referenda in primarily representative systems has an additional consequence: it legitimizes more use of referenda. The 1980 Quebec referendum certainly contributed to the demand for the 1995 referendum. The 1975 vote on EU entry helped legitimize demands for the 2016 "Brexit" vote. The result of the Brexit vote certainly legitimizes demands for another vote on Scottish independence. Legitimacy is like a genie let out of the bottle: once out, it is hard to recork!

It is another feature of democracy, however, that makes it preferable to other systems of social choice: it provides a great check on the abuse of power by giving the electorate the opportunity to overturn onerous policies and remove unpopular decision makers. It is this opportunity to "throw the rascals out" that fundamentally gives democracy its intrinsic social value. Democracy does not always lead to good, let alone the best, policies, but it consistently provides an opportunity to correct the worst mistakes. Therefore, it performs an essential role in the preservation of liberty. It may deny us the full benefits of a truly benevolent and wise government, but it helps protect us from the harm of one that is either evil or foolish.

Problems Inherent in Representative Government

In modern democracies, representatives of the electorate actually make and execute public policy. Although the particular constitutional arrangements vary considerably across countries, in most democratic systems voters choose representatives to legislatures. They sometimes choose executives and judges as well. The legislature typically plays a dominant role in establishing public policy by passing laws, but also often plays an administrative role by providing ministers (in parliamentary systems) and by monitoring budgets and overseeing government operations. The chief executive, usually the prime minister or the president, exercises administrative responsibility, including the appointment of high-ranking officials in government agencies. Members of the executive branch make public policy when they interpret legislation, as, for example, when agency heads issue rules under broadly delegated authority to regulate some aspect of commerce. In addition, they may make proposals that influence the legislative agenda. The members of the judicial branch also interpret laws as well as determine their constitutionality in some political systems. These legislators, executives, and judges, often referred to as "public servants," represent the rest of society in the numerous government decisions that constitute public policy.

Representatives often face the dilemma of choosing between actions that advance their conception of the good society and actions that reflect the preferences of their constituencies. Political philosophers distinguish between the roles of the representative as a *trustee* (who should act on behalf of what he believes to be his constituency's interest) and as a *delegate* (who should act in accordance with the desires of a majority of his constituency). They also distinguish between representation of a local constituency and the entire nation.[9] For example, a legislator may believe that construction of a safe nuclear waste facility is socially desirable and that the best location for society as a whole would be in her district (trustee perspective), but her constituents may oppose it (delegate perspective). On the one hand, one might approve her support of the facility as consistent with increasing aggregate social welfare. On the other hand, one might approve of her opposition as consistent with protecting a segment of society from bearing disproportionate costs of collective action. The very fact that there is no clear-cut way of resolving this sort of dilemma suggests the difficulty society faces in evaluating the behavior of representatives.

Three factors greatly influence the way representatives actually behave. First, representatives have their own private interests. It is naive to believe that they do not consider their own well-being and that of their families.

Elected representatives, motivated perhaps by the status and perquisites of office, usually seek either reelection or election to higher office. Candidates often behave as if they were trying to maximize the percentage of votes they would receive. This strategy, which maximizes the probability that they will actually gain a majority, requires them to pay more attention to the interests of citizens who are likely to vote. Public executives, elected and appointed, similarly seek career security and advancement, as well as circumstances that make it easier for them to manage their agencies. In general, the personal interests of representatives tend to push them toward responsiveness to their constituencies and away from concern for broader social welfare. Legislators tend to emphasize benefits to their districts over social costs; public executives tend to value resources for use by their agencies beyond their contribution to social benefits.

Self-interest also raises the issue of the relationship between the voting behavior of the individual representatives and financial (or in-kind) campaign contributions by interest groups. If financial contributions alter representatives' voting patterns, then most citizens and ethicists would regard such behavior as a source of government failure (although a few have argued that, as long as the purchased votes are bought by constituents, this simply represents the expression of intense preferences). Once vote buying becomes endemic, it will have negative consequences for aggregate (especially dynamic) efficiency.[10] Political corruption threatens confidence in property rights, leading to costly adaptive behavior by citizens. More fundamentally, corruption threatens the legitimacy of all institutions of the state. Yet, determining the actual impact of financial contributions on political behavior is problematic because it is difficult to separate from an alternative, more benign hypothesis: financial contributors support like-minded politicians. Both hypotheses are consistent with a positive correlation between financial contributions and voting behavior.

Second, individuals must incur costs to monitor the behavior of their representatives. Facing financial and time constraints, people usually do not find it in their private interests to articulate policy preferences or to monitor closely the actions of their representatives, exhibiting "rational ignorance."

The broader the scope of government, the more costly is comprehensive articulation and monitoring by citizens. Those who do monitor typically have strong policy preferences, based on either ideology or financial interests. Consequently, representatives tend to be most closely evaluated by groups that may have preferences very different from their broader constituencies. Thus, these *interest groups* enjoy more influence with representatives than they would in a world with perfect information and costless monitoring.

Third, party discipline may constrain the self-interested behavior of individual representatives. Both in terms of the federal and state legislatures, the United States was generally considered to have weak party discipline. Partisanship has increased in the Congress in recent years, however. While about 81 percent of members in each chamber voted with the majority of their party in the 102nd Congress, 89 percent in the House and 84 percent in the Senate did so in the 110th Congress.[11] In many other countries, especially those with parliamentary systems, party discipline is generally strong. This follows from a number of institutional features: centralized control of campaign funds, centralized control of nomination procedures, and the centralizing character of cabinet government in parliamentary systems.

In the following sections, we explore some of the implications of self-interested representatives who are not fully monitored by their constituencies. These implications should be thought of as general tendencies. Not all representatives act in their own narrow self-interests all of the time. Some have strong policy preferences that can lead them to take stands that are politically costly. (Of course, in so doing they run an increased risk of being turned out of office.) Analysts can sometimes make common cause with such representatives on the basis of principle. More generally, however, analysts find allies among representatives whose private interests happen to lead them to what the analysts believe to be good policies.

Rent Seeking: Diffuse and Concentrated Interests

In a world where most people pay little attention to their representatives, the politically active few have an opportunity to wield influence disproportionate to their number. By providing information, activists may be able to persuade representatives to support their positions and advocate for them more effectively in the political arena. By promising to help secure the votes of constituency groups and by providing campaign funds, they may be able to alter the way representatives concerned with reelection view the social good, at least as revealed by the representatives' policy choices.

Who is likely to become politically active? Undoubtedly, some people have a strong sense of duty to promote the social good and, therefore, feel obliged to express their views on public policy issues whether they stand to gain or lose personally. In general, however, private self-interest plays an important role in motivating political participation. If we believe that most people are economically rational, then the greater the expected net benefits one expects to reap from some political activity, the more likely that one will undertake the activity. Policies that would spread large aggregate benefits widely and uniformly among the electorate often do not elicit active political support because, for any individual, the costs of political activity exceed the expected benefits (which are the individual's private gains if the policy is adopted weighted by the probability that political action will result in adoption). Similarly, no individuals may find it in their own self-interests to protest policies that spread costs widely. In contrast, at least some people find it in their self-interest to become politically active when policies involve concentrated costs or benefits. Assuming that representatives respond at least somewhat to political activity, the consequence of individual rationality will be collective choices biased toward policies with concentrated benefits and away from policies with concentrated costs. As a result of these differences in the concentration of benefits or costs, James Q. Wilson proposes four categories of political competition: (1) *interest group* politics, where both costs and benefits are concentrated; (2) *entrepreneurial* politics, where benefits are widely diffused but costs are concentrated; (3) *client* politics, where benefits are concentrated but costs are widely diffused; and (4) *majoritarian* politics, where both benefits and costs are widely diffused.[12]

The bias created by concentrated benefits and diffuse costs, Wilson's client politics, fosters the adoption of policies for which total costs exceed total benefits. Concentrated economic benefits and diffuse economic costs often arise when governments intervene in markets. The interventions generally create economic benefits in the form of rents—payments to owners of resources above those that the resources could command in any alternative use. Lobbying for such interventions is called *rent seeking*.[13]

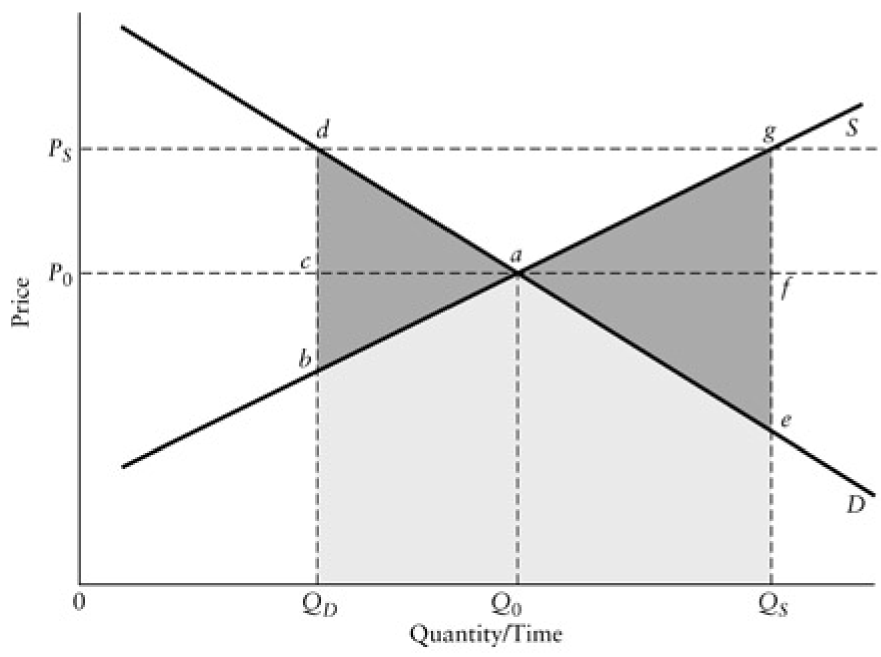
The effort to use government to restrict competition is perhaps the oldest form of rent seeking. (Securing a legal restriction on competition would allow capture of the monopoly rent shown in Figure 4.6.) In earlier times, the king generated revenue by sharing the rents that resulted from monopolies given to favored subjects. In more modern times, bribery by those seeking local monopolies, such as cable television franchises, has contributed to the corruption of local public officials. Professions often attempt to keep wages high by restricting entry through government-sanctioned licensing. If restricted entry does not provide offsetting benefits, such as reductions in information asymmetry (as described in Chapter 5), then the members of the profession gain rents at the expense of consumers. Domestic manufacturers often lobby for tariffs and quotas on foreign imports so that they can sell their products at higher prices. By limiting foreign competition, tariffs and quotas extract rents for domestic manufacturers (and, ironically, for foreign producers) from domestic consumers.

Individual firms within an industry may seek regulation as a way of limiting competition.[14] For instance, each of the three major laws that expanded federal regulation of the U.S. pharmaceutical industry were supported by firms that believed they would be able to meet standards more easily than their competitors.[15] Even when an industry initially opposes regulation, its concentrated interests give it an incentive to lobby the regulators after its imposition; the result may be "capture." For example, the U.S. Interstate Commerce Commission (ICC) was originally created largely in response to anticompetitive practices by the railroads. By the 1920s, however, the ICC so identified with the interests of the railroads that it tried to protect them from the growing competition from trucking.[16]

Sometimes governments generate rents for producers by directly setting prices in markets. For example, many countries set price floors on certain agricultural products like wheat, milk, and honey. Figure 8.2 illustrates the rent transfers and deadweight losses that result. Without a price floor, quantity Q_0 will be supplied and demanded as market clearing price P_0. At an imposed floor price of P_S the quantity demanded falls to Q_D. Those who are able to sell their output extract a rent from consumers equal to the area of rectangle P_SP_0cd. A deadweight loss equal to the area of triangle *abd* also results from the losses in consumer and producer surplus associated with lower consumption levels. Of course, at floor price P_S, farmers will want to supply Q_S. To maintain the price floor, therefore, the government must take the excess supply off the market. If it does so by allocating production quotas to the farmers with the lowest marginal cost of production, then the social surplus loss just equals the deadweight loss, shaded area *abd*. If it purchases and distributes the difference between Q_S and Q_D efficiently to those consumers who value consumption less than the floor price, then the social surplus loss equals the shaded area *aeg*. If the government purchases the excess and destroys it, then the social surplus loss equals the large shaded area Q_DdagQ_S.

The magnitudes of the rent transfers and efficiency losses from price supports in the United States have been at times extremely large. The U.S. price support programs for wheat, corn, cotton, peanuts, and dairy products imposed billions of dollars of annual cost. Gordon Rausser reports that more than 90 percent of sugar, milk, and rice subsidies were predatory government policies (transfers from other sectors), as opposed to productive policies (reductions in transaction costs, provision of information, research, and the like).[17] A more recent study estimates that the quota on imported sugar causes social surplus losses of approximately $500 million per year.[18]

Programs like price supports, however, often produce only temporary rent transfers. If farmers believe that the government will permanently guarantee a price above the market clearing level, then the price of land will rise to completely reflect the higher value of its output. Specifically, the price of land will reflect the present value of future rents. A farmer who sells land after the introduction of the price supports, therefore, captures all the rent. The buyer, who must service the debt on the purchase amount, does not realize any profits above the normal rate of return. In effect, as the "original" farmers extract their shares of the rent by selling their land, the supply curve shifts up to the level of the support price. Attempts to regain efficiency and reduce government expenditures through elimination of the price supports will force many of the current owners into bankruptcy. This so-called *transitional gains trap* is a common occurrence that is by no means limited to agriculture.[19]



Price supports alone "low-cost producers"

Surplus transfer from consumers to producers:	P_SP_0cd
Deadweight loss:	abd

Price supports with government purchases

Surplus transfer from consumers and government to producers:	P_SP_0ag
Surplus loss from overconsumption with efficient distribution:	aeg
Surplus loss from overproduction with destruction:	Q_DdagQ_S

Figure 8.2 Surplus Transfers and Deadweight Losses under Price Supports

Even the initial beneficiaries of the market intervention may fail to realize the full rents because of the costs of the rent-seeking activity. Direct lobbying and campaign contributions can be costly, especially if they must be spread over a large number of representatives to secure adoption of the intervention. Most or all of the rents may be dissipated, leaving the rent seekers no better off than they were in the absence of the intervention. In the extreme, expenditures of resources by competing rent seekers may dissipate more than the rents at stake.

Rents can be realized directly from government as well as through the marketplace. Outright grants can sometimes be secured if a plausible public rationale can be offered. Because they tend to be more hidden from public view, exemptions from regulations and taxes are often more attractive targets for rent seekers. Sometimes tax loopholes provide large benefits to small numbers of firms. For example, the transition rules of the 1986 U.S. tax reform law created about $10 billion in loopholes, some written with restrictions so specific that they benefited single companies.[20]

Despite the advantages concentrated interests enjoy in mobilizing for political activity, they do not always prevail over diffuse interests. If those with similar interests are already organized, then they may be able to use the existing organizational structure to overcome the free-riding problem that would keep them from expressing their interests individually. For instance, many people primarily join the American Association of Retired Persons (AARP) to gain access to private goods such as supplementary health insurance and senior discounts. Although very few members would find it in their personal interests to lobby on their own, many of them also support expenditures by the AARP for lobbying to protect current social security and Medicare entitlements. In effect, organizations like the AARP provide public goods for their members who join primarily to obtain private benefits.

Diffuse interests may also enjoy access to representatives by virtue of their distribution. For example, the National Education Association appears to enjoy great success in lobbying, not only because it has a large, well-educated, and politically active membership, but also because its membership is spread fairly evenly over congressional districts.[21] Similarly, organizations of local governments are often effective at the national level because each member generally has access at least to its own representative. The extent to which such diffuse interests can become politically effective usually depends on the existence of an organization to mobilize the contributions of individual members.

Even without the advantage of prior organization, however, diffuse interests can sometimes overcome highly concentrated interests—entrepreneurial politics in Wilson's typology. Experience suggests that three factors facilitate the success of diffuse interests: attention to the policy issue from a large segment of the electorate, low public trust in the concentrated interests, and political or *policy entrepreneurs* willing to promote and represent the diffuse interests. These factors often hold in situations in which the prices of key goods rise dramatically. For example, rapidly rising rental prices often attract widespread attention in communities with low rates of homeownership and generate animosity toward landlords who appear to be profiting at the expense of tenants. Representatives may respond with rent control programs that provide modest short-run savings to current tenants at the expense of landlords and potential renters. The regulation of petroleum prices in Canada and the United States during the 1970s also shows diffuse interests dominating concentrated ones. The quadrupling of world oil prices during the Arab oil embargo in the 1974 crisis raised widespread suspicion that oil companies were profiteering. Each country instituted extensive regulation of its petroleum industry, including price ceilings on petroleum products.

It is worth noting that policy entrepreneurs who successfully organize diffuse interests may enjoy concentrated private benefits. They may draw salaries as staff members and enjoy access to the political process. These entrepreneurs thus face incentives to search for issues that will enable them to maintain support for their organizations.

Mobilizations of diffuse interests that simply impede rent seeking of concentrated interests probably advance the social good. Sometimes, however, they can also facilitate rent seeking. When they result in price ceilings, for instance, they generally reduce economic efficiency and rarely lead to greater equity. Figure 8.3 illustrates these points. The price ceiling (P_C) lies below the unrestrained market price (P_0). At price P_C, consumers demand Q_D but producers supply only Q_S. In the "best case," consumers gain a surplus equal to the area of rectangle P_0P_Cbc, which corresponds to the producers' loss of rents on the Q_S units they sell; the social cost of the transfer, however, is a deadweight loss of social surplus from reduced consumption equal to the area of triangle *abc*.

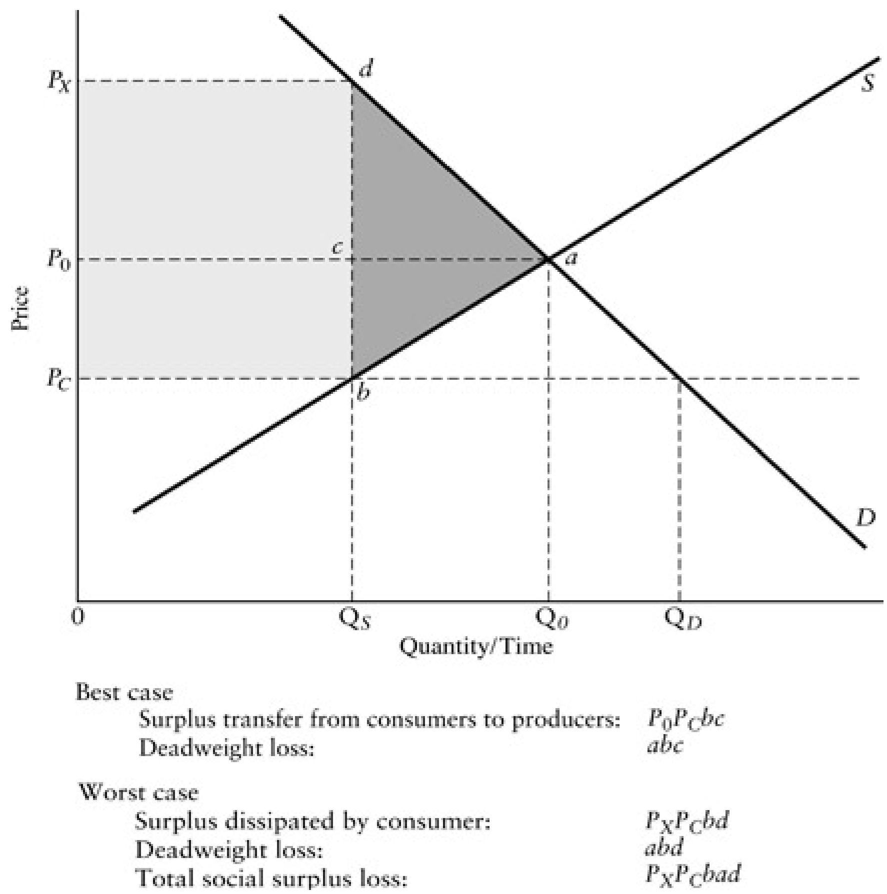


Figure 8.3 Surplus Transfer and Deadweight Losses under Price Ceilings

In the "worst case," consumers dissipate the surplus they gain from producers as well as surplus equal to the area of rectangle P_XP_0cd. When supply is limited to Q_S, consumers value the last unit available at P_X. Therefore, consumers will be willing to pay up to $P_X - P_C$ in nonmonetary costs, such as queuing, searching, and perhaps even bribing, to obtain shares of the limited supply. These costs have the effect of dissipating the benefits consumers enjoy from the lower monetary cost.

In order to achieve the "best case," an efficient rationing scheme is needed. For example, the distribution of coupons entitling consumers to shares of the available supply, coupled with an efficient "white market" for buying and selling coupons, could secure the surplus transfer for consumers without substantial dissipation. Coupon rationing can be administratively costly, however, because it requires creation and distribution of an entirely new currency. In addition, in most markets rationing schemes become obsolete because the shortages induced by ceiling prices eventually disappear as producers reduce product quality and consumers shift their demand to other markets.[22]

Advocates of price ceilings usually defend their positions on equity grounds. They argue that the poor have a better chance at obtaining a "fair share" of the good if allocation is not done solely on the basis of price. However, the mechanisms of non-price allocation do not always favor the poor. Queuing for gasoline, for instance, involves less hardship for two-adult families with flexible professional work hours than for single parents working fixed hours. More important, price ceilings tend to increase a number of big losers. For example, by reducing economic efficiency, price ceilings on petroleum products contribute to higher levels of unemployment and lower wages during oil price shocks. Because the identities of the big losers are generally not revealed until after implementation of the price ceilings, their interests are rarely taken into account at the time of adoption––the "silent losers" problem discussed in Chapter 7.

The accumulation of rent-seeking organizations can have adverse effects on economic growth. Mancur Olson influentially argued that in stable societies rent-seeking "distributional coalitions" shift resources toward cartel activity and lobbying and away from production.[23] Aside from this direct effect on economic efficiency, attempts by coalitions to protect their rents reduce the capacities of societies to adopt new technologies and reallocate resources to meet changing conditions. Consequently, in Olson's view, the greater the accumulation of distributional coalitions, the slower will be the rate of economic growth.

Problems of Geographic Representation: The District-Based Legislature

Legislatures rely on voting to reach collective decisions. As we have already discussed, no method of voting is both fair and consistent. Procedural rules concerning such things as agenda setting and permissible amendments can prevent cyclical voting, but not opportunities for sophisticated voting and agenda manipulation.[24] An additional collective-choice problem arises in legislatures because the members usually represent constituencies with heterogeneous preferences. Specifically, under majority voting, certain distributions of preferences can result in social choices opposed by a majority of the total electorate.[25]

For example, consider a legislature made up of 100 members, each representing the same number of constituents. Assume that representatives vote according to the preferences of a majority of their constituents. Imagine that 51 percent of the voters in 51 of the constituencies favor a particular policy, while none of the voters in the other 49 favor it. Majority voting in the legislature would result in adoption of the policy, even though it is favored by only a little more than one-quarter of the total electorate.

A much more important social choice problem arises from the efforts of representatives to serve the narrow interests of their constituencies. In most countries legislators represent geographically defined districts. Although they may sincerely wish to do what is in the best interests of the entire country, their self-interest in terms of reelection encourages them to pay special attention to the interests of their districts. This often leads them to policy choices that do not contribute to aggregate national welfare.

The problem goes well beyond legislators emphasizing the social benefits that accrue to their own constituencies in deciding how to vote. Certain social costs sometimes appear as benefits to districts.[26] For example, a cost–benefit analysis of a weapons system from the social perspective would count expenditures on components as costs. A legislator, however, might very well count expenditures on components manufactured in her own district as benefits because they contribute to the vitality of the local economy. What from the social perspective might be a cost of, say, $10 million might be viewed by the legislator as a benefit of $5 million. For example, observers have attributed the ultimate political success of the U.S. B-1 bomber program partly to the fact that major subcontractors were located in a large number of congressional districts.[27]

District-oriented valuation of expenditures often leads to adoption of policies with net social costs as legislators bargain with each other to get their share from the *pork barrel*.[28] This process, sometimes called *logrolling*, involves assembling a collection of projects that provide sufficient locally perceived benefits to gain adoption of the packages as a whole.[29] When coupled with representatives' unwillingness to bear the political costs of raising taxes, logrolling for district "pork" contributes to deficit spending.

The legislative process generally favors policies that spread perceived benefits over a majority of districts. Although this tendency may contribute to regional equality, it also makes targeting and policy experimentation difficult. Some programs produce net social benefits only in special circumstances and limited scope. For instance, one or two centers for the study of educational innovation might offer positive expected net benefits; however, because skilled researchers and good ideas are scarce resources, 20 such centers offer net social costs. Nevertheless, legislative approval might be easier to obtain for the more dispersed program because it provides expenditures in more districts. The U.S. Model Cities program fitted this pattern.[30] Originally proposed by President Johnson to demonstrate the effects of concentrated and coordinated federal aid on a small number of cities, over time it was expanded to include a large number of small- and medium-sized cities to gain the support of representatives of rural and suburban districts. As a result, resources were widely disbursed so that the assumed effects of concentration could not be determined.

Shortened Time Horizons: Electoral Cycles

Representatives must often make decisions that will have consequences extending many years into the future. From the perspective of economic efficiency, the representatives should select policies for which the present value of benefits exceeds the present value of costs. In making the comparison, each representative should apply the same social discount rate to the streams of benefits as to costs. Self-interest operating in an environment of imperfect monitoring, however, increases incentives for representatives to discount heavily costs and benefits that will not occur in the short run. For electoral purposes, it is especially tempting for representatives to emphasize immediate benefits and de-emphasize future costs.

Consider a representative who must stand for reelection in, say, two years. Assuming her constituency does not fully monitor her behavior, she faces the problem of convincing the electorate that her actions have contributed to their well-being. She knows that it is easier to claim credit for effects that have actually occurred than for ones expected to occur in the future. Now, imagine choosing between two projects with different benefit and cost profiles. Project A offers large and visible net benefits over the next two years but large net costs in subsequent years, so that, overall, it offers a negative net present value. Project B incurs net costs over the next two years but growing net benefits in subsequent years so that overall it offers a positive net present value. From the social perspective, the representative should select project B; from the perspective of her self-interest, however, she can enhance her chances for reelection by selecting project A and claiming credit for the benefits that are realized prior to the election.

Under what circumstances is such myopic judgment most likely to occur? One factor is representatives' perception of their reelection chances. The more vulnerable representatives feel, the more likely they are to select projects with immediate and visible benefits for which they can claim credit. Those representatives either not standing for reelection or running in "safe" districts are likely to place less value on the short-term political gains and, therefore, are more likely to act as "statespersons" with respect to the time horizon they use in evaluating projects. Another factor is the ease with which an opponent can draw the attention of the electorate to the yet-unrealized future costs. The more easily an opponent can alert the electorate to these costs, the less likely is the incumbent to over-discount the future.

To what extent does the myopia induced by the electoral cycle actually influence public policy? The common wisdom that legislatures rarely increase taxes in the year before an election, for instance, appears to be empirically verified by the pattern of tax changes observed across states within the United States.[31] The U.S. picture with respect to macroeconomic policy at the national level is less clear, suggesting that electoral cycles interact importantly with the partisan preferences of incumbents.[32] These latter findings raise at least some concerns about the ability of national governments to manage macroeconomic activity efficiently, even when reliable policy tools are available.

Posturing to Public Attention: Public Agendas, Sunk Costs, and Precedent

Candidates for public office must compete for the attention of the electorate. Most of us focus our attention on our families, careers, and other aspects of our private lives. While we care about public policy, especially when it directly affects us, we generally do not devote great amounts of our time to learning about issues and about how our representatives handle them. Because gathering and evaluating detailed information on public affairs is costly, we usually rely on newspapers, magazines, radio, and television to do it for us. We should not be surprised, therefore, that representatives consider how their actions and positions will be portrayed in the mass media.

The media offer opportunities for representatives to reach the public. When the news media draw public attention to some undesirable social condition like drug abuse, representatives may be able to share the limelight by proposing changes in public policy. These attempts to convert undesirable and highly visible conditions to public policy problems amendable at least to apparent solution help determine the policy agenda. As newly "discovered" conditions push older ones from the media, the agenda changes. Representatives and analysts who prefer specific policies may have to wait for an appropriate condition to arise to gain a *policy window* into the agenda.[33] For example, advocates of urban mass transit in the United States enjoyed a policy window initially when the media reflected public concern about traffic congestion, another when the environmental movement drew attention to pollution, and a third when the energy crisis raised concern about fuel conservation.[34] They made some progress in gaining attention for their preferred solution when each of these windows opened.

Candidates for elected office view media coverage as so important that they are usually willing, if financially able, to pay large sums for political advertising. In order to be financially viable, U.S. candidates typically spend large amounts of time and effort soliciting campaign contributions from those who share similar political views, those who wish to ensure future access to elected candidates, and those who seek to influence decisions on specific policies. In open seats for the House of Representatives, for example, contributions appear to affect the probability of winning.[35] Campaign finance reforms intended to reduce the role of campaign contributions in politics often have undesirable and unintended consequences. One undesirable consequence is that they may make it harder for challengers to compete against incumbents who use their positions to remain in the public eye. One unintended consequence is the diversion of contributions from the campaigns of specific candidates to any uncovered categories, such as contributions to parties (so-called soft money contributions), and independent spending that indirectly supports or attacks candidates.

A policy agenda strongly influenced by the pattern of media coverage and political advertising is unlikely to be consistent with the concept of public policy as a rational search for ways to improve social welfare. The policy agenda only reflects appropriate priorities if the media happen to cover conditions in proportion to their worthiness as public policy problems, a dubious assumption because undesirable social conditions, no matter how impervious to correction by public policy, may still make good stories. Street crime, for instance, depends most heavily on local rather than national law enforcement policies. Nevertheless, it may elicit a national response when it commands national media attention, as is the case in the United States repeatedly since the late 1960s.[36]

A media-driven policy agenda discourages the careful evaluation of alternatives: representatives who take the lead in proposing policy responses are most likely to benefit from the wave of media coverage. Thus, they face an incentive to propose something before the crest passes. If they wait too long, then the policy window may close. Furthermore, adoption of any policy response may hasten the decline in media coverage by giving the appearance that something has been done about the problem. Consequently, more comprehensive and, perhaps, better-analyzed policy responses may not have time to surface. Even if they are held in reserve for the next policy window, they may no longer be the most desirable in light of new conditions.

Whether or not representatives are attempting to exploit policy windows, they seek to put their actions and positions in a favorable light.[37] Their previous positions often constrain their current options. For example, politicians and economists often view *sunk costs* differently. To an economist, resources committed to a project in the past and no longer available for other uses are “sunk” and irrelevant when deciding to either continue or terminate the project. Politicians who advocated the project initially, however, might be loath to terminate it for fear that opponents will point to abandonment as an admission of a mistake. They may even use the earlier expenditures as a justification for continuation: “If we stop now, then we will have wasted the millions of dollars that we have already spent.” While such statements are true, and perhaps politically significant, they are irrelevant from the social perspective when deciding whether additional resources should be invested in the project.

Precedents often provide opportunities for representatives to gain favorable public reaction to their proposals. By pointing to apparently similar policies adopted in the past, representatives can appeal to people’s sense of fairness: “We gave a loan guarantee to firm *X* last year. Isn’t it only fair that we do the same for firms *Y* and *Z* now?” Without close examination, it may not be clear that the circumstances that made the guarantee to firm *X* perhaps good policy do not hold for firms *Y* and *Z*. Policy analysts should anticipate the possibility that any new policies they recommend may serve as precedents for actions in the future. A subsidy, a waiver, or another action that, when considered in isolation, appears socially desirable, may not actually be so if it increases the likelihood that representatives will inappropriately adopt similar actions in the future.

Posturing also plays a role in competition among representatives over policy choices. As we saw in Chapter 6, the experimental literatures in psychology and economics suggest that the choices people make in experimental settings tend to be sensitive to how alternatives are presented to them.[38] For example, many people show a sufficiently strong preference for certain overly risky alternatives that they appear to be minimizing their maximum possible losses rather than maximizing expected utility. Environmental economists attempting to determine the values people place on wildlife through surveys often encounter unexpectedly large differences from a theoretical perspective between stated values for marginal losses and gains: the amount of money that one would have to give people to just compensate them for a small reduction in the whale population (their willingness to accept) might be an order of magnitude larger than the amount that they would pay (their willingness to pay) to obtain a small increase in the whale population. One source of these differences is probably loss aversion on the part of respondents.

Perceptual biases create opportunities for the effective use of rhetoric in debates over policy choices. In particular, they explain why negative campaigns, which emphasize the costs and risks of opponents' proposals, are so common.[39] To the extent that such rhetorical tactics affect public perceptions of alternative policies, they may result in political choices that do not maximize social value.

Representatives and Analysts

The characteristics of representative government that we have described are summarized in Table 8.2. They suggest that representative political systems often do not base public policy on the careful weighing of social costs and social benefits. To the extent that these failures either result from, or are facilitated by, limited information, policy analysts may be able to contribute to better social choices by providing more objective assessments of the likely consequences of proposed policies. To the extent that inattention to social costs and benefits is tied to the self-interest of representatives, analysts may be able to contribute to the social good by helping to craft politically feasible alternatives that are better than those that would otherwise be adopted. In any event, as discussed in Chapter 11, analysts should try to anticipate the limitations of representative government when they propose and evaluate public policies.

Problems Inherent in Bureaucratic Supply

Governments often create publicly funded organizations to deal with perceived market failures.[40] Armies, court systems, and various other agencies provide public goods that might be undersupplied by private markets. Other agencies regulate natural monopolies, externalities, and information asymmetries. Like private firms, they use labor and other factor inputs to produce outputs. Unlike private firms, however, they need not pass a market test to survive. Consequently, the extent to which their continued existence contributes to aggregate social welfare depends greatly on the diligence and motivations of the representatives who determine their budgets and oversee their operations. The very nature of public agencies, however, makes monitoring difficult and inefficiency likely.

Table 8.2 Sources of Inefficiency in Representative Government: Divergences between Social and Political Accounting

Nature of Interests in Electorate	Concentrated interests have strong incentive to monitor and lobby—too much weight likely to be given to their costs and benefits
	Diffuse interests generally have weak incentives to monitor and lobby—too little weight likely to be given to their costs and benefits
	Organized diffuse interests often overcome collective action problem in monitoring and lobbying—too much weight may be given to their costs and benefits
	Mobilization of diffuse interests through intermittent media attention—too much weight may be given to their costs and benefits when policy windows open
General Electoral Incentives of Representatives	Attempt to realize tangible benefits before elections—too much weight to short-run benefits and too little weight to long-run costs
	Emphasize potential costs of opponents' policy proposals to take advantage of loss aversion—too much weight given to potential costs
Electoral	Seek positive net benefits for district—positive net

Incentives of District-Based Representatives	benefits for majority of districts may result in negative net social benefits
	Seek support from factor suppliers in district—expenditures within district may be viewed as political benefits, despite being economic costs

Agency Loss

The monitoring problem is not unique to public agencies. In fact, a large *principal–agent* literature attempts to explain why principals (such as stockholders) employ agents (such as managers) to whom they delegate discretion.[41] In other words, why do we have firms consisting of hierarchical relationships between principals and agents rather than sets of contracts that specify exactly the services to be provided and the payments to be made by every participant in the production process? The answer lies in the fact that contracting involves costs. Rather than continuously re-contracting, it may be less costly in terms of negotiation and performance-monitoring activities to secure services under fairly general labor contracts. When this is the case, we expect to observe production organized by hierarchical firms rather than by collections of independent entrepreneurs.[42]

The principal–agent problem arises because employers and employees do not have exactly the same interests and because it is costly for employers to monitor the behavior of their employees. For example, although employees generally want their firms to do well, they also enjoy doing things like reading the newspaper on the job. Managers must expend time, effort, and goodwill to keep such shirking under control. More generally, because agents have more information about their own activities than do their principals, agents can pursue their own interests to some extent. The principal faces the task of creating organizational arrangements that minimize the sum of the costs of the undesirable behavior of agents and of the activity undertaken to control it. These costs, which are assessed relative to a world with perfect information, are referred to as *agency loss.*

Public executives generally operate in environments characterized by great asymmetry of information, not just with respect to the general public but with respect to other representatives as well.[43] The executive (agent) is generally in a much better position to know the minimum cost of producing any given level of output than either the public or their representatives (principals) who determine the agency's budget. For example, consider the U.S. Department of Defense. The public tends to pay attention primarily to sensational circumstances—the revelation that some toilet seats were purchased at $600 each, for instance. The Office of Management and Budget and the relevant congressional oversight committees look at the entire budget, but usually do so in terms of such billion-dollar elements as entire weapon systems. Limitations in time and expertise preclude extensive review of the ways available resources are actually used within the agency.[44] Even the Secretary of Defense suffers from an information disadvantage relative to the heads of the many units that constitute the department.

We can see the implications of the asymmetry in information by introducing the concept of the *discretionary budget*: the difference between the budget that the agency receives from its sponsor and the minimum cost of producing the output level that will satisfy its sponsor.[45] Executives enjoy the greatest freedom of action when the size of the discretionary budget is both large and unknown to the sponsor. It would be economically efficient for executives to produce output at minimum cost so that they can return the discretionary budget to the sponsor. If they do so, however, then they reveal information to the sponsor about the minimum cost, information that the sponsor can use in deciding how much to allocate to the agency in next year's budget. The observation that agencies rarely fail to spend their budgets, sometimes exhorting employees to spend faster as the end of the fiscal year approaches, suggests that few executives view the revelation of their discretionary budgets as an attractive management strategy.[46]

If the agency were a private firm with sales revenue as the source of its budget, then the owner-executive would simply keep the discretionary budget as profit.[47] In most countries, however, civil service laws and their enforcement keep public executives from converting discretionary budgets to personal income. Executives not wishing to return their discretionary budgets have an incentive to find personally beneficial ways to use their discretionary budgets within their agencies to make their jobs as managers easier. With extra personnel, executives can tolerate some shirking by employees that would be time consuming and perhaps unpleasant for the executive to eliminate. The added personnel might also make it easier for executives to respond to unexpected demands faced by their agencies in the future. Increased spending on such things as travel and supplies might contribute to morale and thereby make managing more pleasant. These uses of the discretionary budget are the realization of *X*-inefficiency of the sort analyzed in Chapter 5. It is also a source of allocative and dynamic inefficiency as well.[48] This is sometimes summarized as *slack maximization*. Indeed, the concept of the discretionary budget implies that different individuals within the agency might have different objectives. Anthony Downs, for example, suggests that some bureaucrats are self-interested, while others combine self-interest and altruistic commitment to larger values.[49] Similarly, it has been argued that bureaucratic goals vary by level or by role.[50]

The idea that bureaucrats have considerable budgetary discretion that fosters slack maximization has been criticized on a number of grounds. These criticisms have been fueled by the fact that there is quite mixed evidence on the existence of discretionary budget maximization and especially on aggregate budget-maximizing behavior by bureaucrats.[51] A basic assumption of the discretionary budget approach is that the bureau has at least some degree of monopoly power. Yet, at the limit, an agency may not be able to exercise any monopoly power if it faces a monopoly politician (purchaser) with strong bargaining power.[52]

Agency loss is inherent in all organizations, whether private or public. Three factors, however, typically make agency loss a relatively more serious problem for public bureaus than for privately owned firms: the difficulty of valuing public outputs (and, therefore, performance), the lack of competition among bureaus, and the inflexibility of civil service systems.

The Necessity to Impute the Value of Output

In the absence of market failures, the marginal social value of the output of a competitive firm equals the market price. Customers reveal this marginal value by their willingness to pay through purchases. Most public agencies, however, do not sell their output competitively. In fact, many were undoubtedly created in the first place because markets appeared not to be functioning efficiently. Representatives thus face the problem of having to impute values to such goods as national security, law and order, and health and safety. Even when representatives are sincerely interested in estimating the true social benefits of such goods, analysts often cannot provide convincing methods for doing so.

The absence of reliable benefit measures makes it difficult to determine the optimal sizes of public agencies. For instance, most observers would agree that adding an aircraft-carrier battle group to the U.S. Navy would increase national security by some amount. Would it contribute enough, though, to justify its multi-billion-dollar cost? This is a very difficult question to answer. Some progress toward an answer might be made by trying to assess how the additional battle group furthers various objectives of national security, such as keeping the sea lanes open in time of war, preventing piracy, and projecting military capability to regions where the United States has vital interests. Analysts might then be able to find alternative force configurations that better achieve the objectives or achieve them equally well at lower cost. In the end, however, these analyses do little to address the appropriateness of such politically established input goals as the "600-ship navy."

Often distributional goals further complicate the valuation problem. Private firms generally need not be concerned about which particular people purchase their products. In contrast, public agencies are often expected to distribute their output in accordance with principles such as horizontal and vertical equity. Crime reduction, for example, might be thought of as the major output of a police department. At the same time, however, the distribution of crime reduction across neighborhoods is also important—most of us would not favor aggregate reductions in crime gained at the expense of abandoning crime control in some neighborhoods altogether. When goals are multiple and conflicting, the valuation of outcomes requires consensus about how progress toward achieving one goal should be traded off against retrogression toward achieving another goal. Such consensus rarely exists.

Effects of Limited Competition on Efficiency

Competition forces private firms to produce output at minimum cost. Firms that do not use resources in the most efficient manner are eventually driven from the market by firms that do. Because public agencies do not face direct competition, they can survive even when they operate inefficiently.

We have already considered the consequences of an absence of competition in our discussion of natural monopolies in Chapter 5. Natural monopolies can secure rents even if they operate above the minimum achievable average cost curve. The resulting *X*-inefficiency represents some combination of the opportunity cost of wasted resources and rent transfers in the form of excess payments to factor inputs. Because competition is often absent, most clearly at the nation-state level, *X*-inefficiency can also arise in public agencies. At the subnational level, studies show that the presence of greater competition can have some impact in reducing *X*-inefficiency.[53]

Variability in the degree of competition also affects the dynamic efficiency of public agencies.[54] In many circumstances, public agencies can face inappropriate or weak incentives for innovation, and usually this incentive is weaker than for private firms. The profit motive provides a strong incentive for firms to find new production methods (technological or organizational innovations) that will cut costs. When one firm in an industry successfully innovates, others must follow or eventually be driven from the industry. Additionally, firms may be able to capture some of the industry-wide benefits of their innovations through patents. In comparison, public agencies are generally neither driven from existence if they do not innovate nor capable of fully capturing the external benefits if they do.

Some incentives to innovate do operate. Some agency heads seek the prestige and career advancement that might result from being perceived as innovators. Agency heads may also seek cost-reducing innovations at times of budgetary retrenchment to try to maintain output levels to their clientele. These incentives, however, do not operate as consistently as the profit motive and threats to firm survival.

Public agencies that do want to innovate face several disadvantages relative to their private-sector counterparts. They have no direct competitors to imitate. While they can sometimes look to agencies with similar functions in other jurisdictions, they face the valuation problem in deciding whether a new technology has actually produced net benefits elsewhere. Unlike managers of firms, they cannot borrow funds from banks or issue stock to cover start-up costs. Instead, they must seek funds from their budgetary sponsors, who may be unwilling to allocate them for uncertain research and development projects, especially when difficult-to-value improvements in output quality are the anticipated benefits. Finally, as we discuss in the next section, civil service rules may make it difficult to secure specialized expertise needed to implement innovations.

The absence of competition raises the possibility that public agencies can survive even if they fail to operate efficiently. Whether inefficiency actually results depends on the system of incentives that budgetary sponsors actually impose on public executives.[55]

Ex Ante Controls: Inflexibility Caused by Civil Service Protections

The problems principals face in evaluating the ex post performance of public agencies leads to the introduction of ex ante controls that directly restrict the discretion of agency heads in how they use the resources of the agency.[56] For example, agency heads often cannot make expenditures outside of narrow budgetary categories. Civil service rules place especially severe restrictions on how agency heads hire, fire, reward, and punish employees.

The modern civil service provides career opportunities within public agencies. Typically, only high-ranking officials serve at the pleasure of elected leaders. The remaining employees belong to the civil service, which, in theory, determines their employment tenure and salary independent of political parties. This separation of most government employees from partisan politics provides continuity in expertise when ruling parties change, as well as insulation against attempts to use agencies for partisan purposes.

Continuity and nonpartisanship, however, must be purchased at the expense of a certain amount of inflexibility in agency staffing.[57] The same rules that make it difficult to fire employees for political purposes make it difficult to weed out the incompetent and unproductive. Fixed pay schedules, which provide less opportunity for undue political leverage over employees, tend to under-reward the most productive and over-reward the least productive. As the former leave for higher-paying jobs in the private sector, the agencies are left with a higher proportion of the latter.[58]

A civil service system that makes it difficult to fire employees must also take great care in hiring them if quality is to be maintained. However, a slow and complicated hiring system reduces the ability of managers to implement new programs quickly. The problem is particularly severe for the public manager who fears losing allocated slots if they are not filled within the budget period. Rather than wait for the civil service to give approval for hiring the most appropriate employee, the safer course of action may be simply to take someone less qualified who already is in the civil service system. The more often these sorts of decisions are made for expediency, the more difficult it is for agencies to operate efficiently in the long run.

Finally, consider the commonly heard complaints that bureaucracies are unresponsive. Because agencies usually enjoy monopoly status, consumers usually cannot show their displeasure by selecting another supplier. Of course, elected representatives have an incentive to be responsive to their constituent-consumers. Shielding the civil service from undue political interference, however, reduces the role politicians can play in informing agencies about public wants. Threatening budget cuts and badgering agency executives may not be very effective ways of influencing behavior by the employees at the lowest level of the bureau, who, in most cases, actually deal with the public. Again, the necessity for some separation between politics and administration limits the capability of public agencies to meet consumer wants effectively.

Bureaucratic Failure as Intra-Organizational Market Failure

The traditional market failures discussed in Chapter 5 provide useful concepts for understanding bureaucratic failure. We have already discussed how monitoring costs (due to information asymmetry between principals and agents) and limited competition (often making bureaus "natural" monopolies) lie at the heart of bureaucratic failure. Bureaucratic failure often manifests itself as public good and externality problems as well. In situations in which bureaucratic supply is a conceptually appropriate generic policy, or where political constraints force its acceptance as the policy choice, applying these market failure concepts may help analysts better diagnose organizational problems and design ways of mitigating them. In other words, analysts can take the organization as the "market" and look for ways that rules and incentives lead to inefficiency.[59] We next consider two kinds of "internal market failure" with some examples.

"Organizational" Public Goods

As discussed in Chapter 5, a public good is one that is nonrivalrous or nonexcludable. All public agencies must confront intra-organizational public goods problems. The ex ante rules that are generally imposed on public organizations to compensate for lack of competition or contestability contribute to agency costs throughout the hierarchy. We can think of organizational public goods problems as a particular manifestation of agency costs. Furthermore, fully assessing these agency costs is complicated by the reality that most agencies are hierarchies that have more than one level (and type) of agents (summarized as supervisors and agents or "employees"). The presence of hierarchy exacerbates the difficulty of aligning incentives and understanding information asymmetries, partly because these different agents may collude to work against the interest of the principal.[60]

Organizational pure public goods occur if organizational members can consume a good produced by one member. For example, all the members of an organization share in the organization's reputation. However, building and maintaining a favorable reputation may be costly to individual members because it requires them to expend extra effort to meet clientele demands quickly, effectively, and cheerfully. In private organizations, managers can use a variety of rewards and punishments to try to induce employees to contribute to favorable reputations. Civil service rules (already described) make the task much more difficult for the public executive. Indeed, the public goods problem—in this case, under-supply of reputation-enhancing effort—can be seen as resulting from the inability of the executive to match individual rewards to organizational contributions. Bestowing honors and awards to employees or organizational units, or socializing new employees to have strong feelings of loyalty to the organization, may substitute to some extent for more direct incentives.

Consider next organizational open-access resources. Typically, all sorts of equipment and supplies used within organizations are available from a central stock to employees. If employees have unlimited access to the stock at a price below marginal cost to the organization, then we should expect overconsumption. In response, private organizations usually institute internal transfer prices so that employees see prices closer to marginal costs. However, for such a system to be more than an upper limit on total use, employees must be able to react to relative prices by using funds saved by less use in one category for some other purpose, such as greater use of some other resource or profit sharing. In public organizations, however, profit sharing is usually prohibited, and strict line-item budgets often prevent the moving of funds across categories that make relative prices meaningful. Consequently, we generally expect allocations of such resources as travel funds and motor pool use to be fully expended, and often misallocated, because ex ante controls prevent the establishment of effective internal transfer prices.

Ex ante controls also interfere with the organizationally beneficial use of toll goods. Firms can sometimes make themselves more attractive to employees by allowing them the personal use of resources at times when use is nonrivalrous. For example, firms may be happy to have some employees using computing time for personal projects on weekends, when marginal costs are close to zero. In public organizations, such use is almost always prohibited, reducing the capability of executives to reward employees.

"Organizational" Externalities

Production externalities result when organizations do not see the full social marginal costs and benefits of their actions. Several factors make individual members of public agencies prone to impose negative externalities on fellow employees or on members of the public.

Public organizations often have the legal right to use resources without paying their full marginal social costs. For example, citizens generally have a duty to serve on juries for nominal compensation. Because courts pay less than market prices for the time of people on jury panels, we expect them to overuse jurors' time relative to other inputs, such as paid employees to facilitate the efficient use of jurors. Other examples include the compliance costs that tax agencies impose on businesses and individuals, the exemption of many national defense activities from environmental regulations, and waiting time that bureaus in monopoly positions inflict on clientele.

Public organizations are often exempt by law from torts by those who have suffered damages. The exemption means that legal and financial responsibility for large external costs can be avoided. Even when the exemption does not apply, the polity generally bears some of the financial responsibility. In the extreme case of a bureau providing an essential service financed entirely out of general revenue, the entire loss is shifted to taxpayers. Further, civil service protections often make it difficult for overseers to impose penalties on those in the bureau responsible for the external costs, so that deterrence against activity leading to future torts may be insufficient.

Organizations may also undersupply services that have positive externalities. For instance, school districts sometime keep their facilities open after regular hours to provide safe places to their students for study and recreation. The primary motivation of the school district for opening its facilities is likely to be improving the academic performance of its students. However, these openings may also reduce juvenile delinquency and crime, providing savings to law enforcement. They may also provide a benefit to families with working adults who cannot easily supervise their children during after-school hours. If these positive externalities do exist, then a school district motivated only by educational achievement would undersupply after-school access to its facilities.

Analysts and Bureaucracy

Policy analysts who do not anticipate the problems inherent in bureaucratic supply risk giving bad advice about policy alternatives. The risk arises not only in considering such fundamental issues as the scope of government, but also in more instrumental issues such as the choices of organizational forms for providing public goods. Analysts who work in bureaucratic settings must be careful to keep their personal and organizational interests in perspective when called upon to give advice about the social good. At the same time, an understanding of the incentives others face in their organizational setting is often essential for designing policy alternatives that can be adopted and successfully implemented.

Problems Inherent in Decentralization

Canada, the United States, Australia, Germany, and many other democratic countries have highly decentralized and complex systems of government. Separate and independent branches exercise legislative, executive, and judicial authorities. Some powers are exercised by the central government, and others are reserved for the governments of local jurisdictions such as provinces, states, counties, cities, towns, and special districts for specific goods such as schooling. Within the executive branches of these various levels of government, agency executives often enjoy considerable discretion over policy and administration.

Strong normative arguments support several forms of decentralization. Distribution of authority among branches of government provides a system of "checks and balances" that makes abuse of authority by any one official less damaging and more correctable, and reduces the opportunity for tyranny by the majority.[61] Assigning different functions to different levels of government facilitates both the production of public goods at efficient levels and the matching of local public goods to local preferences.[62] Decentralization to lower levels of government is generally desirable because it brings citizens closer to public decisions, usually making it easier for them to exercise "voice" about the quantities and qualities of public good. More importantly, geographic decentralization is generally desirable because it permits citizens to exercise "exit"—those dissatisfied with the policies of a jurisdiction have the opportunity to "vote with their feet" by moving to jurisdictions offering more preferred policies. Adherence to these normative principles, however, leads to governments that are not only decentralized but also complex.

The highly desirable benefits of decentralization come at a price. Decentralization tends to hinder implementation of policies. It also allows for fiscal externalities to occur in association with the supply of local public goods. Thus, while decentralization is a desirable, if not essential, structural feature of government, it sometimes limits the effectiveness of public policies.

The Implementation Problem

Adopted policies gain force through implementation. The passage of a law, the adoption of an administrative rule, or the issuance of a judicial order establishes policy goals and specifies the means thought necessary to achieve them. *Implementation* describes the efforts made to execute the means—efforts that do not always achieve the intended goals. As discussed in greater detail in Chapter 12, Eugene Bardach provides an excellent metaphor for implementation: the process of assembling and keeping in place all the elements needed for a machine.[63] Of course, just as a machine may not work as intended if its design is flawed, a policy based on an incorrect theory may fail to produce desired consequences, or even produce unintended undesirable consequences. Yet, an effective design (correct theory) is only a necessary—but not a sufficient—condition for a working machine (effective policy): if necessary parts (essential policy elements) are either not available or unreliable, then the machine (policy) will not work effectively.

The essence of the implementation problem lies in the distribution of necessary elements. The greater the potential for either persons or organizations to withhold their necessary contributions, the greater is the possibility of failure. Those who oppose the goals of the policy and those who do not view the goals as sufficiently beneficial to justify their own costs of compliance may purposely withhold contributions.[64] Others may do so simply because their resources and competence are too limited to allow them to comply.

In decentralized political systems, many officials have the capability to withhold elements needed for implementation. They thus enjoy bargaining power that may enable them to extract things from those who need the elements. For example, consider the implementation of the U.S. Strategic Petroleum Reserve program, which is intended to reduce the vulnerability of the U.S. economy to petroleum price stocks. Approval from the governor of Louisiana was needed for essential facilities. Before he would give it, the governor secured a number of concessions from the Department of Energy, including a promise that nuclear wastes would not be stored in his state. The Department of Energy was also forced to make concessions to the Environmental Protection Agency, another federal agency, even though the petroleum reserve enjoyed the enthusiastic support of the president and Congress.

The problem of inter-organizational cooperation also arises when central governments must rely on lower levels of government to act.[65] Even when the central government enjoys nominal authority over units of the lower-level government, it faces the problem of monitoring the compliance of multiple jurisdictions that may have very different local conditions. For example, the central government may have the constitutional authority to order local school districts to end racial segregation. Systematic monitoring of compliance, however, may require an extended period of investigation because of the large number of jurisdictions involved. Further, the reasons for noncompliance may vary greatly across jurisdictions, making it difficult for the central government to employ a uniform enforcement strategy.

Implementation of policies requiring cooperation from lower-level governments becomes more difficult when the central government does not have the authority to coerce. In order to induce cooperation successfully, the central government must offer sufficient rewards to secure voluntary compliance. When the lower-level jurisdictions are highly diverse, the central government may offer more compensation than necessary to some and not enough to others, especially if either norms of fairness or legal restrictions force the central government to offer the same opportunities to all. For example, a grant program intended to induce local governments to improve their sewage treatment plants might very well fund some plants that would have been constructed by local governments anyway (the problem of displacement) and some plants that should not be built from the social perspective (the problem of targeting). If the grants are passed through states, then some of the money intended for local use may be captured by the state agencies handling the grants, the so-called *flypaper effect*.[66]

The implementation problem also arises when cooperation among organizations with different responsibilities or in different jurisdictions is required for effectively solving problems. Organizations often operate within narrow mandates that impede comprehensive problem solving. For example, police departments may not have the authority or resources to help substance abusers receive medical treatments that may help them avoid the behaviors that bring them into contact with the criminal justice system. The problem is complicated by an obsolescence that arises as old approaches to problems become entrenched within organizations.[67] Accommodating new policies may require changes in standard operating procedures developed and instilled in employees, disrupting routines and necessitating substantial internal change. Even something as seemingly simple as making the information systems of two or more organizations compatible may be difficult with readily available resources.

Finally, we should note that many of the problems inherent in bureaucratic supply contribute to the implementation problem. The difficulty representatives have in monitoring costs often leaves them uncertain about the size of contributions to implementation that agencies actually can and do make, and sometimes gives public executives opportunities to avoid compliance that is in conflict with their own interests. When executives do wish to comply, they may have difficulty motivating subordinates who face attenuated rewards and punishments within the civil service. These problems introduce uncertainties that complicate inter-organizational bargaining over compliance.

Fiscal Externalities

Decentralization permits the provision of local public goods to be better matched to local demands. It also gives people some opportunity to choose the bundles of public goods they consume through their selections of jurisdictions. The greater the variation in public goods bundles across jurisdictions, the greater is the opportunity for people to exercise choice by voting with their feet.[68] In the extreme, we can imagine migration working to sort people into groups with homogeneous preferences for local public goods. Because the number of jurisdictions is always limited, and because people consider private factors, such as employment, as well as public goods in locational choice, the sorting process is never complete in reality. Furthermore, jurisdictions have an incentive to discourage some migrants and encourage others,[69] a source of government failure resulting from decentralization.

Immigrants who pay above-average tax shares and place below-average demands on public services are particularly desirable because they lower tax shares for everyone else in the jurisdiction without reducing service levels. In other words, they impart a positive fiscal externality to the established residents. In contrast, immigrants who pay below-average tax shares and place above-average demands on public services are particularly undesirable because they impart a negative fiscal externality to established residents. Because tax shares are usually a linear function of property values at the local level, jurisdictions have an incentive to try to exclude those who would have below-average property values. The incentive leads to such local policies as minimum lot sizes, restrictions on multiple-unit dwellings, and restrictive building codes that inflate construction costs. One social cost of these policies is a reduction in housing opportunities for low- and middle-income families.

In competing for wealthier residents, local jurisdictions may inflict fiscal externalities on each other—families with low incomes and high service demands tend to be left behind in poorer jurisdictions. Similarly, local jurisdictions often inflict fiscal externalities on each other in competition for industry. Many local jurisdictions offer incentives, such as tax reductions and site preparation, to lure firms that will provide jobs to residents. These incentives are probably not large enough to influence the level of investment in the country as a whole. Therefore, the jurisdictions compete for a fixed amount of investment. This is especially true for industries, like professional sports, that tightly limit the number of franchises—realistic estimates of the economic benefits provided by municipal subsidies for sports stadiums, for instance, appear to fall far short of their economic costs.[70]

Local jurisdictions face a prisoner's dilemma in their decisions concerning incentives for industrial development. Consider two similar jurisdictions courting a single firm. If neither offered tax abatements to attract the firm, then one of the jurisdictions would get the firm and receive full tax revenues. If one jurisdiction thought it would lose the competition, however, it might offer a tax abatement to change the firm's decision. Of course, the other jurisdiction would be in the same situation. The result is likely to be both jurisdictions offering abatements up to a level that would leave them indifferent between having and not having the firm. As a consequence, neither jurisdiction actually gains from the competition.

Analysts in Decentralized Political Systems

Political decentralization poses challenges for policy analysts. By complicating the implementation process, it greatly increases the difficulty of trying to predict the consequences of alternative policies. Because analysts have clients throughout the political system, they often encounter conflicts between their own conception of what promotes the social good and the personal interests of their clients. Even the most basic question of whose costs and benefits should count often arises. For example, how should an analyst who works for the mayor of a city count costs and benefits that accrue outside of the city? From the social perspective, we would generally regard these spillovers as relevant to the determination of net social benefits, but what would be the consequences of taking them into account when other jurisdictions do not take into account the externalities of their policies? We will address these and other questions that arise as a consequence of political decentralization when we study the art and craft of policy analysis in later chapters.

Table 8.3 Sources of Government Failure: A Summary

Problems Inherent in Direct Democracy	Paradox of voting (meaning of mandate ambiguous)
	Preference intensity and bundling (minorities bear costs)
Problems Inherent in Representative Government	Influence of organized and mobilized interests (inefficiency through rent seeking and rent dissipation)
	Geographic constituencies (inefficient pork-barrel allocations)
	Electoral cycles (socially excessive discount rates)
	Posturing to public attention (restricted agendas and distorted perception of costs)
Problems Inherent in Bureaucratic Supply	Agency loss (*X*-inefficiency)
	Difficulty valuing output (allocative and *X*-inefficiency)
	Limited competition (dynamic inefficiency)
	Ex ante rules including civil service constraints

	(inefficiency due to inflexibility)
	Bureau failure as market failure (inefficient use of organizational resources)
Problems Inherent in Decentralization	Diffuse authority (implementation problems)
	Fiscal externalities (inequitable distribution of local public goods)

Conclusion

Governments, like markets, sometimes fail to promote the social good. We cannot always accurately predict the exact consequences of government failures. (Indeterminacy itself is sometimes a predictable consequence!) We must be aware, however, that they *can* occur if we are to avoid the most ineffective and unwise interferences with private choices. Table 8.3 summarizes the basic sources of government failure. Our next task in developing conceptual foundations for policy analysis is to consider how to frame problems using both market and government failures.

For Discussion

1. The past 20 years have witnessed considerable concern in the United States about the quality of public schooling, particularly in urban areas. Apply the government failure concepts to various aspects of the public education problem.
2. If you are a U.S. citizen, choose some public policy issue at the U.S. federal level that you see as important. Do you know where your representative stands on the issue? Do you know where your senators stand on the issue?

Notes

1 Charles Wolf, Jr., is one of the few social scientists to develop a theory of government ("nonmarket," in his term) failure that could complement the theory of market failure. His classification includes problems of bureaucratic supply and policy implementation, but largely ignores the problems of social choice and representative government. See Charles Wolf, Jr., "A Theory of Nonmarket Failures," *Journal of Law and Economics* 22(1) 1979, 107–39.

2 Duncan Black rediscovered the paradox of voting in the 1940s. For an overview, see William H. Riker, *Liberalism against Populism* (San Francisco: Freeman, 1982), 1–3.

3 Kenneth Arrow, *Social Choice and Individual Values*, 2nd ed. (New Haven, CT: Yale University Press, 1963). For a nontechnical treatment, see Julian H. Blau, "A Direct Proof of Arrow's Theorem," *Econometrica* 40(1) 1972, 61–67.

4 In addition, social choice rules are generally vulnerable to manipulation. See Allan Gibbard, "Manipulation of Voting Schemes: A General Result," *Econometrica* 41(4) 1973, 587–601; Mark Satterthwaite, "Strategy Proofness and Arrow's Conditions," *Journal of Economic Theory* 10(2) 1975, 187–217.

5 In particular, if the agenda setter has full information about the preferences of the voters who vote sincerely, then she can reach any desired alternative through a series of pairwise votes. Richard D. McKelvey, "Intransitivities in Multidimensional Voting Models and Some Implications for Agenda Control," *Journal of Economic Theory* 12(3) 1976, 472–82; Norman Schofield, "Instability of Simple Dynamic Games," *Review of Economic Studies* 45(4) 1978, 575–94.

6 For an introduction to these models, see Keith Krehbiel, "Spatial Models of Legislative Choice," *Legislative Studies Quarterly* 8(3) 1988, 259–319.

7 Riker, *Liberalism Against Populism*, 213–32.

8 Austin Ranney, ed., *The Referendum Device* (Washington, DC: American Enterprise Institute, 1981), xii, 1–18. See also Richard Johnston, Andre Blais, Elisabeth Gidengil, and Neil Nevitte, *The Challenge of Direct Democracy* (Montreal: McGill-Queen's University Press, 1996), 9–41.

9 For a review, see J. Rolland Pennock, *Democratic Political Theory* (Princeton, NJ: Princeton University Press, 1979), 321–34. He argues that the proper role of a representative falls somewhere between that of trustee and delegate (at 325).

10 Pranab Bardhan, "Corruption and Development," *Journal of Economic Literature* 35(3) 1997, 1320–46.

11 *Washington Post*, The U.S. Congress Votes Database, http://projects.washingtonpost.com/congress/.

12 James Q. Wilson, *The Politics of Regulation* (New York: Basic Books, 1980).

13 James M. Buchanan, Robert D. Tollison, and Gordon Tullock, eds., *Toward a Theory of the Rent-Seeking Society* (College Station: Texas A&M University Press, 1980).

14 George Stigler, "The Theory of Economic Regulation," *Bell Journal of Economics and Management Science* 2(1) 1971, 3–21.

15 David L. Weimer, "Organizational Incentives: Safe- and Available-Drugs," in LeRoy Graymer and Frederick Thompson, eds., *Reforming Social Regulation* (Beverly Hills, CA: Sage Publications, 1982), 19–69.

16 Samuel P. Huntington, "The Marasmus of the ICC: The Commission, the Railroads, and the Public Interest," *Yale Law Review* 61(4) 1952, 467–509.

17 Calculated from Table 1, Gordon C. Rausser, "Predatory versus Productive Government: The Case of U.S. Agricultural Policies," *Journal of Economic Perspectives* 6(3) 1992, 133–57.

18 John C. Beghin, Barbara El Ostro, Jay R. Chernow, and Sanarendu Mohanty, "The Cost of the U.S. Sugar Program Revisited," *Contemporary Economic Problems* 21(1) 2003, 106–16.

19 See Gordon Tullock, "The Transitional Gains Trap," in Buchanan, Tollison, and Tullock, eds., *Rent-Seeking Society,* 211–21.

20 Mark D. Uehling and Rich Thomas, "Tax Reform: Congress Hatches Some Loopholes," *Newsweek*, September 29, 1986, 22.

21 Richard A. Smith, "Advocacy, Interpretation, and Influence in the U.S. Congress," *American Political Science Review* 78(1) 1984, 44–63.

22 For a detailed discussion of the impact of price ceilings, see George Horwich and David L. Weimer, *Oil Price Shocks, Market Response, and Contingency Planning* (Washington, DC: American Enterprise Institute, 1984), 57–110.

23 Mancur Olson, *The Rise and Decline of Nations: Economic Growth, Stagflation, and Social Rigidities* (New Haven, CT: Yale University Press, 1982), 69.

24 For an overview, see Kenneth A. Shepsle, *Analyzing Politics: Rationality, Behavior, and Institutions* (New York: W. W. Norton, 2010).

25 James M. Buchanan and Gordon Tullock, *The Calculus of Consent* (Ann Arbor: University of Michigan Press, 1962), 220–22.

26 For a formal development of this idea, see Kenneth A. Shepsle and Barry R. Weingast, "Political Solutions to Market Problems," *American Political Science Review* 78(2) 1984, 417–34; and Barry R. Weingast, Kenneth A. Shepsle, and Christopher Johnsen, "The Political Economy of Benefits and Costs: A Neoclassical Approach to Distributive Politics," *Journal of Political Economy* 89(4) 1981, 642–64.

27 Michael R. Gordon, "B-1 Bomber Issue No Longer Whether to Build It but How Many, at What Price," *National Journal*, September 3, 1983, 1768–72. "… Rockwell has aggressively lobbied for the B-1B program, notifying members of Congress about the jobs at stake, many of which are spread among contractors throughout the United States," at 1771.

28 For a review of the literature, see Eric Kramon and Daniel N. Posner, "Who Benefits from Distributive Politics? How the Outcome One Studies Affects the Answer One Gets," *Perspectives on Politics* 11(2) 2013, 461–74.

29 See, for example, William H. Riker and Steven J. Brams, "The Paradox of Vote Trading," *American Political Science Review* 67(4) 1973, 1235–47. They see logrolling as leading to reductions in social welfare. It is worth noting, however, that logrolling can also lead to socially beneficial results by allowing minorities with strong preferences to form majorities.

30 Bryan D. Jones, *Governing Urban America: Policy Approach* (Boston: Little, Brown, 1983), 410.

31 The year in the governor's term cycle also appears to influence the pattern of tax changes. John Mikesell, "Election Periods and State Tax Policy Cycles," *Public Choice* 33(3) 1978, 99–105; and Michael A. Nelson, "Electoral Cycles and the Politics of State Tax Policy," *Public Finance Review* 28(6) 2000, 540–60.

32 The literature on electoral cycles begins with William D. Nordhaus, "The Political Business Cycle," *Review of Economic Studies* 42(2) 1975, 169–90. The partisan model was introduced by Douglas Hibbs, "Political Parties and Macroeconomic Policy," *American Political Science Review* 71(4) 1977, 1467–87. For overviews showing the importance of partisanship, see Alberto Alesina and Howard Rosenthal, *Partisan Politics, Divided Government, and the Economy* (New York: Cambridge University Press, 1995); and William R. Keech, *Economic Politics: The Costs of Democracy* (New York: Cambridge University Press, 1995).

33 For a discussion of the ideas in this paragraph, see John W. Kingdon, *Agendas, Alternatives, and Public Policies* (Boston: Little, Brown, 1984), 205–18. For a review and critique of policy change, see the special issue of the *Journal of Comparative Policy Analysis* 11(1), 2009, 1–143, edited by Giliberto Capano and Michael Howlett, "The Determinants of Policy Change: Theoretical Challenges," particularly Reimut Zohlnhofer, "How Politics Matter When Policies Change," 97–116.

34 Ibid., 181.

35 James M. Snyder, Jr., "Campaign Contributions as Investments: The U.S. House of Representatives, 1980–1986," *Journal of Political Economy* 98(6) 1990, 1195–227; Robert S. Erikson and Thomas R. Palfrey, "Campaign Spending and Incumbency: An Alternative Simultaneous Equations Approach," *Journal of Politics* 60(2) 1998, 355–73.

36 In the early 1970s, James Q. Wilson worried, "Nearly ten years ago I wrote that the billions of dollars the federal government was then preparing to spend on crime control would be wasted, and indeed might even make matters worse … They were, and they have." *Thinking about Crime* (New York: Basic Books, 1975), at 234.

37 William Riker coined the word *heresthetics* to refer to the art of political strategy. It includes agenda control, rhetoric (the art of verbal persuasion for political purposes), sophisticated voting, and raising new issues to split dominant majorities. William H. Riker, *The Art of Political Manipulation* (New Haven, CT: Yale University Press, 1986),

at ix and 142–52. Limited electoral attention gives representatives ample scope for the use of rhetorical heresthetics. See Iain McLean, *Rational Choice & British Politics: An Analysis of Rhetoric and Manipulation from Peel to Blair* (Oxford, New York: Oxford University Press, 2001).

38 Amos Tversky and Daniel Kahneman, "The Framing of Decisions and the Rationality of Choice," *Science* 211, 1981, 453–58.

39 William H. Riker, *The Strategy of Rhetoric: Campaigning for the American Constitution* (New Haven, CT: Yale University Press, 1996), 49–74.

40 We use the term *public agency* to denote organizations that are staffed by government employees and that obtain the majority of their revenues from public funds rather than the sale of outputs. In Chapter 10 we distinguish public agencies from organizations that are owned by governments but generate the majority of their revenues from sales (usually called state-owned enterprises).

41 See David E. M. Sappington, "Incentives in Principal–Agent Relationships," *Journal of Economic Perspectives* 5(2) 1991, 45–66; and Jeffrey S. Banks, "The Design of Institutions: An Agency Theory Perspective," in David L. Weimer, ed., *Institutional Design* (Boston: Kluwer Academic, 1995), 17–36. For a review of empirical research, see William S. Hesterly, Julia Liebeskind, and Todd R. Zenger, "Organizational Economics: An Impending Revolution in Organization Theory?" *Academy of Management Review* 15(3) 1990, 402–20.

42 Ronald H. Coase has provided much of the underpinning for the theory of economic organization in "The Nature of the Firm," *Economica* 4(16) 1937, 386–405. For a review of the literature that approaches the organization question from the perspective of the distribution of property rights, see Louis De Alessi, "The Economics of Property Rights: A Review of the Evidence," in R. Zerbe, ed., *Research in Law and Economics* 2(1) 1980, 1–47. See also Armen A. Alchian and Harold Demsetz, "Production, Information Costs, and Economic Organization," *American Economic Review* 62(5) 1972, 777–95; and Oliver F. Williamson, *The Economic Institutions of Capitalism* (New York: Free Press, 1985).

43 Political scientists have applied the principal–agent framework to the study of delegation decisions by Congress. See David Epstein and Sharyn O'Halloran, *Delegating Powers: A Transaction Cost Politics Approach to Policy Making under Separation of*

Powers (New York: Cambridge University Press, 1999); and John D. Huber and Charles R. Shipan, *Deliberate Discretion? The Institutional Foundations of Bureaucratic Autonomy* (New York: Cambridge University Press, 2002).

44 See William A. Niskanen, *Bureaucracy and Representative Government* (Chicago: Aldine-Atherton, 1971); Albert Breton and Ronald Wintrobe, "The Equilibrium Size of a Budget-Maximizing Bureau," *Journal of Political Economy* 83(1) 1975, 195–207; and Jonathan Bendor, Serge Taylor, and Ronald Van Gaalen, "Bureaucratic Expertise versus Legislative Authority: A Model of Deception and Monitoring in Budgeting," *American Political Science Review* 79(4) 1985, 1041–60.

45 Niskanen uses the discretionary budget concept to generalize his seminal study of bureaucracy to objective functions beyond simple budget maximization; William A. Niskanen, "Bureaucrats and Politicians," *Journal of Law and Economics* 18(3) 1975, 617–44.

46 On the advantages and disadvantages of not spending allocated funds, see Aaron Wildavsky, *The Politics of the Budgetary Process*, 3rd ed., (Boston: Little, Brown, 1979), 31–32.

47 In firms where executives are not the owners, they usually receive some compensation as stock options or large bonuses to increase their incentives to maximize the present value of the firm. The design of such contractual arrangements in the face of information asymmetry is an important principal–agent issue, as investment bank failures have shown. See, for example, Michael C. Jensen and William H. Meckling, "Theory of the Firm: Managerial Behavior, Agency Costs, and Ownership Structure," *Journal of Financial Economics* 3(4) 1976, 305–60; and Eugene F. Fama, "Agency Problems and the Theory of the Firm," *Journal of Political Economy* 88(2) 1980, 288–307.

48 See William Duncombe, Jerry Miner, and John Ruggiero, "Empirical Evaluation of Bureaucratic Models of Inefficiency," *Public Choice* 93(1–2), 1997, 1–18.

49 Anthony Downs, *Inside Bureaucracy* (Boston: Little, Brown, 1967).

50 On the former, see Patrick Dunleavy, *Democracy, Bureaucracy, and Public Choice* (Englewood Cliffs, NJ: Prentice Hall, 1992). On the latter, see Anthony Boardman, Aidan

Vining, and Bill Waters III, "Costs and Benefits through Bureaucratic Lenses: Example of a Highway Project," *Journal of Policy Analysis and Management* 12(3) 1993, 532–55.

51 Andre Blais and Stephane Dion, *The Budget-Maximizing Bureaucrat. Appraisals and Evidence* (Pittsburgh: University of Pittsburgh Press, 1991). For a detailed case study, see Lee S. Friedman, *The Microeconomics of Public Policy Analysis* (Princeton, NJ: Princeton University Press, 2002), 432–40.

52 See José Casas-Pardo and Miguel Puchades-Navarro, "A Critical Comment on Niskanen's Model," *Public Choice* 107(1–2) 2001, 147–67.

53 For an overview, see Lori L. Taylor, "The Evidence on Government Competition," *Federal Reserve Bank of Dallas: Economic and Financial Review*, 2nd quarter 2000, 2–10. See also Nicholas Bloom, Carol Propper, Stephen Seiler, and John Van Reenen, "The Impact of Competition on Management Quality: Evidence from Public Hospitals," *Review of Economic Studies* 82(2) 2015, 457–89.

54 See David L. Weimer, "Federal Intervention in the Process of Innovation in Local Public Agencies," *Public Policy* 28(1) 1980, 83–116.

55 As with natural monopoly, contestability may limit inefficiency. If privatization, or even transferring funding to another public agency, is a credible threat, then public executives and workers have an incentive to limit *X*-inefficiency so as to make these alternatives appear less attractive. See Aidan R. Vining and David L. Weimer, "Government Supply and Government Production Failure: A Framework Based on Contestability," *Journal of Public Policy* 10(1) 1990, 1–22.

56 Fred Thompson and L. R. Jones, "Controllership in the Public Sector," *Journal of Policy Analysis and Management* 5(3) 1986, 547–71.

57 Ronald N. Johnson and Gary D. Libecap, "Bureaucratic Rules, Supervisor Behavior, and the Effect on Salaries in the Federal Government," *Journal of Law, Economics, and Organization* 5(1) 1989, 53–82.

58 For a brief discussion of the implications of this process for policy analysis offices, see Hank Jenkins-Smith and David L. Weimer, "Rescuing Policy Analysis from the Civil Service," *Journal of Policy Analysis and Management* 5(1) 1985, 143–47.

59 These ideas are developed in Aidan R. Vining and David L. Weimer, "Inefficiency in Public Organizations," *International Public Management Journal* 2(1) 1999, 1–24. See also Michael Brydon and Aidan R. Vining, "Understanding the Failure of Internal Knowledge Markets: A Framework for Diagnosis and Improvement," *Information & Management* 43(8) 2006, 964–74.

60 Jean Tirole, "Hierarchies and Bureaucracies: On the Role of Collusion in Organizations," *Journal of Law, Economics, and Organization* 2(2) 1986, 181–214.

61 James Madison set out the normative arguments for separation of powers in Numbers 47 to 51 of *The Federalist Papers* (New York: New American Library, 1961), 300–325.

62 See Wallace Oates, *Fiscal Federalism* (New York: Harcourt Brace Jovanovich, 1972), 3–20. On the optimal size of local governments, see Joseph Drew, Michael A. Kortt, and Brian Dollery, "Economies of Scale and Local Government Expenditure: Evidence from Australia," *Administration & Society* 46(6) 2014, 632–53.

63 Eugene Bardach, *The Implementation Game: What Happens after a Bill Becomes Law* (Cambridge, MA: MIT Press, 1977), 36–38.

64 Pressman and Wildavsky argue that means, rather than ends, are more likely to be the actual focus of conflicts during the implementation process. Jeffrey L. Pressman and Aaron Wildavsky, *Implementation* (Berkeley: University of California Press, 1973), 98–102.

65 On shared federal-state governance of U.S. health policies, see Simon F. Haeder and David L. Weimer, "You Can't Make Me Do It, but I Could Be Persuaded: A Federalism Perspective on the Affordable Care Act," *Journal of Health Politics, Policy and Law* 40(2) 2015, 281–323.

66 Paul Gary Wykoff, "The Elusive Flypaper Effect," *Journal of Urban Economics* 30(3) 1991, 310–28.

67 Eugene Bardach, *Getting Agencies to Work Together: The Practice and Theory of Managerial Craftsmanship* (Washington, DC: Brookings Institution Press, 1998).

68 The interpretation of locational choice as a method of demand revelation for local public goods was first suggested by Charles M. Tiebout, "A Pure Theory of Local Expenditures," *Journal of Political Economy* 64(5) 1956, 416–24. A similar but more general model of

demand revelation through choice of membership was provided by James M. Buchanan, “An Economic Theory of Clubs,” *Economica* 32(125) 1965, 1–14.

69 See James M. Buchanan and Charles J. Goetz, “Efficiency Limits of Fiscal Mobility: An Assessment of the Tiebout Model,” *Journal of Public Economics* 1(1) 1972, 25–42. For an application to international migration, see Norman Carruthers and Aidan R. Vining, “International Migration: An Application of the Urban Location Model,” *World Politics* 35(1) 1982, 106–20. For an application to metropolitan government in California, see Gary J. Miller, *Cities by Contract: The Politics of Municipal California* (Cambridge, MA: MIT Press, 1981), 183–89.

70 See Roger G. Noll and Andrew Zimbalist, eds., *Sports, Jobs, and Taxes: The Economic Impact of Sports Teams and Stadiums* (Washington, DC: Brookings Institution Press, 1997).

9
Policy Problems as Market and Government Failure

The Madison Taxicab Policy Analysis Example

The theories of market and government failure developed in the previous chapters provide conceptual resources for framing public policy problems. Market and government failures, however, often do not occur in isolation. When they are both present to some degree, they rarely interact in simple and obvious ways. In specific policy contexts, we may observe market failure, with no government response, or we observe no market failure, but extensive government intervention. Quite often, we observe market failure and government failure together. In this chapter, we show how policy analysis can bring together the theories of market and government failure.

Before setting out general guidelines for using market and government failure to frame policy problems, we provide an illustration of a short but comprehensive policy analysis of a pervasive issue—taxicab regulation. This example shows how public policies, purportedly intended to correct market failures, primarily information asymmetry, can become barriers to entry by new firms. The analysis also attempts a brief explanation of why this state of affairs may be difficult to change: the existing taxicab firms are a concentrated interest with a strong incentive to block changes that might benefit consumers and other entrepreneurs.

Box 9.1 Taxi Regulation in Madison

To: Mayor David Cieslewicz
From: Molly Askin and Katie Croake
Date: August 18, 2003
Subject: Taxi Regulation in Madison

Executive Summary

Taxi fares in Madison are well above the national average for cities of comparable size. The primary reason for these high fares appears to be that taxi regulation by the City of Madison restricts entry to the taxi market. Current regulatory policies reflect the fact that city officials are more responsive to the interests of incumbent taxi companies than to the needs and interests of taxi users and potential entrants to the taxi market. This report concludes that the city should remove the primary entry barrier, "the 24/7 rule."

After a discussion of the rationale for regulating the taxi industry, an analysis of Madison's taxi market structure, its taxi operators, and municipal taxi regulations, the current policy and removal of the 24/7 rule are analyzed based on the goals of 24/7 equity, efficiency, fiscal effects on the municipal government, and political feasibility. Currently, Madison requires that taxi companies serve the entire city and that each company provide service 24 hours per day and 7 days per week. Annual license fees are set at $1,000 per company and $40 per vehicle plus $25 per driver.

Based on this analysis, we recommend that the city eliminate the 24/7 rule. Although the Common Council has opposed elimination of the 24/7 rule in the past, the threat of an adverse ruling in federal court may provide an opportunity for change. We recommend that you use this opportunity to propose the removal of the 24/7 rule to the Common Council.

Introduction: Taxi Service in Madison

Taxi service in Madison, Wisconsin, is among the most expensive in the nation.[1] Municipal regulations require all cab companies to operate 24 hours a day, 7 days a week to provide citizens with constant taxi availability. In addition, every cab company must provide citywide service. Firms wishing to enter the market must pay $1,000 to apply for an operating license from the Traffic Engineering Division of the Department of Transportation. Traffic Engineering Division staff members review the application and make a recommendation to the Transit and Parking Commission, and the commission then makes a recommendation to the Common Council. The final decision to approve or deny an application is made by majority rule in the Common Council.

Despite changes in taxi regulations in the past few years, the combination of the current regulations creates service requirements that pose severe barriers to entry into the taxi market. Commentators on Madison's taxi market agree that the city's current regulations create significant barriers to entry.[2] These barriers have led to a situation in which no new taxi companies have entered Madison's market in more than 17 years. Consequently, competition is weak and cab fares are approximately one-third higher than the national average (see the Appendix).

The 24/7 rule, the citywide service requirement, high licensing fees, and the lengthy license approval process all raise the initial capital investment and operational costs of taxi companies. More technically, they artificially raise the minimum efficient scale of operation (meaning taxi companies face higher costs than they would in an unregulated market). Today, high taxicab fares and the absence of market entry over a long period of time indicate a weak competitive environment. This analysis examines whether alternatives for reducing entry barriers would increase competition and benefit taxi riders in Madison.

The Rationale for Taxi Regulation

Because taxis provide door-to-door service, they are a convenient and valuable means of transport for less mobile members of the population. In large- and medium-sized cities, taxi service can eliminate some residents' need for cars, and taxis make travel within cities more convenient for tourists. Many state and city governments have recognized the benefits of taxi service and have adopted various forms of licensing and more extensive regulations to address perceived market failures. London taxis, for example, have been licensed since 1654.[3] Although there has been surprisingly little policy analysis on taxi regulation in the academic literature over the last decade, a review of the extant literature suggests that the most important market failure relating to taxi service results from the information advantage of taxi operators over consumers regarding safety, reliability of service, and prices.[4]

In thinking about potential market failures, it is useful to divide the market for taxis into two submarkets—the cruising market, where customers hail a taxi on the street, and the phone reservation market. When hailing a cruising taxicab, potential users are not in a position to compare cab fares, consider the safety records of drivers, or review the mechanical safety record of the particular cab they are hailing. Instead, consumers accept rides from the first taxi that appears. Because in this situation taxi users have a very limited basis for judging the safety and quality of taxi service (including clear information on pricing), there is asymmetric information in the market.[5] Some analysts have also argued that there may be no competitive market equilibrium in the cruising market. Robert Cairns and Catherine Liston-Heyes argue:

> Modeling this industry as competitive would imply large numbers of firms facing large numbers of customers at a given instance at a given place.… [I]n the cruising market, it is usual for a single customer to hail a single taxi as it goes by. In this situation, where search may be costly to the consumer, customers who are risk-averse may prefer a somewhat higher fixed fare to a fare established through bargaining with operators accustomed to sizing up customers with a high variance across conditions.[6]

Some of these asymmetric information problems and equilibrium issues also occur in the telephone reservation market, but most of them are likely to be less severe. They are less severe because customers can select the cab company they wish to patronize (provided it is not a monopoly, legislated or otherwise) and can rely to some extent on their previous experience. It is important to note that, with the increasing dissemination of mobile telephones, the cruising market is likely to become relatively less important.

Another potential market failure that affects telephone reservations is the presence of network externalities. Holding other attributes of service such as quality constant, potential riders are most likely to be interested in a company that has a network of cabs throughout the city. If they require immediate service, then the availability of multiple cabs is likely to reduce waiting times for customers. If they require pick-up at a specific time, then multiple cabs increase the probability of a booking at the preferred time. The presence of these economies of scale does not necessarily preclude individual operators from entering the market, because such drivers can join, or form, a cooperative. The formation of cooperatives, however, may significantly reduce the number of competing entities in specific markets, especially in smaller jurisdictions.

Types of Taxicab Regulation

Governments around the world have adopted a variety of policies in an effort to correct some of the market failures in the taxi market. To address the information asymmetry in both the hail and phone markets, governments generally require insurance, mandate the posting of price information, inspect vehicles for safety, and conduct driver-background checks. Madison, like many other cities, requires insurance, maintains safety standards, and mandates that companies must file fare information with the city, post fares inside each cab, and charge a consistent rate for service. Typically, in cities that require posting, cab companies must notify the city of price changes. In Madison, companies must notify the city of fare changes 28 days before the changes are effective.

In addition to basic safety regulations and price information, many governments employ two other types of regulation: price regulation and entry regulation.

Price Regulation

Many jurisdictions not only regulate the provision of pricing information, they also directly regulate prices. Some cities create a maximum ceiling fare and allow companies to set prices below the ceiling, while others set uniform fares for all operators. The clearest potential justification for fare regulation occurs in the cruising market. Potential customers are in a relatively weak bargaining position on price during peak demand periods or at other times when demand is inelastic, such as at night or during bad weather.[7] Consumers can, therefore, be subject to a “hold up” problem. In practice, the posting of fares without directly regulating them serves to establish consistent fares for taxi service on a per company basis.

Entry Regulation

Many cities and other units of government directly control the number of cabs that can operate in their jurisdictions. The impact of this control on the number of cabs plying their trade can be most directly observed when jurisdictions change to open entry. Roger Teal and Mary Berglund found that in the United States "the size of the taxi industry in cities freed of entry control has increased by at least 18 percent in every case, and usually by one-third or more."[8] The evidence from other countries that have deregulated entry is similar. For example, when Ireland dropped entry restrictions in October 2000, the number of cabs rose from around 4,000 to about 12,000 by late 2002.[9] New Zealand also experienced a major increase in both cabs and cab companies after entry deregulation in 1989, as did Sweden in 1991.[10] There is also evidence of many illegal service providers in jurisdictions with entry restrictions (responding to the unmet demand).

The impact of entry restrictions is also evident in the sale price of taxi licenses, or "medallions." The more the licensing process restricts the number of cabs below the number that would operate in an open market, the greater will be the license price if it is transferable.[11] In large cities such as New York, where the number of cabs is severely restricted, medallions can sell for several hundred thousand dollars.

Entry Regulation to Ensure Safety?

A market failure argument related to safety may support market entry regulation. Some analysts argue that unrestricted entry results in a flooded market with intense competition, and taxi operators have little incentive to follow safety regulations (because of asymmetric information and the inability of the city to enforce safety requirements and inspect a large number of taxicabs). Profit rates of operators are driven down as the number of cabs increases.[12] By limiting the number of taxis on the road, it is argued that cities are better able to ensure the safety and reliability of taxicabs, and can hold taxi owners responsible for safety violations.

The Australian Productivity Commission considered this argument and concluded that it is difficult to justify restrictions on entry in order to address safety concerns, because safety can be ensured more effectively through direct regulation and taxi inspections.[13] Assuming that a city has (or can generate) sufficient resources to monitor and enforce taxi safety, barriers to entry are likely to be a relatively inefficient method to ensure safe taxi service.

The Distributional Argument for Price and Entry Regulation

Although this analysis has focused on the efficiency-related arguments for government intervention, it is important to note that sometimes a distributional, or equity, argument is put forward. The rationale takes the following form: price regulation (or, more indirectly, entry restrictions) ensures that operators make excess returns during high-demand periods and at high-demand locations that enables them to cross-subsidize and provide service during low-demand periods (for example, the middle of the night) and at low-demand locations (for example, low-density neighborhoods). The need for such cross-subsidy mechanisms, of course, can also be directly linked to meeting service requirements.

Why Governments Regulate: The Effects of Concentrated Interests

Limiting the number of taxis within a jurisdiction concentrates the interests of taxi operators and leads to a situation in which city officials are less responsive to the interests of taxi riders. D. Wayne Taylor offers reasons why certain forms of taxi regulation might persist even though they are inefficient.[14]

The first reason he calls the "producer protection" hypothesis. ("Producers" can be extended to include suppliers of labor, capital, raw materials, and the like.) Producers seek to transfer income from consumers to themselves. This is facilitated in the taxi market by the fact that taxi operators are a small group that receives significant benefits from the status quo, while each individual taxi user each bears only a small cost. Because of the relatively high fixed cost of lobbying and other actions to change the current system, individual users have little incentive to engage in such costly activity for the potentially small individual benefits they would receive under a reformed system. Potential providers of taxi services are also a diffuse interest group because of uncertainty regarding who might receive a license. (In the policy analysis literature, this is often referred to as the "silent loser" problem.)

The second reason Taylor calls the "regulator protection" hypothesis. The argument here is that regulation persists when it benefits the regulator, or government generally. These benefits could include political donations, bureaucratic empire building, and increased government revenues.

The third reason he calls the "social welfare" hypothesis. Relatively invisible industry regulation is often a politically low-cost means of providing cross-subsidies to citizens. Presumably, this would mainly serve the interests of particular political constituencies (hence, it is a variant of the regulator protection hypothesis), but producers would go along if regulatory framework provides them with sufficient expected financial benefits.

It is extremely difficult to determine whether any of these "government failure" hypotheses are true in any particular policy context. It is not in the interest of anyone to advertise, or even admit, such motives. In addition, it is plausible that politicians have been convinced by the market failure arguments and distributional concerns described above. (Note that this could be the case even if one believes that the regulatory instruments they have chosen to use are not particularly well designed to meet the posited problems.) There does, however, appear to be some evidence in support of the producer protection hypothesis for Madison.[15] The city has been more responsive to the interests of current operators than to the interests of both consumers and potential entrants to the taxi market.

Madison's Taxi Market

Madison, Wisconsin, is a city of approximately 220,000 people. It is the state capital and home to a large university with 40,000 students. Madison is surrounded by four lakes, and the center of the city is located on a narrow strip of land between two of the lakes.

Madison's unique geography affects the taxi market because all cross-town traffic must pass through one of three main roads that connect the east and west sides of the city. Because of the geographic separation of people and places, taxi operators generally do not wait at taxi stands (with the exception of stands at the airport, bus station, and downtown hotels), and taxi passengers must generally phone ahead to request taxi service. Consumers usually cannot hail a cab on the streets in Madison.

Madison's Taxi Companies

To request service, taxi riders must phone one of three companies offering taxi service in Madison. Taxi drivers must be employed by a cab company to operate a cab in Madison. Each company serves passengers traveling within Dane County, where Madison is located. Each cab company offers a dispatch service to connect riders with taxis. The three cab companies offering service in Madison and Dane County are Union Cab, Madison Taxi, and Badger Cab.

Union Cab has provided service since 1979 and is a worker-owned cooperative. Cab rides with Union Cab are unshared and direct, and fares are calculated by meter. Union Cab is busiest during weekday rush hours, from 7–9 a.m. and 4–6 p.m. It operates 40 cabs during these hours. Weekend nights are its second busiest time, and 30 cabs are on the road during these hours. Union Cab has 120 drivers and owns 58 cars. Based on 2001 data, the company accrues 34.8 percent of revenue in the taxi market. Dividing the total number of passenger trips by the total number of trips in the market suggests that Union Cab has a 28.4 percent market share.[16]

Pricing: For one rider, a three-mile trip with a three-minute delay costs $9.75. Union Cab monitors its expenditures by maintaining a yearly budget; however, fuel costs often determine if profits are made in any given year. For the past decade, Union Cab rates have increased about once every two years.[17] Union Cab operated in 2001 at a loss and raised fares in 2002.

Madison Taxi is privately owned and began offering cab service in Madison in 1986. In direct competition with Union Cab, Madison Taxi offers unshared, metered rides. Madison Taxi is busiest on weekend nights, when 18 to 30 cabs are scheduled. On weekdays, 12 to 15 cabs are available. In total, 75 drivers are employed and 50 cars are owned. Madison Taxi brought in 27.3 percent of revenue in 2001.[18] The same estimation method used for Union Cab indicates that Madison Taxi has a 21.6 percent market share.[19]

Pricing: For one passenger, a three-mile trip with a three-minute delay costs $9.50. Madison Taxi uses cost forecasting when considering fare increases.

Badger Cab was founded as a cooperative business in 1946 but is now privately owned. Badger Cab, unlike Madison Taxi and Union Cab, charges fares based on a zone system and offers shared-ride service. Rates between locations are predetermined, but there is an additional charge of $1.00 per passenger in a group. Badger Cab estimates its rates to be 30 to 50 percent lower than either that of Union Cab or of Madison Taxi, but suggests passengers allow an extra 10–15 minutes for trips. Badger Cab experiences the highest demand on weekend nights, when 35 cabs are available. On weekdays, the same number of cabs is available. Badger Cab employs 125 drivers and owns 43 cars.[20] Drivers pay a flat fee to the company to lease a cab for their shift, and drivers keep all the money they take in. Based on company revenue (not driver profit), Badger Cab has 37.8 percent of total revenue in the market. Its market share is 49.9 percent.[21]

Pricing: A three-mile trip with one person in the cab costs $5.50, and there is no charge for a delay. Badger Cab's shared, zoned cab service is less expensive than its competitors and a large portion of its riders is relatively low income. This company has not raised fares in the past two years.[22]

The Policy Goals of Taxi Regulation

Madison's Traffic Engineering Department seeks "to provide and manage the environmentally sensitive, safe, efficient, affordable, reliable, and convenient movement of people and goods" around the city.[23] Taxi services are deemed to be an essential part of fulfilling these goals. Because city buses have limited routes and cease operation at midnight, taxis provide transportation services not otherwise available. When considering changes in taxi regulations, the city should consider the effects on community-wide transportation access.

A successful system of cab regulations should promote efficiency, equity, and favorable (or at least neutral) fiscal effects and be politically feasible. An efficient cab market would mean that there are cabs available to passengers who demand cab service at the lowest possible sustainable fares. Equity has two impact categories: the opportunity for new cab operators to enter the industry and the community-wide availability of service to consumers. All qualified entrepreneurs should have access to the market. It is also important that all areas of a city be served by taxi companies (note that we regard this as an equity issue rather than an efficiency issue). Fiscal effects are another essential consideration because changes in policy can affect costs and revenues to the city. Finally, political feasibility is likely to play a role in the successful adoption of any new policy. If new regulations impose significant costs on politicians and concentrated interest groups, then they are less likely to be adopted. As demonstrated below, however, the history of taxi regulation in Madison does suggest willingness to change even when it may not be in the interest of incumbent companies.

Regulation and Deregulation in Madison

The City of Madison took its first steps to deregulate the taxi market in 1979 when the Common Council agreed to revoke the city ordinance that restricted the issuance of taxi permits. Before this decision, the city would approve licenses only to sustain a taxi-to-population ratio of one taxi per 1,000 residents.[24] Since 1979, the city has not considered the ratio of taxis per capita in licensing decisions (there is currently one taxi per 1,400 residents).

The regulation of fares was changed in the early 1980s. The Common Council agreed to remove maximum price limits on fares and changed its policy to require that fares be posted in cabs. Taxi companies and drivers cannot charge an amount different than the posted fare. Since this change in 1982, taxi companies in Madison have been free to determine their own fares. Current ordinances require only that companies notify the city of price changes 28 days before they take effect.

Taxi regulation in Madison did not receive much attention again until the late 1990s, when scholars and researchers argued that Madison's regulations limited entrepreneurship and created unfair barriers to entry.[25] An editorial in an area newspaper illustrated the problems with city regulations that forced new applicants for taxi licenses to prove that their service would address "public convenience and necessity," and also attacked the city's "political [licensing] process." The article proposed that deregulation of Madison's market would encourage entrepreneurship and create jobs for low-income and minority residents.[26] In response to an increased public concern after these articles, the Common Council formed the Ad Hoc Subcommittee on Taxicab Deregulation. This group investigated the efficiency and equity of taxi regulations, and concluded that some policies were not meeting city goals and should be revoked.

After the subcommittee's report was released in August 2000, the Common Council decided to make several changes to regulations in Madison. In October 2000, the Common Council eliminated the need to prove public convenience and necessity, lowered the initial licensing fee from $1,500 to $1,000, and eliminated public hearings for new licensing.

Current Policy in Madison

The following are the current taxi regulations and a discussion of how they are related to the goals of Madison's transportation policy:

1. *Fees:* The city requires a $1,000 licensing fee for new cab companies in Madison. In addition, there is a yearly renewal

fee of $500. Vehicle permits and driver permits must also be obtained from the city. Costs are $40 per vehicle and $25 per driver. The company fees make it less affordable for individuals and small companies to enter the market. Regulations requiring adequate taxi insurance increase the cost of operating a single cab to over $4,900.

2. *Service:* Several requirements address the equity of service in Madison. All cab companies in Madison are required to provide service 24 hours a day, 7 days a week (24/7 rule), and they must serve all parts of the city. No taxi can refuse service to a 24/7 customer traveling within the city limits. These regulations are intended to ensure that consumers are able to receive taxi service at any time, and they prohibit drivers and companies from discriminating against certain areas of the city. At the same time, these regulations, especially the 24/7 rule, prevent single drivers from operating single-cab companies in Madison.
3. *Other regulations:* Several other types of regulation also affect the taxi market in Madison. Safety regulations enforced by the city include vehicle inspections and driver-background checks. The city retains the right to revoke or suspend driver, vehicle, and operating licenses if any city regulations are violated. To protect against tired drivers, the city prohibits taxi drivers from driving more than 12 hours per shift and requires eight hours of time off between shifts. The city also requires companies to post cab fares and to notify the city of their rates. The city must receive notice of any rate changes 28 days before they take effect.

To enforce current regulations and ensure safety, the city divides the responsibilities of taxi regulation among several departments. The Traffic Engineering Department works on cab inspections; the Police Department approves taxi driver licenses; the City Clerk's Office approves company applications for renewal and handles new company applications; the city attorney is responsible for any litigation resulting from the taxi industry; and the Department of Weights and Measures checks taxi meters each year. The Common Council must approve any changes to city taxicab regulations. The most recent city survey of costs estimated that $33,000 was spent on regulating the taxi market in 1998.[27]

Although some of Madison's current taxi regulations are necessary in order to ensure safety standards, existing regulations continue to act as obstacles to entrepreneurship. High entry costs limit the participation of low-income residents and minorities in the taxi industry. Because the same three companies have offered service in Madison for the past 17 years, lack of competition in the market probably causes taxi fares to be higher than they would otherwise be, and this increased cost negatively affects all taxi customers in the city.

Moreover, the 24/7 rule, citywide service requirement, and the 12-hour driving limit make it impossible for individuals to operate single-cab companies, and they make it extremely difficult for small companies to enter the market because of the high start-up costs of operating at all times. By limiting the number of operators in the market, these regulations permit fares to be set above competitive levels. It appears that certain regulations, especially the 24/7 and citywide service requirements, are not fulfilling the efficiency or equity policy goals as well as they could. High entry costs block entrepreneurs wishing to enter Madison's taxi market.

In the past few years, taxi regulation has been hotly debated in the Common Council. A single prospective taxicab operator has repeatedly petitioned the council for an operating license, but the council has repeatedly rejected his application for a license. Though current companies are not directly involved in the licensing process, Badger Cab and Madison Taxi lobby conservative council members to protect the economic interests of existing companies by blocking market entry. Employees of Union Cab also lobby the council and are active in Progressive Dane, the liberal caucus in the Common Council.

Progressive Dane seeks to block entry to the taxi market to protect current taxi drivers.[28]

In the past year, after receiving a consumer complaint, the City Attorney's Office began researching the possible conflict between Madison's taxi regulations and federal antitrust laws. Currently, attorneys representing a city resident are preparing an antitrust case against the City of Madison, and it is expected to be filed in federal court within three months.[29] This case calls into question whether city regulations have created a de facto taxi cartel within the city.

Wisconsin State Statute 133.01 states, "State regulatory agencies shall regard the public interest as requiring the preservation and promotion of the maximum level of competition in any regulated industry consistent with the other public interest goals established by the legislature." The current regulatory regime within Madison may violate this statute because city regulations fail to promote "the maximum level of competition" as required, and the city has made decisions that define public interest goals when the legislature legally must determine such goals. If the Common Council does not change its taxi regulations and the court rules against the city, then the Common Council will be forced to change the current regulations.

Eliminating the 24/7 Rule

As already discussed, the 24/7 rule is the most limiting regulation in the Madison taxi market because, in combination with the 12-hour driving limit, it creates a minimum number of drivers who must be employed by a company to meet city standards. This eliminates the opportunity for single drivers to operate cabs unless they are employed by one of the three existing companies. This policy alternative would remove the 24/7 rule while maintaining all other regulations. We compare its performance to the status quo in terms of each of the goals that we have posited as appropriate:

Efficiency

With the 24/7 rule eliminated, the number of taxi operators would increase. Niche markets would be created as new operators begin to supply cabs during the hours they prefer, whether rush hour, weekend nights, or mid-days. Because Madison does not regulate fares, it is likely that fares are currently at or above the competitive level, so that an increase in the number of taxis operating would result in a decrease in fares.

As discussed earlier, under the current policy few hailing taxis are available, and consumers must phone ahead to request service. An additional efficiency benefit of an increased number of taxi operators would be an increase in the hail market for cabs, which would also allow for decreased waiting time for consumers in those areas where hailed taxis are available. Thus, removal of 24/7 the rule would increase the efficiency of the taxi market.

Equity

The removal of this regulation would directly increase market access because single operators would now be able to offer service. However, start-up costs are still high, so it would still be difficult for low-income residents to establish taxi businesses. Because the citywide rule is still in effect, new operators would be required to pick up any passengers and could not deny passengers service to any part of Madison. Assuming this regulation is enforced, there would be no change in the geographic availability of dispatch service when compared to the status quo policy. Thus, removal of the 24/7 rule would increase equity-of-access for potential operators in the taxi market, but not substantially change equity-of-access to service by consumers.

Fiscal Effects

License fee revenue would increase as more taxis are granted licenses. Regulatory costs would also increase as the number of cabs increases because more time would have to be allotted for city workers to inspect new cabs and to ensure that new drivers meet city safety regulations. Under the retained fee structure, removal of the 24/7 rule would probably result in somewhat higher net costs to the city.

Political Feasibility

As the demand for taxi service appears to be price inelastic, the larger number of taxis available for hire would likely decrease revenue per cab for the companies already in the market, eliciting opposition from them and their drivers. Consequently, a majority of the Common Council would almost certainly continue to oppose elimination of the 24/7 rule under current circumstances. However, the filing of a suit in federal court may create an opportunity to obtain a majority in the Common Council for the removal of the 24/7 rule, because some members may want to avoid a court-imposed change to the current regulatory structure. Thus, although the status quo currently is highly rated in terms of political feasibility, removal of the 24/7 rule might become politically feasible if a successful court challenge is credible.

Overall, removal of the 24/7 rule better promotes the substantive goals of efficiency and equity than the status quo policy. The status quo enjoys at most a small advantage in terms of fiscal effects, and either a large or small advantage in terms of political feasibility, depending on the Common Council's perception of the probability of success of the likely court challenge to the current regulatory regime.

Recommendation

Removing the 24/7 rule would greatly reduce the regulatory barrier to entry into the Madison taxi market. Lower barriers would increase the number of cabs operating, which in turn would result in lower fares and reduced waiting times. It is possible that the prospect of a successful legal challenge to the current regulatory system will create an opportunity for gaining a majority of the Common Council in favor of elimination of the 24/7 rule. As the elimination of the 24/7 rule is desirable relative to the status quo in terms of efficiency and equity-of-access to the market, we recommend that, as mayor, you seize this opportunity to propose this regulatory change to the Common Council, before a potentially costly court case forces a change.

Appendix: Are Madison's Fares Too High?

The *Taxi Division Fact Book* gives comparable taxicab fare data for 252 cities around the United States. Madison's cab fare of $9.50 is 34 percent higher than the national average of $7.08. Of the 252 cities in this study, 75 have populations in a range between 100,000 and 500,000. As the City of Madison has a population of approximately 200,000, but the county population (which cab companies in Madison serve) is closer to 400,000, these cities provide the most appropriate comparison for Madison. Madison's fare is more than $2 above the sample average and 34 percent higher than the comparison group.

To allow for a better comparison, these cities were contacted to determine if and how they regulated taxi fares. Of 75 cities, 11 had company-specific fare regulation (cities approve or reject each operator's fare proposal individually), 49 cities had uniform fare regulations (the city sets a single fare that all operators must follow), and 13 cities did not regulate cab fares. Mean fare information is listed below:

Cities with Population 100,000–500,000

Type of Regulation		*Average Fare*	*Standard Deviation*
Company-specific fare regulation	(n = 11)	$7.11	$1.58
Uniform fare regulation	(n = 49)	$6.98	$1.78
No regulation	(n = 13)	$7.58	$1.20
All	(n = 73)	$7.10	$1.64

Madison's fares are unregulated, and when compared with other cities with unregulated fares, Madison's fare of $9.50 is 25 percent higher and approximately $2 above the average for cities without fare regulation. Furthermore, when population of the sample is limited to 150,000 to 250,000 (n = 27), Madison's fare is 38 percent higher than the average of $6.89.

Postscript: Technology Makes the 24/7 Rule Irrelevant

Neither the threat of legal challenge nor the logic of analysis resulted in reconsideration of the 24/7 rule in the dozen years following the analysis. In 2010 a fourth taxi company, Green Cab, was able to open at sufficient scale to meet the 24/7 rule, offering electric/gas hybrid taxis fitted with bike racks and zone rather than meter fares. Although the established taxi companies, which were providing about 1.5 million rides per year, raised concerns about the adequacy of the market to support a fourth company, all four companies proved to be commercially viable. Nonetheless, concerns about the timely availability and cost of service continued to be raised by taxi customers and the 24/7 rule still excluded single-driver cab companies from entering the market.

Nationally, transportation network companies (TNCs) that used smartphone technology to connect passengers to the drivers of personally-owned vehicles were beginning to operate in major cities. In 2014, the two most prominent TNCs, Uber and Lyft, began operating in Madison. After unheeded warnings to TNC drivers that they were operating in violation of the city taxi ordinances, undercover police issued citations to two drivers. In January 2015, the city attorney issued 42 citations to Uber seeking $42,000 in fines—over half of the citations were based on monitoring of the Uber app by an employee of one of the licensed taxi companies. Uber refused to pay the fines and sought to have the citations adjudicated in U.S. District Court so it would be protected from "local prejudice." (The District Court rejected the argument and remanded the case back to Municipal Court.)

The Milwaukee Common Council took a different approach to the TNCs. Beginning in 1992, it imposed a cap of 320 taxi permits, which resulted in transfers of existing permits at prices as high as $150,000. In May 2013, a state court ruled that the cap violated the state constitution, a decision that was unsuccessfully challenged in federal court by several of the taxi companies. The Milwaukee Common Council initially responded in February 2014 by issuing an additional 100 permits, and in July of 2014 it adopted an ordinance that eliminated the cap altogether and allowed individuals to obtain permits, opening up the market both to independent and TNC drivers. It maintained safety regulations for traditional taxis but did not impose them on TNC vehicles. Several of the existing taxi companies unsuccessfully sought an injunction against the ordinance on the grounds that it violated their right to substantive due process, because it did not take account of their property interest in the secondary market for permits and that the different treatment of traditional and TNC taxis violated the equal protection clause of the U.S. constitution.[30]

In the meantime, there was debate within the Madison Common Council about the appropriate response to the TNCs. A few council members argued for adoption of an ordinance that, like the one adopted in Milwaukee, would accommodate the TNCs, while others wanted to impose requirements similar to those governing taxi companies. The threat of state legislation that would preempt local regulation of TNCs prompted action by the council. On March 31, 2015, Assembly Bill 143 was introduced in the state legislature to impose uniform statewide regulation of TNCs. The next day, the Common Council voted 15 to 5 to adopt an ordinance for TNCs. It maintained the 24/7 rule, required TNCs to offer service in all parts of the city, imposed stronger insurance requirements, and placed a ban on "surge pricing," through which service is offered during peak-demand times at higher fares. A spokesperson for Uber stated that it would not be able to operate under the new ordinance.[31]

Opponents of the ordinance argued that it would make the adoption of Assembly Bill 143 even more likely. The bill had already drawn lobbying from 34 organizations: in addition to Uber, a number of insurance companies and business groups, including the Wisconsin Tavern League, reported lobbying for the bill; the Wisconsin Association of Taxicab Owners, the League of Wisconsin Municipalities, the City of Madison, the City of Milwaukee, and several transportation organizations lobbied against it. The new Madison ordinance was probably irrelevant—Assembly Bill 143, with only a few amendments, passed the Assembly 79 to 19, received concurrence from the Senate, and was signed into law on May 1, 2015.

Assembly Bill 143 requires TNCs to register with the Department of Safety and Professional Services. Drivers, who may only offer services through a licensed TNC, must pass a background check and may not be a convicted sex or habitual traffic offender, or have been convicted of operating a vehicle while under the influence of alcohol. The driver and the TNC must maintain automobile liability insurance. The TNC must have a non-discrimination policy, implement online complaint procedures, suspend drivers subject to complaint until the complaint is resolved, and transmit a photo of the driver and his or her license plate number to the passenger. Local governments cannot impose additional regulations on the TNCs (Wisconsin Statutes, chapter 440.40).

The statewide law seems to provide reasonable protections for passengers from driver malfeasance through the background checks and limitations on drivers, insurance provisions, and the electronically provided driver information. The insurance provisions should also provide some protection against unsafe vehicles. It is not clear, however, if the non-discrimination policies adopted by TNCs will ensure area-wide provision of services, as it will depend not only on their content but also on how vigorously they are enforced in view of the independence of the drivers. Of course, the law does not deal with concerns increasingly being raised about the exploitation of drivers who do not have the rights associated with the status of employee. It also does not require municipalities to open their markets to individual drivers who want to offer services independently of a TNC.

The events in this postscript suggest several general points that policy analysts dealing with regulatory issues should keep in mind. First, as noted in the analysis, rules addressing market failures, such as information asymmetry and platform intermediation in the case of taxicab service, may create barriers to entry that allow established firms to reap rents. Sometimes barriers are obvious, as with Milwaukee's strict limit on taxi permits, but they may be less obvious, as with Madison's 24/7 rule. Second, the regulated firms have a direct and strong interest in preserving such rules to protect their rents. The taxi firms were able to lobby the city councils in Madison and Milwaukee to prevent rule changes, and they attempted to overturn the state court ruling against the Milwaukee permit limit in federal court. Third, the existence of rents creates an incentive for entrepreneurs to create organizational and technological innovations to get around the barriers that keep them from competing for the rent. The spread of smartphones created an opportunity for TNCs to create markets for "ride sharing." Other examples include long-haul trucking to challenge railroad rents, electric hand dryers to challenge organized crime rents in the towel service industry, dish television to challenge cable company rents, and reimportation services to challenge patent-protected pharmaceutical rents. Finally, those affected by rent-creating rules can sometimes change outcomes by changing arenas. Individuals seeking access to the taxi market in Milwaukee by challenging its cap on taxi permits won in state court; taxi companies tried to block the change in federal court. Uber and others supporting its service model were unable to overturn the existing regulatory barrier in Madison through the Common Council, so they moved the issue to the state legislature where they obtained favorable rules and a preemption of local rules that would circumvent them.

The Relationship between Market and Government Failures

The rationale for the analysis of both market and government failure in almost all policy contexts is predicated on the centrality of efficiency to policy analysis. Figure 9.1 summarizes how to approach the question of efficiency through a consideration of the relative contribution of market and government failure to any observed inefficiency.

As Figure 9.1 indicates, the first step involves deciding whether there is market failure. This requires that one decide whether a market operates to accommodate individual preferences. This in itself may be a difficult decision, as there is a complex continuum from free, unfettered markets to the complete absence of markets. We recommend the following working rule: if prices legally exist as signaling mechanisms (no matter how extensively regulated), treat the situation as if it involved an *operational market*. Where prices are not legally permitted (so that only black market transactions take place), start with the assumption that the market is not operational. Of course, if, on closer inspection, the transaction costs in a black market appear to be the result primarily of enforcement, or the absence of legal contract enforcement, then legalization itself may be adequate to create what we call an operational market.

If the market *is* operational (the bottom half of Figure 9.1), then one should next consider the theory, evidence, and facts relevant to market failure. A close familiarity with the content of Chapters 5 and 6 provides a basis for such an investigation. If neither theoretical arguments nor empirical evidence suggest market failure, then it is reasonable to assume that the existing market allocates scarce resources in the most efficient way. Even though there is an operational market, however, there may still be extant government interventions that create inefficiencies; in other words, there may be situations where the elimination of government interventions would improve market allocations.

The housing market illustrates some of these considerations. In any given jurisdiction, there may or may not be an operational market. Though rent control regimes in some jurisdictions are so binding that markets cannot function, in most jurisdictions in most Western countries there are operational housing markets. Economists typically conclude that housing markets are not subject to serious market failures,[32] although they do

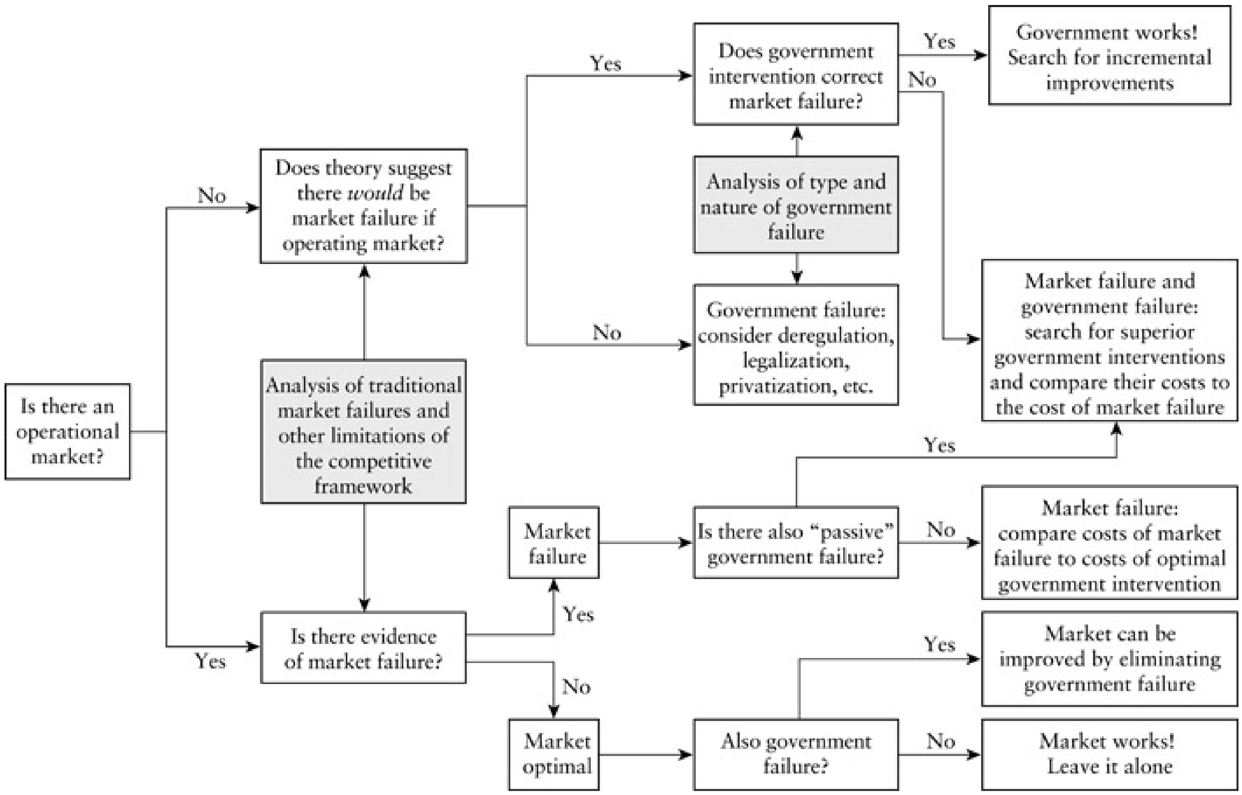


Figure 9.1 A Procedure for Linking Market and Government Failure to Policy Interventions

acknowledge the seriousness of distributional problems.[33] But they also frequently argue that housing prices are artificially inflated by inappropriate government interventions such as large-lot zoning. In this case the operational market can be made more efficient by removing the restrictions. If there is no evidence of market failure and no government interventions, then the market works from the perspective of efficiency.

If theory and evidence suggest market failure in an operational market, then there is a prima facie case for government intervention. Furthermore, one could argue that the presence of market failure is evidence that there must also be government failure: the failure to correct market failure. The failure of government to intervene is best described as *passive government failure*. It can include outcomes that are attributable to the government not diagnosing market failures correctly as well as situations in which the lack of intervention derives from more concrete causes, such as the active influence of organized interest groups that successfully block efforts to correct market failure. For example, the government may either fail to recognize some industrial emission as an externality problem or, recognizing it, fail to intervene because a polluting industry is able to lobby successfully to block taxes or other policies that would internalize the externality.

If a market *is not* operational, then one should work through Figure 9.1, asking if a market could operate efficiently if facilitated by an appropriate framework; in other words, the concept of market failure has relevance here even though no market is currently operational. As this is a "what if" question, direct evidence usually does not exist to inform the answer. In these circumstances, one must draw upon the theory of market failure, evidence available from other jurisdictions (for example, other cities, states, and countries), or analogous problems.

If one suggests that a market could operate without serious flaws, then one can assume a priori that significant efficiency losses are associated with the existing government intervention, whether it be an outright prohibition of market activity, a displacement of private supply by public provision, or simply a failure to establish property rights or other rules needed to facilitate private transactions. Thus, if there is no operational market and no evidence of market failure, one can logically conclude that there is prima facie evidence of government failure. Evidence indicating the cause of government failure, such as interest group rent seeking, further bolsters the case. This line of analysis leads to the examination of such options as deregulation, privatization, and legalization, but keep in mind that real policy alternatives are rarely as generic as these categories might suggest.

Finally, there may be persuasive evidence that there would be market failure if a market were allowed to operate. (This was a concern raised about kidney markets in the analysis presented in Chapter 1.) The question, then, becomes whether the extant government intervention is efficient. If the intervention is of the appropriate kind, then the conclusion must be that "government works." The only question is whether this intervention can be improved through better implementation or management. Though these incremental questions can have great practical importance, they do not raise the same strategic policy issues. If the intervention appears inappropriate, then the working assumption is that both market failure *and* government failure are relevant. This conclusion indicates the desirability of explaining why the current government intervention fails and to search for alternative interventions that may be superior. In both the salmon fishery example in Chapter 16 and the taxi regulation example in this chapter, market failures require government interventions but the interventions evidence government failure.

If one does not identify a government failure, then one is concluding that the existing government intervention corrects the market failure to the greatest extent practical. In other words, if it is not possible to find changes in current policy that could increase efficiency, don't panic—the current policy is at least efficient. Therefore, the question is whether other values are at stake.

Notice the relative simplicity of the analysis if the only goal is the maximization of efficiency. In the absence of market failure, government intervention will almost always be inefficient. Multiple goals make such a simple analysis inappropriate, however. It may be worth bearing losses in net social surplus (in other words, enduring some inefficiency) in order to achieve other social goals.

Most importantly, decision makers usually care about distributional considerations, whether in terms of socioeconomic, racial, or other status characteristics. One should explicitly decide whether distributional concerns are relevant to your particular policy problem. While in most situations one will be familiar with client preferences, other times the client may be unaware of the facts and evidence relating to particular target groups. In the latter case, one may wish to argue to the client that greater equity should be a goal. After making a decision on equity, one should consider other goals and constraints. For instance, does the client expect that public revenue would or should be generated? Thus, the absence of market failure is not, in and of itself, determinative. Government intervention that interferes with well-functioning markets is not necessarily undesirable, as other goals may justify the losses in efficiency.

Before considering how to deal with multiple goals, it is worth reiterating that analysis of market and government (nonmarket) failure plays a vital role in *framing* subsequent solution analysis. John Brandl, an economist and a former member of the Minnesota State Senate, succinctly summarized the value of such an approach for the decision maker:

> To view an issue as an instance of market (or nonmarket) failure is to transform what was bewildering into a solvable problem. An immense variety of issues can be approached in this fashion. Tuition policy becomes pricing policy, airplane noise is an externality, a utility is a natural monopoly, research yields public goods. This application of economic theory is much more than a translation of English into jargon. In each case the economist has advice to offer aimed at rectifying the wrong implicit in market imperfection. (Currently a new economic theory … is being created to explain … nonmarket institutions.)[34]

Conclusion

Theories of market and government failure are valuable for framing public policy problems. Sometimes, as the analysis of Madison's taxicab regulation illustrates, appropriate policy alternatives naturally present themselves. Often, however, a wide range of policy alternatives could reasonably be considered. The next chapter provides a catalogue of generic policy alternatives that can be used as starting points for crafting policy alternatives appropriate for specific contexts.

For Discussion

1. For reasons of brevity and desired pedagogical emphasis, the Madison taxi analysis focused more on the nature of the market and government failure present than on the development and assessment

of policy alternatives. What other policy alternatives might the analysts have considered?

2. Imagine that you are asked to analyze the regulation of taxi service in a city like Madison. Your research, however, leads you to predict that elimination of the 24/7 rule would result in the absence of service during some periods. Design an alternative that permits the entry of single-driver cab companies but also guarantees 'round-the-clock' availability of service.

Notes

1 Taxicab and Limousine and Para-Transit Association, "Chart Ranking of Metered Taxicab Fares in U.S. Cities," *Taxi Division Fact Book* (Washington, DC, 2001).

2 For example, see Peter Carstensen, "Madison's Current and Proposed Taxi Regulation: Bad Public Policy and an Open Invitation to Litigation," Statement for Madison Transit and Parking Commission Meeting on September 12, 2000; and Sam Staley, "Madison's Taxi Regulations Stifle Innovation, Competition," *Capital Times* August 4, 1999.

3 Andrew Fisher, "The Licensing of Taxis and Minicabs," *Consumer Policy Review* 7(5) 1997, 158–61.

4 This rationale is explained in detail in Australian Productivity Commission, *Regulation of the Taxi Industry* (Canberra: Ausinfo, 1999), pp. 9–12.

5 Paul Stephen Dempsey, "Taxi Industry Deregulation and Regulation: The Paradox of Market Failure," *Transportation Law Journal* 24(Summer) 1996, 73–120.

6 Robert D. Cairns and Catherine Liston-Heyes, "Competition and Regulation in the Taxi Industry," *Journal of Public Economics* 59(1) 1996, 1–15, at 5. They also argue that providers may prefer price regulation in the taxi-rank market because there the bargaining advantage is with customers.

7 Cairns and Liston-Heyes provide an efficiency rationale for pricing regulation. For more on demand elasticities for taxis, see Bruce Schaller, "Elasticities for Taxicab Fares and

Service Availability," *Transportation* 26, 1999, 283–97.

8 Roger F. Teal and Mary Berglund, "The Impact of Taxicab Deregulation in the USA," *Journal of Transport Economics and Policy* 21(1) 1987, 37–56.

9 Department of Transport (Eire), "New Regulations Requires Taxis to Issue Printed Fare Receipts to Customers," www.irlgov.ie/tec/press02/Sept1st2002.htm.

10 P. S. Morrison, "Restructuring Effects of Deregulation: The Case of the New Zealand Taxi Industry," *Environment and Planning A* 29(3), 1997, 913–28; and Tommy Garling, Thomas Laitila, Agneta Marell, and Kerstin Westin, "A Note on the Short-Term Effects of Deregulation of the Swedish Taxi-Cab Industry," *Journal of Transport Economics and Policy* 29(2) 1995, 209–14.

11 C. Gaunt, "Information for Regulators: The Case of Taxicab Licence Prices," *International Journal of Transport Economics* 23(3) 1996, 331–45.

12 Paul Dempsey found that this has occurred in some U.S. cities after deregulation. Demsey, "Taxi Industry Deregulation and Regulation."

13 Australian Productivity Commission, *Regulation of the Taxi Industry*, 12.

14 D. Wayne Taylor, "The Economic Effects of the Direct Regulation of the Taxicab Industry in Metropolitan Toronto," *Logistics and Transportation Review* 25(2) 1989, 169–83.

15 Carstensen, "Madison's Current and Proposed Taxi Regulation."

16 City of Madison Traffic Engineering Department, Paratransit Service Survey, 2001.

17 Carl Shulte, General Manager, Union Cab. Telephone interview, November 11, 2002.

18 Rick Vesvacil, General Manager, Madison Taxi. Telephone interview, November 14, 2002.

19 City of Madison Traffic Engineering Department, Paratransit Service Survey, 2001.

20 Tom Royston, Operations Manager, Badger Cab. Telephone interview, October 31, 2002.

21 City of Madison Traffic Engineering Department, Paratransit Service Survey, 2001.

22 Ibid.

23 Traffic Engineering, "Mission, Goals, and Objectives," www.ci.madison.wi.us/transp/trindex.html.

24 U.S. Department of Transportation, *Taxicab Regulation in U.S. Cities*, Vol. 2. Document DOT-I-84-36, October 1983, at 39.

25 Samuel R. Staley, Howard Husock, David J. Bobb, H. Sterling Burnett, Laura Creasy, and Wade Hudson, "Giving a Leg Up to Bootstrap Entrepreneurship: Expanding Economic Opportunity in America's Urban Centers," *Policy Study 277*, February 2001, www.rppi.org/ps277.html.

26 Staley, "Madison's Taxi Regulations Stifle Innovation, Competition," 9A.

27 William Knobeloch, Operations Analyst, Traffic Engineering Department, City of Madison. Telephone interview, November 7, 2002.

28 Peter Carstensen, Law Professor, University of Wisconsin–Madison. Interview, December 18, 2002.

29 Ibid.

30 *Joe Sanfelippo Cabs Inc. et al. v. City of Milwuakee*, U.S. District Court, Eastern District of Wisconsin, Case No. 14-CV-1036, September 12, 2014.

31 Bryna Godar, "Madison City Council Approves Ordinance to Regulate Uber, Lyft," *Capital Times*, April 1, 2015.

32 It does appear that out-of-town buyers may be subject to information asymmetry. See Alex Chinco and Christopher Mayer, "Misinformed Speculators and Mispricing in the Housing Market," *Review of Financial Studies* 29(2) 2016, 486–522.

33 Michael J. Wolkoff, "Property Rights to Rent Regulated Apartments: A Path towards Decontrol," *Journal of Policy Analysis and Management* 9(2) 1990, 260–65.

34 John E. Brandl, "Distilling Frenzy from Academic Scribbling," *Journal of Policy Analysis and Management* 4(3) 1985, 344–53, at 348

Part III
Conceptual Foundations for Solution Analysis

10
Correcting Market and Government Failures

Generic Policies

Our discussion of the ways private and collective actions lead to socially unsatisfactory conditions provides a conceptual framework for diagnosing public policy problems. We now turn our attention to policy solutions. We focus on what we call *generic policies*––the types of actions that government can take to deal with perceived policy problems. Because policy problems are usually complex and always contextual, generic policies must usually then be tailored to specific circumstances to produce viable policy alternatives. Nevertheless, familiarity with generic policies encourages a broad perspective that helps the analyst craft specific solutions.

Policy problems rarely have perfect solutions, but some policies are better than others. A primary task of the policy analyst is to identify those policies that have the best prospects for improving social conditions, as assessed in terms of specific goals and criteria. To facilitate this task, we indicate the market failures, government failures, or equity concerns that each generic policy most appropriately addresses, as well as the most common limitations and undesirable consequences associated with the use of each. In other words, we provide a checklist for systematically searching for candidate policies.

There are two caveats concerning our discussion of generic policies. First, we do not wish to imply that all policy analyses compare and evaluate across generic policies. Many policy analyses involve relatively incremental alternatives. For example, you may be asked to compare the efficiency and equity impacts of a variety of different voucher schemes. Familiarity with a broad range of generic policies helps you *know* that you are examining incremental alternatives; it may also enable you to exploit opportunities for introducing alternatives that are less incremental. Second, there is an enormous literature on each of the generic policies—too much for us to consider in any depth here. In order to prepare you to examine further the issues relating to each generic policy, however, we provide representative references to relevant literatures throughout the chapter.

We group generic policies into five general categories: (1) freeing, facilitating, and simulating markets; (2) using taxes and subsidies to alter incentives; (3) establishing rules; (4) supplying goods through nonmarket mechanisms; and (5) providing insurance and cushions (economic protection). In the following sections we consider specific policies within each of these groups.

Freeing, Facilitating, and Simulating Markets

Market failures, government failures, and distributional concerns underlie perceived policy problems. Markets offer the potential for efficiently allocating goods. Markets, therefore, provide the yardsticks against which to measure the efficiency of government interventions. If we determine that there is no market failure, then we should consider the establishment (or reestablishment) of a market as a candidate solution for our policy problem. Of course, other values besides efficiency may lead us to reject the market solution as a final policy choice. Nevertheless, letting markets work should be among the policies we consider if market failure does not appear inherent in the policy problem.

It is not the case, however, that governments can always create viable markets by simply allowing private transactions. Often, government must play a more affirmative role in enabling the market to function more efficiently. In other circumstances, although an operational market, per se, cannot be introduced, market outcomes can be simulated with the use of market-like mechanisms.

As shown in Table 10.1, we distinguish three general approaches for taking advantage of private exchange (or market-like exchange between private citizens and governments) in dealing with policy problems: freeing markets, facilitating markets, and simulating markets. The second column of the table, which emphasizes that generic policies are responses to diagnosed market and government failures, notes the perceived problems that the generic policies might appropriately address, while the third column presents the typical limitations and collateral consequences.

Table 10.1 Freeing, Facilitating, and Simulating Markets

Generic Policies	*Perceived Market Failure (MF), Government Failure (GF), Distributional Issue (DI), Limitation of Competitive Framework (LCF)*	*Typical Limitations and Collateral Consequences*
	Freeing Markets	
Deregulate	GF: Allocative inefficiency from rent seeking	Distributional effects: windfall losses and gains, bankruptcies.
	LCF: Technological changes	
Legalize	LCF: Preference changes	
Privatize	GF: Bureaucratic supply	Transitional instability.

Generic Policies	*Perceived Market Failure (MF), Government Failure (GF), Distributional Issue (DI), Limitation of Competitive Framework (LCF)*	*Typical Limitations and Collateral Consequences*
	Facilitating Markets	
Allocate through property rights	MF: Negative externalities MF: Public goods, especially open access	Distributional effects: windfall gains and losses.
Create new marketable goods		Thin markets.
	Simulating Markets	
Auctions	MF: Natural monopolies MF: Public goods DI: Transfer of scarcity rents	Collusion by bidders, opportunistic behavior by winning bidder, political pressure to change rules ex post.

Freeing regulated markets should be considered in those situations in which an effective market can be expected to reemerge with relatively minor efficiency distortions—in other words, where there is no inherent market failure. Keep in mind that there may be large windfall—or distributional—gains or losses once the current government intervention is eliminated. In addition, at this general level of discussion we are not considering the possibility that other goals, such as national security, may be relevant. The absence of markets when there is no inherent market failure suggests either government failure or as yet unaccommodated changes in preferences or technology. Accommodation may require the affirmative establishment of property rights by government, an example of facilitating markets. Finally, even when the complete withdrawal of government may be neither feasible nor desirable, there may be opportunities to simulate markets via various auction processes.

Freeing Markets

Unfortunately, a wide range of terminology is used to describe the process of freeing markets, the broadest and most popular being *deregulation*, although the broader term "market liberalization" is also used. We distinguish among deregulation, legalization, and privatization.

Deregulation

It is difficult to justify government interference with private transactions on efficiency grounds in the absence of evidence of market failure. Historically, in the United States and many other countries, in spite of this, governments have engaged in price, entry, and exit regulation of competitive markets. (We consider these various forms of regulation, themselves generic policy solutions, in a later section.) Economists have been almost uniformly critical of the regulation of competitive industries: "[I]f economics has any scientifically settled issues, one is surely that price and entry regulation in perfectly competitive industries generates economic inefficiencies."[1]

We can usually identify various forms of government failure, especially legislators responding to rent seeking by industries, along with sometimes legitimate distributional concerns, as the primary explanations for government regulation of competitive markets. In other cases, changes in technology or patterns of demand may have radically altered the structure of an industry and, therefore, the need for regulation. As we saw in Chapter 5, the natural monopoly characteristics of an industry that justify regulation may erode over time with new technology. Digital technologies have facilitated competition in many telecommunication markets, for instance. In such situations, the efficiency rationale for regulation may no longer hold at some point.

Whatever the putative rationale for regulation, deregulation almost inevitably involves complex efficiency and distributional issues. Indeed, deregulation is often better described as *liberalization* as it usually includes different, although usually less intrusive, new regulatory instruments.[2] This should not be surprising in light of our discussions of market failure in Chapter 5, related failures in Chapter 6, and rent seeking in Chapter 8. One-shot, complete deregulation may be problematic from an efficiency perspective in those industries in which only a small number of firms operate, either because the legacy of regulation may have entrenched incumbent firms with a competitive advantage, or because the industry may be incompletely contestable or otherwise imperfectly competitive. Deregulation in these contexts usually means the removal of formal entry barriers but continuing, and sometimes increased, regulatory oversight of one kind or another. The Federal Communications Commission, for example, issued numerous regulations in its implementation of the Telecommunications Act of 1996, which permits competition in local telephone service.[3] Similarly, while many states allowed entry into local telephony, they generally continued to regulate pricing and service quality.

Efficiency alone is rarely the determinative issue in deregulation. Vested interests—the workers and managers of protected firms, consumers enjoying cross-subsidies, sometimes the regulators themselves—have an incentive to fight to retain the advantages they enjoy under regulation. Consequently, successful deregulation often requires vigorous advocacy that details the failures of the regulatory regime and allays fears about distributional effects.[4] However, in the last three decades the potential efficiency gains from deregulation have become widely known—deregulation, or at least more nuanced regulation, is now a worldwide phenomenon.

Evidence from the deregulations of the U.S. trucking, railroad, airline, and other industries suggests that, as expected, large gains in social surplus result.[5] Similar evidence on the deregulation of competitive industries has emerged from other countries. New Zealand, for example, engaged in a broad range of regulatory reforms, including deregulation, during the 1980s and 1990s. While, not surprisingly, some income groups experienced a lowered living standard, the overall process appears to have substantially raised average living standards.[6] One note of caution: in many countries it is difficult to disentangle the benefits arising from deregulation from those arising from privatization (discussed below), as these often occur simultaneously.[7] There is also evidence of efficiency gains in sectors, such as communications, which may be duopolistic or oligopolistic, rather than competitive, in structure.[8] In sectors with these structural characteristics, however, the changes were rarely simply from regulation to no regulation, even though these changes were often described as deregulation. As we discuss later in this chapter (under price regulation), the switch was more often from highly intrusive regulation to more market-oriented, flexible, and less intrusive regulation. Here, again, it is often difficult to disentangle the benefits that arose from more-efficient regulation from those arising from privatization.[9]

A note of caution is that more permissive regulation has generally allowed further market consolidation in most of these industries. While this has facilitated technical efficiency, especially efficiencies relating to scale and density, the evidence suggests that it has also increased market power.[10] In some cases, such as U.S. banking deregulation, large entities may take on too much risk if they anticipate being bailed out because they are perceived to be "too big to fail"; in such cases the gains from scale economies may be offset by increased risk to the macroeconomy.[11]

It is also apparent that deregulation has been traumatic and costly for many of the *stake-holders* (those with a direct, especially economic, interest) in these industries. Remember, however, that firm or business failure is not synonymous with market failure; the evidence suggests that consumer gains often more than offset employee and shareholder losses.[12] Furthermore, keep in mind that deregulation need not apply to all the activities of the industry−−pricing, entry, and scheduling within the U.S. airline industry were largely deregulated, while safety, traffic control, and landing rights were not.

Legalization

Legalization refers to freeing a market by removing criminal sanctions. There are intermediate steps toward legalization, such as *decriminalization*, through which criminal penalties are replaced by civil penalties, such as fines.[13] Decriminalization reduces the stigma and punishment associated with the activity but does not fully sanction the action as socially acceptable.

The impetus for legalization and decriminalization often stems from changing social attitudes (for example, with regard to sexual behavior and marijuana use). Moves to legalize the market for prostitution that have occurred in the Netherlands and Australia, for example, fall into this category. It may also flow from government's desire for new sources of tax revenue and economic development. The rapid legalization of gambling in North America and elsewhere over the last three decades appears to fall into this category.[14] So too might the legalization of the recreational use of marijuana.[15] Finally, change may flow from the realization that criminalization is an ineffective policy with significant unintended negative consequences.

Privatization

The word *privatization* is used in several different ways: (1) the switch from agency subventions to user fees (discussed below as use of subsidies and taxes to alter incentives); (2) the contracting out of the provision of a good that was previously produced by a government bureau (discussed below as a method of supplying goods through nonmarket mechanisms); (3) denationalization, or the selling of state-owned enterprises to the private sector; (4) demonopolization, the process by which the government relaxes or eliminates restrictions that prevent private firms from competing with government bureaus or state-owned enterprises. Only these latter two types of privatization are directly related to the freeing of markets. Even denationalization, however, may not result in free market outcomes if other private firms are restricted from competing against the newly privatized firm. The restrictions put on competitors was a major criticism of the privatization of British Telecom, for instance.[16]

The presence or absence of market failure is the crucial issue in evaluating privatization. Most public corporations in the United States have been, and are, in sectors for which there is at least some prima facie evidence of market failure. In many other countries, the linkage between market failure and the provision of goods via state-owned enterprises was much more tenuous. However, extensive privatization or partial privatization has now taken place in a wide range of countries, including the United Kingdom, France, Canada, and New Zealand, as well as in the former Soviet bloc countries, South America, and developing countries. The aggregate evidence on privatization is that there have usually been major efficiency gains, both in allocative and technical efficiency.[17] Although the evidence from the former Soviet bloc countries is somewhat more mixed, there is considerable evidence concerning the benefits of privatization.[18]

In sectors that exhibit natural monopoly or oligopoly, such as electricity, gas, water distribution, and railroads, the empirical evidence on privatization suggests that improvements in technical efficiency usually occur.[19] One study concludes that even rail privatization in the United Kingdom, one of most maligned privatizations of all, resulted in significant operating efficiencies, without evidence of lower output quality.[20] However, realization of efficiency gains in these sectors for consumers and governments (especially longer-run gains) appears to be quite dependent on effective post-privatization regulatory structures that encourage ongoing dynamic efficiency improvements.[21]

Facilitating Markets

If a market has not existed previously, then it does not make sense to talk about freeing it. Rather, the process is one of facilitating the creation of a functioning market by either establishing property rights to existing goods or creating new marketable goods.[22]

Allocating Existing Goods

We saw in Chapter 5 that as demand increases a free good can begin to shift into an inefficient open-access situation (a move from the SW1 cell to the SW2 cell in Figure 5.2) if comprehensive and effective property rights are not established. Obviously, it may be very costly and, therefore, infeasible to allocate property rights effectively if the problem is structural in nature, but it may well be possible if the problem is institutional. For example, while a national government cannot allocate effective property rights to internationally migrating fish, it can allocate effective property rights to stationary shellfish.

As we discuss below in this chapter and illustrate in Chapter 16, the allocation of effective property rights is usually extremely contentious, however. Those who previously enjoyed use at below-efficient prices will undoubtedly oppose any distribution of property rights that makes them pay more. Remember, however, that the Coase theorem suggests that from an ex post efficiency point of view it often does not matter who receives a property right, as long as the right is secure and enforceable. From a distributional point of view, however, it does matter who gets a property right. People may, therefore, expend resources on political activity to gain larger allocations (that is, they engage in rent seeking). From an ex ante efficiency perspective, we want allocation mechanisms that limit the political competition for new property rights. Auctions (which we discuss below) and lottery allocation can sometimes serve this purpose.

The allocation of property rights is complex, especially in public arenas, where many different individuals, groups, and organizations have some legitimate claim to some dimension of the property rights to new or contested assets. Property rights are complex because they are potentially divisible along a number of different dimensions. The property rights to assets, including organizational assets, can vary by the degree of fragmentation of ownership, the clarity of ownership rights, the cost of alienating those rights, the degree of security from trespass, the credibility of persistence of the rights over time, and the degree of owner and managerial autonomy.[23] The allocation of property rights to water in the western United States, for example, illustrates the complexities of all of these dimensions of fractionalization. State legislatures have increasingly come to recognize the importance of establishing clearer (and therefore presumably more efficient) property rights as demand for water has increased rapidly with population density and industrialization over the last 50 years. Several studies have documented the many legal, administrative, political, social, and distributional challenges to the establishment of clearer and more efficient property rights regimes.[24]

One way that governments can potentially facilitate markets is by bringing buyers and sellers together more efficiently through *intermediation* or (more colloquially) a *platform*. Frequently, it will be profitable for firms to provide this intermediation function, even when the goods being exchanged vary in quality.[25] The rationale for government intervention should be based on foreseeable market failure or some important fairness concern. Where there is some market failure and it is profitable to provide intermediation, policy analysis mostly focuses on the appropriate form and nature of regulation.[26]

The Creation of New Marketable Goods

In certain circumstances it is possible for the government to create new marketable goods. (By doing so, whether intentionally or not, they create new and valuable property rights as well.) Currently, the most common and important form of these goods are known as *tradable permits*, which convey rights to access resources or discharge pollutants. The programs created to trade permit units are known as "cap and trade" programs because the total number of permits is set at some upper bound (the cap). One recent study found that tradable permits have now been used in 75 applications in fisheries, nine in air pollution control, five in land use control, and three in water allocation.[27] Tradable permits have been successfully used in the United States to allocate the property right to emit pollutants, including sulfur dioxide, into the air.[28] In theory, the allocation of tradable permits for emissions ensures that a specified level of air or water quality is achieved at a minimum total cost (including direct abatement costs and regulatory costs). Under such a tradable permit system, firms maximize profits by restricting emissions to the point where the price of an additional emissions permit equals the marginal cost of abatement. If all firms in the industry can buy and sell permits (including potential entrants), then each firm faces the same price for the last unit of pollution produced, and it would not be possible to find a less costly way of meeting the specified level of total emissions.

Many environmental policy analysts conclude that tradable permits are superior to emissions standards in terms of the informational burden, the speed of compliance, and in making appropriate trade-offs between economic growth and environmental protection.[29] Until 1990, emissions trading had seen limited application, mainly in the Los Angeles air shed. The 1990 Clean Air Act Amendments initiated nationwide trading of permits for emissions of hydrocarbons, nitrogen oxides, particulate matter, sulfur oxides, and carbon monoxide.[30] Commenting on the impact of sulfur dioxide emissions trading, Richard Schmalensee and colleagues are quite positive: "We have learned that large-scale tradable permit programs … can both guarantee emissions reductions and allow profit-seeking emitters to reduce total compliance costs … and to adapt reasonably efficiently to surprises produced elsewhere in the economy."[31] Several critics, however, have emphasized their weaknesses. These include formidable institutional barriers, such as thin markets (few buyers and sellers), to the practical use of tradable permits.[32] From an efficiency perspective, the most telling criticism, though, is that they are inferior to straightforward taxes.

Simulating Markets

In situations in which efficient markets cannot operate, it may be possible for the government to simulate market processes. In 1859, Edwin Chadwick was the first to argue that, even when competition *within* a market cannot be guaranteed, competition *for* the market may be possible.[33] In other words, the right to provide the good can be sold through an auction.[34]

One context in which an auction can often be used to appropriately simulate a market is in the provision of goods with natural monopoly characteristics—cable television, for example. It is not efficient to auction to the highest bidder the right to operate the natural monopoly, however. In a competitive auction, the winning bidder would be prepared to pay up to the expected value of the excess returns from operating the natural monopoly. The winning bidder would then be forced to price accordingly, resulting in the allocative inefficiency we described in Chapter 5. Rather, a more efficient approach is to require bidders to submit the lowest retail price at which they will supply customers. While no bidder will be able to offer to supply the good at marginal cost (as you saw in Chapter 5, this would result in negative profits), the winning bidder should be forced to bid close to average cost.

Oliver Williamson has pointed out a potentially serious problem with the use of auctions to allocate the right to operate natural monopolies: the winning bidder has both an incentive and an opportunity to cheat by reducing the quality of the good. To avoid this outcome, specifications for the good must be fully delineated and enforced. Yet, it is difficult to foresee all contingencies and costly to monitor contract performance. Williamson has documented how many of these specification, monitoring, and enforcement problems actually arose in the case of a cable television network in Oakland, California.[35] Because of this, and probably also for political reasons, there have been few actual cases of the use of franchise auctions.[36] The end result of the interaction between government and franchisee may be quite similar to more traditional regulation.[37]

Auctions are used extensively in the allocation of rights for the exploitation of publicly owned natural resources. As explained in Chapter 4, these resources often generate scarcity rents. If the government simply gives away exploitation rights, then the rents accrue to the developers rather than the public. (Of course, we should not forget that who receives the rent is a distributional issue, rather than an efficiency issue.) Additionally, we should keep in mind that these rents may partly or completely amount to a lottery prize, where unsuccessful explorers hold the losing (costly) tickets. In such cases, expropriating the rent will discourage future exploration.

An auction also has advantages relative to setting a fixed price for the allocation of exploitation rights. Most importantly, selling at a fixed price requires the government to estimate the value of the resource, which, in turn, requires estimates of the quality of the resource, future demand and prices for the resource, and future demand and prices for substitutes. An auction, on the other hand, allows the market, and therefore all information available in the market, to determine the appropriate value. The U.S. government, for example, underestimated the value of airwave spectrum it auctioned.[38] Problems can arise if there are few bidders. If the number of bidders is small, then there is danger that they will collude to limit price. Even if the number of bidders is fairly large, they may not generate competing bids if the number of units being offered is large. In general, the unsurprising lesson is that, unless auctions are skillfully designed, bidders will behave opportunistically.[39] In some cases, the manipulation of auctions may be overt. For example, nineteenth-century squatters in the U.S. Midwest colluded through "claims clubs" to prevent nonmembers from bidding on the land occupied by their members.[40]

Auctions can be useful allocative tools in situations in which governments must allocate *any* scarce resource. They are now being used in a wide variety of policy areas;[41] they might be usefully employed in other areas as well.[42] It is important, however, to design the auctions to ensure there is not systematic underpricing.[43]

Using Subsidies and Taxes to Alter Incentives

Freeing, facilitating, or simulating markets may prove inadequate if market failure is endemic or values other than efficiency are important. In this case, more interventionist approaches may be necessary. The first major class of these more intrusive policies that we examine consists of subsidies and taxes.[44] They aim to induce behavior rather than command it. Subsidies and taxes, therefore, are market-compatible forms of direct government intervention.

Policy analysts, bureaucrats, and politicians have engaged in heated debates about the relative merits of incentives versus other generic policies.[45] While policy analysts, especially those trained in economics, generally view incentives favorably, public managers and politicians have tended to be less enthusiastic. In the United States, this debate has primarily focused on the advantages and disadvantages of incentives relative to direct regulation. This terminology is confusing because incentives also require government intervention and, therefore, involve regulation. To clarify, we distinguish between *incentives* and *rules*.[46]

We are primarily concerned with using taxes and subsidies in situations in which the intention is to correct market failures or achieve redistribution. We are not concerned with taxation intended mainly to raise revenue, even though these taxes induce behaviors that are relevant to policy issues. Indeed, taxes designed to raise revenue inevitably involve economic inefficiency, for example by altering the trade-off between leisure and labor or the choice between savings and consumption. (The efficiency loss of the last dollar raised by the tax is called its *marginal excess tax burden*; if this tax is the marginal source of revenue, it is referred to as the *marginal cost of public funds*.) The net effect of these taxes on efficiency depends on how the revenues are ultimately used. If they help correct market failures, then the combined tax and expenditure programs may enhance efficiency. We also are not concerned with efforts to raise incomes generally (this is dealt with below under the category of cushions). Our concern here is with taxes and subsidies that change incentives by altering the *relative* prices of goods. Put simply, we are considering the use of taxes to raise the private costs of things that are too abundant from the social perspective, and the use of subsidies to lower the private costs of things that are too scarce from the social perspective.

In general, taxes and subsidies can have one of three possible impacts on efficiency. First, where taxes and subsidies are aimed at correcting for externalities in particular markets, their impact may enhance efficiency. It is usually difficult in practice to estimate accurately either social marginal benefits or social marginal costs, measures which are needed to gauge the magnitude of positive and negative externalities. If social marginal costs and benefits are inaccurately assessed, then there may be no efficiency gains and, indeed, net efficiency losses may occur. Second, where taxes and subsidies are not aimed at correcting a market failure, there is inevitably some net deadweight loss. Of course, redistribution may itself enhance efficiency if there is utility interdependence between donors and recipients. (Remember our discussion of preferences in Chapter 6.) Rarely, however, do we have sufficient information to make such assessments confidently. Third, taxes may be used in an attempt to extract scarcity rents that arise in the use of natural resources such as oil. In theory, such taxes can be designed to transfer rents without losses in efficiency; in practice, limited information about the magnitude of rents generally leads to market distortions.

In order to make our discussion of taxes and subsidies more concrete, we divide them into four general categories: supply-side taxes, supply-side subsidies, demand-side subsidies, and demand-side taxes. These categories are summarized in Table 10.2. Note that some policies fall into more than one category. For example, a tax on gasoline can be thought of as either a supply-side or a demand-side tax––perception aside, the effect of the tax does not depend on whether it is collected from refiners or consumers. Nevertheless, we find these categories useful because they emphasize the behavior that is the target of the policy.

Supply-Side Taxes

We consider supply-side taxes under two broad categories: output taxes and tariffs.

Output Taxes

As we saw in Chapter 5, markets with negative externalities overproduce goods from the social perspective (see Figure 5.7). When transaction and coordination costs prevent Coasian market solutions through negotiation among the affected parties, government intervention is desirable to equalize marginal social benefits and costs. Theoretically, the appropriate tax raises price to the level of marginal social cost, thereby internalizing the externality.

The idea that an appropriate per unit tax can lead to an efficient internalization of a negative externality is usually credited to A. C. Pigou and is often referred to as a *Pigovian tax*.[47] The major advantage of using taxes to correct for negative externalities is that they allow firms (or consumers) the choice of how much to reduce production (or consumption) to limit their tax payments. As long as each firm sees the same tax, the industry as a whole reduces the quantity of the externality in the least costly way to society.[48]

Table 10.2 Using Subsidies and Taxes to Alter Incentives

Generic Policies	*Perceived Market Failure (MF), Government Failure (GF), Distributional Issue (DI), Limitation of Competitive Framework (LCF)*	*Typical Limitations and Collateral Consequences*
	Supply-Side Taxes	

Generic Policies	*Perceived Market Failure (MF), Government Failure (GF), Distributional Issue (DI), Limitation of Competitive Framework (LCF)*	*Typical Limitations and Collateral Consequences*
Output taxes	MF: Negative externalities DI: Transfer of scarcity rent	Frequent adjustment of tax levels required.
Tariffs	LCF: Market power of foreign exporters	Deadweight losses for consumers; rent seeking by producers.
	Supply-Side Subsidies	
Matching grants	MF: Positive externalities MF: Public goods DI: Increase equity	Diversion to general revenue by reduction in effort.
Tax expenditures (business deductions and credits)	MF: Positive externalities MF: Public goods	Misallocation of resources across industries; horizontal tax inequity.
	Demand-Side Subsidies	
In-kind subsidies	MF: Positive externalities LCF: Utility interdependence DI: Floors on consumption	Restricts consumer choice; bureaucratic supply failure; lumpiness leads to inequitable distribution.
Vouchers	MF: Positive externalities	Informational asymmetries; short-run

Generic Policies	*Perceived Market Failure (MF), Government Failure (GF), Distributional Issue (DI), Limitation of Competitive Framework (LCF)*	*Typical Limitations and Collateral Consequences*
		supply inelasticities; institutional resistance.
	DI: Increase equity	
	GF: Bureaucratic supply failure	
Tax expenditures (personal deductions and credits)	MF: Positive externalities	Poor targeting of subsidies; vertical and horizontal tax inequities.
	DI: Increase equity	
Demand-Side Taxes		
Commodity taxes and user fees	MF: Negative externalities	Deadweight losses and black markets.
	MF: Information asymmetries	
	MF: Public goods, especially open access	

The implementation of efficient externality taxes has proven to be difficult. The major problem is that the government needs to know the shapes of the social benefit and social cost schedules. The accurate estimation of social benefits from the reduction of a negative externality requires the determination of a damage function—a difficult task because it depends on the impact of a complex set of physical and biological forces upon human beings.[49] Information on the marginal costs of firms, needed to determine the difference between private and social marginal costs, may not be easily determined.[50] Critics have pointed out that, if such information were available for private and social marginal costs and benefits, the government would be able to specify the appropriate level of production directly, without having to deal with taxes at all. (Of course, polluting firms may have this information, but if they have such private information the design of efficient taxes is much more complex.[51]) In practice, these problems, along with objections to firms receiving a "license to pollute," have limited the political acceptability of taxes on pollutants.[52] However, various forms of supply-side carbon (emission) taxes have been implemented, aided by estimates of the relevant social cost and benefit curves. William Nordhaus, for example, has estimated that a 2010 carbon price of around $17 per ton carbon in 2005 prices—rising to $70 per ton in 2050—would maximize the present discounted value of benefits minus costs.[53] A federal government task force has estimated the marginal social cost of carbon emissions as rising to between $26 and $97 in 2007 prices, depending on the assumed rate of time discounting.[54] Similarly, Ian Parry, Margaret Walls, and Winston Harrington have estimated that the optimal Pigovian tax on gasoline in the United States would be approximately $2.10 a gallon rather than the current 40 cents.[55]

Although lack of information usually makes it difficult to set optimal tax rates initially, it may be possible to approximate such rates by trial and error after observing how firms respond. Trial-and-error experiments have drawbacks, however, including uncertainty, opportunistic behavior by externality producers, and political costs.[56] Trial-and-error experiments may also involve substantial monitoring and administrative costs.

Despite these disadvantages, the potential for efficiency gains that should be kept in mind are: (1) lower cost—the same outcome can be achieved but at lower cost than with standards; (2) innovation—taxes encourage appropriate innovation, as innovation continues until the marginal costs of new technology equal the marginal benefits of for-gone taxes; (3) informational requirements—firms face incentives to acquire appropriate information; (4) intrusiveness—government intervention is minimized; (5) administrative complexity—economic incentives require the minimum level of administrative intervention; (6) transaction costs—economic incentives avoid many of the hidden costs of administrative regulation, such as negotiation and lobbying.[57]

Thus far we have discussed supply-side taxes as a way of dealing with externalities. The use of such taxes in transferring rent has been much more common. As noted in Chapter 4 (and later in the policy analysis in Chapter 16), many natural resources generate scarcity rents. The distribution of these rents is often a contentious public issue. Ideally, any efforts to capture these rents for the public would not disturb the harvesting or extraction decisions of the private owners of the resources. Many different types of taxes have been used to transfer scarcity rents: flat-rate gross royalties (on the physical output), profit taxes, resource rent taxes, corporate income taxes, cash flow taxes, and imputed profit taxes.[58] While in theory some of these mechanisms could capture a portion of the scarcity rent without distorting private decisions about the use of the resource, in practice none is likely to be completely neutral because of limited information.[59] For example, a one-time tax on the value of the resource (a resource rent tax) would not interfere with the owners' decisions about future extraction of the existing resource. It would require, however, accurate information about future extraction costs and market prices. Further, its imposition may reduce the incentive for firms to explore for new stocks of the resource.[60]

A potentially important role for output taxes is the preservation of rent in structural open-access situations. Rent capture through unit taxes is an efficiency-enhancing policy if it reduces extraction efforts that are dissipating rents.

Tariffs

A *tariff* is a tax on imported, and occasionally exported, goods. A tariff may either be a percentage of the price (an ad valorem tariff) or a fixed dollar amount per unit, independent of the price. Like any other tax, tariffs generate deadweight losses in the absence of market failures. They usually do so because they impose large consumer surplus losses on domestic consumers, in excess of the gains that accrue to domestic (and foreign) producers. They also impede the development of global comparative advantage and specialization based on increasing returns to scale.

A vast array of arguments have been put forward in favor of tariffs and other barriers to trade, including the “infant industry” argument, international monopsony or monopoly conditions, “strategic trade policy” reasons, revenue enhancement, cultural protection, retaliation against another country’s tariffs, and domestic redistribution.[61] Here we review two of the most important arguments that are often used to justify tariffs.

The most common argument used to justify tariffs asserts that protecting a fledgling industry (the so-called infant industry argument) provides positive externalities. The idea is that a domestic industry may have a potential comparative advantage that cannot be initially internalized by specific firms because some factor of production, such as skilled labor, is mobile. Another version of the argument is that domestic firms have the opportunity to achieve minimum efficient scale under tariff protection. This argument assumes that investors are myopic in that they are not prepared to accept short-term losses for long-term profits. Many studies have argued, however, that the major impetus in the United States for tariffs, and protectionism in general, has stemmed from redistributional politics and is best understood in terms of government failure. One study from the 1980s estimates that the annual welfare losses caused by U.S. import restraints was approximately $8 billion, with the cost to consumers per job saved ranging from a low of approximately $25,000 to a high of approximately $1 million.[62]

Monopsony effects have also been used to justify import tariffs. If a country accounts for a large share of world demand for some good, then that country may be able to affect world price by restricting its demand through tariffs (or quotas).[63] Depending on the magnitude of the domestic and world elasticities of supply and demand, the imposition of a tariff may actually increase domestic social surplus by depressing the world price.[64] The monopsony effect is one of the rationales advanced in support of a U.S. oil import tariff, although supporters from oil-producing states seek the higher prices for domestic oil that would result.

A similar argument has been used to justify export tariffs where a country produces enough of a good to potentially have monopoly power in the world market. Individual domestic producers competing against each other will not be able to extract monopoly rents, but a government imposing an export tariff might be able to raise the world price and so extract the rent. (Governments may also use more subtle mechanisms, such as commodity pools and joint marketing organizations, for the same purpose.) Export taxes are not very common because producers prefer export quotas (which allow them to capture the rent more directly by selling the quotas). However, national governments sometimes preemptively impose export tariffs when another national government is threatening import tariffs. (Developing countries with limited inland taxing capabilities may impose export tariffs at the border as a more feasible way of collecting revenue.)

Generally, the importance of tariffs is decreasing under global international agreements administered by the General Agreement on Tariffs and Trade (GATT) and its successor, the World Trade Organization (WTO).[65] Tariffs have also been reduced under regional trade agreements, such as the North America Free Trade Agreement (NAFTA), although it is not as clear that such regional agreements are efficiency enhancing.[66] Although tariffs are being reduced, in some cases countries are replacing them with less transparent barriers to trade (collectively, these are known as *nontariff barriers*), such as the imposition of standards that favor domestic producers. For example, the European Union is seeking to prevent the marketing of U.S.-produced cheeses through geographical indication, the requirement being that names with geographic origins, such as Parmesan cheese, be used only for products actually from that geographic area. The Uruguay Round of GATT tried with limited success to reduce nontariff barriers with a process called *tariffication*, which assesses all trade barriers in terms of their equivalent tariffs so that national trade policies could be more easily compared and negotiated.

Supply-Side Subsidies

One way to increase the supply of goods is to give direct subsidies to the suppliers of the goods. The subsidies may be directed at either private firms or lower levels of government. Intergovernmental subsidies are usually referred to as *grants in aid*, or simply *grants*.[67] In most respects, subsidies to internalize positive externalities are analytically symmetrical with the taxes on negative externalities described above. Broadly speaking, therefore, if there is a positive externality, an appropriately designed per unit subsidy to the supplier generates an increased supply of the good, reducing the undersupply caused by the externality and thereby increasing social welfare.

Matching Grants

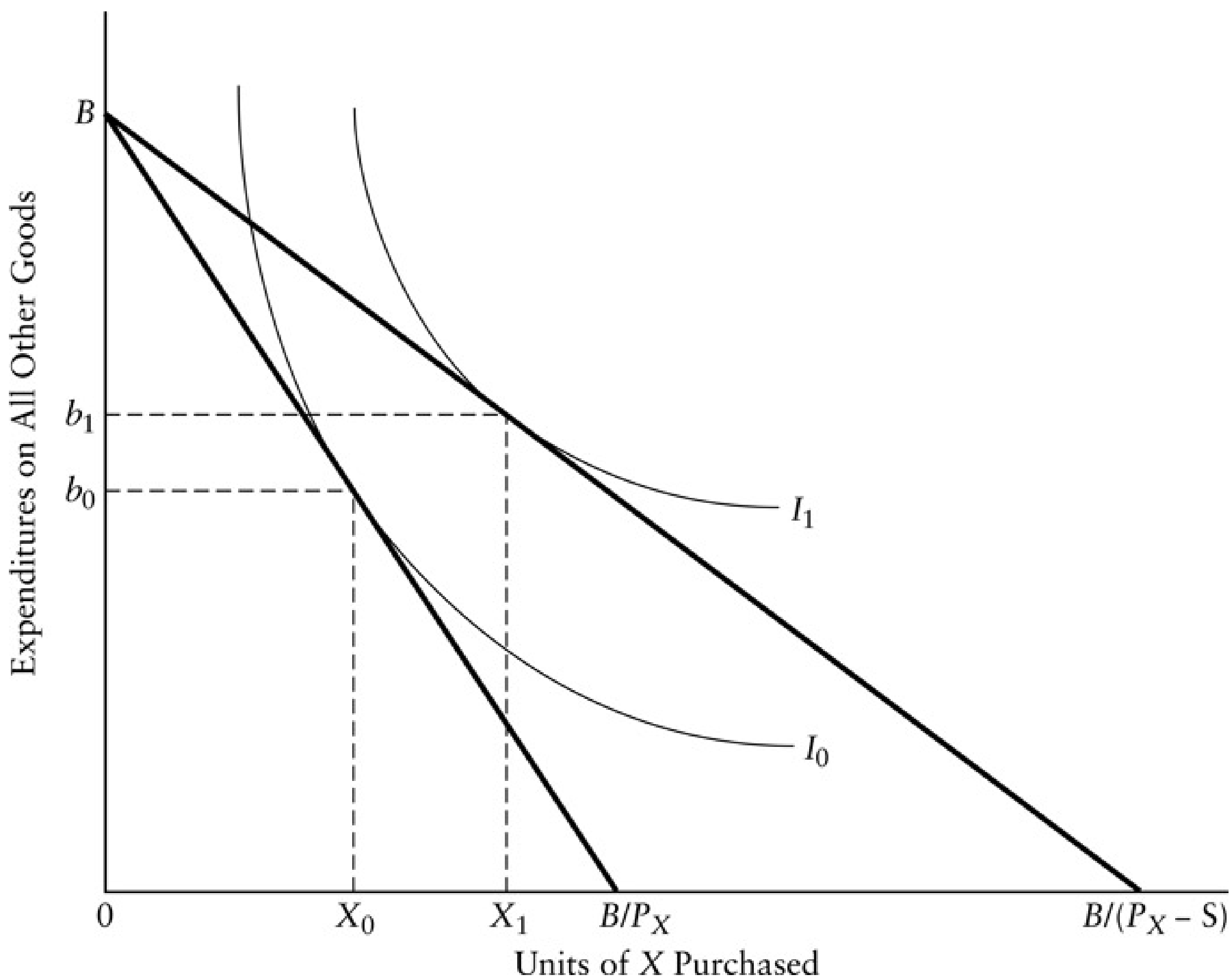


Figure 10.1 The Effect of a Matching Grant on Provision of Targeted Good

In Figure 10.1, we illustrate how a subsidy might be used by a central government to induce a local government to supply more of some public good (X). The vertical axis measures the local government's expenditure on all goods other than X. The horizontal axis measures the quantity of X that the local government provides. For example, X might be remedial classes for children who are slow learners. Given a total budget of B, the local government could spend nothing on X and B on other services; nothing on other services and purchase B/P_X of X, where P_X is the price of X; or any point on the line connecting these extremes. Given this budget line, assume that the local government chooses to provide X_0 units of X. The indifference curve labeled I_0 gives all the combinations of X and expenditures on other goods that would be as equally satisfying to the local government (say, the mayor) as X_0 and b_0 spending on other goods.

Now imagine that the central government offers to pay S dollars to the local government for each unit of X that the local government provides. (We refer to this as a *matching grant* because it matches local expenditures as some fixed percentage. It is also *open ended* because there is no cap on the total subsidy that the local government can receive.[68]) The effective price that the local government sees for X falls from P_X to $P_X - S$ because of the subsidy. As a result, the budget line for the local government swings to the right. The local government now buys X_1 units of X, reaching the higher level of satisfaction indicated by indifference curve I_1. Note that the local government also spends more on other goods as well—some of the subsidy for X spills over to other goods. In the terminology of grants, part of the categorical grant has been *decategorized.*

To counter this spillover, the subsidy could be given with a *maintenance-of-effort requirement:* only units beyond X_0 would be subsidized. The budget line under the maintenance-of-effort requirement follows the original budget line up to X_0, and then rotates to the right so that it becomes parallel to the budget line for the subsidy without maintenance-of-effort requirements. The local government responds by purchasing more units of X and by spending less on other goods than it would in the absence of the subsidy. Maintenance of effort is useful in targeting subsidies so that given expenditure levels have the greatest desired impact. Unfortunately, analysts in central governments do not always have sufficient information to design effective maintenance-of-effort requirements.

Subsidies can also be used to deal with negative externalities.[69] Instead of taxing the good generating the externality, firms can be paid for reducing the level of the externality itself. Such subsidies, however, are vulnerable to opportunistic behavior on the part of suppliers. Unless the government knows how much of the externality the firms would have produced in the absence of the subsidy, the firms have an incentive to increase their externality levels in anticipation of the subsidy. For example, if a firm expects a subsidy program to be implemented for reducing emissions of some pollutant, then it may temporarily increase emissions to qualify for greater levels of future subsidies. It would be particularly difficult to monitor such behavior if the firms simply slowed the introduction of reductions that they were planning to adopt anyway.

Subsidies also have different distributional consequences than taxes. While taxing goods with externalities usually generates revenue, subsidies must be paid for with other taxes. When negative externalities are internalized with taxes, consumers of the taxed good share some of the costs of the reduction in output; when negative externalities are reduced through subsidies, consumers of the good generating the externality generally bear a smaller burden of the reduction. Indeed, sufficiently large subsidies may induce shifts in technology that lead to reductions in the externality without reductions in the supply of the good.

Subsidies can sometimes be effective mechanisms for dealing with market failures other than externalities. For instance, an alternative to auctioning the right to operate a natural monopoly is to induce the natural monopolist to price efficiently (where price equals marginal cost) by providing a subsidy that gives the monopolist a positive rate of return. While this approach has some appeal, it has not been utilized much in practice. One reason is that it requires information about the firm's marginal cost schedule. Another reason is that government has to pay for the subsidy with revenue raised from elsewhere.

Skeptics argue that most supply-side subsidies are provided for inappropriate distributional, rather than efficiency enhancing, reasons. As Gerd Schwartz and Benedict Clements note, “Subsidies are often ineffective (i.e., they fail to benefit their intended target group) and costly (they have adverse real welfare and distributional implications).”[70] While such direct help is relatively uncommon at the federal level in the United States, it is extremely common at the state level and in other countries. Such help may take the form of direct grants, loan forgiveness, loan guarantees, or tax expenditures. Such subsidies are frequently directed at declining, or “sunset,” industries. They are likely to be especially inefficient if they simply slow the exit of unprofitable firms from the industry.

Tax Expenditures

Probably the most common form of supply-side subsidy is through *tax expenditures*, such as deductions to taxable income and credits against taxes otherwise owed under corporate income taxes.[71] One can best understand the nature of tax expenditures by realizing that a cash gift and a forgiven debt of the same amount are financially equivalent. If we assume that there is some benchmark tax system (or comprehensive tax base) that treats all taxpayers similarly, no matter what their expenditures, then we would have a system without tax expenditures. If one is forgiven a tax payment (that is, debt) from this benchmark rate, it is equivalent to being given a subsidy of the same amount. We classify tax expenditures as subsidies because they change relative prices by making certain factor inputs less expensive. For example, allowing firms to fully deduct investments in energy-saving equipment from their current tax liabilities makes such equipment appear less expensive than if they had to depreciate it over its useful life, as they must for other capital goods. Again, to the extent that the tax expenditures do not correct for market failures, inefficiency results, including both interindustry and intraindustry misallocation of resources.

The public good nature of certain aspects of *research and development (R&D)* may also serve as a rationale for subsidies. The U.S. government, as well as the governments of nearly every other industrial nation, directly and indirectly provides R&D assistance. Indeed, these subsidies are viewed as the cornerstones of industrial policies in several countries.

R&D investments will be supplied at inefficient levels to the extent that they have the characteristics of public goods. First, where exclusion of competitors from access to findings is not possible, private firms tend to underinvest in R&D.[72] No market mechanism can ensure that some of the benefits of the innovation will not be captured by other firms. Some of the benefits will accrue to users as consumer surplus, while some of the producer surplus will go to other producers. The typical policy approach has been to subsidize private R&D in an attempt to raise it to socially optimal levels. Second, where exclusion is possible—through patents or effective industrial secrecy—firms tend to restrict the distribution of information concerning their R&D to suboptimal levels. Consequently, private firms may wastefully duplicate research.

There may be underinvestment because of problems hindering the adequate spreading of risks. Research is an inherently risky activity. However, if the returns on various projects are independent, as the number of projects approaches infinity, the risk of a portfolio of projects approaches zero. Research should be conducted to the point where marginal social benefits equal marginal social costs, regardless of the risk of individual projects. Yet, in a competitive market private firms generally cannot hold an adequate portfolio of independent projects.

The underinvestment argument is not without its critics. Gordon McFetridge has questioned whether the public sector is more effective than the private sector in dealing with risk. He argues that individuals in the market demand risk avoidance and that there are mechanisms, such as venture capital, available for pooling such risks. He also points out that the stock market itself is a mechanism for risk pooling (of projects) and risk spreading (for individuals).[73]

Do public subsidies actually affect the efficiency and dissemination of R&D? Richard Nelson and a team of researchers conducted case analyses of seven industries: aviation, semiconductors, computers, pharmaceuticals, agriculture, automobiles, and residential construction. Their study addresses allocative efficiency, as well as distributional and implementation issues. Nelson concludes that government supply or funding of basic and generic research incrementally increases the aggregate of R&D activities and encourages the wide dissemination of information.[74] Empirical research in recent years generally finds that government subsidies and tax credits increase net investment in R&D, as do investments in university research.[75] That is, public subsidies appear to induce rather than fully crowd-out private R&D investments.

Demand-Side Subsidies

Demand-side subsidies aim at increasing the consumption of particular goods by reducing their price to final consumers. There are two basic methods of providing demand-side subsidies: subsidies (vouchers), and personal deductions and credits (tax expenditures). Two major efficiency rationales for intervention, each involving arguments about positive externalities, may be offered for demand-side subsidies. In other cases, demand-side subsidies are justified primarily on redistributional grounds. In many debates about such subsidies, confusion often arises because the efficiency and equity dimensions are not clearly distinguished.

The distributional argument alone for such subsidies is weak because the recipients would always be better off (from their own perspective at least) with straight cash transfers (which would not alter relative prices directly). Therefore, the rationale for such transfers is often put in terms of *merit goods*.[76] While the term has no precise and generally accepted meaning, it usually incorporates a mix of redistribution and market failure arguments (most frequently positive externalities, but also information asymmetry and nontraditional market failures such as unacceptable or endogenous preferences).

In-Kind Subsidies

In-kind grants subsidize the consumption of specific goods. Strictly speaking, in-kind grants refer to the direct provision of a commodity to consumers. For example, the government may purchase food and distribute it directly to people. U.S. programs to distribute surplus agricultural products, like cheese, are literally in-kind grants. So, too, are public housing programs (in contrast to rent and construction subsidies)—recipients receive housing services directly from the government at a subsidized rent. In the United States and Canada, however, most "in-kind grants" are distributed through vouchers that allow recipients to purchase the favored goods in markets. For example, the U.S. food stamp program distributes food vouchers to those meeting income requirements.

Figure 10.2 illustrates the impact of a lump-sum in-kind subsidy of Z^* units of some good Z. The budget line initially connects the points B and B/P_Z. The introduction of the subsidy shifts the effective budget line to the right by an amount Z^*. If it is possible for the recipient to sell the subsidized good to others, then the effective budget line also includes the dashed line from point a to point b which, for purchases of Z greater than Z^*, is identical to the budget line resulting from a cash grant of P_Z times Z^*, where P_Z is the price of the subsidized good. Even if resale were not possible, the subsidy level is sufficiently small so that the consumption level Z_1 is the same as would result from a cash grant of P_Z times Z^*. Note that, in this particular illustration, the equivalence results even though the in-kind subsidy is larger than the presubsidy consumption level. Put bluntly, the disconcerting conclusion under these circumstances is that one may as well give the recipient booze as soup.[77] Giving the grant in-kind rather than as money would only be relevant to consumption if the indifference curve I_1 were tangent to the extension of the budget line (the line segment *ab*).

If the subsidy is sufficiently large and nonmarketability can be enforced, then consumption of the subsidized good can be increased above the level that would result from an equivalent cash grant. What rationales might justify large in-kind subsidies and the necessary enforcement efforts to prevent trading of the subsidized good in black markets? One putative rationale for effective (large and nontradable) in-kind subsidies based on efficiency grounds is that the givers of the subsidies derive utility from the specific consumption patterns of recipients. (Another way of describing this interdependence is that taxpayers receive a positive consumption externality from seeing recipients consume particular goods.[78]) For example, many people place a positive value on knowing that the children of the poor are fed adequately. In the presence of utility interdependence, in-kind transfers may be more efficient than unconstrained (cash) transfers. This should not be surprising in light of our discussion of internalizing externalities—cash grants do not internalize the externality associated with the specific goods. Another possible rationale is that the good in question generates positive externalities. There has been considerable debate about the existence, and magnitude, of positive externalities associated with the consumption of housing, education, health services, and food. These are empirical issues that can only be considered on a case-by-case basis.

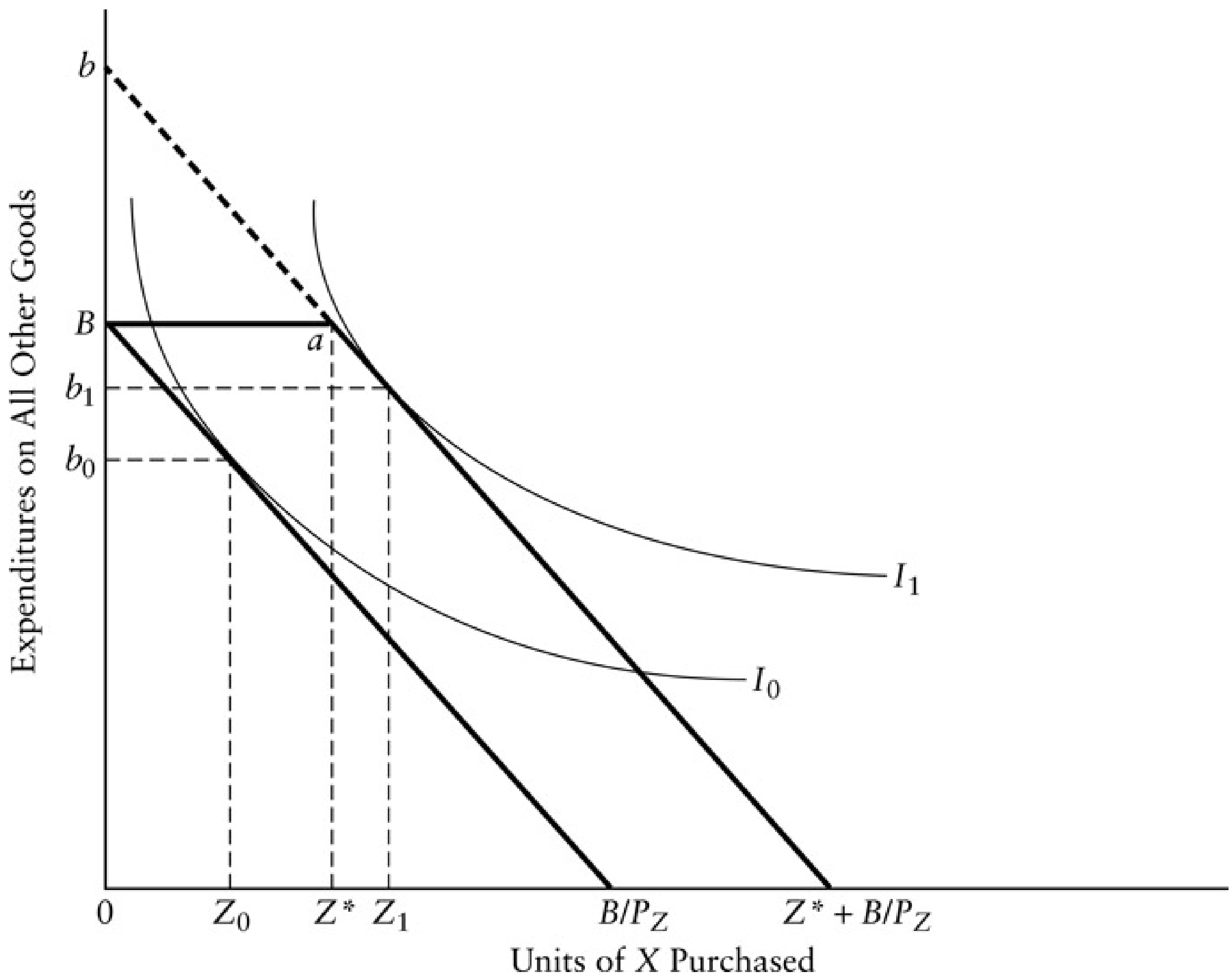


Figure 10.2 The Effect of an In-Kind Subsidy on Consumption

Vouchers

In-kind grants are often administered through *vouchers*, which allow consumers to purchase marketed goods at reduced prices. Typically, the vouchers are distributed to selected consumers at prices lower than their face value. Those suppliers who sell the favored goods for vouchers, or consumers with receipts for purchase, then cash the vouchers at their face value. If the vouchers are distributed in fixed quantities at zero prices to consumers, then they are conceptually identical to the lump-sum in-kind grants analyzed in Figure 10.2.

The voucher system in the United States with the greatest participation is the food stamp program. Controlled experiments that replaced food stamp coupons with their cash equivalents have found either no statistically significant reductions in total food purchases (Alabama Food Stamp Cash-Out Demonstration) or very small reductions (San Diego Food Stamp Cash-Out Demonstration).[79] Food stamps, therefore, may currently be little more than an administratively costly form of cash transfer that nonetheless enjoys political popularity.

Vouchers are used for subsidizing a wide range of goods, such as primary and secondary education, day care, food and nutrition, environmental protection, and housing.[80] If the vouchers provide large subsidies, then they will stimulate market demand for these goods. Because these goods are usually purchased in local markets where short-run supply schedules are upwardly sloping, they may cause the prices of the subsidized goods to rise. For example, if the short-run supply schedule of housing in a locality were perfectly inelastic, the introduction of housing vouchers would simply drive prices up for all renters without increasing supply. These higher prices would, however, eventually induce new construction and the splitting of structures into more rental units.

The price effects of housing vouchers were tested through the Housing Allowance Program, a more than $160 million experiment sponsored by the U.S. Department of Housing and Urban Development (HUD).[81] Unfortunately, researchers disagree with the experimental findings that the housing vouchers did not increase local housing prices.[82] Nevertheless, HUD now assists more households through vouchers or certificates than through more traditional public housing.[83]

Many analysts advocate vouchers as devices for increasing access to private primary and secondary education, for increasing parental choice, and for introducing competition among public schools.[84] The primary rationale for public support of education is that individuals do not capture all the benefits of their educations—others benefit from having educated citizens and workers around them. Many see the direct provision of education through public institutions, however, as suffering from the government failures that result from lack of competition. Advocates see vouchers as simultaneously permitting public financing and competitive supply. Critics counter that competitive supply suffers from information asymmetry (parents may not discover the quality of the education their children receive until they have fallen far behind), so that vouchers could not be effectively used without the direct regulation of quality that might reduce the advantages that private schools now enjoy relative to public schools.[85] Yet, less intrusive measures, such as regular reporting of average test score gains for students, might deal with the information asymmetry problem without subjecting private schools to excessive regulation.

There have been a number of experiments with educational vouchers in U.S. cities, including public programs in Milwaukee and Cleveland, and privately funded programs in Dayton, San Antonio, New York, and Washington. Although the results from evaluations of these experiments have been controversial, especially with respect to assessing differences in educational outcomes, it seems to be the case that parents whose children are participating in voucher programs generally have higher levels of satisfaction than they did when their children were in public schools.[86] There is some evidence, although again controversial, that vouchers may be helpful in closing the gap between the educational achievement of African American and white students in urban school districts.[87] Reviewing the overall school voucher evidence, Isabel Sawhill and Shannon Smith conclude: "Whether the education their children receive is of higher quality remains somewhat unclear, but the results to date are modestly encouraging."[88] In a survey of school vouchers globally, Edwin West reported that educational vouchers have been adopted in many countries, including Sweden, Poland, Bangladesh, Chile, and Colombia—usually in the form of a "funds follow the children" model in which governments fund schools of choice in proportion to enrollment.[89] The mere existence of competition may create stronger incentives for public schools to improve.[90]

Tax Expenditures

Tax expenditures are commonly used to stimulate individual demand for housing, education, medical care, and childcare. Other tax expenditures stimulate demand for goods produced by certain kinds of nonprofit agencies, such as charitable, cultural, or political organizations. Tax expenditures have their effect by lowering the after-tax price of the preferred good. For example, being able to deduct the interest payments on mortgages from taxable income makes homeownership less expensive. In some cases, they may be intended to stimulate aggregate demand during recessions or periods of slow growth. For instance, depreciations rules under individual and corporate taxes may be temporarily reduced in an effort to speed up capital investment.

Tax expenditures are an important source of subsidy in the United States. For example, the forgone revenue from allowing mortgage interest and property tax deductions in fiscal year 2015 was estimated to be $71 billion.[91] Additionally, the taxation of imputed income (an increase in wealth that does not directly accrue as a money payment) is not normally treated as a tax expenditure. For example, federal estimates of tax expenditures do not include the imputed income on equity in owner-occupied housing.[92] Most Western countries now recognize the importance of the magnitude of tax expenditures. The United States, Germany, the United Kingdom, Japan, and Canada all require the annual presentation of tax expenditure accounts.[93]

Critics argue that tax expenditures are less desirable than direct subsidies for two reasons. First, because tax expenditures emerge from relatively hidden debates about the tax code, their role as subsidies is not rigorously analyzed. Consequently, various forms of government failure, like rent seeking, are encouraged. Second, tax expenditures are notorious for their inequitable distributional consequences. Higher-income individuals are much more able to take advantage of tax expenditures than members of lower-income groups, who, not surprisingly, pay little or no tax in the first place.[94] Deductions, which reduce taxable income, avoid tax payments at the payer's marginal tax rate and thus favor higher-income persons in progressive tax systems. Credits, which directly reduce the tax payment, are worth the same dollar amounts to anyone who can claim them. Therefore, credits generally preserve progressivity more than deductions costing the same amount in forgone revenue.

Demand-Side Taxes

We divide demand-side taxes into two major categories: commodity taxes and user fees. Note, however, that other analysts may use different terminology, and even the distinction between these two categories is often unclear.

Commodity Taxes

The terms *commodity tax* and *excise tax* are frequently used interchangeably. We can think of commodity taxes as internalizing the impacts of goods with negative externalities. The most common applications are to reduce consumption of so-called demerit goods like alcohol (see our earlier discussion of merit goods under demand-side subsidies). The use of taxes in these contexts often appears to display a certain amount of schizophrenia. Will taxes only minimally affect demand and raise lots of revenue, or will taxes substantially decrease demand and generate much less revenue? Presumably, the idea in this context is to dampen demand, but the revenue goal often appears to displace it. The price elasticity of demand determines the balance between reduced consumption and revenue generation in each particular policy context.[95]

User Fees

The technical terms often used by policy analysts for user fees include *congestion taxes, peak-load pricing, marginal social cost pricing* (or often just *marginal cost pricing*), and *optimal tolls*. In more common bureaucratic parlance, they are usually called *license fees, rental charges*, or *fares*.

There are two efficiency rationales for user fees. The first rationale is to internalize externalities. In this context, they are simply a demand-side form of Pigovian tax. The second rationale is to price public goods appropriately, specifically in the context of congested public (toll) goods like bridges (see NE2 in Figure 5.2), and open-access resource public goods such as fishing grounds (see SW2 in Figure 5.2). We already reviewed the appropriate price for toll goods in Chapter 5 (see Figure 5.3). Efficient allocation requires that the price charged equal the marginal costs users impose on other users (and more generally all other members of society), implying a zero price during zero marginal cost time periods (for example, when a road is uncongested) and a positive price during positive marginal cost time periods (when the road is congested). Marginal costs may vary on any temporal dimension, including by time of day, time of week, time of year, or gradually increase over time. In North America, for example, the social marginal cost of water is typically much higher in summer than it is in winter.[96]

There are four problems with the practical implementation of efficient user fees. First, "price should equal marginal cost" is only the appropriate rule when all other prices in the economy also equal marginal cost. For example, this implies that tolls on a given bridge are only efficient if all other congested bridges in a transportation system are tolled as well; otherwise, use will be inefficiently distorted as some drivers seek out the untolled alternatives. When this condition is not met, there is a *second-best* pricing problem. In such cases, the optimal price under consideration is in general not exactly equal to the marginal cost, and in some cases there can be a substantial gap between the two.[97]

Second, because marginal cost is often less than average cost, marginal cost pricing may involve prices that do not cover the costs of service provision. While this is not problematic from an efficiency perspective, it does mean that service provision would require a subsidy. In the absence of subsidies, one approach is a *two-part tariff*. A two-part tariff can simultaneously satisfy the goals of efficiency and governments' revenue requirements. An efficient two-part tariff imposes both a fixed price on each user for access to service provision (an *access charge*) and additionally imposes use charges that equal marginal cost. The fixed charge paid by the customer is independent of the volume consumed and is akin to a lump-sum tax. It therefore does not influence the consumers' level of use or consumption.[98]

Third, when users are accustomed to receiving a service at zero, or below, marginal cost prices, they do not like to start paying. In other words, policymakers must deal with the political feasibility of such peak-load prices.

Fourth, the transaction costs of implementing user fees may be too high to make them practical. Historically, for example, when demand for water was low it did not make sense to install costly water meters in every household. One context where the feasibility of such pricing is changing rapidly because of technological advances is road congestion. Although road pricing had been considered in many jurisdictions,[99] it was not widely implemented until low-cost electronic monitoring became feasible.[100] It has now been introduced in a number of jurisdictions, including San Diego, Orange County, California, and Houston in the United States, and in a number of other urban areas around the world, such as London and Singapore.

Establishing Rules

Rules pervade our lives. Indeed, they are so pervasive that we tend not to think of them as instruments of government policy. Our objective here is to emphasize that rules are like other generic policies, with advantages and disadvantages. Government uses rules to coerce, rather than induce (through incentives), certain behaviors. Policymakers may enforce compliance by either criminal or civil sanctions. We cannot always clearly distinguish between rules and incentives in terms of their practical effect, however. For instance, should one regard small fines imposed by the criminal courts as rules or as implicit taxes (in other words, negative incentives)? Although it is fashionable to focus on the disadvantages of rules (for example, relative to incentives), rules provide the most efficient method for dealing with market failures in some contexts.

We divide rules into two major categories: (1) framework rules, encompassing both civil and criminal law; and (2) regulations, including restrictions on price, quantity, quality, and information, as well as more indirect controls relating to registration, certification, and licensing of market participants. Table 10.3 sets out these generic policies.

Frameworks

We should not forget that it is meaningless to talk of competitive markets *except* within a rule-oriented framework. Lester Thurow makes this point strongly:

> There simply is no such thing as the unregulated economy. All economies are sets of rules and regulations. Civilization is, in fact, an agreed upon set of rules of behavior. An economy without rules would be an economy in a state of anarchy where voluntary exchange was impossible. Superior force would be the sole means for conducting economic transactions. Everyone would be clubbing everyone else.[101]

The genesis of this idea goes back to Adam Smith, who recognized the need for frameworks when he pointed out that the first inclination of people in the same line of business when they gather together is to collude and subvert the operation of a competitive market.[102] The competitive market itself, then, can be thought of as a public good that will be undersupplied if left to private actors to organize. Contract law, tort law, commercial law, labor law, and antitrust law can all be thought of as *framework rules.*

Table 10.3 Establishing Rules

Generic Policies	*Perceived Market Failure (MF), Government Failure (GF), Distributional Issue (DI), Limitation of Competitive Framework (LCF)*	*Typical Limitations and Collateral Consequences*
	Frameworks	
Civil laws (especially liability rules)	MF: Negative externalities MF: Information asymmetries MF: Public goods DI: Equal opportunity LCF: Thin markets	Bureaucratic supply failure; opportunistic behavior; imbalance between compensation and appropriate deterrence.
Criminal laws	MF: Negative externalities MF: Public goods LCF: Illegitimate preferences	Costly and imperfect enforcement.
	Regulations	
Price regulation	MF: Natural monopolies DI: Equity in distribution of scarcity rent DI: Equity in good distribution	Allocative inefficiency; X-inefficiency.
Quantity regulation	MF: Negative externalities MF: Public goods, especially open access	Rent seeking; distorted investment; black markets.
Direct information provision	MF: Information asymmetries	Cognitive limitations of consumers.

(disclosure and labeling)		
	MF: Negative externalities	
Indirect information provision (registration, certification, and licensing)	MF: Information asymmetries	Rent seeking; cartelization.
	MF: Negative externalities	
	GF: Bureaucratic supply failure	
Regulation of the circumstances of choice	LCF: Cognitive limitations to rationality	Few applications discovered so far beyond opt-out versus opt-in.

Although few dispute that a system of criminal and civil rules is in and of itself efficiency enhancing, there is considerable debate over the most efficient structure of such rules. In civil law, for example, the exact nature of optimal liability rules (say, negligence versus strict liability standards) is contentious. Much of the literature on law and economics is concerned with the specification of such optimal rules in various contexts.[103]

One of the most basic public goods is the establishment and enforcement of property rights, including the rights to health and safety. In the United States, Canada, and other countries with roots in English common law, tort systems allow those who have suffered damages to seek compensation through the courts. Depending on the particular rules in force, the possibility of tort lowers the expected loss that consumers face from collateral damage and deters some risky behavior by producers in situations involving information asymmetry. Because tort often involves substantial transaction costs, however, it may not work effectively as a deterrent or compensation mechanism when the damage suffered by individual consumers is relatively small. (Small claims courts and class action suits are attempts to deal with this problem.) Because the liability of corporations is generally limited to their assets, tort may be ineffective when large losses result from products produced by small corporations. In addition, we expect tort to be least effective in limiting the inefficiency of information asymmetry in cases of post-experience goods, because of the difficulty of establishing links between consumption and harmful effects.

Contract law can also be formulated to reduce the consequences of information asymmetry. For example, insurance law places the burden of informing the insured of the nature and extent of coverage on the insurer.[104] Even when contracts state explicit and contradictory limits, the reasonable expectations of the insured are generally taken by the courts in the United States as determining the extent of coverage. Indeed, most insurance agents carry their own "errors and omissions" insurance to cover their liability when the impressions they convey to clients diverge from the coverage implied by contracts. In many jurisdictions, other types of agents, such as those selling real estate, have a duty to disclose certain product characteristics related to quality. Such rules, however, may create an incentive for firms not to discover negative characteristics of their products. To be effective, therefore, they may have to be coupled with "standard of care" rules.

Antitrust law, which can be designed to employ either criminal or civil procedures, seeks to prevent firms from realizing rents through collusive efforts to restrict competition. As briefly discussed in Chapter 6, the opportunity for collusion usually arises when only a few firms dominate an industry. Firms may form a cartel and attempt to allocate output among members in an effort to achieve the monopoly price. The allocations may be through such mechanisms as explicit quotas, geographic divisions of markets, and bid rigging to rotate contract awards. Where collusion is illegal, these efforts often fail because cartel members cannot make enforceable contracts to prevent cheating. The possibility of entry by new firms and the availability of substitutes also limit the ability of cartels to sustain prices above market levels. Nevertheless, active enforcement of antitrust laws may be necessary to preserve competition in some concentrated industries. One approach is investigation and prosecution under criminal laws by a government agency, such as the Antitrust Division of the U.S. Justice Department. Another is the creation of incentives, such as multiple damages, to encourage those who have been harmed by collusion to bring civil suit.

Framework rules can also be used to counter some of the problems associated with government failures. For example, individual rights given as constitutional guarantees can protect minorities from tyranny of the majority under direct and representative democracy. Similarly, restrictions on gifts and perquisites that can be given to representatives may help avoid the most blatant abuses of rent seeking.

Regulations

While framework rules facilitate private choice in competitive markets, *regulations* seek to alter choices that producers and consumers would otherwise make in these markets. Regulations generally operate through *command and control*: directives are given, compliance is monitored, and noncompliance is punished. Regulation can be extremely costly. The budgetary costs of U.S. federal regulation (only a relatively small part of the total costs) have been estimated to exceed $40 billion.[105] Similarly, the costs of industry-specific economic regulation have been massive.[106] Of course, high cost alone does not imply that regulation is not worthwhile, but it reminds us that it should be rigorously justified. Additionally, the evidence suggests that, perhaps unsurprisingly, those subjected to regulation tend to overestimate the cost of regulatory compliance.[107]

Price Regulation

In Chapter 8 we analyzed the efficiency costs of imposing either price ceilings or price supports on a competitive market (see Figures 8.2 and 8.3). The conclusion that price regulation leads to inefficiency can be generalized to wage and price controls and income policies, which have frequently been adopted by Western governments, including the United States.[108] The extent of the inefficiency is often difficult to determine, however.

If the quality of a good is variable, then the imposition of price floors often results in firms competing on the basis of quality rather than price. The result may be excessive quality and higher prices, but only normal rates of return. In contrast, price ceilings often lead to declines in product quality, enabling firms to sell at a price closer to the competitive level for the lower-quality product. In other words, social surplus analysis that assumes that quality remains constant does not always tell the full story of the impact of price regulation.

Price regulation is frequently used as a method of preventing monopolies from charging rent-maximizing prices. For example, the Cable Television Consumer Protection and Competition Act of 1992 gave the Federal Communications Commission (FCC) authority to regulate cable TV rates. The FCC ordered a 10 percent rate rollback in 1993, and a further 7 percent reduction in 1994. The FCC estimated the competitive price by comparing prices of monopoly cable systems to those with duopoly systems (called "overbuilt" systems in industry parlance). It found that duopoly systems had prices that were, on average, approximately 16 percent lower than those of monopoly systems. It is estimated that these price reductions reduced industry revenues by $3 billion.[109]

As we saw in Chapter 5, if a natural monopoly can be forced to price at average cost, then deadweight losses are much lower than under rent-maximizing pricing. Many regulatory regimes attempt to force natural monopolies to price at average cost by regulating prices. Historically, the usual approach has been *rate-of-return* regulation. Large sectors of our energy, transportation, communication, and urban services sectors are, or have been, regulated in this way. Typically, the statutes authorizing regulations speak of "reasonable" prices and profits. In practice, however, it is often not clear what is the relative importance of efficiency and equity in defining reasonableness.

There have been two major lines of criticism to the use of this type of price regulation to limit undersupply by natural monopolies. First, in line with various forms of government failure, George Stigler and others have argued that the regulators are quickly "captured" by the firms that they are supposed to be regulating, such that the outcome may be worse than no regulation at all.[110] Second, such regulation induces inefficient and wasteful behavior. Two well-documented outcomes of such incentives are *X*-inefficiency and an overuse of capital under rate-of-return regulation.[111]

To counteract these inappropriate incentives, a newer form of price regulation, labeled *price cap* regulation, has emerged; it is also referred to as *yardstick competition* regulation. Although largely developed in the United Kingdom to regulate newly privatized and entry-deregulated industries, it is now being adopted in many countries, including the United States.[112] Over the last 15 years, almost all states switched their regulation of local exchange telephone companies from rate-of-return regulation to price cap regulation.[113] Price cap regulation is frequently used to regulate industries with natural monopoly characteristics, in conjunction with the removal of legal entry barriers, vertical disintegration, and sometimes market share restrictions. In this form of price regulation, the regulatory agency sets an allowed price for a specified time period, such as four years. The allowed price is adjusted annually to take into account inflation but reduced by a percentage to reflect cost reductions that the regulator believes the firm can achieve, based on some notion of the extent of previous *X*-inefficiency and the extent that new technology may allow cost reductions (that is, the potential for technical efficiency improvements, not related to *X*-inefficiency). For example, if the rate of inflation is 3 percent annually and the required annual productivity gain is 4 percent, the nominal price allowed would decrease 1 percent annually. Typically, the regulated firm is allowed to reduce prices if it wishes to do so (in other words, there is a ceiling but no floor); as price cap regulation is often accompanied by open entry, incumbents may, in fact, be forced to lower prices more quickly than the cap price.

An advantage of such regulation is that it focuses on dynamic efficiency—managers have an incentive to improve productivity over time because they get to keep the surplus if they can reduce costs faster than the cap price declines. However, price cap regulation requires the regulator to estimate the achievable cost reductions periodically. Ideally, the estimate should provide incentives for firms to reduce cost, but not to allow them to capture too-high profits. This draws the regulator into a process that has some similarities to old-style rate-of-return regulation, where the regulators can be influenced by the regulated firms or other interest groups.[114] One important difference, however, remains between price cap regulation and rate-of-return regulation: price cap regulation requires investors in the firm to carry the risk of profit variability.[115]

There is evidence that price cap regulation has led to significant efficiency improvements for firms with natural monopoly characteristics. For example, one study found efficiency gains at the British Gas corporation, even though the firm remained a (vertically integrated) monopoly during this period.[116] Again, though, these regulatory changes have been accompanied by other important changes, such as privatization, so that it is difficult to attribute efficiency improvements to one cause.

Notice that using price regulation to correct for market failure caused by natural monopoly is only one alternative. We have already considered the alternative possibilities of dealing with natural monopoly through auctions and subsidies; we will consider government ownership below. Here we see the substitutability of generic policies. Analysis of the specific context is needed to determine which generic policy is most appropriate.

Price ceilings are sometimes used in an attempt to transfer scarcity rents from resource owners to consumers. For example, during the 1970s ceilings kept the wellhead price of U.S. and Canadian crude oil well below world market levels. While these controls transferred rents from the owners of oil reserves to refiners and consumers, they reduced the domestic supply of oil, increased demand for oil, and contributed to higher world oil prices.[117] In general, the use of price ceilings to transfer scarcity rents involves efficiency losses.

Quantity Regulation

We have already mentioned quantity regulation as a means of controlling negative externalities, in our discussion of taxes and subsidies. Although quantity regulation is less flexible and generally less efficient than market incentives, it usually provides greater certainty of outcome. Therefore, it may be desirable in situations where the cost of error is great. For example, if an externality involves a post-experience good with potentially catastrophic and irreversible consequences, directly limiting it may be the most desirable approach. Should we risk using economic incentives to control the use of fluorocarbons, which have the potential of destroying the ozone layer? If, in the first taxing iteration, we overestimate the price elasticity of demand, the cost and difficulty of achieving reductions later may be great. Other potentially planet-threatening dangers, such as global climate change, raise similar issues.

The relative steepness of the marginal benefit and marginal abatement (cost) curves should determine whether a price or quantity instrument should be used. Suppose that the marginal benefit curve of reductions in a given pollutant is quite steep, and the marginal costs are constant. In this situation, the benefits from changes in environmental quality vary greatly with changes in pollution levels: the costs from too-high levels of the pollutant will be high. Quantity controls are appropriate. If, on the other hand, the marginal benefit curve is flat but the marginal cost curve is steep, then the potential policy problem is the deadweight loss from too-tight regulation. Pollution taxes are appropriate.

In the United States, quantity regulation of pollutants has often taken the form of specifications for the type of technology that must be used to meet standards. For example, the 1977 Clean Air Act Amendments required that all new coal-fired power plants install flue gas scrubbers to remove sulfur emissions, whether the plant used low- or high-sulfur coal.[118] This approach was appealing to representatives from states that produce high-sulfur coal, to owners of old plants whose new competitors would enjoy less of a cost advantage, and to many environmentalists who saw it as a way of gaining reductions with minimal administrative costs.[119] By raising the cost of new plants, however, the scrubber requirement may actually have slowed the reduction of aggregate sulfur emissions by reducing the speed at which electric utilities replace older plants. In general, regulating how standards are to be met reduces flexibility and thus makes the standards more costly to achieve.

Workplace health and safety regulation in the United States under the administration of the Occupational Safety and Health Administration (OSHA) has also focused on technology-based standards. This is partly because the U.S. Supreme Court explicitly rejected the use of cost–benefit analysis by OSHA in the administration of workplace safety and health.[120] The standard-based approach of OSHA has been controversial, especially among economists. A review of the empirical literature concluded with the rather mixed message that "… there is no evidence of a substantial impact of OSHA [but] … the available empirical results for the overall OSHA impact … suggest that OSHA enforcement efforts may be beginning to enhance workplace safety."[121]

It is worth noting that tradable emissions permits (discussed earlier in this chapter under simulating markets) also contain an element of quantity regulation. Before trading begins, a total quantity of allowable pollution must be established. Tradable permits, therefore, combine some of the advantages of quantity control with those of more market-like mechanisms, such as taxes and subsidies.

The use of quotas in international trade is an illustration of quantity regulation, albeit one that has been vigorously criticized.[122] As we have already discussed, tariffs and quotas differ little in theory: in the case of tariffs, prices are raised until imports are reduced to a certain level; in the case of quotas, the government limits the import level directly, and prices adjust accordingly. In practice, tariffs have the advantage of being more accommodating than quotas to changes in supply and demand. Additionally, tariffs are generally easier to implement than quotas because the latter require the government to distribute rights to the limited imports. There are also important distributional differences between quotas and tariffs: quotas transfer rent to foreign producers, while domestic governments capture tariff revenue.

Jose Gomez-Ibanez, Robert Leone, and Stephen O'Connell have examined the impact of quotas on the U.S. automobile market.[123] They estimate that the short-run consumer losses are in excess of $1 billion per annum and that "it is apparent that under most assumptions the economy as a whole is worse off because of restraints."[124] They also point out that the longer-run "dynamic" efficiency costs may be even more significant than the short-run costs, as Japanese automobile producers adjust their behavior in the face of quotas.

The most extreme form of quantity regulation is an outright ban on usage or ownership, usually enforced via criminal sanctions. As discussed in Chapter 1, in 1984 the U.S. National Organ Transplant Act banned the sale or purchase of human organs.[125] Many countries are currently considering whether to legally ban human cloning. More familiar prohibitions include bans on gambling, liquor, prostitution, and drugs such as heroin and cocaine. Again, countries have chosen to use this policy instrument differently. Many countries ban the private ownership of handguns, while most jurisdictions in the United States allow sale and ownership. However, no country that we know of has banned the use or sale of tobacco.[126] There has not been a consensus on these types of prohibitions among the analytical community. This is not surprising because many of the issues relate to strongly held ethical values. As we saw in our discussion of legalization, changes in such values often lead to calls for freedom of choice. If there is extensive consumer demand for illegal products, then the usual result is black markets, which generate deadweight losses and negative externalities.

One context in which criminal sanctions are now being used more extensively is the regulation of toxic chemicals. In one well-known case, several Illinois businessmen were convicted of murder and sentenced to 25 years' imprisonment for the toxic-chemical-related death of an employee.[127] While such severe criminal sanctions may provide a powerful deterrent, they may also discourage firms from reporting accidents where they may have been negligent.

The design of efficient standards requires that noncompliance be taken into account. If we want those subjected to standards to make efficient decisions, then they should see an expected punishment of an order of magnitude of the external costs of their behavior. For example, if dumping a toxic substance causes $10,000 in external damage, and the probability of catching the dumper is 0.01, then the fine should be on the order of $1 million to internalize the externality. The political system, however, may not be willing to impose such large fines. Nor may it be willing to hire enough inspectors to increase the probability that noncompliance will be detected. In general, the problem of noncompliance limits the effectiveness of standards.[128]

Direct Information Provision

We discussed information asymmetry extensively in Chapter 5. We noted that the quality of many goods cannot be evaluated until long after they are consumed (for example, post-experience goods like asbestos insulation and certain medical treatments). As products become more technologically complex, product quality is likely to become an increasingly important area of policy concern.

The presence of information asymmetry suggests a relatively simple policy: require the provision of the information! An important question is whether it is more effective for government to supply such information to consumers or to require the suppliers of goods to provide the information. Few studies directly address this important question. In practice, governments tend to engage in both strategies. For instance, the U.S. government, through the Centers for Disease Control and Prevention, provides information concerning the health impact of cigarette smoking. It also requires cigarette manufacturers to place warning labels on their products.

The practice of requiring firms to supply information about various attributes of product quality has become commonplace.[129] Examples include appliance energy efficiency labeling, automobile mileage ratings, clothing care labeling, mortgage loan rate facts, nutrition and ingredient labeling, octane value labeling, truth-in-lending provisions, and warranty disclosure requirements.[130] Additionally, federal, state, and local governments have devoted much effort to instituting so-called plain language laws to make contracts readable.[131]

Another variation on the theme of requiring suppliers of goods to provide information is to facilitate the provision of information by employees free from fear of reprisal (whistle-blower protection).[132] Currently, airline employees can communicate safety problems to federal regulators, presumably without being exposed to employer reprisals. Clearly, airline pilots have a more direct incentive to report safety problems than do airline executives. Of course, this policy approach can be applied to government itself by facilitating whistle-blowing protection to employees who report inefficiency or corruption.

Organizational report cards, which provide consumers comparative information on the performance of organizations such as schools, health maintenance organizations, and hospitals, are becoming increasingly common.[133] When organizations have a strong incentive to participate, organizational report cards may be provided by private organizations. For example, colleges willingly provide information to *U.S. News & World Report* for its annual report card on colleges, because not being listed would hurt their efforts to compete for students. Many health maintenance organizations participate in report cards at the insistence of large employers that purchase their services. When organizations do not have an incentive to participate, then the government may intervene, either to require the publication of information necessary for private report card makers or to produce the report cards directly. For example, many states now publish report cards on school districts or require the districts to provide parents with comparative information about particular schools, and the U.S. federal government provides report cards on nursing homes, dialysis centers, and hospitals.[134]

Direct information provision is likely to be a viable policy response to pure information asymmetry problems. It is an attractive policy option because the marginal costs of both providing the information and enforcing compliance tend to be low. It may be less viable when information asymmetry is combined with other market imperfections, such as limited consumer attention, misperception of risk, endogenous preferences, or addictive products. For example, providing information about the adverse health effects of smoking may not be an adequate policy response because of the addictive properties of tobacco. If we believe that many smokers are incapable of rationally evaluating the health risks, then further regulation may be appropriate, including, perhaps, the imposition of quality standards.

Standards provide information to consumers by narrowing the variance in product quality. For example, the Food and Drug Administration (FDA) requires that scientific evidence of efficacy and safety be presented before a new drug can be sold to the public. These quality standards provide as least some information to consumers about marketed drugs.

The effective use of quality standards may be limited by government failure. Regulatory bureaus often lack the expertise to determine appropriate quality standards. Also, they may operate in political environments that undervalue some errors and overvalue others. For example, when the FDA allows a harmful product to reach the market, it is likely to come under severe criticism from members of Congress who can point to specific individuals who have suffered. In contrast, until the 1990s the FDA received virtually no congressional criticism when it prevented a beneficial product from reaching the market, even though large numbers of (generally unidentified) people forwent benefits.[135] Increasingly, however, the FDA is subject to pressure from organizations created to increase the availability of treatments for sufferers of particular disease in the wake of the successful efforts of HIV/AIDS advocates to speed up the release of experimental drugs —over 3,000 disease-specific advocacy groups, many supported by pharmaceutical companies, engage in at least some political activity.[136]

Programs of quality regulation and information disclosure run the risk of regulatory capture. Firms in the regulated industry that can more easily meet quality and disclosure standards may engage in rent seeking by attempting to secure stringent requirements that inflict disproportionate costs on their competitors and increase barriers that make it more difficult for new firms to enter the industry. In addition to reducing competition, stringent standards may also impede innovation by forcing excessive uniformity.[137] The retarding of innovation is likely to be especially serious when the standards apply to production processes rather than to the quality of final products. Delegating standard-setting authority to private organizations may facilitate incremental standard setting that better reflects changing technology.[138]

Indirect Information Provision

Unfortunately, direct information about the quality of services, as opposed to physical products, usually cannot be provided. Typically, service quality is not fixed; it may change over time, either with changes in the level of human capital or the amount of input effort. Because the quality of services may vary over time, providing reliable information about their quality directly may be impractical. The infeasibility of providing such direct information has led policymakers and analysts to search for indirect ways of providing information. A common policy approach is to register, license, or certify providers who meet some standard of skill, training, or experience.

Licensure can be defined as "a regulatory regime under which only the duly qualified who have sought and obtained a license to practice from an appropriate agency or delegate of the state are legally permitted to perform or to take responsibility for given functions."[139] It can be distinguished from *registration*, which allows those seeking to practice to do so through a simple declaration. It can also be distinguished from *certification* "under which qualified practitioners receive special designations or certifications which other practitioners cannot legally use; however, uncertified practitioners are legally permitted to provide the same functions, provided they do so under some other designation. Certification involves exclusive rights to a professional designation but not to practice."[140]

Milton Friedman has succinctly summarized the weakness of using these approaches: "The most obvious social cost is that any one of these measures, whether it be registration, certification, or licensure, almost inevitably becomes a tool in the hands of a special producer group to obtain a monopoly position at the expense of the rest of the public."[141] Because certification usually does not forestall entry, this criticism has focused particularly on licensure.[142]

While the rationale for providing indirect information through licensure does not necessarily imply *self*-regulation, this is the route that almost all states have taken for occupational licensure. The typical steps are members form a professional association, the association sets up a system of voluntary licensure, and the profession petitions the legislature for mandatory licensure under the auspices of its association.[143]

Occupational licensure suffers from a number of weaknesses: the correlation between training or other measurable attributes and performance may be low; the definition of occupations may "lock in" skills that become outmoded in dynamic markets; high entry standards deny consumers the opportunity to choose low-quality, but low-price, services; and when professional interests control the licensing standards, the standards may be set unduly high to restrict entry so as to drive up the incomes of the existing practitioners.[144]

While all of these problems deserve serious attention, economists have tended to concentrate on the social costs of entry barriers and the resulting monopoly pricing—perhaps because it is one of the easier licensure impacts to measure. There is certainly strong empirical evidence that professional cartels do indeed raise prices and restrict competition.[145] In light of the existence of these excess returns and what you have already learned about rent seeking, you should not be surprised to hear that many professional and quasi-professional groups continue to seek licensure.[146] We should, therefore, be cautious in advocating licensure as a policy alternative. When we do, we should consider alternatives to professional *self*-regulation. Yet, we believe it is fair to say that the critics of professional regulation have not been particularly imaginative in proposing alternative policies.[147]

Licensure is now spreading beyond its traditional boundaries. Numerous paraprofessional occupations are winning the right to license and self-regulate. One interesting proposal even suggests that prospective parents should be licensed before having children![148]

Regulating the Circumstances of Choice

In Chapter 6, we noted that behavioral economists seek to incorporate the findings of cognitive psychology into models of economic behavior. The potential usefulness of behavioral economics in policy design is suggested by the large differences usually observed between opt-in and opt-out systems—the latter usually produces higher levels of participation than the former in applications like savings plans. Richard Thaler and Cass Sunstein call policy design that takes advantage of such findings "choice architecture."[149] This approach responds to the evidence that people often do not invest heavily enough in making choices what might nonetheless be important for their well-being, and that self-beneficial choices may be encouraged by changing the circumstances surrounding them.

Changing the circumstances of choice often involves making what the regulator believes to be the best choice an easier one for people to make. It may operate through changes in visibility or convenience. For example, to encourage children to eat healthier school lunches, foods may be arranged by order, height, and other factors affecting prominence to favor fruits and vegetables over desserts and fatty foods. Some college cafeterias have attempted to reduce waste in "common property" cafeterias by removing trays so that it is less convenient to take large quantities the first time through the line.

A number of countries have created units to explore opportunities for using nudges to advance social policy goals. For example, in 2010 the United Kingdom government established the Behavioral Insights Team, often referred to as the "Nudge Unit," to consider applications of behavioral economics. It has issued reports on the possible uses of nudges in the areas of health, energy conservation, and tax compliance.[150] Although regulating the circumstance of choice is generally less intrusive than more direct quantity regulation, it may not always be viewed as such. For example, a recent proposal by British economist and public figure Julian Le Grand to require an annual license, issued at no charge, to buy cigarettes set off considerable controversy.[151]

Summary: The Wide World of Rules (and Some Checks)

As emphasized in the introduction to this section, rules are the most pervasive form of government policy in all our lives. National and subnational (state or provincial, local) regulations continue to multiply. The U.S. federal government has required the application of cost–benefit analysis to major new rules since the Reagan administration.[152] Some states have followed suit.[153]

Supplying Goods through Nonmarket Mechanisms

Surprisingly, policy analysts have had relatively little to say about when government provision of goods through public agencies is an appropriate response to market failure. Peter Pashigian puts it bluntly: "Public production of goods and services is somewhat of an embarrassment to most economists. It exists, and will in all likelihood increase in importance, but it is difficult to explain. An acceptable theory of public production has not yet appeared."[154]

Indeed, the choice between government provision and other generic policies is one of the least well-understood policy issues. Perhaps the reason is that while we have a convincing theory of market failure and an increasingly convincing theory of government failure, we have no overarching theory that delineates the efficiency trade-offs between market failure and government failure.[155] As we discuss more fully in Chapter 13, perhaps the best hope for such an ovearching theory is transaction cost economics.

It is tempting to believe that the theory of market failure itself resolves the dilemma of the choice between public and private supply: direct government provision is appropriate when there is endemic market failure. This is a weak argument because it ignores the fact that market failures can be addressed with other generic policies. Therefore, market failure provides a rationale generally for government intervention, rather than specifically for direct provision by government. For example, consider the possible responses to natural monopoly: auctions to take advantage of competition for markets, subsidies to facilitate marginal cost pricing, and direct regulation. These approaches are alternatives to government ownership of the monopoly.

William Baumol provides the beginnings of a moral hazard theory of government production.[156] His insight is best introduced with an example: the provision of national defense. The common rationale for the provision of national defense by government rests on the argument that it is a public good. But this is only an argument for government intervention, not a public army, per se. Baumol's case for a public army is made in terms of moral hazard. A government intends to use its armed forces only when needed for its own purposes. An army provided by the market to the government under contract may have very different incentives that bring into question its loyalty and reliability.

We suggest that the moral hazard problem can be viewed more broadly in the context of transaction costs and, more specifically, opportunism (see Chapter 13 for more detail). Again, we can illustrate the crux of the problem using the example of national defense: private armies cannot be trusted to carry out the contracts because the government does not have an independent mechanism of enforcement. In other words, the armies are in a position to engage in opportunistic behavior, like the soldiers in the Thirty Years' War who were lured from one side to the other with bonuses and higher wages.[157] Even if a particular private army does not engage in such behavior, the state will inevitably be forced into costly monitoring. For instance, the government would at least have to establish an office of inspector general to make sure that the contractor can actually muster the promised forces.

Although the danger of opportunistic behavior is most starkly revealed in the area of national defense, it is by no means unique to it. The collection of taxes, the printing of money, the administration of justice—all could potentially face serious agency problems if supplied by private firms.

Our analysis suggests a *double market-failure test* for the use of public agencies: first, evidence of market failure or a redistributive goal; and second, evidence either that a less intrusive generic policy cannot be utilized or that an effective contract for private production cannot be designed to deal with the market failure (that is, opportunism cannot be reasonably controlled). While we believe that this double market-failure approach provides a rationale for government production, only modest progress has been made in delineating the range of circumstances in which government production is likely to be the best generic policy.[158] The task is especially difficult because public agencies themselves, as we saw in Chapter 8, are prone to serious principal–agent problems.

In spite of these theoretical difficulties, we can at least provide an overview of alternative forms of nonmarket supply. Broadly speaking, once a decision has been made to supply goods publicly, governments can do so either directly or indirectly. Direct supply involves production and distribution of goods by government bureaus. As outlined in Table 10.4, the major means of indirect supply are independent agencies (usually government corporations or special districts) or various forms of contracting out. Although these distinctions are theoretically clear, in practice they are often less so.

Direct Supply by Government Bureaus

The direct production of goods by government bureaus is as old as government itself. Karl Wittfogel argues that the stimulus for the development of many early civilizations in semiarid regions (for example, Egypt, Mesopotamia, and parts of China and India) came from the need to construct and manage water facilities collectively. This, in turn, tended to generate direct government supply of nonhydraulic construction: "under the conditions of Pharaonic Egypt and Inca Peru, direct management prevailed."[159]

A note of caution: there is no bright line separating government bureaus from government corporations, an issue which we consider next. The U.S.

government provides a vast array of goods through agencies, such as the Army Corps of Engineers, the Bureau of the Mint, the National Forest Service, and the Cooperative Extension Service of the Department of Agriculture, that vary in their formal independence. Even in the early days of the Republic, direct supply was extensive when the activities of the states are included.[160] Christopher Leman finds that domestic government production can be divided into ten functional categories: facilitating commerce; managing public lands; constructing public works and managing real property; research and testing; technical assistance; laws and justice; health care, social services, and direct cash assistance; education and training; marketing; and supporting internal administrative needs.[161] The "nondomestic" functions of government include national defense and the administration of foreign policy.

Table 10.4 Supplying Goods Through Nonmarket Mechanisms

Generic Policies	*Perceived Market Failure (MF), Government Failure (GF), Distributional Issue (DI), Limitation of Competitive Framework (LCF)*	*Typical Limitations and Collateral Consequences*
	Direct Supply	
Bureaus	MF: Public goods	Rigidity; dynamic inefficiency; and X-inefficiency.
	MF: Positive externalities	
	MF: Natural monopolies	
	DI: Equity in distribution	
	Independent Agencies	
Government corporations	MF: Natural monopolies	Agency loss.
	MF: Positive externalities	

Generic Policies	*Perceived Market Failure (MF), Government Failure (GF), Distributional Issue (DI), Limitation of Competitive Framework (LCF)*	*Typical Limitations and Collateral Consequences*
Special districts	DI: Equity in distribution	Agency loss; insensitivity to minorities with intense preferences.
	GF: Bureaucratic supply failures	
	MF: Natural monopolies	
	MF: Local public goods	
	MF: Negative externalities	
	DI: Universal provision	
Contracting Out		
Direct contracting	MF: Public goods, especially local public goods	Opportunistic behavior by suppliers; lock-in and low-balling.
	GF: Bureaucratic supply failures	
Indirect contracting (nonprofits)	MF: Positive externalities	Weak coordination of services.
	GF: Bureaucratic supply failures	
	DI: Diversity of preferences	
	LCF: Endogenous preferences (behavior modification)	

A perusal of Leman's categories suggests that all could be justified by some market failure or redistributive rationale. Yet this does not address the question of whether these goods could be efficiently provided by other generic policies or provided *indirectly* by government. Clearly, some of these goods (or at least close substitutes) could be supplied using other generic policies; for instance, elements of health care can be variously provided in free markets, through market incentives, and under rules.

Leman also points out that other interventionist generic policies inevitably involve some level of *direct* government provision.[162] Some examples: the provision of in-kind subsidies requires a bureau to dispense the goods and vouchers; the use of rules requires agencies to monitor and enforce compliance; and even the creation of tradable property rights may require agencies to act as banks and clearinghouses.

Our discussion in Chapter 8 of the various forms of government failure, especially those of bureaucratic supply, suggests that government supply is not a panacea. The policy pendulum appears to have swung against the use of public agencies, as shown by the continuing interest in privatization and contracting out, as well as a growing interest in supply through networks of nongovernmental organizations.[163]

Independent Agencies

The range and types of (more) independent agencies than bureaus are enormous (and are discussed at more length in Chapter 13). The British use the term *quango* (quasi-autonomous nongovernmental organization) to describe bodies that are not explicitly government departments. The terms *corporatization* and *agencification* are also widely used to describe these entities.[164] Such agencies, many of which are "off budget" with their own funding sources, have grown explosively in almost all countries around the world, including the United States. Ann-Marie Walsh identified more than 1,000 state and interstate authorities, as well as more than 6,000 subnational authorities.[165]

These regional, state, and local agencies construct and operate a wide range of facilities, such as dams, airports, industrial parks, and bridges, and provide a wide range of services, including water, gas, and electricity. Because there is no universal nomenclature, however, even identifying and classifying such entities is problematic. Many authorities have considerable independence but are not corporate in form, while others are corporate but are formally included in government departments. Even more confusingly, formal inclusion in a bureau does not necessarily subject the corporation to departmental supervision or regulation. With these caveats in mind, here we briefly review two forms of agency: government corporations and special districts.

Government Corporations

Government corporations are found everywhere, especially in developing countries.[166] Government corporations generally operate with their own source of revenue under a charter that gives them some independence from legislative or executive interference in their day-to-day operations. For example, electric utilities owned by municipalities generate revenue from the electricity that they sell. Usually, their charter requires them to do so at the lowest prices that cover operating costs and allow for growth.

Although government corporations are less important in the United States than in many other countries, their presence is not insignificant. The Tennessee Valley Authority generated $11 billion in revenue in 2015, making it one of the largest electric utilities in North America; other important federally owned electricity producers include the Bonneville Power Administration, the Southeastern Federal Power Program, and the Alaska Power Administration. Major federal corporations include the Federal National Mortgage Association, the Postal Service, the Federal Deposit Insurance Corporation, and the Corporation for Public Broadcasting.[167] At the subnational level, major enterprises include the Port Authority of New York and New Jersey, the New York Power Authority, and the Massachusetts Port Authority.

What is the rationale for government corporations? Most government corporations, at least in the United States, meet the *single market-failure test* —that is, they are in sectors in which natural monopoly or other market failure suggests the need for government intervention. In our view, the critics of government corporations are implicitly arguing that few meet the double market-failure test. The superficial appeal of government corporations is that they can correct market failures while presumably retaining the flexibility, autonomy, and efficiency of private corporations. Are government corporations less prone to principal–agent problems than government bureaus?

Louis De Alessi and other critics contend that government corporations are more prone to principal–agent problems than private firms: "The crucial difference between private and political firms is that ownership in the latter effectively is nontransferable. Since this rules out specialization in their ownership, it inhibits the capitalization of future consequences into current transfer prices and reduces owners' incentives to monitor managerial behavior."[168] The theoretical force of this argument is somewhat weakened by the growing realization that large private firms are also subject to principal–agent problems, especially if shareholding is dispersed.[169] The majority of empirical studies do, indeed, find that dispersed shareholding results in lower profitability and higher costs.[170]

Are public firms less efficient than private firms? Most surveys of empirical studies that compare private and public performance find superior private-sector performance in reasonably competitive sectors.[171] Anthony Boardman and Aidan Vining compared the performances of the largest non–U.S. public, private, and mixed firms operating in competitive environments. After controlling for important variables, such as industry and country, they found that public and mixed firms appeared to be less efficient than the private firms.[172] These findings, of course, are closely related to the findings concerning the effects of privatization discussed earlier in this chapter.

The cross-sectional evidence, however, is more mixed in sectors such as electricity, gas, and water collection and distribution where there is little competition (at least historically) and usually extensive (rate-of-return) regulation.[173] In some parts of these industries this may be due to natural monopoly conditions.[174] Here, *X*-inefficiency among private firms resulting from limited competition and rate-of-return regulation is not surprising. Performance differences appear to be small, suggesting that the degree of competition in a given market is a better predictor of efficient performance than ownership, per se.

From the 1990s on, however, a new type of evidence became available on comparative performance: "before and after" (that is, time-series) evidence on the privatization of public firms. This evidence strongly suggests large efficiency gains, both *X*-efficiency and allocative efficiency, and is broadly consistent over industrialized countries, former Soviet bloc countries, and developing countries.[175] However, the evidence from the Soviet bloc countries and from developing countries suggests that governments must create effective institutional environments for private corporate entities to create wealth. In other words, the production of framework rules, considered earlier in this chapter, are a fundamental requirement for efficient capitalism.[176] Without effective governance institutions, the corporation is unlikely to be superior to the bureau, even in the absence of market failure.[177]

Certainly, many countries have, at least until recently, utilized government corporations in an extremely broad range of sectors where there is little evidence of underlying market failure. In France, for example, state firms have produced machine tools, automobiles, and watches. Until 1987, the French government even controlled one of the largest advertising agencies in the country! In Sweden, there is a government corporation that makes beer. Following the financial sector meltdown beginning in 2008, many countries have now renationalized at least some of their banks.

Special Districts

Single-purpose government entities, called *special districts*, are usually created to supply goods that are believed to have natural monopoly, public goods, or externality characteristics. Such goods are typically local in nature, but may extend to the state or region. By far the most common use of special districts in the United States is to provide primary and secondary schooling. Other examples include air pollution, water, and transportation districts.

What are the advantages and disadvantages of special districts relative to cities and counties? Indeed, we might ask why we have cities at all. One advantage of special districts is that they allow consumers to clearly observe the relationship between service provision and tax price. Another advantage is that they can be designed to internalize externalities that spill across the historically evolved boundaries of local governments. A major disadvantage is the costs that consumers face in monitoring a whole series of "mini-governments" for different services: "… only for the most important collective functions will wholly independent organization be justified on cost grounds…. [T]he costs of organizing each activity separately would be greater than the promised added benefits from alternative organization."[178]

An important consideration is that collections of special districts prevent logrolling across issue areas. As we discussed in Chapter 8, logrolling often leads to inefficient pork barrel spending. But it also permits minorities to express intense preferences that are not registered in majority voting on single issues where the social choice only reflects the preferences of the median voter. The more issues that are handled independently in special districts, the fewer that remain for possible inclusion in logrolls. Whether the reduction in logrolling contributes to efficiency and equity depends on the distribution of preferences in the population.

Contracting Out

In the past, private firms mostly provided services *to* government, while government itself provided the final product or service to the public (for example, private firms built the ships, but the government provided the navy), but increasingly the private sector provides services directly to consumers. The U.S. government, for example, purchases regular commercial services like maintenance. Contracting where the service is delivered to consumers is an important component of the health care system and is becoming increasingly important in some sectors of government activity traditionally associated with direct supply, such as corrections (if the service was previously delivered by government directly, this is a form of privatization).[179] An International City Management Association survey found that contracting for a wide range of services increased rapidly in the 1980s, especially at the local level.[180] The trend continues.[181]

What is the evidence on efficiency of direct supply vis-á-vis contracting out? Here, the empirical evidence does suggest that contracting out is frequently more efficient than either market delivery (private subscription) or direct government supply: "… the empirical studies have found that contracting out tends to be less costly than government provision … cover[ing] a number of distinct services and pertain[ing] to a variety of geographical areas."[182] The limitations of these findings, however, should be kept in mind. First, the services examined usually have tangible, easily measurable outputs, such as are found in the provision of highways, garbage disposal, transportation, and food services. Consequently, monitoring the quality of output is relatively straightforward. Second, for these kinds of services contestability is high. And third, neither party to the contract has to commit assets that become sunk (asset specificity is low). While opportunistic behavior is possible in such circumstances, even at its worst, it is likely to be limited in nature. (All three concepts—contestability, asset specificity, and opportunism—are explained in Chapter 13.) Additionally, almost all studies have only compared the production costs of contracting out versus in-house production.

Further, contracting out imposes new, more complex contract-monitoring responsibilities on governments; this is costly. Scholars are just beginning to try to measure these costs. As policy analysts increasingly must make recommendations concerning the choice between in-house production and contracting out.

Contracting out is not limited to common goods and services. The federal government effectively delegates responsibility for the development of rules to stakeholder organizations in a number of areas where it would be difficult for an agency to maintain the expertise needed to gather and interpret necessary information on a timely basis. For example, the federal government delegates responsibility for the development of rules for the allocation of cadaveric organs to the Organ Procurement and Transplantation Network. These kinds of arrangements are often politically attractive to legislators.

Legislators wish to avoid having to intervene with regulatory agencies in situations where the potential costs from being blamed by unsatisfied constituents is greater than the potential benefits from receiving credit from those that are satisfied.[183]

Providing Insurance and Cushions

Some government interventions provide shields against the "slings and arrows of outrageous fortune." We divide these into two general categories: *insurance* and, for want of a better term, *cushions*. Table 10.5 presents a summary of these generic policies: mandatory and subsidized insurance, stockpiles, transitional assistance, and cash grants.

Table 10.5 Providing Insurance and Cushions

Generic Policies	*Perceived Market Failure (MF), Government Failure (GF), Distributional Issue (DI), Limitation of Competitive Framework (LCF)*	*Typical Limitations and Collateral Consequences*
	Insurance	
Mandatory insurance Subsidized insurance	LCF: Adverse selection	Moral hazard.
	MF: Information asymmetries	
	DI: Equity in access	
	LCF: Myopia	
	LCF: Misperception of risk	
	Cushions	

Generic Policies	*Perceived Market Failure (MF), Government Failure (GF), Distributional Issue (DI), Limitation of Competitive Framework (LCF)*	*Typical Limitations and Collateral Consequences*
Stockpiling	LCF: Adjustment costs	Rent seeking by suppliers and consumers. Inequity in availability.
	GF: Price controls	
Transitional assistance (buy-outs, grandfathering)	LCF: Adjustment costs	
	LCF: Macroeconomic dynamics	
Cash grants	DI: Equality of outcome	Reduction in work effort, dependency.
	LCF: Utility interdependence	

Insurance

The essence of insurance is the reduction of individual risk through the pooling of the independent risks borne by members of a group. Insurance can be purchased in private markets to indemnify against loss of life, damage to property, cost of health care, and liability for damages imposed on others. As we discuss in Chapter 6, factors such as moral hazard, adverse selection, and limited actuarial experience can lead to incomplete insurance markets. Furthermore, people do not always make optimal decisions about purchasing insurance coverage because of biases inherent in the heuristics commonly used to estimate and interpret probabilities. Therefore, there may be an appropriate role for public intervention in some insurance markets. More generally, insurance can be used in conjunction with liability laws to deal with problems caused by information asymmetry.

In designing public insurance programs, care must be taken to limit moral hazard. By making certain outcomes less costly, insurance may induce people to take greater risks or bear greater compensable costs than they otherwise would.[184] One way to limit moral hazard is to invest in monitoring systems. Another is to design payment structures that reduce the incentive for beneficiaries to inflate compensable costs. For example, requiring beneficiaries to pay a fraction of their claimed costs through copayments reduces their incentive to incur unnecessary expenses. The general point is that in analyzing proposals for public insurance programs, it is necessary to anticipate how participants will respond to the availability of benefits.

Mandatory Insurance

When people have better information about their individual risks than insurers, adverse selection may limit the availability of insurance. For example, health insurance is sold primarily through group plans because they are less prone to adverse selection than individual policies. Because the premiums for individual policies are set on the assumption that the worst risks will self-select, good risks that do not have access to group coverage may decide not to insure. Government can use its authority to mandate universal participation in insurance plans to prevent adverse selection from operating.[185]

Does the mere existence of an incomplete insurance market justify mandatory insurance? Usually, economists base the rationale for mandatory insurance on the argument that the losses suffered by those left uncovered involve negative externalities. For example, the personal assets of many drivers are insufficient to cover the costs of property damage and personal injury that they inflict on others in serious accidents. Therefore, many states mandate liability coverage for all drivers so that those injured have opportunities to obtain compensation through tort law.

One step beyond nudge is paternalism. Paternalism may serve as a rationale for mandatory insurance. For example, one function of the Old Age, Survivors, Disability and Health Insurance Program, the U.S. Social Security System, is to insure against the possibility that people have insufficient savings for retirement because of myopia, misinformation, miscalculation, bad luck, or simple laziness.[186] People who do not save an adequate amount will either consume at low levels in retirement or receive assistance from private or public charity. It is worth noting that social security is not a pure insurance program but, rather, involves transfers from high-income workers to low-income workers and from current workers to current retirees.[187] Mandatory insurance programs typically combine actuarially based risk pooling with income transfers. Of course, if the programs were not mandatory, then those paying the subsidies through higher premiums or lower benefits would be unlikely to participate.

Mandatory insurance can also be used to privatize regulation. One reason why legal liability alone cannot fully remedy information asymmetries and negative externalities is that firms often have inadequate assets to pay damages. This is especially likely for the short-lived fly-by-night firms that often exploit information asymmetries. By requiring that they carry minimum levels of liability insurance, the government can guarantee that at least some assets will be available for compensating those who have been harmed. Perhaps more important, the insurance companies providing coverage have an incentive to monitor the behavior of the firms with an eye to reducing liability.[188]

It is important to note that mandatory insurance does not require that individuals be restricted from purchasing additional, supplementary insurance. Conceptually, mandatory insurance only requires that everyone, no matter what his or her wishes, acquire the core insurance. Mandatory insurance supplied through a government monopoly is normally based on the argument that if individuals are allowed to purchase additional private insurance their support for the publically financed system will erode, or that health care is so important that everybody should be required to utilize approximately the same amount.[189]

Subsidized Insurance

Rather than mandating insurance, the government can provide it at subsidized premiums when myopia, miscalculation, or other factors appear to be contributing to underinsuring. For example, in the United States, the Federal Emergency Management Agency provides subsidized flood insurance under the Flood Disaster Protection Act of 1973. As might be expected, restrictions on coverage are necessary to discourage those who suffer losses from rebuilding in floodplains.[190]

Fairness often serves as a rationale for subsidizing premiums. For example, residents and businesses in poor neighborhoods may find actuarially fair premiums for fire insurance to be so high that they forgo coverage. A perception that they are unable to move to other locations with lower risks may serve as a rationale for subsidy. Rather than pay the subsidies directly, governments often force private insurers to write subsidized policies in proportion to their total premiums. Such *assigned risk pools* have been created to provide subsidized premiums for fire insurance in many urban areas. They are also commonly used in conjunction with mandatory insurance programs to reduce the otherwise high premiums that the highest risks would face.

Cushions

While insurance schemes reduce the variance in outcomes by spreading risk, *cushions* reduce the variance in outcomes through a centralized mechanism. From the perspective of beneficiaries, the consequences may appear identical. The key difference is that with insurance individuals make ex ante preparations for the possibility of unfavorable outcomes, while with cushions they receive ex post compensation for unfavorable outcomes that occur.

Stockpiling

In an uncertain world, we always face the potential for supply disruptions or "price shocks," whether as a result of economic or political cartelization, unpredictable supply volatility, or man-made or natural disasters. These shocks are unlikely to cause major problems unless they involve goods without close substitutes that are central to economic activity. In practice, the most serious problems arise with natural resources and agricultural products.

With respect to natural resources, if they are geographically concentrated then the owners may be able to extract monopoly rents (as well as scarcity rents) from consumers. In a multinational context, cartels, whether they have primarily economic or political motives, may attempt to extract rents. While the long-run prospects for successful cartelization are limited because higher prices speed the introduction of substitutes, cartels may be able to exercise effective market power in the short run. In addition, the supply of concentrated resources may be subject to disruption because of revolution, war, or natural disaster.

The adverse consequences of these supply disruptions can be dampened with *stockpiling programs* that accumulate quantities of the resource during periods of normal market activity so that they can be released during periods of market disruption. For example, the United States has long maintained government stockpiles of critical minerals like chromium, platinum, vanadium, and manganese, for which production was concentrated in South Africa, China, the former Soviet Union, and other countries of questionable political reliability or friendliness.[191] Experience with the program suggests that political pressure from domestic producers tends to keep items in stockpiles long after they have lost their critical nature and that there is a reluctance to draw down stocks at the beginning of supply disruptions when they are most valuable to the economy.[192]

The most prominent U.S. program that stockpiles resources is the Strategic Petroleum Reserve (SPR), which stores about 700 million barrels of crude oil for use during oil supply disruptions.[193] Because much of the world's low-cost reserves of crude oil are located in the politically unstable Middle East, the possibility exists for sharp reductions in supply that would cause steep rises in the price of oil and consequent losses to the U.S. economy.

Drawdown of the SPR would offset a portion of the lost supply, thereby dampening the price rise and reducing the magnitude of economic losses.

Because private firms could make an expected profit by buying low during normal markets and selling high during disruptions, we might ask why the government should stockpile.[194] One reason is government failure: based on past experience, firms might anticipate the possibility that the government will institute price controls, preventing them from realizing speculative profits. Another reason is market failure: acquisition and drawdown decisions by private firms have external effects on the economy.

In ancient times, agricultural stockpiles served primarily as hedges against scarcity; in modern times, they serve mainly as hedges against abundance. For example, the U.S. Commodity Credit Corporation in effect buys surpluses of grain from farmers to help support prices.[195] While stockpiles of agricultural commodities provide security against widespread crop failures, they are costly to maintain. Further, modest price swings can probably be adequately accommodated by private stockpiling and by risk-diversification mechanisms like private futures markets.

State and local governments, which typically must operate with a balanced budget, can prepare for revenue shocks by stockpiling financial resources. These reserves are sometimes called *rainy day funds*.[196] They serve as a form of self-insurance for governments that have difficulty borrowing on short notice. As with stockpiles of physical commodities, an important question is whether political interests permit efficient decisions concerning acquisition and use. In particular, we might expect the pressures of the election cycle to lead to inefficient decisions concerning use.

Transitional Assistance

Policy analysts and politicians have increasingly come to realize that efficiency-enhancing policy changes are strongly resisted by those who expect to bear net costs. This may result from the elimination of existing benefits or the imposition of new costs. Indeed, if there have only been transitional gains (as discussed in Chapter 8), the elimination of a benefit can impose real economic loss. Resistance to the elimination of benefits is likely to be especially bitter in these circumstances. Government may also wish to pay compensation when its projects would increase aggregate welfare but would impose disproportionate costs on particular individuals or localities.

Compensation may take either monetary or nonmonetary forms. Monetary compensation is typically in the form of a *buy-out,* whereby government purchases, at a fixed price, a given benefit. Examples of cash payments include relocation assistance payments to homeowners and renters displaced by federal urban renewal and highway construction projects.[197] Nonmonetary payment usually takes the form of either a *grandfather clause* or a *hold harmless provision*, both of which allow current right holders to retain their benefits that will not be available in new rights holders in the future. Grandfather clauses can increase political feasibility, but they can also reduce effectiveness.[198]

Cash Grants

The most direct way to cushion people against advance economic circumstances is through *cash grants.* This generic term covers such programs in the United States as Aid to Families with Dependent Children before 1996 and Temporary Assistance for Needy Families after 1996 (welfare), and Supplemental Security Income (aid to the blind and disabled). The 1974 Health, Education, and Welfare "megaproposal" summarizes the situations in which cash grants are preferable to vouchers: "the provision of cash benefits is an appropriate public action when the objective is to alter the distribution of purchasing power in general and when the recipient of the benefit is to be left free to decide how to spend it …," while "… the provision of vouchers is appropriate when the objective is to alter the distribution of purchasing power over specific goods and services when the supply of these can be expected to increase as the demand for them increases."[199]

The advantage (or disadvantage) of cash grants is that they do not interfere with the consumption choices of a target population. This is a valuable attribute when the goal is simply to raise incomes. However, if the objective is to *alter* consumption patterns, then cash grants are less effective instruments. Cash grants do not restrict consumption, but they may affect other kinds of behavior, most importantly the "leisure–labor" trade-off. Increases in unearned income, whether through cash grants, in-kind grants, or other subsidies, generally increase the demand for all goods, including leisure, so that the amount of labor supplied falls.[200]

A fundamental trade-off between transferring income and discouraging work effort arises in designing cash grant programs. For example, consider the goals of a negative income tax: (1) to make any difference, the transfer should be substantial; (2) to preserve work incentives, marginal tax rates should be relatively low; and (3) the breakeven level of income (the point at which the transfer reaches zero) should be reasonably low to avoid high program costs. Unfortunately, at least to some extent, the three are incompatible. For example, if a negative income tax guaranteed every family a floor on income of $4,000 through a cash grant, then a tax rate of 20 percent on earned income would imply that only when earned income reached $20,000 would net payments to families be zero. Raising the tax rate to, say, 50 percent would lower the breakeven earned income to $8,000. The higher tax rate, however, would reduce the after-tax wage rate and thereby probably reduce work effort.

If recipients reduce their work effort, then their chances of becoming dependent on the cash grants increase. For some categories of recipients, such as the permanently disabled, the increased risk of dependence may not be a serious concern. For others, however, there may be a trade-off between the generosity of short-term assistance and the long-run economic vitality of recipients. Work requirements may be one way to improve the terms of the trade-off.[201] Unfortunately, they are typically costly and difficult to implement, especially for single-parent families with young children.

Cash grants can also influence choices about living arrangements and family structure. For example, Aid to Families with Dependent Children often made it financially possible for young unmarried mothers to set up their own households. The less generous are state benefit levels, the more likely it is that these mothers will stay with their parents.[202] Indeed, the opportunity to gain independence may have encouraged some teenage girls to have children. The general point is that the availability of cash grants may influence a wide range of behaviors.

Conclusion

Table 10.6 Searching for Generic Policy Solutions

	Market Mechanism	*Incentives*	*Rules*	*Nonmarket Supply*	*Insurance and Cushions*
Traditional Market Failures					
Public goods	S	S	S	P	
Externalities	S	P	P	S	
Natural monopolies	S	S	P	P	
Information asymmetries			P	S	S
Other Limitations of the Competitive Framework					
Thin markets			P		
Preference-related problems	S	S	P		
Uncertainty problems			P		S
Intertemporal problems			S		P
Adjustment costs					P
Macroeconomic dynamics		P			S
Distributional Concerns					
Equity of opportunity		S	P		S
Equality of outcomes			S	S	P
Government Failures					
Direct democracy			P		
Representative government	P		S		
Bureaucratic supply	P	S	S	S	
Decentralization	S	P		S	

Sources for solutions (though not necessarily most often used!): P — primary, S — secondary

More than one generic policy can often be used to address a market or government failure. Table 10.6 indicates the generic policy categories that are most likely to provide candidate solutions for each of the major market failures, government failures, and distributional concerns. In many cases, more than one generic policy can provide potential solutions for the same problem. But the solutions are never perfect—they must be tailored to the specifics of the situation and evaluated in terms of the relevant goals.

For Discussion

1. Consider a developing country that has no policies regulating the discharge of pollutants into its surface waters. In the past, there has been some degradation of water quality from agricultural waste. Increasingly, however, factories have discharged industrial waste into lakes and rivers, raising environmental and health concerns. What are the generic policy alternatives the country might adopt to limit or reverse the reductions in water quality from industrial pollution?
2. Your state or province is concerned that too few children are being immunized against childhood illnesses. What are some of the generic policy alternatives that could be adopted to increase immunization rates?

Notes

1 Paul L. Joskow and Roger G. Noll, "Regulation in Theory and Practice: An Overview," in Gary Fromm, ed., *Studies in Public Regulation* (Cambridge, MA: MIT Press, 1981), at 4.

2 See Martin Cave and Robert Crandall (eds.), *Telecommunications Liberalization on Two Sides of the Atlantic* (Washington, DC: AEI-Brookings Joint Center for Regulatory

Studies, 2000).

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11 Adoption

The adoption and execution of collective decisions inherently involve cooperation. Collective decisions begin as proposals in political arenas and culminate in effects on people. We can divide this process into two phases: *adoption* and *implementation*. The *adoption phase*, the focus of this chapter, begins with the formulation of a policy proposal and ends, if ever, with its formal acceptance as a law, regulation, administrative directive, or other decision made according to the rules of the relevant political arena. The *implementation phase*, the focus of the next chapter, begins with the adoption of the policy and continues as long as the policy remains in effect. Although policy analysts typically contribute to the policy process by formulating and evaluating proposals during the adoption phase, they cannot do so effectively without anticipating the entire process, from proposal to effect.

Yet, the distinction between adoption and implementation does not fully explain the complexity that typically characterizes the policy process. Adopted policies, especially laws, rarely specify exactly what is to be done. They may require complementary policy decisions to be made in other arenas. For example, a county legislature might adopt an ordinance that prohibits smoking in certain public places, makes violations punishable by fines, and delegates responsibility for enforcement to the county health department. Now the county's health department must adopt an enforcement policy. Should it simply wait for complaints about violations from the public, or should it institute spot checks? Should it issue warnings or immediately impose fines? Although the head of the health department may have legal authority to answer these questions as she sees fit, her decisions probably would be influenced by the (perhaps conflicting) advice of her staff. If spot checks are to be instituted, then the chief inspector must design a sampling procedure that his deputies would execute. Thus, the smoking policy that the public actually encounters results from a series of decisions following adoption of the ordinance.

We naturally think of values and interests coming into play during the adoption phase, and we are not surprised when those who oppose policies attempt to block their adoption. By envisioning the implementation phase as a series of adoptions, we prepare consciously for the intrusion of the values and interests of those whose cooperation is needed to move the adopted policy from a general statement of intent to a specific sequence of desired impacts. In other words, we begin to think strategically about process as well as substance: How can we take advantage of the interests and values of others to further our ends? This chapter guides strategic thinking about adoption; the next chapter guides strategic thinking about implementation.

Policy analysts do not always have to think in an explicitly strategic manner. We can often make predictions about the consequences of adopted and fully implemented policies without considering the responses of particular individuals. For example, demand schedules usually allow us to make plausible predictions of the aggregate consequences of taxing goods. When we attempt to predict the prospects for adoption and successful implementation, however, we must think strategically in the sense of anticipating the responses of others.

Formal models of strategic behavior usually take the form of games that unfold through a series of moves by players. For example, imagine that you are the player who begins by selecting a strategy (move 1). In the case of policy adoption, your strategy choice may be to convince your client to propose a particular policy. Another player adopts a strategy in response to your first move (move 2), say by opposing or supporting your client's policy proposal or by suggesting some other policy. Strategic thinking requires you to anticipate the responses that each of your possible strategy choices for move 1 would elicit from the other player in move 2. The prisoner's dilemma presented in Chapter 5 to model the common property problem is an example of a very simple but informative formal game.

Keep four points in mind about strategic thinking. First, as we discuss in Chapter 15, we cannot make accurate predictions of the consequences of candidate policies without paying attention to their likelihood of adoption and their successful implementation.

Second, clients often want their analysts to consider the political feasibility of adoption of policy alternatives. Sometimes political feasibility is explicitly treated as one of multiple goals in a policy analysis. Other times clients ask analysts to prepare confidential analyses of the political feasibility of getting particular policy alternatives adopted. In either case, good analysis requires strategic thinking.

Third, designing effective policy alternatives requires a certain amount of creativity. Strategic thinking helps us to be more creative by drawing our attention to ways of taking advantage of the self-interested behaviors of others. Similarly, it helps us to be more perceptive about potential problems and opportunities in designing implementation plans.

Fourth, analysts themselves often participate in adoption and implementation efforts. Clients sometimes solicit advice from analysts about day-to-day maneuvering in political arenas, invite them to participate as "technical experts" in negotiations, or send them as representatives to meetings where issues are discussed and perhaps even decided. Analysts are sometimes called upon to advise, direct, oversee, or even manage implementations. These sorts of hands-on activities require analysts to put strategic thinking into practice.

In this chapter, we present several perspectives on policy adoption. We begin with an overview of the big picture by briefly reviewing the most prominent *policy process* frameworks with an eye to their implications for policy analysts seeking to influence the prospects for adoption of policy alternatives. These frameworks and theories offer some useful insights about political feasibility but were not originally intended as practical guides for policy analysts. Therefore, we also provide a practical guide for analysts that they can use to assess the political feasibility of specific policies. We then consider some general strategies analysts can employ to increase the political feasibility of policy proposals.

The Big Picture: Policy Process Frameworks and Theories

As one might expect, many political scientists do research relevant to the politics of adoption. Those who study specific institutions, such as the Canadian parliament or the U.S. Congress, often consider the adoption of legislation. Other researchers consider the regulatory and judicial processes that also make public policy. Although these lines of research are often valuable resources for policy analysts trying to understand the political feasibility of proposed policies in particular contexts, we focus our attention here on the frameworks and theories that seek to describe policy processes more generically.

Table 11.1 Policy Process Frameworks and Theories: Implications for Policy Analysts

Framework/Theory	*Essential Elements*	*Implications for Policy Analysts*
Interest Group Theory	Concentrated interests advantaged over diffuse interests; organized interests advantaged over unorganized interests	Mobilize or create organizations to represent diffuse interests; anticipate possible post-adoption mobilization of diffuse interests

Framework/Theory	*Essential Elements*	*Implications for Policy Analysts*
Institutional Rational Choice	Institutions shape the circumstances of choice facing individuals pursuing their own interests	Understand the institutions that make and implement policies; look for changes in institutions to change policy outcomes
Path (State) Dependence	Accumulated policies create externalities that affect the viability of policy alternatives in the future	Recognize that quick fixes or failed proposals may limit future options; anticipate the creation of policy constituencies
Advocacy Coalition Framework	Relatively stable networks of actors sharing core and policy beliefs oppose each other to make major change difficult in policy subsystems	Produce evidence to change policy beliefs; take advantage of external shocks to coalitions; seek negotiation to end hurting stalemates
Social Construction Framework	Both political power and the social construction of targeted groups as deserving or undeserving affect the distribution of benefits and burdens	Frame policies to take advantage of favorable social constructions; look for ways to change social constructions
Multiple Streams Framework	Political, policy, and problem streams join at critical times to create opportunities, or policy windows, for major policy change	Prepare for policy windows by having well-analyzed alternatives on hand; look for ways to help open policy windows

Framework/Theory	*Essential Elements*	*Implications for Policy Analysts*
Punctuated Equilibrium Theory	Limitations in information processing hinder proportionate responses so that policy tends to be stable except during major catch-ups	Anticipate stability occasionally punctuated by major policy change.

We identify seven prominent approaches to understanding the policy process: (1) interest group theory; (2) institutional rational choice; (3) path dependence; (4) the advocacy coalition framework; (5) the social construction framework; (6) the multiple streams framework; and (7) punctuated equilibrium theory. Our treatment of each is necessarily brief.[1] Some of these approaches provide frameworks that identify factors generally related to policy adoption; others offer more explicit theories about how these factors are related. As summarized in Table 11.1, we identify the essential elements of each model and consider its implications for policy analysts.

Interest Group Theory

Diffuse interests become politically effective in two ways. First, as suggested by Wilson's label of entrepreneurial politics introduced in Chapter 8, politicians or other participants in the policy process seek to gain public attention by framing the diffuse benefits (or costs) in terms of widely held values, such as environmental quality or protection of public safety. To overcome the advantage of the concentrated interests in opposition, entrepreneurs often require some sensational public event to attract the attention of those with diffuse interests and perhaps discredit those with concentrated interests. Second, diffuse interests may become more politically effective through organization. Sometimes people are willing to contribute to an organization to engage in political activity even though others will free ride in the sense of potentially gaining from successful political action even though they did not make contributions to it. However, a more common way that this public good for the diffuse interests gets provided is when organizations already exist for other purposes, such as providing specific private goods, so that they can "tax" their members through higher dues to pay for it.[2] For example, the National Rifle Association (NRA) provides information about guns and hunting, discounts, and other individualized, or selective, benefits that engage members who might not be willing to join solely to support lobbying.

Analysts wishing to advance policies that favor diffuse interests should identify and attempt to mobilize organizations that either represent those interests or related ones. For example, analysts attempting to secure more stringent enforcement of driving-while-impaired laws to protect the general public might elicit the support of Mothers Against Drunk Driving (MADD), which has a direct interest in the issue, or perhaps the state medical society, which has a general interest in promoting public health and safety. Taking a longer-term perspective, analysts might even think of policies to encourage the formation of organizations that might represent diffuse interests in the future. For instance, the "maximum feasible participation of residents" provision of the Economic Opportunity Act of 1964 sought to create more effective representation for the urban poor.[3]

Analysts often perform a valuable service by identifying those who would bear diffuse costs from policies if they were adopted. Although these diffuse interests usually do not actively oppose a policy prior to adoption, they may very well do so after costs are actually realized. Policy or political entrepreneurs may then be able to mobilize these interests for repeal of the policy. For example, the Medicare Catastrophic Coverage Act had overwhelming support in both houses of Congress on its passage in 1988. However, equally large majorities voted to repeal the Act in 1989, when strong opposition from the wealthier elderly to progressive fees became apparent.[4] The failure to take account of diffuse costs prior to adoption may sometimes have dramatic electoral consequences. For instance, U.K. Prime Minister Margaret Thatcher had relatively little difficulty getting the poll tax adopted, but this almost certainly contributed to her grossly underestimating the political unpopularity of the policy once in place. Ultimately, the poll tax was a major factor in driving her from office.[5] The electorate may have other such "sleeping giants" that can be awakened by political entrepreneurs who focus their attention on adverse policy impacts.[6]

Institutional Rational Choice

Many lines of research in the social sciences focus on the role of institutions, which can be defined as "fairly stable sets of commonly recognized formal and informal rules that coordinate or constrain the behavior of individuals in social interactions."[7] Research on institutions includes the principal–agent models discussed in Chapter 8 and the transaction cost analyses of contracting presented in Chapter 13, as well as models of norms presented at the end of the next chapter. Institutional rational choice draws on institutional theories to view the policy process from the perspective of multiple levels of rules.

Three levels of rules are commonly identified.[8] First, the consumption and production decisions of individuals are shaped by rules at an *operational level.* The operational rules may be formal, such as rules and regulations set by the state, or informal, such as conventions and norms about acceptable behavior. Second, the operational rules are themselves set according to rules at the *collective choice level.* Most obviously, legislative and executive authorities formulate operational rules. Third, the *constitutional level* sets the rules about collective choice. In the case of governments, there is literally a constitution, whether written or unwritten. In the case of organizations, such as not-for-profits, a government may serve as the constitutional level, setting rules about existence and governance.

The most systematic application of the institutional rational choice framework has been to the understanding of the governance of common property regimes.[9] However, its applicability to understanding public policy problems is much broader. Indeed, *identifying the relevant rules at the three levels often provides an excellent starting point for framing policy problems and thinking about rule changes as policy alternatives.* Recognizing the interrelationship among the levels also encourages analysts to think about a broader range of policy alternatives. Sometimes problems at the operational level can be addressed by changing rules at the collective choice level. For example, creating regional transportation authorities with taxing power may overcome cooperation problems among jurisdictions within the same metropolitan area.

In terms of assessing political feasibility, *analysts should anticipate that opponents of current operational rules will seek to bypass them by changing rules at the other levels.* For example, developers facing difficulty in gaining approval from local planning boards for windmills may seek to have authority for approval transferred to a state agency. Or, for example, those opposed to same-sex marriage in some states attempted to constrain their legislatures through constitutional bans.

Path (State) Dependence

Does the market always rationally sort out new technologies so that the superior one comes to dominate? Some economists argue that increasing returns to scale can result in an arbitrary selection among technologies when chance events give one of them a head start so that its initial advantage becomes magnified.[10] This process is generally referred to as *path dependence*. Political scientists argue that a similar process operates with respect to public policies.[11] *Policy feedback*, including changes in administrative capacity and the identities and resources of social groups, makes political processes dependent upon previous policy adoptions. It may also affect public opinion.[12]

The concept of path dependence as applied to the policy process can be sharpened in several ways.[13] The term *path dependence* should be reserved for situations in which the sequence of policy outcomes and opportunities affect the political feasibility of future policy alternatives. If only the current set of policies, and not their sequence of adoption, affects the feasibility of future policy alternatives, then a better term would be *state dependence*.[14] Path dependence calls for historical analysis; state dependence only requires an assessment of the status quo. Although increasing returns to scale can create path or state dependence, so too can the adoption of any policy that affects the distribution of costs and benefits of future policies. Most prominently, policies that create constituencies of suppliers or beneficiaries also make future policies that adversely affect these constituencies more difficult to adopt. For example, a loophole in Second World War wage and price controls in the United States allowed firms to compete for workers by offering health insurance. A subsequent ruling by the Internal Revenue Service excluded the health insurance benefit from income for purposes of taxation. As a consequence of these policies, an employer-provided private health insurance became the primary form of health insurance. The large private health insurance industry became a major political force in health policy, reducing the viability of alternatives, such as single payer systems, that would challenge their position in the market.

A path may also include failed efforts to adopt policy changes that influence future opportunities. Failures both reveal the strength of the opposition and the public perception of what is desirable and possible. Memories of the failures, which tend to be more salient than memories of successes, can become a barrier to subsequent efforts. For example, the failure of the Clinton administration's Health Security Act of 1993 effectively took efforts to achieve universal health insurance coverage in the United States off the political agenda for 15 years.

The path dependence framework suggests that *analysts should invest in learning the policy history relevant to current issues* and offers several cautions to policy analysts. First, *beware of quick fixes*, or proposing policies simply for the sake of doing something. Such policies may create externalities of various sorts that, although seemingly minor, alter the feasibility of future options. Second, at the other extreme, *beware of radical proposals that are likely to fail.* Although they may prepare the ground for future adoption by educating policymakers and the public, their immediate failure may create perceptions and mobilize opposition that reduces prospects for future adoption. Third, *anticipate that policies create constituencies of beneficiaries and providers.* Once created, these constituencies are likely to become politically active in seeking to defend or expand the policy, making change, especially termination, politically difficult.

Advocacy Coalition Framework

The complexity of modern society requires specialization. With respect to public policy, this usually occurs within *policy subsystems*, collections of individuals and organizations that seek to influence public policy within a substantive policy domain. The *advocacy coalition framework* (ACF) considers learning and policy change within policy subsystems over long periods of time.[15] The key elements of the ACF are the policy subsystem as the unit of analysis, the focus on two or more relatively stable and opposed coalitions of actors within the subsystem, the belief systems of the coalitions, and the processes through which learning by the coalitions and policy change can occur.

The ACF shares with several other frameworks and theories the importance of external events, such as major economic or social changes, policy changes in other subsystems, and electoral changes, in contributing to major policy change. The ACF also recognizes shocks within the policy subsystem, such as catastrophic policy failures or events affecting important coalition members, that may contribute to major policy change by altering the various political resources available to coalition members. However, its value is primarily its focus on the relatively stable belief systems of the coalitions and the extent to which policy learning can affect these beliefs.

Coalitions share *deep core beliefs* about the relative importance of fundamental values and human nature. *Core policy beliefs* span the issues within the policy subsystem and concern the appropriate roles of markets and government, the seriousness and causes of policy problems, and the desirable generic policy instruments for dealing with problems. *Secondary policy beliefs* are narrower in scope, concerning either specific issues with the policy subsystem or policies and processes in specific locales. For example, a coalition may have core policy beliefs about the causes of and proper responses to drug abuse in general, but secondary beliefs about the specific policies appropriate in a particular urban area.

Within this hierarchy of beliefs, the ACF hypothesizes that secondary policy beliefs are more amenable to change than core policy beliefs, which are more amenable to change than relatively immutable deep core beliefs.[16] Changes in secondary beliefs do not require rethinking across all issues within the policy subsystem. Changes may also occur initially within specific locales where subsets of the coalition can alter beliefs as a starting point for broader change. For example, those involved in substance abuse policy in a particular city, including those in a coalition initially opposed to a generic policy intervention, may agree that a particular application of the intervention has net desirable effects. These members may bring the local experience to their coalitions, perhaps eventually facilitating a change in core policy beliefs.

More recent versions of the ACF draw on the dispute resolution literature to consider factors that contribute to negotiated agreements between opposing coalitions.[17] A so-called *hurting stalemate*, where neither side is satisfied with the status quo policy, is fertile ground for successful negotiation. A *negotiation forum* that allows participation by major coalition members, long-term interaction, the building of trust, a focus on important empirical questions, and the engagement of a negotiator or broker who is perceived by both sides as neutral facilitates negotiated change.

The ACF offers three lessons for policy analysts. First, *producing evidence relevant to policy problems and policy interventions can potentially change beliefs*. However, analysts require patience and a willingness to focus on secondary policy beliefs as a path for eventually changing core policy beliefs. Second, *be prepared to exploit external or internal shocks to the policy subsystem that change the resources of the coalitions and potentially create an opportunity for major policy change.* Third, *look for opportunities to encourage negotiation in the presence of hurting stalemates*. In particular, search for a neutral policy broker to help the parties identify critical empirical questions and develop methods for answering them. For example, contracting with the National Academy of Sciences or some professional association for studies to resolve empirical issues may provide a basis for narrowing the range of dispute between coalitions.

Social Construction Framework

Just as social scientists require frameworks to help them understand phenomena amid complexity, so too do we all employ frames to help us interpret our observations of the world. To the extent that these frames can be altered by argument, they provide an avenue for changing the political support for various policies.[18] The social construction framework applies this notion to the distribution of benefits and burdens among the groups that are the targets of public policies.

Specifically, the social construction framework identifies four target groups in terms of their social construction and political power.[19] The *advantaged* consist of groups with positive social constructions and relatively strong political power, such as small businesses, homeowners, and the military. Policies targeted on the advantaged tend to include benefits but not burdens. *Dependents* include groups with positive social constructions but weak political power, such as children and mothers. Policies targeted at dependents tend not to include burdens, but the benefits provided tend to be smaller and less secure than those provided to the advantaged. *Contenders*, such as big business and labor unions, have negative social constructions but strong political power. Contenders tend to receive both benefits, but in ways that are not always obvious to the public, and burdens, but often ones less severe than the rhetoric surrounding them would suggest. Finally, *deviants*, such as welfare recipients and substance abusers, have negative social constructions and weak political power. They tend to be subjected to burdens.

One can also think of social constructions of policies, which can change over time and vary across place. For example, policies to encourage the use of ethanol as a fuel began with a very favorable public image but, with discussion of life-cycle environmental effects, impacts on food prices, and links to big agribusiness, the image is no longer so favorable. Or, consider perceptions of genetic engineering. Both the North American and European publics seem to have embraced "red" genetic engineering aimed at improving clinical interventions, but the North American public has largely accepted "green" genetic engineering aimed at improving agriculture while the European public has largely rejected it.[20]

Maintaining and changing social constructions depends on *framing*, through which one selects "some aspects of a perceived reality and make them more salient in a communicating text, in such a way as to promote a particular problem definition, causal interpretation, moral evaluation," or policy alternative.[21] At the individual level, framing refers to the meaning structures and schema individuals use to process and interpret information; at the public level, framing refers to the use of organizing concepts or storylines by the media or politicians to create specific meaning for events.[22] Those seeking to frame issues in public discourse choose language and metaphors that resonate with the frames used by others. For example, gaining support for public policies that impose burdens to reduce risk appear to be most effective when the issue is framed as involving risk that is involuntary (imposed rather than sought out), universal (affecting all of us rather than some others), environmental (external rather than internal to the individual), and knowingly created (there is a culpable party to which burdens can be assigned).[23]

What lessons does social construction offer? Most obviously, *analysts should pay attention to how they frame policies in terms of target populations*. For example, a program that provides services (benefits) to children in poor families receiving public support might have greater political feasibility if it is framed as targeted at children rather than at welfare families. *Analysts should recognize opportunities for mobilizing support when constructions are not universal*. That is, some segments of the population may have more or less positive social constructions of some group. Analysts may be able to portray polices in ways that differentially mobilize these segments of the population in favorable ways. Finally, *analysts may be able to contribute to changes in social constructions through the evidence they bring to the policymaking process*. For example, convincing reviews of the scientific literature on the physiologically addictive properties of some substance may contribute to a more positive social construction of those who abuse the substance by changing perceptions about their culpability in the abuse.

Multiple Streams Framework

Motivated by observation of the governance of universities, the *garbage can model* sees organizational decisions as resulting not from some logical weighing of alternatives to solve well-defined problems, but rather as the confluence of ambiguous problems and often unrelated solutions in "organized anarchies … characterized by problematic preferences, unclear technology, and fluid participation."[24] In the spirit of the garbage can model, John Kingdon proposed a framework of multiple streams to understand the patterns of agenda setting he observed in his study of federal policymaking in the areas of health and transportation.[25] He describes three distinct streams. The *problem stream* consists of the various conditions that the public and politicians want addressed. At any particular time, only a few problems can command public attention. Focusing events, such as the 9/11 terrorist attacks or the financial crisis in late 2008, can push specific issues to the forefront. The *policy stream* consists of various policy ideas, such as generic policy alternatives, that circulate within the relevant policy subsystem. It includes policy analysts and policy researchers. The *politics stream* consists of the public officials, national mood, and organized interests that affect legislative and administrative decisions. Changes in elected and appointed officials, swings in the national mood toward a different ideology, or interest group mobilizations can dramatically affect the politics stream.

As already noted in Chapter 8, the opportunity for major policy change occurs when the three streams combine in a *policy window*.[26] Changes in the problem stream, such as a focusing event or accumulation of evidence creating a compelling problem, or changes in the politics stream, such as a new administration taking office, open a window for advocates within the policy stream to push for their favored alternatives. Only one window can be open at a time and it will not stay open long. New focusing events may open a new window, closing the old one. Any political response, even if token or unsuccessful, may close the window. If entrepreneurs fail to bring immediately forward an alternative widely viewed as viable, then the window will close before there is an opportunity to consider substantial policy change.

The most important implication for the multiple streams framework for policy analysts is to be prepared: *analysts seeking change should have developed policy alternatives ready to push through policy windows when they open.* Analysis that moves generic alternatives to concrete proposals or that promotes acceptance of alternatives within the policy subsystem, although not immediately relevant to political choice, may increase the chances of successfully exploiting future opportunities created by the opening of policy windows. *Analysts may also contribute to the opening of policy windows by helping to assemble evidence and argument that makes a convincing case that some condition should be viewed as an important public policy problem.*

Punctuated Equilibrium Theory

In the United States, the separation of the legislative and executive functions, the bicameral legislature, the necessity for supermajorities to stop filibusters in the Senate, and the interpretation of statutes by the courts introduce veto points that contribute to policy stability. Policy entrepreneurs attempt to overcome this stability by creating policy images (frames) to appeal to the normally inattentive electorate. They also search for the most favorable venues (so-called venue shopping). In this context, the interaction between the creation of policy images and venues creates the possibility for substantial change. Frank Baumgartner and Bryan Jones labeled this process of general stability interrupted by large changes *punctuated equilibrium*.[27]

The punctuated equilibrium framework has evolved to be a characterization of the general problems of information processing facing all political systems.[28] Interestingly, it does provide a testable implication about the output of any political system over time: information processing produces distributions of policy changes that are leptokurtic (more very small and very large changes, but fewer moderate changes than would occur if policy changes followed a normal distribution). Empirical tests based on outputs such as budget changes appear to favor this hypothesis. Unfortunately, aside from *anticipating stability with occasional punctuations*, this generalization does not appear to have implications for policy analysts.

Practical Approach to Assessing and Influencing Political Feasibility

The policy process frameworks and theories offer general insights into political feasibility useful for policy analysts. However, policy analysts must assess the political feasibility of specific policy proposals in specific contexts.

Arnold Meltsner provides a checklist of the information needed to assess the political feasibility of adoption: Who are the relevant actors? What are their motivations and beliefs? What are their political resources? In which political arenas will the relevant decisions be made?[29] Although we discuss these questions sequentially, in practice we answer them iteratively, moving among them as we learn more about the political environment. After considering these questions that are relevant to the assessment of feasibility, we turn to strategies that can be used within political arenas to increase feasibility.

Identifying the Relevant Actors

Which individuals and groups are likely to voice an opinion on an issue? Two, usually overlapping, sets of actors need to be identified: those individuals and organized groups with a substantive interest in the issue, and those with official standing in the decision arena. For example, imagine a proposal before a city council that would prohibit firms within the city from subjecting their employees to random drug testing. We expect unions to support the measure and business groups to oppose it. Further, we expect unions and business groups that have been politically active in the past, such as the Labor Council and the Chamber of Commerce listed in Table 11.2, to be the ones most likely to become active on this issue. At the same time, we identify the members of the city council as actors by virtue of their rights to vote on legislation and the mayor by virtue of her veto power.

We expect union and business leaders to be active because their direct interests are at stake. Civil libertarians might become involved because values they perceive as important are at issue. (Some may see the ordinance as necessary to protect the right of workers to privacy; others may view it as an unwarranted interference in the decisions of firms.) Perhaps certain community groups such as the Urban League will become active, either because they have a direct interest or because they usually ally themselves with one of the interested parties. As noted in the discussion of the interest group theory, we expect concentrated and organized interests to become involved.

Other public figures besides the members of the city council may also be relevant. For example, although the city attorney does not have a vote on the council, his opinion on the legality of the proposal may carry considerable weight with council members. The opinion of the director of public health on the accuracy of testing may also be influential. The editor of the local newspaper has no official standing whatsoever, yet may be an important participant by virtue of the editorials she writes and the influence she exerts over the coverage of the news and how it is framed.

Table 11.2 A Political Analysis Worksheet: Feasibility of a Ban on Random Workplace Drug Testing

Actors	*Motivations*	*Beliefs*	*Resources*
Interest groups			
Labor Council	Protect workers from harassment	Testing would be used unfairly	Large membership; ties to Democrat Party
Chamber of Commerce	Protect firms' rights to weed out dangerous and unproductive workers	Testing may be necessary to detect and deter employee drug use	Influential membership; ties to Republican Party
Civil Liberties Union	Protect rights of individuals	Testing infringes on right to privacy	Articulate spokesperson

Actors	*Motivations*	*Beliefs*	*Resources*
Libertarian Party	Protect right of contract	Testing limits should be a matter of negotiation between labor and management	Vocal membership
Urban League	Protect minority employees	Testing disproportionately hurts minorities	Can claim to speak for minority interests
Daily newspaper	Support business environment	Testing ban not appropriate at city level	Editorials
Unelected officials			
City attorney	Support mayor and protect city from law suit	Ban probably legal	Professional opinion
Director of public health	Fight drug abuse	Testing probably desirable, if not punitive	Professional opinion; evidence on effectiveness of tests
Elected officials			
Council member A (Democrat)	Support labor	Ban desirable	Vote

Actors	*Motivations*	*Beliefs*	*Resources*
Council member B (Democrat)	Support labor	Ban probably desirable	Vote
Council member C (Democrat)	Support community groups	Ban probably desirable	Vote; agenda control
President of Council Council member D (Republican)	Support business	Ban probably undesirable	Vote
Council member E (Republican)	Support business	Ban undesirable	Vote
Mayor (Democrat)	Maintain good relations with labor and business	Ban probably undesirable	Veto power; media attention

How should analysts compile lists of potential actors? Obviously, *assume that anyone with a strong interest, whether economic, partisan, ideological, or professional, will become an actor.* Also, include those in official positions. If you are new to the issue or arena, then try to find an experienced person to be your informant. Use newspapers or other written accounts to discover who participated in public debates on similar issues in the past. Finally, when doing so will not adversely affect your client or the future prospects for your proposals, contact potential actors directly to question them about their views and to assess the likelihood that they will become active participants.

Understanding the Motivations and Beliefs of Actors

The motivations and beliefs of organized interest groups will often be apparent. When they are not, you can usually make a reasonable guess about how they will view particular proposals by comparing the costs and benefits that the leaders of the groups are likely to perceive. If their perceptions are based on what you think are incorrect beliefs, then you may be able to influence their positions by providing information. For example, the president of the local Urban League chapter may support a ban on random drug testing because he believes that the ban would protect minority workers, one of his constituencies. He might decide to oppose the ban, however, if he comes to believe that the ban will result in a net loss in minority jobs because some firms will leave the city to avoid its restrictions.

It may be more difficult to determine the relevant motivations and beliefs of those in official positions. Elected officials, political appointees, and members of the civil service all have a variety of motivations. As we discussed in Chapter 8, elected officials are likely to be concerned with reelection or election to higher office, as well as with representing the interests of their constituencies and promoting the social good. Political appointees may be motivated by their substantive values as well as by their loyalties to their political sponsors, by their desire to maintain their effectiveness in their current positions, and by their interest in opportunities for future employment. In addition to substantive values, civil servants are often motivated by their sense of professionalism and by their desire to secure resources for their organizational units.

It should not be surprising that it is often difficult to predict which motivations will dominate. Indeed, such conflicting motivations can lead the officials themselves to the sort of personal ethical dilemmas that we described in Chapter 3. How can we understand the relative importance of officials' various motivations concerning a particular issue? We can begin by heeding the insight *where you stand depends on where you sit.*[30] In other words, put yourself in the position of the relevant officials: What would you want if you were in their place? What actions would you be willing to take to get what you want?

Obviously, the more you know about particular officials, the better you will be able to answer these questions. If an official holds a strong substantive value relevant to the issue, for instance, then she may be willing to act against even vocal constituent interests. On the other hand, she may be willing to go against her own substantive values if the issue is of fundamental concern to one of her important constituencies. Of course, such factors as the electoral competitiveness of her district and the nearness to an election can also affect the position that she takes on the issue.

Assessing the Resources of Actors

Actors have a variety of political resources. Interest groups can claim to speak for a constituency. They may have financial resources that can be used to pay for lobbying, analysis, publicity, and campaign contributions. Their leaders may have ongoing relationships with relevant officials. By virtue of their memberships, analytical capacity, or track record, the information they provide may command attention or carry weight. All these resources can be thought of as potentially relevant. Whether they actually come into play depends on the motivations of the groups and their leaders.

Officials have resources based on their positions and relationships. Legislators can vote, hold hearings, and influence the agenda; elected executives, like mayors, usually have veto power as well as considerable discretion in interpreting laws that are adopted; unelected executives often have influence by virtue of their professional status, programmatic knowledge, and ties to clientele groups. Any of these actors may be able to influence others through personal relationships based on trust, loyalty, fear, or reciprocity.

Table 11.2, which presents a simple worksheet that identifies the actors who may be relevant in predicting the political feasibility of a city ordinance to ban random drug testing of employees, also notes the likely motivations, beliefs, and resources of the actors relevant to the city ordinance to ban rand. Many of the entries may be little more than guesses, especially for analysts new to the particular political arena. The entries in the worksheet should be updated as more information becomes available. For example, the actors' actual responses to the proposal once they learn about it may very well change your assessment of their beliefs and their willingness to use resources.

Choosing the Arena

Each political arena (the collective and constitutional choice levels in the institutional rational choice framework) has its own set of rules about how decisions are made. The basic rules are generally written—legislatures have "rules of order" and agencies have administrative procedures. Nonetheless, unwritten traditions and standard practices may be just as relevant to understanding how decisions are typically reached. Becoming familiar with these rules, both formal and informal, is important for political prediction and strategy.

To use the information in Table 11.2 to make a prediction about the likelihood of adoption of the drug testing ban, we must first determine the arenas in which the proposal will be considered. As suggested by the entries under "elected officials," we expect that the city council will be the primary arena. If the council members vote according to their constituencies' apparent interests, then the ban would pass on a party-line vote of three to two. Passage would put the mayor in a difficult situation. If she vetoes the ordinance, then she may alienate her fellow Democrats. If she does not veto it, then she will alienate business interests.

The mayor might be able to get out of this difficult political position by trying to change the arena. She might argue that, although restrictions on testing are desirable, they are more appropriately imposed at the state level.

If she can find a state assembly member or senator to propose the ban in the state legislature, then she could argue that a vote by the city council should be delayed until the prospects for state action become clear. Perhaps she would ask the council to pass a resolution urging state action. If the council agrees, then she will have been successful in changing arenas.

More generally, we should expect that actors who lose, or anticipate losing, in one arena to try to move the issue to another. As illustrated by the mayor's maneuver, unfavorable outcomes at one level of government can sometimes be avoided by shifting the issue to another level. For example, one reason why labor unions pushed for the federal legislation that created the Occupational Safety and Health Act of 1970 was their dissatisfaction with their ability to influence the setting and enforcement of health standards at the state level.[31] The arena can also shift from one branch of government to another. For instance, consider the controversy over the role of geography in the allocation of cadaveric livers for transplantation.[32] In 1984, Congress created the Organ Procurement and Transplantation Network (OPTN) to develop rules for organ allocation with oversight by the Department of Health and Human Services. In 1998, the department attempted to finalize a rule requiring the OPTN to eliminate local priority in liver allocation. Opponents were able to obtain from Congress a moratorium on implementation of the rule and a requirement for a study of the issue by the Institute of Medicine, a component of the National Academy of Sciences. Additionally, opponents were able to get some states to adopt laws intended to blunt the rule.

Often those who lose in the legislative and executive branches try to shift the arena to the courts. For instance, during the 1970s, advocates of busing to reduce the racial segregation in schools caused by housing patterns routinely achieved their objectives through the courts. Regulatory agencies that fail to establish strong cases for rule making invite court challenges by those who oppose the rules. Of course, the ability to make a credible threat to move the issue to another arena can itself be a political resource.

Political Strategies within Arenas

Trying to shift issues to more favorable arenas is not the only political strategy that can be used to achieve desired policy outcomes. Policy analysts acting as political entrepreneurs can often use one or more of four other general strategies: co-optation, compromise, her-esthetics, and rhetoric. We briefly consider each of these in turn.

Co-optation

People, especially the types with strong egos who typically hold public office (and teach at universities), tend to take pride in authorship. Indeed, we are sometimes reluctant to admit the weaknesses in what we perceive to be *our* ideas. Getting others to believe that your proposal is at least partly their idea is perhaps one of the most common political strategies. In legislative settings, it often takes the form of co-sponsorship, across aisles and chambers. In administrative settings, it often involves creating an advisory group that is constituted so as to arrive at the desired recommendation. Potential adversaries who believe that they have contributed to the recommendation as committee members may be less likely to become active opponents. Public officials with strong interpersonal skills can sometimes successfully co-opt potential opponents with as little as a seemingly heart-to-heart conversation or other gesture of personal attention. (Did you ever wonder why the dean invited you and the other student leaders to that dinner?)

Co-optation may be useful when your proposal infringes on the turf of other political actors.[33] Politicians and bureaucrats often stake out areas of interest and expertise. Although some of them may be natural allies by virtue of their interests and beliefs, they may, nevertheless, feel threatened by proposals originating from other people that appear to fall within their areas. Unless you involve them to some extent in the design of policy proposals, they may voice opposition without seriously considering substantive merits. Furthermore, other actors who take their cues on the issue from the recognized experts may decline to give the proposals close attention. For example, council members may not give serious attention to a proposal for the establishment of a particular type of drug rehabilitation program unless the director of public health says that it is worthy of consideration.

Of course, you cannot use co-optation as a strategy unless you are willing to share credit. As an analyst, the nature of your job demands that you allow your client to take credit for your good ideas (and that you save any discredit for bad ideas for yourself). Remember, you earn your keep by providing good ideas. Co-optation strategies, therefore, should be based on your client's explicit willingness to give credit in return for progress toward policy goals.

Compromise

We use the word *compromise* to refer to substantive modifications of policy proposals intended to make them more politically acceptable. When our most-preferred policy lacks adequate support, we can consider modifying it to gain the additional support needed for adoption—in terms of policy analysts seeking to promote multiple goals, we trade progress toward some goal to achieve greater political feasibility. Compromise is desirable if we prefer the resulting adoptable proposal over any others that are also feasible. Generally, desirable compromise involves making the smallest modifications necessary to attract the minimal number of additional supporters required for adoption.[34]

One approach to compromising is to remove or modify those features of the proposal that are most objectionable to its opponents. Do any features appear as red flags to opponents? Sometimes offending features can be removed without great change in the substantive impact of the proposal. For example, imagine that you have made a proposal that the county hire private firms to provide educational services to inmates in the county jail. Some opponents, who object on ideological grounds to profit-making activity in connection with criminal corrections, may be willing to support your proposal if you specify that only nonprofit organizations can be service providers. With this restriction, you may be able to gain adoption of a policy with most of the benefits of your original proposal.

Another approach to compromising is to add features that opponents find attractive. In Chapter 8 we discussed logrolling, a form of compromise that is characterized by the packaging together of substantively unrelated proposals. The engineering of logrolls is usually the purview of clients rather than analysts. More commonly, analysts are in a position to give advice about the composition of a single proposal. For example, consider again the proposal to ban random drug testing in the workplace. Matched against the status quo, it appears that the ban would be adopted by the city council. If you opposed the ban, then you might try to stop it by proposing that firms be banned not from testing but from firing, demoting, or disciplining an employee on the basis of a single test. Assuming that the two council members (D and E) who are opposed to the ban would support this compromise position, it might get the necessary votes by attracting the support of one of the less enthusiastic supporters of the stronger ban (B or C). You might be able to get the mayor and the director of public health to support the compromise position and help secure the third vote.

In organizational settings, compromise often takes the form of *negotiation*, whereby interested parties attempt to reach agreement through bargaining. For example, local implementation of a state law requiring teachers and health professionals to report suspected cases of child abuse may require an agreement to be reached between the police chief and the director of social services on how reports are to be investigated. We have already noted the opportunity for negotiation provided in hurting stalemates, but the opportunity for potentially beneficial negotiation often arises in less extreme circumstances.

What factors are likely to influence the character and outcomes of negotiations?[35] One factor is the frequency with which the participants must deal with each other. If they must deal with each other often, then it is likely that they will be more flexible and amicable than they would be in an isolated negotiation. (We consider this idea more formally in the discussion of repeated games at the end of Chapter 12.) Another factor is the political resources that each brings to the negotiations. Is either one in a better position to appeal to outside parties, such as the county executive or the mayor? Can either one use precedent as an argument? Can either party inflict costs on the other by delaying agreement?

Roger Fisher and William Ury provide a useful practical guide for negotiating effectively.[36] We note here two of their themes that can be thought of as general strategies for successfully negotiating compromise.

First, *remember that you are dealing with people who have emotions, beliefs, and personal interests.* The presentation as well as the content of proposals can be important for reaching mutually satisfactory agreements. For example, if your counterpart in the negotiation has gone on record as opposing a tax increase, then, even if you convince him that additional revenue is needed, he is unlikely to agree to something called a tax increase. Look for a compromise that allows him to save face. Perhaps what we would normally describe as a gasoline tax could be called a "road user's fee" instead. Maybe just substituting the euphemism "revenue enhancement" will be an adequate face saver, perhaps even if everyone knows that it is just a different phrase for the same thing! The general point is to always keep in mind that you are dealing with people, who, like yourself, want to feel good about what they are doing.

Second, *try to negotiate interests, rather than positions, so that mutually beneficial compromises can be found.* For example, Fisher and Ury point to the negotiations between Egypt and Israel over the Sinai, which was captured by Israel in the Six-Day War of 1967.[37] As long as both sides approached the issue territorially, Egypt demanding return of the entire Sinai and Israel demanding that the boundary line be redrawn, there was little possibility of agreement. By looking at interests, however, a solution became possible. Although Egypt, after centuries of domination, was unwilling to cede sovereignty to any of its territory, Israel did not desire the territory per se. Rather, it wanted to keep Egyptian military forces with offensive capabilities away from its border. The solution was to return the Sinai to Egyptian sovereignty but to demilitarize those areas that could be used as staging grounds to threaten Israel.

The general point to keep in mind is that the ultimate purpose of negotiations should be to satisfy the interests of the parties involved. In other words, approach negotiations as a sort of on-the-spot policy analysis, get agreement on the facts in the dispute (define the problem), identify interests (determine goals), agree on criteria for determining a fair solution (specify impact categories), brainstorm for possible solutions (develop policy alternatives), and only then consider specific solutions (evaluate alternatives in terms of goals and impact categories).[38]

Heresthetics

William Riker has coined the word *heresthetics* to refer to strategies that attempt to gain advantage through manipulation of the circumstances of political choice.[39] Heresthetical devices fall into two categories: those that operate through the agenda and those that operate through the dimensions of evaluation.

In Chapter 8 we illustrated the paradox of voting with the hypothetical example of pairwise votes on school budgets. The preferences of the voters were such that, if everyone voted his or her true preferences, the order in which policies were considered—the agenda—determined the policy that would be selected. Thus, *manipulation of the agenda can be a powerful political strategy.*

The holders of specific offices often have the opportunity to manipulate the agenda directly. Such officials as the speaker of the legislature, the committee chair, or the administrator running a meeting may be able to influence the policy outcome by the way they structure the decision process. Sometimes they can bluntly structure the decision process to their advantage by refusing to allow certain alternatives to be considered, perhaps justifying their actions by a call for "further study." Other times they can achieve their ends by determining the order in which the alternatives will be considered.

Obviously, agenda setters are especially important actors in any political arena. If your client is an agenda setter, then you are in an advantageous position for seeing your accepted recommendations adopted as policy. If your client is not an agenda setter, then you must find ways to increase the chances that your client's proposals receive favorable agenda positions. One approach is to design your proposals so that they appeal to the agenda setters.

Another approach is to mobilize other political actors on your behalf to make it politically costly for agenda setters to block your proposals. For example, in 1962 Senator Estes Kefauver found that the legislation he introduced to amend the laws governing the regulation of pharmaceuticals was stalled in the Judiciary Committee; the chairman of the committee, Senator James Eastland, opposed it. Only by generating criticism of inaction in the press and mobilizing the support of the AFL-CIO did Kefauver succeed in getting President John F. Kennedy to pressure Eastland to report out a version of Kefauver's bill.[40] Obviously, such tactics must be used with care, especially when the goodwill of the agenda setter may be important on other issues in the future.

Once a proposal gets a place on the agenda, its prospects for adoption may be improved by *sophisticated voting*—that is, by someone not voting his or her true preferences at some stage in the voting agenda to achieve a better final outcome. For example, in 1956 the Democratic leadership of the U.S. House of Representatives proposed that federal aid be given directly to local school districts for the first time. If everyone had voted his or her true preference, then the Democratic majority would have passed the bill. Representative Adam Clayton Powell introduced an amendment to the bill, however, that would have prevented funds from going to segregated schools. Northern Democrats supported this amendment; however, southern

Democrats opposed it. Riker argues that a number of Republicans, who preferred no bill to the unamended bill and the unamended bill to the amended bill, nevertheless voted for the amendment because they expected that southern Democrats would vote against the amended bill.[41] Their expectations were correct and the amended bill did not pass.

Assuming that the northern Democrats preferred the unamended bill to no bill, why didn't they counter the Republicans' sophisticated voting with their own by voting against the Powell amendment so that the southern Democrats would join with them to pass the unamended bill? Perhaps, as Riker suggests, the northern Democrats saw taking a stand against segregation as more important to them in terms of constituent support than obtaining a more preferred policy outcome. In other words, their position was complicated by the need to consider a second dimension of evaluation.

The situation of the northern Democrats suggests the second category of heresthetical devices: those that alter the *dimension of evaluation.* When you are in a minority position, you can sometimes introduce a new consideration that splits the majority. For example, in 1970 several west coast senators proposed amendments to the pending military appropriations bill that would have prohibited the Department of Defense from shipping nerve gas from Okinawa to the United States. Although they initially did not have enough votes to prevent the shipments, Senator Warren Magnuson helped them gain a majority by arguing that what really was at stake was the dignity of the Senate. The Senate had earlier passed a resolution saying that the president could not change the status of any territory involved in the peace treaty with Japan without consulting the Senate. Magnuson asserted that the gas shipments were part of preparations to return Okinawa to Japan, an unconstitutional usurpation of the Senate's right to ratify treaties. By casting the issue in terms of the rights of the Senate, Magnuson was able to gain passage of the amendment by attracting enough votes from those who would otherwise have favored the gas shipments.[42]

Some rather surprising political outcomes appear to result from heresthetical maneuvers. For example, despite domination of both the House of Commons and the House of Lords by landed interests opposed to agricultural free trade in the middle of the nineteenth century in the United Kingdom, the Corn Laws were repealed. It appears that this result was achieved by the successful efforts of Prime Minister Sir Robert Peel to inject the issue of public order, especially salient with respect to the onset of famine in Ireland.[43] What was originally a one-dimensional issue with a stable outcome of keeping the Corn Laws became a two-dimensional issue that provided an opportunity for repeal.

Rhetoric

Perhaps the most common political strategy is *rhetoric*, the use of persuasive language to frame issues in favorable ways. At one normative extreme, rhetoric provides correct and relevant information that clarifies the probable impacts of proposed policies. At the other normative extreme, it provides incorrect or irrelevant information that obfuscates the probable impacts of proposed policies. As a policy analyst, you are likely to face ethical questions concerning the extent to which you should participate in your client's use of rhetoric that obfuscates rather than clarifies.

As an illustration of the ethical use of rhetoric, return to the proposal to ban random testing for drug use. In Table 11.2, council member C is listed as likely to vote for the ban because she sees her major constituency as community groups, which favor the ban. The director of the Urban League, for instance, favors the ban because he believes that testing might be used unfairly against minority workers. If you opposed the ban, then you might provide information to the Urban League director indicating that the ban would discourage some employers from staying or expanding in the city, the net effect on minority employment may actually be negative. If convinced, the director might then convince council member C to change her position so that the ban would not pass.

Often the most effective rhetoric influences political actors indirectly through public opinion rather than directly through persuasion. As we discussed in the context of the multiple streams framework, there are policy windows, periods when public opinion and media attention are sensitive to political initiatives in specific policy areas. Public figures may be moved to act by the opportunity to gain publicity.[44] To keep policy windows open, or perhaps even to create them, policy advocates can provide information to the media through such mechanisms as press conferences, news releases, hearings, planted stories, and leaks. Obviously, information that can be portrayed as sensational will be more likely to be reported by the media than information that appears mundane.

A common, but ethically questionable, rhetorical strategy is to emphasize very negative, but unlikely, consequences.[45] As noted in Chapter 8, this strategy takes advantage of the findings of psychologists that people have a tendency to overestimate the likelihood of certain kinds of very low probability events and to be more sensitive to potential losses than to potential gains. For example, in discussions of health and safety, opponents of nuclear power tend to emphasize the consequences of a meltdown in combination with a containment structure failure. Now such an accident would indeed be catastrophic. However, it is hard to imagine peacetime scenarios that would result in the failure of a containment structure. Despite the very low probability of such a disaster, the public tends to focus on it in discussions of nuclear safety to the exclusion of more mundane risks inherent in the mining and transportation of fuels and the disposal of wastes.

Telling stories is often a very effective way to communicate complicated ideas.[46] Stories capture people's attention when they have victims and villains—the deserving and undeserving in the social construction framework. However, few public policy issues have such unambiguous characters. Indeed, there may be a conflict between constructing an effective story that captures attention, and one that appropriately conveys the multiple values at stake and the inherent trade-offs among policies to promote them.

In Chapter 3 we discussed analytical integrity as one of the values that should be considered by policy analysts. We argued that the professional ethics of policy analysts should generally preclude participation in the injection of false or grossly misleading information into public debates. If you decide to place responsibility to your client or the furtherance of your conception of the good society above analytical integrity, then you might still want to avoid rhetorical dishonesty in most circumstances so that you will be credible when you do sacrifice analytical integrity. As Machiavelli suggests, for political actors the appearance of honesty is more important than the virtue itself.[47] Yet, generally, *the best way to appear honest is to be honest.*

Conclusion

Analysts who want to influence public policy should know the political environment. Clients are more likely to embrace policy proposals that promote their political interests. Strategies that take account of the interests of actors in the relevant political arenas can influence the likelihood that policies will be adopted. Thus, if analysts are to provide useful advice, then they cannot avoid paying attention to the political environment.

For Discussion

1. Imagine that a construction project will close a major parking lot on your campus for the next three years. As a consequence, some people currently parking on campus for a nominal fee will no longer be able to obtain permits. A member of the economics faculty has suggested raising the price substantially from $25 per year to $200 per year, so that the number of permits demanded at the higher price will equal the number available. This proposal would have to be approved by the transportation planning

committee, which includes representatives from the faculty, the administration, unionized employees, and students. How would you assess the prospects for adoption of the proposal? What ideas do you have about how to improve the prospects?

2. Discuss the social construction of the lesbian, gay, bisexual, and transsexual (LGBT) community. Has it changed or remained the same over the past 40 years? Does it currently vary by demographic group? Discuss the implications of your answers for public policy.

Notes

1 Fuller treatments of most of these models can be found in Paul A. Sabatier and Christopher M. Weible, eds., *Theories of the Policy Process*, 3rd ed. (Boulder, CO: Westview Press, 2014).

2 Mancur Olson, *The Logic of Collective Action* (Cambridge, MA: Harvard University Press, 1973), at 47.

3 Paul E. Peterson, "Forms of Representation: Participation of the Poor in the Community Action Program," *American Political Science Review* 64(2) 1970, 491–507.

4 Richard Himelfarb, *Catastrophic Politics: The Rise and Fall of the Medicare Catastrophic Coverage Act of 1988* (University Park: Pennsylvania State University Press, 1995).

5 See Connor McGrath, "Policy Making in Political Memoirs, The Case of the Poll Tax," *Journal of Public Affairs* 2(2) 2002, 71–85.

6 Vincent L. Hutchings, *Public Opinion and Democratic Accountability* (Princeton, NJ: Princeton University Press, 2003).

7 David L. Weimer, *Institutional Design* (Boston: Kluwer Academic, 1995), at 2.

8 Larry L. Kiser and Elinor Ostrom, "The Three Worlds of Action: A Metatheoretical Synthesis of Institutional Approaches," in Elinor Ostrom, ed., *Strategies of Political Inquiry* (Beverly Hills, CA: Sage, 1982), 179–222. On the relationship of the institutional

rational choice framework to other prominent frameworks, see José Real-Dato, "Mechanisms of Policy Change: A Proposal for a Synthetic Explanatory Framework," *Journal of Comparative Policy Analysis* 11(1) 2009, 117–43.

9 Elinor Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action* (Cambridge: Cambridge University Press, 1990).

10 Brian Arthur, *Increasing Returns and Path Dependence in the Economy* (Ann Arbor: University of Michigan Press, 1994).

11 See, for example, Paul Pierson, "Dependence, Increasing Returns, and the Study of Politics," *American Political Science Review* 94(2) 2004, 251–67.

12 Joe Soss and Sanford F. Schram, "A Public Transformed? Welfare Reform as Policy Feedback," *American Political Science Review* 101(1) 2007, 111–27.

13 Scott E. Page, "Path Dependence," *Quarterly Journal of Political Science* 1(1) 2006, 87–115.

14 Page also identifies a third process, which he calls "phat-dependent" ("phat" is an anagram for path), if the outcome in any period depends on the collection, but not the strict order, of the outcomes and opportunities that previously occurred. Ibid., 96–97.

15 Paul A. Sabatier and Hank Jenkins-Smith, "An Advocacy Coalition Model of Policy Change and the Role of Policy Oriented Learning Therein," *Policy Sciences* 21(2–3) 1988, 129–68. See also Paul A. Sabatier and Hank Jenkins-Smith, eds., *Policy Change and Learning: An Advocacy Coalition Approach* (San Francisco: Westview Press, 1993).

16 Paul A. Sabatier and Hank Jenkins-Smith, "The Advocacy Coalition Framework: An Assessment," in Paul A. Sabatier, ed., *Theories of the Policy Process* (Boulder, CO: Westview Press, 1999), 117–66.

17 Paul A. Sabatier and Christopher M. Weible, "The Advocacy Coalition Framework: Innovations and Clarifications," in Paul A. Sabatier, *Theories of the Policy Process*, 2nd ed. (Boulder, CO: Westview Press, 2007), 189–220.

18 Deborah Stone, *Policy Paradox and Political Reason* (Glenview, IL: Scott Foresman, 1988).

19 Anne Schneider and Helen Ingram, "The Social Construction of Target Populations," *American Political Science Review* 87(2) 1993, 334–46.

20 Frédéric Varone and Natalie Schiffino, "Regulating Red and Green Biotechnologies in Belgium: Diverging Designs of Biopolicies," *Archives of Public Health* 62(1–2) 2004, 83–106.

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24 Michael D. Cohen, James G. March, and Johan P. Olsen, "A Garbage Can Model of Organizational Choice," *Administrative Science Quarterly* 17(1) 1972, 1–25, at 1.

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26 Ibid., 174–80.

27 Frank R. Baumgartner and Bryan D. Jones, *Agendas and Instability in American Politics* (Chicago: University of Chicago Press, 1993).

28 Bryan D. Jones and Frank R. Baumgartner, *The Politics of Attention: How Government Prioritizes Problems* (Chicago: University of Chicago Press, 2005).

29 Arnold Meltsner, "Political Feasibility and Policy Analysis," *Public Administration Review* 32(6) 1972, 859–67.

30 Rufus E. Miles, Jr., "The Origin and Meaning of Miles' Law," *Public Administration Review* 38(5) 1978, 399–403.

31 John Mendeloff, *Regulating Safety: An Economic and Political Analysis of Occupational Safety and Health Policy* (Cambridge, MA: MIT Press, 1979), 15–16.

32 David L. Weimer, "Public and Private Regulation of Organ Transplantation: Liver Allocation and the Final Rule," *Journal of Health Politics, Policy and Law* 32:1 (2007), 9–49.

33 For a discussion of the use of co-optation at the organizational level, see Philip Selznick, *TVA and the Grass Roots* (Berkeley: University of California Press, 1949), 13–16.

34 In arenas where the gains of winners come at the expense of losers (*zero-sum games*), Riker's *size principle* predicts such behavior: "In social situations similar to *n*-person, zero-sum games with side-payments, participants create coalitions just as large as they believe will ensure winning and no larger." He calls these *minimal winning coalitions*. William H. Riker, *The Theory of Political Coalitions* (New Haven, CT: Yale University Press, 1962), 32–33. In some circumstances, however, compromising more to obtain a broader coalition may be desirable if the greater consensus will deter opponents from seeking to overturn the policy at a later time or in another arena. For example, Justice Felix Frankfurter successfully delayed the Supreme Court decision in *Brown v. Board of Education,* 347 U.S. 483 (1954), so that a unanimous opinion could be achieved. Bernard Schwartz with Stephan Lesher, *Inside the Warren Court* (Garden City, NY: Doubleday, 1983), 21–27.

35 For a more comprehensive listing, see Howard Raiffa, *The Art and Science of Negotiation* (Cambridge, MA: Harvard University Press 1982), 11–19.

36 Roger Fisher and William Ury, *Getting to Yes: Negotiating Agreement Without Giving In*, 2nd ed. (New York: Penguin, 1991).

37 Ibid., 40–42.

38 Ibid., 10–14.

39 William H. Riker, *The Art of Political Manipulation* (New Haven, CT: Yale University Press, 1986), ix.

40 Richard Harris, *The Real Voice* (New York: Macmillan, 1964), 166.

41 Riker, *Art of Political Manipulation*, 114–28.

42 Ibid., 106–13.

43 Iain McLean, *Rational Choice & British Politics: An Analysis of Rhetoric and Manipulation from Peel to Blair* (Oxford, New York: Oxford University Press, 2001).

44 For a general overview on how federal officials interact with the press, see Martin Linsky, *Impact: How the Press Affects Federal Policymaking* (New York: W. W. Norton,

1986), 148–68.

45 Riker labels this strategy *induced dread* and explains how it operates in "Rhetoric in the Spatial Model," a paper presented to the Conference on Coalition Government, European Institute, Florence, Italy, May 28, 1987. For fuller development of these ideas, see William H. Riker, *The Strategy of Rhetoric: Campaigning for the American Constitution* (New Haven, CT: Yale University Press, 1996). See also R. Kent Weaver, "The Politics of Blame Avoidance," *Journal of Public Policy* 6(4) 1986, 371–98.

46 Emery Roe, *Narrative Policy Analysis: Theory and Practice* (Durham, NC: Duke University Press, 1994), at 9.

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12
Implementation

While adoption is a process that occurs primarily in political arenas, such as legislatures or regulatory agencies, implementation is almost always a broader, more complex, and more sequential process involving actors of more than one kind: political and organizational, not to mention those actors whose behaviors adopted policies seek to change. Furthermore, because in some sense implementation never ends in the absence of explicit termination, those seeking to implement must continue to guard against subversion of the adopted policy by its opponents. Unexpected external events (shocks) may give opponents opportunities to stop policies by legitimizing their initial opposition. For all these reasons, anticipating implementation problems and incremental opportunities to improve the policy is an important but challenging task for policy analysts.

This chapter provides a framework for understanding the implementation process as a whole. It also provides frameworks and techniques for assessing the implementation prospects of proposed policies.[1] Policy alternatives with substantial risk of implementation failure should be redesigned to have better prospects, or at least should only be recommended with explicit consideration of their inherent risk. We begin by discussing preliminary concepts. Most fundamentally, success requires a *plausible chain of hypotheses linking adoption to desired outcomes*. To help identify the links in chain, we liken implementation to an *assembly process* in which one must rely on others to provide the necessary parts, or elements.

After these prerequisites, we consider three questions to help guide implementation analysis. First, what roles need to be filled during implementation? We focus on three major types of implementation actors—implementation managers, "doers," and "fixers"—and their incentives, resources, and capabilities. One way or another, these actors must design and execute the assembly of elements needed to implement the policy. Yet, to do implementation well, they must have appropriate incentives, adequate resources, and necessary competencies.

Second, how can analysts take account of implementation in their design of policy alternatives? We present two techniques, forward and backward mapping, that analysts can use to help them anticipate implementation problems and redesign policy alternatives to mitigate them.

Third, how can analysts improve their policy designs? We consider a number of concepts that help analysts think strategically about designing policies that have good prospects for producing desired outcomes. One important idea is that of building error correction mechanisms into policy designs to address the endemic uncertainty that characterizes complex implementation. Some error correction mechanisms, or ways of anticipating "Murphy's Law," include banking redundant resources and organizational slack, designing in the requirement for early and regular evaluation, and facilitating the potential for termination and cutting one's losses through sunset laws and similar administrative rules. Another important idea is anticipating, or even taking advantage of, spatial and temporal variability or other kinds of heterogeneity by using bottom-up processes to get others to communicate potential implementation problems and by phasing in implementation over time to even out resource requirements.

Finally, we want to emphasize the importance of considering the potential impact of implementation on *corporate culture*. Implementation of a new policy, or indeed any form of synoptic change, can disrupt an existing culture, either positively or negatively. The risk of disruption, however, is often underestimated or ignored. A functioning corporate culture operates on the basis of repeated and well-understood interpersonal interactions between organizational members; these interactions, in turn, are underpinned by a foundation of social and organizational norms. Ideally, both policy designers and implementation managers should anticipate this kind of disruption. If they can do this well, new policy may strengthen and enrich organizational culture.

Prerequisite: Sound Logic

What theory underlies the connection between policy and intended outcomes? Is the theory reasonable? The characteristics of the policy, and the circumstances of its adoption, determine the hypotheses underlying implementation and the likelihood that they are true. Specifically, we can think of the logic of a policy as a chain of hypotheses.[2]

Even a seemingly simple policy may have a long logic chain. Consider a state program to fund locally initiated experiments that are intended to identify promising approaches to teaching science in high schools. For the program to be successful, the following hypotheses need to be true: first, that school districts that have good ideas for experiments are those that apply for funds; second, that the state department of education selects the best applications; third, that the funded school districts actually execute the experiments they proposed; fourth, that the experiments produce valid evidence on the effectiveness of the approaches being tested; and fifth, that the department of education be able to recognize which successful approaches can be replicated in other jurisdictions.

It is quite possible to imagine that any one of these hypotheses could be false, or at least not universally true. For instance, applicants may be predominantly school districts with experience in applying for grants rather than school districts with good ideas; the education department may face political pressure to distribute funds widely rather than to the school districts with the best ideas; school districts may divert some of the funds to other uses, say, paying for routine classroom instruction; school districts may lack skilled personnel to carry out evaluations or they may be reluctant to report evaluation results that do not support the efficacy of their approaches; or the department of education may not have sufficient personnel to look closely at the evaluations to determine which ones are valid. The more likely it is that these hypotheses are false, the less likely it is that the program will produce useful information about how to do a better job teaching science in high schools.

In sum, a policy alternative is likely to be *intrinsically* unimplementable if we cannot specify a logical chain of behaviors that leads to the desired outcomes. Later we discuss scenario writing as a way of clarifying and evaluating plausible versus implausible logic chains.

Identifying the Links in the Chain: The Assembly Metaphor

Eugene Bardach uses an assembly metaphor to frame implementation: assembly involves securing the essential elements from those who control them.[3] In general, *the more numerous, the more heterogeneous, and the more tightly linked are the elements that must be assembled, the greater is the implementation complexity*. The link between the number of elements and complexity is obvious. The link between heterogeneity and complexity is straightforward: the more diverse the elements, the more varied the connections between elements. More tightly coupled elements increase implementation complexity by making it more difficult either to select or replace coupled elements independently.[4] Tight coupling also increases the risk of overall failure by necessitating major redesign when any one element fails. Some implementation scholars take this issue so seriously that they call it the "domino syndrome."[5]

For example, an agency that has an ongoing contractual relationship with an information system vendor may find it difficult to arrange for other services required for implementation from a more appropriate contractor. *Increased complexity increases the probability of nonlinear and unpredictable interactions among the elements and raises the risk of implementation* *failure.* This suggests important questions to ask when considering the prospects for successful implementation of a policy alternative: Exactly what elements (the hypotheses linking policy to desired outcomes) must be assembled? Who controls these elements? What are their incentives? What resources does the implementer have available to induce them to provide the elements? What consequences will result if the elements cannot be obtained either on a timely basis or at all?

Roles in the Implementation Process

Implementation of all but the simplest public policies requires the efforts of at least two and often three types of actors: always implementation managers and doers; and often fixers. Either implementation managers or doers may sometimes decide that they need to interact with policy designers. This is usually an extremely sensitive process, especially when doers wish to do so, as this almost always treads on the sensitivities and sensibilities of implementation managers. Fixers are sometimes called on to perform this role. In this role, fixers may clarify the intentions of designers or the concerns of implementers, or even negotiate with designers to obtain incremental redesign in light of the concerns of implementation managers and doers. Fixers may also directly mediate between implementation managers and doers, especially when they are in different organizations.

Policy alternatives rarely specify in detail the tasks of particular doers, so someone must direct the implementation effort. *Implementation managers* provide the detailed designs for implementation. For example, implementation of a law by a public agency typically requires a manager in the agency to assign tasks to agency personnel and motivate their completion. Their initial efforts can be thought of as implementation design; their continuing efforts to complete the implementation can be thought of redesign. For example, the manager may specify a specific task as a modification of a standard operating procedure (SOP) well understood by agency personnel. However, if the modification does not provide the needed element for implementation, then it may be necessary to create an entirely new routine that does deliver it.

Implementation almost always requires actions by those who provide any of the elements needed for the desired policy outcome: the *doers.* Most often they are organizational members who directly provide services to clients, so-called street-level bureaucrats. But they may also direct and assign tasks to personnel in their own or other organizations that play a role in the implementation. For example, they may be organizational employees who provide licenses for some type of private-sector vendor or information needed by employees in their own or other organizations, staff doctors who inform patients about the health benefits of lifestyle changes, or individuals who are expected to follow some law.

Implementation often requires major and rapid changes in organizational routines that pose novel challenges for both implementation managers, who may have little experience with synoptic change, and doers, who often must take on additional responsibilities during implementation. Other actors with an interest in the implementation may be available to contribute to the implementation process. Bardach calls such actors *fixers*—those who can intervene in the assembly process to help gain needed elements that are being withheld.[6] We limit the use of the term fixer to someone who provides either specialized expertise to implementing organizations in a formal role or information useful either to the implementers or to the policy's designers or its proponents in an informal role.[7] Typically, fixers are *boundary spanners*, who facilitate communication between and within organizations.[8]

Figure 12.1 summarizes the relationships between policy designers, implementation managers and doers. It emphasizes that formal authority flows "downhill" from design, through management to frontline action, but influence and information—primarily emanating from professional and craft knowledge—often flows in the opposite (or some other) direction. Figure 12.1 also shows that fixers can span a variety of boundaries and that they can intermediate between the various other actors in a number of different ways and contexts.

Effective implementation depends on who will actually play these roles. Sometimes those who adopt policies name names, at least in terms of assigning responsibilities to individuals holding particular organizational roles. Otherwise they delegate responsibility for filling roles to the heads of agencies, or, in the case of assigning tasks to doers, the implementation managers. The first step in anticipating the implementation prospects of any policy alternative is a plausible inventory of who will play the various implementation roles and their incentives, resources, and competencies. In practice, assessing the incentives, resources, and competencies of each role requires analysis of both the particulars of the policy alternative, and the organizational and political environment in which its implementation would occur.

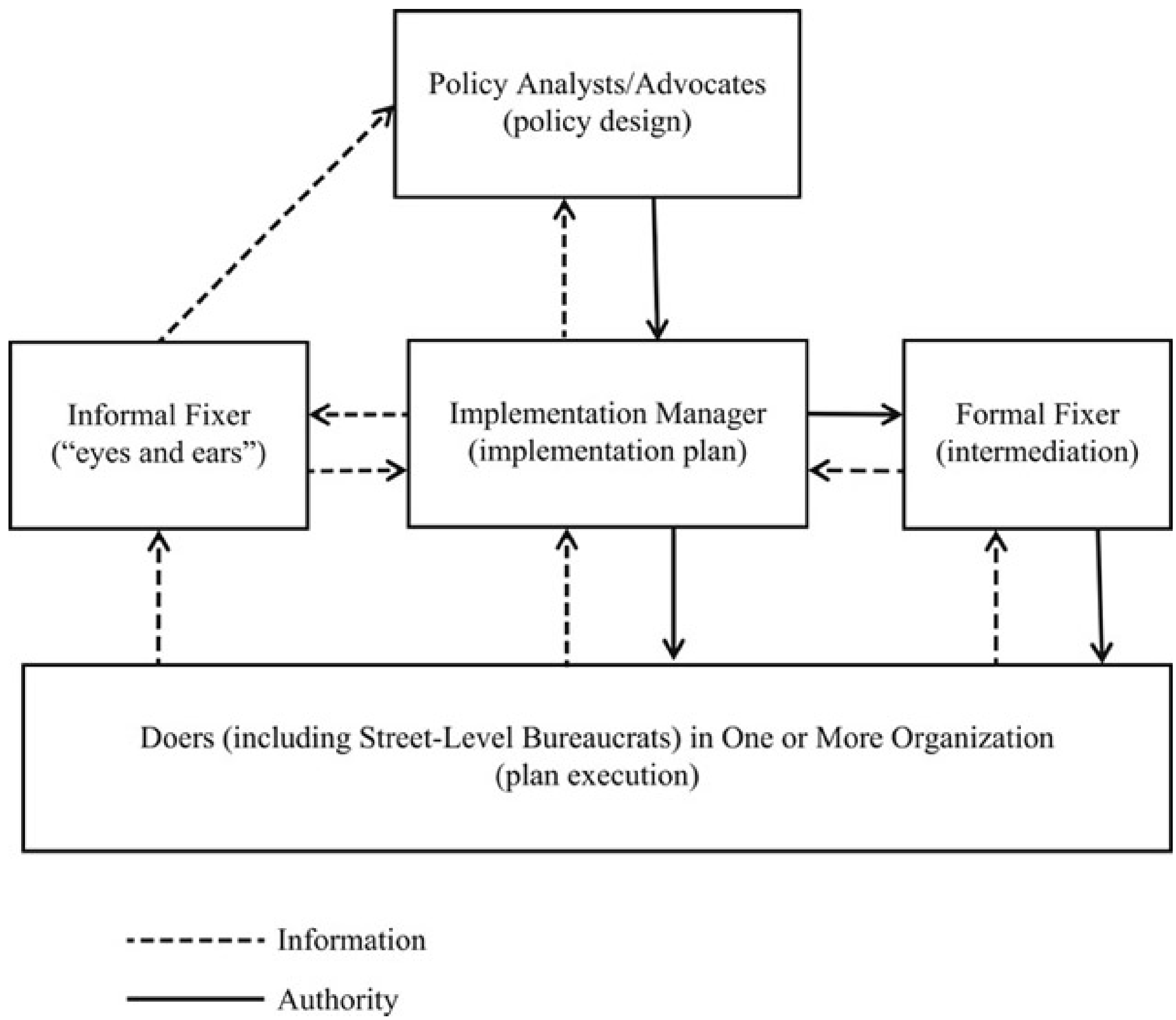

Figure 12.1 Implementation Roles

Implementation Managers

Policies rarely specify all the elements needed for their implementation. Someone or some team must identify the necessary elements and oversee their procurement into a working assembly. The person who plays the role of implementation manager is typically an employee in the organization given responsibility for implementation, usually someone given the task by the chief executive of the organization.

In light of the importance of politics in the implementation process, understanding the motivations (most importantly, the incentives) and political resources of the implementation manager is obviously important for predicting the likelihood that the policy will produce the intended consequences. A manager who views the policy as undesirable or unimportant is less likely to expend personal and organizational resources during the assembly process than one who views the policy more favorably. One reason that organizational units are sometimes created to implement new policies is the perception that the administrators of existing units will not be vigorous implementers because of their commitments to already-existing programs.

How does one determine if possible implementation managers have adequate incentives for playing their roles effectively? It usually begins by assessing the extent of congruence between the goals that the policy seeks to promote and those of the manager's organization. A high degree of goal congruence provides at least the initial motivation for vigorous efforts to complete the assembly. However, the process of implementing the new policy will almost always compete with routine activities of the manager that also promote organizational goals—so goal congruence alone does not guarantee that that the manager will have the appropriate incentives to pursue implementation vigorously. Beyond goal congruence at the organizational level, the clarity and transparency of the responsibility for successful implementation may also affect managerial effort: sharing the credit for success and the blame for failure typically creates weaker incentives than visible and personal responsibility.

Often observers will not be able to assess the outcomes of implemented policies immediately, or, indeed, even after considerable delay. In these cases, managers facing difficult implementation tasks may decide to reduce their effort by overseeing the assembly of a nominally implemented program that does not actually produce the putative outcomes of the policy. For example, a school principal with responsibility for managing the implementation of a program to improve the nutritional content of the lunches served in the school cafeteria may decide to consider the policy implemented once appropriate menu changes have been made, rather than after students actually begin to eat healthier food.

The effectiveness of implementation managers at shepherding successful implementation also depends on the resources they have available to assemble the necessary elements for the policy. Resources are multifaceted and include authority, political support, and "treasure" (additional resources that it is in the power of the implementation manager to bestow).

In general, the greater the legal authority adopted policies give to implementation managers, the greater is their coercive capacity to compel required behavior. Similarly, the stronger the political support for the adopted policy and its putative goals, the greater the capacity of those implementing it to secure required behavior. For example, a manager in a state department of education is in a better position to get school districts to provide competent evaluations of their experiments if it can require selected grant recipients to hire external evaluators. With respect to political support for policy goals, this department of education is in a better position to ward off pressure for widespread distribution of grants if most of the supporters of the program in the legislature hold the putative goal of improving high school science education as more important than using program funds to provide general aid to school districts.

Having more treasure, including funding and commitment of staff time, almost always makes it easier to assemble the necessary elements. Adequate funding usually allows managers to secure needed staff time, whether internally or through contracting, to help manage a complex implementation. Managers will likely be more effective if they have sufficient treasure in both the short run, to support the assembly process, and in the longer run, to ensure that any new activities that are required to maintain the policy detract less from ongoing activities. Public agencies are sometimes unable to use funding effectively on a timely basis, either because of bureaucratic failure as discussed in Chapter 8 or the problems of effective contracting discussed in Chapter 13. Nonetheless, access to more treasure generally increases the effectiveness of implementation managers; too little treasure can doom their efforts.

Even good intentions and adequate resources, however, are not sufficient to guarantee success. The implementation managers must have the capacity to design plausible assembly plans and to redesign them when miscalculation or various shocks make them ineffective, perhaps by overwhelming a previously plausible logic chain. Such capacity typically depends on skills, tacit knowledge, and relevant experience.[9] Although the particular skill set needed in any particular context varies, both an ability to see the big picture and to attend to detail are almost always as valuable as generic managerial skills, such as being able to effectively communicate with doers and motivate them to deliver their elements to the assembly. Tacit knowledge of organizational routines gained through experience in the organization usually facilities more realistic designs. Learning through involvement in prior implementation efforts is also likely valuable. Lack of necessary capabilities can hinder both design and assembly.

Doers

In assessing incentives, it is important to emphasize that frontline employees, sometimes called *street-level bureaucrats*, actually implement almost all policies.[10] We treat the term "street-level" broadly; they may be highly professionalized individuals or groups, such as public health physicians, legal aid lawyers, or university teachers (so they may have higher professional status than their implementation managers). Furthermore, in many policy implementation contexts, government may not directly employ these street-level implementers: they may be employed by either not-for-profit or for-profit entities and are often spread across multiple organizations.[11] To return to our example, even if school district-level administrators can be incentivized to implement by the threat of withholding future grants, "street-level" teachers may be totally unmoved, or even dis-incentivized, by this kind of threat.

Many public policies have faced severe implementation problems or even collapsed because policy designers and implementation managers did not recognize and address the complexity of the incentives of frontline workers. Consequently, they failed to ensure the realignment of frontline employee incentives to a new program or policy. However, we should be careful to avoid a reductionist approach to the incentives of frontline workers. A commitment to the policy goals can often motivate them as much as intrinsic incentives.[12]

It is obviously easier to incentivize both implementation managers and frontline employees if the resources to do so are being made available. *Monetary resources are one important source of incentives, but a realistic time frame, staff to train frontline personnel, and adequate, reliable, and robust information technology systems are other important resources.* The absence of any these resources can foster frontline apathy, passive resistance, or even outright guerilla warfare against the policy.

As we have already discussed, clear legal authority is almost always a valuable resource for implementers. It may not be sufficient by itself to guarantee cooperation, however. For example, imagine a mayor opposed to the transportation of nuclear wastes on the railroad running through her city. Assume that she is required by law to prepare a plan for the evacuation of her city in the event of an accident involving the release of nuclear waste into the environment. Also, assume that the transportation through the city cannot begin until an evacuation plan is accepted. The mayor might hinder implementation of the shipment program by using one of three tactics: *tokenism*, *delayed compliance*, or *blatant resistance*.[13]

The mayor might make a token response by intentionally having her staff prepare an evacuation plan that meets the letter of the law but falls short substantively. Because the mayor has complied in form, the implementer may have difficulty gaining the political or legal support needed to force the mayor to comply in spirit. Indeed, tokenism is so difficult to deal with because the implementer generally bears the burden of showing that the compliance was inadequate. Instead of making a token response, the mayor might simply refuse to allow her staff to prepare an evacuation plan. Such blatant resistance may be costly for the mayor if the implementation manager decides to seek legal sanctions. However, taking the mayor to court may be politically costly for the implementer and would probably involve considerable delay. Yet, not challenging the mayor's noncompliance might encourage other mayors also to refuse to prepare evacuation plans, throwing the whole shipment program into jeopardy.

Massive resistance is rare because it takes fortitude and works best from a position that makes it costly for the implementer to force compliance. Tokenism and purposeful delay, however, are common resistance tactics. Using them, frontline employees, especially those with civil service protection, can often slow, or even stop, implementation with halfhearted or leisurely contributions.[14] Their ability to impede implementation is enhanced when they must be relied upon to make ongoing contributions to implementation maintenance. For instance, management information systems typically require a steady and accurate flow of input data to generate information benefits: delays or inaccuracy by even a few data providers may be enough to degrade the information and undermine the value of the entire system.

So, even when policy designers and implementation managers have the legal authority to demand compliance, they will not necessarily get enough, or of sufficient quality, for successful implementation. Implementation managers should expect that their efforts to secure program elements will require political effort: allies must be mobilized and agreements must be reached with people who have contrary interests. Therefore, implementers must be prepared to use strategies that are essentially political in nature, especially co-optation and compromise, to assemble program elements and keep them engaged. For example, to increase the chances that mayors will produce acceptable and timely evacuation plans, the implementation managers might offer to discuss limitations on the frequency and circumstances of the shipments before the mayors commit themselves to noncompliance through their public stands. Perhaps a concession on an entirely different issue of mutual concern can be offered.

Noncompliance need not be purposeful to hinder the implementation process. Someone holding a necessary program element may have every intention of providing it but fail to do so because of incompetence or an inability to get others to provide necessary support. For example, the mayor may believe that the preparation of an evacuation plan as required by law is a duty that should be discharged. However, her staff may lack the necessary skills to produce an adequate plan. Or, perhaps members of the staff are competent but cannot complete their task quickly because local procedural rules require that the plan be discussed in public hearings and reviewed by the city attorney, the city and county planning boards, and the city council. Even without strong opposition to the plan, scheduling problems and routine delays may prevent the plan from being approved in what the implementation manager considers a reasonable period of time.

Personal, value, and role heterogeneity among those providing seemingly similar elements make it difficult to predict how much inadvertent noncompliance will be encountered. For example, imagine that the shipment program requires that evacuation plans be prepared by 20 cities. Perhaps most have staff with adequate compliance skills. Yet, even if only a few are incapable of complying, the implementation may be jeopardized because, say, the authorization for the program requires that all evacuation plans be filed before shipments can begin.

In summary, *legal authority alone is insufficient to guarantee the compliance of those who control necessary program elements*. If they view the program, or their specific contributions to it, as contrary to their interests, then the implementation manager should expect them to try to avoid full compliance through tokenism, delay, or even blatant resistance. Furthermore, the implementer should hold realistic expectations about the capabilities of those who nominally control the elements to provide them.

Fixers

Implementation managers may not have sufficient information or expertise to assess whether doers are accomplishing their required tasks for successful implementation, especially when doers are not members of the managers' own organizations. In these situations, fixers serve as the "eyes and ears" of the implementation manager. However, they also can serve as the eyes and ears of program designers or analysts. When fixers observe that implementation managers are either trying to implement a faulty plan or not trying hard enough to implement a sound plan, they can make the policy's advocates aware of the impending failure. Often the fixer plays a crucial role within the implementing agency by facilitating communication, and even negotiation, between implementation managers and doers. For example, they may mediate negotiations over redesign or provide information to help implementation managers better align doer incentives to provide elements needed for the policy.

The failings of the implementation manager may be countered by having a competent fixer. For example, the staff of the legislative sponsor of the policy may oversee the implementation process, perhaps helping to negotiate compromises with those who are not complying or inducing compliance through complaints to the implementation manager's superiors or threats of hostile hearings. Oversight by the staff may also help motivate a less-than-enthusiastic implementer.

The availability of fixers at the local level can be especially helpful in adjusting centrally managed policies to heterogeneous local conditions and the needs and incentives of front-line employees. Sometimes local fixers can be found among interest groups that support the policy; other times local administrators may serve as effective fixers. In their evaluation of the implementation of youth employment programs, for instance, Martin Levin and Barbara Ferman found that the most successful local programs usually had executives who were willing to intervene in administrative detail to correct problems. Some of these fixers were especially effective in using incentives to turn the mild interest of other local actors into active support. Other fixers had official and personal connections that enabled them to facilitate inter-organizational cooperation.[15]

Allies at the local level can play the role of fixers by being a source of intelligence for either implementation managers or policy sponsors.[16] Information about local contexts is often costly for a centrally located implementation manager to gather in the absence of willing informants.

Local supporters of the policy may be able to provide information that is useful in anticipating and countering noncompliance tactics. For example, if the state attorney general's office is attempting to implement more stringent medical care standards for local jails, it might find local groups interested in judicial reform or county medical societies as sources of valuable information about whether the standards are actually being followed. Such information could help the attorney general's office target limited enforcement resources on the worst offenders.

Policy analysts may facilitate fixing through aspects of their policy designs. Increasing the transparency of the implementation process by requiring implementation managers to provide progress reports or create advisory bodies may enable interest groups to play the role of fixers. Such provisions may transform their interest in successful implementation into the capability to contribute to it. However, the more the goals of the interest group deviate from the goals of the policy, the greater is the danger that the fixers will hinder, rather than help, implementation. An interest group sharing only one of the goals of the policy may divert resources from overall success to further its primary goal. For example, designing a state program to improve mental health services in schools might require community advisory boards that could potentially play the role of fixers. If the boards attracted parents with children suffering from a particular problem, such as mild learning disabilities, or providers of particular types of mental health services, their fixing may divert the program away from intended improvements toward those they favor.

Earmarking resources for hiring or borrowing staff unburdened by routine organizational responsibilities to play the role of troubleshooters may enable implementation managers to identify and address bottlenecks and noncompliance more quickly and effectively. Implementation managers must be able to identify possible fixers and secure their services quickly and without large investments of already scare time for their contributions to be useful. In organizations that do not have flexibility in hiring, enabling implementation managers to borrow staff from other units within the organization may be more effective in supplying fixers than resources to hire.

Implementation Analysis Techniques

Two general approaches provide useful ways of systematically assessing implementation prospects in practical situations. *Forward mapping*, which moves from policy to outcomes, employs *scenario writing*. Scenarios involve specifying and questioning the chain of behaviors that link policies to desired outcomes. In contrast, *backward mapping* begins with the desired outcomes, determines the most direct ways of producing them, and then maps actions from effects to causes through the organizational hierarchy to the highest-level policy that must be adopted to realize the desired outcomes. Forward mapping is most useful for anticipating the problems that are likely to be encountered during the implementation of already formulated policy alternatives. Backward mapping is most useful for generating policy alternatives that have good prospects for successful implementation. This distinction recognizes that analysts often face implementation situations where policy is already formulated, for example situations where the policy is federally designed and mandated, but implementation will be carried out at the state level. Sometimes, however, analysts have the luxury of being able to design policy to minimize or at least mitigate potential implementation problems.

Apart from backward mapping, the other implementation factors we consider relate to strategic thinking primarily in the context of predicting and influencing the organizational and inter-organizational feasibility of specific policy proposals or policies. After considering forward mapping and backward mapping, we look at broader ways of designing better policies that are less likely to suffer from serious implementation problems. Specifically, we recognize three common problems: policy outcome uncertainty, heterogeneous circumstances, and the reality that policy change also "disturbs" the organizations that must implement change.

Forward Mapping: Writing Implementation Scenarios

Forward mapping is the specification of the chain of behaviors that link a policy to desired outcomes. We emphasize the “specific” in specification: Exactly what must be done and by whom for the desired outcomes to occur? Like Bardach, we recommend scenario writing as a method for analysts to organize their thinking about the behaviors that must be realized for successful implementation.[17] Writing implementation scenarios helps analysts discover implicit assumptions that are unrealistic. It may also help them discover alternative approaches to implementation with better prospects for success.

Effective forward mapping requires cleverness and a certain amount of courage. Research helps as well. The forward mapper must think about how those involved in the implementation will behave and how their behavior might be influenced. It requires what has been referred to as *dirty mindedness*—the ability to think about what could possibly go wrong and who has an incentive to make it go wrong.[18] In other words, one must be able to visualize the worst case. Research on outcomes in other jurisdictions is a valuable tool for making such predictions. Decentralized countries, such as the United States and Canada, provide many opportunities for comparative outcomes research.[19] Other things equal, observing the successful implementation of a particular type of policy in a number of jurisdictions makes that policy a more promising candidate elsewhere.

The analyst engaged in forward (and backward) mapping must be bold about making predictions. Many predictions will be incorrect, especially in their details. Much of the value of forward mapping comes from the thinking that goes into being specific. Therefore, the forward mapping analyst must not become paralyzed by fear of being wrong.

Table 12.1 Thinking Systematically about Implementation: Forward Mapping

Scenario	Write a narrative that describes all the behaviors that must be realized for the policy to produce desired outcome. Be specific about who, what, when, and why.
Critique	Is *the scenario plausible?*

1	For each actor mentioned:
	1. Is the hypothesized behavior consistent with personal and organizational interests?
	2. If not, what tactics could the actor use to avoid complying?
	3. What counter-tactics could be used to force or induce compliance?
Critique 2	*Thinking of the direct and indirect effects of the policy, what other actors would have an incentive to become involved?*
	For each of these actors:
	1. How could the actor interfere with hypothesized behaviors?
	2. What tactics could be used to block or deflect interference?
Revision	*Rewrite the narrative to make it more plausible.*

We recommend a three-step approach to forward mapping: (1) write a scenario linking the policy to outcomes; (2) critique the scenario from the perspective of the interests of its characters; and (3) revise the scenario so that it is more plausible. Table 12.1 highlights these steps, which we discuss in turn.

1. Write the Scenario

Scenarios are stories. They are narratives about futures as you see them. They have beginnings and ends that are linked by plots involving essential actors. Plots should be consistent with the motivations and capabilities of the actors. Plots should also be "rich" in the sense that they convey all the important considerations that are relevant to implementation.[20]

The plot consists of a series of connected actions. Your narrative should answer four questions for each action: What is the action? Who does the action? When do they do it? Why do they do it? For example, suppose that you are writing a scenario for the implementation of a Neighborhood Sticker

Plan (NSP) that would limit long-term on-street parking in specified city neighborhoods to residents who purchase stickers annually to identify their automobiles. An element of your plot might read as follows:

> Upon passage of the NSP [the previous "what?"], the director of parking in the police department [the "who"?] designs procedures for verifying that applicants are bona fide residents of participating neighborhoods and that their out-of-town guests qualify for temporary permits [the "what?"]. As requested by the police chief [the "why?"], he submits an acceptable procedure to the planning department within one month [the "when?"].

By putting together a series of such elements, you can connect the adopted policy (the NSP) with the desired outcome (neighborhood residents and their out-of-town guests having convenient access to on-street parking).

Writing scenarios that read well may enable you to get the attention of clients and colleagues who might otherwise be distracted; most people like stories and learn easily from them. You may be able to make your scenarios more convincing and lively by inserting quotes from interviews and noting past behaviors that are similar to the ones you are predicting. For example, if the director of parking has told you in an interview that his staff could easily design verification procedures, weave his statement into the plot. Even if you are preparing the scenario for your own exclusive use, write it out. The discipline of writing a coherent story forces you to consider the obvious questions about what must happen for the implementation to be successful.

2. Critique the Scenario

Is the scenario plausible? Are all the actors capable of doing what the plot requires? If not, your plot fails the basic test of plausibility and should be rewritten. Of course, if you cannot specify a plausible plot, then you can be fairly certain that the policy is doomed to fail.

The more interesting test of plausibility lies in considering the motivations of the actors. Will they be willing to do what the plot requires? For each actor mentioned in the plot, ask yourself if the hypothesized behavior is consistent with personal and organizational interests. If not, what tactics could the actor use to avoid complying? For example, the director of parking may view the verification procedures as an undesirable burden on his already overworked staff. He might, therefore, produce a token plan that will not adequately exclude nonresidents from purchasing stickers. The result could be that parking congestion will remain a problem in the participating neighborhoods. Perhaps he opposed the NSP in the first place and sees inadequate verification procedures as a way of undermining the whole program.

Can you think of any ways of countering the noncompliance? More generally, this raises the issue of providing either positive or negative incentives for compliance. For example, would a memorandum from the police chief clearly stating the director's responsibility for designing effective verification procedures be sufficient to get the director to do a good job? Would he be more cooperative if he were allocated additional overtime hours for his staff? If inducements such as these appear ineffectual, then you should consider writing him out of the plot by assigning the task to some other organizational unit.

After considering the motivations of actors who are in the plot, think about the "dogs that didn't bark." Who not mentioned in the plot will likely view the policy, or the steps necessary to implement it, as contrary to their interests? How might they interfere with the elements of the plot? What tactics could you employ to block or deflect their interference?

For instance, what will the commuters who have been parking in the participating neighborhoods do when they are excluded? Will they park in other residential neighborhoods or on commercial streets closer to the downtown? This might create pressure for the creation of additional sticker parking areas or lead adversely affected city residents to oppose continued implementation. Perhaps arranging more convenient bus services from parking lots on the outskirts of the city could help absorb the displacement of nonresident parking from the participating neighborhoods.

3. Revise the Scenario

Rewrite the scenario in light of the critique. Strive for a plot that is plausible, even if it does not lead to the desired outcome. If it still does lead to the desired outcome, then you have the basis for a plan of implementation. If it does not lead to the desired outcome, then you can conclude that the policy is probably unfeasible and should be replaced with an alternative that can be implemented.

Backward Mapping

Richard Elmore describes the logic of backward mapping as follows:

> Begin with a concrete statement of the behavior that creates the occasion for a policy intervention, describe a set of organizational operations that can be expected to affect that behavior, describe the expected effect of those operations, and then describe for each level of the implementation process what effect one would expect that level to have on the target behavior and what resources are required for that effect to occur.[21]

In other words, begin thinking about policies by looking at the behavior that you wish to change. What interventions could effectively alter the behavior? What decisions and resources are needed to motivate and support these interventions? The candidate policies are then constructed from alternative sets of decisions and resources for achieving the interventions.

Backward mapping is really nothing more than using your model of the policy problem to suggest alternative solutions and iteratively returning to adoption. It adds something to our thinking about policy alternatives, however, by drawing our attention to the organizational processes that give them form. Also, by focusing initially on the lowest organizational levels, it may help us discover less centralized (that is, more "bottom-up") approaches that we might have otherwise overlooked.

The development of an implementation plan for neighborhood sticker parking in San Francisco provides an example of the usefulness of backward mapping. By the early 1970s, parking congestion had become a serious problem in many San Francisco neighborhoods, as single-family homes were split into multiple units and per capita rates of car ownership increased. The congestion was aggravated in many neighborhoods by commuters who parked near access points to public transportation and by users of public facilities like hospitals. Several neighborhood associations and the City Planning Department favored the introduction of neighborhood sticker parking (like the NSP discussed in the previous section). One of the major issues was determining which neighborhoods would participate in the plan. The City Planning Department originally thought in terms of a top-down approach: planners in the department would designate the neighborhoods that would participate as part of the policy proposal. An analyst convinced them to use a bottom-up approach instead: establish a procedure for neighborhoods to self-select.[22]

The analyst reached his recommendation through a backward mapping of the political problem faced by the City Planning Department. Many residents wanted to participate in the sticker plan and expressed their support for it through their neighborhood associations. Some residents, however, preferred the status quo to having to pay for the privilege of using off-street parking in their neighborhoods. Until they realized that their neighborhoods were to be included in the plan, these latent opponents would probably have remained silent and unknown to the City Planning Department. Therefore, the City Planning Department was faced with the prospect of some local opposition when it made public the neighborhoods designated for participation. Viewing dissatisfaction with the status quo expressed by the currently vocal residents as the behavior that would be the policy target, the analyst thought of a procedure that would allow them to seek participation. That way the City Planning Department would not be seen as imposing its preferences on the neighborhoods. Because there was already a procedure in place that allowed neighborhoods to self-select for designation as two-hour parking zones, the analyst specified an adaptation of that procedure as the way to structure the implementation of the NSP.

Policy Outcomes: Uncertainty and Error Correction

Policy analysis inherently involves prediction. Because the world is complex, we must expect to err. Our theories about human behavior are simply not powerful enough for us to have great confidence in most of our predictions. Changing economic, social, and political conditions can make even our initially accurate predictions about the consequences of adopted policies go far astray as time elapses. Our design of policies should acknowledge the possibility of error. In particular, can we design policies that facilitate the detection and correction of poor policy outcomes?

Redundancy and Slack

Duplication and overlap generally carry the negative connotation of inefficiency. In a world with perfect certainty, this connotation has validity: Why expend more than the minimum amount of resources necessary to complete a task? In our uncertain world, however, some duplication and overlap can be very valuable. We do not consider redundant safety systems on airplanes as necessarily inefficient—even the best engineering and assembly cannot guarantee perfect performance from all primary systems. We should not be surprised, then, that redundancy can be valuable in many organizational contexts. However, redundancy is most valuable, and most justifiable, for new organizations carrying out new policies. In novel implementation situations, redundancy provides a safety margin, serves as a resource for dealing with unforeseen situations, and makes available slack resources for organizational experimentation and learning.[23] In sum, *designing various forms of redundancy into new policies makes them more feasible and robust.*

Consider the problem of critical links in the implementation process. If one link in a chain of behaviors connecting the policy to desired outcomes seems especially tenuous, then you should look for ways to make the link redundant, recognizing the trade-off between costs and the probability of successful implementation. For instance, return to our earlier example of the program to ship nuclear wastes on railroads running through the state. Rather than simply ask mayors of cities along the most-preferred route to prepare evacuation plans, you might simultaneously have evacuation plans prepared along a few alternative routes as well. If a mayor along the preferred route failed to comply, then you might be able to switch to one of the alternative routes without great loss of time. Whether the higher administrative (and probably political) costs of such a parallel approach were warranted would depend on the value of the reduction in expected delay it would provide.

Redundancy can also be useful in facilitating experimentation and the reduction of uncertainty. For example, imagine you are designing a county-funded program to purchase counseling services over the next year for 100 indigent drug abusers. Conceivably, a single firm could provide all the services. While the administrative costs of contracting with one firm for the entire 100 participants might be lower, hiring more than one firm would offer the potential for comparing performance. It also would enhance competition by increasing the chances that more than one firm will bid to provide services in future years.

Using redundant suppliers also makes it easier to cope with the unexpected loss of one supplier. For instance, one of the counseling firms may unexpectedly go bankrupt early in the year and default on its contract. It would probably be faster, and perhaps more desirable, to shift participants to other firms already under contract than to hire new firms.

Finally, building redundant resources into programs may be desirable if supplementing initial allocations during implementation will be difficult. For instance, if the credibility of the state transportation department's proposal to replace a series of bridges hinges critically on completing the first replacement in the promised six months, then the department would probably be wise to pad the budget for this first project with some reserve funds to deal with such unlucky circumstances as exceptionally bad weather or construction accidents. The political trade-off is between the lower initial support because of the higher budget and a reduced risk of lower future support due to loss of credibility.

Designing in Evaluation

Often the effectiveness of adopted policies is not directly observable. For example, consider a program that provides intensive supervision for certain parolees. How would we determine whether the participants had lower recidivism rates? We could fairly easily observe the number of arrests and the number of months of street-time (time in community after release from prison) for the group. We could interpret the ratio of total arrests to total street-time as a crude measure of the recidivism rate for the participants. But how would we interpret this rate? We might compare it to the recidivism rate for a sample of nonparticipants. The comparison could be misleading, however, if participants differed substantially from the controls in terms of age, criminal history, education level, or some other factor potentially relevant to recidivism. We would have to worry that one of these factors may be confounding our results.

To guard against confounding, we might have designed the intensive supervision program as an experiment, randomly selecting participants and controls to increase the chances that the only systematic difference between the groups is the intensive supervision. We could then more confidently attribute any measured differences in recidivism rates between the participant and control groups as due to the intensive supervision. Of course, the random selection of participants and controls must be part of the original program design.

When randomized experiments are not possible, evaluators may turn to nonexperimental designs, such as *before/after comparisons*, which take measures on variables of interest before and after the policy is implemented, or *nonequivalent group comparisons*, which take measurements of the variable of interest for a group subjected to the intervention and a group not subjected to the intervention.[24] Even nonexperimental evaluations of policy effects generally require some forethought, however. For example, if we want to know whether improvements at a public park have increased use, then we had better plan on measuring use levels *before* the improvements are made so that we have a benchmark for comparing measurements made *after* the improvements. If we are worried that changes in the local economy over the period of the improvements may influence use, then we would be wise to take contemporaneous measurements at another park not being improved to help us spot any area-wide changes.

Building the prerequisites for evaluation into policy designs is not without cost. Such preparations as random selection of participants and collection of baseline data can involve substantial effort, perhaps consuming scarce managerial resources needed for implementation. Preparations for future evaluations can also delay implementation. So, before bearing these costs (or inflicting them on administrators), it is important to think carefully about the feasibility of actually carrying out the evaluation and the value of the resulting information.

Several questions are relevant to determining whether planning for evaluation is likely to be worthwhile. First, can you imagine the results influencing a future decision? If the costs of the park improvements are sunk (there are no ongoing costs) and if no other park improvements are likely to be considered in the near future, then an evaluation is unlikely to be worthwhile. Second, will personnel with appropriate skills be available to conduct the evaluation? The parks department may be unable or unwilling to provide staff to collect and analyze data. Third, could the evaluation be completed soon enough to influence any decisions? Perhaps the county legislature will consider making similar improvements in another park next year; if an evaluation of the current improvements cannot be ready in time, then it will be irrelevant. Fourth, will the results of the evaluation be credible? If the county legislature does not trust the parks department to provide reliable information, then the evaluation is not likely to have much effect.

Short of evaluation requirements, it is sometimes desirable to build reporting requirements into policy designs. Program administrators are commonly required to provide accounting data on expenditures and revenues. If activity levels (types of clients served, for instance) are routinely reported as well, then executive and legislative oversight may be better able to spot anomalous performance and perhaps take corrective action.

Reporting requirements may also have educational value—they let the administrator know of factors that are likely to be given attention by overseers.[25] For example, if you are concerned that the state employment office avoids trying to help parolees from prison, you might require that, along with total clients, it report the number of clients with serious criminal records. If the administrator of the employment office believes that the budget committee or the governor's office is paying attention to the reports, then it is likely that more clients with criminal records will be served. Reporting requirements may also induce dysfunctional behavioral responses, say, by inappropriately shifting effort from highly valued but uncovered activities to less valued but covered ones. Consequently, reporting requirements themselves should be the subject of analysis and not imposed willy-nilly.

Facilitating Termination

Once policies are adopted, it is often difficult, but not impossible, to repeal them.[26] Policies with large net social costs often have vocal constituencies who benefit. Policies embodied in programs typically enjoy the support of their employees, clientele, and political sponsors. The public organizations that house programs tend to enjoy great longevity. Even if we are fairly certain that a proposed policy is desirable, and will remain desirable for a long time, the inherent persistence of policies should give us some pause. When the consequences of a policy are very uncertain, we should design it with the possibility of termination in mind.[27]

The most common way for policy designs to anticipate the possibility of termination is through *sunset provisions*, which set expiration dates for policies. Unless policies are legislatively renewed prior to their expiration dates, they automatically terminate. In this way, sunset provisions force explicit reconsideration of policies at specified times. We expect sunset provisions to be most effective when coupled with evaluation requirements that provide evidence on effectiveness during the consideration of renewal.[28]

Support for continuation of policies should be expected from employees and other people whose livelihoods or careers depend on them. Furthermore, others may oppose termination because they feel a moral repugnance against disrupting arrangements upon which such people have come to rely.[29] By designing organizational arrangements that keep the cost of termination to employees low, we reduce the likelihood of strong opposition. One approach to keeping costs of termination low is to avoid designing permanent organizations to implement and administer policies. Instead, consider using ad hoc groups like task forces that temporarily assign employees from elsewhere in the organization to the implementation of the policy.[30] If these employees see termination as simply a return to their former duties without loss of career status, then they may be less likely to oppose it than if it involves the loss of jobs or status. Note that contracting out for services, rather than producing them in-house, also reduces the direct cost of termination to public organizations.

The use of temporary organizational structures is not without its disadvantages, however. Ad hoc groups of employees may not be willing to invest the extra effort needed for successful implementation if they do not associate their personal interests with the programs. Also, at some point ad hoc groups in effect become permanent as people become accustomed to new positions and lose interest in the old. Therefore, it probably makes sense to use ad hoc groups for implementing new programs only if any attempts at termination are likely to come in months rather than years.

What can we say about designing termination policies? A general strategy is to try to buy off the beneficiaries of the policy you hope to terminate. As noted in Chapter 10, grandfather provisions allowing current beneficiaries to continue receiving benefits is one approach. Another approach is to provide direct compensation to those who will bear costs as a consequence of the termination. For example, say we wanted to remove an agricultural price support that was becoming increasingly expensive and socially costly. We might be able to lessen opposition by proposing to pay current recipients at current levels for some period following removal of the price support.

A word of warning: designing in such features as evaluation and termination usually makes for better policies, but it does not necessarily make for more politically feasible policies. Partisan proponents of policies usually dislike policies that require explicit evaluation or that contain sunset provisions.

Responding to Heterogeneity

Analysts often face the problem of designing policies for heterogeneous circumstances. Uniform policies that usually accomplish their goals may nevertheless fail with respect to specific localities, communities, categories of people, or firms. Recognizing this heterogeneity by designing in flexibility and decentralization often reduces implementation failures. Central governments typically face the problem of heterogeneity in connection with policies that influence the provision of local public goods or regulate local externalities. Sometimes such heterogeneity can be accommodated by decentralizing certain aspects of the policy. Other times, heterogeneity can be exploited in the design of implementation strategies.

Bottom-up Processes

One way to design in flexibility is to allow self-selection. Rather than mandate participation in centrally established policies, it is sometimes advantageous to allow for self-selection. The implementation of sticker parking in San Francisco illustrates the use of a selection process as part of a policy design: neighborhoods had to apply for designation as sticker parking areas. This bottom-up process avoided the opposition that might have arisen if the areas had been designated by the City Planning Department.

More generally, processes that permit the participation of interest groups in aspects of implementation may be valuable in forestalling political attacks on the entire policy. The basic idea is to provide a mechanism that permits the policy to be altered somewhat to accommodate local interests. Requiring public hearings, establishing local advisory boards, and providing resources for discretionary uses may offer opportunities for making policies more attractive to local interests. From the perspective of the policy designer, the trick is to structure such mechanisms so that they promote acceptable program adaptations and co-opt potential political opponents without creating bottlenecks or veto points that can threaten implementation.

Where local governments and organizations lack the capability to deal with local policy problems, higher levels of government may find it worthwhile to adopt long-run strategies of *capacity building*—that is, facilitate improvements in the managerial and analytical capabilities of local organizations so that they will be better able to initiate bottom-up policies. For example, if a state department of transportation were concerned that cities were failing to consider improved traffic control technologies, then it might adopt policies to make local personnel more aware of available technologies (through state-sponsored training, information networks, personnel exchanges, and direct information) and more capable of implementing them (through funding and technical assistance programs). Such programs are intended to increase the capability of local organizations to choose improvements that best fit local conditions.

Phased Implementation

Limited resources, including the time and attention of implementers, often make phased implementation desirable or necessary. How should the sites for the first phase be selected? One strategy is to select a representative set of sites so that the full range of implementation problems will likely be encountered. Implementing the policy in a representative sample in the first phase makes sense when the implementers are confident of their ability to deal with the problems that do arise. Assuming that the program survives the first phase, knowing the full range of problems that are likely to be encountered will permit reevaluation of its desirability and redesign of its implementation plan.

An alternative strategy for selecting sites for the first phase of implementation is to stack the deck in favor of the implementers by picking sites where the conditions for success seem most favorable. By starting with easier sites, the implementer is better able to avoid failure and build an effective staff for dealing with the later stages of implementation. Stacking the deck in the first phase makes sense when an early failure will make the program vulnerable to political attack. The trade-off is between more realistic information about future problems and better prospects for immediate success. Make a conscious choice.

Understanding the Implications of Repeated Interaction

Our discussions of bureaucratic supply as a source of government failure (Chapter 8) and as a generic policy alternative (Chapter 10) emphasized formal incentives. However, formal structures do not always fully explain organizational performance. Organizations with very similar formal structures can vary greatly in the way they perform—students of management and public administration have long recognized the importance of norms and leadership to organizational performance.[31] The absence of a theoretical framework for understanding these concepts has made it difficult to generalize about them. In recent years, however, the *rational choice theory* of institutions,[32] which models social interactions as repeated games, has provided insight into social norms, leadership, and corporate culture. The rational choice theory of institutions offers some useful general insights into policy design.[33] Two important insights relate to the role of social norms in conditioning behavior, and the role of leadership and corporate culture in shaping organizational norms. Both are potentially important at any time in an organization's life cycle, but they are particularly important during change. Implementation of a new policy is one important form of change.

Social and Organizational Norms

People often exhibit behavior that appears to be consistent with certain *social norms* that involve people cooperating in ways that are immediately costly to them. For example, police officers often share a norm that requires them to rush to the aid of any fellow officer who appears to be in danger, even though doing so involves risks for those responding. Thus, if one adopts a myopic view of only the isolated situation, social norms appear to be irrational. From a less myopic perspective that incorporates future consequences, however, social norms can be seen to be individually rational.[34] Modeling social interaction as a repeated game provides a way of exploring the rational basis of social norms.

Consider Figure 12.2, which displays an abstraction of the prisoner's dilemma used to illustrate the problem of open access in Figure 5.6. The payoffs for combinations of the players' strategies, cooperate (C) and defect (D), are shown such that $a > 1$, $b > 0$, and $a - b < 2$, which defines the payoffs in a prisoner's dilemma game. The payoff from receiving unreciprocated cooperation, a, is larger than 1, the payoff from mutual cooperation; the payoff from providing unreciprocated cooperation is $-b$, which is less than the payoff of zero that results if neither player cooperates. In a single play of the game, sometimes referred to as the *stage game*, the only equilibrium is for both players to defect and receive a payoff of zero.

Now imagine that the players anticipate possibly playing the game against each other repeatedly. Specifically, imagine that each player believes that having completed a play of the game, there is a probability of δ of playing yet again. A δ close to 1 implies that the players believe that there is a very high likelihood of the game continuing for a long time; a low δ implies that they believe the game is likely to end soon.[35] As the players discover themselves in play, they anticipate the probability of playing two times is δ (the current play for certain and a probability of d of a second play), the probability of playing three times is δ^2, the probability of playing four times is δ^3, and so on.

Both players defecting in each round is one equilibrium in the repeated game. Another possible equilibrium involves both players using the following strategy: cooperate on the first play and continue to cooperate as long as the other player cooperated in the previous round; if the other player ever defects, then defect in all subsequent plays. Both players following this strategy will be in equilibrium if neither player can obtain a higher expected payoff from defecting. The expected payoff to each player of following the strategy if the other is playing it is $1 + \delta + \delta^2 + \delta^2 + \ldots = 1/(1 - d)$. In terms of deviating from this strategy, the largest possible gains come from defecting in the first play, so that if the other player were playing the strategy the payoffs would be $a + 0 + 0 + \ldots = a$. In other words, gaining the payoff a in the first play of the game would result in zero in all subsequent plays. Now neither player will wish to defect from the strategy if $a < 1/(1 - \delta)$, or, rearranging, if the probability of playing again, δ, is larger than $(a - 1)/a$. Thus, as long as this inequality holds, then both players following the strategy is an equilibrium that would appear as full cooperation between the players.

Even this simple example has some implications for the emergence of social norms like cooperation. The larger δ, and, therefore, the greater possible gain from mutual cooperation in the future, the larger can be the payoff from unilateral defection, a, without precluding the possibility of a cooperative norm. In other words, the more likely it is that the interaction will continue, the stronger the temptation for defection that can be possibly controlled by the social norm. In the context of organizational design, situations in which people interact more frequently (have a higher probability of playing another round of the game together) are thus more likely to support cooperative norms. Thus, building in the expectation of repeated interaction among organizational members offers the prospect for more cooperation. Sometimes, of course, organizational designers wish to discourage cooperation that may not be socially desirable, as, say, between regulators and those being regulated. In such cases, inspection systems that avoid repeat interactions may be desirable from the perspective of discouraging the norm of corruption.[36]

Stage Game

Player 2

Player 1		Cooperate (C)	Defect (D)
	Cooperate (C)	(1, 1)	($-b$, a)
	Defect (D)	(a, $-b$)	(0, 0)

$a > 1$ and $a - b < 2$

Payoffs: (player 1, player 2)

Repeated Game: Stage game repeated with probability δ

Some possible equilibria in repeated game:

(1) Both players defect every play (only equilibrium in stage game repeated as long as game played)

Observed strategy pairs in equilibrium: (D, D), (D, D), ...

(2) Both players cooperate on first play and continue to cooperate as long as other player cooperated on previous play; if one player ever defects, then the other defects on all subsequent plays. Equilibrium if $\delta > (a-1)/a$

Observed strategy pairs in equilibrium: (C, C), (C, C), ...

Figure 12.2 Equilibria in a Repeated Game

This simple model suggests the importance of recruitment and socialization. Suppose that members of an organization did not know which specific person with whom they were playing the game, but they did know that they were all following strategies that involved cooperation in equilibrium. Now imagine that a person joined the organization and that this new person assumed that everyone played the strategy of defecting, which is also an equilibrium. Individuals playing the game with the new member would encounter defections and thus themselves defect in future rounds. The result would be the movement from an equilibrium in which everyone cooperated on every play to one in which no one ever cooperated. Clearly, if the organization were able to do so, it would benefit from recruiting new members who would be predisposed to assuming that others played strategies consistent with cooperative equilibria. Socialization, through training or less formal interaction, would be valuable for increasing the chances that new members do not disrupt desirable equilibria.

Leadership and Corporate Culture

There are two obvious equilibria in the simple repeated game presented in Figure 12.2: each player defects on every play (the equilibrium in the stage game) and, assuming that the probability of repeated play is high enough relative to the payoff from defection, each player cooperates until encountering a defection and then defects in all future plays. There are, however, other equilibria. For example, one might cooperate until one encounters a defection and from then on play the strategy played by the other player in the previous round (the so-called *tit-for-tat strategy*). Indeed, repeated games generally have an infinite number of possible equilibria.[37] Those playing in a repeated game thus face the problem of coordinating on one of the desirable equilibria from among the many that are possible.

From this perspective, one function of *leadership* is to provide solutions to this type of coordination problem. Leaders may try to create *focal points*, or distinctive outcomes that attract attention, to help members coordinate.[38] For example, leaders may set clear examples by their own actions, by creating symbols, by identifying and rewarding exemplary behavior, or by making examples of those who violate social norms the leader is trying to promote. Creating and communicating a shared mission for the organization may also help the leader create expectations among members that facilitate coordination.

Yet leaders come and go. Members of the organization may be unreceptive to the coordinating efforts of the current leader, if they fear that their cooperative efforts will not be rewarded by reciprocation from others if new organizational leadership results in a less cooperative equilibrium in the future. Here is where *corporate culture*, which is a set of norms associated with the organization itself rather than any particular leader, may come into play.[39] Organizations that have been able to establish reputations for consistently treating members in certain ways can draw on these expectations to maintain social norms among members even when leaders change.

Fully analyzing the impact of policy changes on organizations requires that attention be given to corporate culture. Functional corporate cultures are assets that should not be squandered. Externally imposed changes on organizations that force them to violate, and thereby undermine, valuable corporate cultures may result in reductions in performance that would not be predicted on the basis of changes in resources alone. For example, a unit of a public agency that allows its employees to take time off as reward for intense or off-hours periods of work might be able to maintain a high level of morale and productivity that would be impossible if employees were required to work in strict compliance with the general rules of the agency.

Conclusion

Successful implementation, like adoption, requires analysts to be strategic. Good policy design is based on realistic predictions of the responses of people who must provide essential elements during implementation. Such prediction is inherently difficult because the holders of necessary elements typically work in a variety of organizations with different missions, constraints, incentives, and norms. Consequently, prudent policy design anticipates implementation problems by including policy features to generate information, resources, and fixers to solve them.

For Discussion

1. Imagine that the transportation planning committee adopts the policy of raising the parking fee, as presented in the first discussion question for Chapter 11. Sketch a brief implementation scenario.
2. Your agency is considering how to deliver a new program that has been mandated by the state legislature. There is a specified level of funding, but the legislation does not specify how, or by whom, the program will be delivered. Your agency has narrowed the delivery alternatives down to two: either deliver the program directly through the agency or deliver it through an inter-organization network that includes several other state agencies and two not-for-profit agencies. Provide a review of some of the implementation issues that are likely to arise under each alternative.

Notes

1 The seminal work is Jeffrey L. Pressman and Aaron Wildavsky, *Implementation* (Berkeley: University of California Press, 1973). For a recent review, see Peter Hupe and Harald Sætren, "Comparative Implementation Research: Directions and Dualities," *Journal of Comparative Policy Analysis* 17(2) 2015, 93–102.

2 In program evaluation, this process is called *logic modeling*. See John A. McLaughlin and Gretchen B. Jordan, "Using Logic Models," in Joseph S. Wholey, Harry P. Hatry, and Kathryn E. Newcomer, eds., *Handbook of Practical Program Evaluation* (New York: John Wiley & Sons, 2015), 55–80.

3 Eugene Bardach, *The Implementation Game: What Happens after a Bill Becomes a Law* (Cambridge, MA: MIT Press, 1977), 57–58.

4 The concept of tight linkage is prominent in the literature on technological failures. See Charles Perrow, *Normal Accidents: Living with High Risk Technologies* (Princeton, NJ: Princeton University Press, 2011).

5 Dominic Cooper, *Improving Safety Culture: A Practical Guide* (New York: Wiley, 1998).

6 See Bardach, *The Implementation Game*, 273–78.

7 In the context of policy implementation, we see fixers as primarily positive contributors to the process. In more general contexts, the term is often used pejoratively. See Yaron Zelekha and Simcha Werner, "Fixers as Shadow 'Public Servants': A Case Study of Israel," *International Journal of Public Administration* 34(10) 2011, 617–30.

8 Paul Williams, "The Competent Boundary Spanner," *Public Administration* 80(1) 2002, 103–24.

9 Different implementation literatures use different terminologies for competencies: "capabilities" is one such term. In the context of organizational change, the role of "dynamic capabilities" is stressed. See David J. Teece, "Explicating Dynamic Capabilities: The Nature and Microfoundations of (Sustainable) Enterprise Performance," *Strategic Management Journal* 28(13) 2007, 1319–50.

10 Michael Lipsky, *Street-Level Bureaucracy: Dilemmas of the Individual in Public Services* (New York: Russell Sage Foundation, 1980).

11 See Kenneth J. Meier and Laurence J. O'Toole, Jr., "Managerial Strategies and Behaviors in Networks: A Model with Evidence from U.S. Public Education," *Journal of Public Administration Research and Theory* 11(3) 2001, 271–93.

12 See Peter J. May and Søren C. Winter, "Politicians, Managers, and Street-level Bureaucrats: Influences on Policy Implementation," *Journal of Public Administration Research and Theory* 19(3) 2009, 453–76. On the role of street-level bureaucrats in policy change, see Anat Gofen, "Mind the Gap: Dimensions and Influence of Street-level Divergence," *Journal of Public Administration Research and Theory* 24(2) 2014, 473–93.

13 Ibid., 98–124.

14 On the resources available to frontline employees and lower-level managers, see David Mechanic, “Sources of Power of Lower Participants in Complex Organizations,” *Administrative Science Quarterly* 7(3) 1962, 348–64.

15 Martin A. Levin and Barbara Ferman, *The Political Hand: Policy Implementation and Youth Employment Programs* (New York: Pergamon, 1985), 102–4. On approaches for improving inter-organizational cooperation, see Eugene Bardach, *Getting Agencies to Work Together: The Practice and Theory of Managerial Craftsmanship* (Washington, DC: Brookings Institution Press, 1998).

16 Policies are sometimes designed by legislatures to enable interest groups to play the role of fixers even beyond initial implementation. See Mathew D. McCubbins and Thomas Schwartz, “Congressional Oversight Overlooked: Police Patrols versus Fire Alarms,” *American Journal of Political Science* 28:1 (1984), 165–79.

17 Bardach, *The Implementation Game*, 250–67.

18 Levin and Ferman, *The Political Hand*, at 322.

19 There are many examples of such comparative research; for example, Daniel Eisenberg, “Evaluating the Effectiveness of Policies Related to Drunk Driving,” *Journal of Policy Analysis and Management* 22(2) 2003, 249–74.

20 On policy narratives, see Thomas J. Kaplan, “The Narrative Structure of Policy Analysis,” *Journal of Policy Analysis and Management* 5(4) 1986, 761–78; and Emery Roe, *Narrative Policy Analysis: Theory and Practice* (Durham, NC: Duke University Press, 1994).

21 Richard F. Elmore, “Backward Mapping: Implementation Research and Policy Design,” *Political Science Quarterly* 94(4) 1979–80, 601–16, at 612. For an application, see Marcia K. Meyers, Bonnie Glaser, and Karin McDonald, “On the Front Lines of Welfare Delivery: Are Workers Implementing Policy Reforms?” *Journal of Policy Analysis and Management* 17(1) 1998, 1–22.

22 Arthur D. Fulton and David L. Weimer, “Regaining a Lost Policy Option: Neighborhood Parking Stickers in San Francisco,” *Policy Analysis* 6(3) 1980, 335–48.

23 See Martin Landau, “Redundancy, Rationality, and the Problem of Duplication and Overlap,” *Public Administration Review* 29(4) 1969, 346–58; and Jonathan B. Bendor,

Parallel Systems: Redundancy in Government (Berkeley: University of California Press, 1985).

24 Nonexperimental designs attempt to draw inferences about program effects when the use of randomly selected intervention and control groups is not possible. For a thorough guide to the analysis of data from nonexperimental designs, see Laura Langbein, *Public Program Evaluation: A Statistical Guide* (Armonk, NY: M. E. Sharpe, 2006).

25 Janet A. Weiss and Judith E. Gruber, “Deterring Discrimination with Data,” *Policy Sciences* 17(1) 1984, 49–66.

26 Christopher R. Berry, Barry C. Burden, and William G. Howell, “After Enactment: The Lives and Deaths of Federal Programs,” *American Journal of Political Science* 54(1) 2010, 1–17.

27 See Mark R. Daniels, “Policy and Organizational Termination,” *International Journal of Public Administration* 24(3) 2001, 249–62.

28 On the effectiveness of sunset provisions, see Russell S. Sobel and John A. Dove, “Analyzing the Effectiveness of State Regulatory Review,” *Public Finance Review* 44(4) 2016, 446–77.

29 Eugene Bardach, “Policy Termination as a Political Process,” *Policy Sciences* 7(2) 1976, 123–31.

30 Robert P. Biller, “On Tolerating Policy and Organizational Termination: Some Design Considerations,” *Policy Sciences* 7(2) 1976, 133–49. He offers ideas about how to make policy termination more acceptable to organizational members using: savings banks (allow organizations to keep some of the savings from termination); trust offices (keep a skeleton crew to preserve institutional memory and provide a home for program employees); and receivership referees (encourage organizations to voluntarily eliminate ineffective programs).

31 Chester I. Bernard, *Functions of the Executive* (Cambridge, MA: Harvard University Press, 1938); Herbert Kaufman, *The Forest Ranger* (Baltimore: Johns Hopkins University Press, 1960). For overviews of leadership, mission, and corporate culture in the public sector, see James Q. Wilson, *Bureaucracy: What Government Agencies Do and Why They Do It* (New York: Basic Books, 1989).

32 Andrew Schotter, *The Economic Theory of Social Institutions* (New York: Cambridge University Press, 1981).

33 Randall Calvert, "The Rational Choice Theory of Institutions: Implications for Design," in David L. Weimer, ed., *Institutional Design* (Boston: Kluwer, 1995), 63–94.

34 Michael Taylor, *Community, Anarchy, and Liberty* (New York: Cambridge University Press, 1982); Russell Hardin, *Collective Action* (Baltimore: Johns Hopkins University Press, 1982); Robert Axelrod, *The Evolution of Cooperation* (New York: Basic Books, 1982).

35 Alternatively, we can interpret d as a discount factor for time preference in a game with an infinite number of plays. In terms of our discussion of discounting in Chapter 17, we would interpret d as $1/(1 + d)$, where d is the individual's discount rate.

36 Having small numbers of professional inspectors may necessitate repeated interaction. It may be desirable to increase the number of inspectors by recruiting volunteers. For a discussion of the use of volunteers from charitable organizations to monitor money flows in casinos, see Aidan R. Vining and David L. Weimer, "Saintly Supervision: Monitoring Casino Gambling in British Columbia," *Journal of Policy Analysis and Management* 16(4) 1997, 615–20.

37 This is the implication of the so-called *folk theorem*. For a formal proof, see Drew Fudenberg and Eric Maskin, "The Folk Theorem in Repeated Games with Discounting or with Incomplete Information," *Econometrica* 54(3) 1986, 533–54.

38 On the role of focal points in coordination games, see Thomas C. Schelling, *Strategy of Conflict* (Cambridge, MA: Harvard University Press, 1960).

39 David M. Kreps, "Corporate Culture and Economic Theory," in James E. Alt and Kenneth A. Shepsle, eds., *Perspectives on Positive Political Economy* (New York: Cambridge University Press, 1990), 90–143; Gary J. Miller, *Managerial Dilemmas: The Political Economy of Hierarchy* (New York: Cambridge University Press, 1992).

13
Government Provision

Drawing Organizational Boundaries

Public policies require an organization to deliver the promised product or service, even when it is intangible in nature or when the organization only ensures that another organization provides it. These organizations range from government bureaus funded by lump-sum budgets and staffed by civil servants, to privately owned for-profit firms producing under contract. The range of potential organizational forms emphasizes that government delivery of goods or services does not necessarily imply direct government production, only some form of *government provision.*

From a practical perspective, decisions on the nature of provision require the drawing of organizational boundaries. The analysis of the most efficient form in specific circumstances and their boundaries requires that attention be paid to the rules that govern various sorts of transactions. *Neoinstitutional economics* (NIE) engages directly with this question by focusing on the specific nature of the transaction. NIE retains most core assumptions of economic theory, but explicitly considers the rules and information relevant to different sorts of transactions. NIE concepts are, therefore, particularly useful in considering the appropriate boundaries for public organizations: that is, in deciding "who should do what."[1]

Provision or Production?

A rationale for public provision, or financing, based on the presence of market failure or the achievement of social goals beyond efficiency, does not necessarily imply a rationale for government production or direct service delivery.[2] Government can finance provision of services so that they are delivered by other public organizations, for-profit firms, private not-for-profit organizations, or *hybrid* organizations that combine features of public and private organizations. Furthermore, even the delivery of public services through public organizations does not necessarily imply doing so through bureaus. Public organizations or agencies separated to some degree from traditional hierarchies, and therefore (at least formally) more insulated from political influence, are increasingly common. Somewhat confusingly, these public organizations have been described as being either autonomous or semi-autonomous agencies! In some countries, the creation process is generically known as *corporatization*; so, such government agencies are collectively and generically labeled as *corporatized* entities. In other countries, the move from bureau to agency (or the *de novo* creation process) is better known by the inelegant terms *agencification* or *quangofication*.

A defining feature distinguishing between public and private for-profit organizations is that public managers cannot legally retain any fiscal residual (that is, revenues in excess of expenditures, or profits).[3] The delivery of public services through either for-profit firms or private not-for-profit organizations is typically described as *contracting out* or *outsourcing*. It is now common in the United States, the United Kingdom, Australia, and many other countries in Europe and Asia, indeed globally.[4]

Rather than production by bureaus, contracted provision under bureau supervision, or production by autonomous government-owned entities, governments increasingly now employ more complex hybrid organizational forms. A bewildering variety of types and labels are found. *Mixed enterprises* involve both public and private share ownership, often with special privileges attached to the government shares.[5] *Social enterprises* distribute ownership between for-profit and not-for-profit private-sector organizations.[6] *Government sponsored enterprises* grant special legal privileges to for-profit owners.[7] *Public–private partnerships*—known either as *P3s* or *PPPs*—typically involve a private entity that finances, constructs, and manages a project in return for a promised stream of payments directly from government or indirectly from users of the project's services over its projected life or some other specified period. These hybrids raise both principal–principal and principal–agent problems that we consider from a property rights perspective later in the chapter.

What factors should analysts consider in deciding whether the delivery and implementation should be pushed outside the traditional organizational boundary of government? The answer to this question warrants more in-depth treatment than provided in Chapter 10 in the discussion of bureaucratic supply as a generic policy alternative. As we explain in more detail here, the major purpose of redrawing organizational boundaries of public agencies should be to improve the efficiency with which public services are delivered.[8] As we discuss, the key challenge for policy analysts is to take account of all costs and benefits in making that efficiency assessment.

In order to address the relative efficiency of different organizational forms, we draw from *transaction cost theory*, also known as *transaction cost economics*, a core field of NIE that seeks to explain why some economic activities are organized as markets, while others are organized as hierarchical entities. Ronald Coase pioneered transaction cost theory. Oliver Williamson applied it to organizations, and especially organizational boundaries.[9] Transaction cost theory recognizes that production costs are only part of the total costs of organizing the supply of a good or service. In addition, transaction costs are associated with making, executing, monitoring, and renegotiating contracts among the entities engaged in the production process. *In drawing organizational boundaries for policy alternatives, analysts should take account of transaction costs as well as production costs, because both are components of social cost.* The challenge is to find institutional arrangements, including the arrangement of organizational responsibilities and incentives, that minimize the total costs of provision, including those arising from making, executing, and monitoring contracts.

Before examining issues relevant to determining the total costs of some of these alternative institutional designs, it is useful to clarify the core ideas behind transaction cost theory. First, the theory uses "the transaction," whatever that may be in the particular context, as the unit of analysis. For example, the transaction could be a contemporaneous exchange in a spot market, the exercise of delegated authority in a hierarchy, an exchange between employees, or the execution of a contract specifying a sequence of actions to complete a series of related exchanges occurring over a long period of time. As we discuss later in this chapter, a transaction can only be understood in the context of the initial allocation of property rights. We are primarily concerned with the allocation of these property rights to and within organizations. Second, transaction cost theory recognizes that almost all production relationships involve contracts. Hierarchies, such as bureaus, nonprofits, or firms, are bundles of long-term contracts that define and at least partially lay out the responsibilities of employers and employees to each other. Markets involve contracts among hierarchies and individuals. In either case, the contracts can be formal, as in legal documents, or informal, as in, say, customary practice. Third, it recognizes that contracts are rarely, if ever, *complete*; the world is complex, and unforeseen contingencies can almost always arise to make existing contracts incomplete. Fourth, it recognizes the ubiquity of informational asymmetries. As noted in the discussion of insurance markets in Chapter 6, asymmetries may arise either as hidden information, in which someone has relevant information unavailable to others, or as hidden action, in which someone can take relevant actions that are unobservable by others.

In summary, "transaction cost economics is concerned mainly with the governance of contractual relations."[10] The optimal organization of governance and, concomitantly, of appropriate organizational boundaries, depends on the nature of the goods and services that need to be produced and the characteristics (and especially incentives) of the parties engaged in contracting. We now turn to the set of cost factors that should, and do, influence the choice of institutional arrangements, or the governance, of contracted supply of goods to citizens.

Production Costs, Bargaining Costs, and Opportunism Costs in Contracting

Three types of costs are relevant to the choice between direct production by public agencies and contracting out: production costs, bargaining costs, and opportunism costs.[11] Production costs are the costs associated with directly creating the good or service. Bargaining and opportunism generate the transaction costs of governance. Table 13.1 provides an overview of these costs and how they relate to the contracting decision.

Production Costs

Production costs are the *opportunity costs* of the real resources—land, labor, and capital—actually used to produce something, measured in terms of the value of the things that these resources would have produced in their next best alternative use. Economic theory suggests that competition pushes production costs to their lowest possible levels in *competitive environments*. The empirical evidence suggests that competition promotes such technical efficiency.[12] Furthermore, in a competitive environment free from market failures, profit-oriented firms will generally have lower costs than public or mixed-ownership organizations. Production costs are likely to be lower with competitive contracting out, or with provision by the private partner in a public–private partnership, for three reasons.

Table 13.1 When Are Incremental Costs Likely to Favor Contracting Out?

Production Costs	*Bargaining Costs*	*Opportunism Costs*
• Unrealized economies of scale, so opportunities for gains in allocational efficiency. • Lack of competition for agency product, so opportunities for reductions in X-inefficiency. • Inflexibility in use of inputs in-house, so opportunities for reductions in X-inefficiency.	• Low task complexity, so relative certainty and less costly monitoring. • High contestability, so less bargaining leverage for contractee. • Low asset specificity, so less need to compensate contractee for risk.	• Low task complexity, so relatively little information asymmetry and more effective monitoring. • Low task complexity, so production externalities less likely. • High contestability, so less post-contract leverage for contractee. • Low asset specificity, so less risk of hold-up problem.

First, in-house production may entail production at too low levels to be efficient;[13] that is, the public organization does not use enough of the good itself, or supply enough of the good to clients, to be able to produce it at the minimum efficient scale. Thus, in-house production may involve allocational inefficiency. An independent producer selling to multiple buyers may be able to achieve minimum efficient scale. These production costs should be conceived broadly—the most significant economies of scale might occur for intangible factors such as administrative systems and knowledge and learning. Scale economies alone, however, are not a sufficient argument for private-sector production. Although public organizations could engage in cooperative production that takes advantage of economies of scale, in practice it is often difficult to design government organizations that can span political jurisdictions to achieve economies of scale. In addition, this cooperation also engenders bargaining and opportunism costs. However, large, jurisdiction boundary-spanning not-for-profit organizations may be able to achieve economies of scale.[14] Contracting out or using a P3 may provide a way of benefiting from private-sector economies of scale.

Second, as we discussed in Chapter 8, in-house production may not achieve the minimum costs that are technically feasible because of the absence of competition, resulting in *X*-inefficiency. Lack of competition usually blunts efficiency incentives because it eliminates comparative performance benchmarks for customers and overseers. Furthermore, the marginal costs of services are obscured, and monitoring by representatives and taxpayers difficult, when they are funded through lump-sum budget allocations to multiservice agencies rather than through sales to consumers. Contracting out may create efficiency incentives by competing with in-house production. It may also focus attention on the costs of particular goods and services.

Third, in-house production may be hampered by inflexibility in the choice of factor inputs, which contributes to *X*-inefficiency. Public agencies usually operate under civil service systems that make it difficult to hire and fire workers to obtain employees with the most appropriate skills for producing goods and services, although corporatized entities may be less subject to these constraints. Also, public agencies often do not have direct access to capital markets, making it difficult to obtain or adjust equipment and other physical resources. Thus, even when the managers of public agencies have strong incentives to produce in the most technically efficient way, they may face constraints that prevent them from doing so. Contracting out or employing P3s may facilitate more technically efficient uses of inputs.

There is considerable empirical evidence from a range of government activities that contracting out by government to private suppliers generally lowers production costs. One survey of the contracting-out experience of the 66 largest cities in the United States found that the annual cost savings were between 16 and 20 percent and also estimated that contracting out improved service quality by between 24 and 27 percent.[15] A comprehensive meta-analysis based on "around three dozen studies" with "reasonable research integrity," concluded:

> This analysis has shown that contracting studies have typically reported real cost savings … These effect sizes were found to be highly significant and were interpreted as being equivalent to an average cost saving of 8 to14 percent. Overall, the reliability of this average saving was beyond doubt. Cost savings differed between services and a general rule could not be applied to all.[16]

Two important caveats to the general findings of cost savings deserve note. First, studies that have examined the relative production costs of internal provision versus contracting out have not included bargaining and opportunism costs, which a priori might be expected to be higher with contracting out. Researchers are just beginning to turn their attention to this topic. One study finds that governments engage in more monitoring, and bear more monitoring costs, when the risks of opportunism are greater.[17] Additionally, not all forms of contracting out can be expected to lower production costs. In particular, *cost-plus contracts*, which guarantee the contractee a profit no matter how high the measured costs of production, are unlikely to lead to lower-cost production.[18] Second, cost comparison studies should control for quality differences; a few studies have done so,[19] but most have not.

Bargaining Costs

Bargaining costs include the following components: (1) the costs arising directly in negotiating contract details, including identifying potential contractors, gathering information about their likely performance, and writing mutually acceptable contract provisions; (2) the costs of negotiating changes to the contract in the post-contract stage when unforeseen circumstances arise; (3) the costs of monitoring whether performance is being adhered to by other parties; and (4) the costs of disputes that arise if neither party wishes to utilize pre-agreed resolution mechanisms, especially "contract breaking" mechanisms. Although only the first category of costs is experienced in the pre-contract period, the other categories of costs should be anticipated.

As an example, consider a social service agency attempting to contract out drug rehabilitation services to private organizations. After identifying relevant organizations, the agency would have to decide how many contracts to offer. For each contract, it would have to specify the quantities and characteristics of the services to be provided, the payments to be made for the services, and the circumstances that would allow the agency to terminate the contract. After the contract is put into effect, a contracted organization may discover that it cannot actually recruit enough clients of the types specified in the contract. The agency and organization must then reach an agreement on how to deal with the changed circumstances by, say, allowing substitution of client types or adjusting payments. Throughout, the agency must monitor the performance of the organization to make sure that it is providing the services specified in the contract. If the agency believes that the organization is not providing the specified services, then it must take actions to enforce or break the contract, actions that the organization may resist legally or politically if it believes that either the agency has made an incorrect assessment or unforeseen circumstances have arisen that preclude it from meeting contract terms.

Bargaining costs arise when both parties act with self-interest, but in good faith. The incremental bargaining costs of contracting out are relevant because an advantage of not contracting out is that the distribution of costs within the organization can be resolved by the existing arrangements of hierarchical authority. Nonetheless, bargaining within organizations, for example over wages, bonuses, internal transfer prices, or areas of responsibility, may also be costly;[20] thus, it is incremental bargaining costs of contracting out relative to agency production that are relevant.

Opportunism Costs

Opportunism is behavior by a party to a transaction designed to change the agreed terms of that transaction to be more in its favor. Opportunism costs arise when at least one party acts in *bad faith*. Opportunism is more likely in the context of contracting out than in agency production because the question of who gets the rents (payments in excess of costs) is more relevant in the nonhierarchical relationships between organizations. Additionally, employees within organizations have better, and usually more frequent, opportunities to "pay back" (and, therefore, discourage) opportunistic fellow employees within the same organization. Nonetheless, just as there are bargaining costs between organizations, opportunism can occur to some extent within organizations. Therefore, as with bargaining costs, it is incremental opportunism costs that are relevant in choosing organizational arrangements.

Opportunism is more often considered to occur *after* contracting has taken place, but some behaviors prior to contracting have opportunism-like characteristics.[21] One party may advance contract provisions that appear neutral but actually facilitate behavior inconsistent with the purposes of the contract. For example, the organization offering drug rehabilitation services may write a contract provision that ambiguously defines drug abuse so that it will be reimbursed for services to clients who the sponsoring agency does not intend to be the targets of treatment.

Opportunism can manifest itself in many ways. Returning to the contracting out of drug rehabilitation services, one can imagine ways that the contractee could take advantage of imperfect monitoring. For example, it may inappropriately count group sessions as individual services, treat canceled appointments as services delivered, shave time from sessions, use less-qualified personnel than specified in the contract, or even simply not provide services, perhaps in collusion with clients who are receiving them involuntarily. If it would be very difficult for the agency to replace the contractee on short notice, then the contractee may be tempted to assert falsely that changed circumstances not covered in the contract make it unable to continue delivering services without additional payments or other more favorable contract terms.

Although it is possible to make a sharp analytical distinction between bargaining and opportunism costs, they are difficult to distinguish in practice. It is in the interest of opportunistic parties to claim that their behavior results from an unexpected change in circumstances. Frequently, the other side of the contract cannot assess whether the claim has genuinely arisen from unforeseen circumstances or not. Contracting-out costs are increased by the difficulty of distinguishing between legitimate bargaining and opportunism.

In summary, public policy should seek the governance arrangements that minimize the sum of production, bargaining, and opportunism costs across government, their contractees, and third parties (citizens) for any given level and quality of service. This objective emphasizes that governments and policy analysts should be as concerned with the real (opportunity) costs that governments impose on those they contract with and on third parties as with the budgetary costs that government bears. However, at the organizational level (for example, an individual hospital), it is unrealistic to expect even nonprofit organizations to act with this degree of public spiritedness (or altruism) without appropriate incentives. Consequently, higher levels of government may have to impose appropriate contractual conditions (that is, design meta-contracts). For example, requirements for cost–benefit analysis to support decisions force attention to be given to social costs, including the costs borne by third parties.

Predicting Bargaining and Opportunism Costs

Three major factors are likely to determine the sum of bargaining and opportunism costs in a specific contracting-out context: task complexity, degree of contestability, and degree of asset specificity. We discuss each of these in turn.

Task Complexity

Task complexity is the degree of difficulty in specifying and monitoring the terms and conditions of a transaction. For example, specifying and measuring the quality of food served by a contractee is relatively easy. Specifying and measuring the quality of complex medical services for which patients have widely varying risk factors is relatively difficult. The degree of task complexity largely determines both the uncertainty surrounding the contract (this affects both contracting parties equally) and the potential for information asymmetry, or hidden information, in which one party to the contract has information that the other party does not have.

Complex tasks involve uncertainty about the nature and costs of the production process itself. Additionally, complex activities are more likely to be affected by various external "shocks," or unforeseen changes in the task environment. Greater uncertainty raises bargaining costs, during both contract negotiations and execution. For example, a study of the aerospace industry found that more complex components were considerably more likely to be produced internally by firms than to be contracted out.[22]

Although information asymmetry does not always raise costs, usually it does, especially if a contract involves services for which quality is only revealed considerably after the transaction occurs. High task complexity raises the probability that there will be information asymmetry, because it implies specialized knowledge or assets whose characteristics are only initially known to contractees or other external experts.[23] Consequently, information asymmetry is likely to create circumstances whereby a transaction party can behave opportunistically. Opportunism arising from information asymmetry can occur either at the contract negotiation stage or after the contract has been signed (the post-contract stage), but is most likely to be significant after contract signing. Either governments or contractees may generate these costs. Higher task complexity also increases the potential for production externalities; that is, the potential for serious disruption to the rest of the public organization if the contracted service is withdrawn or degraded.[24]

Contestability

As discussed in Chapter 5, a *contestable market* is one in which, even if only one organization is immediately available to supply a service, many other suppliers would quickly become available if the offered price exceeds the average cost incurred by potential contract providers. For example, markets for psychological assessment services are highly contestable, as many organizations have the capabilities to supply these services even if they are not currently doing so. The degree of *contestability* may, in some cases, be more important than the number of currently competing organizations providing the service.[25]

When the market for the service in question is highly competitive, with a large number of organizations in the relevant (usually geographic) market producing the service or very close substitutes, the public agency will likely be able to realize lower production costs without substantially higher negotiation and opportunism costs. Public agencies, however, may still be able to reduce production costs by contracting out activities even where there is no direct competition. Such goods may involve substantial scale economies and, as a result, there may be some degree of local, regional, or even national natural monopoly. If providers are able to switch production to the good without sinking large upfront costs, however, then there is contestability.

The degree to which the activity being contracted for is contestable affects opportunism costs. If the market for the activity is contestable, then opportunism is reduced at the contract stage and potentially also at the execution stage. Low contestability raises different issues in the two phases, however. During contract negotiations, a potential supplier in a market with limited contestability is tempted to offer services at a price above marginal cost (or average cost in circumstances in which average cost is declining for the demanded good). This higher price can be thought of as a bargaining cost to the agency, because it must be paid to achieve the contract. If rent is fully realized by the private-sector contractee as profit, then it will represent a true social cost only to the extent that its financing from tax revenues involves a marginal excess burden (inefficiency related to the collection of revenue from the marginal tax source). It is more likely, however, that some of the rent will be hidden by the contractor in nonproductive factor expenditures, such as excessive use of capital equipment, which does increase social costs.

Even after the contract is in force, low contestability increases the risks of opportunism facing the other party and possibly third parties, for two reasons. First, from the government perspective, the contract supplier cannot be quickly replaced (temporal specificity). Second, the government also faces a heightened risk of "contract breach externalities."[26] This risk is especially relevant when the contract involves services related to a network of some kind. For example, a firm carrying out payroll operations might threaten to withdraw service, jeopardizing the payment of all government paychecks. This could effectively shut down a government. Core services that public organizations fear may be subject to breach externalities should normally be retained in-house. However, it is useful to emphasize that public bureaus or agencies do not eliminate such externality problems by carrying out production themselves. Government employees can also opportunistically "hold up" the government (their employer) by withdrawing essential services (passive breach) or by picketing and various forms of sabotage (active breach), as the French government often experiences.

Imperfect competition in the supply of many services attenuates the potential efficiency gains from contracting out. In particular, the presence of small and geographically dispersed populations, combined with the existence of economies of scale, typically limits the potential number of local service suppliers that can compete for government contracts,[27] while high sunk costs act as a barrier to de novo entry of new suppliers. Further, evidence suggests that governments often exacerbate the contestability problem. If potential contractees perceive that public organizations are soliciting unreasonably low bids or are arbitrarily requiring them to rebid at lower-than-originally-contracted prices, a competitive market may not emerge. Such behavior can thwart competition even in potentially competitive markets.

It may be possible for governments to enhance competition by expanding the size of the relevant geographic market through exchange agreements among themselves, whereby consumers could cross geographic jurisdictional boundaries to acquire services, with payments made to their suppliers from the agencies serving their home jurisdictions. For example, some states allow parents to send their children to school districts in which they do not live. A number of practical political obstacles can arise, however. In particular, governments are often loath to fund services that are provided outside their administrative boundaries, especially if doing so threatens the financial viability of provider organizations in their own localities.

Another potential approach to mitigating competition problems is for government contractors to own the (sunk cost) assets associated with service provision and for providers to own relatively fungible assets. As a practical matter, this requires governments to maintain ownership of buildings and relatively specialized and expensive equipment that would be leased to providers who successfully win the right to provide service. This strategy mitigates the need for entrants to make large sunk-cost investments and enhances contestability, though it requires that contracts be designed to ensure appropriate use and maintenance of the assets.

Asset Specificity

An asset is *specific* if it makes a necessary contribution to the production of a good or service and has much lower value in alternative uses. There are various kinds of *asset specificity*, including physical, locational, human,[28] and temporal specificity.[29] Whatever the kind of asset specificity, contracts requiring any party to employ assets (usually physical capital assets, but in some circumstances human capital assets) with much less value in alternative uses, or even no alternative uses (that is, they are "sunk"), raise the potential for opportunism. Either contracting party that commits assets is vulnerable to holdup[30]—no matter what prices are specified in the contract, the non-committing party may be able to act opportunistically. The government may do so by reneging and offering lower prices that only cover incremental costs; the service provider by reneging and demanding higher prices. Asset specificity also reduces contestability at the time of contract renewal by creating a barrier to entry for potential bidders who have not accumulated specific assets.

Although the effects of asset specificity have not been studied systematically in the public-sector context, extensive analogous private-sector outsourcing evidence suggests that high asset specificity reduces contracting out.[31] The public-sector evidence does suggest that governments engage in more monitoring when they contract out services that involve higher asset specificity.[32]

From a social efficiency perspective, government opportunism is as undesirable as opportunism by any other contracting party. Where firms provide and own specific assets, such as highly specialized buildings, they have to worry about the potential for opportunistic behavior by public agencies. Governments can behave opportunistically once other parties have made asset-specific investments.

It is important to emphasize that the costs that potentially arise from asset specificity are not totally outside the control of the parties. This point can be illustrated by thinking of this issue as a multiple period game. Although bargaining and opportunism costs can occur either during contract negotiation (period 1) or after the contract is in place (period 2), government contractors can address both these costs at the contracting stage (that is, in period 1). The government player can anticipate what the optimal strategy in each period of the game will be for the contractee and then, by backward induction, identify its own optimal strategy in each period.

For example, suppose the government contractor is playing a game in which contestability is high in period 1, but expected to be low in period 2 and subsequent periods, once an initial bid winner makes sunk investments. Governments, therefore, should be able to predict that the other contracting party will behave opportunistically or generate bargaining costs in period 2 or some subsequent period. Governments should, therefore, incorporate this expectation into their period 1 strategy. To do so, governments engaging in contracting out must think through the factors influencing opportunism and bargaining costs, as well as strategies to mitigate those costs.

In principle, then, contractual provisions should be implemented such that the expected sum of (social) production, opportunism, and bargaining costs are minimized, even though they will only be triggered by (contingent) events that occur post-contractually. Thus, it is useful to distinguish between *ex ante mechanisms* and *ex post mechanisms*, emphasizing that in the case of the latter it is only the trigger that is ex post.

Can Opportunism Be Controlled by the Use of Not-for-Profits?

The not-for-profit sector includes many hospitals, museums, human service organizations, and, in the United States, many universities. The double market-failure rationale, introduced in Chapter 10, has been used to explain the role of not-for-profit organizations in the economy (although using somewhat different language). Not-for-profits typically provide public goods (the first rationale) in a context in which potential contributors (governments or private individuals) do not have the information to assess whether their contribution is actually used to produce the public good (the second, or principal–agent, rationale). Henry Hansmann argues that, as a result of this principal–agent problem, profit-making firms are able to divert funds to owners because contributors have neither the incentives nor the information to monitor diversion. In addition, because of the difficulty of monitoring performance (that is, information asymmetry), the profit-making firm may provide inferior goods at excessive prices. In other words, donors may "trust" not-for-profits because they are not allowed to make a profit. Perhaps donors also see certain not-for-profits as having a strong sense of mission that leads them to maximize the output of desired goods. Against this must be set the possibility that not-for-profits, given the lack of information, may transfer or dissipate (for example, through higher salaries or less work) the excess of revenue over costs.[33]

One additional potential advantage of contracting with not-for-profits is that their form allows for voluntary contributions—those who value the supplied good can provide it or, where some level is provided by government, make additional contributions to expand its supply. Another is that it allows for great flexibility in the mix of services provided, a feature especially important when the recipients of the services have heterogeneous preferences.[34] Of course, the relative independence of not-for-profits sometimes makes it difficult for the government to mandate the exact set of services that will be supplied. Indeed, it may be difficult for government to stimulate the creation of not-for-profits where they do not already exist.

Governments may also wish to consider the consequences of contracting with not-for-profits on the not-for-profits themselves. Some opponents to providing publicly funded schooling through vouchers redeemable at private schools worry that regulations imposed on participating schools will destroy the characteristics that make them attractive alternatives to public schools. Participation in contracting may also lead not-for-profits to change their organizational strategies, such as reducing their emphasis on fund-raising from nongovernment sources.[35]

More Autonomous Public Supply

In reaction to the perceived problems of bureaucratic supply, there has been a global trend toward the creation of more autonomous public organizations. These types of agencies account for an increasingly large share of public expenditures and public employment in some countries. Governments retain formal ownership and politicians retain hierarchical control, but managers have more discretion than their bureaucratic counterparts. Their primary organizational goal remains something other than profit maximization so that they are distinct from purely private organizations. These organizations typically place more emphasis on operational efficiency, and often on greater revenue generation as well.[36] Managers who are not members of the civil service are generally given considerably more discretion over strategic direction as well as day-to-day management. Public ownership with more autonomous management represents a somewhat awkward balance between government and market.

In the United States, corporatized entities encompass a vast array of public agencies that are labeled independent agencies, boards, commissions, or even committees. In the United Kingdom, such corporate agencies are usually called *nondepartmental public bodies* (NDPBs). In the United States, the Public Company Accounting Oversight Board (PCAOB) created by Title I of the Sarbanes–Oxley Act of 2002 illustrates the kinds of functions carried out by corporatized agencies. The PCAOB's autonomy was established in the enabling legislation, which explicitly states that the PCAOB's employees are not employees of the federal government and that they may be paid compensation equivalent to private-sector self-regulatory bodies. Additionally, highly unusually, in order to foster the agency's financial autonomy, all publicly traded private corporations are subject to a mandatory "accounting support fee" that is payable directly to the PCAOB and provides sufficient revenue to cover all of its expenditures. The PCAOB illustrates that some of these agencies have a high degree of formal independence.

Increased managerial autonomy presents a trade-off between two kinds of principal–agent problems. On the one hand, greater control over hiring, firing, and rewarding employees gives managers more opportunity to employ the sorts of business practices that contribute to private-sector efficiency. Additionally, higher-powered incentives, such as performance bonuses, potentially incentivize managers to efficiently use organizational resources. On the other hand, an increased emphasis on efficiency by managers may conflict with more varied public goals. Further, autonomy typically weakens the authority of government oversight mechanisms normally used to discipline bureaus.

The literature on agency autonomy finds both behavioral and performance effects. Managers and employees generally perceive greater autonomy as contributing to their performance. Limited empirical evidence specifically on performance related to technical efficiency suggests both short- and long-term improvement.[37] Evidence on high-powered extrinsic incentives, such as large salary bonuses, is quite mixed; intrinsic incentives, such as public recognition, do seem to have a positive impact.[38]

Complex Public Provision

Public–private hybrid organizations combine public and private "ownership" of physical assets or financial residuals. This intermingling, or fractionalization, of ownership rights greatly increases their complexity. Ownership across sectors often introduces principal–principal problems because public purposes usually go beyond the profit maximization that motivates private owners. Even when public and private goals are fairly closely aligned, the inherent complexity of the contractual relationships defining such hybrids often invites opportunism. Anticipating opportunism requires systematic consideration of the property rights that define ownership.

The fractionalized property rights (FPR) framework, which label the rows in Table 13.2, sets out six features of property rights to help guide systematic consideration of ownership. First, and most relevant to the distinctive feature of hybrids, is the degree of ownership fragmentation. The more widely dispersed is ownership, the higher are transaction costs of forming a voting majority able to adjust the current strategy and the higher are the costs of effectively monitoring the agents who manage the hybrid. For example, in the relatively simple case of a mixed enterprise, ownership is generic with the government and private investors owning stock in the firm. In the more complex case of P3s, the ownership may be differentiated with the government owning assets but the private partner owning shares of revenue streams that the assets generate. Fragmentation of ownership creates potential principal–principal problems and these, in turn, complicate principal–agent relationships.

The second feature is the clarity of allocation of ownership rights. Lack of clarity in the allocation of ownership rights opens the door for opportunism. Especially in cases of differential ownership across sectors, changing circumstances may obscure the duties that one party owes to another. Mixed enterprises usually have clear ownership rights in terms of stock ownership. However, P3s' ownership rights are likely to be ambiguous in the face of unanticipated events, such as natural disasters that damage assets, changing economic circumstances that change revenue streams, or technological change that reduces costs or makes assets obsolete. The need to resolve ambiguities arising because of the inevitable incompleteness of contracts invites opportunism, especially by the private partner.

The third feature is the transaction costs of alienating assets. Transaction costs interfere with efficiency when they prevent trading that would move assets to more valued uses. Stock ownership in mixed enterprises typically allows parties to alienate their ownership shares. However, transaction costs for alienating organizational assets may be high if one of the owners enjoys veto power over the transactions by virtue of special share rights. Differentiated ownership rights may also increase the costs of alienation. The magnitudes of alienation costs differ widely across property rights systems. In developed countries, the transaction costs associated with selling individual property rights are normally relatively low. Principal–principal bargaining over alienation when ownership rights are fragmented can raise transaction costs to a level that precludes otherwise mutually beneficial transfers.

Security from trespass, the fourth feature, ensures that rights holders can exercise them. An effective property rights system makes de jure rights de facto. Fragmentation and lack of clarity in allocation create opportunities for one party to expropriate assets formally owned by another. Trespass may occur when one party diverts assets to its own use, such as when managers divert assets to private purposes. More commonly, it occurs when a party with responsibility for maintenance of assets fails to provide sufficient resources for the task. Weak compliance norms, imperfect enforcement, and costly self-protection set the stage for trespass. In hybrid organizations, public sector and private sector norms may clash. The opaqueness of organizational property rights may make enforcement within existing legal frameworks, which often assume private owners, difficult. Principal–agent problems may make self-protection by the public partner particularly difficult, especially if the private partner provides the hybrid's management.

The fifth feature, the credibility of the persistence of rights, largely determines the willingness of partners to make investments. A central concern in the study of political economy is whether or not governments can credibly commit to maintaining property

Table 13.2 Six Dimensions of Property Rights: Differences between Public, Private, and Public-Private Hybrid Organizations

	Public Organizations	*Public-Private Hybrids*	*Private Organizations*

	Public Organizations	*Public-Private Hybrids*	*Private Organizations*
Degree of Fragmentation of Ownership	Usually single owner: minimal principal–principal problems, but “networked” public organizations introduce multiple “owners” with distinct strategies and interests	Two owners at a minimum with different goals: serious principal-principal problems	Multiple owners (usually shareholders) with mostly same goal (profit): main principal-principal problem is free-riding in joint stock companies
Clarity of Allocation	Clear because ex ante rules make internal allocation clear; allocation between public organizations may be unclear	Unclear from difficulty of writing complete inter-owner contracts; exacerbated for long-life projects	Clear internal and external allocation, but legal and regulatory change over time can create ambiguity
Cost of Alienation	Very high cost for organizational assets; generally politically high cost for government principals	Generally high cost because of need for principal-principal negotiations among multiple owners	Low cost

	Public Organizations	*Public-Private Hybrids*	*Private Organizations*
Security from Trespass	High security in good institutional environments Low security in weak, fragile, or corrupt institutional environments	High cost of monitoring by public principal(s) creates opportunities for diversion to private owners	High, if strong rule of law and enforcement institutions Low, if weak rule of law or weak institutionals
Credibility of Persistence	Depends on political and institutional environments	Depends on regime stability, commitment to rule of law, and political commitment to arrangement	Depends on regime stability, commitment to rule of law, and political commitment to current regulatory policy
Autonomy	Little managerial discretion over internal allocation of rights because of strong ex ante controls	High managerial discretion over internal allocation of rights because of goal conflict	Some managerial discretion over internal allocation of rights; greater with more share dispersion

Source: Adapted from Aidan R. Vining and David L. Weimer, "The Challenges of Fractionalized Property Rights in Public-Private Hybrid Organizations: The Good, the Bad, and the Ugly," *Regulation & Governance* 10(2) 2016, 161–78, Table 1.

rights over time.[39] In the P3 case, in addition to the usual risk that the government will act opportunistically and arbitrarily change the terms of the contract at some time in the future, there is also the risk that the private partner will abrogate its responsibilities by declaring bankruptcy. The lack of credibility arising from the private partner is especially great if the liability of the private partner does not extend beyond the assets of the hybrid.

The sixth feature is the extent of autonomy given to the managers. Greater managerial autonomy potentially increases efficiency by allowing managers to deploy assets to their most valuable uses. It also raises principal–agent problems. Unlike in single-sector hierarchies, however, the monitoring and incentivizing of agents is complicated by principal–principal problems. In P3s, for example, the private owner usually provides the management services so that, from the perspective of the public partner, there is a principal–principal–agent problem.

The entries in Table 13.2 compare public bureaus, private firms, and public–private hybrids in terms of the six property rights dimensions. Whether the public–private hybrids are the "best of both worlds," as claimed by their advocates, or the "worst of both worlds," as suggested by a number of prominent cases of failure, depends on the specific features of its design.

Public–Private Partnership Case

In order to illustrate the centrality of property rights and transaction costs to the question of alternative institutional designs for government provision of goods, we consider public–private partnerships (P3s) in more detail.[40] The evidence suggests that P3s involve heightened transaction costs because of fragmented ownership. The public- and private-sector partners usually have different and conflicting goals. Ideally, governments seek to minimize all social costs that arise, whether they are borne by government, the private sector, or third parties. However, we have to consider the possibility of government failure and recognize that representative governments seek reelection. This can lead them to be primarily concerned with minimizing their current, on-budget expenditures. Private-sector partners, whether equity providers or debt holders, desire to maximize profits. Clearly, the public-sector goal of social cost minimization can conflict with private-sector goal of profit maximization. Yet, vote seeking can also, in many circumstances, conflict with profit maximization.

The fragmentation of property rights raises transaction costs and invites opportunism. Many P3s are large, complex, and unique projects, if not in terms of technology then at least in terms of topography. They also often involve capital-intensive infrastructure projects with considerable asset specificity. This can encourage opportunism by either party to the P3 contract.

We briefly illustrate how property rights fractionalization and the resulting transaction costs manifest themselves in P3s by drawing on the evidence from a recent major infrastructure project in the United Kingdom known as Metronet Rail. Metronet was initiated in 2003 as a P3 project to refurbish and upgrade the London Underground (subway) track and stations; it went into receivership in 2007.[41] Similar kinds of transaction costs have also been observed in P3s in the United States, such as the Dulles Greenway (formerly the Dulles Toll Road extension); State Route 91 in Orange County, California; and the Tampa Bay Desalination Project. Both the Dulles Toll Road and the Tampa Bay Desalination Project went through bankruptcy, and the government partner eventually purchased State Route 91 from the private-sector partner.[42]

London Underground's rail system (the "Tube") is the oldest in the world, dating back to 1863. It is now made up of 12 interconnected lines and over 250 miles of track. It is one of the largest systems in the world and carries more than a billion passengers per year. Various levels of government have been involved in its management. In 2003, formal management of the Tube was transferred to London's metropolitan government, but the national government remained heavily involved.

Metronet consisted of two separate P3s that were jointly responsible for upgrading nine of the lines: Metronet Rail BCV, responsible for the upgrade of the Bakerloo, Central, Victoria, and Waterloo & City lines, and Metronet Rail SSL, responsible for the Circle, District, Metropolitan, East London, and Hammersmith & City lines. In essence, the contracts required Metronet to spend approximately £17 billion on the upgrade work in return for an annual payment of £600 million pounds for 30 years. Under the contract, any risk of expenditures beyond the annual payment was transferred to the private-sector equity partners.

As in almost all major P3 infrastructure projects, the contractees borrowed most of the required capital from third parties, in this case (as in many others) major European banks, including Deutsche Bank and the European Investment Bank. By 2007, Metronet Rail BCV had incurred almost a billion pounds in costs above those that it had projected for the time period. Metronet claimed that this extra cost was caused by added requirements from London Underground. However, an arbiter ruled under the "extraordinary review" process that Metronet was only entitled to a relatively small reimbursement on this basis. After the review, Metronet BCV went into receivership and Metronet Rail SSL soon followed.

Transaction costs of almost all kinds appear to have been particularly high in this P3. First, bargaining costs were high because the parties disagreed in practice on the fundamental nature of the contract: The government (finally) acted as if it had purchased an output-based, fixed-price contract. The private sector acted as though it had agreed to a series of cost-plus contracts. Not surprisingly, this was a source of significant bargaining costs during the relatively short period that the contract was actually operational. As with almost all cases of complex contracting, it is difficult, if not impossible, to distinguish between genuine disagreement and opportunism, because even if the parties had not really genuinely disagreed, they had incentives to act as though they did.

At first blush, fundamental disagreement on the nature of the contract would seem implausible in an enterprise of this magnitude, but it is not impossible. The U.K. government was anxious to have most of the upgrades in place for the summer Olympics scheduled for London in 2012. As a result, the project started before the contract was finalized. Until the government triggered the extraordinary review in 2007, it partially acted as if it agreed with the private partners' interpretation of the contract. Additionally, the U.K. government, and especially the then the Chancellor of the Exchequer, Gordon Brown (subsequently the prime minister), was determined to use a P3 for the project and to keep the expenditures off the budget. In the Metronet P3, the political commitment of the Chancellor of the Exchequer was widely known. As the *London Times* reported:

> Gordon Brown is the driver. The Chancellor was determined that Ken Livingstone [London's directly elected mayor] would not have his way and finance the much needed revamp of the Underground through a bond issue. As far as the Chancellor was concerned, it was PPP or nothing and eventually he fought London's mayor through the courts to get his way. Mr. Brown's devotion to the PPP was not wasted on those who might put it into practice. They saw a desperate customer coming and, as the National Audit Office relates, pitched their charges accordingly.[43]

Second, Metronet illustrates the difficulty of actually achieving risk transfer, even though it was initially a major rationale for selecting a P3. In 2005, the U.K. House of Commons' Committee of Public Accounts noted that, by the time the contract was actually finalized, there was quite limited risk transfer to the private sector (that is, to the "Infracos" or infrastructure companies):

> There are caps, caveats and exclusions to project risks borne by the Infracos. The risk of cost overruns in repairing assets of unknown condition, such as tunnel walls, is excluded because knowledge of their residual life and associated costs is incomplete. In the case of assets whose condition has been fully identified against specific engineering standards, the cost overruns that the Infracos have to bear are capped, so long as the Infracos can demonstrate that they are acting economically and efficiently. In the case of Metronet the limit in each 7½ year period is £50 million…. There is no definition of economic and efficient behavior in the contracts; an independent arbiter can make a ruling if asked. Exclusions to the risks borne by the Infracos include passenger demand, lower income with fewer users and capacity constraints in the face of increased use. These are borne by London Underground.[44]

Third, the project had a high debt-to-equity ratio: approximately 88.3 percent debt to 11.7 percent equity. In the event of bankruptcy, the costs would largely have been passed on to debt holders—major banks including Deutsche Bank and the European Investment Bank—rather than the equity participants. This alone would have presented the private-sector equity participants with the potential for opportunistic behavior. However, because the government had become politically committed to the use of a P3, it eventually guaranteed 95 percent of the banks' £3.8 billion loans in order to secure financing. This further raised the potential for opportunistic behavior by equity participants. It obviously reduced the incentives of the bondholders to inject more capital rather than see the project go bankrupt. Even with loan indemnities, the House of Commons' Transport Committee estimated that the banks charged £450 million more than they would have for debt issued directly by the government. Apart from the political motivation, it is also probable that the U.K. government had to provide loan guarantees for this project because it had acted opportunistically in the near collapse of a previous P3 project, Railtrack! After the insolvency of Metronet, the U.K. government agreed to reimburse the banks £1.7 billion.

Fourth, the five equity participants, all large corporations, split the equity requirement of £350 million among them, approximately £70 million each. However, this only amounted to approximately £250 million after the tax write-offs that arise upon bankruptcy (the eventual outcome). For these corporations, this was certainly not a huge after-tax loss (the relevant number). Furthermore, these firms were also major suppliers to the project. Indeed, Metronet private-sector equity participants billed and received £3 billion in service charges from 2003 to 2007, or approximately 60 percent of all their capital expenditures. One commentator wryly concluded after the bankruptcy that: "It is most likely that overall the shareholders may not have lost any money on the PPP at all (e.g. 20% of £2 billion is £400 mn.)!! It will be just that they—the shareholders—have made less money on the PPP than they had originally hoped!"[45]

Fifth, the monetary bargaining costs associated with negotiating the project were high. The House of Commons' Public Accounts Committee estimated that: “Transaction costs for the deal were £455 million, or 2.8 percent of the net present value of the deal. London Underground’s own costs were £180 million. It also reimbursed bid costs of £275 million. The department said that it had learned a lesson about controlling bid costs.”[46]

The Metronet case makes clear the fundamental challenge posed by fractionalized property rights, most particularly fragmented ownership, in the use of P3s. Without anticipation and careful design, governments risk creating P3s with high risks of failure.[47]

Assessing and Building Public Agency Capacity

Public agencies may be able to make intra-organizational investments to increase the range of services for which contracting is feasible. Adding some personnel with relevant expertise may increase capacity for writing effective contracts for complex services and monitoring the performance of the services. Thus, the first step in upgrading to a more sophisticated computer system through contracting, for instance, may be to hire someone familiar with candidate systems. Building organizational capacity to monitor service quality may reduce opportunism once the contract is in place. More generally, the appropriate first steps for effective contracting out of services may be adding capacity to the central functions of public agencies. An agency is likely to be better able to steer effectively if it has personnel with rowing experience.

Managers may also be able to influence their external environments to promote more effective contracting. In situations in which the initial awarding of a contract may confer an advantage to incumbent service providers, it may be desirable to multisource the task among several different suppliers to preserve competition in renewals. In other words, it might be worthwhile to bear higher contract administration costs and perhaps forgo some economy of scale to reduce future opportunism.[48]

Conclusion

All public agencies contract out the production of some goods—none that we know of make their own pencils. Deciding whether to contract out for more complex inputs to production, such as computer systems or highly specialized labor, or for final goods and services, such as refuse collection or special education, requires analysts to consider governance and associated costs, as well as production costs. These aggregates include the incremental costs of bargaining and opportunism over those that would result from in-house production. Incremental costs are likely to be high when the tasks to be completed by contractees are complex, contestability for the contracts is low, and production requires specific assets.

For Discussion

1. Your city's highway department currently operates sidewalk snowplows. During major snowstorms, the department gives top priority to clearing major streets. Consequently, sidewalks are often not plowed until several days after major storms. What factors should the mayor consider in deciding whether to contract out for sidewalk snowplowing?
2. You are the mayor of a medium-sized city in a developing country. A company is seeking a contract that would give it rights of way for installing cables to provide television and modem services within the city limits. The contract would specify subscription rates

for the first three years of the contract. Should you be concerned about opportunism?

Notes

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Part IV
Doing Policy Analysis

14
Gathering Information for Policy Analysis

Very few policy problems are truly unique. Invariably there is *some* information available somewhere that will assist you in some aspect of your policy analysis. Sometimes it is difficult to find relevant information. Other times, especially when gathering information from searches on the World Wide Web, it is difficult to extract relevant and reliable information from among a great abundance of facts, data, theories, claims, and policy advocacy. Often one must tap data originally collected for some purpose that is very different from your own. In dealing with some policy analysis problems, however, you will have a mandate, time, and resources to conduct field research to gather directly relevant data. Your field research may involve taking a firsthand look at some policy problem, or it may be limited to gaining expert advice about relevant theories and data sources. We provide advice about how to begin gathering information in these various contexts.

In conducting policy analyses, you face the crucial tasks of developing explanations or models of what is going on and what will happen if any of the considered alternatives is implemented. What, then, is the relevance of facts? Theories and models can tell you a great deal about broad trends and the general directions of expected impacts, but they can rarely tell you about magnitudes. For example, at any given moment, it is possible to observe only a single point on a demand schedule—almost always the current price, but sometimes not even the current quantity. It is relatively straightforward to predict that the imposition of a tax will decrease consumption of a good. It is basically an empirical question, however, as to whether the reduction in the quantity demanded will be large (that is, if demand is elastic) or small (if demand is inelastic). Yet, predicting the size of impacts is essential for evaluating alternative policies. In criminal justice, for instance, policymakers may be correct in assuming that increasing penalties for drug possession will decrease consumption of a given drug; but if demand for the drug is considerably more inelastic than they predict or hope, the policy may be ineffective, perhaps leading to more street crime to support more expensive habits.

Facts are relevant, therefore, in estimating the extent and nature of existing market and government failures and in predicting the impacts of policy alternatives. Data can often help us discover facts. For example, the total budgetary costs of a program might be calculated by identifying and summing all the expenditures for program elements. Data can also enable us to make inferences that we are willing to treat as facts. Returning to the drug example, we might statistically infer the magnitude of the elasticity of demand with respect to punishment by analyzing data on drug consumption in jurisdictions with different levels of punishment. Using standard statistical techniques to make our inferences, we can generally say something about their probable accuracy through standard errors or confidence bounds. The facts we assemble, either through direct observation or inference, and then organize by theories constitute our evidence for supporting our assertions about current and future conditions.

Two points warrant note. First, what one classifies as a fact will often depend upon the theory one brings to bear. The elasticity estimate we treat as a fact will depend on the assumptions embedded in the statistical model employed. A different, but perhaps equally plausible, model might have led to a different inference. Second, virtually all the facts we bring to bear will be, to some extent, uncertain. Therefore, we are almost never in a position to prove any assertion with logic alone. Rather, we must balance sometimes inconsistent evidence to reach conclusions about appropriate assertions.

Gathering evidence for policy analysis can be usefully divided into two broad categories: document research and field research. "In policy research, almost all likely sources of information, data and ideas fall into two general classes: documents and people."[1] Field research includes conducting interviews and gathering original data (including survey research). Document research includes reviewing relevant literature dealing with both theory and evidence, and locating existing sources of raw (primary) data.

Document Research

Relevant literature is generally more identifiable and usable when the policy problem is more universal than particular, more national than local, more strategic (major) than tactical, and more inherently important than unimportant. Thus, broadly speaking, the bigger the problem, the more likely it is that there will be an existing literature of potential usefulness. Unfortunately, we note one major caveat to this comforting assertion: often little relevant literature will be available on major, but newly emergent, problems. For example, one would have had difficulty finding literature on the policy aspects of HIV/AIDS until the late 1980s. When confronting such new problems or others with little literature, you will be forced to be bolder in the application of theory and more creative in the search for analogous policy problems.

Literature Review

Five general categories of documents deserve consideration in the search for policy-relevant information: (1) systematic reviews and meta-analyses; (2) scholarly work reported in journal articles, books, and dissertations; (3) organizational sources such as publications and reports of think tanks, interest groups, and consulting firms; (4) government publications and research documents; and (5) the popular press and blogs. Almost all these sources are directly accessible to students through electronic database searches in university or college libraries, and in other institutional libraries, as well as through standard search engines.

Systematic Reviews

When they exist, well-done systematic reviews of research on specific topics provide analysts with summaries of the best available evidence. In addition to overall assessments of research findings, the reviews identify individual studies that may be valuable beyond their contribution to the overall assessment of evidence. For example, a review that synthesizes evidence on the impact of a particular type of policy intervention may identify studies that also consider implementation issues. Meta-analyses statistically combine findings from quantitative studies of particular interventions into an overall effect size, a procedure that increases the precision of estimates by effectively pooling the samples of individual studies into a single large sample.

Several organizations conduct and make available systematic reviews. The Campbell Collaboration Library of Systematic Reviews provides reviews relevant to social policy issues (www.campbellcollaboration.org/lib/). Another valuable source of systematic reviews and meta-analyses of the impacts of social policies is the cost–benefit analyses conducted by the Washington State Institute for Public Policy (www.wsipp.wa.gov/benefitcost). Systematic reviews of medical and health policy interventions are provided by the Cochrane Collaborative (www.cochrane.org/evidence), and the Cost-Effectiveness Registry maintained by Tufts Medical Center catalogues numerous cost-effectiveness studies of medical interventions that often provide information, such as quality-of-life measures, useful in other types of analyses (http://healtheconomics.tuftsmedicalcenter.org/cear4/). The National Research Council (www.nationalacademies.org/publications/index.html), which conducts commissioned studies for government agencies and other clients, often includes systematic reviews relevant to policy issues in its reports.

Academic and other researchers often publish systematic reviews as stand-alone articles or to place their own research into contexts. Increasingly, they publish meta-analyses. Dissertations also can be a source of systematic reviews of research relevant to particular policy issues. Finding these reviews involves the same search techniques used to locate relevant scholarly research more generally.

Scholarly Sources

There are three approaches to accessing relevant journal articles and books: first, by topic, sector, or field (for example, housing, energy, or education); second, by discipline (for example, economics, political science, or sociology); and third, through literature with a direct public policy focus. This overtly public policy literature overlaps with each of the other two approaches. In the section that follows, we consider these sources as separate categories, but it is often initially easier to initiate a search using various social sciences databases.

Databases can be searched using appropriate keywords. For example, if one is conducting a policy analysis relating to the addictiveness of cigarettes, one might try various combinations of "nicotine," "cigarettes," "addiction," "genetics," and "dependence." There are a number of rules, heuristics, and tricks of the trade that are useful for most databases searches; these include knowledge of such things as Boolean logic, controlled vocabulary, and truncation.[2] The most critical tool is an understanding of Boolean logic and how to use Boolean operators. If you are not already familiar with Boolean operators, make sure you take a tutorial of some kind before searching. (Most libraries provide guides of some sort.)

One can also initiate an electronic search by placing relevant keywords in Google, Google Scholar, or other search engines. In general, these engines lead to a much broader, but less reliable, range of sources. Google Scholar (http://scholar.google.com), however, focuses on scholarly journals and related academic publications. As most academics and scholars at research institutions now at least publicize (advertise!) their peer-reviewed (that is, journal) output on the Web (if only by placing articles and working papers on their institution's Website), this can be a useful indirect route to journal sources, as well as as-yet-unpublished academic papers.

For most policy research purposes, it makes sense to look first in policy-oriented journals such as *Journal of Policy Analysis and Management, Journal of Benefit–Cost Analysis, Policy Sciences, Policy Studies Journal, Canadian Public Policy, Journal of Comparative Policy Analysis, Journal of Public Policy, Journal of European Public Policy*, *Public Administration Review, Journal of Public Administration Research and Theory*, and *Economic Analysis & Policy*. Many professional policy analysts find it worthwhile to subscribe to several of these journals. However, an individual subscription is often a convenience rather than a necessity because most university libraries now provide extensive e-access to these journals.

An increasing number of policy journals deal with public policy in particular substantive areas. The *Journal of Health Politics, Policy and Law*, for instance, concentrates on health policy. Similarly, many disciplinary journals have acquired a strong policy emphasis. For example, the *Yale Journal of Regulation*, the *Harvard Journal of Law and Public Policy*, and *Law and Contemporary Problems*, normally found in law libraries, are legal journals that have a particularly strong emphasis on public policy.

An obvious starting point for a literature review is an electronic search in the particular topic area of your problem: housing, energy, criminal justice, health, or transportation, for example. The major strength of topic-oriented journals and periodicals is also their weakness. They tend to be concerned with what is unique about the topic. In doing so, they can sometimes leave the impression that the topic is indeed unique, thereby blinding the policy analyst to the possibilities of examining analogies and similarities in other policy areas. Keep in mind that this text—with its emphasis on market and government failure—asserts that many aspects of policy analysis are common across substantive areas. Another problem is that many topic-oriented journals have a specific disciplinary perspective that may be relatively hidden. It is important for you to understand whether a particular journal is primarily concerned with, say, the efficiency of employment issues rather than some other aspect.

One must carefully distinguish between journals and periodicals. Journals are usually run by editorial boards with a high percentage of academics with scholarly norms. Articles in journals are usually refereed by experts or professional peers. While this does not eliminate, and may even contribute to, disciplinary bias, it does ensure that the vast majority of articles meet basic standards of competence and honesty. Periodicals, on the other hand, may be put out by individual firms, industry associations, or other interest groups, so that one must be cautious in using them as credible sources. Another major advantage of using journals is that they, unlike most periodicals, usually provide extensive references. Thus, articles lead to other articles. A reality of the Web, though, is that it is now much easier for a periodical to masquerade as a journal (because the cost of presenting the "look and feel" of a scholarly journal has gone down dramatically over time). We are currently experiencing the rise of purely electronic journals, some of which are also open access. Some of these journals are highly reputable, and others are much less so: caveat emptor.[3]

Simultaneously, the literature can be approached from a disciplinary perspective. With the growth of specialized professional schools devoted to such topics as education, social welfare, and criminology, the distinction between disciplinary and topical research has, to some extent, blurred. Nevertheless, the distinction is still relevant for major disciplines such as economics, political science, sociology, psychology, and anthropology.

Because of the importance of efficiency considerations in almost any type of policy analysis, economics journals are an obvious starting point. Two of the most useful sources are the *Journal of Economic Literature* (JEL) and the *Journal of Economic Perspectives* (JEP).

The JEL provides three useful services for the policy analyst. First, it provides literature reviews of both specific fields and theoretical issues. These reviews can save an analyst a tremendous amount of work in getting on top of an issue quickly and preparing comprehensive surveys of relevant literature. JEP also provides useful review articles, usually on more applied topics. Second, JEL provides the titles of articles in recent issues of economic journals. Third, it provides a subject or topic index of recent articles in economic journals (fortunately for the analyst, broadly defined).

The American Economic Association, the publisher of the JEL and JEP, produces an electronic index to economics journals, books, dissertations, and selected working papers called ECONLIT (www.aeaweb.org/econlit). Most research libraries provide online access. Business Source Complete (EBSCO) (www.ebscohost.com) and ABI/INFORM (ProQuest) (www.proquest.com) provide good coverage of business and applied economics sources with extensive full-text coverage. Securing the full text of articles identified through searches is also becoming more convenient. Articles from all but the most recent five years of most major economics and policy journals are available through JSTOR (www.jstor.org).

There are numerous specialized sources. In the criminal justice field, for example, the National Criminal Justice Research Service (NCJRS) of the National Institute of Justice provides an index and electronic access through its Website (www.ncjrs.org). The Educational Resource Information Center of the Department of Education provides an index to educational sources called ERIC (www.eric.ed.gov). Although less policy oriented, the National Agricultural Library of the Department of Agriculture provides an index to agricultural sources called AGRICOLA (www.nal.usda.gov). And the U.S. National Library of Medicine provides an index to international journals of medicine called PubMed (www.nlm.nih.gov).

Analogous to journal articles are master's theses and doctoral dissertations. Theses are often extremely useful information sources because they usually delve into a problem in considerable detail and they often contain extensive bibliographies. Information on dissertations can be found through ProQuest (www.proquest.com/products-services/dissertations).

Apart from journals and theses, a large number of books are published annually on a wide range of public policy topics. Books published by academic presses, such as Cambridge University Press and the University of Chicago Press, are subject to the same sort of refereeing used by scholarly journals. Most trade presses, which publish books aimed at specific markets, usually also employ referees, though the rigor of the process varies by press. Take special care in relying on books published by the mass market presses, as sensationalism may trump accuracy. If a book of interest has been in print for several years, then book reviews published in journals may be available to help assess its quality.

Organizational Sources

The third source of literature is interest groups, think tanks, and consulting firms. We consider the written output of these organizations together because they perform overlapping functions. *Interest groups* usually provide unsolicited information on policy issues, but they occasionally do contract research. *Consulting firms* mainly produce narrowly focused analyses, but sometimes they do policy research of more general scope. MDRC (www.mdrc.org), RAND (www.rand.org), and Mathematica (www.mathematica-mpr.com), for example, do much government-funded research that is both highly credible and useful for policy analysis. *Think tanks* tend to emphasize broader policy research, but many of the newer ones are closely tied to specific interest groups.

Even experienced analysts face difficulty in assessing the value of the mass of potentially relevant material produced by interest groups, think tanks, and consulting firms. Almost all of this written material is reasonably accessible on the Web. As a result, the distinction between “published” and "unpublished" is no longer as useful a proxy for credibility, a crucial consideration for the policy analyst.

The Think Tank and Civil Society Program provides information on over 1,800 U.S. think tanks.[4] A search of the Web almost always turns up pages maintained by think tanks and interest groups that provide information about, and sometimes electronic access to, their publications. Most think tanks provide access to working papers and other analysis through their Websites. In the United States, these include the American Enterprise Institute for Public Policy Research (www.aei.org), the Brookings Institution (www.brookings.edu), the Cato Institute (www.cato.org), the Center on Budget and Policy Priorities (www.cbpp.org), Resources for the Future (www.rff.org), and the National Bureau of Economic Research (NBER) (www.nber.org). In Canada, important think tanks include the C. D. Howe Institute (www.cdhowe.org), the Canadian Centre for Policy Alternatives (www.policyalternatives.ca), and the Fraser Institute (www.fraserinstitute.org). In the United Kingdom, there are a large number of policy-related think tanks, including the Institute for Fiscal Studies (www.ifs.org.uk), the Policy Studies Institute (www.psi.org.uk), the Institute of Economic Affairs (www.iea.org.uk), RAND Europe (www.rand.org/randeurope), and the Regulatory Policy Institute (www.rpieurope.org).

Think tanks can be a major source of policy information, research, and analyses. In the United Sates, a small number of think tanks, including the Brookings Institution, the RAND Corporation, and the American Enterprise Institute for Public Policy Research, once dominated policy debate. The number of think tanks, including those at universities, has mushroomed over the last two decades. There has been a proliferation of more specialized and more overtly ideological think tanks.[5] Before this proliferation, analysts could generally presume that research from think tanks met the standards of accuracy and validity found in peer-reviewed scholarly research. Now that presumption is much weaker. Indeed, until one becomes familiar with the research standards employed by a particular think tank, it is probably prudent to treat it as one would an interest group.

Many organizations have sections that essentially function as interest groups. Corporations, labor unions, trade and professional associations, and consultants often provide information to the political process. Often various government agencies operate in the manner of interest groups. For example, foreign embassies often do so in matters of international trade, and commerce offices at the state and local levels do so with respect to economic development.

Many interest groups—corporate, consumer, professional, regional, political, issue-specific, and others—produce various kinds and quantities of policy analysis, or at least policy-relevant information. These organizations range from Common Cause (www.commoncause.org) and the Sierra Club (www.sierraclub.org) to the Conservative Caucus (www.conservativeusa.org) and the National Rifle Association (http://home.nra.org). Professional and trade associations also provide often useful information on their Webpages.

Why should an analyst consult these sources if their objectivity is suspect? A substantive reason is that these analyses almost always propose public policy goals and policy alternatives, whether explicitly or implicitly. Therefore, they provide valuable sources of *potential* goals and alternatives for an analysis, even if you ultimately decide to reject them. Another reason is that these sources may help you prepare for political opposition. If a particular interest group does not agree with the policy recommendations of your analysis, then it is likely to be a major critic. It is usually more effective to deal preemptively with these disagreements in your own analysis rather than to attempt to deal with them later.

Governmental Sources

The fourth major source of information is government publications: national, state, regional, and local. The U.S. government is one of the world's most prolific publishers, producing on the order of 100,000 documents yearly. These documents are mainly published by either the Government Publishing Office (GPO) (www.gpo.gov), which deals with congressional and agency materials, or the National Technical Information Service (NTIS) (www.ntis.gov), which acts as a clearinghouse of government-funded research, including scientific, economic, and behavioral sciences material. In addition, many state and federal agencies publish materials directly and often advertise their availability on Webpages. Studies from a variety of policy areas can be found on the Websites of the Congressional Budget Office (www.cbo.gov), the Council of Economic Advisers (www.whitehouse.gov/administration/eop/cea), and the Government Accountability Office (www.gao.gov). With a little effort, the Webpages of other government agencies can be located with any of the popular search engines. You may find FedWorld (http://fedworld.ntis.gov) helpful.

There are a number of valuable sources for tapping information about and from the U.S. Congress. The annual legislative products of the Congress are published in the *United States Statutes at Large*, and integrated into the *United States Code* by broad categories, or titles, about every six years. An excellent source of timely information about proposed legislation before the current and most recent Congress is www.congress.gov, a service provided by the Library of Congress that allows one to search by keyword and retrieve the full text of bills. Other useful sources include *Congressional Quarterly Weekly Report*, for an overview of ongoing policy debates and politics; *Congressional Quarterly Almanac*, for yearly overviews and roll call votes; and *Digest of Public General Bills* (annual) and *Major Legislation of the Congress*, for information on new legislation.

Rulemaking by federal executive agencies, especially those with explicit regulatory roles, such as the Environmental Protection Agency (EPA) and the Food and Drug Administration (FDA), is likely at some time to be relevant to analysts working on policy issues at all levels of government. The official record of executive agency rule making is provided by the *Federal Register*, which is published each business day. Agencies announce proposed rules along with justifications, histories of related regulations already in force, and an invitation for interested parties to submit comments to the agency's docket on the proposed regulation. After the comment period, the agency publishes the regulation in final form, typically with responses to the comments it received on its proposed rule making. As the agencies must show reasonableness in rule making to avoid court challenges, the material they publish in the *Federal Register* is often detailed and therefore is a valuable source for analysts working on related issues. The *Federal Register* can be accessed online (www.federalregister.gov). Most federal agencies make their proposed rules and supporting information available through a common portal (www.regulations.gov) that facilitates searching.

Analysts dealing with issues at all levels of the U.S. government are also likely to confront federal constitutional issues on occasion. Decisions of the U.S. Supreme Court are published annually in *United States Reports: Cases Adjudged in the Supreme Court* (www.supremecourt.gov/opinions/boundvolumes.aspx). Recent Supreme Court decisions, as well as information about cases decided in other federal and state courts, are made available by the Cornell University Law School (www.law.cornell.edu). For those without training in legal research, it is usually best to search for articles in law school journals that put specific decisions into broader legal context. As articles in these journals customarily cite profusely, they often provide leads to other sources relevant to policy issues. Law journals and related sources can be searched through the *Index to Legal Periodicals*. A broader index to legal material is provided by the commercial service LexisNexis, which also provides full text access to legal journals and case law.

Selected state and local government materials can be researched through the *Index to Current Urban Documents*, the *Municipal Yearbook*, the *County Yearbook*, the *Council of Planning Librarians Bibliographies*, the *Monthly Checklist of State Publications*, and the National Conference of State Legislatures (www.ncsl.org). Almost all state agencies have established Webpages that provide electronic access to reports, contact information for key personnel, and general information on administrative structure. Larger cities and counties now usually provide Webpages as well.

Popular Media

Finally, the popular press can be a valuable source of background information, especially when the analyst is confronting a new issue. Newspaper and magazine articles rarely provide detailed information and analysis, but they often mention and quote experts, stakeholders, organizations, documents, and other sources of potential value. These leads are particularly valuable because they often appear in the popular press long before they do in other published sources. They also may be the only published references to many local issues. For these reasons, it is often useful for an analyst to begin any new investigation with a quick search through the popular press.

Data and Statistical Sources

In many analyses it is useful to present and analyze new data. A major source of both raw and analyzed data is likely to be the articles, books, and documents described above. However, you also may wish to examine primary data sources. Once again, an excellent source is the U.S. government. A very valuable desk reference is the *Statistical Abstract of the United States* (www.census.gov/library/publications/time-series/statistical_abstracts.html), which is published annually by the U.S. Bureau of the Census. FedStats provides convenient access to statistical information from many agencies (http://fedstats.sites.usa.gov).

The U.S. Census provides much useful demographic data. Three publications of the Census Bureau are especially useful for making comparisons across state and local jurisdictions: the *County and City Data Book*, the *Congressional District Data Book*, and the *State and Metropolitan Area Data Book*. The Website maintained by the Census Bureau provides online access to most census data (www.census.gov).

Often universities and think tanks are important sources of data. For example, the Inter-University Consortium for Political and Social Research (ICSPR) at the University of Michigan archives data from surveys and other studies (www.icpsr.umich.edu), and the Urban Institute has established a data base with over 700 variables describing state social service systems through its Assessing the New Federalism Project (www.urban.org).

Finally, many social science journals now require authors to deposit the data they used for their statistical analyses in accessible archives. These data may be useful for addressing questions not directly addressed in the published research.

A Note on Documentation in the Web Age

The sources listed here should make clear that the World Wide Web has become a major source of information for policy analysts. This has dramatically increased the amount of information that analysts can gather quickly; very few electronic tools were available when the first edition of this book was written. This raises important issues about documentation of sources.

There are three main purposes for documenting sources. First, it is right and necessary to give credit to others for the information and ideas that they have made available. This is especially important in academic settings, where people are often judged by their intellectual contributions. Second, documentation tells others how to find the sources that your analysis has used—sometimes to verify your claims, but more often to explore in depth the provided information. Third, documentation provides an indication of the credibility and authority of the cited documents: Why should readers believe the sources?

The Web has complicated all three of these purposes. As it is sometimes difficult to determine the original source of material found on Webpages, it is often difficult to give appropriate acknowledgment to those who produced the information. Clearly, for those following in the analyst's footsteps, it is important to have access to Webpage addresses as part of the citation.[6] Although Webpages come and go, providing the address that worked for analysts is at least a starting point for their search.

The most difficult issue concerns the role of citations in communicating the credibility of used sources. *Remember that anyone can put anything on a Webpage.* Appearance on a Webpage lends no particular credibility to a source. It is important, therefore, that a citation provide as much information as possible about the source. Is it an electronic version of published material? If so, *provide a complete citation to the published source and also provide a Web address for material not readily available in other format than electronic format.* For example, there is no need to cite the Web address of a journal unless it is *only* published electronically. If it is unpublished, is there any information relevant to assessing the author's credibility? For example, is the author writing as an individual or under the auspices of an organization? If as an individual, can one say anything about the author's professional status? If under the auspices of an organization that might not be well known, what type of organization is it?

Field Research

Field research consists of talking to people, gathering raw data, or finding unpublished reports, memoranda, or other organizational documents. These tasks are often related because it is usually difficult to find data without interviewing, and impossible to assess its reliability, validity, and comprehensiveness without talking to those who actually gathered it. Similarly, putting unpublished documents into perspective usually requires contextual information from those involved in their preparation.

How should an analyst decide with whom to talk? A literature review often suggests some key people. The discussion in the previous section, however, necessarily concentrated on the federal branches of government. For state or local problems, one must make sure that relevant directories and organizational charts are consulted. Such directories for the legislative and executive branches of government are usually readily available. Most analysts quickly develop their own directories of local interest groups, professional bodies, regulatory agencies, consulting firms engaged in policy research, quasi-public agencies, and law firms that deal with public issues. They also network with others working in the policy area.[7]

Analysts should try to avoid limiting sources to only people who are currently in relevant organizations. In particular, recently retired employees are often valuable sources. Retired employees offer several advantages: they usually have time for interviews, they have had some time to reflect on their experiences, and they no longer have to worry about agency politics or retribution. Consequently, they may well be more forthright and more analytical. The only problem is that the "shelf life" of their information may be short.

Interviews need not be in person. Use the telephone or e-mail to check whether someone who appears to be an appropriate source (judging by title, department, or organizational role) is likely to be helpful. On projects with short time frames, you may be forced to do both preliminary spadework and interviewing over the telephone.

We cannot provide here a comprehensive guide to interviewing; several complete books deal with this topic alone.[8] Instead, we give some basic advice on the major issues relating to interviewing: What kind of information does interviewing elicit most effectively? How can you judge the efficacy of the information you do get? How do you get interviewees to talk? How should you decide when to interview someone?

In the following list we reproduce, with some modification, the advice offered by other writers:[9]

1. *What information will interviewing elicit most effectively?*

- **(a)** Historical background and context. The narrative of what has happened.
- **(b)** Basic facts, whether directly through interviews or the raw data provided by interviewees.
- **(c)** Political attitudes and resources of major actors. (These may not be written anywhere. Interviews may be the only source for this kind of information.)
- **(d)** Projections about the future; extrapolations of current trends.
- **(e)** Other potential interviewees and written materials.

2. *How can the efficacy of an interview be judged?*

- **(a)** The plausibility, reasonableness, and coherence of answers.
- **(b)** The internal consistency of answers.
- **(c)** The specificity and detail of answers.
- **(d)** Correspondence to known facts.
- **(e)** Firsthand familiarity of the interviewee with the facts described.
- **(f)** The interviewee's motivation, bias, and position.
- **(g)** Reasons that the interviewee might withhold information.
- **(h)** The self-critical nature of the interviewee.

3. *How do you get interviewees to talk?*

- **(a)** The "energy" must come from the analyst, not the interviewee. Be prepared to ask questions.
- **(b)** The analyst should not pretend to be neutral, but should avoid dogma and hostility.
- **(c)** Demonstrate that you have other sources of information that interpret events differently from the current interview.
- **(d)** Demonstrate reasonable tenacity on important questions.
- **(e)** Point out other views on a topic and indicate that this is the interviewee's chance to tell her side of the issue.

4. *How should you decide when to interview someone?*

Approach *early* in the interview sequence:

- **(a)** Those who are likely to be rich sources of information.
- **(b)** Individuals who have power. (They can provide access, either directly or indirectly, to other sources of information.)
- **(c)** Knowledgeable persons who can give you information that will induce others to talk more freely.
- **(d)** Friendly expert interviewees who will contribute to the credibility of the analysis.
- **(e)** Potential opponents, to the extent that they can be assessed.
- **(f)** Retired employees.[10]

Approach relatively *late*:

- **(a)** Those interviewees who are likely to be hostile or defensive. (Use earlier interviews to acquire leverage.)
- **(b)** Interviewees whom you may not be able to speak to again because they are busy, remote, or otherwise difficult to reach. (Especially if you want their reactions to policy alternatives that you cannot fully specify until the later stages of your project.)
- **(c)** Powerful political opponents who could prevent you from gaining access to other interviewees. (Bardach is aware of the contradiction of recommending both early and late interviews for potential opponents.)
- **(d)** Administrators who may not identify critical issues even though they have the requisite knowledge.
- **(e)** "Expert" interviewees, especially academics, who may be more theoretically oriented.[11] (If you interview them too early, you may not know enough to frame questions that take full advantage of their expertise.)

Most of these points speak for themselves, yet some additional considerations deserve mention. The responses to question 1 suggest that interviews are likely to be most useful for facts, history, and projections. As a corollary, they are likely to be less useful for either a theoretical explanation of what is going on (that is, a model) or a clear and reasoned delineation of goals. These must usually come from your own familiarity with theory and your literature review. There are two exceptions to this general observation, however. First, academic interviewees are usually more comfortable with models and theory. Indeed, they are unlikely to be familiar with program budgets, organizational structures, and institutional history. Second, many government agencies have their own analysts and researchers. These government employees are often more comfortable dealing with theoretical issues than are either program administrators or regular staff.

A major point to note with respect to the second question is that the reliability and value of an interview can often be judged only by conducting apparently redundant interviews. To the extent that the information from a particular interview is vital to your analysis, you should attempt to verify it with other interviews when possible.

Undoubtedly, the hardest issue to deal with is getting interviewees to talk (question 3). One general hint is not to demonstrate that you don't know what you're talking about. Why do we use the double negative? We do so because *demonstrating that you are not ignorant is sometimes not the same as demonstrating knowledge*. Put another way: if you know something, flaunt it subtly! If done skillfully, your knowledge can be one way of introducing some reciprocity into the interview, something especially useful if you are likely to need additional interviews. This is just one aspect of the art of interviewing. More broadly, the trick, to paraphrase Bardach, is to be fair-minded, discreet, intelligent, and self-possessed, and to avoid appearing to be partisan, a tale-bearer, a dope, or a dupe.[12]

In situations in which you think you have discovered information that is particularly important or controversial, you may want to write a follow-up e-mail to the interviewee restating the information. Conclude with a request that the interviewee let you know if you have misinterpreted anything.

Interviewees will sometimes respond with useful elaborations as well as corrections. The record of communication can also be useful in avoiding arguments about what was said. Always, common courtesy suggests that you send a brief note of thanks to people who have given you more than a few moments of their time.

Final practical tips on interviewing: Always do your homework—plan a sequence of questions but be flexible in following it. Don't ask questions that you can answer from documents or from interviews with more accessible people. Make sure that you write up interview notes as soon as possible after an interview. When practical, do not schedule too many interviews in a given day. If you do, then you are likely to appear flushed and rushed. It also makes it less practical to write up notes, and interviews start to merge into each other in your memory. Don't schedule lunch meetings if you expect to take extensive notes. Ask about other potential interviewees. And ask about potentially useful documents and data.

Conclusion

In general, the more relevant information you can bring to bear on a policy problem, the greater will be your capability for producing good analysis. With the increasingly rich sources of information available, especially through the Web, it is important to be thoughtful in your information gathering efforts. Robert Behn and James Vaupel have pointed out that most students spend 99 percent of their time gathering and processing information (or conducting the literature review and field research).[13] They suggest instead that students should spend at least half their time thinking. The heuristic they suggest is *model simple*: think *complex*.[14] We agree. Yet, this is easier said than done. A critical point is to make sure that you don't get into a position where you can't see the forest for the trees. Keep in mind that the mass of data that you are collecting is useful only if it can be placed into an analytical framework. Continually ask yourself where the facts you have fit in and what facts you would like to have.

For Discussion

1. Academic researchers in the United States must follow increasingly stringent guidelines for the protection of human research subjects. Although much research involving interviews with elected or appointed officials is often declared exempt, it must nevertheless be reviewed by institutional review boards at universities and research organizations. The work of policy analysts is usually exempt from direct human subject regulation. Nonetheless, analysts have a responsibility to those who they rely on for information. What ethical principles do you think should guide analysts' information gathering from people?
2. The authors have heard policy analysts claim that, "the telephone is one of the most important tools of a policy analyst." Should "the Internet is one of the most important tools of the policy analyst" replace this claim? Cite your reasons.

Notes

1 Eugene Bardach, "Gathering Data for Policy Research," *Urban Analysis* 2(1) 1974, 117–44, at 121.

2 For a useful review of these and other tricks of the searching trade, see Suzanne Bell, "Tools Every Searcher Should Know and Use," *Online* 31(5) 2007, 22–27.

3 For a discussion about open-access issues in economics journals, see John Willinsky, "The Stratified Economics of Open Access," *Economic Analysis & Policy* 39(1) 2009, 53–

70.

4 James G. McGann, *2015 Global Go To Think Tank Index Report*, Think Tanks and Civil Society Program, University of Pennsylvania, 2016, February 9 (gotothinktank.com).

5 See R. Kent Weaver, "The Changing World of Think Tanks," *PS: Political Science & Politics* 22(3) 1989, 563–78; and Thomas Medvetz, *Think Tanks in America* (Chicago: University of Chicago Press, 2012).

6 On Internet citations, see Melvin E. Page, "A Brief Citation Guide for Internet Sources in History and the Humanities (Version 2.0)," *PS: Political Science and Politics* 29(1) 1996, 83–84. See also Xia Li and Nancy Crane, *Electronic Styles: A Handbook for Citing Electronic Information*, 2nd ed. (Medford, NJ: Information Today, 1996).

7 Networks work best when those involved are willing to share, not just information but camaraderie as well. On effective networking, and more generally working well with colleagues, see Michael Mintrom, *People Skills for Policy Analysts* (Washington, DC: Georgetown University Press, 2003), 231–45.

8 See Lewis A. Dexter, *Elite and Specialized Interviewing* (Evanston, IL: Northwestern University Press, 1970); and Jerome T. Murphy, *Getting the Facts: A Fieldwork Guide for Evaluators and Policy Analysts* (Santa Monica, CA: Goodyear, 1980). For an excellent treatment of gathering information through direct observation, see Richard F. Fenno, Jr., *Watching Politicians: Essays on Participant Observation* (Berkeley: IGS Press, University of California, 1990).

9 Questions 1 and 2 are drawn with modifications from Carl V. Patton and David Sawicki, *Policy Analysis and Planning* (Englewood Cliffs, NJ: Prentice Hall, 1986), 67–70; questions 3 and 4, again with modifications, are from Bardach, "Gathering Data for Policy Research."

10 Added by authors to Bardach's original list.

11 Added by authors to Bardach's original list.

12 Bardach, "Gathering Data for Policy Research," at 131.

13 Robert D. Behn and James W. Vaupel, "Teaching Analytical Thinking," *Policy Analysis* 2(4) 1976, 661–92, at 669.

14 Behn and Vaupel, “Teaching Analytical Thinking,” at 669.

15
Landing on Your Feet

Organizing Your Policy Analysis

The previous chapters have been concerned primarily with the conceptual foundations of policy analysis: how to diagnose problems, how to identify possible policy alternatives, how to think about efficiency and other policy goals, and how to assess some of the costs and benefits of either intervening in markets or altering existing public interventions. On these foundations the analyst must build a structure, one that is useful and appropriate for the context. In this chapter we focus on the construction process: how should one plan and execute a policy analysis? In answering this question, we keep central the notion of policy analysis as a process that involves *formulating* and *communicating* useful advice to a client.

Analyzing Yourself

Especially if you are a novice policy analyst, before you begin trying to do analysis, we suggest that you analyze yourself. Your self-analysis should influence the way you go about doing policy analysis. You may base the self-analysis either on your first attempt at a policy analysis (the preferred method) or on your experience in writing academic papers (less preferable).

Students (and most others as well) tend to fall into one of two broad categories of thinkers and writers: linear and nonlinear. Linear thinkers tend to solve problems by moving sequentially through a series of logical steps. Nonlinear thinkers tend to view problems iteratively, moving back and forth over steps as various pieces of the puzzle become apparent and begin to fall into place. We should stress that, for most purposes, neither is better or worse; each has its strengths and weaknesses.

Your particular weakness (or strength) may not have been revealed in other courses that were more structured and dealt with greater substantive certainty. This point is crucial. Your formal schooling has made you familiar with course assignments, especially problem sets in mathematics, statistics, and economics, where right and wrong answers can be specified.

Policy analysis is never so certain. This does *not* mean that there are not good or bad analyses, but that your answer, the recommendation you make to a client, rarely by itself determines the quality of your analysis. Good analysis asks the right questions and creatively, but logically, answers them. Your selected approach should allow you to eliminate, minimize, or at least mitigate your particular weaknesses in thinking and writing.

How can you diagnose your weaknesses? We have found that students who are linear thinkers and writers tend to suffer from "analysis paralysis." Linear thinkers, not surprisingly, like to start at the beginning of an analytical problem and then work step by step through to the end, following what is sometimes called a *rationalist approach.* If they cannot complete these steps sequentially, however, they tend to become paralyzed. In contrast, many others do not like to approach analysis sequentially. They have many ideas that they wish to get down on paper, yet they often have difficulty communicating these ideas in a well-organized presentation.

The first self-analysis rule is that *linear writers should adopt nonlinear **thinking** strategies, while nonlinear thinkers should adopt linear **writing** strategies.* The format of this book should assist linear thinkers in adopting a nonlinear thinking approach because it compartmentalizes the analytical process. For example, you do not need to understand the problem fully to sketch out some generic policy alternatives. The previous chapters of the book, together with the next section, should also assist nonlinear thinkers in organizing analyses so that they can be communicated more clearly. Nonlinear thinkers can, and should, continue to think nonlinearly, but they must *write* linearly and comprehensively. Linear thinkers and writers, on the other hand, will find that they will be more productive and less vulnerable to analysis paralysis if they also adopt nonlinear work strategies. Therefore, the second self-analysis rule is that *analysts should simultaneously utilize linear and nonlinear modes when conducting policy analyses.*

Subsequent sections of this chapter lay out the eight steps in the rationalist, or linear, mode. We label it the rationalist mode, rather than the rational mode, because *only* following it is unlikely to be rational. Nonlinear, creative thinking can be very rational! Later, we return to how nonlinear strategies can be translated into practical techniques for conducting analysis. Before proceeding through these analytical steps, we wish to reiterate the importance of the client orientation in framing the analytical steps we subsequently discuss.

The Client Orientation

In the introductory chapters we emphasized that policy analysis is client driven, and we considered some of the ethical issues related to the relationship between analysts and clients. The concern here is more with the practical consequences of having a client. The first client heuristic may seem obvious, but it is often neglected: *you must address the issue that the client poses.* Academic experiences (especially in advanced courses) often do not prepare one for this reality because one has considerable discretion as to topic and approach. This is reasonable because when a professor is the client, he or she is most interested in your intellectual development. Real clients are only interested in getting their questions answered. An important heuristic flows from this unpleasant fact: *it is almost always better to answer with uncertainty the question that was asked than to answer with certainty an easier or less important question that was not asked.* Another heuristic follows as a corollary: *good analysis does not suppress uncertainty*, whether with respect to evidence, predictions, or theories.

We all like neat solutions, and most of us have been rewarded for unambiguous answers. In policy analysis, however, it is more effective to highlight ambiguities than to suppress them. Remember that if clients do not hear of these ambiguities from you, they will normally hear of them from analytical, or political, opponents—a much more unpleasant experience. As an analyst, you bear an essential responsibility to keep your clients from being blindsided as a result of your advice.

Highlighting ambiguity should not be seen as an excuse for vague, wishy-washy, or poorly researched analysis. Indeed, you will have to work harder to arrange the competing theories and evidence intelligently. Additionally, highlighting ambiguity does not absolve an analyst from drawing analytical conclusions. For example, if, for a given policy problem, it is unclear whether there is a market failure, then you should succinctly summarize evidence on both sides of the issue and *only then* reach your conclusion. Your client will then be aware of both the relevant arguments and your conclusion. It is particularly important to make your client aware of the weaknesses of the relevant data and evidence. Although the "facts" used in policy debates are often inaccurate or at least unverified, they nonetheless can remain unchallenged. This has been referred to as the "vitality of mythical numbers."[1] As one example, Douglas Besharov has demonstrated the wholesale deceptive use of statistics in analyses of the child abuse problem.[2] As another example, Cary Coglianese documents how for 20 years prominent practitioners and academics readily accepted and repeated the wildly exaggerated and apocryphal claim that 80 percent of rules produced by the Environmental Protection Agency resulted in litigation.[3]

What should you do if you become convinced that your client has asked the wrong question? We offer no definitive advice on how to deal with this difficult situation. One clear rule, however, is that you *must fully explain to your client why you believe that he or she has asked the wrong question.* Clients are often ambiguous about their goals, and they sometimes appear to have goals that you may consider inappropriate. You may be able to help your client ask a better question by identifying ambiguity and by indicating why you believe that certain goals are inappropriate. Almost always, you should try to do so at the early stages of your analytical effort, rather than waiting until you deliver what your client expects to be the answer to the original question.

Clients often ask questions that are not wrong, but rather are poorly formulated. Many times you will be presented "symptoms" that your client finds troubling. ("My constituents are complaining about the rising cost of day care.") Other times you may be presented with a policy alternative rather than a policy problem. ("Should the state subsidize liability insurance for day-care centers?") Your task as an analyst is to reformulate expressions of symptoms and statements of policy alternatives into coherent analytical frameworks. ("Does the day-care industry under current regulations provide an efficient and equitable supply of services? If not, why not?") The subsequent discussion of problem analysis provides guidance for doing so.

Steps in Rationalist Policy Analysis

The term *analysis* comes from the Greek word meaning "to break down into component parts." Teachers of policy analysis usually specify the components of the analytical process as a sequence of steps along the lines of the following: define the problem, establish goals and valuation criteria, identify alternative policies, display alternatives and select among them, and monitor and evaluate the policy outcomes. Such lists usually begin with "defining the problem" so that all the following steps can be described as "solving the problem." This formulation incorrectly suggests to the novice analyst that defining, or explaining, the problem is a relatively short and simple part of the analytical process. In practice, analysts usually encounter the greatest difficulty and often expend the most time trying to define, explain, and model the problem in a useful way.[4] This task is important because the framing of the problem drives the rest of the analysis, including which goals and methods should be used to judge the desirability of alternatives and the recommendation of a policy alternative.

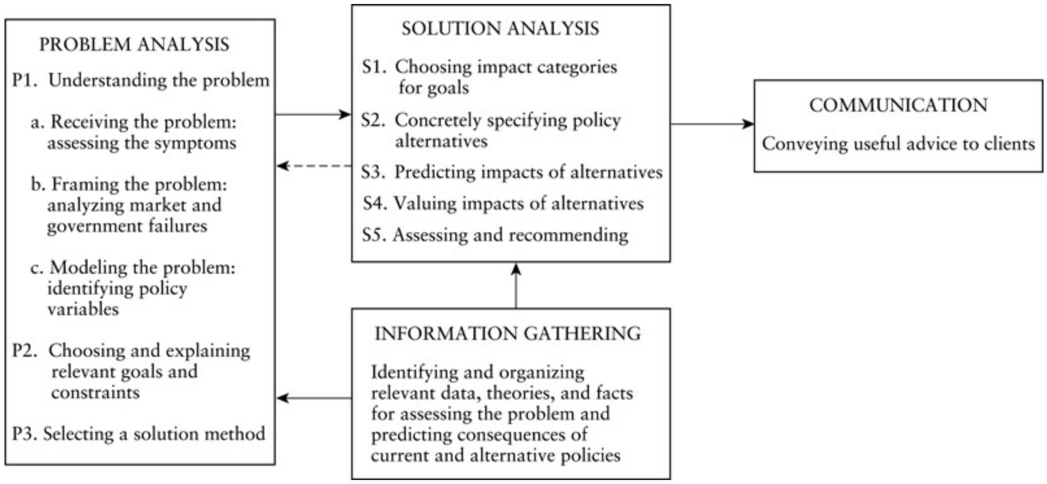

Figure 15.1 A Summary of Steps in the Rationalist Mode

Figure 15.1 provides a perceptual picture of the policy analysis process. It initially (over) simplifies the process by dividing it into two major components—problem analysis and solution analysis—both of which are vital and approximately equal in importance. For example, an analysis that devotes the great majority of its pages to analyzing the problem inevitably will not be credible in terms of the policy alternatives it presents or the reasons for choosing among them. Such an analysis may portray the nature of the problem convincingly, but its solution analysis will be deficient. Because most clients seek solutions, such an imbalance typically diminishes the value of an analysis. Conversely, your solution analysis and the resulting recommendation will carry little weight unless you convince the client that you have framed the problem correctly, thought carefully about the potentially relevant goals, and considered a range of alternatives.

Our experience suggests that quite a few students (and analysts) suffer in the extreme from looking only at solutions. They suffer from “recommendationitis”—trying to cram their complete analysis into their recommendations. If you suffer from this syndrome (or the tendency to cram all your analysis into any one step of the analytical process), then you are probably a nonlinear thinker who should take especially seriously the need to structure your policy using the steps in the rationalist mode.

Problem Analysis

Problem analysis consists of three major steps: (P1) understanding the problem, (P2) choosing and explaining relevant policy goals and constraints, and (P3) choosing a solution method.

Understanding the Problem

Understanding a policy problem involves assessing the conditions that concern your client, framing them as market or government failures, broadly defined, and modeling the relationship between the conditions of concern and variables that can be manipulated through public policy.

Assessing Symptoms

Clients generally experience problems as conditions that some group perceives as undesirable. They tend to specify problems to analysts in terms of these undesirable symptoms, or impacts, rather than as underlying causes. The analyst's task is to assess the symptoms and provide an explanation (model) of how they arise.

Assessing symptoms involves determining their empirical basis. In a narrow sense, this means trying to locate data that put the symptoms in quantitative perspective. For example, if your client is concerned about automobile accidents in your county caused by drunk drivers, then you might try to locate data to help you estimate the number of such accidents, how the number has changed over time, what percentage of total accidents they comprise, and other measures that help you determine the magnitude, distribution, and time trend of the symptom. In a broader sense, you should become familiar with current public discussion about the symptom (read the relevant newspapers!) and the history of existing policies that are generally perceived as being relevant to it. For example, you may find that, although there has been a steady decline in the number of alcohol-related accidents in recent years, a particularly tragic accident has focused public attention on the consequences of drunk driving. Viewed in the perspective of the favorable trend, drunk driving may seem less deserving of policy attention than other conditions of concern to your client.

Your description and assessment of symptoms generally appear as background in your problem analysis. It conveys the relative importance and urgency of the problem, and it begins to establish your credibility as someone who is knowledgeable about it. Nonetheless, the assessment of symptoms alone provides an inadequate basis for your analysis. You must argue that there is a *policy problem* and identify causal relationships that link the symptoms to factors that can be changed by public policy. In other words, you must frame and model the problem.

Framing the Problem

Potentially any social science model can be used as the basis for problem analysis. The major focus of explanation here should be on a specification of the expected deviation between individual self-interest (utility maximization by individuals) and aggregate social welfare. Although this focus is usually the best starting point for framing policy problems, several caveats should be noted.

First, we must avoid the danger of reductionism in such an approach. Although for many purposes we can treat wealth maximization and utility maximization as synonymous, saying that people maximize utility is *not* the same as saying that people care only about money. Clearly, they care about many other things as well. Also, as we saw in the examination of nontraditional market failures, economics has tended to treat preferences as fixed and, therefore, deals primarily with utility articulation rather than utility formation. Other social sciences have devoted considerably more effort to examining how preferences are formed. Consequently, much room exists for the other social sciences, including anthropology, psychology, and sociology, to play a part in framing the issues of market and government failures.

Second—and obviously related—we have not claimed that efficiency is the only appropriate goal of public policy. Therefore, any realistic problem analysis framework must enable the analyst to integrate other goals or constraints into the analytical process. The framework should also allow the analyst to set out the goals in a coherent way: "… good policy analysis … requires a clear statement and defense of the value judgments that combine with the analysis to lead to specific conclusions."[5] In the next section we suggest how to incorporate these goals into policy analysis. We believe, however, that you should begin by considering efficiency.

The concepts of market and government failures provide the basis for framing policy problems in terms of efficiency. Chapter 9 offered a logical procedure for application of market and government failure concepts. As outlined in Figure 9.1, the starting point is a determination if there is an operational market. If there is, and there is no evidence of either market or government failure, then there is no basis for intervention in the market to increase efficiency. If there is no operational market, or there is evidence of either market or government failure, then there is the opportunity for a policy change to increase efficiency. Framing in this way is likely to suggest some generic policy alternatives (review Table 10.6), as well as provide a starting point for more explicit modeling of the policy problem.

Modeling the Problem

The framing of problems in terms of market and government failures often leads directly to models linking policy variables to the conditions of concern. For example, consider a mayor worried about the rising cost of land-filling solid waste. An analyst who viewed the problem in terms of market and government failures might frame the problem as one of an institutionally created negative externality: because residents pay for refuse collection based on the assessed value of their property rather than on the volume of refuse that they generate, they perceive the price that they pay to dispose of an additional unit of garbage (their marginal private cost) to be virtually zero rather than the actual cost that the city must bear to collect and landfill it (the marginal social cost). A model of the problem follows: the larger the marginal private cost of disposal seen by residents, the smaller the amount of garbage generated. Further elaboration of the model involves specifying the ways that marginal private cost can be increased and identifying effects of the increases. Simply charging fees in proportion to the amount of garbage residents generate, for instance, would directly internalize the externality but perhaps involve a number of undesirable impacts: high administrative costs and residents dumping illegally to avoid fees. Recognizing that these undesirable impacts might be less serious for some types of garbage, such as yard wastes, the analyst may decide to specify different models for different categories of garbage.

Sometimes modeling must go well beyond framing to identify important policy variables. Consider, for example, the problem of drug abuse. Many aspects of it can be framed as a negative externality problem. Some of the relevant impacts include street crime by addicts, the spread of diseases through the sharing of needles, child neglect and abuse by addicted parents, and crime associated with the distribution system. Yet, identifying and evaluating the full range of policy variables requires a more explicit explanation of why people experiment with, and become addicted to, mood-altering drugs. Several hypotheses have been advanced: *contagion*—experimentation results from observation of, and pressure from, members of the peer group who are enjoying pleasures of early stages of use; *delinquency*—use by present-oriented young because it is risky, precocious, and pleasurable; *environmental escape*—use to escape reality of undesirable physical or social situation; and *occupational hazard*—experimentation because of availability to those in the distribution system. Obviously, these models imply different policy variables. The contagion model suggests that anti-drug commercials by celebrities admired by the young may help counter peer pressure; the environmental escape model suggests a need for more fundamental changes in conditions and opportunities. The analyst seeks to determine the circumstances in which each of these behavioral models is most likely to apply.

In summary, understanding the problem involves investigating the symptoms that prompt client interest, framing the undesirable conditions in terms of market and government failures, and developing models that relate the conditions of concern to variables that can be manipulated by public policy. Each of these three tasks, but especially modeling, can only be done provisionally at the outset of an analysis. As we discuss below, each of the other steps in the rationalist mode forces reconsideration of the initial problem definition.

Choosing and Explaining Goals and Constraints

Probably the most difficult step in any policy analysis is deciding on appropriate goals. Even with the advice we offer, all analysts are likely to find goal selection and delineation an inherently difficult task, as most of you will have had little experience in your academic careers in choosing goals. In most contexts you will have been given one or more goals to achieve. In policy analysis, you face a much more difficult problem because you must determine the relevant goals. Specifying goals requires you to be normative: you often must decide what *should* be wanted. This is both difficult and inherently controversial. It is controversial because it means being overt about value systems. Nonetheless, we can suggest some ways that you may be able to assist your clients in establishing appropriate goals—ones that make trade-offs among policy alternatives apparent. Our advice falls under three headings: first, accepting that goals are outputs of analysis as well as inputs to analysis, or, in other words, dealing with the vagueness, multiplicity, and conflict among goals as part of the analytical process; second, clarifying the trade-offs among goals; and third, clarifying the distinction between goals and policies (and especially between goals and policy alternatives).

Goal Vagueness: Goals as Outputs of Analysis

The problem analysis framework laid out in Figure 15.1, as well as the examples of actual policy analysis presented earlier in the book, should warn you to expect goal vagueness at the beginning of a policy analysis. Even if a particular client has clear goals, there may be good reasons why the client will not reveal them. Obviously, where clients do not have clear goals, they find it hard to structure the solution analysis for the analyst, whether the goals themselves or policy alternatives that might satisfy those goals. The realization of this reality probably generates more analysis paralysis among novice analysts than anything else.

Novices often try to elicit goals from clients at the beginning of their efforts. Our advice: *resist this temptation.* You will notice that, for example, goals are not presented at the beginning of the kidney shortage analysis (Chapter 1) or the Madison taxicab policy analysis (Chapter 9). After you have provided your own initial explanation of the problem, eliciting goals from your client will probably be valuable and perhaps essential, but it is rarely so at the outset. There are two reasons why it is usually futile to ask for goals before starting the analysis. First, the client may not have decided on the appropriate goals in a particular policy context.[6] As explained earlier, clients are usually driven by symptoms. Even if the client has decided on the appropriate set of goals, he or she will almost certainly not have decided on the appropriate trade-offs between pairs of these goals. Second, the client may have goals but be unwilling to reveal them. Let us deal with each in turn.

Many observers argue that individuals preparing to make complex decisions do not have predetermined goals.[7] In fact, it may be better if decision makers do not have rigid goals when dealing with complex, unstructured policy problems until after the problems have been analyzed. Put another way, it does not make sense to want something until you have some understanding of what is going on. For example, your client may not recognize the efficiency costs (and, therefore, the importance of efficiency as a goal) of a particular government intervention until you have explained it. However, clients are aware of specific impacts of the current situation or policy that concern them: "waiting times for taxis are too long," "too many young people are being killed in auto accidents involving alcohol," or "people are avoiding garbage pick-up charges by dumping at abandoned industrial sites." It is useful to get clients and others interested in the policy question to describe as many of these impacts as possible. However, it is usually the analyst's job to translate these impacts into goals.

How should you formulate goals in these situations? Whether or not your client has suggested a particular goal or set of goals, you should explicitly consider the relevance of efficiency and equity. Clearly, this book's focus on both market and government failures suggests that policy analysis should always be centrally concerned with aggregate social welfare. Keep in mind that there is wide misunderstanding as to the meaning of efficiency, even among politicians and senior public managers. If asked directly whether they care about efficiency, some decision makers answer negatively. However, if those same individuals are asked if they are concerned that a current policy is resulting in death or injury or in extensive queuing by citizens, they would emphatically answer yes. Yet, these are efficiency impacts.

Some policy analysts (mostly economists) argue that, in general, aggregate efficiency should be the primary concern of policy analysis, and that equity and other goals are rarely appropriate in evaluating alternative policies. They do so because they argue that seeking efficiency leads to the largest supply of goods and therefore provides the greatest opportunity for any desired subsequent redistribution. They advocate that distributional goals be met through explicitly redistributional programs, such as a progressive tax system. Although we are very sympathetic to the view that policy analysts should provide a voice for efficiency, especially because there is rarely an organized constituency in the political arena for maximizing aggregate social welfare, we assume that other goals are also usually important. We urge you to assume, a priori, that other goals as well as efficiency are relevant. This approach forces you to present reasoned arguments for either *including* or *excluding* a particular goal.

It is important to take seriously the question of goals beyond efficiency, whether substantive or instrumental. Thus, if your analysis is not going to include goals that various stakeholders in the policy environment hold important, you should explain why. In the next section, we argue that equity, or any other goal, can be usefully viewed in terms of trade-offs with efficiency. Yet, the fact remains that analytical technique cannot tell your client, or you, what you should want. Ultimately, the client, the analyst, and the political process must decide how much efficiency should be given up to achieve a given amount of redistribution or some other goal.

As introduced in Chapter 7, goals can be broken down into two broad categories: substantive and instrumental. *Substantive goals* represent values, such as equity and efficiency, that society cares about for their own sake. These include considerations of human dignity, self-perception, and self-actualization. *Instrumental goals* are conditions that increase the prospects for achieving substantive goals. Commonly relevant instrumental goals include political feasibility, increasing government revenues, and decreasing government expenditures (sometimes, more specifically, balancing budgets). As we explain later in this chapter, such instrumental goals often enter solution analysis as constraints. A constraint is simply a goal that must be satisfied up to some specified level.

Political feasibility is usually an appropriate policy goal. "The motive may be defensive or offensive—to prevent the abuse of their analysis, or to make their analytic voices more influential—but analysts still need to increase their political sophistication."[8] A clear example of political feasibility as an instrumental goal (or even constraint) arose in debates over the 1986 U.S. tax reform bill. Although many analysts argued that mortgage interest deductibility is both inefficient *and* inequitable, it was undoubtedly retained because any attempt to eliminate the deduction would have made the whole concept of tax reform politically unfeasible.

Policy analysis is the art of the possible. Resource requirements, and especially resource constraints, therefore, are of central importance. Resource constraints such as administrative infrastructure and availability of skilled personnel are often critical in determining the viability of policy alternatives. Often this can be considered as a specific efficiency impact. Sometimes, it makes more sense to treat a resource constraint as a separate (instrumental) goal/constraint. Generally, *the constraint set should include any resources that are essential for either maintaining current policy or implementing alternative policies.*

Your list of goals for a given problem might include efficiency, equity, human dignity, political feasibility, and budget availability. It is impossible, however, to determine in the abstract the relevant goals for all policy problems. As you research a problem, you should always be aware of the importance of identifying potential goals. As you become familiar with specific policy areas, you will become aware of the typical goals that are advocated.

It should be becoming clear that an important heuristic is that the *specification of goals is an important output of policy analysis.* A clear elucidation of goals is fundamentally important to good policy analysis. Yet, in practice, it is delineating the *trade-off* between goals, as implied by their relative weights, that is especially difficult and controversial.

Clarifying the Trade-offs between Goals

Selecting goals, per se, may be relatively noncontroversial. After all, it is easy to agree that distributional considerations should play *some* role in the analysis of almost any policy problem. One complicating factor in assessing goal trade-offs is that the process of explaining relationships between goals and policies may itself alter the choice of goals. The set of feasible policies identified determines the marginal trade-offs that are possible between any pair of goals, such as equity and efficiency. The focus of much public policy research and analysis is devoted to clarifying trade-offs. For example, in the area of welfare policy, an important consideration is the trade-off between benefit levels and work effort by recipients. Notice that in engaging in this process, the provision of such information does not, per se, change preferences. Rather, the levels of efficiency and equity selected do change, revealing different rates at which the decision maker would be willing to make marginal trades between them.

The lesson is that the set of feasible policies, which usually must be identified by the analyst, determine the possible trade-offs among goals. Therefore, we have an important heuristic: *the weights placed on goals (or impact categories) are more commonly an output of, rather than an input to, policy analysis.*

Clarifying the Distinction between Goals and Policies

Probably the most confusing semantic difficulty you will come across in policy analysis is making the distinction between goals (the values we seek to promote) and policies (the alternatives and strategies for promoting them). This semantic confusion arises because, in everyday language, policies (concrete sets of actions) are often stated as goals: "Our goal is to build 100,000 low-income housing units," or "Our goal is to reduce class size to 18 students." Although this everyday use of such language is harmless, it derails many neophyte analysts. Goals should be used to *assess* alternative policies, but if a policy is stated as a goal, how can one assess it? Indeed, stated this way, any policy is self-justifying.

In order to avoid this confusion, one must keep in mind a clear separation between goals and policies. We suggest that you do this by following another heuristic: *start by formulating goals as abstractly as possible and policy alternatives as concretely as possible.* Keep in mind that goals must ultimately be *normative*, a reflection of human values. Policy alternatives, on the other hand, are specific actions to achieve these goals. A policy alternative should seek to promote progress toward all the relevant goals, although, of course, in most cases a particular alternative will do better on some goals, but worse on others, than some other alternative. The "Holy Grail" of policy analysis is the dominant alternative that does better than all other alternatives on *all* goals. It has been found somewhat more often than the Holy Grail, but not very often!

In the context of choosing implementation strategies, it is often reasonable to take already-decided policies as goals. For example, if the head of the state health department has already made a final decision that 90 percent of school-age children will be vaccinated against some disease this year, then this policy may be reasonably taken as a goal by those in the department who must decide *how* the vaccinations will be accomplished. Other goals, such as minimizing cost and maximizing population immunity, should be raised in connection with the choice of implementation strategies. Indeed, one might question the wisdom of the policy if it were made without consideration of these goals.

As an analysis proceeds, goals typically become less abstract as we identify specific impacts that help us to predict the overall effect each alternative would have in terms of goal achievement. For example, in cost–benefit analysis we are interested in the efficiency of each of the alternatives relative to current policy. Nevertheless, the impacts that are relevant for efficiency may be quite varied. For example, the benefits of a new freeway might include both reduced driving time (time saved) and reduced accidents (lives, injuries, and property damage avoided). Similarly, the costs may include the (opportunity) cost of the construction workers' labor and the cost of the macadam and blacktop. Ultimately, each benefit impact category can be quantified (more specifically *monetized*) in terms of willingness to pay, and each cost category can be quantified and monetized in terms of dollars of opportunity cost—in other words, dollars that can then be aggregated by algebraic addition. However, if we are not performing a cost–benefit analysis in which we actually do aggregate, it is easy to forget that these various impacts are, in fact, ways at getting at predicting the extent to which alternatives will contribute to the achievement of a goal. These impacts can become goals in their own right. What is the problem? Well, certainly a "goal" of a highway project may be to reduce travel time. However, it is dangerous not to realize that this is only one dimension of the efficiency impact of the project––accident risk is also relevant. Of course, often policymakers tend to focus on highly visible and, therefore, salient impact categories. Indeed, these may turn out to be the only relevant categories, or at least the highly dominant ones, but they may not.

As we have stressed earlier in the chapter, you must overtly ask yourself what values (e.g. efficiency, equity, human dignity) lie behind the impacts that you or your client believe are important. As you become more experienced, you will begin to get better at doing this. When you observe impacts, you will be able to backward-map to a goal. For example, if a program uses real resources (employee time, building space, etc.), efficiency is inevitably relevant. If "who gets what" is an issue, as it almost always is, equity is *likely* relevant, although inclusion should usually follow a considered analysis of the legitimacy of the equity claims. This usually requires identification of the groups (for example, license holders, Native fishers, and taxpayers in the salmon fishery analysis in Chapter 16) and their characteristics.

A note of warning: getting, and then keeping, a focus on "big picture" goals is often difficult in addressing nuts and bolts issues such as sanitation, police patrol, and emergency services often found at the municipal and county levels. Clients at these levels of government sometimes claim that these sorts of issues are purely technical in nature and do not involve such value-laden considerations as efficiency and equity. Ultimately, good garbage collection is about efficiency (and perhaps equity and other goals in specific circumstances), even though it is easy to get mired in, well, the garbage. For example, technical efficiency as commonly measured includes the budgetary costs to the municipality but not the compliance costs of residents, a relevant component of efficiency as a social value.

Our *experience* suggests that policy problems require an explicit goal framework to ensure that the impact categories comprehensively correspond to, and capture, appropriate values. *Developing an appropriate and comprehensive set of goals is an important part of any analysis.*

Choosing a Solution Method

Figure 15.1 shows that goal selection is a component of problem analysis. In other words, you must decide which goals are relevant to your analysis before you can begin to consider solutions systematically. This is because the nature and number of goals largely determine the appropriate solution method. Choosing *how* you will do the solution analysis is also a component of problem analysis.

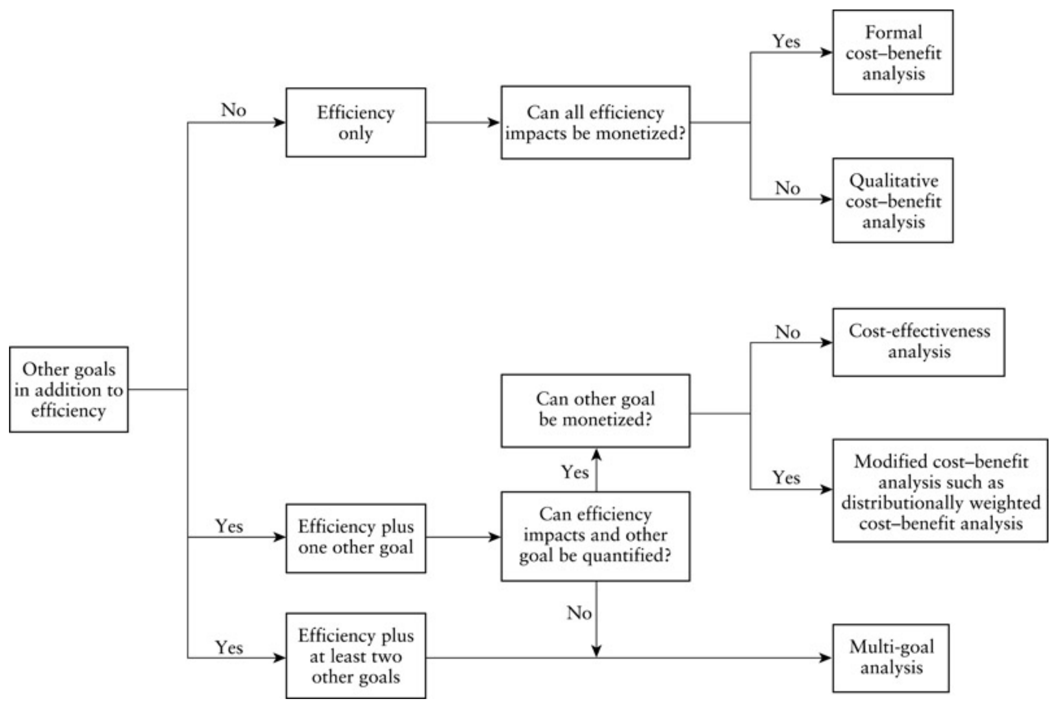

Figure 15.2 Choosing a Solution Method

Remember, policy analysis is an ex ante exercise. That is, we are interested in assessing policies *before* they are adopted. Policymakers are also interested in ex post analysis, that is, the performance of policies *after* they have been put into operation. Assessments of such implemented policies are usually known as *program evaluations*. These evaluations may use a wide variety of techniques, including cost–benefit analysis, cost-effectiveness analysis, and qualitative multi-goal analysis. (As we see, these same techniques are used in ex ante analysis). Policy analysis and program evaluation are distinct, although related, activities. They are related for several reasons. First, if an ex ante policy analysis has compared alternatives in terms of clearly specified goals and impact categories, it usually makes sense to conduct the ex post program evaluation of the alternative actually implemented using the same goals and impact categories. Second, ex post evaluations of programs that are similar to the alternatives your analysis is considering are extremely useful inputs for ex ante analysis because they provide credible information on projected impacts. The crucial point is that we are interested in analytical techniques in an ex ante context where we have to predict, as well as value, impacts.

Figure 15.2 distinguishes three general goal circumstances: First, efficiency is the only relevant goal. Second, efficiency and one other goal are relevant. Third, efficiency and two or more other goals are relevant. The number of relevant goals determines the appropriate solution method.[9]

There are five basic approaches to policy analysis: first, formal ex ante cost–benefit analysis; second, qualitative ex ante cost–benefit analysis; third, modified ex ante cost–

benefit analysis; fourth, ex ante cost-effectiveness analysis; and fifth, ex ante multi-goal policy analysis. From now on in this section we drop the ex ante prefix, but it is important to remember that it is always present in this context. Figure 15.2 indicates when each approach is most appropriate.

Cost–Benefit Analysis

As indicated in Figure 15.2, *formal cost–benefit analysis* is the preferred solution method when you believe that efficiency is the only relevant goal. Conceptually, cost–benefit analysis is relatively simple. If you read Chapter 17, or have already been exposed to cost–benefit analysis elsewhere, then you may be surprised to hear it described as being relatively simple. Consider, however, what cost–benefit analysis attempts to do. It reduces *all* the impacts of a proposed alternative to a common unit of impact, namely dollars. Of course, once all impacts have been measured in dollars, they can be algebraically summed. If individuals would be willing to pay dollars to have something, then presumably it is a benefit; if they would pay to avoid it, then it is a cost. Once all the impacts have been reduced to dollars, the evaluation rule is relatively straightforward: recommend the alternative that generates the largest aggregate net benefits (in dollars). Thus, in cost–benefit analysis, although we have different "goals" in the conventional use of the word, they can all be reduced to positive efficiency impacts (benefits) or negative efficiency impacts (costs), which, in turn, can all be measured in dollars, or monetized.

Sometimes prices revealed in markets provide a basis for *monetization*. Often, however, market prices often do not reflect marginal social costs or marginal social benefits because of the distortions caused by market failures and government interventions. There are also many classes of impacts, such as willingness of people to pay to preserve wilderness (existence value), that usually cannot be monetized through estimations based on direct observation of markets. Considerable skill and judgment must be exercised to assess the costs and benefits of these impacts plausibly.

Keep in mind that you will need some practice before you will be comfortable deciding if all the relevant impacts that are called goals can really be viewed as elements of efficiency. Consider an example. You are confronted with a crowded freeway. Let us suppose for the sake of simplicity that your client states that her goals are to reduce commuting time and fuel consumption. Superficially, these appear to be different goals (and they certainly may differ in importance to different clients), but both can be translated into a common impact: dollar cost savings.

Qualitative Cost–Benefit Analysis

As shown in Figure 15.2, even when you decide that efficiency is the only relevant goal, you must still determine whether all efficiency impacts can be reasonably monetized. If not, then *qualitative cost–benefit analysis* is the appropriate solution method. Like standard cost–benefit analysis, it begins with a prediction of impacts. Some of the impacts may be expressed in natural units (say, for example, hours of delay or tons of pollutants), while others may be qualitative (for example, a despoiled scenic view). If you are unable to monetize one or more of these impacts, then you cannot directly calculate the dollar value of net benefits. Instead, you must make qualitative arguments about the orders of magnitude of the various nonmonetized impacts.

Sometimes it is not practical to monetize impacts because of the technical difficulties in making such valuations. When standard procedures do exist for inferring such values, limitations of time, data, and other resources frequently make monetization impractical. Even highly skilled professional economists sometimes resort to qualitative cost–benefit analyses when they write about policy issues. Rather than attempt difficult and time-consuming valuations, they fall back on theoretical arguments to assess orders of magnitude of efficiency impacts.

If you cannot confidently monetize important efficiency impacts to within even orders of magnitude, then you may find it useful to work with the nonmonetized impacts as if they were separate goals. For example, you may have to decide how to compare certain program costs with highly uncertain benefits, such as environmental or catastrophic failure risks. Thus, your qualitative cost–benefit analysis takes the form of the multi-goal analysis we describe below.

Modified Cost–Benefit Analysis

It may seem reasonable to assume that if a client is only concerned with equity, or any other single, nonefficiency goal, then efficiency is irrelevant. Yet, some thought should convince you that efficiency is almost certainly relevant. This can be most clearly illustrated for the case where equity is putatively the only goal. Any intervention in the market to finance redistribution (absent utility interdependence and market failure) must inevitably result in some deadweight loss. Even where we are primarily concerned with achieving a given redistribution, we should seek to minimize the deadweight loss; in other words, we should attempt to carry out the redistribution as efficiently as possible. In our conception of policy analysis, therefore, *analysis almost never involves a single goal unless that goal is efficiency.*

In a particular analysis you may conclude that only efficiency and one other goal (most frequently, equity) are appropriate. As indicated in Figure 15.2, you can employ modified cost–benefit analysis if you are able and willing to monetize impacts on both efficiency and the other goal. In other words, you must be willing to assign dollar values to various levels of achievement of the other goal. For example, if the other goal is equality of the income distribution, modified cost–benefit analysis would involve weighting costs and benefits accruing to different income groups, resulting in *distributionally weighted cost–benefit analysis*. The advantage of such an approach should be clear. By incorporating distributional issues into a cost–benefit analysis, you can come up with a single metric for ranking alternatives. This is obviously attractive. The major disadvantage should also be clear from our discussion above—namely, that the metric is achieved only by forcing efficiency and equity to be commensurable.

Modified cost–benefit analysis, by assuming a particular set of distributional weights, also risks diverting attention from the trade-off between efficiency and distributional values in the ranking of alternatives, which merges both the distributional and efficiency assessments. When using modified cost–benefit analysis, it is usually desirable to present both the distributionally weighted net benefits and the unweighted net benefits of the policy alternatives.[10] A comparison of the weighted and unweighted net benefits makes clear the importance of the particular weights selected.

Cost-Effectiveness Analysis: Identifying Technical Efficiency

Cost-effectiveness analysis is appropriate when both efficiency and the other goal can be quantified but the other goal cannot be monetized (see Figure 15.2). In contrast to modified cost–benefit analysis, where both goals are measured in dollars and thus commensurable, the two goals are treated as non-commensurable.

We can approach cost-effectiveness analysis in one of two ways. The first method, the *fixed budget approach*, is to choose a given level of expenditures (say, $10 million) and find the policy alternative that will provide the largest benefits (that is, the greatest achievement of the nonefficiency goal). The second method, the *fixed effectiveness approach*, is to specify a given level of benefit (however defined) and then choose the policy alternative that achieves the benefit at the lowest cost. Both are cost-effectiveness procedures. In both, it is important to consider incremental changes in costs and the nonefficiency goal when comparing alternatives. In some cases the increments are assessed relative to current policy, but in others they are assessed relative to another alternative that is logically between it and current policy. For example, in assessing class sizes of 20 and 15 relative to the status quo of 25 students, the incremental impacts of the 20-student alternative should be assessed relative to the status quo, but the incremental impacts of 15-student alternative should be assessed relative to the 20-student alternative.[11]

The crucial distinction between cost–benefit analysis and cost-effectiveness analysis is that cost–benefit analysis can assess both whether any of the alternatives are worth doing (that is, whether social benefits exceed social costs) and how alternatives should be ranked if more than one generates positive net social benefits. Cost-effectiveness cannot tell the analyst whether a given alternative is worth doing, but, once a decision has been made to redistribute or achieve some other goal, it can help in deciding which policy alternative will do so most efficiently (with minimum losses of social surplus). It thus identifies technically efficient policies that may or may not be socially efficient.

Edward Gramlich and Michael Wolkoff provide an excellent illustration of both this distinction and how to do cost-effectiveness studies. They suppose that the goal is equity: specifically, to raise the income of some group of people. They compare a negative income tax alternative, minimum wage legislation, and a public employment plan. Each intervention involves costs that may or may not exceed benefits. For example, the negative income tax is likely to discourage some individuals from working who would otherwise do so. The public employment plan is likely to *attract* low-income individuals. In short, all *may* generate net costs (that is, fail the cost–benefit test). If we are determined to go ahead with some program for redistributional purposes, however, which would make most sense? Gramlich and Wolkoff adopt the fixed budget approach and seek to find the most beneficial policy alternative, given a fixed $5 billion expenditure constraint. (Notice, in this context, *benefit* is used in a special sense. It does not refer to aggregate social welfare, but rather counts only benefits going to specific low-income groups.) The authors developed a weighting scheme that ranks these impacts. They found that the negative income tax was much more successful in redistributing income per $5 billion expenditure than either the higher minimum wage or public employment.[12]

Multi-Goal Analysis

When three or more goals are relevant, multi-goal analysis is the appropriate solution method. As indicated in Figure 15.2, it is also the appropriate method when one of two goals cannot be quantified. It is usually the most appropriate solution method, and *it should be the assumed approach until the explicit conditions set out in Figure 15.2 for one of the other methods are confirmed.*

Because all of the other solution methods can be viewed as special cases of multi-goal analysis, the remaining steps in the rationalist mode (all within the solution analysis phase) set out how to conduct multi-goal analysis. Specifically, they indicate how goal achievement can be estimated using impact categories, how alternatives can be formulated, and how the alternatives can be systematically compared in terms of the goals.

Solution Analysis

When there are multiple goals, solution analysis requires five steps: (S1) selecting impact categories for the relevant goals; (S2) generating a mutually exclusive set of policy alternatives; (S3) predicting the impact that each alternative would have in terms of achieving each goal; (S4) valuing the predicted impacts, either using qualitative, quantitative, or monetized measures; and (S5) assessing the alternatives in terms of the goals and making a recommendation. At the heart of multi-goal analysis is the systematic comparison of alternative policies (step S2) in terms of goals (step S4). As explained in the rest of this chapter, we believe that a *goals/alternatives matrix* that displays the impacts of alternatives relevant to goals greatly facilitates this process.

Table 15.1 illustrates the usual structure of a goals/alternatives matrix. The leftmost column presents the goals determined during problem analysis. Step S1 involves *specifying the relevant impact categories* for assessing how well each policy alternative contributes to each of five goals. The column to the immediate right of the column of goals lists the impact categories determined in step S1. In an actual application, the goals and impacts should be given descriptive labels, rather than the generic ones shown in Table 15.1.

Step S2 in solution analysis is the detailed *specification of policy alternatives* that can potentially promote the policy goals. In Table 15.1, the three policy alternatives label the leftmost columns. Note that policy I is identified as the current policy. In general, current policy should be included as an alternative to avoid the risk of recommending a "best alternative" that is actually worse than the current policy. Rather than number alternatives, it is usually better to give them names that help the reader easily distinguish among them.

Table 15.1 The Simple Structure of a Goals/Alternatives Matrix

		Policy Alternatives		
Goals	*Impact Category*	*Policy I (current)*	*Policy II*	*Policy III*
Goal A	*Impact A1*	Medium	Low	High
	Impact A2	$1.2 million	$4.5 million	$3.3 million
	Impact A3	35 units	65 units	80 units
Goal B	*Impact B1*	Poor	Excellent	Very good
	Impact B2	Meets minimum standard	Meets minimum standard	Exceeds minimum standard
Goal C	*Impact C1*	Very likely	Likely	Somewhat likely

Step S3 is *prediction*: What is your prediction of the impact of the alternative on the goal or impact category? Step S4 is *valuation*: How desirable is the impact from a policy perspective? These two steps are often combined in a single goals/alternatives matrix by filling in all the cells of the matrix with valuations of the predictions of the impact of each policy alternative on each impact category. (However, it is often useful for the analyst to do *separate* prediction and valuation worksheet matrices.) These predictions and subsequent valuations are sometimes quantitative. For example, the row for impact A3 in the table shows quantitative predictions, which are also directly interpreted as positive (negative) valuations as long as we always want more (fewer) units of the impact. Row A4 shows a prediction in terms of dollars, say, resulting from a social surplus analysis, and also serves as an immediate valuation. Other times, the predicted impacts and their valuation are qualitative and made in such terms as "high," "medium," or "low" as shown for impact A1, or "poor," "good," "very good," or "excellent" as shown for impact B1. The qualitative impacts can also be in terms of the extent to which impacts will fall above or below some cutoff (impact B2) or the likelihood that some particular impact will occur (impact C1).

Step S5 involves *assessment*: recommending one of the alternatives and explaining the basis for the choice. As one alternative rarely dominates the others in terms of all the goals, the explanation for the recommendation almost always requires an explicit discussion of trade-offs among goal achievements predicted for the recommended alternative.

Table 15.2 shows an actual goals/alternatives matrix prepared by the Congressional Budget Office (CBO) as part of its analysis of broad alternatives for allocating radio spectrum. The analysis was done in response to a request from the House Committee on the Budget. It contributed to a five-year experiment in auctioning set out in the Omnibus Budget Reconciliation Act of 1993 (P.L. 103-66). The simultaneous ascending auction that was eventually implemented in 1994 and 1995 elicited almost $18 billion dollars in winning bids and set the stage for the personal communications revolution that followed.[13]

Goals/alternatives matrices emphasize a crucial point: *an analysis cannot be competent unless it comprehensively reviews all of the alternatives in terms of all of the goals. In terms of the goals/alternatives matrix, every cell must have an entry.* The following sections elaborate on the five solution analysis steps.

Choosing Impact Categories for Goals

The first step in solution analysis involves moving from the goal or goals developed in problem analysis to more specific impact categories for evaluating the desirability of alternative policies in terms of the goals. For example, the general goal of equality in the distribution of some service might be operationalized as the impact category "the variance in service consumption across income groups." Sometimes a single impact category, perhaps even the initially specified goal itself, will provide an adequate basis for assessment. In Table 15.1 this is illustrated by goal C, which has only one impact category associated with it. For example, the goal might be "efficiency" and, if all efficiency impacts can be monetized, its associated impact category would be "net benefits." If the original goal were "pass constitutional test," then it might reasonably serve as its own impact category. In many cases, however, more than one impact category is required to assess progress toward a goal adequately—goal A, for example, is shown as having three impact categories.

Table 15.2 License Assignment Methods Compared

Method	*Efficiency*	*Fairness*	*Revenues*
Comparative Hearing	• Might not assign the license directly to the user who values it most. • Secondary markets allow license sales to the users who value them most. • Consumes substantial private resources in license-seeking activity and inflicts high administrative and delay costs on society.	• Can ensure a specific distribution of licenses. • Legal and administrative costs of the process give larger financial interests an advantage.	• Revenues limited to license application fees. • Total FCC fees for 1991 were $46.6 million, including renewals and fees for lotteries. New license fees range from $35 to $70,000. • Comparative hearing fee for a new applicant for land-mobile services was $6,760 in 1991.

Method	Efficiency	Fairness	Revenues
Lottery	• A random process unlikely to assign the license directly to the user who values it most. • Secondary markets allow license sales to the users who value them most. • Less prone to delay than hearings, less prompt than auctions.	• Allows applicants equal opportunity if they can pay the application fee. • By awarding licenses to applicants who do not intend to provide services, grants lottery winners a windfall not shared by the public.	• Lottery revenues are included in totals noted above. • Fees for specific lotteries can be substantial. • The digital electronic message service lottery, the 220–222 MHz filing of 1991, drew 60,000 applicants and total fees of $4.4 million.

Method	*Efficiency*	*Fairness*	*Revenues*
Auction	• Is likely to assign the license directly to the user who values it most. • Should assign licenses more quickly and at a lower cost to society than alternatives.	• Gives taxpayers a share of spectrum rents. • Can be structured to accommodate small bidders.	• CBO estimates an auction of 50 MHz of spectrum for two additional land-mobile licenses could generate between $1.3 billion and $5.7 billion in fiscal years 1993 through 1995.

Source: Congressional Budget Office, *Auctioning Radio Spectrum Licenses,* March 1992, Table 2, at 18.

As efficiency is almost always one of the relevant goals, it is worth noting some of the common impact categories that arise in different substantive contexts. Table 15.3 lists the sorts of efficiency impacts that typically arise in several important policy areas. Of course, the particular set of impacts will depend on the specific policy alternative being considered. For example, if the policy alternatives were to include programs to reduce substance abuse, then it may be appropriate to expand the list under the category of productivity changes to include absenteeism and "presentism," coming to work impaired. Specifically with respect to alcohol abuse programs, the risks of driving under the influence of alcohol to both the individual and others would almost certainly be included.

For presentational purposes, it is sometimes helpful to use the headings in the table as goals and the subheadings as the impact categories. In such cases, the discussion of goals can be reformulated so that, say, improving productivity, reducing crime-related costs, improving health, and minimizing program costs (the headings in the first column of Table 15.3) are advanced as separate goals related to efficiency. Keep in mind that this approach involves narrower goals than the very general goals of efficiency, equity, and the like that we previously discussed. Therefore, caution is needed to ensure that these narrower goals do not foreclose any potentially desirable alternatives—doing the analysis initially with the broader goals and then making the switch for presentational purposes is probably the safest approach. The subheadings would then become the relevant impact categories.

Table 15.3 Typical Impact Categories for Efficiency

Social Policies	*Infrastructure Investments*	*Medical Interventions*	*Educational Investments*
Productivity changes: • Employment related • Household production Crime changes: • Victim costs • Criminal justice system costs Health changes: • Medical care • Mental health care • Fertility choices Program costs: • Labor and other resources • Participant time • Volunteer time	Accident reduction: • Fatalities • Injuries • Property damage Real resource use: • Fuel savings • Maintenance • External effects Time savings: • Leisure • Work Project costs: • Construction • Maintenance	Medical system costs: • Hospital care • Outpatient care • Long term care Patient effects: • Morbidity • Mortality • Productivity • Time Public health effects: • Contagion	Productivity changes: • Employment related Consumption changes: • Behaviors related to health • Cultural goods Social participation: • Citizenship • Community service Schooling costs: • Labor and other resources • Student time • Parental time

How can the appropriate set of impact categories be determined? A starting point should be to think of all the possible impacts that might be relevant to the goal. You should then make two adjustments to your initial list. First, if you have an impact category for which you predict the same outcome for all alternatives, then you can eliminate that impact category unless there is some expository reason for keeping it. Second, if you have identified an impact of one of your alternatives that is potentially valued by someone but does not fit into one of the initial impact categories, then you must add an impact category to take account of it in your goals/alternatives matrix. Usually, this can be done by adding the new impact category under one of the goals. However, if you cannot do so, then you almost certainly must rethink your initial set of goals—most likely you will have to add a new goal to take account of the effect. These two adjustments should make clear that your final set of impact categories may depend on the particular set of alternatives you ultimately select for analysis.

Thinking of your impact categories as the structure of a scorecard, the important heuristic is to *make sure that your scorecard provides a place to record every valued impact predicted for your alternatives.*

Specifying Policy Alternatives

We have already presented a set of generic policy alternatives in Chapter 10. They provide templates for beginning the construction of policy alternatives. The specification of policy alternatives is one component of policy analysis where nearly everyone agrees one can, and should, be creative. When told this, many of our students have replied, "Give us a hint." Here we attempt to do so.

There are a variety of sources for developing policy alternatives: existing policy proposals; policies implemented in other jurisdictions; generic policy solutions, as outlined in Chapter 10; and custom-designed alternatives.

Existing policy proposals, including the current policy, should be taken seriously, not because they necessarily represent the best set of alternatives, but rather because advocates or other analysts have found them to be plausible responses to that policy problem. Proposals already on the table sometimes are the product of earlier analyses; other times they represent attempts by interest groups or by policy entrepreneurs to draw attention to policy problems by forcing others to respond to concrete proposals. (Indeed, you can sometimes work backward from policy proposals to infer some of the perceptions and goals of the proposers.)

In some contexts we can borrow, or at least consider, policy alternatives from other jurisdictions.[14] How have other cities, states, or countries handled policy problems similar to the one that your jurisdiction faces? In particular, have similar jurisdictions appeared to have handled a problem particularly well? If so, the policies adopted by those jurisdictions can be a source of policy alternatives.

Once you have identified a policy that appears to have been successful elsewhere, you can create many additional alternatives through the process of tinkering.[15] The idea behind tinkering is to decompose an alternative into its essential components, identify different designs for the components, reassemble the designs into alternatives, and then select those combinations that look most promising. For example, imagine that you found a recycling program in a neighboring city that appeared to be working well. It involves four components: (1) residents separating newspapers and metal cans from their regular refuse; (2) residents putting these materials at the curbside in a standard container provided by the city on regular refuse-collection days; (3) the city refuse department collecting these materials; and (4) the refuse department packaging and selling them to recyclers. It is easy to imagine variations on each of these components. Residents, for instance, could be required to separate only newspaper, or perhaps plastic and glass as well, from their regular refuse. They could be required to drop them off at a recycling center rather than placing them at the curb. The city could contract with a private firm to collect the materials. Reassembling the variations on the components, a new alternative would be to require residents to deliver only newspapers to a central location operated by a private recycler.

As discussed in Chapter 12, efforts to predict the course of implementation of a policy alternative may require you to tinker in order to avoid problems that are likely to be encountered. Thus, in the course of your analysis, you are likely to identify new alternatives that result from tinkering with alternatives that you have subjected to substantial analysis.

It is rare, although not impossible, that an analyst will end up considering a set of purely generic policy alternatives. Nonetheless, generic policy solutions often provide a good starting point for design. For example, if you frame the apparent overexploitation of a resource as an open-access problem, then it is natural to look first at such generic policy solutions as private ownership, user fees, and restrictions on access. Although the particular technical, institutional, political, and historical features of the problem may complicate their direct applicability, the generic solutions can provide a framework for crafting and classifying more complex alternatives.

Once you develop a portfolio of generic alternatives, you can begin to modify them to fit the particular circumstances of the policy problem. For example, an open-access resource such as salmon has value to both sport and commercial fishers. Selling exclusive harvesting rights to a private firm might create appropriate incentives for efficient long-term management of the salmon. The firm would probably find it difficult to control sport fishing, however, because it would face high transaction costs in using the civil courts to protect its property rights against individual poachers. A hybrid policy that reserved an annual catch of fixed size for licensed sportsmen might deal more effectively with the poaching problem by bringing the police powers of the government to bear. Modified alternatives of this sort can often be formed by combining elements of generic solutions or by introducing new features.

Finally, in the course of your analysis, you may come up with a unique, or custom-made, policy alternative. Its elements may be lurking in the literature or it may be the product of your imagination.[16] As we discussed in Chapter 12, “backward mapping” can sometimes be used to produce such custom-made designs. Certainly this is one area of policy analysis in which you should stretch your imagination. Much of the intellectual fun of policy analysis arises in trying to come up with creative alternatives. Be brave! You can always weed out your failures when you begin your systematic comparison of alternatives. Indeed, you may not be able to identify your failures until you begin your comparative assessment.

Be warned that your creative alternatives are likely to be controversial, so be prepared to take the heat. (Sometimes you can launch trial balloons in order to get a sense of how hostile the reception will be. For example, you might try to get informal reactions to alternatives during interviews with people interested in the policy area.)

Keeping these sources in mind, we can suggest some heuristics for crafting policy alternatives.[17] First, *you should not expect to find a perfect, or even a dominant, policy alternative*. Policy analysis generally deals with complex problems and, most importantly, multiple goals. It is unlikely that any policy is going to be ideal in terms of *all* goals. Rarely is the best policy also a totally dominant policy.

Further, *do not contrast a preferred policy with a set of "dummy" or "straw person" alternatives*. It is often very tempting to make an alternative, which for some reason you prefer, look more attractive by comparing it to unattractive alternatives—almost anyone can look good if compared to Frankenstein's monster. This approach usually does not work and, moreover, it misses the very point of policy analysis. It rarely works because even relatively inexperienced clients are usually aware of the extant policy proposals advanced by interested parties. Your credibility can be seriously eroded if the client realizes that the alternatives have been faked. It misses the point of policy analysis because such an approach assumes that the critical component of analysis is the recommended alternative. As we have stressed, the *process* of policy analysis is equally important. You achieve the process goal of policy analysis by considering the best possible set of alternatives. Of course, comparing viable candidates makes determination of the best policy alternative more difficult and less certain, but, as we have already pointed out, it is better to recognize an ambiguous reality than fake certainty.

Another heuristic may help you to avoid dummy alternatives: *don't have a "favorite" alternative until you have assessed all the alternatives in terms of all the goals*. This may seem too obvious to state. Yet many neophyte analysts sprinkle their analyses with hints that they have accepted or rejected a policy alternative before they have formally assessed it. The heuristic is, *make your primary ego and intellectual investments in the analysis and only marginally in the particular recommendation.*

Having ensured that your policy alternatives are not straw persons, you should *ensure that your alternatives are mutually exclusive*; they are, after all, *alternative* policies. Alternatives are obviously not mutually exclusive if you could logically adopt more than one of them. That is, you could combine all the features of alternative A with all the features of alternative B and come up with a viable alternative C. In such circumstances, A and B may be too narrow and perhaps should be eliminated from the set of alternatives. For example, imagine a series of alternatives that specify fees for different classes of users of a public facility. If the adopted policy is very likely to set fees for all the classes, then it probably would be appropriate to combine the set of fees into a single alternative that can then be compared to other combinations that also cover all the classes of users.

You almost always face an infinite number of potential policy alternatives. If one of your policy alternatives is to build 10,000 units of low-income housing, mutually exclusive alternatives include building 9,999 units or 10,001 units. An infinite number of policy alternatives is a few too many. Given clients' limited attention span (and analysts' limited time), *somewhere between three and seven policy alternatives is generally a reasonable number*.[18] Keep in mind that one of the alternatives should be current policy—otherwise you introduce a bias for change.

It is preferable to provide a reasonable contrast in the alternatives examined. Comparing alternatives that are too similar to each other is likely to be both analytically and presentationally wasteful. The *alternatives should provide real choices*.

You should *avoid "kitchen sink" alternatives*—that is, "do everything" alternatives. Such alternatives are often incomprehensible and unfeasible. If you find yourself proposing a kitchen sink alternative, then take a close look at all the constraints your client faces. Does your client have the budgetary, administrative, and political resources it requires? If not, then it is probably not a valid alternative. You failed to recognize this because you omitted instrumental goals that encompass these constraints.

More generally, *alternatives should be consistent with available resources, including jurisdictional authority and controllable variables*. If your client is a mayor, then there is usually little point in proposing alternatives that require new federal resources. If you believe that such an alternative should be formulated, then you must recast it as a call for mobilization, coordination, or lobbying. In other words, it must be oriented around a set of steps that your client can take to generate the appropriate federal action.

Remember that *policy alternatives are concrete sets of actions*. (Remind yourself of the distinction between goals and policies.) Generic policy solutions are abstract statements. Thus, while it is useful for analytical purposes to think of a given alternative as the "demand-side subsidy alternative," this abstraction should be converted to a concrete proposal (for example, housing voucher worth *X*, going to target population *Y*) in your policy analysis.

You will not be able to predict consequences unless you provide clear and detailed specifications of your alternatives. The alternatives should be clearly specified sets of instructions so that the client knows exactly what she is choosing and how it will be executed. To prepare these instructions, you must determine what resources will be needed during implementation and how these resources are to be secured from those who control them. In effect, as discussed in Chapter 12, you must be able to create an implementation scenario that shows how the policy can be moved from concept to reality.

Prediction in Terms of Goals and Their Impact Categories

Once you have specified the relevant impact categories and policy alternatives, you must bring them together in a way that facilitates choice. The first task is to predict, or forecast, how each alternative will perform in terms of each goal and, if a goal has multiple impacts associated with it, all of these impacts. You should confront this task explicitly. By doing so, you make clear the predictions inherent in your analysis. This step is necessarily explicit in cost–benefit analysis: a prediction of impacts underlies the estimation of the stream of future costs and benefits. For example, you may need to predict how many AIDS patients will use a clinic over the next 15 years.

Your model of the policy problem (in problem analysis) is crucial to the prediction of impacts. Your model helped you understand and explain current conditions, which are observable. It should also help you predict what will happen in the future under current policy. For example, assume that the policy problem is rush-hour traffic congestion in the central business district (CBD), and that your model shows that crowding results because people base their commuting decisions on the private costs and benefits of the various transit modes. Because drivers do not pay for the delay costs that their presence inflicts on everyone else driving in the CBD during rush hour, too many people commute by automobile from the perspective of total social costs and benefits.

Your model suggests that changing conditions, such as growing employment in the CBD, which alter the costs and benefits of various transit modes, will affect future congestion. By projecting changes in conditions, therefore, you can predict future congestion levels under current policy. You would make predictions about congestion under alternative policies by determining how they would alter the costs and benefits of different transit modes. Higher parking fees, for instance, would raise the private costs of commuting by automobile.

Consider how you might actually go about making the link between higher parking fees and congestion. Ideally, you would like to know the price elasticity of demand for automobile commuting in your city. That is, by what percentage will automobile use for commuting change as a result of a 1 percent change in price? Starting with estimates of the current price and level of automobile commuting, you could use the elasticity to predict levels under various parking fee increases. It is unlikely that you would have the authority, time, or resources to do an experiment to determine the elasticity. You may be able to take advantage of natural experiments, however. For example, your city may have raised parking fees for other reasons in the past. What effect did this have on automobile use? Do you know of any other cities that raised parking fees recently? What happened to their congestion levels?

If you cannot find answers to these questions, then you might be able to find some empirical estimates of the price elasticity of automobile commuting in the literature on transportation economics. If a number of estimates are available, they may be systematically assessed through a meta-analysis.[19] As a last resort, you might ask some experts to help you make an educated guess, or simply make a best guess yourself. Chapter 17 considers these approaches in more depth in the context of cost–benefit analysis.

Policies almost always have multiple impacts. Use your model, your specification of the alternative, and your common sense to list as many different impacts as you can. For example, with respect to a parking fee increase: What will be the impact on automobile use for commuting? On the price and quantity of private parking in and near the CBD? On city revenues from parking fees and parking tickets? On the use of other transit modes? On off-peak driving in the CBD? On resident and commuter attitudes toward city hall? Sometimes, by thinking about one impact, you identify others. For instance, once you start thinking about how commuters might respond to the higher parking fees, you may realize that some will park in nearby residential neighborhoods and ride public transit to the downtown. If you had not already considered on-street parking congestion in nearby residential neighborhoods as an impact category, then you should add it to your list under the appropriate goal.

Once you have a comprehensive set of impacts associated with goals, you must link them comprehensively to the alternatives. The key point, as already noted, is that *you should predict and value the effects of each alternative on* every *impact category.* After you have worked through all the alternatives, you will be able to compare them, either qualitatively or using explicit valuation criteria.

Do not try to suppress the uncertainty in predictions. You need not fill in cells with single numbers (point estimates). Instead, ranges (perhaps confidence intervals) may be appropriate. For example, you may be fairly confident that the average number of vehicles entering the CBD at rush hour on workdays will be very close to 50,000 under current policy over the next year because this has been the average over the last two years. In contrast, you may be very uncertain about the average number of vehicles that would enter if parking fees were doubled. Perhaps you believe it unlikely that the number would be either less than 45,000 or greater than 48,000. Rather than fill the appropriate cell with a specific number, you should indicate this range. Later you can use these upper and lower bounds to come up with "best" and "worst" cases for each alternative.

Remember that predictions always depend on assumptions about the future (even predictions about what will happen tomorrow). Often, there are possible different "states of the world" that it is useful to include explicitly in the analysis. We can think of these sets of assumptions as *scenarios*, which we already encountered in Chapter 12 in the context of implementation analysis. The most likely state of the world is known as the *baseline scenario*, or the *baseline case*. For example, Robert Hahn makes a number of different assumptions about the costs of technology and fuel costs to compare the cost-effectiveness of a variety of policies intended to reduce emissions from vehicles.[20] For each set of assumptions (that is, for each scenario), he calculates the dollar cost of reducing the sum of emissions of organic gases and nitrogen oxides by one ton. As fuel costs are directly relevant to most of the alternatives he considers, it makes sense to assess the cost-effectiveness of each alternative in terms of an assumption about fuel prices. More generally, *when uncertainty about the future requires more than one scenario, it is necessary to produce a goals/impact matrix for each scenario considered*. This ensures that comparisons among alternatives are made under the same set of assumptions.

Valuing Alternatives in Terms of Goals and Their Impact Categories

Once you have made predictions of the consequences of each alternative for each impact category, the next step is to value the predictions in terms of their contributions to the goals. In some cases, the predictions themselves may already be appropriate valuations. For example, in an analysis with multiple impact categories for efficiency, many of which cannot be monetized, "lives saved" is an impact category that has a natural evaluative interpretation —more lives saved are better than fewer. In other cases, however, you must convert the impact categories into more evaluative forms. So, for example, if all other impacts can be monetized, it may be desirable to translate "lives saved" into a dollar estimate of how much an average member of society would be willing to pay for the corresponding reductions in risk.

If efficiency is the only relevant goal and monetization is possible, then the choice rule is simply to select the alternative that gives the greatest excess of benefits over costs. If you have constructed a separate prediction matrix, then it will typically express impacts in units that are not readily comparable. By introducing a common metric for several impacts, you can make them directly comparable. In this way, impacts can be aggregated into a smaller number of categories. Cost–benefit analysis serves as an extreme example—it requires that all impacts be valued in dollars. More generally, some, but not all, impacts can be expressed in the same units. *You should try to make the impact categories as comparable as possible without distorting their relationships to the underlying goals.* By combining impacts that are truly commensurate, you are likely to be able to better make manageable comparisons across alternatives.

When monetization is not appropriate or feasible, valuation must be applied to each impact category. For example, the impact category "variance in service consumption across income groups" can be operationalized for valuation as "minimize the variance in service consumption across income groups." Valuation can sometimes be best done by treating the achievement of some level of the impact category as a constraint. This same impact category might be operationalized as a constraint, such as "families with incomes below the poverty line should be given full access to the service."

You should exercise considerable care in selecting and valuing impact categories to measure the achievement of goals. Always ask yourself: How closely do high scores in the impact categories correspond to progress toward goals? Asking this question is especially important because analysts and clients tend to focus attention on those impact categories that can be easily measured. A sort of Gresham's law operates: *easily measured impact categories tend to drive less easily measured ones from analytical attention.* This tendency may lead us astray when the easily measurable impacts fail to cover all the important dimensions of the goal.

For example, the number of casualties inflicted on the enemy is one impact category for measuring success in war. It may be secondary to other impacts—the morale of each of the opposing sides, their respective capabilities for protracted struggle, or the control of disputed populations—for measuring progress toward the goal of ultimate victory. Yet, during the height of the Vietnam War, body counts became the primary measure of U.S. success because they could be easily reported as a number on a weekly basis. This emphasis made search-and-destroy missions appear relatively more effective than efforts to establish stable political control over the population, even though the latter might very well have contributed more to the chances of victory.[21]

To illustrate the use of a goals/alternatives matrix to summarize complex information, consider an analysis of policies to address global climate change performed by the Congressional Budget Office (CBO). Since the industrial revolution, the concentration of carbon dioxide (CO_2) in the atmosphere has increased substantially from about 280 parts per million (ppm) at the beginning of the industrial revolution to 379 ppm in 2005.[22] In recent years, this concentration increased at an average annual rate of about 1.9 ppm. Most scientists now believe that the historical increase in concentration CO_2 contributed to a warming of the average surface temperature of the Earth; most climate models predict that the projected increase in CO_2 concentration under current policies will lead to continued warming over the century, with attendant rising ocean levels and more extreme weather events (such as heavy precipitation). The increasing CO_2 concentration poses some risk of a catastrophic rise in ocean level and dramatic climate changes.

In *Policy Options for Reducing CO_2 Emissions*, CBO analysts present comparisons of a CO_2 tax and several versions of cap-and-trade programs to create incentives for firms and consumers to reduce their CO_2 emissions.[23] A CO_2 tax could be imposed either upstream (on energy producers) or downstream (on energy users). As there are already excise taxes on coal and petroleum, CBO analysts discuss an upstream tax.

As discussed in Chapter 10, the generic cap-and-trade program sets a cap on total emissions and issues a corresponding number of permits. Permits may be used by their holders or traded in a market. Permit holders who can reduce emissions at a cost lower than the permit price have an incentive to do so to free up a permit for sale to firms that can only reduce emissions at a cost higher than the permit price. A generic cap-and-trade program has two potential advantages over a tax. First, assuming effective enforcement, it guarantees that emission targets are met. Second, it offers the political advantage of allowing an initial allocation of permits based on historical emissions to cushion the initial financial impact on firms.

The flip side of certainty about the quantity of emissions under cap-and-trade is uncertainty about the price at which permits trade. The CBO analysts identified three modifications of the generic (inflexible) cap-and-trade program to reduce price volatility. First, government could set ceiling (safety valve) and floor prices. It could make the ceiling and floor prices effective by selling permits at the ceiling price and buying them at the floor price. Second, firms could be allowed to engage in banking, saving unused permits to loan to themselves or other firms in the future. Third, government could operate a "circuit breaker" that delayed cap reductions or managed banking to increase temporarily the quantity of permits if permit prices rise above a ceiling.

As shown in Table 15.4, the analysts compared a carbon tax, an inflexible cap, a cap with a safety valve and either banking or a price floor, and a cap with banking and either a circuit breaker or managed borrowing. (Note that the absence of current policy as an

Table 15.4 Comparison of Selected Policies tor Cutting CO_2 Emissions

Policy	*Efficiency Considerations*	*Implementation Considerations*	*International Consistency Considerations*

Policy	*Efficiency Considerations*	*Implementation Considerations*	*International Consistency Considerations*
Carbon Dioxide Tax	A tax would avoid significant year-to-year fluctuations in costs. Setting the tax equal to the estimate of the marginal benefit of emission reductions would motivate reductions that cost less than their anticipated benefits but would not require reductions that cost more than those benefits. Research indicates that the net benefits of a tax could be roughly five times as high as the net benefits of an inflexible cap. Alternatively, a tax could achieve a long-term target at a fraction of the cost of an inflexible cap. *Efficiency ranking: 1*	An upstream tax would not require monitoring emissions and could be relatively easy to implement. It could build on the administrative infrastructure for existing taxes, such as excise taxes on coal and petroleum.	A U.S. tax could be set at a rate consistent with carbon dioxide taxes in other countries. Consistency would require comparable verification and enforcement. If countries imposed taxes at different points in the carbon supply chain, special provisions could be needed to avoid double-taxing or exempting certain goods. Setting a U.S. tax that would be consistent with allowance prices under other countries' cap-and-trade systems would be somewhat more difficult because it would require predicting allowance prices in different countries.
Cap with Safety Valve and Either Banking or a Price Floor	A cap-and-trade program that included a safety valve and either banking or a price floor could have many of the efficiency advantages of a tax. The safety valve would prevent price spikes and could keep the costs of emission reductions from exceeding their expected benefits. Banking would help prevent the price of allowances from falling too	An upstream cap would not require monitoring emissions. It would require a new administrative infrastructure to track allowance holdings and transfers. Implementing a safety valve	Either a safety valve or banking would become available to all sources of CO_2 emissions in a linked international cap-and-trade program. Some countries could object to linking with a U.S. program that included those features, because linked countries could not ensure that their emissions would be below a required level in a given year. Linking would also create concerns about inconsistent monitoring and enforcement among countries and

Policy	*Efficiency Considerations*	*Implementation Considerations*	*International Consistency Considerations*
	low, provided that prices were expected to be higher in the future. A price floor, however, would be more effective at keeping the cost of emission reductions from falling below a target level. *Efficiency ranking: 2*	would be straightforward: the government would offer an unlimited number of allowances at the safety-valve price. Banking has been successfully implemented in the U.S. Acid Rain Program. A price floor would be straightforward to implement only if the government chose to sell a significant fraction of emission allowances in an auction.	international capital flows (as described below in the inflexible cap policy). Countries with different cap-and-trade programs could capture many of the efficiency gains that would be achieved by linking—while avoiding some of the complications—if they each included banking (or set a similar price floor) and agreed on a safety-valve price.
Cap with Banking and Either a Circuit Breaker or Manage Borrowing	Allowing firms to bank allowances would help prevent the price of allowances from falling too low, provided that prices were expected to be higher in the future. Including a circuit breaker—or increasing the ability of	An upstream cap would not require monitoring emissions. It would require a new administrative infrastructure to	Including banking and either a circuit breaker or borrowing in the U.S. program could reduce the likelihood of linking because it would cause uncertainty about the stringency of the U.S. cap relative to other countries' caps and about the total supply of allowances in the global trading market.

Policy	Efficiency Considerations	Implementation Considerations	International Consistency Considerations
	firms to borrow allowances — would help keep the price of allowances from climbing higher than desired, but would be significantly less effective at doing so than a price ceiling. *Efficiency ranking: 3*	track allowance holdings and transfers. Banking has been successfully implemented in the U.S. Acid Rain Program. Determining when to trigger a circuit breaker, or modify borrowing restrictions, would require judgment about current and future allowance prices. Such interventions could aggravate price fluctuations if those judgments were incorrect.	

Policy	*Efficiency Considerations*	*Implementation Considerations*	*International Consistency Considerations*
Inflexible Cap	Allowance prices could be volatile. An inflexible cap could require too many emission reductions (relative to their benefits) if the cost of achieving them was higher than anticipated, and could require too few reductions if the cost of meeting the cap was lower than policy makers had anticipated. *Efficiency ranking: 4*	An upstream cap would not require monitoring emissions. It would require a new administrative infrastructure to track allowance holdings and transfers.	Linking an inflexible U.S. cap with other countries' cap-and-trade systems would create a consistent global incentive for reducing emissions. However, inconsistent monitoring and enforcement in any one country could undermine the entire linked trading system. Further, linking would alter allowance prices in participating countries, create capital flows between countries, and possibly encourage countries to set their caps so as to influence those flows.

Note: An "upstream" tax or cap would be imposed on suppliers of fossil fuel on the basis of the carbon dioxide (CO_2) emitted when the fuel was burned. A "safety valve" would set a ceiling on the price of allowances. "Banking" would allow firms to exceed their required emission reductions in one year and use their extra allowances in a later year. Under a "circuit breaker," the government would stop a declining cap from becoming more stringent if the price of allowances exceeded a specified level.

Source: Congressional Budget Office, *Policy Options for Reducing CO_2 Emissions* (Washington, DC: Congress of the United States, February 2008), Summary Table 1, xi–xii.

alternative does not allow the analysis to address directly the question of whether any of these alternatives should be adopted.)

The analysts compare the four alternative programs qualitatively in terms of three goals, which they called criteria.[24] The first criterion is *efficiency considerations*, or "efficiency in maintaining a balance between the uncertain benefits and costs of reducing CO_2 emissions." The second criterion, *implementation considerations*, refers to the "ease or difficulty of implementation." The third criterion, *international consistency considerations*, is "possible interactions with other countries' policies for curbing CO_2" emissions. In addition, the analysts note other criteria that policymakers may wish to consider in the future: the certainty of future emission levels and the distributional consequences.[25]

The entries in the table are brief discussions of the alternatives in terms of the criteria. The analysts assess the carbon dioxide tax as providing the most efficient balancing of costs and benefits. They list this alternative first, and follow with the three cap-and-trade alternatives in descending order of efficiency. The discussion of implementation points to some of the issues that would have to be resolved with the two modified cap-and-trade alternatives. The discussion of international consistency takes account of the carbon dioxide cap-and-trade program already implemented by the European Union. The report includes an appendix describing both the European program and the older cap-and-trade program for sulfur dioxide implemented in the United States to address the acid rain problem, and draws on experience from these programs to motivate the various cap-and-trade modifications it discusses.

Assessment: Comparing Alternatives across Incommensurable Goals

Choosing the best alternative is trivial when you have either a single goal with commensurate impact categories or an alternative that ranks highest on all impact categories. Unfortunately, reality is rarely so kind. Although you may sometimes be pleasantly surprised, you should expect to find different alternatives doing best in terms of achieving different goals. Your task is to make explicit the trade-offs among goals implied by various choices, so that your client can easily decide the extent to which he or she shares the values you brought to bear in choosing what you believe to be the best alternative. In other words, *you should continue to be overt about values in the final phase of assessment.*

You should also be explicit about uncertainty. Rarely will you be able to predict and value impacts with great certainty. The predictions of impacts often constitute your best guesses. If your predictions are based on statistical or mathematical models, then your best guesses may correspond to sample means or expected values, and you may be able to estimate or calculate variances as measures of your confidence in them. More often your best guesses and your levels of confidence in them will be based on your subjective assessment of the available evidence. In situations in which you are generally confident about your best guesses for the major assessment criteria, a brief discussion of the range of likely outcomes may suffice.

We have already discussed some ways of organizing an assessment when you are not very confident about your best guesses. When lack of confidence springs from uncertainty about relevant conditions in the future, one can construct a number of scenarios that cover the probable range. You can then choose the best alternative under each scenario. If one appears to dominate under all scenarios, then you can choose it with some confidence. If no alternative dominates, then you can make your choice, either on the basis of the best outcomes under the most likely scenarios or on the basis of avoiding the worst outcomes under any plausible scenario. In either case, you should discuss why you think your approach is the most appropriate one.

Sometimes confidence in best guesses will vary greatly across alternatives. You may be very certain about your valuations of some alternatives, but very uncertain about others. One approach is to conduct a "best case," "most likely case," and "worst case" assessment for each of the alternatives with very uncertain outcomes. You must then decide which case is most relevant for comparisons with other alternatives. (Chapter 17 presents more appropriate ways to convey uncertainty when most or all impacts can be monetized.) Another approach is to create a new goal, perhaps labeled "minimize likelihood of regret," that gauges how probable it is that the actual outcome will be substantially less favorable than the best guess. You could then treat this new goal as just another incommensurable goal.

No matter from what source, as the number of goals or impact categories deserving prominence rises, the complexity of comparison becomes greater. In the face of such complexity, it may be tempting to resort to a more abstract decision rule. For instance, you might begin by scoring the alternatives on a scale of 1 to 10 for each of the criteria (say, 10 points for fully satisfying the criterion, zero points for not satisfying it all). One possible decision rule is to select the alternative that has the highest sum of scores; another is to select the alternative with the highest product of scores. Although rules such as these can sometimes be useful, *we recommend that you* not *use them as substitutes for detailed comparisons of the alternatives*. Simple decision rules tend to divert attention from tradeoffs and the values implied in making them. Also, they invariably impose arbitrary weights on incommensurate goals. In other words, we urge caution in their use because they tend to obscure, rather than clarify, the values underlying choice.

In the decisions we make in our everyday lives, we often predict, value, and choose implicitly and incompletely. Indeed, our goals and alternatives often remain unspecified. When decisions are routine, our experience allows us to take these shortcuts with little risk of serious error. When the decision problems are novel or complex, however, we run the risk of missing important considerations when we do not explicitly value all of our alternatives in terms of all of our goals.

Presenting Recommendations

The final step in the rationalist mode of analysis is to give advice. Specifically, you should clearly and concisely answer three questions: First, what do you believe your client should do? Second, why should your client do it? And third, how should your client do it? Answers to the first two questions should come directly from your assessment of alternative policies. Your answer to the third should include a list of the specific actions that your client must take to secure adoption and implementation of the recommended policy.

We offer several heuristics to help guide your presentation of recommendations. First, *your recommendations should follow from your assessment of the alternatives (step S4).* While this may seem obvious, we think it is worth stating. Sometimes what seem to be good ideas for policy solutions take form only as your deadline gets near. Resist the temptation to introduce these new alternatives as recommendations. The proper approach is to redo your specification and assessment of alternatives so that the new candidate is systematically compared with the others. Otherwise you risk giving advice that you may later regret. One reason that we advocate working in a nonlinear way toward completion of the steps in the rationalist mode is that doing so increases the chances that good ideas arise early enough to be treated seriously.

Second, *you should briefly summarize the advantages and disadvantages of the policy that you recommend.* Why should your client accept your recommendation? What benefits can be expected? What will be the costs? Are there any risks that deserve consideration? By answering these questions, you appropriately draw your client's attention to the consequences of following your advice.

Finally, *you must provide a clear set of instructions for action.* Exactly what must your client do to realize the policy that you recommend? Sometimes the set of instructions can be very short. For example, if your client is a legislator, then the instruction "Vote for bill X" may be adequate. Often, however, adoption and implementation of your recommendation require a much more complex set of actions by your client. For example, imagine that you recommend to the director of the county social services department that funds be shifted from one day-care vendor to another. When and how should approval be secured from the county manager? Is it necessary to consult with the county's legal department? When and how should the vendors be notified? When and how should families be notified? Should any members of the county legislature be briefed in advance? Which staff members should be assigned to monitor the transition? These questions may seem mundane. Nonetheless, with a little thought, you should be able to imagine how failing to answer any one of them might jeopardize the successful implementation of the recommended policy. To do so systematically, you should prepare an implementation scenario as outlined in Chapter 12.

Communicating Analysis

The format of your policy analysis plays an important part in determining how effectively you communicate your advice to your client. Clients vary greatly in their levels of technical and economic sophistication; you should write your analysis accordingly. Generally, however, clients share several characteristics: they usually want to play some role in shaping the analysis (but they do not want *to do* the analysis); they are busy and they face externally driven timetables; and they are nervous about using the work of untested analysts when they have to "carry the can" for it in the policy arena. These generalizations suggest some guidelines on how to present your work.

Structuring Interaction

Often you can productively involve your client in the analysis by sharing a preliminary draft. Do so early enough so that you can make use of your client's comments, but not so early that you appear confused or uninformed. *By trying to prepare full drafts of your analysis at regular intervals over the course of your project, you force yourself to identify the major gaps that you must yet fill.* Giving your client the opportunity to comment on one of these intermediate drafts will usually be more effective than ad hoc interactions. Of course, if you believe that your client is a better listener than reader (perhaps because you can only claim your client's time and attention through an appointment), you may find oral progress reports, perhaps structured by a prepared agenda, to be more effective. Be flexible. Use whatever type of communication that seems to work best in the particular context.

You can improve the effectiveness of your written interaction by carefully structuring your draft. You should follow two general guidelines: First, decompose your analysis into component parts; and second, make the presentation within the components clear and unambiguous. These guidelines are not only appropriate for your final product, but they also promote effective communication at intermediate stages by allowing your client to focus on those components that seem weak or unconvincing. Decomposition and clarity also tend to crystallize disagreement between you and your client. Although this may seem like a disadvantage, it usually is not. By crystallizing disagreement at an early stage in your project, your draft analysis helps you determine which of your client's beliefs might be changed with further evidence and which are rigid. In this way, your preliminary drafts and other structured interaction with your client reduce the chances that your analysis will ultimately be rejected.

The steps in the rationalist mode, shown in Figure 15.1, provide a general outline for decomposing your analysis. While your final analysis must be written as if you began with the problem description (step P1) and moved sequentially to your recommendations (step S5), you should not necessarily try to write (as opposed to present) the components of your preliminary drafts in strict order lest you encounter the "analysis paralysis" we mentioned earlier. Obviously, the steps cannot be treated as if they were completely independent. For example, the impact categories you choose for evaluating your alternatives (step S1) cannot be finalized until you have specified the relevant goals (step P2). Nonetheless, very early in your project you should try to write a draft of each of the components as best you can. This effort forces you to think configuratively and anticipate the sort of information you will need to make the final draft effective. This may be particularly valuable in helping you move from problem to solution analysis so that you do not end up with an overdeveloped description of the status quo and an underdeveloped analysis of alternative policies.

Keeping Your Client's Attention

Clients are typically busy people with limited attention spans. Reading your analysis will be only one of many activities that compete for your client's attention. You bear the burden of producing a written analysis that anticipates your client's limited time and attention.

While most of our suggestions stress presentational issues, timeliness is by far the most important element. If you are trying to inform some decision, then you must communicate your advice before the decision must be made. Sometimes clients can delay decisions. Often, however, the need to vote, choose a project, approve a budget, or take a public stand places strict deadlines on clients and, therefore, on their analysts. While you should always strive for excellence, *keep in mind that an imperfect analysis delivered an hour before your client must make a decision will almost always be more valuable to your client than a perfect analysis delivered an hour after the decision has been made.*

You can facilitate more effective communication with busy clients by following a few straightforward rules: provide an executive summary and a table of contents; set priorities for your information; use headings and subheadings that tell a story; be succinct; and carefully use diagrams, tables, and graphs.

Your analysis should not read like a mystery. Rather than holding your client in suspense, tell her your recommendations at the very beginning in an *executive summary.* The executive summary should be a concise statement of the most important elements of your analysis including a clear statement of your major recommendations. An analysis of more than a few pages should have a separate executive summary that stands on its own. *It should generally be structured so that the last sentence of your first paragraph presents your recommendation and the subsequent paragraphs rehearse the analysis that supports it.* Your objective should be to produce a concise statement that clearly conveys the essence of your advice and why you are offering it.

Especially in large organizations, you may be asked to produce policy memoranda with strict length limits, often one or two single-spaced pages. Preparing these memoranda can be challenging as they should include the basic elements of a policy analysis—explicit goals, concrete alternatives, systematic comparison, and clear recommendation. A common way to accommodate these demands within the length constraint is to limit attention to current policy and a single alternative to it. Indeed, the demand for these policy memoranda often comes in the form of a request to assess a particular policy proposal. As in an executive summary, the first paragraph of the memorandum should set out the issue being addressed and conclude with a clear statement of the recommendation. The second paragraph typically frames the policy problem and states the relevant goals. The third paragraph specifies the policy alternative. Subsequent paragraphs compare the alternative to the status quo policy in terms of the goals. Based on the comparisons, the final paragraph supports the recommendation set out at the end of the first paragraph.

A *table of contents* enables your client to see at a glance where your analysis is going. It presents the structure of your decomposition so that your client can focus on aspects of particular interest. Together with the executive summary, the table of contents enables your client to skip portions of your analysis without losing the major points. While everyone wants people to read what they write, you should consider yourself successful (at least in a presentational sense) if your client takes your advice on the basis of your executive summary and table of contents alone.

You should arrange your material so that a client who reads sequentially through your analysis encounters the most important material first. Usually a ten-page analysis is not nearly as useful to the busy client as a five-page analysis with five pages of appendices. Doesn't the client still have to read ten pages of material? Only if she wants to! By breaking the analysis into five pages of text and five pages of appendices, you have taken the responsibility for prioritizing the information. As you and your client develop an ongoing relationship, your client may find it unnecessary to check the background facts and theoretical elucidations provided in the appendices.

Headings and subheadings allow your client to move through an analysis much more quickly. As a general rule, headings should approximately correspond to the steps in the rationalist mode. They should, however, be concise *and* tell a story. For example, rather than the heading "Market Failure," a section of your analysis might be titled "Smokers Do Not Bear the Full Social Costs of Smoking." Similarly, rather than "Government Failure," a preferable heading might be "Why State Price Ceilings Are Leading to Undersupply and Inefficiency." *Use headings and subheadings freely*—even a few pages of unbroken text can lose your client's attention––*but logically*.

Other presentational devices are also helpful in making your analysis readily useful to your client. Indenting, selective single spacing, numbering, and judicious italicizing can all be used to highlight and organize important points in your analysis. The key is to make sure that they draw attention to the material that deserves it. A long series of points, or "bullets," especially if expressed in sentence fragments without verbs, not only denies the reader adequate explanation but also fails to highlight the really important points.[26] Favor well-constructed paragraphs over bullets—your written report should *not* resemble a PowerPoint presentation. Even if you occasionally use bullets or other highlights for emphasis, continue to rely on the paragraph as the main unit of presentation.

Diagrams, graphs, and *tables* can be very useful for illustrating, summarizing, and emphasizing information. Use them, but use them sparingly, so that they draw attention to important information. Like headings, their titles should tell a story. All their elements should be labeled completely so that they can be understood with little or no reference to the text. Be sure you appropriately *round all numerical information*; the number of significant figures should be determined by the least accurate data rather than the number mechanically calculated by your computer. Strive to make the diagrams and graphs as clean as possible, avoiding what Edward Tufte calls "chartjunk"—ink on the page that does not tell the reader anything new.[27] Convey as much information as you can with as little ink as possible. Also, document your diagrams and graphs with sources at the bottom. If you created the diagram or graph yourself, then list "author" as the source.

Be succinct. Keep the text focused on the logic of your analysis. Relegate tangential points and interesting asides to your files, or, if you think they might be useful in some way to your analysis, put them in footnotes or appendices. Be careful not to cut corners with jargon; if you use technical terms, use only those that your client will understand. (Sometimes you might purposely include unfamiliar technical terms to prepare your client for debate with others—your task then is to explain the terms in the clearest possible manner.)

Try to write crisp text. Start paragraphs with topic sentences. Favor simple over complex sentences. Use the active rather than passive voice to keep your text lively: "I estimate the cost to be …" rather than "The cost was estimated to be …" Allow yourself time to edit your own text, especially if you tend to be wordy.

We offer Table 15.5 as a summary of our major suggestions on communication.

Establishing Credibility

Until you have established a track record as a reliable analyst, you should expect your clients to be somewhat nervous about relying on your analysis. After all, they are the ones who will bear the political and career risks from following your advice. Therefore, if you want your advice to be influential, then you must establish the credibility of your analysis.

You can enhance the credibility of your analysis in several ways. *First, make sure that you cite your sources completely and accurately.* Of course, you will find this easier to do if you have kept clear notes on your document and field research. Appropriate citation, as noted in Chapter 14, enables you to give credit to others for their ideas (something you have discovered is important to academics), and provides a good starting point in the event you or some other analyst has to address this, or a related issue, in the future.

Second, flag uncertainties and ambiguities in theories, data, evidence, and predictions. You do a great disservice to your client by hiding uncertainty and ambiguity, not just because it is intellectually dishonest but also because it may set your client up to be blindsided by others in the policy arena who have more sophisticated views. After flagging uncertainties and ambiguities, you should resolve them to the extent necessary to get on with the analysis. (Perhaps a “balance of evidence” will be the best you can do.) You should always check the implications of your resolution of uncertainty for your recommendations. Where your recommendations are very sensitive to the particular resolution, you should probably report on the implications of making a range of resolutions.

Table 15.5 Communicating Policy Analyses

Do	*Remember the client.* Keep in mind that your task is to provide useful advice.
	Set priorities. Organize your information carefully (essential material in the text, supporting material in appendices).
	Decompose your analysis into component parts.
	Use headings that tell a story. Avoid abstract headings such as “Market Failure.”
	Be balanced. Give appropriate coverage to problem analysis and solution analysis.
	Acknowledge uncertainty but then provide your resolution of it. (Support your resolution with sensitivity analysis where appropriate.)
	Be credible by documenting as extensively as possible.
	Be succinct.
	Avoid jargon and clearly explain any technical terms.
	Be “value overt.” Make explicit arguments for the importance of goals.
	Write crisp text. Favor short and direct sentences; use the active voice.
	Provide clear tables and figures that stand alone. Use them to emphasize important information or support critical arguments.
	Round numbers appropriately. False precision undercuts your credibility.
Don’t	*Write an essay.* The difference between an essay and a well-structured policy analysis should be clear to you by now.
	Tell the client everything that you know as it comes into your head. It’s fine to think nonlinearly, but write linearly.
	Write a mystery. Instead, state your important conclusions and recommendations up front in an executive summary.
	Create a new alternative when recommending. Your recommendation should follow from the analysis.
	Use the format of PowerPoint slides. Avoid bullets in favor of well-structured paragraphs.

Finally, as we have already argued, you should be “value overt.” Clearly set out the important goals and explain why you believe that they are important. Also, explain why you have rejected goals that others might believe important. Your explication of goals is especially important if you wish to argue to your client that she should alter her goals or give them different emphases.

Self-Analysis Once Again: Combining Linear and Nonlinear Approaches

What we have called the rationalist mode of policy analysis consists of eight sequential steps, beginning with understanding the policy problem (step P1) and ending with presenting recommendations (step S5). These steps promote logical and comprehensive analysis. Rather than viewing them as the sequence that you should follow to produce analysis, however, you should think of them more as the rough outline for your final product. Someone who just reads your final report should be left with the impression that you followed the steps in sequence, even if someone else following your efforts from start to finish observed you jumping and iterating among them. Indeed, we believe that you generally *should* jump and *must* iterate, working nonlinearly toward your linear product.

A brief reflection on our discussions of gathering information, specifying goals, and designing alternatives should make apparent why we urge you to work nonlinearly. Rarely do you know what information is available before you start gathering it—as noted in the previous chapter, one source leads to another. You may not be able to specify realistic goals until you have considered the range of feasible policy alternatives. Ideas for new alternatives may not emerge until you start to evaluate the ones you have initially designed. As we have noted, the policies advocated by interested parties can help you determine how they view "the problem," perhaps helping you to understand it better yourself.

We offer a practical hint for helping you combine the linear and nonlinear approaches: begin your analysis by starting a file (actually, both an electronic and a hard copy file usually help) for each of the steps in the rationalist mode: descriptions and models of the problem, goals, solution methods (though often this file can be closed very early), impact categories, alternatives, predictions, valuations, and recommendations. As you gather information, insights will come to you. Write them down in the appropriate file. Even if they do not survive in your final analysis, they not only help you get started but also provide a record of how your own thinking has progressed, something that may be useful when you think about how to communicate your analysis effectively to others.

Working with parallel files may have the added advantage of reducing the anxiety you face in writing to meet deadlines. If you already have a start on each of the components, then putting together the complete analysis will be less traumatic.[28] You can help yourself even more by occasionally going through the files to convert your insights and information into paragraphs. Doing so forces you to confront the weak links in your arguments, thereby helping you focus your attention on critical questions and required information. Also, some of the paragraphs may survive to your final draft, thus sparing you the anxiety of facing a blank page as your deadline approaches.

Once you have had some experience in doing policy analysis, take some time to reanalyze yourself. If you find yourself becoming paralyzed at one step or another, then you should probably try to force yourself to work more configuratively. (As already suggested, you may find it helpful to try to draft a full analysis about midway to your final deadline.) If you have trouble organizing and presenting your analysis, then you should probably try to follow the steps in the rationalist mode more closely.

Conclusion

Our focus in this chapter has been on policy analysis as the process of providing useful written advice to a client. We set out a series of sequential steps, what we called the rationalist mode, that should help you structure your written product. We emphasized, however, that you should think configuratively about the steps as you gather information and work toward a final product. Although we have provided considerable practical advice, the analytical process cannot be reduced to a simple formula. That is why doing policy analysis is so interesting and doing good policy analysis is so challenging.

For Discussion

1. Construct a goals/alternatives matrix for the Madison taxi regulation analysis presented in Chapter 9. How would your goals have to change if an alternative were added that involved the city organizing a common dispatching system for all taxi companies?
2. What goals might be relevant for assessing allocation systems for transplant organs?

Notes

1 See Max Singer, "The Vitality of Mythical Numbers," *Public Interest* 23, 1971, 3–9. See also Peter Reuter, "The Social Costs of the Demand for Quantification," *Journal of Policy Analysis and Management* 5(4) 1986,

807–12.

2 Douglas Besharov, "Unfounded Allegations—A New Child Abuse Problem," *Public Interest* 83, 1986, 18–33.

3 Cary Coglianese, "Assessing Consensus: The Promise and Performance of Negotiated Rulemaking," *Duke Law Review* 46(6) 1997, 1255–349.

4 For a thoughtful treatment of problem definition in the organizational context, see David Dery, *Problem Definition in Policy Analysis* (Lawrence: University Press of Kansas, 1984).

5 Helen Ladd points out that economists doing policy analysis often forget this: "The failure of some of the authors to spell out and defend their value judgments in some cases leaves the misleading impression that the policy conclusion follows logically from the analysis alone." Review of John M. Quigley and Daniel L. Rubinfeld, "American Domestic Priorities: An Economic Appraisal," *Journal of Economic Literature* 24(3) 1986, 1276–77, at 1277.

6 The problem is not limited to decision makers in the public sector. A study of private-sector decision makers found that they consistently omitted nearly half of the objectives that they later identified as personally relevant to the decision. Samuel D. Bond, Kurt A. Carlson, and Ralph L. Kenney, "Generating Objectives: Can Decision Makers Articulate What They Want?" *Management Science* 54(1) 2008, 56–70.

7 See Henry Mintzberg, Gurev Raisinghani, and Andre Theoret, "The Structure of Unstructured Decision Processes," *Administrative Science Quarterly* 21(2) 1976, 246–75. As James March puts it: "It seems to me perfectly obvious that a description that assumes goals come first and action comes later is frequently radically wrong. Human choice behavior is at least as much a process of discovering goals as for acting on them." James March, "The Technology of Foolishness," in James C. March and J. P. Olsen, eds., *Ambiguity and Choice in Organizations* (Bergen, Norway: Universitetsforlaget, 1976), at 72.

8 Robert D. Behn, "Policy Analysis and Policy Politics," *Policy Analysis* 7(2) 1981, 199–226, at 216.

9 One can also compare methods in terms of their comprehensiveness of goals and the willingness to monetize impacts relevant to efficiency. See Aidan R. Vining and Anthony E. Boardman, "Metachoice in Policy Analysis," *Journal of Comparative Policy Analysis* 8(1) 2006, 77–87.

10 Arnold C. Harberger, "On the Use of Distributional Weights in Social Cost–Benefit Analysis," *Journal of Political Economy* 86(2) 1978, S87–S120.

11 Boardman et al., 464–68.

12 Edward M. Gramlich and Michael Wolkoff, "A Procedure for Evaluating Income Distribution Policies," *Journal of Human Resources* 15(3) 1979, 319–50.

13 The concept for the auction drew on both formal economic modeling and review of the experiences of other countries with spectrum auctions––the rules developed to guide the auctions ran to 130 pages. See R. Preston McAfee and John McMillan, "Analyzing the Airwaves Auction," *Journal of Economic Perspectives* 10(1) 1996, 159–75.

14 See Anne Schneider and Helen Ingram, "Systematically Pinching Ideas: A Comparative Approach to Policy Design," *Journal of Public Policy* 8(1) 1988, 61–80.

15 See David L. Weimer, "The Current State of Design Craft: Borrowing, Tinkering, and Problem Solving," *Public Administration Review* 53(2) 1993, 110–20.

16 For some examples of sources for such custom design, see David L. Weimer, "Claiming Races, Broiler Contracts, Heresthetics, and Habits: Ten Concepts for Policy Design," *Policy Sciences* 25(2) 1992, 135–59.

17 Several of these ideas are drawn from Peter May, "Hints for Crafting Alternative Policies," *Policy Analysis* 7(2) 1981, 227–44. See also Ernest R. Alexander, "The Design of Alternatives in Organizational Contexts," *Administrative Science Quarterly* 24(3) 1979, 382–404.

18 On the question of attention span, see George A. Miller, "The Magical Number Seven, Plus or Minus Two: Some Limits on Our Capacity for Processing Information," *Psychological Review* 63(2) 1956, 81–97.

19 For an illustration of meta-analysis, see Stephanie Lee, Steve Aos, and Marna Miller, *Evidence-Based Programs to Prevent Children from Entering and Remaining in the Child Welfare System: Benefits and Costs for Washington,* Document No. 08-07-3901 (Olympia: Washington State Institute for Public Policy, 2008).

20 Robert W. Hahn, "Choosing among Fuels and Technologies for Cleaning Up the Air," *Journal of Policy Analysis and Management* 14(4) 1995, 532–54.

21 Alain C. Entoven and K. Wayne Smith, *How Much Is Enough? Shaping the Defense Program, 1961–1969* (New York: Harper & Row, 1971), 295–306.

22 The figures cited in this paragraph are drawn from the Intergovernmental Panel on Climate Change, "Summary for Policymakers," in Susan D. Solomon, Dahe Qin, Martin Manning, Zhenlin Chen, M. Marquis, K. B. Averyt, M. Tignor, and H. L. Miller, eds., *Climate Change 2007: The Physical Science Basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change* (New York: Cambridge University Press, 2007), 1–18.

23 Congressional Budget Office, *Policy Options for Reducing CO_2 Emissions* (Washington, DC: Congress of the United States, February 2008).

24 Ibid., vii.

25 Subsequently, CBO predicted the financial impacts of the cap-and-trade program in the American Clean Energy and Security Act (HR 2454) by income quintile. Letter from Douglas W. Elmendorf, CBO Director, to Dave Camp, Ranking Member, House Committee on Ways and Means, June 19, 2009.

26 More generally on the dangers of rigid PowerPoint style in presentation, see Edward R. Tufte, *The Cognitive Style of PowerPoint* (Chesire, CT: Graphics Press, 2003).

27 Edward R. Tufte, *The Visual Display of Quantitative Information* (Chesire, CT: Graphics Press, 1983), 107.

28 If you suffer from writer's block, then you might want to look at Martin H. Krieger, "The Inner Game of Writing," *Journal of Policy Analysis and Management* 7(2) 1988, 408–16.

16
Case Study

The Canadian Pacific Salmon Fishery

In late 2015, Canadians elected a new federal government. Suppose that the new Minister of Fisheries, Oceans and the Canadian Coast Guard (FOCCG) asked you to conduct an analysis of the commercial "small boat" salmon fishery on the Pacific coast in the province of British Columbia (BC). In his appointment letter, the Prime Minister of Canada specifically mandated the minister to "act on recommendations of the Cohen Commission on restoring sockeye salmon stocks in the Fraser River." More generally, suppose the minister asked you whether the current policy promotes FOCCG's effective management of the fishery. Because he is a new minister, he would particularly like you to provide an assessment of current policy in comparison with possible alternative policies.

The term "small boat" fishery is used to distinguish this fishery from the commercial aquaculture (or farmed) salmon fishery which is regulated by the provincial, rather than the federal, government. The minister has instructed you to exclude the salmon sports fishery from your analysis except to the extent that it directly affects the commercial fishery. The minister has also instructed you to ignore for the purposes of this analysis negotiations with the United States over the division of salmon stocks between the two countries. He has also instructed you to treat the current constitutional law on the Aboriginal (or "First Nations") salmon food fishery as a given minimum, but does not wish to preclude you from considering policy alternatives that might transfer a greater share of the commercial fishery harvest to Aboriginal groups.

As the minister is considering proposing a new policy, he has given you one month to complete your analysis. Although you are new to the ministry and have no specific background in fisheries, your training as a policy analyst should enable you to gather relevant information quickly from already available research and organize it effectively to provide useful answers to the minister's questions. An example of the sort of report you might produce follows.

Box 16.1 Increasing the Social Value of the BC Salmon Fishery

Prepared for Minister of the Department of Fisheries, Oceans, and Canadian Coast Guard 2016

Executive Summary

The salmon fishery in British Columbia (BC) continues to face serious challenges, even though its economic importance has declined in relative terms over the last decade. The evidence suggests that, as currently organized, the fishery continues to be a net drain on the wealth of Canadians despite a series of costly reforms. Although some of the problems associated with the fishery can be attributed to inherent market failures, some are attributable to ineffective, or indeed counterproductive, government interventions. Although the current regime as now managed probably does not threaten the viability of most salmon runs, it continues to endanger some of the smaller runs and subspecies. On balance, this analysis argues it is desirable to replace the current regulatory regime with a policy based on exclusive ownership of salmon fishing rights for specific rivers.

This analysis examines four policy alternatives: (1) current policy; (2) harvesting royalties and license auction; (3) river-specific, exclusive ownership and harvesting rights (with cooperative ownership on the Fraser River); and (4) individual transferable quotas (ITCs). It assesses these four alternatives in terms of their capability for meeting the following four goals: *efficiency* (primarily reduce rent dissipation, the unnecessary expenditure of resources to secure the catch); *biodiversity preservation* (primarily maximize the number of viable river runs); *equitable distribution* of the economic value of the harvest (to current license holders, to Aboriginal fishers, and to taxpayers); and *political feasibility*.

Based on this assessment, the analysis concludes that the minister should adopt river-specific, exclusive ownership and harvesting rights (with cooperative ownership on the Fraser River). The adoption of ITQs also has the potential to improve significantly the efficiency and equity of the fishery. Although it would probably not be as efficient as the preferred alternative, it is probably more politically feasible, as indicated by the recent implementation of ITQs for other BC fisheries and their recent endorsement by the Cohen Commission.

Allocating river-specific exclusive ownership rights would ensure efficient resource use through incentives to use the lowest-cost fishing technology. Further, it would eliminate most public costs and ensure biodiversity protection. With compensation to incumbents (as described), it should be equitable to the major relevant constituencies.

Experience in Canada and other countries over the last decade suggest that boat-based ITQs would largely eliminate rent dissipation (although it is not the lowest-cost harvesting method for catching salmon). Effective implementation of the quotas would probably be the greatest challenge (poor implementation would reduce many of the benefits of the structural efficiency improvements). Nevertheless, the evidence, particularly from New Zealand and Iceland, suggests that ITQs can be implemented in such a way that efficiency is substantially improved. This alternative could also be designed and implemented so that salmon biodiversity is maintained (provided share catches rather than fixed quotas are utilized), and it would be equitable to relevant groups.

Introduction

A review of commercial fisheries around the world concludes: "The world's fisheries face the crises of dwindling stocks, overcapitalization, and disputes over jurisdiction."[1] In Canada, the collapse of the Atlantic cod fishery dramatically illustrated the risks of mismanagement: losses to the Canadian economy, personal hardship to those in the fishing and related industries, and the adverse fiscal impacts of adjustment expenditures by the government. Does current policy provide an adequate basis for effective management of the BC salmon fishery? Do alternative policies offer the prospect for better management? In addressing these issues, this analysis draws on numerous previous analyses of the BC salmon fishery, as well as on theoretical and empirical analyses of salmon fisheries and other fisheries worldwide.

The BC Salmon Fishery

Despite decline, the commercial west coast salmon fishery remains an important Canadian industry. In the five-year period ending in 2014, the average annual market value of the BC salmon catch was $42.3 million (2014 Canadian dollars).[2] The fishery, if effectively managed, has the potential to generate benefits considerably in excess of the costs required to harvest the fish. This value is known as economic, or scarcity, rent. However, mismanagement of the fishery dissipates some, or even all, of this rent by encouraging too much investment in fishing effort. The evidence suggests that dissipation of the economic rent offered by fisheries is a common worldwide phenomenon.[3] Indeed, mismanagement combined with other misguided policies could lead to social costs in excess of total economic rent.[4]

Over the last 25 years, the tonnage of the salmon catch has declined substantially. Figure 1 shows the annual catch in metric tons from 1951 to 2014. The catch peaked in the late 1980s. From 1986 to 1990, the annual value of the catch was $463.4 million (2014 Canadian dollars), more than ten times the value over the most recent five-year period.[5] Although some of the decline can be attributed to policies that reduced the capacity of the commercial fleet, it also reflects a declining stock, raising concerns about the long-term sustainability of the fishery.[6]

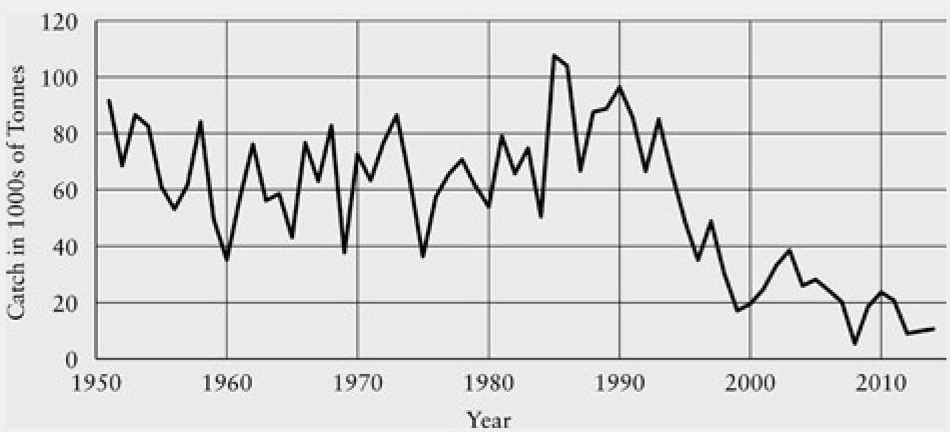

Figure 1 British Columbia Salmon Catch (1951–2014)

Source: Based on data available from Fisheries and Oceans Canada, www.pac.dfo-mpo.gc.ca/stats/comm/index-eng.html

Economics of the Fishery: The Open-Access Case

It is useful to contrast the current state of the salmon fishery with what would likely occur if it were to operate as an unregulated resource. In the unregulated fishery, we assume that individual fishers cannot be excluded from access to fish until the fish are actually appropriated in the process of fishing—in other words, there is competitive "open access" and a "law of capture" applies. An open-access salmon fishery can be best understood and analyzed as a public goods problem; that is, one in which the good is rivalrous in consumption (if one individual captures and consumes a salmon, that particular fish is not available to any other individual) but in which exclusion is not economically feasible. Extensive theoretical and empirical research on open-access resources in general, and on fisheries in particular, characterizes the consequences of open access: most importantly, too much of the resource will be harvested from the social perspective.[7] Fishers will find it in their individual self-interest to fish until the costs they see of catching an additional fish (their marginal cost) just equal the price of the fish (their marginal benefit). Their individual efforts, however, increase the marginal costs of all other fishers by reducing the stock of fish available for catching.

Before the arrival of Europeans, open access of the salmon fishery as just described may not have presented an economic problem because supply exceeded demand at zero price. It is certainly plausible to posit that there was an Aboriginal fishery before the arrival of Europeans in which salmon were essentially "free goods," so that the catches of individuals did not increase the marginal costs borne by other fishers. In such circumstances, the indigenous peoples of the northwest coast would have been free to exploit the salmon fishery fully without depleting the resource in any meaningful economic sense.[8]

Yet considerable evidence suggests that, even before the European incursion, increasing demand for fish had moved much of the Aboriginal salmon fishery beyond the status of a free good. However, rather than allowing open access, the indigenous population had developed a system of private and common property ownership.[9] The northwest coast Aboriginal salmon fishery was generally a terminal, or river, fishery that caught fish in an efficient manner using weirs and traps.[10] Most importantly, the fishery was efficient in a temporal sense; the Aboriginal population had both the incentives and social organization to allow sufficient spawning fish through to replenish the stock.[11]

The influx of Europeans, who, in practice, were not subject to the extant indigenous property rights, created an open-access regime. Not only were the new users not excluded from the river-mouth fishery, they also introduced technology that allowed them to leapfrog Aboriginal riverine fisheries by fishing in the open sea.[12] This leapfrogging created an open-access environment that introduced less cost-effective fishing methods. One estimate is that, in the 1930s, traps in neighboring Washington State were approximately two-thirds more cost-effective than the emerging small-boat fishery, and that, without the interception of fish by the small-boat fishers, the traps could have been five-sixths more cost-effective.[13] The history of the regulation of the fishery makes clear that the incentive to leapfrog with new technology has remained strong.

A Brief History of Government Regulation of the Fishery

As we demonstrate below, the history of the salmon fishery in BC is well documented.[14] For this analysis, it is sufficient to review briefly five eras of policy development: the early fishery, the Sinclair Report/Davis Plan, the Pearse Commission, the 1996 Mifflin Plan and subsequent buy-backs up to the year 2001, and the 2009–2012 Cohen Commission to the present day.

Early History: Open Access

Even by the 1880s, there were signs of overfishing in BC, especially on the Fraser River, which paralleled problems experienced on the Columbia and Sacramento rivers.[15] It appears that open access had begun to lead to reductions in returning stocks. Licenses were introduced, but as much for their revenue potential as to deter entry. In 1887, the first formal attempts were unsuccessfully made to restrict entry. The first restrictions on entry were made in 1908, but new licenses were issued in 1914 and limitations on licenses were suspended completely in 1917. Effective entry restrictions were not implemented again until the late 1960s.

Between the end of the First World War and the 1950s, pressure on the fish stocks increased both from new entrants and increased investments in capital and labor by both incumbents and new entrants. The growing problem of overcapitalization had long been recognized. The 1917 report of the BC Commissioner of Fisheries noted: "The solution of this problem would not seem to be found in encouraging or permitting the employment of more capital or more labour than can efficiently perform the work.... If the cost of production becomes too great all hope of advantage to the public as consumers will disappear."[16]

The Sinclair Report/Davis Plan

With pressure on the fish stocks mounting, the federal government commissioned Sol Sinclair, an economist, to provide a detailed report on the fishery and to make recommendations for improving its management. Sinclair identified three policy alternatives: creating exclusive property rights to the resource through a monopoly, using taxes to discourage overcapitalization, and closing entry.[17] In spite of these recommendations, the federal government was reluctant to stop entry and chose to restrict fishing incrementally, especially through restrictions on fishing times and locations, which still dissipated the resource rent but reduced the risk of physical exhaustion of the stocks. The other alternatives were even less politically popular as they would directly harm incumbent fishers—the most organized and vocal group. License limitation was introduced in 1968 under the Davis Plan (named after the then Minister of Fisheries) to little opposition: "incumbent vessel owners generally approved of the plan … the revocation of rights provoked no public outcry … the interest group adversely affected (potential participants) was dispersed, unorganized, and undoubtedly incapable of calculating the loss."[18]

The Davis Plan introduced the first effort to reduce fleet size by a voluntary buy-back. But this aspect of the plan suffered from an ongoing "vicious circle": the buy-back raises the market value of all remaining licenses (the anticipated fishing effort reduction "capitalizes" into the remaining licenses). In turn, this raises the cost of the buy-back and quickly exhausts the buy-back budget. Clearly, even after the buy-back, there remained many more than the number of vessels that would maximize rent.[19] A final important consequence of the Davis Plan was that it effectively "froze" vessel types to seine, gillnet, troll, and gillnet-troll boats, which have very different capabilities and efficiencies.

Apart from failing to eliminate excess incumbent boats, the Achilles heel of the Davis Plan was that it did not curb overcapitalization unrelated to entry. Indeed, it worsened the problem. Incumbents were now relatively fixed in number and therefore only had to compete with each other for the rents. They did this by replacing older boats with newer ones and purchasing ancillary technology, a process known as *capital-stuffing*. This is a widely occurring problem in fisheries with entry restrictions.[20] The buy-back was somewhat effective through most of the 1970s as salmon prices rose, but landed values dropped substantially between 1978 and 1983. New entrants who gained access by purchasing licenses from incumbents found themselves facing a financial crisis (many "old" incumbents who had held licenses when the Davis Plan was implemented had become rich enough to retire on their windfall gains). These conditions led to the 1982 Pearse Commission.

The Pearse Commission

The Pearse Commission again noted the disastrous state of the fishery: "the economic circumstances of the commercial fisheries are exceptionally bleak … [and] there is growing concern about the precarious condition of many of our fish stocks."[21] In spite of this situation, economist Peter Pearse pointed out that the primary problem was *not* the size of the harvest, which had remained relatively stable for approximately 50 years, whether measured in terms of the number of fish or landed weight. The combination of restricted openings, gear restrictions, riverine protection, and salmon enhancement were generally succeeding in protecting the viability of the stocks. However, Pearse argued that the catch was still considerably below the potential sustainable yield. This was partly because of the degradation of fish habitat by the forest industry and other land-based developments that are subject to provincial jurisdiction.

The Pearse Commission concluded that previous efforts to limit boat length and tonnage alone had not sufficiently restricted capital-stuffing: "when one or more inputs in the fishing process are restricted, the capacity of the fleet can continue to increase by adding other unrestricted inputs."[22] Although regulators could continue to restrict more input dimensions, "such restrictions would have to be so numerous and diverse (covering vessel size, power, crew, time spent fishing, gear for finding, catching and holding fish, and so on) that they would be virtually impossible to administer and enforce."[23]

The Pearse Commission's most important recommendation was to impose royalty taxes on the salmon catch: "a realistic portion of revenues derived from commercial fisheries should now be directed away from excess catching capacity toward recouping these costs to the public, reducing bloated fleets and enhancing the resource."[24] Note that although the distributional consequences of the tax are emphasized in this quote, such a tax would be efficiency enhancing because the revenues that would now go to government previously represented economic waste. Thus, such a tax both preserves and transfers the rent. Pearse recommended that a royalty rate of between 5 and 10 percent of the gross value of the harvest be imposed on buyers (that is, mostly on the fish processors). Pearse did not discuss the incidence of this royalty tax, which would be spread among foreign consumers, domestic consumers, processors, and fishers.

Pearse also recommended reducing the fishing capacity of the fleet by approximately 50 percent by buying back licenses and by auctioning licenses for ten-year periods. During the first ten-year period, only existing license holders would be allowed to bid. Bids would be in sealed envelopes and would be ranked from the highest offer downward. The bid value of the lowest accepted bid would determine the price to be paid by all successful applicants. Each license would specify the gear, vessel capacity, and zone from which fish could be taken. No individual or corporation would be allowed to acquire new licenses if, as a result, they would hold more than 5 percent of the total available licenses. Finally, during the initial ten-year period, incumbent fishers could sell their license to a buy-back authority at market value. Pearse acknowledged that the proposal to reduce fishing capacity by 50 percent was greater than apparent excess capacity, but argued that "many underestimate the *potential* capacity of the fleet unencumbered by many of the restrictions on time, location and gear that have been imposed to constrain fishing power."[25]

The major Pearse Commission recommendations relating to the salmon fishery were not implemented.

1996–2001: The Mifflin Plan and Subsequent Restructuring

At the end of 1995, the size of the fleet was much as it had been in 1982, at the time of the Pearse Commission.[26] However, fishing power had increased enormously over this period because of continued capital-stuffing. The fishing power of the fleet meant that the major mechanism available for protecting fish stocks was strong restrictions on the time in which the fishery was open. While this approach generally protected the viability of stocks, for many river runs there was little margin for error because of the immense fishing capacity that could be deployed. This, in turn, put tremendous pressure on marine scientists in the then Department of Fisheries and Oceans (DFO) to get their estimates of specific run sizes right.

The federal government introduced the Pacific Salmon Revitalization Strategy in 1996, known as the Mifflin Plan after the then DFO minister. The plan proposed to reduce the fleet in two ways. First, $80 million was allocated to a voluntary license buy-back program. Under this buy-back, the total fleet was reduced by approximately 27 percent of the extant capacity. Second, "area licensing" was imposed, consisting of two geographic regions for seine vessels and three regions for gillnet and troll vessels. Existing license holders had to select one geographic area. However, they could "stack" licenses by purchasing them from other license holders. The plan did not address other aspects of capital-stuffing.[27] Area licensing was an attempt to reduce congestion externalities but, in view of the large number of vessels remaining in each area, the impact was small. In 1998 and 1999, the federal government committed an additional $400 million to the fishery, with approximately half the budget for further buy-backs.[28] By 2002, the fleet was down to 1,881 vessels, a 57 percent reduction.[29] The Mifflin Plan and its immediate successors largely represent the current salmon fishery regime.

The 2012 Cohen Commission of Inquiry

The $37 million Cohen Commission, which commenced in 2009, was the most expensive review of the Pacific coast salmon.[30] It focused primarily on the Fraser River salmon fishery. Once again, the commission commented on the well-known weaknesses of "derby" fisheries and noted that the ministry was running experimental share-based ITQs. However, Commissioner Cohen did not explicitly recommend moving to an ITQ, but rather that, though "I support in principle DFO's commitment to moving to share-based management, it is not realistic to do so without first completing its analysis of the socioeconomic implications of implementing the various management models."[31] The most charitable assessment of the commission is that it only delayed the potential for serious reform of the salmon fishery by four or five years.

The next section of this report summarizes the evidence on the current state of the fishery, and projects the probable impacts these changes will have in the future.

The Current and Expected State of the Fishery

Will fleet reductions eliminate or substantially reduce rent dissipation? Will they adequately protect salmon stocks? Buy-backs have certainly resulted in a substantial short-run reduction in capacity. Yet the historical experience shows that the remaining incumbents will be tempted into a new round of capital-stuffing. In order to assess the current state of the fishery, it is useful to ask: What is the economic cost of rent dissipation, and how might buy-backs and other forms of cost reduction mitigate it?

A cost–benefit analysis (CBA) of the BC salmon fishery addressed some of these questions, although it was conducted based on data before both some buy-backs and declines in salmon volume and value of landings. The analysts estimated annual costs and benefits for specific elements of the fishery: value of the harvest, harvesting costs of participants in the industry, and government expenditures on management, enhancement, enforcement, and unemployment insurance payments.[32] The *net present value* (NPV), a measure of the stream of net benefits of the fishery expressed in terms of an equivalent amount of current consumption, under various assumptions, was estimated using a 25-year time horizon and a base-case (real) discount rate of 5 percent.

The analysis assumed that the fishery was being managed close to its maximum sustainable yield. The initial estimate of the NPV of the fishery assumed no Mifflin Plan. Under this assumption, the NPV was *negative*, indicating net social losses in terms of current consumption equivalents. The NPV of the fishery was estimated at between −$784 million and −$1,573 million. Thus, its operation prior to the Mifflin Plan appeared to involve a substantial net loss to the Canadian economy.

The Mifflin Plan was primarily a buy-back plan, but it also increased locational constraints through area licensing. Subsequent buy-backs involved some other relatively minor policy changes. One optimistic assumption is that all of the boat retirements induced by the buy-backs will go straight to "the bottom line" *and stay there*. In other words, the boat retirements will represent capital permanently removed from the fishery. The CBA authors present a number of alternative estimates of the impact that the Mifflin buy-back (but not the subsequent buy-backs) might have on both private and public costs. They estimate that if private costs are reduced by 27 percent, approximately equivalent to the percentage of the fleet that has been retired under Mifflin, and public costs are unchanged, then the NPV would be $101 million.[33] This is a significant improvement over the pre-Mifflin NPV and indicates modest net economic benefits from operation of the fishery.

However, significant economic rents would still be forgone. In order to estimate the forgone rents, the projected net benefits of the current fishery should be compared to the potential benefits of an efficient fishery. In order to do this, one must make a number of assumptions about what an "efficient salmon fishery" means. Schwindt, Vining, and Globerman argued that, even organized as a post-Mifflin small-boat fishery, it could be fully exploited with approximately half the extant resources that were employed at that time. They based this estimate on the costs of the more-efficient Alaskan salmon fishery, which had a lower-cost fleet. Assuming that the fishery was managed in this way, and also assuming that public costs remained the same, they estimated that the NPV of the fishery would increase to $1,140 million.[34]

In 2003, Schwindt, Vining, and Weimer provided NPVs that take into account the probable impact of subsequent buy-backs on private costs. They concluded that "if private costs are reduced (approximately equivalent to the percentage of the fleet that has been retired through buy-backs or stacking) and public costs remain the same … the NPV of the fishery would be about $1,258 million."[35] Although this NPV represents some improvement over $1,140 million, it is not much higher because the increased benefits of further fleet buy-backs are offset by new, less-optimistic price and volume predictions. Under these less-optimistic assumptions, the NPV is likely to be negative, and possibly as substantial as −$951 million, a major loss to the Canadian economy.

It is useful to have some handle on the sources of rent dissipation in thinking about policy alternatives. Fleet composition is probably the major source of rent dissipation, with excess boats second and other margins of input substitution third.[36] Buy-backs did not directly address these issues, although more troll and gillnet boats were retired than the more-efficient (in terms of effort-to-catch ratio) seiners. This suggests that incremental reform that does not alter, or eliminate, boat types will not be able to realize the largest potential rent gains.

In spite of the introduction of area licensing, current policy is also unlikely to protect vulnerable fishing runs adequately, especially if there is renewed capital-stuffing. There is still excess capacity hovering over particular runs. The number of salmon can vary greatly from year to year, from area to area, and from species to species—sometimes by twenty- or thirtyfold. Under these conditions, incorrect volume and escapement estimates by Fisheries and Oceans Canada could be catastrophic for preservation of runs.

Policy Goals

What policies should govern the BC salmon fishery? An answer to this question requires the specification of policy goals that provide an appropriate basis for comparing current policy with possible alternatives. The preceding discussion of problems inherent in the status quo immediately suggests a number of important goals.

First, the salmon fishery is a potentially valuable resource, but because of rent dissipation it is likely a net drain on the wealth of British Columbians and Canadians. A primary policy goal, therefore, should be *economically efficient use* of the fishery. The primary measure of the predicted impact of each alternative in terms of this goal is the *extent of rent dissipation.* As the prediction of rent dissipation will be based on the logic of common property resources, it is important to also consider the feasibility of implementation in assessing economic efficiency. The two impacts used for assessing implementation feasibility are the *ease of enforcement* of regulations, and the *degree of flexibility* to allow the policy to accommodate the dynamic and cyclical nature of salmon stocks.

Second, the current policy poses a risk to the preservation of salmon runs. Most Canadians place a positive value on preserving the fishery for multiple uses and future generations.[37] If these amounts could be measured, then they could be included in economic efficiency. However, as they cannot be measured for this analysis, *preservation of the fishery* is included as a separate policy goal. The primary impact category for measuring progress toward this goal is the impact on the *number of viable salmon runs.*

Third, the costs and benefits of the fishery should be fairly distributed among stakeholders. *Equitable distribution* should be a policy goal. Fairness requires that current license holders, who have made investment decisions based on a reasonable expectation that current policy will continue, receive explicit consideration. As current license holders, who may or may not themselves be fishers, are likely to be highly attentive to proposed policy changes and very vocal in opposition to changes they view as harmful, consideration of their interests is likely to contribute to the political feasibility of any policy alternative, an instrumental value beyond fairness itself. Aboriginal peoples have cultural, economic, and constitutionally protected stakes in the use of the salmon fishery. Any policy change should at least protect their interests. As alternative policies have implications for government revenues and expenditures, fairness to Canadian taxpayers should be a concern. (In this analysis, therefore, we treat impact on government revenues as a fairness issue, although it is certainly reasonable to consider it as a goal in its own right.) The federal government still faces a large accumulated debt that burdens Canada with a high ratio of national debt to gross domestic product. Therefore, reducing subsidies from taxpayers to the fishery, or even capturing some fishery rent for the public treasury, is desirable. These considerations suggest that for assessing progress toward achieving an equitable distribution, we are interested in the impact on three groups: *fairness to current license holders, fairness to Aboriginal fishers*, and *fairness to taxpayers.*

Fourth, *political feasibility* is always relevant to some degree. Radical alternatives are usually more politically difficult to achieve than more incremental change. There is more uncertainty as to the impacts of more radical policy alternatives, especially distributional impacts.

It is important to note that these goals are often in conflict. For example, while allowing small salmon runs to be extinguished might be consistent with a narrow definition of economic efficiency, it would conflict with the goal of preserving the fishery. Therefore, selecting the most socially desirable policy involves making tradeoffs among the goals.

Some New Ways of Organizing the Salmon Fishery

The analysis presented in the remainder of this report compares current policy to the following three alternatives.

The Pearse Commission Proposal: Harvesting Royalties and License Auction

This alternative revives the Pearse Commission recommendations. Specifically, it includes a royalty tax of 10 percent, the ten-year rolling license auction, and the 50 percent buy-back, as described above. The Pearse Commission also proposed providing financial assistance to First Nations bands to further their participation in license purchases ($20 million over five years) and a ban on licenses that were held by Aboriginal fishing corporations from being sold to non-Aboriginals. As described earlier, extensive buy-backs have now taken place (although not in the manner recommended by Pearse), but most of the other recommendations of the Pearse Commission were never adopted. Many fisheries experts regard the Pearse Commission recommendations as exemplary.

River-Specific Exclusive Ownership (with Cooperative Ownership on the Fraser)

This alternative largely follows the proposal put forward by Schwindt, Vining, and Weimer.[38] Under this alternative, the right to catch all salmon entering each river system would be allocated to a single owner for a period of 25 years ("single" ownership could include corporate or nonprofit ownership). During the last five years, incumbents would only be allowed to catch the same average number of fish as in the first five years of their tenure. The small-boat fishery would be bought out and disbanded. Owners would almost certainly adopt fish traps and weirs, the lowest cost method of catching salmon.[39] Current incumbents of the small-boat fishery would receive compensation for their withdrawn licenses from the government equal to license market value plus a 50 percent premium. Payments would be annualized over 15 years; the payments would include interest at market rates.

Compensation necessarily has some element of arbitrariness. The rationale for a level of compensation greater than license value is that market value would fully compensate marginal fishers (who, with the payments, would be indifferent between staying in or leaving the fishery) but not fishers who would not be willing to sell at market prices (whether because of superior skill, utility from the fishing lifestyle, or other reasons). First Nations would receive a percentage of the ownership rights at least equal to their current (approximate) 19 percent of commercial fishing licenses. Fish hatcheries would be included in the ownership rights, and owners would have the right to continue using them or closing them down.

The Fraser River is the only river system in BC for which sale of rights to a single owner is not practical. The Fraser accounts for more than 50 percent of the sockeye, pink salmon, and chinook originating from Canadian Pacific waters.[40] Sockeye (approximately 30 percent of the total catch) and pink (approximately 48 percent of the total catch) are the most important species in the commercial fishery by a considerable margin. Although other rivers are important for particular species (for example, the Skeena accounts for approximately 12 percent of Pacific sockeye runs), no other river presents insurmountable "single owner" problems. On the Fraser, the federal government would establish a cooperative.[41] It is essential that this cooperative be organized so that it is not subject to its own set of open-effort problems. In view of the number of appropriators involved and the other characteristics of the Fraser fishery, a reasonably efficient fishery should be achievable.[42]

Individual Transferable Quotas

One widely recommended regime for fisheries is ITQs. ITQs are now widely used in New Zealand, Iceland, Australia, South Africa, the United States, and Canada.[43] Specifically, they have been used with some evidence of success in both the BC halibut fishery[44] and the BC sablefish fishery.[45] A tremendous amount has been learned from the experiences in New Zealand and Iceland as to the appropriate structure of ITQs in fisheries with high stock variability.[46]

ITQs would be allocated to percentage shares of the salmon total annual catch (TAC). Government biologists would continue to predict the TAC and would have the power to adjust it throughout the harvesting season. Fixed quotas offer more certainty and, therefore, value to fishers, but allocation of shares rather than a number or weight of fish ensures that fishers bear the risks associated with the large temporal fluctuations in the salmon harvest.[47] The fishery would remain a small-boat fishery. Once allocated, quotas would be transferable to other incumbents or to potential new entrants. The quotas would be allocated to incumbents in the small-boat fishery based on previous average catch history.[48] The government would also collect a tax from each share-owner based on the amount of quota held, rather than on fish caught. The proposed fee would equal 5 percent of the value of the TAC. While it is not possible to discuss all the details of this alternative here, it would operate very much like the current New Zealand regime, which is explained in depth elsewhere.[49]

A Comparison of the Alternatives

The following discussion compares current policy, harvesting royalties and license auction, river-specific auctioned monopolies, and individual transferable quotas in terms of economically efficient use, preservation of the fishery, equitable distribution, and political feasibility.

Current Policy: The Mifflin Plan and Subsequent Restructurings

Efficiency: Current policy will perform poorly in terms of efficiency, especially over the longer term. Over time, new capital-stuffing will negate any short-term benefits in regard to capacity reduction. Within ten years, there will probably be little difference in effective capacity compared to that at the time of the 1996–2001 buy-backs. At best, there would be a small decrease in capacity. Also, as with all the small-boat alternatives, current policy does not move the fishery toward the lowest-cost catch technology. It is an inflexible regulatory regime. Experience with it helps counter its inherent difficulties to some extent. Therefore, in NPV terms, current policy will likely produce efficiency outcomes essentially no different from those occurring in the pre-Mifflin period.

Preservation: This is probably the most difficult consequence of current policy to predict. Much depends on specific decisions made by DFO biologists. The outcome could range from "poor" (where small, variable stocks are continually under threat) to "reasonable" (where these stocks are protected). However, it is impossible to be optimistic under current policy because of the tremendous fishing power that continues to "hover" at every river opening and the continued incentive to "race to fish" (although the situation has been somewhat improved by the introduction of area licensing that reduces excess fishing capacity at particular river openings). Any miscalculation by DFO, whether by the biologists, by more senior officials succumbing to political pressure, or from enforcement difficulties, could lead to small-run extinctions.[50] Thus, at best, current policy must be rated as medium in terms of this goal.

Equity: On one hand, current policy can be thought of as being very generous to incumbents: it provides them with de facto property rights to the resource at nominal prices. On the other hand, if the rent is largely dissipated, this is not the bargain it seems to be. The situation is approximately the same for Aboriginal fishers. The current system, however, is very inequitable to taxpayers: as documented earlier in the analysis, rather than collecting resource rents, taxpayers end up paying a large net subsidy to the fishery.

Political feasibility: Absent reductions in viable runs or a Supreme Court of Canada–imposed reallocation of fishing rights to First Nations, keeping the current regulatory regime in place is politically feasible.

The Pearse Commission Proposal: Harvesting Royalties and License Auction

Efficiency: A major advantage of this policy alternative relative to current policy is in terms of efficiency: it would preserve more of the potential rent. The originally proposed license buy-back would have considerably reduced excess fishing capacity. Although almost 20 years later, the percentage reduction recommended by Pearse has now been achieved. It is, of course, impossible to know what further reductions

Pearse would now regard as appropriate in view of technological progress in the intervening 20 years. Equally important, this regime would eliminate incentives (through royalties and the license auction) to begin another round of capital-stuffing to replace the retired capacity.

Because the Pearse proposal retains the small-boat fishery, it probably only has the potential to reduce fishing costs approximately halfway to their lowest potential level (that is, costs under an efficiently organized terminal fishery). However, how much rent is preserved crucially depends on the extent to which the combination of royalty rate and auction would actually extract rents. A 5 percent royalty rate would not preserve and extract much of the rent, but the auction license, if competitive and appropriately designed, should preserve and extract most of it. We know a great deal more about the practical issues involved in running public-sector auctions than was known in 1982, at the time of the Pearse Commission, so they can be more confidently implemented.[51] This proposal would also encourage the gradual transfer of licenses to fishers who can exploit the resource most efficiently. The primary efficiency weakness of this alternative is that it takes the small-boat fishery as a given, so that the harvest cannot be taken at lowest cost (similarly, see discussions of status quo above and ITQs below).

Enforcement under this regime would be more complex than under current policy, as taxing on the basis of catch volume would require extensive monitoring of all boats because incentives for "highgrading" (discarding overboard of less-valuable fish and the substituting of more-valuable fish to stay within catch limits—it is individually rational but socially wasteful) and "quota busting" (smuggling) would be much greater. Threat of license loss should be the major deterrent to such behaviors.[52] Nonetheless, the need for more extensive monitoring makes enforcement more difficult than under current policy.

Preservation: A weakness of this alternative is that it can endanger specific fish stocks and thus it is weak in terms of resource conservation. Indeed, this is probably its major weakness. The reason is that a royalty tax is inherently a price mechanism rather than a quantity (restriction) mechanism. There are risks in using price mechanisms when there is uncertainty over stocks and the costs of being wrong are potentially high in terms of species, or subspecies, extinction.[53] Quantity mechanisms, such as quotas and opening restrictions, lower the risk of catch "overshooting" that can threaten stock viability. Of course, it is possible to design an intermediate alternative that incorporates opening (that is, quantity) restrictions, but this raises (public) expenditures on the fishery. Such a proposal is not examined here.

Equity: The major concession to existing fishers in the Pearse proposal was the ongoing buy-back of licenses. This would allow those who were prepared to leave the industry to avoid capital losses. Remaining fishers would have greatly reduced incentives to engage in overcapitalization because the government would be extracting the rent from the fishery, partly in royalties but largely through the license auction. However, if the license auction process is reasonably competitive, all expected rents would be transferred to the government. It is, therefore, remaining fishers who would suffer the loss of economic rents. There is some justice in this in that they are presumably those who would get the most utility from actually fishing.

Pearse proposed some financial assistance to Aboriginals to acquire licenses over a five-year period (see above). Both perception and law have changed since 1982 as to what would be fair to Aboriginals, and many would now argue that Pearse's subsidy to the First Nations population was too low. The Supreme Court of Canada might well take that view. Nonetheless, it would still place Aboriginals in a considerably better position than does current policy.

This alternative would generate extensive (new) government revenues (probably more through the auction than the royalties) and might reduce government expenditures slightly. The reduced government expenditures should occur because fishing capacity would be reduced, thus reducing fleet monitoring costs. In general, the taxpayers' position would be much improved. Estimating the size of the benefit is difficult, though, because auction receipts are hard to predict.

Political feasibility: The loss of rents by remaining fishers would likely mobilize them in opposition to the plan.

River-Specific Exclusive Ownership Rights (with Cooperative Ownership on the Fraser)

Efficiency: The advantage of this alternative is that it is the *lowest-cost* fish-catching regime and, therefore, offers the potential of being the most efficient at considerably lower cost than the small-boat fishery alternatives.[54] As owners would have long-term property rights, they would have the correct conservation incentives. It is important to emphasize that the term "monopoly" as used here is only a monopoly to a single river system and, therefore, does not imply the inefficiencies society would normally incur from monopoly supply of a good. This form of monopoly would be efficiency enhancing because much of the open effort associated with the small-boat fishery would be eliminated, without creating any effective market power. Indeed, even if a single owner controlled all of the BC monopolies, that owner would be unable to exercise much monopoly pricing power, because salmon are sold into a competitive, global market. (Provided, of course, the government continues to allow free trade in salmon.)

The major problem with this alternative, from an efficiency perspective, is that, as discussed above, it would not be possible, or indeed appropriate, to create a single monopoly owner on the Fraser River. It is important that the regime proposed for the Fraser eliminate capital-stuffing and other forms of open effort. In order to ensure this, there would have to be a fixed, relatively small number of license holders.[55]

Preservation: This alternative also is likely to do well in terms of resource conservation, for two reasons. First, the monopoly owners have strong and unambiguous incentives to maximize the NPV of their stocks. Second, information is clearer as to the link between escapement and specific stock preservation. This link may be somewhat more difficult to establish for the Fraser River, potentially requiring a more extensive government oversight role.

Equity: Under this alternative, the small-boat fishery is retired; on the face of it, this appears to be a major loss for incumbent fishers. However, because this alternative is so efficient, it offers the potential for considerable rent, presumably well in excess of the $2 billion NPV estimated as being the best-case scenario under an efficiently organized small-boat fishery. Some of this rent could be used to compensate existing fishers—either in the form of direct payments, as described above, or by offering them preferential bidding rights (for example, by allowing them to bid with a 50-cent-per-dollar subsidy).

This alternative allocates a fixed percentage of river monopolies to First Nations. It is, therefore, equitable to them relative to current policy. The alternative would also enhance government revenues considerably, especially after small-boat license holders have been reimbursed. Overall, it would probably generate the largest benefit to taxpayers. The reason is that the river monopolies would be extremely valuable resources because they offer a low-cost regime for catching the fish. Therefore, licenses would likely be auctioned for high prices; the total revenue from all river auctions should exceed considerably the total revenue from auctions of small-boat fishery licenses as proposed by Pearse.

Political feasibility: This alternative is the most radical departure from the existing regulatory regime. It would induce a complete restructuring of the industry, which, along with its novelty, would make it politically difficult to achieve.

Individual Transferable Quota Scheme

Efficiency: Provided that fishers are reasonably certain that they can make their quota using current levels of inputs, ITQs should reduce incrementally overcapitalization incentives, although not congestion externalities at specific river mouths. While, as yet, there have been no reported evaluations of ITQ fisheries based on actually realized costs and benefits, detailed reviews of both the New Zealand and Icelandic experiences with ITQs are now available. Annala concludes that the efficiency impacts have been positive and that "the economic performance of the industry has improved."[56] Árnason has similarly examined the impact of ITQs on the Icelandic fisheries. In general, he finds that total catches have increased and capital investment has been reduced, resulting in greatly improved "catch per unit effort."[57] Given that the small-boat fishery would remain, and given the other factors discussed here, this alternative would reduce, but not minimize, rent dissipation, because it would not induce a switch to the lowest-cost technology.

There is some disagreement on the administrative costliness of ITQs. Incentives to engage in both quota-busting and highgrading are inherent to ITQs.[58] However, Annala, based on the New Zealand experience, considers one of the major gains from ITQs to be "minimal government intervention."[59] In contrast, Grafton concludes on the basis of a multi-country review that "an essential component to the success of ITQs, therefore, is adequate monitoring and enforcement."[60] On balance, the evidence from other countries suggests that these problems should be solvable in the salmon fishery without substantial increases in public costs.

Preservation: ITQs regimes have generally been successful at preserving or even enhancing stocks. Resource conservation problems have occurred when total allocated catch has been set too high: this is an endemic problem that has little to do with ITQs, per se. In these cases biological preservation (or recovery) has remained a problem.[61] The salmon fishery is probably more vulnerable in this regard than most other fisheries where ITQs are being utilized (although some of these fisheries, such as the Icelandic capelin, also experience high stock variability). One advantage that ITQs would have in the salmon fishery is that the TAC under the new regime would not be very different from the current catch. In many other fisheries, ITQs have been introduced in open-access contexts or other situations in which the resource was in serious danger and resource-conserving TACs needed to be set considerably lower than current catches.[62]

Equity: Because the proposed regime allocates ITQs to incumbents with relatively low tax rates, it would be generous to those currently in the fishery. As rent dissipation would be substantially curbed, the rents would largely accrue to incumbents who would be considerably enriched. This has been the main criticism of the ITQ system in Iceland, where quota fees only amount to approximately 0.5 percent of estimated catch value: "Why, it is asked, should a relatively small group of fishing firms and their owners be handed, more or less free of charge, the extremely valuable property rights to the Icelandic fisheries?"[63] The proposal here is a considerable improvement on the Icelandic regime, as it would introduce a 5 percent quota fee.

This alternative does nothing for Aboriginal fishers, except to the extent that they hold commercial fishing licenses. Additionally, this alternative does relatively less for taxpayers, as the only rents transferred to them would result from the 5 percent quota fee. It is not clear that current government expenditures would be reduced either, because of the potential for high administrative costs.

Political feasibility: The generosity of this plan for incumbent fishers, the most attentive stakeholder group, makes it the most politically feasible alternative to the status quo. The Cohen Commission implied that it supported ITQs, as does the DFO.

Assessment and Recommendation

The issues discussed in this section of the report are summarized in a simple matrix (Table 1 in the Appendix) that presents policy alternatives on one dimension and the goals for assessing them on the other. It should be stressed that these are predictions, based on the extant research, of how each of the alternatives would perform in terms of the stated goals. Of course, the preferred alternative depends on how the Minister weights the goals described above. However, because the Minister has instructed us to make a recommendation, we do so based on the following comparisons.

As indicated in Table 1, all three alternatives are superior to current policy terms of almost all the goals except political feasibility. River-specific exclusive ownership rights (with cooperative ownership on the Fraser) is the highest-ranked alternative in terms of reversing and eliminating rent dissipation, because it would result in the small-boat fishery being replaced with fish traps and weirs, the lowest-cost fishing technology. The Pearse Commission proposal and ITQs probably do equally well in preserving rent, but offering more modest gains in the reduction of rent dissipation than river-specific ownership rights, because they each maintain the small-boat fishery. They would encourage license utilization by those who could exploit the resource most efficiently.

None of these three alternatives to current policy stand out as being clearly superior. Our recommendation is that the minister should adopt river-specific exclusive ownership rights (with cooperative ownership on the Fraser). This is, however, a radical departure from current policy, and it would be politically difficult to implement. ITQs are a close second in terms of efficiency and preservation. It is especially important, however, that share quotas (rather than fixed number of fish or weight of fish) be adopted in the salmon fishery because of the cyclical stock variability. The only weakness of the particular ITQ scheme outlined here is that it does not generate a high degree of equity for taxpayers: rent transfers are modest and reduced by monitoring and enforcement costs. However, it would be the most politically feasible of the three alternatives.

Appendix

Table 1 A Summary of Fishery Alternatives in Terms of Policy Goals

Goals	Impact Category	Policy Alternatives: *Current Policy: Continued Implementation of the Mifflin Plan*	*Harvesting Royalties and License Auction (Pearse Plan)*	*River-Specific Auctioned Monopolies (Schwindt Plan)*	*Individual Transferable Quotas to Current License Holders*
Economically Efficient Use	*Impact on Rent Dissipation*	Poor—large negative net present value	Good—considerable improvement over status quo	Excellent—major improvement over status quo, lowest-cost technology	Good—considerable improvement over status quo
	Ease of Enforcement	Medium—inherent difficulties offset by much experience	Low	High, except on Fraser	Medium, but requires close attention to incentives
	Flexibility	Low	Medium—good price flexibility	Medium—good price flexibility	Medium, with "shares" system
Preservation of the Fishery	*Impact on the Number of Viable Runs*	Very poor—high risks	Poor—continued risk for vulnerable runs	Very good except for Fraser River where good if coop supervised well	Good, provided "share" quotas used
Equitable Distribution	*Fairness to Current License Holders*	Nominally fair, but not so in practice because rent is dissipated	Good—generous to existing fishers but less so to remaining incumbents	Very good	Excellent
	Fairness to Aboriginal Fishers	Nominally fair, but not so in practice because rent is dissipated	Good	Good	Excellent for Aboriginal incumbents
	Fairness to Taxpayers	Poor, large net costs	Excellent	Excellent	Acceptable, improvement over status quo
Political Feasibility	*Likelihood of Successful Adoption*	High—in place	Low—not attractive to remaining fishers	Medium—large potential gains, but radical change	High—attractive to incumbent fishers

For Discussion

1. The analysis of the British Columbia salmon fishery included considerable historical background. What are the advantages and disadvantages of including this kind of information?
2. Fisheries around the world are being depleted, either because they involve open access or common property. As should be clear from the analysis of the British Columbia salmon fishery, good analysis requires an understanding of the specific context. For practice, consider a simpler common property problem: Imagine you are on the director's staff of a business licensing bureau and the director has expressed concern about excessive photocopying. Currently, any employee can make copies without an access code. Employees have some discretion over how much photocopying to do to support their reviews of applications. They also have discretion over the extent to which they make copies of key documents for the convenience of those applying for licenses. The bureau's photocopying machine seems to always be in use. Employees sometimes have to queue up to make copies, and the photocopying supply and maintenance budgets always seem to be exceeded. You have a meeting with the director in one hour to discuss ways of dealing with the photocopying problem. Sketch an analysis to help structure your meeting.

Notes

1 R. Quentin Grafton, Dale Squires, and James Kirkley, "Private Property Rights and Crises in World Fisheries: Turning the Tide?" *Contemporary Economic Policy* 14(4) 1996, 90–99, at 90.

2 Fisheries and Oceans Canada, www.pac.dfo-mpo.gc.ca/stats/comm/index-eng.html.

3 Jeffrey Sachs and Andrew Warner, "The Curse of Natural Resources," *European Economic Review* 45(4–6) 2001, 827–39.

4 Robert Deacon, "Incomplete Ownership, Rent Dissipation, and the Return to Related Investments," *Economic Inquiry* 32(4) 1994, 655–83; Richard Schwindt, Aidan Vining, and Steven Globerman, "Net Loss: A Cost–Benefit Analysis of the Canadian Pacific Salmon Fishery," *Journal of Policy Analysis and Management* 19(1) 2000, 23–45.

5 Ibid.

6 On the difficulty analysts face in sorting out these effects, see Daniel Pauly, Ray Hilborn, and Trevor A. Branch, "Fisheries: Does Catch Reflect Abundance?" *Nature* 494(7437) 2013, 303–6.

7 See H. Scott Gordon, "The Economic Theory of a Common-Property Resource: The Fishery," *Journal of Political Economy* 62, 1954, 124–42; and Anthony Scott, "The Fishery: The Objectives of Sole Ownership," *Journal of Political Economy* 63(2) 1955, 116–24.

8 Robert Higgs, "Legally Induced Technical Regress in the Washington Salmon Fishery," 247–79, in Lee Alston, Thrainn Eggertsson, and Douglass North, eds., *Empirical Studies in Institutional Change* (New York: Cambridge University Press, 1996).

9 Russell Barsh, *The Washington Fishing Rights Controversy: An Economic Critique* (Seattle: University of Washington Graduate School of Business Administration, 1977).

10 Richard Schwindt, "The Case for an Expanded Indian Fishery: Efficiency, Fairness, and History," in Helmar Drost, Brian Lee Crowley, and Richard Schwindt, *Market Solutions for Native Poverty* (Toronto: C. D. Howe Institute, 1995); Higgs, "Legally Induced Technical Regress in the Washington Salmon Fishery."

11 Higgs, "Legally Induced Technical Regress in the Washington Salmon Fishery."

12 Fay Cohen, *Treaties on Trial: The Continuing Controversy over Northwest Indian Fishing Rights* (Seattle: University of Washington Press, 1986).

13 Higgs, "Legally Induced Technical Regress in the Washington Salmon Fishery," 273–74.

14 These sources include: Schwindt, "The Case for an Expanded Indian Fishery"; Higgs, "Legally Induced Technical Regress in the Washington Salmon Fishery"; and Richard

Schwindt, Aidan R. Vining, and David L. Weimer, "A Policy Analysis of the BC Salmon Fishery," *Canadian Public Policy* 29(1) 2003, 1–22.

15 Fraser, *Licence Limitation in the British Columbia Salmon Fishery*, Technical Report Series No. PAC/T-77 (Vancouver: Department of the Environment, Fisheries and Marine Services, 1977).

16 Ibid., at 5.

17 Sol Sinclair, *Licence Limitation—British Columbia: A Method of Economic Fisheries Management* (Ottawa: Department of Fisheries, 1960), 101–6.

18 Schwindt, "The Case for an Expanded Indian Fishery," at 106.

19 See Peter H. Pearse, "Turning the Tide"; see also Diane Dupont, "Rent Dissipation in Restricted Access Fisheries," *Journal of Environmental Economics and Management* 19(1) 1990, 26–44.

20 Ralph Townsend, "Entry Restrictions in the Fishery: A Survey of the Evidence," *Land Economics* 66(4) 1990, 359–78.

21 Pearse, "Turning the Tide," at vii.

22 Ibid., at 83; see also R. Rettig, "License Limitation in the United States and Canada: An Assessment," *North American Journal of Fisheries Management* 4(3) 1984, 231–48.

23 Pearse, "Turning the Tide," at 83.

24 Ibid., at 93.

25 Pearse, "Turning the Tide," at 111. See also Frank Asche, Trond Bjorndal, and Daniel V. Gordon, "Resource Rent in Individual Quota Fisheries," *Land Economics* 85(2), 2009, 279–91.

26 Schwindt, Vining, and Globerman, "Net Loss."

27 R. Quentin Grafton and Harry Nelson, "Fisher's Individual Salmon Harvesting Rights: An Option for Canada's Pacific Fisheries," *Canadian Journal of Fisheries and Aquatic Sciences* 54(2) 1997, 474–82.

28 See Schwindt, Vining, and Weimer, "A Policy Analysis of the BC Salmon Fishery," Table 1, at 5.

29 Ibid., 6 and Table 2.

30 *Commission of Inquiry into the Decline of the Sockeye Salmon in the Fraser River: The Uncertain Future of Fraser River Sockeye: Volume 1: The Sockeye Fishery; Volume 3: Recommendations-Summary-Process*, Ottawa, Minister of Public Works and Government Services Canada, October 2012.

31 Ibid., at 39.

32 Schwindt, Vining, and Globerman, "Net Loss."

33 Ibid., Table 9.

34 Ibid., 38.

35 Schwindt, Vining, and Weimer, "A Policy Analysis of the BC Salmon Fishery," at 7.

36 Dupont, "Rent Dissipation in Restricted Access Fisheries."

37 This is known as *existence value*. See Aidan R. Vining and David L. Weimer, "Passive Use Benefits: Existence, Option, and Quasi-Option Value," 319–45, in Fred Thompson and Mark T. Green, eds., *Handbook of Public Finance* (New York: Marcel Dekker, 1998).

38 Schwindt, Vining, and Weimer, "A Policy Analysis of the BC Salmon Fishery," 10–20.

39 Some analysts have argued that in-river fish are inherently of lower quality than those caught out to sea, but this is unlikely to be an issue with modern technology (Schwindt, "The Case for an Expanded Indian Fishery").

40 Pearse, "Turning the Tide", Appendix D.

41 Schwindt, Vining, and Weimer, "A Policy Analysis of the BC Salmon Fishery," at 11–12, describe how such a cooperative could be organized.

42 See Elinor Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action* (Cambridge: Cambridge University Press, 1990) for the institutional characteristics of long-enduring common property arrangements.

43 Grafton, Squires, and Kirkley, "Private Property Rights and Crises in World Fisheries"; see also R. Quentin Grafton, "Individual Transferable Quotas: Theory and Practice," *Reviews in Fish Biology and Fisheries* 6(1) 1996, 5–20.

44 Keith Casey, Christopher Dewees, Bruce Turris, and James Wilen, "The Effects of Individual Vessel Quotas in the British Columbia Halibut Fishery," *Marine Resource Economics* 10(3) 1995, 211–30.

45 Grafton, "Individual Transferable Quotas."

46 See John Annala, "New Zealand's ITQ System: Have the First Eight Years Been a Success or Failure?" *Reviews in Fish Biology and Fisheries* 6(1) 1996, 43–62; and Ragnar Árnason, "On the ITQ Fisheries Management System in Iceland," *Reviews in Fish Biology and Fisheries* 6(1) 1996, 63–90.

47 For a discussion of the trade-offs between shares and fixed quota under ITQs, see Carl Walters and Peter Pearse, "Stock Information Requirements for Quota Management Systems in Commercial Fisheries," *Reviews in Fish Biology and Fisheries* 6(1) 1996, 21–42.

48 For example, see Árnason, "On the ITQ Fisheries Management System in Iceland," 73.

49 For an analysis of the New Zealand model, see Annala, "New Zealand's ITQ System."

50 On these problems in the Atlantic fishery, see A. Bruce Arai, "Policy and Practice in the Atlantic Fisheries: Problems of Regulatory Enforcement," *Canadian Public Policy* 40(4) 1994, 353–64.

51 R. Preston McAfee and John McMillan, "Analyzing the Airways Auction," *Journal of Economic Perspectives*, 10(1) 1996, 159–75.

52 Rick Boyd and Christopher Dewees, "Putting Theory into Practice: Individual Transferable Quotas in New Zealand's Fisheries," *Society and Natural Resources* 5(2) 1992, 179–98; and Annala, "New Zealand's ITQ System."

53 See Wallace Oates and Paul Portney, "Economic Incentives and the Containment of Global Warming," *Eastern Economic Journal* 18(1) 1992, 85–98.

54 See Michael C. Healey, "Resilient Salmon, Resilient Fisheries for British Columbia, Canada," *Ecology and Society* 14(1) 2009, online, www.ecologyandsociety.org/vol14/iss1/art2/.

55 See Ostrom, *Governing the Commons*; and Elinor Ostrom and Roy Gardner, "Coping with Asymmetries in the Commons: Self-Governing Irrigation Systems Can Work,"

Journal of Economic Perspectives 7(4) 1993, 93–112.

56 Annala, “New Zealand’s ITQ System.”

57 Árnason, “On the ITQ Fisheries Management System in Iceland,” 77–80, 83–84.

58 Lee Anderson, “Highgrading in ITQ Fisheries,” *Marine Resource Economics* 9(3) 1994, 209–26.

59 Annala, “New Zealand’s ITQ System,” at 46.

60 Grafton, “Individual Transferable Quotas,” at 18.

61 Annala, “New Zealand’s ITQ System”. Árnason (“On the ITQ Fisheries Management System in Iceland,” at 83) comments on the Icelandic cod TAC: “The catches have simply been excessive compared with the reproductive capacity of the fish stocks … [because] the TACs have been set too high.”

62 R. Francis, D. Gilbert, and J. Annala, “Rejoinder—Fishery Management by Individual Quotas: Theory and Practice,” *Marine Policy* 17(1) 1993, 64–66.

63 Annala, “New Zealand’s ITQ System,” at 88.

17
Cost–Benefit Analysis

Assessing Efficiency

Cost–benefit analysis (CBA) is a technique for systematically assessing the efficiency impacts of policies. CBA came into common use in the United States in the evaluation of flood-control projects in the 1930s.[1] As economists developed new techniques for predicting the costs and benefits of proposed policies, its application grew from infrastructure projects and economic regulation to environmental and social policies. When there is some consensus that efficiency should be the dominant goal in assessing a set of policy alternatives, CBA provides an appropriate decision rule: adopt the policy that offers the largest net benefits. When efficiency is one of the goals in a multi-goal analysis, as is more commonly the case, CBA alone does not provide the decision rule, but it does contribute an important input to it by providing a way to assess efficiency quantitatively. Thus, CBA concepts and techniques have wide application in policy analysis.

Governments seek to promote efficiency by requiring that many types of policy changes use CBA to assess policies. In the United States, a series of executive orders made CBA an important feature of federal rulemaking. In 1981 President Ronald Reagan introduced Regulatory Impact Analysis (RIA) as the framework for contemporary requirements for the use of CBA by cabinet agencies in their rulemaking (Executive Order 12291). Agencies proposing a rule that would have annual effects on the economy of $100 million or more in costs, benefits, or transfers must prepare an RIA that assesses its costs and benefits, considers alternative rules, and selects the one offering the largest net benefits. President Bill Clinton expanded the rules subject to the RIA requirement in Executive Order 12866. In addition to reaffirming the major provisions of the Clinton order, President Barack Obama required that agencies conduct retrospective analyses to assess the accuracy of predicted costs and benefits in prior RIAs to determine if existing rules were economically justified (Executive Order 13563). Subsequently, President Obama required agencies to inform the Office of Management and Budget of their plans for reviews (Executive Order 13610). The president recommended that independent agencies (those led by commissioners with fixed terms of appointment and therefore not subject to executive orders like cabinet agencies) also conduct RIAs (Executive Order 13579).

The resulting volume of CBA performed by agencies is substantial: over the 10 fiscal years from 2004 through 2013, the Office of Management and Budget reviewed 3,040 final rules−−116 of the most significant rules involved estimated benefits of between $261.7 billion and $1,042.1 billion and costs of between $68.5 billion and $101.8 billion (2010 dollars).[2]

CBA is not just a requirement in American governments. The Cabinet Directive on Streamlining Regulation requires Canadian departments and agencies to assess a variety of policies with CBA.[3] The U.K. Treasury Department recommends CBA for the appraisal of central government projects.[4] The Australia Productivity Commission, which advises the Australian government on microeconomic policy issues, employs CBA and advocates its broader use.[5] CBAs must be included in applications to the European Union for funding of transportation and environmental projects proposed for inclusion in its program to promote infrastructure investment in less wealthy member states.[6]

The use of CBA, or at least its protocols, can increasingly be found outside of central governments. Most U.S. states have statutes that require the application of CBA in specific circumstances.[7] Efforts by nongovernmental organizations to value their outputs in terms of the social rate of return on investment employ, or should employ, CBA protocols for assessing their efficiency impacts.[8] Doing so is especially important for social impact bonds: these instruments should tie payments to service providers to the social value of their outputs if they are to be valuable as efficient policy instruments.[9] Properly placing a dollar value on environmental services, either to guide efficient policy choice or to support efficient payment schemes to resource users, also requires application of CBA.[10]

The application of CBA in rulemaking and other contexts demonstrates the value that analysts gain from understanding its core concepts and basic techniques. These concepts and techniques, such as opportunity costs and time discounting, often have application in other types of analysis. Although a full treatment of CBA requires its own book, this chapter provides a broad overview.[11] After previewing an application of CBA and introducing the rationale for focusing on net benefits, it lays out CBA in nine steps: (1) specify current and alternative policies; (2) specify whose costs and benefits count; (3) catalogue relevant impacts; (4) predict impacts over the time horizon of polices; (5) monetize all impacts; (6) discount benefits and costs to obtain present values; (7) compute the present value of net benefits; (8) perform sensitivity analysis; and (9) recommend.

Preview: CBA of a Juvenile Justice Program

The budgetary costs of social programs are apparent to the elected officials of fiscally strapped subnational governments. Less apparent are the aggregate benefits that these programs provide directly to their participants and indirectly to nonparticipants. Recognizing this greater saliency of costs relative to benefits, in 2007 the John D. and Catherine T. MacArthur Foundation launched The Power of Measuring Social Benefits initiative. It funded a review of the application of CBA in ten areas of social policy, ranging from early childhood interventions to prisoner reentry programs. It identified the Washington State Institute for Public Policy (WSIPP) as an organization that routinely produces high-quality CBAs that then influence state policy.[12] In 2011, the MacArthur Foundation partnered with the Pew Charitable Trusts to assist other states and local jurisdictions wishing to adopt the WSIPP approach. After vetting and documenting the computerized WSIPP CBA tool, the Results First Initiative began providing technical assistance to help other jurisdictions adapt it to their local circumstances. By the end of its first five years, the initiative was supporting use of the WSIPP CBA tool in 21 states and four counties.[13]

WSIPP takes several approaches to predicting policy impacts in its CBAs. When the Washington legislature requests reviews of alternatives for addressing some policy problem, WSIPP conducts a *meta-analysis* of evaluations of relevant programs.[14] For example, one of its influential analyses identified community-based programs that could reduce the need to build expensive new state prison capacity.[15] Its meta-analyses seek to identify all relevant evaluations of each type of relevant program, assess the validity of these evaluations, and combine their results to obtain estimates of the likely impacts of the programs were they to be implemented in the state.

Another WSIPP approach involves assessing the impact of ongoing programs. The following excerpt describes its CBA of the King County education and employment training (EET) program. EET was a promising but untested program that was implemented just in King County in 2010. It provides employment counseling and modest financial incentives to increase high school completion and employment readiness for moderate- to high-risk juvenile offenders 15 years old and older. The analysts developed a matched sample of juvenile offenders from two similar counties to serve as a comparison group. The primary outcome was the rate of recidivism over the two years following EET participation for the King County sample and two years for the comparison group. The period of observation was not long enough to estimate the effect of EET on schooling or health related impacts directly. To take account of these effects, the analysts imputed them by drawing on research showing their relationship to recidivism.

Box 17.1 The King County Education and Employment Training (EET) Program

Outcome Evaluation and Benefit–Cost Analysis

Over the past two decades, the Washington State Legislature has focused its operating budget on increasing the use of evidence- and research-based interventions in the juvenile justice system. The Washington State Institute for Public Policy (WSIPP) has evaluated several juvenile justice interventions to determine whether these programs reduce recidivism and whether the benefits outweigh the program costs.

In this report, we evaluate the effect of the Education and Employment Training (EET) program for moderate- to high-risk juvenile offenders. EET was originally established by the King County Juvenile Court in 1982.[16]

The report is organized as follows:

- **Section I** provides background on research-based interventions in juvenile justice and EET, specifically.
- **Section II** outlines WSIPP's methodology.
- **Section III** summarizes the key findings from our evaluation.
- **Section IV** presents our benefit–cost analysis of EET.

A **Technical Appendix** is provided for supplemental analysis and technical detail.

Summary

Education and Employment Training (EET) is a program, currently operating exclusively in King County, for juvenile offenders at moderate- to high-risk to reoffend.

For youth in school, the program provides job readiness training and connects youth with jobs—King County pays the wages. Youth who are not in school must reengage in school, or the program provides assistance to prepare for General Equivalence Diploma (GED).

In 2010, EET was designated a “promising program” by the Community Juvenile Accountability Act oversight committee. At that time, the Washington State Institute for Public Policy agreed to evaluate the program when enough time had passed to measure the program’s effect on recidivism.

This study compares recidivism rates for youth served by EET to that of similar juvenile offenders served by other court programs in Pierce and Snohomish Counties.

We find that EET reduces overall recidivism by 12 percentage points, from 51 percent to 39 percent, compared to youth who participated in typical juvenile court programs.

The overall economic benefits of EET exceed the cost of providing the program to eligible youth. We estimate EET produces $34 in benefits per $1 of costs.

Table 1 Benefits and Costs per Participant for EET vs. Comparison Group (in 2014 Dollars)

Program cost	
EET participants	
Per youth expenditures reported in *King County Children, Youth and Young Adult Services*[*]	$2,857
Comparison group costs	
Weighted average cost of programs provided to youth in comparison group	$2,006
(1) Net EET cost	–$851
Recidivism effects	
Decreased taxpayer costs due to decreased recidivism	$6,276

Decreased crime victim costs due to decreased recidivism	$13,834
Decreased deadweight cost of taxation due to decreased taxpayer criminal justice cost	$3,123
Healthcare-related effects	
Increased healthcare insurance costs to participants due to moving from public to employer or private insurance	−$97
Decreased healthcare insurance costs to taxpayers due to movement from public to employer or private insurance	$353
Decreased deadweight cost of taxation due to decreased taxpayer health care costs	$176
Increased costs to private or employer-sponsored insurance programs	−$388
High school graduation effects	
Increased income to participants due to increased labor market participation	$3,011
Increased tax revenue to taxpayers due to increased labor market participation	$1,284
Positive externalities to society due to greater number of high school graduates	$1,488
Deadweight cost of taxation for program costs	−$426
(2) Total benefits	$29,361
Bottom line	
Total net benefits (cost) per participant (3) Net (benefits - costs)	$28,510
Benefit-to-cost ratio	34.50
Probability of positive net benefits (risk analysis)	100%

Note: * Downloaded from www.kingcounty.gov/~/media/operations/DCHS/2012_KC_Children_Youth_YA_Services_Rev8_31_12.ashx

The table in the extract summarizes the costs and benefits of the EET relative to the mix of programs provided to juveniles in the comparison groups. They are expressed in 2014 dollars for a single participant. The EET costs on average $851 dollars more than the weighted average cost of conventionally provided programs. (As we discuss later, it is this incremental cost of the EET relative to the cost of current programs that is relevant for CBA.) As is commonly the case in social policy applications, benefits often arise from avoided costs. In this CBA, the largest benefit—$13,834—results from the avoided costs to victims resulting from reduced recidivism. The next largest benefits—$6,276—results from savings to the criminal justice system. As expenditures require taxes that inflict efficiency losses on the economy, these savings produce an additional benefit to the economy that is offset somewhat by the higher cost of EET per participant. Algebraically summing all the estimated costs and benefits results in positive net benefits of $28,510 per participant, or a ratio of benefits to costs of 34.5.

These estimates involve a number of uncertainties. Most fundamentally, the statistical estimate of the reduction in recidivism that drives the various benefits is uncertain. The analysts took account of these various uncertainties through a *Monte Carlo simulation* (explained later in this chapter under step 8) in which they specified distributions of values for all uncertain parameters and randomly drew values from these distributions to calculate net benefits many times. The numbers presented are means resulting from these simulation "trials." As might be expected because of the large net benefits per participant, all trials produced positive net benefits, as indicated by the 100 percent probability of positive net benefits in the last row of the table––if, say, only three-fifths of the trials had produced positive net benefits, then 60 percent would have been reported and replication of the program would be much riskier in terms of increasing efficiency.

Net Benefits and Potential Pareto Improvement

The basic principle underlying CBA is the Kaldor–Hicks criterion: a policy should be adopted only if those who will gain could fully compensate losers and still be better off.[17] Such compensation is possible if the policy would produce *positive net benefits*. In other words, when efficiency is the only relevant goal, a *necessary* condition for adopting a policy is that it offers positive net benefits and therefore has the potential to be Pareto improving. As we discussed in Chapter 4, policies that increase social surplus are potentially Pareto improving and, therefore, meet the Kaldor–Hicks criterion. Further, when considering mutually exclusive policy alternatives, the one that produces the greatest increase in social surplus should be selected because, if adopted, it would at least be possible through the payment of compensation to make everyone at least as well off as they would have been under any of the alternative policies.

Many economists treat the move from the Pareto-improving criterion to the *potentially* Pareto-improving criterion as if it were a minor step. It is not. Actual Pareto-improving exchanges are voluntary and, by definition, make someone better off without making anyone else worse off. Potential Pareto-improving moves cannot guarantee that no one is made worse off—just that everyone could be made better off with appropriate redistribution following the move. Thus, while the Kaldor–Hicks criterion is ingenious in that it is "Pareto like," it is also controversial. As Richard Posner, among others, argues, CBA based on the Kaldor–Hicks criterion contains elements of utilitarianism ("the surplus of pleasure over pain-aggregated across all of the inhabitants of society") that the pure Pareto criterion avoids.[18] Most fundamentally, the aggregation underlying the Kaldor–Hicks criterion implicitly compares utility across individuals in terms of a common dollar measure, or *money metric*. Nevertheless, the criterion provides a good test for determining if policies could be Pareto efficient, if not a fully acceptable rule for choosing among policies. Of course, distributional and other values may be relevant to the choice of good policy in any particular case. However, if the Kaldor–Hicks criterion is applied consistently, and the winners and losers of policies differ, it is likely that the resulting portfolio of adopted policies would actually be Pareto improving.

The following nine steps guide application of the Kaldor–Hicks criterion.

Step 1: Specify Current and Alternative Policies

Although CBA focuses on the goal of efficiency, it nonetheless shares a number of features with multi-goal analysis, including the necessity for specifying policy alternatives in sufficient detail to allow prediction of their impacts.[19] An important starting point is a clear statement of current policy: What policy will be in place in the absence of the adoption of an explicit alternative? Current policy provides the baseline against which the impacts of alternative policies should be compared.

The identification and clear description of a current policy is often straightforward. However, it can be complicated in a number of respects. First, current policy in any policy area is usually a complex bundle of laws, rules, and norms. Analysts must identify which elements of the bundle are relevant for assessing alternatives. For example, the CBA of a proposal to reduce class size for a school district's elementary students would begin with a specification of current class sizes. However, current district rules on classroom standards may also be relevant, because smaller class sizes would require the use of more classrooms that might not currently meet the standards.

Second, the impacts of current policy may change over time. For example, demographic changes may bring more or fewer students to the district's elementary schools in coming years. An increase in the number of students would increase the total costs to the district of a class-size reduction; a decrease in the number of students would decrease the total costs. Such cost changes may not just be proportional: a declining number of students might allow accommodation of smaller class sizes without the addition of new classroom space if the policy were phased in over time.

Third, policy changes at superior levels of government may be relevant to specifying "current policy" over time. For example, imagine that the state government has adopted a law that all school districts must provide elementary classes of less than some specified size within five years. If this state requirement is binding on the school district, then the specification of current policy with respect to class size follows district policy only for five years and then becomes the state-imposed maximum.

In each of these cases—changing impacts or exogenously imposed policy change—it may be analytically necessary to specify multiple baselines for predicting policy impacts.[20] For instance, increasingly closer votes in the state legislature on bills imposing a class size limit make such a change possible but not certain. One relevant scenario would be the school district's current policy throughout the relevant time period; another scenario would be a state imposition of a class size limit in the next legislative session. Impacts would then be assessed for each alternative relative to each scenario for "current policy." Valuation of the impacts would eventually lead to an estimate of net benefits for each alternative relative to each current policy scenario. Assigning probabilities to the scenarios then allows analysts to compute expected net benefits for each alternative as the weighted average of the net benefits across scenarios. An alternative analytical approach would be to build the scenario uncertainty into a Monte Carlo simulation of the sort we discuss under sensitivity analysis, a task beyond the basics covered in this introduction to CBA.[21]

Unlike multi-goal analysis, CBA does not display estimates of net benefits for current policy and for each alternative to it. Rather, it presents the net benefits of each alternative policy relative to current policy. These net benefits are *incremental*—they are based on differences in the impacts relative to current policy that the policy alternative would produce. Returning again to the class size example, the reduction in class size might be predicted to result in an increase in the average academic achievement of students. This increase is an incremental impact of the smaller class size policy that would be the basis for a predicted incremental benefit. Predicting in terms of incremental impacts permits focusing on changes that can more confidently be predicted as impacts and monetized as costs or benefits. What are the net benefits of the current class size? Answering this question would require specification of an alternative to current policy, such as “no schooling.” Estimating the net benefits of the current policy relative to no schooling would be much more difficult than predicting the change in net benefits of reducing the current class size. In terms of basic economic analysis, incremental analysis usually requires that we only need to know the position of supply and demand schedules near the equilibrium price and quantity under current policy to predict the prices and quantities at the equilibrium under the alternative policy—in other words, we usually can predict without knowing the positions of the supply and demand schedules over the full range of quantity and price. In summary, the net benefits that CBA reports for alternatives to current policy are actually incremental net benefits predicted relative to current policy.

When CBA is used to assess efficiency within a multi-goal analysis, comparison of alternatives requires setting out current policy as a distinct alternative. In these cases, the assessment of efficiency should still be carried out in terms of incremental net benefits, typically labeled as “net benefits relative to current policy.” Consequently, the cell capturing the efficiency of current policy will show zero. Positive net benefits indicate there are alternatives that are more efficient than current policy; negative net benefits that there are those that are less efficient than current policy.

Step 2: Specify Whose Costs and Benefits Count

“Social” often precedes CBA as a description to emphasize that it seeks to take account of the impact of policies on all members of society. The definition of society for purposes of CBA borrows the legal terminology of “standing,” which means having an interest in a dispute that entitles one to seek judicial relief.[22] In the context of CBA, determining standing means to specify whose costs and benefits count.[23]

The issue of standing arises in the determination of the jurisdictional boundaries of society. Usually, national boundaries define society for purposes of CBA. Unless countries have ceded some authority to supranational bodies like the European Union, they have their own monetary and fiscal policies that make them natural units for economic analyses like CBA. In addition, countries have written or unwritten constitutions that set out rules for their exercise of sovereign authority. While national-level clients readily embrace national standing, clients who are officials in subnational governments often wish to restrict standing to residents of their own jurisdictions.

The choice between national and subnational standing can be important to the assessment of efficiency if policy impacts spill across subnational borders or if they involve fiscal transfers across jurisdictions. For example, consider a subnational government assessing a policy that would reduce the quantity of a pollutant being discharged into one of its rivers. If it adopted standing just for its residents, then it would not include the benefits of the reduction in the pollutant accruing to residents in other jurisdictions along the course of the river. Imagine that the national government offers a grant to the subnational government to undertake a project. From the perspective of subnational standing, the grant is a pure benefit; however, from the perspective of national standing, it represents both a benefit realized by the subnational residents and an offsetting cost realized by national taxpayers—a transfer whose cost and benefit net out ignoring the costs of raising funds through taxation. In situations in which subnational clients demand subnational standing, analysts seeking to promote good public policy should predict costs and benefits from the perspective of national standing, as well as from the perspective demanded by the subnational decision makers.

Determining societal membership for purposes of CBA may also require determining who located within a jurisdiction should have standing. Legality, morality, and common sense tend to align in giving standing to citizens and legal residents of jurisdictions. Considering the standing of other residents is more complicated. Based on the argument that legal constraints should be treated like technical constraints in CBA,[24] undocumented immigrants probably should not be given standing. Yet, in a country that grants citizenship by birth, the children of the undocumented immigrants born in country would have standing by virtue of being citizens. However, as the welfare of children is largely determined by the circumstances of the households in which they live, can the standing of these children be taken into account without also granting standing to their parents? Assessing the costs and benefits of policies that affect those incarcerated by the criminal justice system also raises difficult issues of standing. Does increased consumption by those in prison count as a social benefit? What if the consumption also contributes to increases in human capital? As with issues of appropriate jurisdictional boundaries, absent a consensus on resolving such issues of standing, prudence suggests conducting the CBA from each of the competing perspectives on standing.

Sometimes analysts must also determine the standing of particular preferences. Illegality can often help rule out particular preferences. One early controversy in CBA along these lines was whether or not to count the value of stolen goods to thieves. Some older CBAs dealing with criminal justice policies did impute a value to these goods.[25] However, the recognition of legal constraints as a basis for guiding standing has resulted in almost universal rejection of this imputation. The legal constraint perspective can raise some interesting issues when policies change legal status. For example, consumption of marijuana would not have standing in jurisdictions in which it is illegal, but it would have standing in jurisdictions where it is legal, so that legalization itself produces substantial consumption benefits, even if the quantity consumed remains unchanged!

Controversial and analytically consequential issues of standing often arise in assessing policies that affect addictive consumption. For example, cigarettes are widely recognized as being addictive in that people have difficulty stopping consumption that poses a health risk. Nonetheless, people get some pleasure from smoking. Do policy-induced reductions in smoking result in consumption losses as well as health gains for smokers? This question arose in the context of the CBA supporting the Food and Drug Administration's proposed rules requiring graphic cigarette labeling.[26] The somewhat arbitrary decision by analysts to reduce health benefits to take account of the lost value of consumption to smokers implicitly gave standing to addictive consumption. Many health analysts reject giving standing to addictive consumption in general, as well as to the particular ad hoc adjustment made in the RIA.[27] These objections have halted efforts by the Department of Health and Human Services to develop guidelines for giving standing to demand for harmfully addictive goods in RIAs.[28] Consequently, analysts must resolve this issue of standing in particular applications. As with other issues of standing that involve controversy, they should do so explicitly, perhaps by predicting net benefits for alternative specifications of standing.

Step 3: Catalogue Relevant Impacts

Policies use various kinds of resources to produce impacts. Costs arise from the use of these resources; benefits, which may be positive or negative, arise from the impacts. Cataloguing these sources of costs and benefits sets out the scorecard for conducting CBA. The scorecard should comprehensively take account of all impacts on everybody with standing. Subsequent steps require that appropriate units for resource use and impacts be specified to facilitate prediction and monetization. If one stopped the analysis at this step, a well-constructed scoresheet would comprise an informative qualitative CBA.

The complexity of impacts makes truly comprehensive cataloguing of impacts impractical. Deciding which impacts are large enough to be meaningful requires some judgment. On the one hand, including every possible relevant impact reduces the chance of excluding a significant impact. On the other hand, agonizing over impacts unlikely to be meaningful risks wasting scarce analytical resources. Unfortunately, there is no clear definition of "meaningful." Impacts that advocates and other analysts see as relevant should almost certainly be considered whether or not they are likely to affect net benefits. Other impacts can reasonably be excluded if they are likely to provide benefits or involve costs orders of magnitude smaller than the more important impacts.

As an example, consider the basic microeconomic analyses in Chapters 4 and 5. These analyses measured induced social surplus changes in primary markets, ignoring effects in markets for substitutes and complements. Effects in these secondary markets are generally small, and anyway their contribution to benefits is often implicitly captured in conventional measures of social surplus in the primary market.[29] Therefore, it is generally reasonable to limit attention to the primary market. However, impacts in some markets do have large impacts in others. For instance, in the United States access to public school systems based on residential location means that school quality and changes to it can have significant impacts on housing markets. Consequently, a comprehensive cataloguing of impacts of a policy that significantly affected school quality should include impacts on the housing market.[30]

It may turn out that it is not feasible to monetize some catalogued impacts and therefore these impacts do not enter the eventual calculation of net benefits. If these impacts cannot be confidently excluded based on their likely impact, being orders of magnitude smaller than net benefits, then they should be flagged as potentially relevant, but not quantified, impacts. For example, it may not be practical to assess the value people who do not use public transportation place on the option of using an expanded public transportation system, so that a CBA of the expansion should indicate this option value as an unmeasured benefit. As discussed in the sensitivity analysis step, one can then ask the question: How large would the benefit, positive or negative, of the excluded impact have to be to change the sign of net benefits?

Step 4: Predict Impacts over Time Horizon of Policies

All policy analysis requires prediction. CBA assumes the extra, but valuable, burden of quantitatively predicting impacts over the relevant time horizon.

Determining the Relevant Time Horizon

The guide for determining the time horizon for a CBA is to identify a realistic period over which the policy being assessed has incremental impacts relative to current policy. In the case of an infrastructure project, the expected life of the facility, determined by engineers based on historical experience, is typically used as the time horizon. However, even after the useful life of the facility, the project may still produce impacts. Thus, while engineering practice may specify the useful life of a roadway as 40 years, the CBA could recognize that after 40 years there would still be some value from the project in providing the foundation for a replacement. In such a situation, the time horizon would typically be set at 40 years, but some terminal, or scrap, value would be specified for the final year of the project.[31]

A similar approach can often be applied to other types of policies. For example, an early childhood intervention may produce cognitive gains that affect educational achievement and attainment. A plausible time horizon would be the number of years from the intervention to the normal age of high school graduation. Behavioral impacts may occur during this period, but primary impacts for the analysis would be predicted for the end of the period: an increased probability of attaining a high school diploma and increases in the opportunity for higher education from increased achievement. The valuation of these impacts might implicitly depend on impacts even further in the future, such as higher productivity or reduced adult crime.

Many policies involve the repeated use of the same resources to produce the same impacts. For example, health inspections of restaurants may produce the same expected reduction in the number of foodborne illnesses year-over-year. In such cases, a CBA based on a single year will indicate whether or not the polices are efficient even if they are intended to be ongoing—positive (negative) net benefits in the one year promises positive (negative) net benefits regardless of how many years the program continues. However, if resource use or impacts are expected to change over time, then a longer time horizon is needed. For example, the first year of inspections may induce changes in restaurant hygiene that may make inspections more effective in future years; or, perhaps inspectors eventually become less vigilant and therefore become less effective in future years!

Sources of Evidence for Prediction

In an analytically ideal world, experimental evidence would be available as a basis for predicting the impacts of proposed policies. The impacts of policies exactly like the one being analyzed would have been assessed through comparison of randomly assigned treatment and control groups. The similarity of the programs studied to the one being assessed would guarantee external validity (the appropriateness of application of study results in prediction of the impacts of the policy) and random assignment would guarantee internal validity (the methodological soundness of these results).[32] In the real CBA world, however, we are rarely blessed with such experimental evidence. Relevant experimental evidence is often not available for a number of reasons: the policy being assessed is unique; it applies to units, like subnational jurisdictions, that cannot be reasonably randomized; or, because experiments are costly, none have been done and resources do not permit one to be done.

Sometimes it is possible to get strong evidence from natural experiments, or situations in which fortuitous random assignment provides treatment and comparison groups for inferring impacts.[33] For example, consider the Head Start program established by the U.S. federal government in 1965. The Office of Economic Opportunity provided technical assistance to the 300 poorest counties in the United States to help them prepare applications for participation in the program. This assistance resulted in participation rates that were substantially higher in counties with poverty rates just above the cutoff than in counties with poverty rates just below it. Consequently, researchers were able to identify the effect of Head Start on children's mortality by comparing mortality rates across these counties under the reasonable assumption that their participation rates were effectively random—an increase in Head Start funding by from 50 to 100 percent decreased mortality from program-relevant causes by 33 to 50 percent.[34] As a validity check, reductions were not seen in mortality rates for adults or for injuries to children, which were not expected to be affected by Head Start. A study taking advantage of this sort of boundary effect, which can be assumed to be unrelated to the outcome of interest, is called a *regression discontinuity* design. In this case, the approximately linear relationship between poverty and participation rates jumped near the cutoff for counties qualifying for technical assistance.

Absent the random assignment provided by true experiments, or the near-random assignment of natural experiments, reasonable comparison groups can nonetheless often be constructed for estimating impacts. Another evaluation of the long-term effects of Head Start, for example, compared participating and nonparticipating siblings within the same families.[35] Or sometimes a plausible comparison group can be constructed by using propensity scoring, which identifies nonparticipants who are statistically identical to participants in terms of observable characteristics.[36] WSIPP analysts used propensity scoring to construct a comparison group for their King County EET participants from juveniles in two other counties that did not have EET programs. In some cases, changes in outcomes before and after treatment can be compared to changes in outcome over a similar period for the constructed comparison group. This so-called *difference-in-differences* design helps guard against observed effects of treatment resulting from extraneous factors under the assumption they would similarly affect members of the comparison group.[37] For example, researchers seeking to assess the impact of converting vacant urban lots into greenspaces assessed changes in gun violence, vandalism, and resident stress in the adjoining neighborhoods of converted lots with those in neighborhoods with unconverted lots matched through propensity scoring.[38]

When policies change the effective prices seen by consumers or producers of goods, it is reasonable to use predictions based on elasticities available in the economic literature. Some of these estimates are based on experiments. For example, predicting the effect of a price change on the utilization of medical care can make use of the price elasticity of demand estimated in the RAND Health Insurance Experiment.[39] Other useful elasticity estimates are based on meta-analyses. For example, meta-analyses are available for price elasticities of demand for cigarettes,[40] gasoline,[41] illicit drugs,[42] electricity,[43] alcohol,[44] and residential water,[45] as well as tax elasticities for economic development[46] and various elasticities for public transportation.[47] When the specific elasticity desired for prediction is not available in the literature, and it is not practical to conduct a study to estimate it for purposes of the CBA, finding the most similar elasticity may be the best that the analyst can do. For example, in the next section we reference an RIA that used a price elasticity of demand for general recreation when a price elasticity of demand for swimming by those who are wheelchair bound was the elasticity needed.

When quantitative evidence to support predictions is unavailable, it may be necessary to rely on expert opinions. Experts draw on their personal experience and on rules of thumb that encapsulate collective professional wisdom. If analysts have access to multiple experts, then they can employ the *Delphi Method* or some other systematic way of combining the expert opinions.[48] For example, researchers relied on expert opinion to predict the cost reduction for nuclear power that would result from increases in research and development expenditures.[49] Absent access to experts, analysts may be forced to draw on their own experience to make "guestimates," which are most appropriately expressed as plausible ranges.

No prediction can be made with certainty. Analysts should take account of this uncertainty in their sensitivity analyses. To do so, they will need not just *point estimates*, the particular values of impacts viewed as most likely, but also estimates of their distributions. Most quantitative studies facilitate recovery of distributions by providing standard errors, or confidence bounds, for point estimates. Depending on the statistical model employed, the point estimates and their standard errors usually allow analysts to make a plausible assumption about a distribution. For example, the common assumption of normally distributed standard errors in ordinary least squares regression models, imply that the point estimates have a Student's t-distribution. Expert elicitations and guestimates require analysts to specify plausible distributions, such as assuming that all values within the range are equally likely.

Step 5: Monetize All Impacts

The fundamental concepts guiding monetization in CBA are *opportunity cost* and *willingness to pay (accept).*

Valuing Inputs: Opportunity Cost

Public policies usually require resources (factor inputs) that could be used to produce other goods. Some examples: public works projects such as dams, bridges, and highways require labor, materials, land, and equipment; social service programs typically require professional services and office space; and wilderness preserves, recreational areas, and parks require at least land. The resources used to implement these policies cannot be used to produce other goods. The values of the forgone goods measure the costs of policies. In general, the *opportunity cost* of a policy is the value of the required resources in their best alternative uses.

The nature of the market for a resource determines how one goes about measuring its opportunity cost. If the resource is purchased in an efficient market (no market failures) and the purchase does not increase price (i.e., there is a flat supply schedule), then its opportunity cost is just the expenditures on the resources. In this case, fiscal and opportunity costs are equal. Consider a proposed remedial reading program for a school district.

The additional textbooks are purchased in a national market. They represent only a small addition to total demand for textbooks and, hence, result in a negligible increase in their price. However, if the resource drives up price (i.e., there is an upward-sloping supply schedule), then purchases at the resulting market price overestimate the opportunity costs because some of the expenditure is simply a producer surplus, or just a transfer from the government to producers.[50] For example, the factor may be qualified teachers for the remedial reading program who may demand higher salaries than those being paid to the reading teachers already employed.

Assessing opportunity costs in the presence of market failures or government interventions requires a more explicit accounting of social surplus changes. For example, take the problem of estimating the opportunity costs of labor in a market in which minimum wage laws, union bargaining power, or simply custom keeps the wage rate above the market clearing level. In Figure 17.1, the pre-project demand schedule for labor D and the supply schedule for labor S intersect at W_E, the equilibrium wage in the absence of the wage floor, W_M. At the wage floor, L_S units of labor are offered but only L_D units are demanded, so that $L_S - L_D$ units are "unemployed."

Now imagine that L' units are hired for the project. This shifts the demand schedule to the right by L'. As long as L' is less than the quantity of unemployed labor, price will remain at the floor. The total expenditure on labor for the project is W_M times L', which equals the area of rectangle abL_TL_D. But trapezoid *abcd* represents producer surplus enjoyed by the newly hired and, hence, should be subtracted from the expenditure to obtain an opportunity cost equal to the shaded area inside trapezoid cdL_DL_T. Alternatively, we can think of the shaded area as the value of the leisure time (a good) given up by the newly hired workers.

The interpretation of cdL_DL_T as the opportunity cost of the labor depends on the assumption that the workers hired by the project place the lowest values on leisure among all those unemployed. (They have the lowest *reservation wages*, the minimum wages at which they will offer labor.) Note that this may not be a reasonable assumption, because at wage W_M all those offering labor between L and L_S will try to get the jobs created by the project.

Consequently, the shaded area in Figure 17.1 is the minimum opportunity cost; the maximum opportunity cost would result if the L' workers hired were the unemployed whose reservation wages were near W_M. Actual opportunity cost would likely be some average of these two extremes. In any event, the opportunity cost of the project labor would be below its budgetary cost.[51]

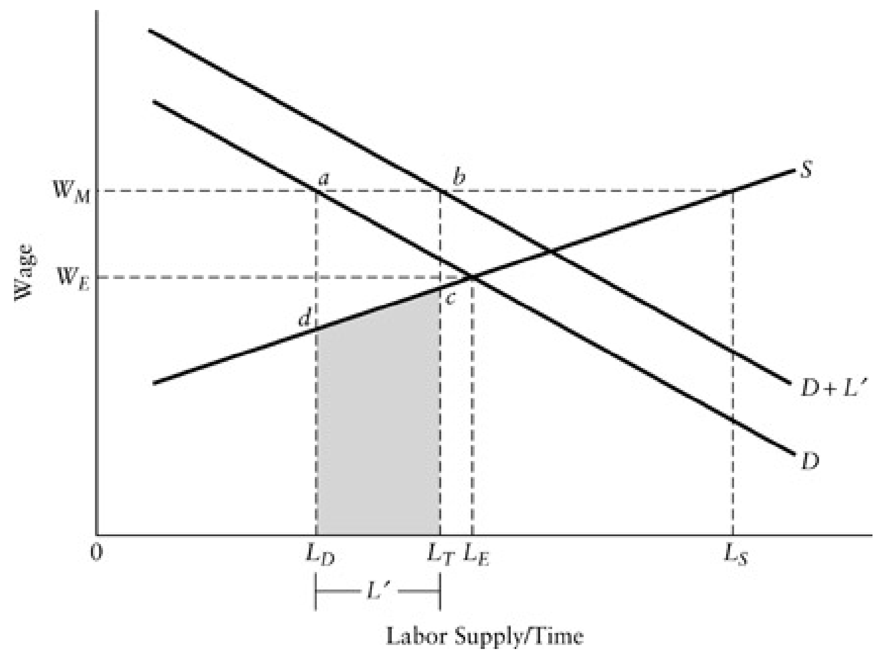

Figure 17.1 Opportunity Cost in a Factor Market with a Price Floor

Other market distortions affect opportunity costs in predictable ways. In factor markets where supply is taxed, expenditures overestimate opportunity costs; in factor markets where supply is subsidized, expenditures underestimate opportunity costs. In factor markets exhibiting positive supply externalities, expenditures overestimate opportunity costs; in factor markets exhibiting negative supply externalities, expenditures underestimate opportunity costs. In monopolistic factor markets, expenditures overestimate opportunity costs. To determine opportunity costs regardless of whether or not the factor is provided through an efficient or distorted market, apply the general rule: *opportunity cost equals expenditures on the factor minus (plus) gains (losses) in social surplus occurring in the factor market.*

If factors are not traded in markets, the general rule cannot be directly applied. Consider a proposal to establish more courts to hold more criminal trials. Budgetary costs would include the salaries of judges and court attendants, rent for courtrooms and offices, and perhaps expenditures for additional correctional facilities (because the greater availability of trial capacity may permit more vigorous prosecution). The budget may also show payments to jurors. Typically, however, these payments just cover commuting expenses. Compensation paid to jurors for their time is typically not related to their wage rate but, rather, set at a nominal per diem rate. Thus, the budgetary outlay for payments to jurors almost certainly understates the opportunity cost of the jurors' time. A better estimate of opportunity cost would be commuting expenses plus the number of juror-hours times either the average or the median hourly wage rate (including benefits) for the locality. The commuting expenses estimate the resource costs of transporting the jurors to the court; the hourly wage rate times the hours spent on jury duty estimates the value of goods forgone because of lost compensated labor or lost household production.

A final point on opportunity costs: the relevant question is what must be given up today and in the future, not what has already been given up, which are *sunk costs*. For instance, suppose that one is asked to reevaluate a decision to build a bridge after construction has already begun. What is the opportunity cost of the steel and concrete that are already in place? It is not the original expenditure made to purchase them. Rather, it is the value of these materials in the best alternative use—most likely measured by the maximum amount that they could be sold for as scrap. Conceivably, the cost of scrapping the materials may exceed their value in any alternative use, so that salvaging them would not be justified. Indeed, if salvage is necessary, say for environmental or other reasons, then the opportunity cost of the material will be negative (and thus counted as a benefit) when calculating the net benefits of *continuing* construction. In situations in which resources that have already been purchased have exactly zero scrap value (the case of labor already expended, for instance), the costs are fully sunk and are not relevant to our decisions concerning future actions.

Valuing Impacts: Willingness to Pay (Accept)

Public policies can either increase or decrease the available quantities of valued goods. In the case of increases, the guiding principle is *willingness to pay* (WTP): *the sum of the maximum amounts that people would be willing to pay to obtain the increases in consumption of the good.* In the case of decreases in valued goods, the guiding principle is *willingness to accept* (WTA): *the sum of the minimum amounts people would be willing to accept as compensation for the decreases in consumption of the good.* In assessing changes in market goods with close substitutes, it is reasonable to assume that the WTP for an increase of some quantity will be very close to the WTA for a decrease of the same quantity, differing primarily because of income effects: receiving (making) a payment increases (decreases) demand for all normal goods including the one being valued.

However, when the good does not have close substitutes, these WTP and WTA amounts could differ substantially, so that they cannot be used interchangeably.[52] Even when there are relatively close substitutes, a large gap consistent with endowment effects may occur.[53] In these cases, the measures are not close approximations for each other, so that it is important to use whichever of these measures is logically appropriate. For example, removing some quantity of spilled oil from a waterway would likely have a WTP much smaller than the WTA an additional spill of the same magnitude as the quantity being removed.[54] Properly valuing a cleanup would require the use of WTP; properly valuing a deregulation that would increase spills would require the use of WTA.

When markets exist for the goods whose availabilities are affected by public policies, changes in consumer surplus provide good estimates of WTP and WTA.[55] Estimating benefits when markets are missing, however, poses a challenge. Indeed, the success of economists in developing clever ways of estimating WTP and WTA when markets are missing has allowed CBA to be applied in an ever-widening range of policies.

Constructed Markets

In some cases, the market for a good does not exist, but a demand schedule for the good can be plausibly specified and estimated. Interpreting the constructed demand schedule as the marginal valuation schedule for the good provides a basis for valuing changes in prices or quantities resulting from policy changes.

The RIA of rules implementing the Americans with Disability Act of 1990 (PL 101–336) illustrates this approach.[56] Analysts estimated demand schedules for various activities and then measured the social surplus that would result from reducing the time costs of use by the disabled. In assessing the benefits of Requirement 79, which required easier access to privately owned swimming pools for those with ambulatory disabilities, the analysts relied on an expert-panel estimate that this and related rules would save the wheelchair-disabled person 43 minutes per use of the facility. Valuing the time of this population at $10 per hour, these rules can be modeled as if they reduced the use price by about $7.10 per visit. Drawing on a variety of data sources, they estimated that current use by wheelchair-disabled people was 2.02 million per year. As the analysts were unable to find estimates of the price elasticity of demand for swimming, they used an estimate of the price elasticity of demand for general recreation as the basis for specifying a demand schedule with a slope of 0.274 million additional uses per dollar reduction in price. The area of the shaded trapezoid in Figure 17.2 gives the estimate of the annual benefits of the access rules for the wheelchair disabled of $21.2 million.

Hedonic Price Models

Levels of nonmarketed goods sometimes affect the prices of goods that are traded in markets. For example, a housing price not only reflects the characteristics of the house but also such locational factors as the quality of the public school district, the level of public safety, and accessibility.

(Remember the old saying: "There are three important factors in real estate—location, location, and location.") If we could find houses that were identical in all these factors except, say, the level of public safety, then we could interpret any price difference as the value the market places on the difference in safety levels.

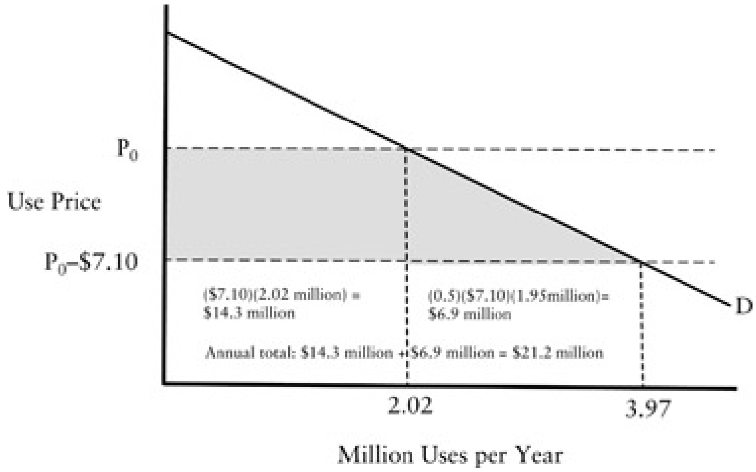

Figure 17.2 Consumer Surplus Gain of Increased Swimming Pool Access for the Wheelchair Disabled

Source: Based on data from Department of Justice, Disability Rights Section of the Civil Rights Division, *Final Regulatory Impact Analysis of the Final Revised Regulations Implementing Titles II and III of the ADA, Including Revised ADA Standards for Accessible Design, Final Report*, July 23, 2010.

In practice, we are rarely fortunate enough to find goods that are identical in every way except for the particular characteristic of interest. Nevertheless, statistical techniques can often be employed to identify the independent contribution of specific characteristics on price. The theoretical foundation for such estimation is known as the *hedonic price model*.[57] Later in this chapter we discuss use of the hedonic price model to estimate the value of life implied by observed risk–wage trade-offs. Other applications include using intercity salary differences to estimate the benefits of air quality improvements,[58] and using housing prices to estimate the value of air quality improvements,[59] values placed on health risks,[60] and the quality of public schools.[61]

Lack of appropriate data severely limits the widespread use of the hedonic price model. Unless data describing all the major characteristics affecting price are available, one cannot make a reliable estimate of the independent contribution of the characteristic of interest. Even when data on all major characteristics are available, it may still be difficult to separate the independent effects of characteristics that tend to vary in the same pattern.

Stated Preferences (Contingent Valuation Surveys)

A direct approach to estimating the benefits of public goods is to ask a sample of people how much they would be willing to pay to obtain them in *contingent valuation surveys*.[62] By comparing the demographic characteristics of the sample to those of the general population, one can estimate the aggregate willingness to pay for specific levels of public goods. This approach permits estimation of the benefits of a wide range of public goods, including national ones like defense that do not vary at the local level. It also permits estimates of the benefits people derive from the provision of public goods to others. In some contexts, where there are no direct or easily observable indirect *behavioral traces*, contingent valuation surveys may be the only feasible way to estimate willingness to pay.[63]

Contingent valuation suffers from all the well-known problems of survey research: answers are sensitive to the particular wording of questions; nonrandom sampling designs or nonresponses can lead to unrepresentative samples; respondents have limited attention spans; and respondents often have difficulty putting hypothetical questions into meaningful contexts. There are also problems specific to valuing public goods.[64] First, there is the practical difficulty of adequately explaining the good being valued to respondents. If respondents do not understand what is being valued, then it is not clear how to interpret estimates of their willingness to pay. Second, respondents may not treat the hypothetical choice they face as an economic decision. If this is the case, then they may ignore budget constraints and respond to gain *moral satisfaction*, or what has been more colloquially called a *warm glow* feeling.[65] Third, respondents may answer strategically. For example, because respondents know that they will not actually have to pay the amount that they state, they may inflate their true willingness to pay for public goods they value in the survey. Although a threat to surveys in theory, *strategic behavior* is probably not that serious in practice.[66] Further, strategic behavior can be reduced by using referendum-type formats for valuation—typically with each respondent being given a random dollar amount and asked if he or she would vote for provision of the public good if it were to cost that amount.

Activity Surveys: The Travel Cost Method

Some of the problems associated with opinion surveys can be avoided by surveying people about their actual behavior. The most common application of this approach is the estimation of the value of recreation sites from people's use patterns.[67] For example, imagine that one wishes to estimate the value people place on a regional park. Researchers could survey people who live various distances from the park about how frequently they visit it. One then would statistically relate the frequency of use to the travel costs (the effective price of using the park) and the demographic characteristics of the respondents. These relationships enable researchers to estimate the population's demand schedule for park visits and apply standard consumer surplus analysis. Of course, the accuracy of the estimated demand curve depends on how well wage rates measure the opportunity cost of travel time; it will not be a good measure for people who view the travel itself as desirable.

Valuing Common Impacts: Shadow Prices

Some types of policy impacts commonly arise in doing CBA. Policies in the areas of health, environment, and transportation often affect mortality risk. Many criminal justice and other social policies have impacts on the number and types of crimes committed. Because they are common, many researchers have invested in developing generic shadow prices. As their name implies, *shadow prices* are marginal valuations of impacts not revealed in the "light" of markets. Consequently, analysts can borrow, or transfer, these shadow prices from their original sources to the CBA at hand.[68] Three important shadow prices are the statistical value of life, the cost of crime, and the marginal excess tax burden.

Consider, first, the *value of statistical life* (VSL).[69] A variety of methods have been used to estimate the willingness of people to pay for small reductions in mortality risk, including hedonic price methods applied to wage and risk trade-offs in labor markets, and contingent valuation surveys. An influential meta-analysis of over 60 wage-risk studies found that the VSL depends on income and base mortality risk: studies with higher-income subjects or lower base mortality risks reported higher estimated VSLs.[70] U.S. federal agencies use a variety of values in their regulatory analyses.[71] For example, in 2015 the Department of Transportation used a VSL of $9.4 million as its base estimate and $5.8 million and $13.0 million as values in its sensitivity analyses.[72] The Environmental Protection Agency uses a VSL of $7.4 million (2006 dollars) or $8.7 million (2015 dollars).[73]

Second, consider the cost of crime (and, therefore, the benefit of avoided crime). There are various estimates of the average cost of various crimes for the United States. The most common estimates, called bottom-up measures because they add up various costs, typically include both the costs of the measurable harm suffered by victims (medical expenses, loss of property, and "pain and suffering") as well as the opportunity cost of the criminal justice system resources required to arrest, prosecute, and punish offenders.[74] Direct estimates of willingness to pay for crime avoidance through stated preference methods also take account of the costs of fear of crime.[75] The stated preference methods tend to produce larger values. For example, a comparison found the following costs (2015 dollars) based on the bottom-up and stated preference methods, respectively: homicide ($5.72 million and $13.49 million), rape ($171.5 thousand and $331.5 thousand), armed robbery ($57.2 thousand and $320.1 thousand), and burglary ($5.7 thousand and $40.1 thousand).[76]

Third, consider the *marginal excess tax burden* (METB). Most CBAs do not include the opportunity cost of government expenditures. However, many policy interventions, although not all, require net public expenditures to implement them. Either now or in the future, these expenditures must be funded from taxation. This use of taxes involves some deadweight loss. Additionally, it requires real resources to collect the taxes in the first place. The ratio of these additional costs of taxation to the amount of revenue collected is the marginal excess tax burden. Each dollar of expenditure funded through tax revenue costs society (1+METB) dollars in real resources. The U.S. METB appears to be on the order of between 15 to 35 percent for income taxes.[77] Note that a policy that reduces net government expenditure provides a benefit of (1+METB) per dollar of saving.

Thorough searches of the economics and policy literatures often yield potentially useful shadow prices. In deciding whether a shadow price can be reasonably transferred from its original source to a particular application, analysts should answer a number of questions: First, how conceptually close is the available shadow price to the ideal one sought? Analysts generally accept VSLs based on employment risk to value other sorts of mortality risks, such as automobile fatalities. But is an adult VSL of this sort a reasonable estimate of how parents value mortality risks to their children? At least some research suggests that parents place a higher cost on negative impacts on their children than the costs of the same impacts on themselves—perhaps twice the value for very young children, but closer to parity for teenagers.[78]

Second, is the shadow price based on a study carried out in a similar jurisdiction? Ideally, analysts for the State of Wisconsin would find shadow prices based on empirical studies done in Wisconsin. Studies based on data from Minnesota, a similar state on many dimensions, would likely be a quite appropriate alternative source. Studies using data from other states, or the United States as a whole, may also be reasonable sources, but borrowing studies using data from other countries requires more care as cultural differences may result in different valuations and legal differences may impose different constraints. In borrowing VSLs from studies done in the United States for use in developing countries, for instance, adjustments would be required to take account of differences in income and base levels of mortality risk.

Third, at what date was the shadow price estimated? Historical shadow prices may not be appropriate for contemporary analysis, depending on the extent of relevant changes in income, technology, demographics, or even preferences. In any event, available research provides shadow prices in current dollars for the year of publication or earlier, which will almost always be earlier than the year in which the CBA is being done. Consequently, these shadow prices must be adjusted to take account of general inflation using a price index. This converts the nominal dollars reported in the study to real dollars for purposes of the contemporary analysis. In the United States, common practice is to use the consumer price index (CPI) estimated by the Bureau of Labor Statistics, which conveniently provides an online calculator to convert the purchasing power of a dollar in one year to that in another.[79]

Step 6: Discount Benefits and Costs to Obtain Present Values

Most of us would be unwilling to lend someone \$100 today in return for a promise of payment of \$100 in a year's time. We generally value \$100 today more than the promise of \$100 next year, even if we are certain that the promise will be carried out and that there will be no price inflation. Perhaps \$90 is the most that we would be willing to lend today in return for a promise of payment of \$100 in a year's time. If so, we say that our present value of receiving \$100 next year is \$90. We can think of the \$90 as the future payment *discounted* back to the present. Discounting is the standard technique for making costs and benefits accruing at different times commensurate so that they can be summed to obtain present values.

Present Values

Imagine that you are the economics minister of a small country. The curve connecting points X_1 and X_2 in Figure 17.3 indicates the various combinations of current (period 1) and future (period 2) production that your country's economy could achieve. By using available domestic resources to their fullest potential, your country could obtain any combination of production on the curve connecting X_1 and X_2. The locus of these combinations is what economists call the *production possibility frontier* for your economy. If all effort is put into current production, then X_1 can be produced and consumed in period 1, but production and consumption will fall to zero in period 2. Similarly, if all effort is put into preparing for production in period 2, then X_2 will be available in period 2, but there is no production in period 1.

Any minister would undoubtedly find neither of these extremes attractive. Indeed, let us assume that you choose point *i* on the production possibility frontier as the combination of production and consumption in the two periods that you believe makes your country best off. The indifference curve labeled I_1 gives all the other hypothetical combinations of production and consumption in the two periods that you find equally satisfying to combination (Q_1, Q_2). You consider any point to the northeast of this indifference curve better than any point on it. Unfortunately, combination (Q_1, Q_2) is the best you can do with domestic resources alone.

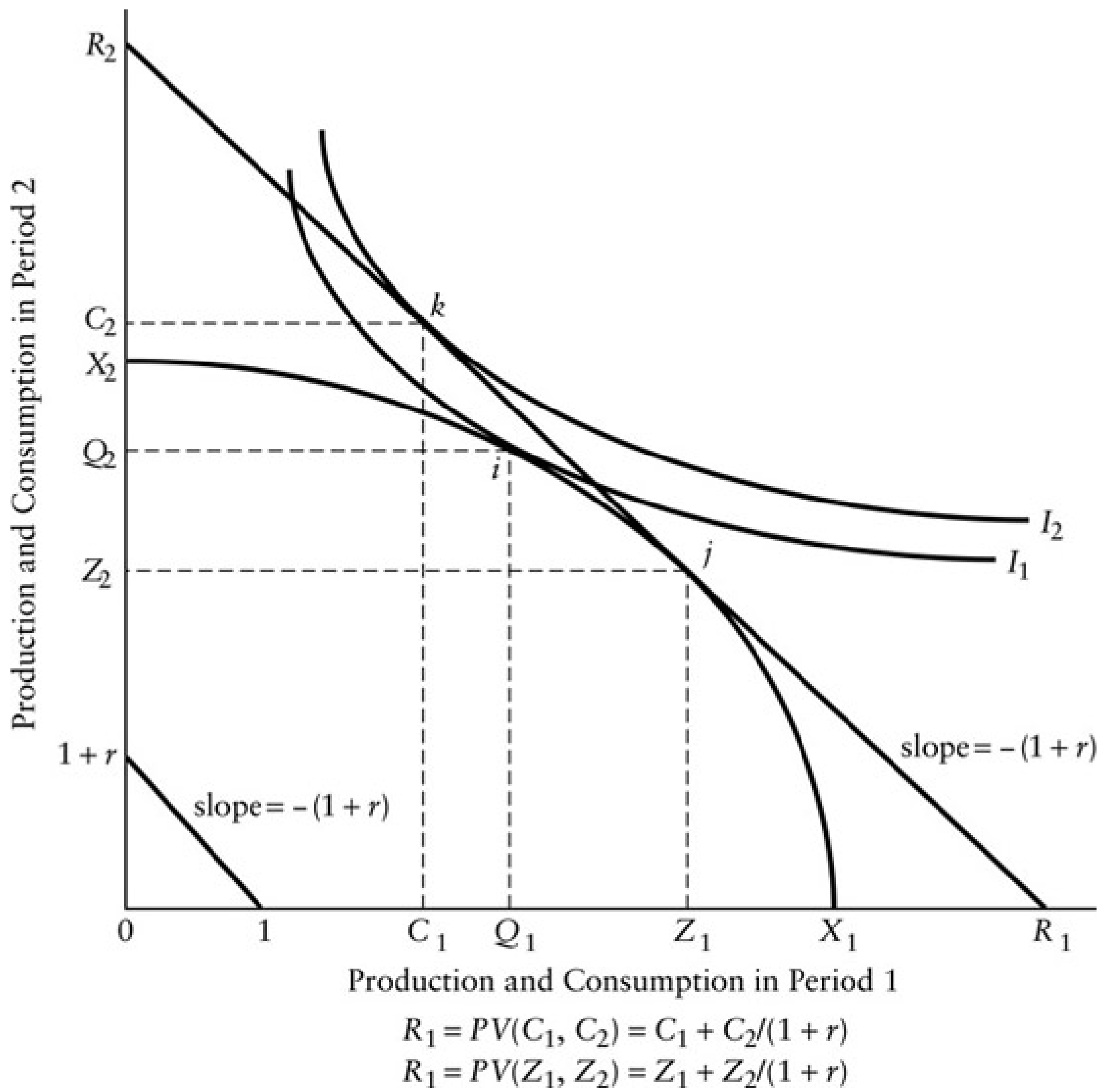

Figure 17.3 The Present Value of Production and Consumption

Now suppose foreign banks are willing to lend or borrow at an interest rate r. That is, the banks are willing to either lend or borrow one dollar in period 1 in return for the promise of repayment of $(1 + r)$ dollars in period 2. You realize that, with access to this capital market, you can expand your country's consumption possibilities. For instance, if you choose point j on the production possibility frontier in Figure 17.3, then you could achieve a consumption combination anywhere along the line with slope $-(1 + r)$ going through point j by either borrowing or lending. This line is shown as connecting points R_1 and R_2, where R_2 equals $(1 + r)R_1$. Each additional unit of consumption in period 1 costs $(1 + r)$ units of consumption in period 2; each additional unit of consumption in period 2 costs $1/(1 + r)$ units of consumption in period 1. Once these intertemporal trades are possible, your country's consumption possibility frontier expands from the original production possibility frontier connecting X_1 and X_2 to the discount line connecting R_1 and R_2.

The most favorable consumption possibility frontier results from choosing the production combination such that the discount rate line is just tangent to the production possibility frontier. This occurs at point *j*. The most desirable consumption opportunity is shown at point *k*, which is on indifference curve I_2, the one with the highest utility that can be reached along the consumption possibility frontier (the line connecting R_1 and R_2). Thus, the best policy is to set domestic production at Z_1 and domestic consumption at C_1 in period 1, lending the difference $(Z_1 - C_1)$ so that in period 2 the country can consume C_2, consisting of Z_2 in domestic production and $(C_2 - Z_2)$, which is equal to $(1 + r)$ times $(Z_1 - C_1)$ in repaid loans.

Note that the most desirable point on the production possibility frontier gives the country the largest possible value for R_1. But R_1 can be interpreted as the *present value* of consumption, which is defined as the highest level of consumption that can be obtained in the present period (period 1). We write the present value of consumption as $PV(C_1, C_2)$ and the present value of production as $PV(Z_1, Z_2)$. Each of these present values equals R_1.

One can arrive at an algebraic expression for the present value of consumption in a future period as follows: First, note that the present value of C_1 is just C_1. Second, note that the maximum amount that could be borrowed against C_2 for current consumption is $C_2/(1 + r)$. Once one is in the second period, the present value of consumption in the third period is $C_3/(1 + r)$, the maximum quantity of consumption in the second period that can be obtained from C_3. Now step back to the first period. The present value of consumption equals $C_3/(1 + r)^2$. In general, we can write the present value for a cost (or benefit) accruing N in the future as periods as $C_N/(1 + r)^{N-1}$, where C_N accrues at the end of the N^{th} period.

When doing CBA, we apply the discounting procedure by converting benefits and costs to present values. So, for instance, if benefits B_t and costs C_t occur t periods beyond the current period, then their contribution to the present value of net benefits equals $(B_t - C_t)/(1 + d)^{t-1}$, where d is the social discount rate.

Determining the Social Discount Rate

In a world with perfect capital markets, illustrated in Figure 17.3, every consumer is willing to trade between marginal current and marginal future consumption (called the *marginal rate of pure time preference*) at the market rate of interest. At the same time, the rate at which the private economy transforms marginal current consumption into marginal future consumption (called the *marginal rate of return on private investment*) also equals the market rate of interest. Thus, the market rate of interest would be an appropriate *social discount rate* (SDR) because it would represent both how consumers trade intertemporal consumption and how firms trade current for future production.

The situation is more complicated when one relaxes the assumption of a perfect capital market. First, because individual consumers have finite lives, they may not sufficiently take into account the consumption possibilities of future generations. Those yet unborn do not have a direct voice in current markets, yet one may believe that they should have standing in the CBA. The willingness of people to pass inheritances to their children (*bequest value*), and to pay to preserve unique resources, gives indirect standing to future generations. Furthermore, future generations will almost certainly inherit a growing stock of knowledge that will compensate at least somewhat for current consumption of natural resources. Nevertheless, to the extent that the interests of future generations remain underrepresented in current markets, an argument can be made for using a discount rate lower than the market interest rate.[80] The issue is particularly relevant when evaluating projects, like the storage of nuclear wastes, with consequences that persist far into the future. Even very low positive discount rates make negligible the present value of costs and benefits occurring in the far future.

Second, taxes and other distortions lead to a divergence between the rate of return on private investment and the rate at which consumers are willing to trade current and future consumption. Suppose that consumers are willing to trade marginal current and marginal future consumption at a rate of 6 percent, their marginal rate of pure time preference.[81] If they face an income tax of 25 percent and firms face a profits tax of 50 percent, then they will invest only in projects that earn a return of at least 16 percent. (The firm returns 8 percent to the investor after the profits tax is paid; the investor retains 6 percent after paying the income tax.)

Which, if either, of these rates should be interpreted as the SDR? Economists generally answer this question in terms of the opportunity costs of the public project.[82] If the public project is to be financed entirely at the expense of current consumption, then the marginal rate of pure time preference is the appropriate discount rate. If the public project is to be financed at the expense of private investment, then the rate of return on private investment is appropriate. The *shadow price of capital approach* allows for financing at the expense of both consumption and private investment by first converting reductions in private investment to the streams of benefits that they would have produced, and, once all financing is expressed in terms of lost consumption, discounting at the marginal rate of pure time preference.[83]

Unfortunately, analysts rarely have enough information to apply the shadow price of capital approach. Instead, they must choose a particular value for the SDR, a task complicated by the often large gap between the marginal rate of pure time preference and the marginal rate of return on private investment. Estimates for the U.S. economy suggest that, in real terms (after adjusting for expected inflation), the former appears to be between about 1 and 4 percent and the latter between 5 and 8 percent,[84] which is quite a big difference for projects with long time horizons when costs precede benefits, which often is the case for investments in physical or human capital.

Using the marginal rate of return on private investment as the basis for the SDR is consistent with resources required for public policies being funded from borrowing that crowds out private investment through higher interest rates.[85] Using the marginal rate of pure time preference as the basis for the SDR is consistent with resources required for public policies being funded by taxes that primarily reduce consumption.[86] Recognizing that all government borrowing must eventually be repaid with tax revenue, we see the marginal rate of pure time preference as the appropriate basis for the SDR in intragenerational discounting.[87]

What can analysts do in the absence of a consensus on the appropriate SDR? One approach is to report net benefits for a range of discount rates. Analysts should also explain why they believe the range they have used is reasonable. A related approach is to report the largest SDR that yields positive net benefits. Clients can then be advised to adopt the project if they believe the correct rate is smaller than the reported rate.

Another approach is to use the same SDR for all projects being considered by the decision-making unit. With this approach, at least all projects are evaluated using the same standard. The U.S. Office of Management and Budget (OMB) recommends real discount rates of 3 and 7 percent in RIAs.[88] The United Kingdom, Germany, and France recommend 3.5 percent, 3 percent, and 4 percent, respectively.[89] The lower of the two U.S. rates and the rates of these European countries are consistent with an SDR based on the marginal rate of pure time preference.

Finally, there are some situations in which market interest rates represent the opportunity cost of public investment and, therefore, are the appropriate discount rates. For example, if we are doing a CBA from the perspective of a local government, then the opportunity cost of public investment is the rate at which the local government can borrow. A similar argument can be made for small countries that borrow in international capital markets.

Step 7: Compute the Present Value of Net Benefits

To compute the present value of net benefits, we simply discount all future costs and benefits back to the present. Typically, policies involve some immediate cost, C_0, such as for the purchase of equipment and space required to begin implantation of the project. Then, in year n of N subsequent years, B_n in benefits and C_n in costs accrue. The annual benefits and costs almost always accrue continuously over the course of the year—a road improvement provides continuous time savings to drivers or an additional police patrol provides continuous reductions in crime. Such continuously realized costs and benefits are typically discounted at the mid-year point, rather than at either the beginning or end of the year. Finally, when the project ends after N years there may be some scrap value of freed-up resources, S_N. For a discount rate d, the present value of this time pattern of costs and benefits would be calculated with the following formula:

$$PV = C_0 + \sum_{n=1}^{N} \frac{B_n - C_n}{(1+d)^{(n-0.5)}} + \frac{S_N}{(1+d)^N}$$

A simple example, shown in Table 17.1, illustrates the calculation of present values using either nominal or real values for costs and benefits. Consider a city that uses a rural landfill to dispose of solid refuse. By adding larger trucks to the refuse fleet, the city would be able to save $100,000 in disposal costs during the first year after purchase, and a similar sum each successive year of use. The trucks can be purchased for $500,000 and sold for $200,000 after five years. (The city expects to open a resource-recovery plant in five years that will obviate the need for landfill disposal.) The city can currently borrow money at an interest rate of 10 percent. Should the city buy the trucks? Yes, if the present value of net benefits is positive and there is no other alternative to current policy offering a larger present value of net benefits.

Table 17.1 Present Value of Net Benefits of Investing in New Garbage Trucks

		Costs and Benefits in Real Dollars	*Costs and Benefits in Nominal Dollars Assuming 4 Percent Inflation Rate* (i = *0.04)*
Year 0	Purchase	−500,000	−500,000
Year 1	Savings	100,000	101,980
Year 2	Savings	100,000	106,060
Year 3	Savings	100,000	110,302
Year 4	Savings	100,000	114,714
Year 5	Savings	100,000	119,303
	Liquidation	200,000	243,331
Discount Rate Present Value of Net Benefits		Real: d = 0.057692 $87,040	Nominal: r = 0.10 $87,040

Note: $d = (r-i)/(1 + i) = (0.10 - 0.04)/(1 + 0.04) = 0.057692$

To calculate the present value of net benefits, we must decide whether to work either with *real* or *nominal* dollars—either approach will lead to the same answer, as long as it is followed consistently.

In most CBA, we are looking into the future. Obviously, we cannot know precisely how price inflation will change the purchasing power of future dollars. Nevertheless, if we take the nominal market interest rate facing the city as its discount rate, then we must estimate future costs and benefits in inflated dollars. The reason is that the market interest rate incorporates the expected rate of inflation—lenders do not want to be repaid in inflated dollars. Thus, *one appropriate approach to discounting is to apply the nominal discount rate to future costs and benefits that are expressed in nominal dollars.*

The column of Table 17.1 on the extreme right shows nominal benefits for a 4 percent annual rate of inflation. Assuming that the projected annual savings of $100,000 in years 2, 3, 4, and 5 represent a simple extrapolation of the savings in the first year, and that the $200,000 liquidation benefit is based on the current price of used trucks, one can interpret these amounts as being expressed in constant dollars. (This implicitly assumes that wage rates, gasoline prices, and other prices that figure into the benefit calculations increase at the same rate as the general price level.) The purchase price is already in nominal dollars. To put the liquidation benefit, which accrues at the end of the fifth year, in nominal dollars, one multiplies it by $(1 + i)^5$, where i is the 4 percent assumed rate of inflation. As the annual savings accrue continuously during the year, it makes sense to convert them to nominal dollars at the middle of the year. For example, multiplying the annual savings by $(1 + i)^{0.5}$ gives the nominal value of first year savings; multiplying by $(1 + i)^{4.5}$ gives the nominal value of fifth year savings valued mid-year. To obtain the present value of the investment, one would discount each nominal value to the present using the nominal discount rate. The savings in year n would be discounted by multiplying by $1/(1 + r)^{(n - 0.5)}$ where r is the nominal discount rate and n is the year the savings accrue. The liquidation value would be discounted by multiplying by $1/(1 + r)^5$. Applying these discount factors results in a net present value of $87,040.

An alternative method of discounting leads to the same present value of net benefits: *apply a real discount rate to future costs and benefits that are expressed in constant (real) dollars.* Projecting costs and benefits in constant dollars (that is, ignoring inflation) is natural. The difficulty arises in determining the appropriate discount rate. Nominal interest rates are directly observable in the marketplace; real interest rates are not. So if we decide that the interest rate facing the decision maker is the appropriate discount rate, then we must adjust the observable nominal rate to arrive at the real rate. This requires us to estimate the expected inflation rate just as we must if we decide to work with nominal costs and benefits. The real interest rate approximately equals the nominal interest rate minus the expected rate of inflation; more precisely $d = (r - i)/(1 + i)$ where d is the real discount rate, r is the nominal discount rate, and i is the expected rate of inflation.[90]

For the city, which sees a nominal discount rate of 10 percent and expects a 4 percent inflation rate, the real discount rate is 5.7692 percent.[91] Applying this real discount rate to the yearly real costs and benefits in the left-hand column of Table 17.1 yields a present value of net benefits equal to $87,040. Thus, discounting real costs and benefits with the real discount rate is equivalent to discounting nominal costs and benefits with the nominal discount rate. As long as one uses either real dollars and real discount rates or nominal dollars and nominal discount rates, one will arrive at the same present value.

Generally, predicting costs and benefits in real dollars, and therefore discounting with a real rate, is most convenient. When we predict impacts, we typically do so in natural units that we then monetize. When we monetize these impacts with current prices, the resulting predictions of costs and benefits accruing in future years are real because we have not taken account of general price inflation.

The discussion so far applies to an analysis with a single alternative to current policy. If there is more than one alternative, but they all have the same time horizons, then their present values of net benefits are still directly comparable. For example, if we considered two different garbage trucks with the same useful life (for example, the five-year time horizon in Table 17.1), then the truck with the largest present value of net benefits would be the most efficient relative to the current equipment. However, we cannot directly compare alternatives with different time horizons, because the alternative with the longer time horizon has more opportunity to accumulate costs or benefits. To compare alternatives with different time horizons, one must annualize their present values of net benefits, that is, convert each alternative to a stream of constant net benefits over its time horizon by dividing by the *annuity factor*:[92]

$$a_d^N = \frac{1-(1+d)^{-N}}{d}$$

Here, d is the discount rate and N is the time horizon of the policy. (The same d is used across alternatives but the N is specific to each alternative.) Conceptually, one could repeat each alternative indefinitely, so that if one alternative has a larger annualized present value of net benefit than another, it must also have a larger present value of net benefits if the alternatives were repeated indefinitely.

Step 8: Perform Sensitivity Analysis

Every CBA involves prediction and valuation, neither of which can be done with certainty. Good analysis conveys the degree of uncertainty that clients face in the reported net benefits of policy alternatives. In the original applications of CBA to infrastructure projects, the sign of net benefits often hinged on a small number of assumptions. For example, the benefits of a highway improvement might depend critically on the predicted growth in usage over the time horizon of the project. Focusing on the implications of alternative assumptions through *partial sensitivity analysis* was often reasonable, especially in view of the high cost of computation. (Young readers, please consult the oldest professor you can find about submitting programs to mainframe computers coded on punch cards.)

As the application of CBA has expanded to include environmental, health, and social policies, the number of uncertainties has also expanded and it is normally not reasonable to treat only one assumption as crucial to the sign of net benefits. The cost of computing is so low, however, that more comprehensive sensitivity analysis can be readily performed using Monte Carlo simulations.

Partial Sensitivity Analysis

Analysts commonly focus on point estimates, or specific values, for uncertain parameters in arriving at a prediction of net benefits. Partial sensitivity analysis involves systematically varying the value of a single point estimate, or a small number of point estimates, to assess how such variation affects net benefits. It is partial because most CBAs involve quite a number of uncertain parameters. For example, consider CBA of an immunization program in which reduced mortality monetized with VSL is an important benefit. A straightforward partial sensitivity analysis would be to calculate net benefits over the plausible range of the VSL. If another important parameter for calculating benefits is the probability of a relevant epidemic, then it could be varied across its plausible range. The sensitivity of net benefits to the two parameters could be displayed for different combinations of their values, say by sketching a curve of net benefits as a function of epidemic probability for the low, point, and high estimates of VSL.

As noted, there remains debate over the appropriate SDR. Investigating the impact of alternative values of the SDR on predicted net benefits is one of the most common examples of partial sensitivity analysis. RIAs often report net benefits at the 3 and 7 percent discount rates that the Office of Management and Budget recommends. Predicting net benefits for more values of the SDR enables analysts to sketch the former as a function of the latter. If the sign of net benefits does not change over the plausible range of the SDR, than one can be confident that it does not make a major difference. If the sign switches at some plausible discount rate, then the analyst can convey this information to the client as a breakeven value: the SDR at which net benefits would be predicted to be zero. Higher or lower values produce net benefits of opposite signs.

Some CBAs have potential impacts that cannot be monetized. For example, fears that radiation sources used in the blood irradiators found in most U.S. hospitals and blood banks could be diverted for use in a radiological dispersion device, or "dirty bomb," has prompted consideration of policies to subsidize or require their replacement with x-ray devices that have no use in terrorism.[93] Analysts can predict the differences in capital and operating costs of the two technologies; monetizing the reduced risk of a terrorist attack cannot be done with any confidence. One analytical approach is to estimate net benefits exclusive of the benefit of terrorist risk to answer the question: How large would these excluded benefits have to be to produce positive net benefits for the induced change in technology? The answer is commonly referred to as a *breakeven* value.

Monte Carlo Simulation

Partial sensitivity analysis has the advantage of focusing attention on a particular assumption, but the disadvantage of doing so without taking account of other uncertainties relevant to the prediction of net benefits. *Monte Carlo simulation* provides a more comprehensive approach. Named for the European gambling center, it involves four steps: First, specify a probability distribution for each uncertain parameter, such as a predicted impact or a shadow price. Second, randomly draw values from each distribution to calculate a predicted net benefit. Third, repeat the second step numerous times, typically thousands of times by an uncomplaining computer. And fourth, present and analyze the resulting distribution of net benefits.

The specified distributions for parameters should capture uncertainty about the parameters. At one extreme, statistical theory suggests an appropriate distribution for an estimated parameter (typically a Student's t-distribution or normal distribution). At the other extreme, analysts may only be willing to put upper and lower bounds on a parameter, assuming that any values between these extremes are equally likely (a uniform distribution). In between, analysts may be willing to specify upper and lower bounds, but believe that values closer to some central value are more likely than values further away (typically a triangular distribution).

Both spreadsheets and statistical packages generally allow random draws of values from specific distributions, such as those mentioned above to implement the second and third steps. (These types of software also may have add-ons specifically designed for doing Monte Carlo simulations.) Brute force methods, such as repeating the net benefits formula in rows of a spreadsheet or as observations in a created data set in a statistical program, have the benefit of facilitating transparency, not just for others but for the analysts themselves, who may be asked to redo the CBA sometime in the future.

The final step interprets the distribution of net benefits created by the repeated trials. The mean of the trials usually is presented as the best point prediction of net benefits—note that the mean may differ substantially from the prediction of net benefits based on the point estimates of parameters, especially when the calculation of net benefits depends on the product or ratios of uncertain parameters. Reporting the standard deviation and the percent of trials providing positive net benefits conveys the degree of uncertainty in the predicted net benefits. Displaying the entire distribution of trials in a histogram is also informative.

As an illustration of the use of Monte Carlo simulation in CBA, consider the predicted costs and benefits of Helping Educate and Link the Homeless (HEALTH). This hospital outreach program increased the access of homeless people to health care in order to improve their health and reduce long hospital stays resulting from lack of primary care.[94] The program, which was sponsored by the research foundation of UnityPoint Health–Meriter, the hospital's corporate owner, included weekday access to a nurse at a facility near the city's largest food pantry and daytime drop-in shelter, as well as access to physicians and a rotating clinic staffed by hospital employees and volunteers. An evaluation to assess the impact of the program on health care utilization compared health care costs before and after participation for 160 program participants, as well as for a group of 594 nonparticipants matched through propensity scoring to facilitate a difference-in-differences estimation. The results suggest that, although HEALTH did not reduce emergency room use, it did substantially reduce very costly hospital admissions.

Table 17.2 Mean Costs and Benefits of HEALTH from Monte Carlo Simulation in Thousands of 2011 Dollars (Standard Deviations in Parentheses)

	Social	*UnityPoint Health-Meriter*	*Replication*	*Adjusted Replication*

	Social	*UnityPoint Health-Meriter*	*Replication*	*Adjusted Replication*
Operating Cost	403 (16.5)	403 (16.5)	403 (16.5)	403 (16.5)
Value of Donations	87 (3.6)	-	87 (3.6)	87 (3.6)
Volunteer Cost	83 (16.0)	-	83 (16.0)	83 (16.0)
Cost of Primary Care	258 (52.8)	-	258 (52.8)	258 (52.8)
Total Cost	**830 (57.9)**	**403 (16.5)**	**830 (57.9)**	**830 (57.9)**
Avoided Hospital Costs	1,397 (907)	1,397 (907)	1,397 (907)	931 (605)
Quality of Care Benefits	837 (537)	-	-	-
Avoided Crime Costs	40 (11.5)			
Total Benefit	**2,274 (1,049)**	**1,397 (907)**	**1,397 (907)**	**931 (605)**
Net Benefit	**1,444 (1,050)**	**981 (895)**	**554 (897)**	**101 (608)**
Percent of Trials with Positive Net Benefit	**92**	**86**	**74**	**57**

Source: Bethany Ackeret, David L. Weimer, Erik A. Ranheim, Lisa Urban, and Elizabeth Jacobs, “Net Benefits of Hospital-Sponsored Health Care for the Homeless: Cost-Benefit

Analysis of a Demonstration Project," Robert M. La Follette School of Public Affairs Working Paper No. 2014-010, April 14, 2014, Table 3.

The column of Table 17.2 labeled "Social" shows the costs and benefits of HEALTH giving standing to the homeless (quality of care benefits), benefactors (value of donations and volunteer cost), taxpayers (cost of primary care), the hospital (operating cost and avoided hospital costs), and the rest of society (avoided crime costs). A Monte Carlo simulation treated each of these categories as uncertain. For example, the distribution of avoided hospital costs was assumed to follow a normal distribution, with the estimated monthly mean savings of $209 per participant with a standard error of $135; the distribution of operating cost was assumed to be a triangular distribution centered at the measure program cost, but with upper and lower bounds 10 percent larger and smaller, respectively. Figure 17.4 shows the resulting distribution of net benefits resulting from a simulation with 10,000 trials. The mean net benefit across the trials was $1,444 million with a standard deviation of $1,050 million, as noted in Table 17.2. Note the vertical line in Figure 17.4 indicating zero net benefits. The percent of trials to the right of the line gives the estimated probability, 92 percent, that HEALTH had positive net benefits. (The 100 percent reported in the WSIPP analysis of the EET was based on a Monte Carlo simulation in which all the trials yielded positive net benefits.)

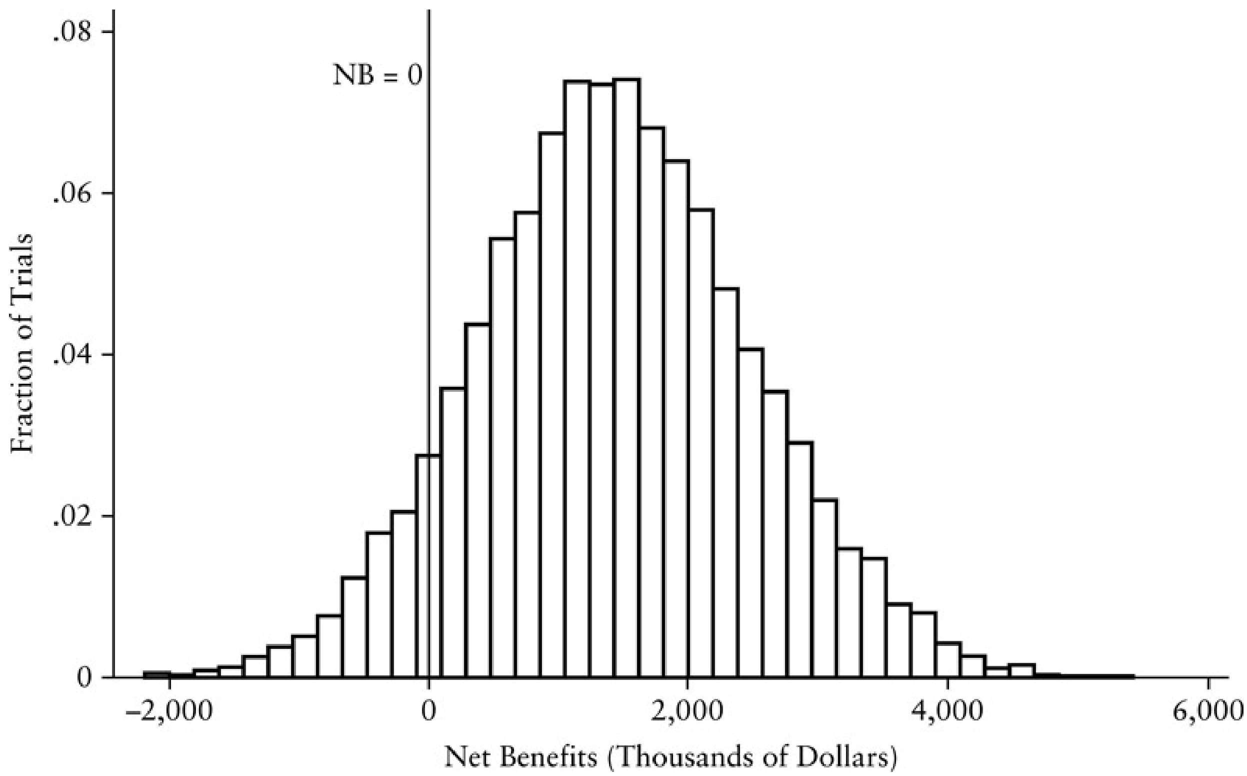

Figure 17.4 Net Social Benefits of HEALTH

Monte Carlo simulation helps accommodate very imprecise estimates of nonetheless needed parameters. Social science convention is to treat statistically insignificant estimates of parameters as if they were indistinguishable from zero. This logic might lull an analyst into leaving an estimated impact that is statistically insignificant out of the analysis entirely, effectively treating it and its standard error as zero! However, the statistically insignificant value, rather than zero, is typically the most likely value.[95] Leaving it out biases net benefits and leads to an underestimation of its uncertainty. A better approach is to take account of the distribution of the impact in the Monte Carlo simulation—its large standard error (relative to the parameter's point estimate) will translate into a larger standard deviation in the distribution of net benefits, indicating less certainty.

It is sometimes informative to combine partial sensitivity analysis and Monte Carlo simulation. So, rather than treating the SDR as a draw from a distribution, analysts conducting an RIA might do Monte Carlo simulations at each of the two OMB recommended SDRs. Also, consider the last two columns of Table 17.2, which predict the impacts on adopters of HEALTH under two different scenarios. The first assumed that adopters will realize the same magnitude of avoided hospital costs as UnityPoint Health–Meriter. (It differs from the second column, showing the fiscal costs and benefits to the sponsoring hospital, by the value of donations and volunteer time, which adopters would likely have to fund.) The second scenario assumed that other adopters might not realize the same magnitude of avoided hospital costs because they would not get the same level of effort as exercised by the doctor who initiated HEALTH as a demonstration of its feasibility and desirability. Specifically, the second scenario drew on results from meta-analysis research that suggested that, on average, effect sizes from nondemonstrations are only about two-thirds of the effect sizes for demonstrations.[96] (Note that WSIPP routinely makes a similar adjustment to demonstration studies when doing its meta-analyses.) The Monte Carlo simulations showed a 74 percent chance of realizing positive net fiscal benefits for the first scenario, but only a 57 percent chance of positive net fiscal benefits for the second scenario.

Step 9: Recommend

Our discussion so far has largely considered the evaluation of single policies in isolation. If a policy offers positive net benefits, then it satisfies the Kaldor–Hicks criterion and should be adopted when efficiency is the only relevant goal. A more general rule applies when we confront multiple policies that may enhance or interfere with each other: *choose the combination of policies that maximizes net benefits.* Physical, budgetary, and other constraints may limit the feasibility of generating such combinations.

Physical and Budgetary Constraints

Policies are sometimes mutually exclusive. For example, it is not possible to both drain a swamp to create agricultural land and to also preserve it as a wildlife refuge. *When all of the available policies are mutually exclusive, we maximize efficiency by choosing the one with the largest positive net benefits.* For example, consider the list of projects in Table 17.3. If we can recommend any combination of projects, simply recommend all those with positive net benefits—namely, projects A, B, C, and D. Instead, however, assume that all the projects are mutually exclusive except C and D, which can be built together. By taking the combination of C and D to be a separate project, it is feasible to consider all the listed projects to be mutually exclusive. Looking down the column labeled "Net Benefits"—one can see that project B offers the largest net benefits and therefore should be the one recommended.

Analysts sometimes compare programs in terms of their benefit–cost ratios. Note that project B, which offers the largest net benefits, does not have the largest ratio of benefits to costs. Project A has a benefit–cost ratio of 10, while project B has a benefit–cost ratio of only 3. Nevertheless, project B offers larger net benefits than project A. This comparison shows that the benefit–cost ratio sometimes confuses the choice process. Another disadvantage of the benefit–cost ratio is that it depends on how we take account of costs and benefits. For example, consider project B. Suppose that the costs of $10 million consist of $5 million of public expenditure and $5 million of social surplus loss. We could take the $5 million of social surplus loss to be negative benefits, so that costs would then be $5 million and benefits $25 million. While net benefits still equal $20 million, the benefit–cost ratio increases from 3 to 5. Therefore, *we recommend that you avoid using benefit–cost ratios altogether in comparing mutually exclusive alternatives.* However, once all projects have been assessed in terms of net benefits, reporting benefit–cost ratios may be a more effective way of communicating—the WSIPP report of a benefit–cost ratio of 34.5 more dramatically conveys the relative efficiency of the EET to conventional juvenile programs.

Table 17.3 Choosing among Projects on the Basis of Economic Efficiency

	Costs Relative to No Project (millions of dollars)	*Benefits Relative to No Project (millions of dollars)*	*Net Benefits (millions of dollars)*	*Benefit-Cost Ratio*
Project A	1	10	*9*	10
Project B	10	30	20	3
Project C	4	8	4	2
Project D	2	4	2	2
Projects C and D	7	21	14	3
Project E	10	8	−2	0.8

No constraints:	Recommend Projects A, B, C, and D (net benefits $43 million).
All projects mutually exclusive:	Recommend Project B (net benefits $20 million).
Costs cannot exceed $10 million:	Recommend Projects A, C, and D (net benefits $23 million).

Also note that projects C and D are shown as synergistic: the net benefits from adopting both together exceed the sum of the net benefits from adopting each independently. Such might be the case if project C were a dam that created a reservoir that could then be used for recreation and D were a road that increased access to the reservoir. Of course, projects can also interfere with each other. For instance, the dam might reduce the benefits of a downstream recreation project. The important point is that care must be taken to determine interactions among projects so that the combination of projects providing the greatest net benefits in aggregate can be readily identified.

Returning to Table 17.3, interpret the listed costs as public expenditures and the listed benefits as the monetized value of all other effects. Then assume that while none of the projects are mutually exclusive in a physical sense, total public expenditures (costs) cannot exceed $10 million. If project B is selected, the budget constraint is met and net benefits of $20 million result. If projects A, C, and D are selected instead, the budget constraint is also met but net benefits of $23 million result. No other feasible combination offers larger net benefits. Thus, under the budget constraint, choosing projects A, C, and D maximizes net benefits.

These recommendations ignore the analytical uncertainty in estimating of net benefits. In other words, they ignore the risk that clients bear in accepting the recommendations. In practice, a more appropriate approach would be assess combinations of policies in Monte Carlo simulations.

Distributional Constraints

The Kaldor–Hicks criterion requires only that policies have the potential to produce Pareto improvements; it does not require that people actually be compensated for the costs that they bear. One rationale for accepting potential, rather than demanding actual, Pareto improvements for specific policies is that we expect different people to bear costs under different policies, so that over the broad range of public activity few, if any, people will actually bear net costs. Another rationale is that, even when some people do end up as net losers from the collection of policies selected on the basis of efficiency, they can be compensated through a program that redistributes income or wealth.

These rationales are less convincing for policies that impose high costs on small groups. If we believe that the losers will not be indirectly compensated, then the polity may wish to consider designing policies so that they either spread costs more evenly or provide direct compensation to big losers. These limits on the size of losses could be codified as a constraint that must be met in applying the Kaldor–Hicks criterion. Alternatively, we can imagine applying weights to costs and benefits accruing to different groups. Using distributional constraints and weights obviously introduces values beyond efficiency.[97] It is important to make these values explicit. Indeed, the best approach would be to treat net benefits as the measure of efficiency within a multi-goal analysis.

Of course, the inclusion of distributional values requires that costs and benefits be disaggregated for "relevant" groups. This entails doing separate CBAs from the perspective of different income classes, geographic regions, ethnic groups, or other categories that have some legitimate distributional relevance. When you begin your analyses, we recommend erring on the side of disaggregating too much. It may not be possible to gather necessary information to do distributional analysis at the end of a study. In addition, it may be that a distributional value will become important to you or your client only after you have seen the estimated distribution of net benefits. If efficiency does turn out to be the only relevant value, then aggregation is no more difficult than addition.

Disaggregation of net benefits by pertinent interest groups may be valuable in anticipating political opposition. Estimation of aggregate net benefits enables us to answer the normative question "Is the policy efficient?"; estimation of net benefits by interest groups helps us answer the positive question "Who will oppose and who will support the policy?"

Disaggregation of net benefits may also be relevant to whether or not a policy needs to be mandatory. Returning to Table 17.2, note that the Replication and Adjusted Replication columns take the perspective of potential adopters, showing their fiscal impacts from adopting HEALTH. If they anticipate positive net fiscal benefits, then a policy that simply makes them aware of the HEALTH program and its fiscal implications may be sufficient to induce replications. However, if social net benefits are positive but net fiscal benefits to adopters are negative, then the adopters would have to be required, or otherwise induced, to adopt the program.

Analytical Honesty

CBAs produce clear “bottom lines” that can mask complexity and uncertainty. Analytical good practice requires their revelation in two ways. First, the CBA should be well documented. The objective should be to provide sufficient explanation of methods, procedures, and assumptions to enable someone to replicate the predictions of costs and benefits. This typically requires extensive use of appendices to provide detail without getting in the way of the presentation of substantive findings to the client.

Second, the CBA should make the degree of certainty about the predicted net benefits clear. We believe that this is almost always best done by predicting the probability of positive net benefits through Monte Carlo simulation and presenting it clearly to the client along with the bottom line.

Conclusion

CBA systematically assesses the efficiency of proposed policies. If efficiency were the only relevant policy goal, then CBA would be an appropriate decision rule for making policy choices. Although efficiency is rarely the only relevant policy goal, it is usually one within a multi-goal analysis. In practice, therefore, CBA most commonly makes a contribution by providing protocols for systematically and quantitatively assessing the relative efficiency of alternative policies.

Completing a CBA that provides reasonably confident predictions of monetized net benefits usually demands considerable amounts of information and analytical effort. Consequently, the decision about whether or not to complete a CBA requires a weighing of its costs and benefits! Even when the CBA steps cannot be completed to obtain a monetary value of net benefits, attempting to follow them may help analysts anticipate impacts across organizational and policy domain boundaries and over appropriate time horizons.

For Discussion

1. Consider a regulation that would ban the use of cell phones while driving automobiles. List the effects of the regulation and classify them as costs and benefits.
2. Cost–benefit analysis provides a method for identifying the most efficient alternative. Consider its relationship to distributional values from both a theoretical and a practical perspective.

Notes

1 The U.S. Flood Control Act of 1936 required that water resource projects be evaluated on the difference between estimated benefits and costs. In 1952, the Bureau of the Budget set out its own criteria in *Budget Circular A-47.* See Peter O. Steiner, "Public Expenditure Budgeting," in Alan S. Blinder and Robert M. Solow, *The Economics of Public Finance* (Washington, DC: Brookings Institution, 1974), 241–357.

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3 Treasury Board, *Canadian Cost–Benefit Analysis Guide: Regulatory Proposals* (Ottawa, Canada 2007).

4 H.M. Treasury, *The Green Book: Appraisal and Evaluation in Central Government* (London, U.K.: The Stationery Office, 2003), at 3.

5 Productivity Commission, *Productivity Update* (Australia, July 2015).

6 Massimo Florio, "Cost–Benefit Analysis and the European Union Cohesion Fund: On the Social Cost of Capital and Labour," *Regional Studies* 40(2) 2006, 211–24.

7 Darcy White and Gary Van Landingham, "Benefit–Cost Analysis in the States: Status, Impact, and Challenges," *Journal of Benefit–Cost Analysis* 6(2) 2015, 369–99.

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11 We recommend Anthony E. Boardman, David H. Greenberg, Aidan R. Vining, and David L. Weimer, *Cost–Benefit Analysis: Concepts and Practice*, 4th ed. (Upper Saddle River, NJ: Pearson, 2011).

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13 See "Where We Work," www.pewtrusts.org/en/projects/pew-macarthur-results-first-initiative.

14 For an overview of meta-analysis techniques, see Mark W. Lipsey and David B. Wilson, *Practical Meta-Analysis* (Thousand Oaks, CA: Sage, 2001).

15 Elizabeth K. Drake, Steve Aos, and Marna G. Miller, "Evidence-Based Public Policy Options to Reduce Crime and Criminal Justice Costs: Implications in Washington State," *Victims and Offenders* 4(2) 2009, 170–96.

16 The program was initially called the Vocation, Education, Service, Training (VEST) Program and has evolved since it was established. Personal communication, Steven Noble, King County EET/Community Programs Supervisor.

17 Nicholas Kaldor, "Welfare Propositions of Economics and Interpersonal Comparisons of Utility," *Economic Journal* 49(195) 1939, 549–52; John R. Hicks, "The Valuation of the

Social Income," *Economica* 7(26) 1940, 105–24. The principle as stated by Hicks is: adopt a policy only if it would not be in the self-interest of those who will lose to bribe those who will gain not to adopt it.

18 Richard Posner, *The Economics of Justice* (Cambridge, MA: Harvard University Press, 1983) at 49. For a discussion of the same point from a very different perspective, see Steven Kelman, "Cost–Benefit Analysis: An Ethical Critique," *Regulation* 1981(January/February), 33–40.

19 Consistent with its initial application to infrastructure investments, most early treatments of CBA referred to its application to "projects." As its application expanded, the object of analysis became the program. We use the broadest term—"policy"—to encompass projects, programs, and other government interventions.

20 For a thorough treatment of the challenges of addressing uncertainty in policy analysis, and avoiding "incredible certitude," see Charles F. Manski, *Public Policy in an Uncertain World: Analysis and Decisions* (Cambridge, MA: Harvard University Press, 2013).

21 For an example in which the baseline scenarios differ by individual program participant, see David L. Weimer and Mark A. Sager, "Early Identification and Treatment of Alzheimer's Disease: Social and Fiscal Outcomes," *Alzheimer's & Dementia* 5(3) 2009, 215–26.

22 The term *standing* has its origins in the legal doctrine "standing to sue": plaintiffs have standing if they have a legally protectable and tangible interest at stake in the litigation. See *Black's Law Dictionary*, 5th ed. (St. Paul, MN: West, 1979), at 1260.

23 This terminology was introduced by Dale Whittington and Duncan MacRae, "The Issue of Standing in Cost–Benefit Analysis," *Journal of Policy Analysis and Management* 5(4) 1986, 665–82.

24 See William N. Trumbull, "Who Has Standing in Cost–Benefit Analysis?" *Journal of Policy Analysis and Management* 9(2) 1990, 201–18; and Richard O. Zerbe, "Comment: Does Benefit–Cost Analysis Stand Alone? Rights and Standing," *Journal of Policy Analysis and Management* 10(1) 1991, 96–105.

25 See, for example, David A. Long, Charles D. Mallar, and Craig V. D. Thornton, "Evaluating the Benefits and Costs of the Job Corps," *Journal of Policy Analysis and Management* 1(1) 1981, 55–76.

26 U.S. Food and Drug Administration, "Required Warnings for Cigarette Packages and Advertisements," *Federal Register* 75(218) 2010, 69523–65.

27 Frank J. Chaloupka, Jonathan Gruber, and Kenneth E. Warner, "Accounting for 'Lost Pleasure' in Cost–Benefit Analysis of Government Regulation: The Case of Food and Drug Administration's Proposed Cigarette Labeling Regulation," *Annals of Internal Medicine* 161(1) 2015, 64–65. For an alternative view that advocates that loss resulting from reductions is only the fraction of consumption that is non-addictive, see David L. Weimer, Aidan R. Vining, and Randall K. Thomas, "Cost–Benefit Analysis Involving Addictive Goods: Contingent Valuation to Estimate Willingness-to-Pay for Smoking Cessation," *Health Economics* 18(2) 2009, 181–202.

28 Sharon Begley and Toni Clarke, "Exclusive: U.S. To Roll Back 'Lost Pleasure' Approach on Health Rules," *Reuters* March 18, 2015.

29 See Boardman et al., *Cost–Benefit Analysis*, chapter 5.

30 Models that allow simultaneous analysis of related markets are called *computable general equilibrium* models. For an overview, see H. Allen Klaiber and V. Kerry Smith, "Developing General Equilibrium Benefit Analyses for Social Programs: An Introduction and Example," *Journal of Benefit–Cost Analysis* 3(2) 2012, 1–52.

31 An appropriate comparison of multiple alternatives to current policy predicts their impacts over the same time horizon. When the useful lives of the alternatives differ, present values should be converted to annualized net benefits as discussed under step 7. See Boardman et al., *Cost–Benefit Analysis*, chapter 6.

32 Internal validity is the primary focus of courses on research design and statistical inference. Policy analysts must also pay attention to external validity. For an overview, see Thomas D. Cook, "Generalizing Causal Knowledge in the Policy Sciences: External Validity as a Task of Both Multiattribute Representation and Multiattribute Extrapolation," *Journal of Policy Analysis and Management* 33(2) 2014, 527–36.

33 True experiments involve random assignment into treatment and control groups, as well as experimental control over the treatment. Natural experiments have fortuitously occurring random, or as "good as random," assignment into treatment and control groups, but without experimental control of the treatment. For a thorough treatment, see

Thad Dunning, *Natural Experiments in the Social Sciences: A Design-Based Approach* (New York: Cambridge University Press, 2012).

34 Jens Ludwig and Douglas L. Miller, "Does Head Start Improve Children's Life Chances? Evidence from a Regression Discontinuity Design," *Quarterly Journal of Economics* 122(1) 2007, 159–208. For a discussion of the use of Head Start evidence on CBA, see Jens Ludwig and Deborah A. Phillips, "Long-Term Effects of Head Start on Low-Income Children," *Annals of the New York Academy of Sciences* 1136(1) 2008, 257–68.

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81 This example is from Robert C. Lind, Kenneth J. Arrow, Gordon R. Corey, Partha Dasgupta, Amartya K. Sen, Thomas Stauffer, Joseph E. Stiglitz, and Jacob A. Stockfisch, *Discounting for Time and Risk in Energy Policy* (Washington, DC: Resources for the Future, 1982), 28–29.

82 Ibid., 32–33.

83 Randolph M. Lyon, “Federal Discount Policy, the Shadow Price of Capital, and Challenges for Reforms,” *Journal of Environmental Economics and Management* 18(2), Part 2, 1990, S29–S50.

84 Mark A. Moore, Anthony E. Boardman, and Aidan R. Vining, “More Appropriate Discounting: The Rate of Social Time Preference and the Value of the Social Discount Rate.” *Journal of Benefit–Cost Analysis* 4(1) 2013, 1–16.

85 Burgess and Zerbe, “Appropriate Discounting for Benefit–Cost Analysis”; and David F. Burgess and Richard O. Zerbe, “The Most Appropriate Discount Rate,” *Journal of Benefit–Cost Analysis* 4(3) 2013, 391–400.

86 Mark A. Moore, Anthony E. Boardman, and Aidan R. Vining, “The Choice of the Social Discount Rate and the Opportunity Cost of Public Funds,” *Journal of Benefit–Cost Analysis* 4(3) 2013, 401–409.

87 Moore et al., “‘Just Give Me a Number!’”

88 U.S. Office of Management and Budget, *Circular A-4, Regulatory Analysis*, 2003, September 17.

89 Moore et al., “More Appropriate Discounting.”

90 The real discount rate equals the nominal rate less the expected inflation rate when discounting is done continuously, rather than by discrete period. (Think of the distinction between annual and continuous compounding of interest.) In discounting by discrete period, we can separate the nominal discount factor $(1 + r)$ into the product of the real discount factor $(1 + d)$ and the constant dollar correction $(1 + i)$. From the expression $(1 + r) = (1 + d)(1 + i)$, we can solve for $d = (r - i)/(1 + i)$.

91 Of course, in practice, one would round the rate. However, to make the present value of net benefits in the example exactly equal for the real and nominal approaches, it is necessary to retain this many decimal places.

92 See Boardman et al., *Cost–Benefit Analysis*, chapter 6.

93 National Research Council, *Radiation Source Use and Replacement* (Washington, DC: National Academies Press, 2008).

94 Bethany Ackeret, David L. Weimer, Erik A. Ranheim, Lisa Urban, and Elizabeth Jacobs, "Net Benefits of Hospital-Sponsored Health Care for the Homeless: Cost–Benefit Analysis of a Demonstration Project," Robert M. La Follette School of Public Affairs Working Paper No. 2014-010, April 14, 2014.

95 Including the statistically insignificant estimate and its standard error is almost always better than assuming both are exactly zero. An even better approach is often to account for possible regression toward the mean by shrinking both the estimate and its standard error toward zero; see David L. Weimer, "The Thin Reed: Accommodating Weak Evidence for Critical Parameters in Cost–Benefit Analysis," *Risk Analysis* 35(6) 2015, 1101–13.

96 Mark W. Lipsey, "Those Confounded Moderators in Meta-Analysis: Good, Bad, and Ugly," *Annals of the American Academy of Political and Social Science*, 587 (May) 2003, 69–81.

97 For a full discussion, see Boardman et al., *Cost–Benefit Analysis*, chapter 18.

18
Public Agency Strategic Analysis

Identifying Opportunities for Increasing Social Value

Policy analyses typically respond to perceived problems. Although we have demonstrated how analytical skills are central to anticipating and exploiting future opportunities for increasing social value, the scarcity of analytical resources relative to the number of problems confronting society often diverts attention away from a comprehensive social value perspective. Elected representatives, who create the broad outlines of public policy, should ideally sponsor policy analyses to anticipate opportunities. But they face political pressures to respond to immediate problems that lead them to deploy analysts to more expedient purposes. Agency managers also face pressures to deal with those urgent problems that directly confront their agencies. However, they often have more certain job tenure, deeper substantive interests, and concerns about the sustainability of their agencies that make them more amenable to devoting resources to structured analysis to assess future opportunities and challenges. We label this kind of agency-oriented analysis as *public agency strategic analysis* (PASA). Can PASA be conducted so that it has the potential to increase the social value produced by agencies?

PASA can contribute to social value in two ways. First, it can enable agencies to use available resources more effectively, perhaps simply by reorganizing to lower the costs of delivering its services. Further, public agencies typically accumulate programs over time as they take responsibility for implementing new ones. Once implemented, the programs may operate indefinitely with minimal external oversight. Because the social value of these programs can change with changing circumstances, agencies may be able to reallocate resources among the programs in their portfolios to increase the social value they create. Sometimes agencies can exploit these opportunities for favorable reallocations within their existing mandates and authorizations; other times they may seek new or expanded mandate to do so, and perhaps even recommend the termination of ineffective programs within their purview.

Second, PASA can help agencies preserve or enhance their value as organizational resources for implementing public policies. Existing agencies can often implement new policies more quickly than could a newly created agency, especially when strong civil service rules make it difficult to hire personnel quickly. More fundamentally, existing agencies often have human and social capital that give them capacities that are difficult for other potential suppliers of the program to replicate from scratch. However, the value of particular configurations of human, social, and physical capital can change over time. Anticipating these changes can enable agencies to add personnel and routines to increase their future capacity.

In this sense, PASA is a form of policy analysis aimed at maximizing the instrumental value of an agency as an organizational resource. However, it has become increasingly common in the public management literature to label this form of analysis as *strategic analysis* rather than policy analysis, and to perform the analysis explicitly from the perspective of a specific public agency.[1] It fits within the purview of policy analysis as developed in the previous chapters provided it takes social value seriously and goes beyond simply developing "a valid livelihood scheme" for the agency based on an assessment of organizational resources and external constraints.[2]

Private firms routinely conduct strategic analyses in an effort to preserve rents they currently enjoy or to identify and exploit new opportunities for gaining rents in the future. The difference between maximizing private rents and maximizing social value make the appropriateness of the direct transfer of strategic analysis techniques from the private sector to public agencies inappropriate––the private purpose of exploiting barriers that restrict competition conflicts with the public purpose of promoting social value through competition when markets function efficiently, as explained in Chapter 4. Public and private organizations also differ in their legal and political environments. Consequently, effective PASA differs in important ways from private-sector strategic analysis, although we show that it can use some of the same ideas and techniques if they are suitably modified.

In this chapter, we present a brief overview of PASA. We begin by stressing the importance of explicitly considering social value in the particular context of the agency. Not surprisingly to those who have internalized the central message of earlier chapters, we see economic efficiency as an important component of social value. The tendency of agencies to focus exclusively on impacts relevant to their missions, and to those impacts occurring within their subnational jurisdictions, often results in important impacts and geographic spillovers being ignored—PASA can contribute to social value by at least acknowledging or even countering these tendencies. We then turn to two of the most important aspects of the PASA perspective. First, we consider how to assess the external environment of the agency, including the determinants of political influence, the importance of competition (if it exists), the threat of substitutes to the agency's services, and the extent of both supplier and client bargaining power. Second, we turn to the implications of the internal production processes for assessing value creation, distinguishing between three generic organizing metaphors: chain, modular, and platform technologies. Finally, we consider how the combined external and internal components of the analysis drive the formulation of *feasible* agency alternatives.

Clarifying Social (Public) Value

Policy analysis assesses and compares the social values that would result from the adoption of concrete policy alternatives. Rather than focusing on the policy alternatives that directly produce social value, PASA focuses on alternative strategies for developing and using organizational resources that affect the availability and value of policy alternatives in the future. For instance, an agency might conduct a policy analysis to choose among modes of delivering mental health services to the homeless. Its PASA would instead focus on changes in the availability of resources, such as the capacity of agency personnel to deliver mental health services directly, or to contract with vendors and evaluate the services vendors provide. The resulting agency strategy would change the mix of new hires or provide additional training so that, over time, the human capital of the organization supports policy alternatives expected to be desirable in the future.

Public administration scholars have broadly framed the appropriate objective of agency managers as *creating public value*.[3] Some of them share our view that, absent relevant social goals other than economic efficiency, the assessment of public value is best guided by cost–benefit analysis.[4] Other proposals for assessing public value, however, include achieving political mandates, meeting professional standards, and satisfying stakeholder interests or improving client satisfaction.[5] Each has some measure of legitimacy as proxies for social value.

Seeking to achieve political mandates harks back to an earlier conception of public administrators as passive recipients of policy, ignoring the practical necessity for them to make design choices during implementation. Achieving professional standards does contribute to social value to the extent that these standards bring competence or desirable norms to policy choice and implementation. It is, however, a minimal process threshold for assessing social value because organizational resources go well beyond meeting professional standards. Thus, this conception of public value is much too narrow. Maximizing client and stakeholder satisfaction may sometimes be consistent with public value, but either one is rarely completely synonymous with it. Indeed, the meaning of "stakeholder" is often more likely to be synonymous with concentrated interest groups that, as we discussed in Chapters 8 and 11, seek advantages inconsistent with social value. Stakeholder groups quite commonly value the securing of rents, for example by demanding protection from, or the erection of barriers to, competition. As we have discussed throughout the text, these rents are normally undesirable from the perspective of society as a whole. Even when viewed from a local public agency perspective, focusing on the impacts on (organized) stakeholder groups can diminish, rather than augment, social value. A focus on client satisfaction, in and of itself, is rarely problematic from a strategic perspective, but an exclusive focus on what only impacts clients is problematic. Agency clients nearly always prefer to pay less or nothing, even when doing so imposes costs on other residents of the jurisdiction, and whether or not doing so decreases social value.

If efficiency is the only relevant goal, then ideally a PASA would first identify all the possible policies that could be pursued with alternative configurations of organizational resources, and then conduct cost–benefit analyses of each of the possible future policies. Strategies could then be compared in terms of the net benefits, discounted for time and risk, that the policies they facilitate would produce. Unfortunately, one cannot imagine any agency ever having the analytical resources to attempt, let alone execute convincingly, such an analytic approach. PASA requires a more practical approach.

Analogous to private-sector strategy analysis, a feasible PASA assesses both the external and internal resources (often called for symmetry the internal environment or more generally social or public value) of an agency. Together, they inform decisions about the deployment of organizational capital that creates the potential for increasing social value. However, the techniques of private firm strategic analysis cannot be directly transferred to PASA for two reasons. First, unlike the relatively clear bottom line of maximizing the present value of the firm, PASA must recognize that social value is multifaceted. Even when economic efficiency can be taken as the appropriate basis for assessing social value, predicting net present value, even qualitatively, is demanding and difficult work. This estimation becomes even more difficult and less certain when distributional impacts are relevant to social value. Second, the external environments of firms and agencies differ greatly. Indeed, these environments often vary markedly across programs within an agency: one program may face no competition or even a threat of it, while another may have numerous nonprofit and for-profit competitors.

One of the most important differences between PASA and private-sector strategic analysis is that political influence plays a more central, and sometimes more legitimate, role in the external environments of most agencies. Of course, political influence can also be a very important external force for firms and industries that are directly regulated. Nonetheless, in spite of some important differences, the value of a general assessment of the external environment is common to both agencies and firms, and some features of the assessment are very similar.

The External Forces on an Agency

Systematic consideration of the external economic and political environment is a key component of almost all kinds of strategic analysis.[6] Lack of fit between agency resources and the opportunities or constraints imposed by the external environment (and resulting poor strategies) has been identified as one important reason for strategic failure among public agencies.[7] The empirical evidence also suggests that strategic attention to an identified lack of fit can help to improve public agency performance on a number of dimensions.[8]

The approach used here to analyze the external environment derives from the industrial organization perspective, which also underlies the "five competitive forces" framework used extensively in firm (specifically, business unit) external analysis.[9] There are, however, three important differences between PASA external analysis and competitive forces analysis. First, many public agency environments are characterized by a *lack* of competition, and consequently a greater importance of other factors. So, in many cases it is more accurate to describe the relevant influences as "external forces" rather than as "competitive forces." Second, public agency external analysis differs because of the quite different goals of the private and public sectors: profit maximization versus economic efficiency and social value more generally. Third, the public analyst has to recognize that political influence is an important external force, and to incorporate a consideration of the factors that determine the barriers to political influence across public agencies and programs.

Figure 18.1 summarizes the relevant public agency forces. It retains the competitive forces convention of placing the extent of competition (rivalry) in the central box, even though it may not be the most important force in the public agency context. In the public agency context, it is useful to think of the agency itself as being located in the central box and has all five forces impinging on it. However, where the agency is in a monopoly position (including facing contestability, because there would still only be a single agency), the agency usually has considerable reciprocal bargaining power or ability to respond strategically to threats. In other cases, the agency has some capacity to negotiate and propose (threaten!) some strategic response; we acknowledge this reality by showing bargaining power as flowing bidirectionally (that is, the arrows point both ways). We show the force arrows flowing in as solid and those flowing out as dotted, but to identify the degree of agency power requires specific strategic analysis. We do not show the threat of entry and substitutes as a bidirectional force because it is very unlikely that a public agency has any real power or influence over this force (the emergence of a substitute technology will generally be exogenous to any strategic behavior by the agency). We do show political influence as a bidirectional force because we argue below that some agencies *can* take strategic actions that would affect the political influence to which they are subject.

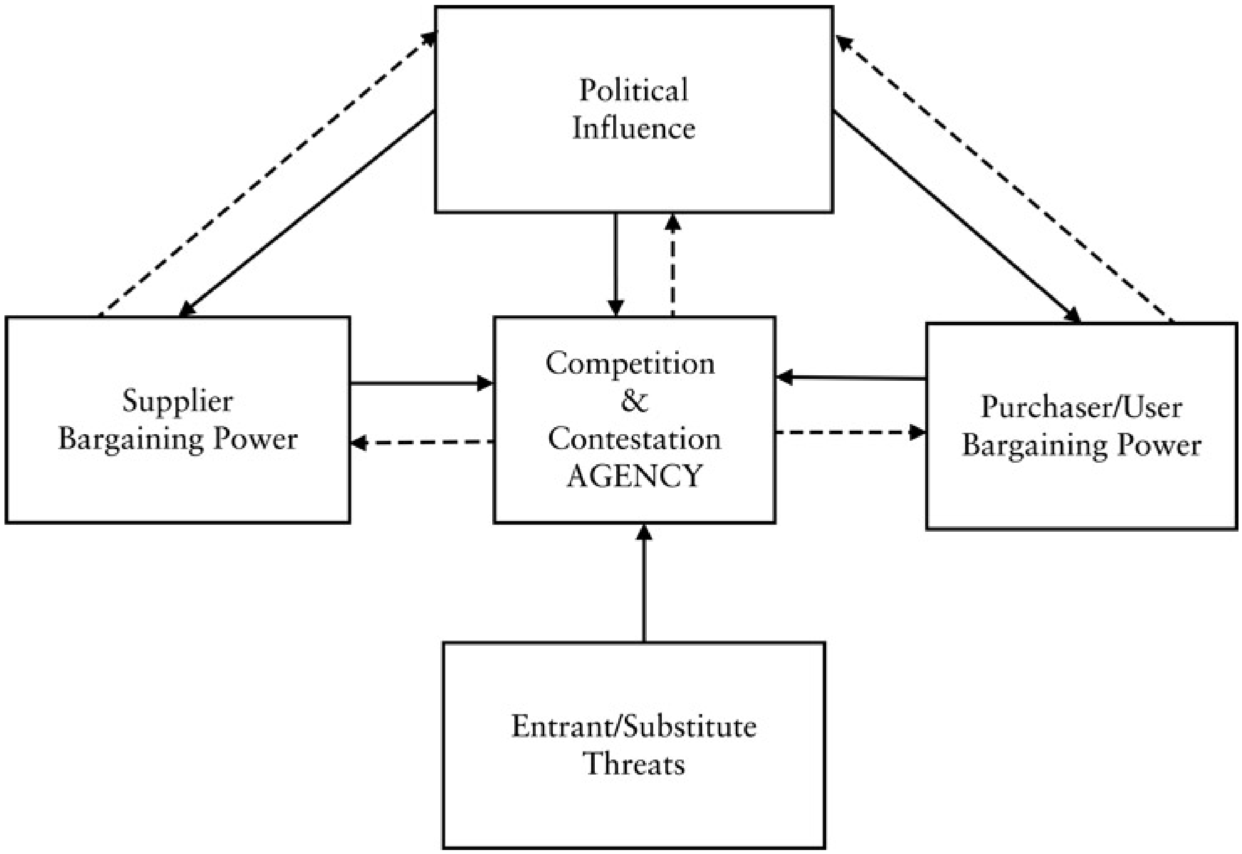

Figure 18.1 PASA External Analysis: Five Forces that Shape the Strategic Situation

Source: Adapted from Aidan R. Vining, "Public Agency External Analysis Using a Modified 'Five Forces' Framework," *International Public Management Journal* 14(1) 2011, 63–105.

Before considering those forces that differ significantly, though, we review those forces that are analytically quite similar across the sectors: supplier bargaining power, the threat of either substitutes or newly competing entities (new entrants), and "customer" bargaining power.

Supplier Bargaining Power

Supplier bargaining power is directly relevant to public agency programs. It is relevant because public agency suppliers have, and try to exercise, bargaining power to their advantage: "Powerful suppliers capture more of the value for themselves by charging higher prices, limiting quality or services, or shifting costs to industry participants."[10] Public executives face numerous input suppliers with whom they must bargain. The concept of "power" and "powerful" in the public agency context inevitably encompasses both economic and political dimensions. The strength of the economic bargaining power of suppliers matters because nearly all public agencies purchase most inputs, including often labor, land, and equipment, from profit-maximizing supplier markets. It is a strategic issue because evidence suggests that at least some public agencies bargain poorly with suppliers over input prices and consequently pay too-high prices.[11] A program-specific analysis of supplier bargaining power is necessary because supplier markets (and the resulting bargaining power of participants in the supply chain) vary widely in their competitiveness. Assessing the economic bargaining power of suppliers is usually relatively straightforward because most agency transactions with suppliers are monetized, whether in the form of wages and salaries, computer and equipment purchases, or contracted services. As a result, the (social) opportunity cost of these inputs generally can be estimated relatively easily from their market prices.

Yet, some supplier transactions may not be monetized. A separate government entity, for example, may allocate land and buildings to the agency by fiat, and without rent or formal accounting of the opportunity cost of these resources. In these cases, bargaining is over the quality and quantity of inputs rather than their price and quality. Because nonmonetized inputs are typically more common in the public agency context, the nature of the bargaining is more multidimensional and complex than in most private-sector markets. Most importantly, in the absence of monetization, bargaining has political as well as economic dimensions, and it is more difficult to figure out the distribution of bargaining power. In view of these realities, to get the best input prices, agencies have to put thought and effort into elements of their strategies that would reduce the bargaining power of suppliers. Centralizing purchasing across agencies may counter supplier advantage, but it risks making purchasing less flexible.

Understanding supplier bargaining power can enable agencies to increase social value. Where agencies do purchase inputs in competitive (and fully monetized) supplier markets, an understanding and effective use of bargaining power can help ensure that the agency extracts the best (lowest) price from suppliers. Where agencies purchase in concentrated supplier markets, agencies should consider strategies that over time might offset the effects of concentration, for example by seeking exemptions from government-wide “buy domestic” rules.

Threat of Substitutes/Entry Barriers

Many public agencies are subject both to the emergence of substitute outputs and to the emergence of direct competition. Substitution is often the greatest threat to public agencies because, although governments and their regulatory agencies can anticipate conventional entry and consider whether they should seek to forestall it by law or regulation, they usually cannot anticipate and forestall radical technologically-based substitution threats. If realized, such threats can erode or even destroy the viability of a public program. Radical substitute technologies, for example, potentially allow current clients to bypass an agency through “self-service.” In many technologically dynamic markets, distinguishing between substitutes and new entrants is analytically irrelevant. Public library systems, for example, have experienced a multipronged entry/substitutes threat, with the rapid diffusion of the Internet, the World Wide Web, digitization, and electronic publishing.

In an increasing number of cases, even conventional entry is occurring as governments remove or lower legal barriers to the provision of services. For example, U.S. states have removed a number of the barriers that have historically given school districts monopolies over the provision of publicly funded primary and secondary education. In other cases, extant legal barriers may continue, but international agreements may limit their longer-term viability. Of course, not all public agencies have an explicit legal and enforceable monopoly. Entry may simply have not been viable under the previously existing cost-and-demand regime (for example, production may have had natural monopoly characteristics). With the advent of new cost-lowering technology, a threat of entry becomes real. Entry, and even the threat of entry (a form of contestability), can seriously erode strategic autonomy.

Anticipating substitution threats can help agencies decide whether they should make investments to prepare to fight to remain a provider, flee the market to release resources for other uses, or adopt some other strategy. For example, city agencies with a mission to make parks available to residents may face competition from grassroots environmental organizations. The agencies may decide to invest in additional staff to maintain their monopoly over the use of public spaces (fight), delegate development of pocket parks to local organizations (flee), or invest in community outreach to involve neighborhood organizations in the implementation of agency plans (co-optation).

Purchaser and User Bargaining Power

In private-sector strategic analysis, the degree of customer (or buyer) bargaining power is obviously crucial: "Powerful customers—the flip side of powerful suppliers—can capture more value by forcing down prices, demanding better quality or more service (thereby driving up costs), and generally playing industry participants off against one another, all at the expense of industry profitability."[12] This force is equally crucial in PASA, although the interests of the agency and the "customer" are not always in conflict. Analysis of this bargaining power is complicated because there are frequently two distinct types of customers in the public sector.[13] The first are those who use the product or service, and the second are those who make the actual purchase decision (users themselves or third-party payers). Additionally, because of this customer bifurcation, there can be a high degree of variability in the extent to which public agencies charge users for any given service.

In the public sector, customers (whether users or purchasers) come in many different guises. These labels vary depending on the agency, program, or service. For purchasers, these labels include: citizens, taxpayers, residents, "sponsors," or (generically) purchasers. Users may be called clients, patients, inmates, respondents, or what makes sense in the particular program context. In most private-sector contexts, users make purchasing decisions, although in markets with significant asymmetric information, purchasers often rely on agents.[14] For some public agencies, the distinction between users and purchasers is stark and the differences in the interests of the two usually presents a strategic challenge, especially if agency managers take efficiency seriously. In the most extreme form of bifurcation between users and purchasers, users of the service (say prison inmates) do not participate in the purchase decision at all. Although sometimes it is useful to lump public agency users and purchasers together as consumers for semantic simplicity, it is almost always a bad idea to do so for analytic and strategic purposes.

Often in the presence of complete user–purchaser bifurcation, the strategic reality is that purchasers are more important to the agency than are users.[15] In most circumstances, purchasing authority confers greater direct bargaining power than does use along the lines of, "He who pays the piper picks the tune." The bargaining power of non-purchasing users is typically indirect and depends on a number of factors: their motivation to interact with agency managers, the nature and extent of their resources, cohesion, and organizational skills, and their ability to influence the political forces relevant to the agency. For programs in which users do not value the service on all the same dimensions as purchasers, they only have the motivation to try to exercise bargaining power on those dimensions that they do value. For example, drivers convicted of DUI offenses participating in a detoxification program may not value the rehabilitative quality of their treatment, although they are likely to value the cleanliness of the treatment facility. In contrast, purchasers (such as a department of health) are likely to value rehabilitation highly, but may be prepared to sacrifice high levels of cleanliness to keep down costs. In practice, the overlap of users and purchasers varies widely on a program-by-program basis.[16]

Purchasers have differing interests and, therefore, differing focus in what they are motivated to bargain about. In some cases, these purchasers are central budget control agencies. In other cases, purchasers are other government line agencies. So, it is not surprising that the leverage points in these bargaining relationships vary widely. Budgetary and fiscal control agencies are typically primarily concerned with limiting or reducing their total payments to the agency, and are not as interested in bargaining about altering per unit prices, missions, or policies. Line-agency purchasers, on the other hand, often seek to influence program policies. To the extent that particular purchasers are concerned with the agency's strategies, their concerns may mirror those of users or represent an independent agenda. For the agency, an assessment of the bargaining power of purchasers is strategically more complex in the presence of multiple purchasers with differing demands and differing demand elasticities. It is also more complex because, beyond purchasers and users, various political actors may be interested in various dimensions of services. A specific legislative committee with a strong interest in the quality of service provided to some kinds of users, for example, often has greater bargaining leverage than do purchasers.

Programs vary widely in the degree to which users pay the cost of the services they receive. Road usage, for example, demonstrates this variability. Most road users pay nothing directly. Toll road users, though, may pay above or below the social marginal cost of use (the social efficiency criterion). So, users may either contribute none, some, most, or all of a regional transportation agency's total revenue. Full-cost (or more) pricing is consequently also highly variable: it is quite common for state-owned enterprises (SOEs), corporatized entities, and for the services of special districts in the United States. However, where users do pay a significant percentage of service costs, they often do so through, or with the advice of, agents. These agents typically have stronger bargaining power than would the individual purchasers they represent, because of greater concentration and less information asymmetry (including more knowledge of other government agencies or nonprofits that could supply the equivalent service or provide a viable substitute).

Where users contribute a high percentage of revenue to a program's expenditures and the agency faces competition, even individual user-purchasers have some bargaining power. Nonpaying users may still try to influence both the agency and purchasers by lobbying their political representatives. Astute and organized users quickly recognize when their service dimension preferences do not match those of purchasers and they can exercise greater indirect bargaining power through lobbying elected or politically appointed officials. In general, though, individual users have only weak incentives to invest effort in interacting with agency service providers, except with respect to those specific quality dimensions that matter to them. This lowers the bargaining power of individual users, even when they are numerous. The emergence of social media has lowered users' cost of organizing. However, organizing costs are less debilitating in political markets than in bureaucratic contexts. In political markets, influence effort can involve little individual disutility. As Aidan Vining notes, "[I]t is usually more fun to harangue vote-seeking politicians than to complain to stonewalling bureaucrats."[17]

Understanding the bargaining power of those who use agency services can promote social value by helping agency managers to recognize that bargaining might is not (social) right. Note that a stakeholder perspective can actually feed this misperception! Again, agency managers may have to develop strategies to mitigate or offset user or purchaser bargaining power that diminishes social value.

Extent of Actual Competition or the Threat of Contestability

In competitive forces analysis, the intensity of rivalry is usually the most important factor. In PASA, competition is quite often *not* the most important force, most obviously when an agency has a de jure or de facto monopoly. Many public programs deliver pure public goods that would not be supplied at optimal levels by the market. Further, rivalrous services often have declining average costs that make them potentially natural monopolies. In the presence of natural monopoly, it is most efficient to have only one supply source. The strategic problem from an agency's perspective is that it could be some other organization! In other words, the *contestability* of the monopoly then becomes a crucial issue.[18] Sometimes the program's monopoly status is secure and competition is an unimportant force. In other circumstances, however, contestability is a credible and serious threat and deserves the "central box" role (see Figure 18.1), even though there is no threat of direct competition. Various political influencers, for example, may prefer another agency to deliver the service, and this external force largely drives a high threat of contestability.

Many programs, however, do not have an unassailable monopoly, simply because they do not have natural monopoly or have public-good characteristics so that it is feasible to have multiple suppliers. Structural cost factors alone do not foreclose competition that might enhance efficiency from not-for-profit or even for-profit competitors. The United Kingdom, for example, mandates compulsory competitive tendering (CCT) for many public services (especially at the local level). This can result in direct competition or spatially contiguous contestability that essentially amounts to the same thing.[19] Governments in some countries at least encourage competition between public organizations.[20] In other sectors, such as nursing home provision, governments may allow or encourage even more extensive competition between public-sector agencies, for-profit firms, and not-for-profits.

One reason that many public agencies, especially corporatized ones, face increased competition is that they are increasingly being required to charge prices for their services that meet or exceed the cost of provision. If agency prices are forced to approximate the marginal costs of delivery, then agencies usually face a greater threat of competition, whether from for-profit firms or nonprofits.[21] Increased competition can also result from technological change that reduces minimum efficient scale and erodes entry barriers, or from broader socioeconomic change that alters demand conditions. For example, public schools face increasing competition from private schools because of changing preferences, wealth increases, and a variety of other factors. Competition may or may not promote social value,[22] but it is always strategically important because it reduces the feasible alternatives available to an agency. Multiple competitors provide alternative supply sources for both purchasers and users; their presence increases the bargaining power of at least purchasers. Furthermore, competition usually provides relevant political institutions with information about the agency's behavior, output, and performance. This gives these political institutions greater influence and leverage.

The presence of contestability and the threat of replacement usually affects agency strategy preemptively. Many subnational government agencies face quite credible threats of replacement by alternate agencies.[23] In some circumstances, customers may have quite low relocation costs and mobility functions as a form of contestability.[24]

In sum, both competitive and contestable environments limit the autonomy of agencies and affect strategies—for either (social) good or evil. An agency with a monopoly and low contestability has the opportunity to pursue social value, including the pursuit of goals beyond economic efficiency. For instance, a publicly owned utility with a local monopoly may be able to use rents to subsidize supply to low-income residents, or to serve low-density areas where marginal cost pricing would make service prohibitively expensive. Such cross-subsidization to promote distributional goals is likely to be less feasible when an environment is competitive or contestable.

Political Influence

Political influence should be viewed as a distinct force in PASA external analysis. This distinguishes PASA from business unit analysis.[25] Two factors largely determine the aggregate extent and importance of the political influence force. The first factor is the institutional structure surrounding the agency's establishment (essentially, the formal legal and administrative structure that lays out the degree of insulation of the agency). Some agencies are established with high degrees of protection from many forms of political influence. For example, civil service rules may limit political influence over hiring, and authorizing legislation may limit special pleading for constituents by clearly specifying who can receive agency services. The political sponsors of programs or new agencies may purposely create barriers to political influence to protect them from changes in the political environment.[26] Especially in stable and rule-bound institutional environments, such formal barriers insulate agencies to a greater or lesser extent from overt political influence on a continuing basis. However, potential influencers may be able to circumvent these formal barriers, and thus the second factor that requires analysis is the degree to which these formal barriers are permeable in practice.

Many different legislative and executive branch entities or individuals may seek to influence the policies and budgets of public agencies within their jurisdiction. Politicians, at various jurisdictional levels, have a wide range of indirect and informal methods to try to permeate, circumvent, or simply overpower the formal barriers to their influence.[27] There is evidence, for example, that U.S. state-level politicians exercise considerable influence over a number of federal agencies, even though they have *no* formal authority over them.[28]

As with other external forces, agencies can preemptively engage in strategies that blunt this influence, unless they perceive it as reinforcing their preferred strategies. Agency managers can only implement these strategies if they can exercise some reciprocal influence or bargaining power. Politically salient program outputs tend to attract attention that undercuts insulation. Of course, saliency may be intermittent and variable over time. For example, emergency response services are likely to be more salient when disasters threaten or strike: their saliency has risen consistently since 9/11. Salience may also rise when users can effectively lobby or otherwise organize. Another factor that affects the level of influence is the scientific, technical, or professional complexity of the agency's task mandate. A high degree of complexity tends to reduce permeability and discourage political influencers.[29]

Political influence can directly affect the level of competition or contestability. Political institutions may either foster or block programs that provide competing services, impose or lift barriers to private-sector supply, or change the availability of critical resources. Understanding the political environment is important for all organizations; it is especially important for agencies that rely on politically granted budgets and authorities. Strategic analysis helps agencies assess not only the direct effects of political influence on competition and contestability, but also its indirect effects on the bargaining power of suppliers and purchasers/users.

Internal Analysis: The Value Creation Process

Conceptual frameworks for performing internal analysis of private-sector organizations are well developed.[30] However, as is the case with external analysis, many of these frameworks do not transfer well to public agencies without modification. Again, this lack of transferability fundamentally relates to goals. A primary focus of most internal analyses of private firms is on how to increase profit through more-efficient internal processes. The VRIO model (resources that are valuable, rare, and imperfectly imitable, and that the organization can capture), for example, focuses on the more efficient use of assets to create rents for owners.[31] Although these models offer many useful insights that can be applied to public-sector agencies, the dominant focus on rent extraction generally conflicts with the social value perspective.[32] The value chain model and related models that disaggregate this process into activities also offer many useful insights to structure thinking about the public agency value creation process. Again, though, these components of the value creation process must be adjusted to the goals and strategic needs of public agencies.

An emerging literature explicitly addresses the internal analysis of public organizations.[33] For strategic purposes, the process of *categorizing and assessing assets* can be conceptualized as performing an inventory analysis, also referred to as "stock," "hardware," or "balance sheet" analysis. The complementary task of assessing the *value creation process* is best conceptualized as a "flow," "software," or (social) "income statement" analysis. For many reasons, an analysis of public agency social value creation is more analytically challenging and time consuming than merely doing an inventory. Many agencies deploy unique bundles of evolving knowledge and problem-solving expertise.[34] As a result, agencies cannot simply engage in mimetic behavior with respect to trying to improve their value creation process, as they possibly could by observing asset inventories in other organizations. (This is not to say that considering learning from "best practice" is wrong; however, it is rarely enough.) Nonetheless, understanding the social value creation process is central to good PASA because it is the foundation for assessing the need for normatively based strategic change in public agencies.

For some public programs, the value creation (or destruction) process can often be best disaggregated and analyzed by thinking of it as a sequential chain of steps. Such processes involve so-called long-linked technologies,[35] often referred to in the business strategy literature as *value chains*.[36] This metaphor often applies when the production process has a clear temporal sequencing of activities. For example, many surgical procedures can be straightforwardly modeled as a chain. Patients enter the system via a number of preparatory administrative and medical steps ("inbound logistics"), then are treated through a number of sequential operating theater steps such as sedation and surgery ("operations"), and then exit the hospital through another series of "post-op" administrative and medical steps, such as recovery and discharge ("outbound logistics"). The division of the chain into inbound logistics, operations, and outbound logistics represents the simplest disaggregation of the value chain into three activity steps.

Two other kinds of technological processes may better model value creation for other programs, especially those with important service or intangible elements. These processes have been subsequently adapted for strategic management purposes.[37] Where appropriate, the value creation process can be analyzed using either the value chain, a modular assembly process (the "job shop") that varies somewhat for each assembly iteration, or a platform or intermediation production (the "value network") process, or some combination of the three. However, although all of these value creation models are used in private-sector analysis, the latter two are less frequently employed in the assessment of production within public agencies, although that is changing.[38]

In view of the uniqueness and complexity of some public agency production, it is also quite possible that additional production metaphors would be useful in specific agency contexts. For example, some agencies primarily create social value by overseeing and regulating third-party organizations that deliver services. This regulatory production process does not fall neatly within the chain, modular assembly, or platform production class.

The chain metaphor works best with sequential and linear activity processes that are divisible into inbound logistics, production, and outbound logistics activities. The modular assembly metaphor is based on the idea of the auto repair shop where each "repair" problem is unique, but many of the same modular elements can be used to address the specific production problem. That is, although each production problem is unique, solutions can be entirely or mostly built from standard analytic components and assembly heuristics. For example, an agency responsible for disaster response might have routines for rescue and recovery that vary by type of disaster—any particular disaster would elicit combinations of rehearsed responses.

The use of the platform metaphor places the producing agency at the nexus of some exchange process (such as between the supply and the demand for some good or service) where it acts as an intermediary or broker. In the public agency context, intermediation is most likely to be socially valuable when the desired exchange would not sustain market-like prices (that is, there is some market failure) or price-based exchange is not feasible or disallowed on equity grounds.[39] Thus, the analysis focuses on the set of activities required to analyze the platform for the intermediation value creation process.[40] Consequently, the analysis is potentially concerned with providing the platform, reconfiguring an existing one to better serve the needs of the intermediate agency or the exchangers, or eliminating it where technological or other changes have ameliorated the market failure.

The platform production process is different from either the chain or modular assembly process because by its nature the public agency platform must provide value to both sides of the platform, so that each side has an incentive to voluntarily join and actively use the platform. For example, a platform that matches housebound elderly with volunteer caregivers must do so in a way that provides value to both parties, say, services to the elderly and "warm glow" to the volunteers.

Assessing Value Creation: Either Current or Under Feasible Alternatives

The analysis of value creation is difficult because it requires both an analysis of current value creation and the potential for greater value creation when considering feasible alternatives. The former (current value creation) is considered at this stage of the strategic analysis, while the latter (potential value creation) under the alternatives is considered in the next section.

The first way of thinking about value creation in the public agency focuses on the cost of an activity and assessing the potential for cost reduction, holding the current level of output constant. Although this cost focus can be criticized as being reductionist, it has the advantage of usually being assessable. The minimalist justification for activity–cost analysis from a social efficiency perspective is that cost reduction is rarely normatively inappropriate if output is really held constant.[41] The ideal objective is cost minimization, but as there is seldom a convincingly knowable minimum cost, cost reduction is often a practical proxy. Sometimes, cost reduction can be analyzed in terms of simply eliminating an unnecessary activity step (a link in a chain or a modular element) in an existing service. More frequently, however, it means reconfiguring the current activity steps: that is, some kind of "business process reengineering" (BPR) as applied to public agencies.[42] For example, the introduction of electronic communication systems may facilitate BPR cost reductions, ranging from large reductions from moving to paperless offices to small reductions from speeding up communication among employees.

The second way to assess value creation is to observe quality or quantity changes (more generally, social benefits changes) while holding cost constant. This is often most plausible and convincing to third parties when these benefit changes can be monetized, or at least quantified. For instance, a parole department may identify one of its programs as being more effective than others for a particular type of parolee. Shuffling parolees among programs to take advantage of the greater effectiveness may allow more beneficial outcomes without the investment of additional resources.

Third, an analysis could assess value creation within each agency program, holding neither costs nor outputs constant. Ideally, with sufficient information and analytical resources, cost–benefit analysis would be applied to each ongoing program in order to identify those producing the largest social net benefits. Those with negative net social benefits and no offsetting distributional values would be reduced, eliminated, or investigated in greater depth to look for opportunities for improving their value creation processes. Limited data and analytical resources often—indeed, usually—make monetization of all impacts and resources impractical. Switching to qualitative cost–benefit analysis, of the sort discussed in Chapter 17, typically provides useful information for helping managers identify the weakest programs so that analytical resources can be focused on their improvement if not replacement.

Feasible, Rather than Ideal, Alternatives

A central role of those engaged in policy research is the presentation and discussion of superior policy alternatives, including those that require additional resources. Public managers engaged in PASA, however, are, and must be, more concerned with alternatives that are feasible within the constraints of existing agency resources and external environmental constraints. Of course, not all constraints are necessarily immutable: it may be possible, for example, to consider some alternatives that are only feasible with some greater degree of user pay.[43] Programs and agencies vary tremendously in their strategic degrees of freedom, and, therefore, in their ability to change strategies. What determines agency autonomy? Simplifying somewhat, autonomy is a function of the combined absence of policy and fiscal constraints.[44]

The analysis of external forces provides the basis for not only assessing the current constraints on agency autonomy, but also predicting how they are likely to change in the future. The assessment of current constraints helps determine the feasible alternatives for changing the value chain. Anticipating changes in the constraints helps identify alternatives that may become feasible in the future. For example, although the number of people without Internet access may still be too large to make a switch to fully electronic tax filing politically feasible, anticipating that this number will shrink quickly may make it a feasible alternative in the near future. Further, the time at which it would become feasible may be speeded up by an investment in the capability of the agency, either to assist non-Internet users directly or to contract with vendors to do so reliably. Indeed, the identification of constraints and current investments to change them may be the largest payoff from PASA.

Conclusion

Que PASA? Strategic analysis to increase the social value of public agencies is conceptually and practically difficult. It requires all the skills of policy analysis, as well as the capacity and willingness to identify future opportunities and challenges, envision appropriate policy responses, and predict and value their likely outcomes. Because it is much broader in scope than policy analysis, it cannot be easily set out as a series of steps that can be iteratively applied to reach a final product. Rather, it requires managers to assess the external and internal environments of their agencies with an eye not only to making better use of existing resources, but also to investing in resources that will create future opportunities to increase social value.

For Discussion

1. In 1972 the U.S. Congress created the Office of Technology Assessment (OTA) to provide advice to Congress about complex scientific and technical issues. By 1995, when it was defunded by Congress, it had grown to have a staff of over 140. Imagine that in 1990 you were asked to apply PASA to the OTA. What factors would you consider in assessing the challenges it would face over the next decade?
2. Consider a municipal sanitation department that collects and disposes of all household refuse in a small city. What factors would you consider in doing PASA for the department?

Notes

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Part V
Conclusion

19
Doing Well and Doing Good

The preceding chapters set out what we believe to be the important elements of an introduction to policy analysis. As a faithful reader, you should now be ready to begin to produce your own analyses with some success. As you gain practical experience, we hope you will find it useful to return to some of the chapters for reference and reinterpretation. We conclude with a few thoughts about how you should view your role as a policy analyst.

Our first exhortation: do well! Keep your client in mind. Policy analysts earn their keep by giving advice. Always strive to give useful advice by keeping in mind the range of actions available to your client. But, like a doctor, one of your guiding principles should be: do no harm. This demands skepticism and modesty. Be skeptical about information provided by others and be modest about the accuracy of your own predictions, so as to avoid leading your client into costly or embarrassing mistakes. As a professional advice giver, you should be prepared to allow your client to take credit for your popular ideas while keeping the blame for unpopular ones for yourself. Beyond the satisfaction of knowing that you have done a job well, you are likely to be rewarded with the trust of your client and the opportunity to continue to be heard.

Yet, as we tried to make clear in Chapter 3, your duties as a policy analyst go beyond responsibility for the personal success of your client. You have responsibilities to the integrity of your craft and to contributing to a better society. Thus, our second exhortation: do good! Doing good requires some basis for comparing alternative courses of action. At the most general level, promoting human dignity and freedom are values widely shared in Western societies. Unfortunately, these values sometimes conflict, and they often do not provide direct guidance for dealing with the majority of issues that most analysts confront in their day-to-day work. As a practical matter, analysts must employ less abstract values in their search for good policy.

Throughout this book we have emphasized economic efficiency as an important policy goal, which often serves as a valid proxy for social, or public, value. Aside from the intuitive appeal of efficiency—if we can make someone better off without making anyone else worse off, then shouldn't we do so?—we believe that it often receives too little attention in political arenas because it generally lacks an organized constituency. As a society, we might very well be willing to sacrifice considerable efficiency to achieve distributional or other goals, but we should do so knowingly. Introducing efficiency as a goal in the evaluation of policies is one way that policy analysts can contribute to the "public good" (for once not used in the technical sense!).

Beyond raising efficiency as a goal, you can contribute to the public good by identifying other values that receive too little attention in political arenas. Are there identifiable groups that consistently suffer losses from public policies? Is adequate consideration being given to the future, and to the interests of future generations? Are dangerous precedents being set? Your social responsibility as an analyst is to make sure that questions such as these get appropriately raised.

Is raising politically underrepresented values consistent with providing useful, presumably politically feasible, advice? Unfortunately, the answer is often no: self-interest often dominates. Yet at other times people, including politicians, respond to analysis in a public spirit, putting aside their narrow personal and political interests for the greater good.[1] Also, ideas that are not immediately politically feasible may become so in the future if they are repeatedly raised and supported with substantively sound analysis; witness the role of economic analysis in setting the stage for the eventual deregulation of the U.S. telephone, trucking, and airline industries.[2] Sometimes doing good simply requires analysts to advise their clients to forgo some current popularity or success to achieve some important value. You are doing exceptionally well when you convince your client to accept such advice!

Notes

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